

FRANK J. REGAN

Re-Entry Vehicle Dynamics



**AIAA
Education Series**

J.S. PRZEMIENIECKI / SERIES EDITOR-IN-CHIEF



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FRANK J. REGAN

Re-Entry Vehicle Dynamics

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Frank J. Regan

Naval Surface Weapons Center

White Oak Laboratory

Silver Spring, Maryland

AIAA EDUCATION SERIES

J.S. Przemieniecki

Series Editor-in-Chief

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Foreword

This book, *Re-Entry Vehicle Dynamics*, by Frank J. Regan, inaugurates the AIAA Education Series.

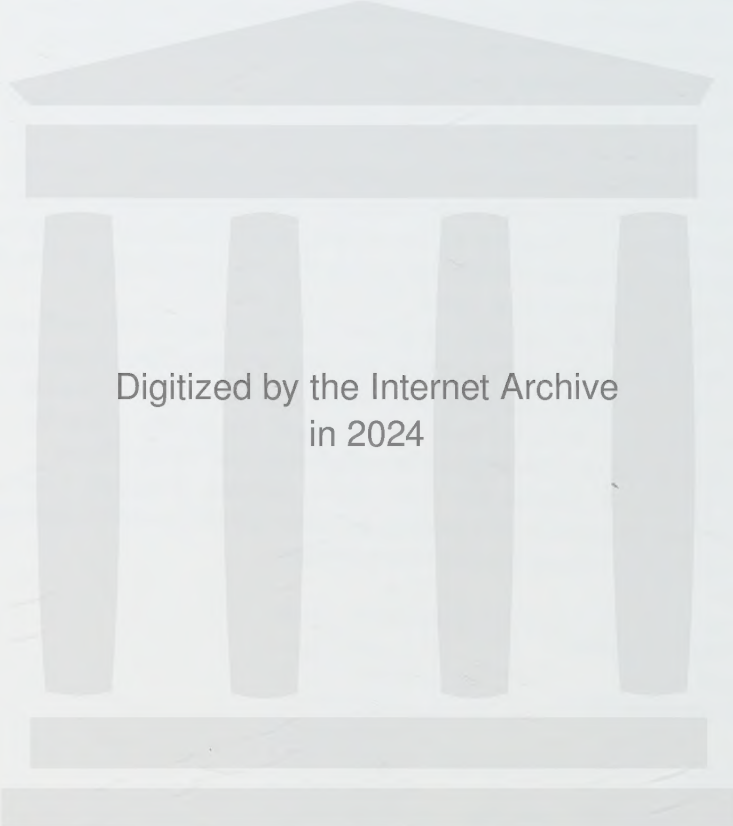
The Series represents the Institute's response to a need for textbooks and monographs in highly specialized disciplines of aeronautics and astronautics. The Institute's Publications Committee identified this need in a study spanning nearly three years.

Books in the AIAA Education Series will be designed for students and researchers at colleges and universities and for scientists and engineers in aerospace industries and government agencies. Thus a typical book in the Series will present subject material tutorially, discussing fundamental principles and concepts, and additionally will give perspective on the state-of-the-art; it can be used as a textbook or as a supplement to professional work.

That last point deserves emphasis. Learning new fields and new skills in engineering science and technology will be a lifelong activity for most aerospace scientists and engineers. The AIAA Education Series will serve well, we believe, this quest for greater competence. This present text by Dr. Regan sets a high standard for the Series.

The AIAA thanks the U.S. Naval Surface Weapons Center, White Oak Laboratory, for helping to make this text possible through page charges covering the cost of typesetting.

J.S. PRZEMIENIECKI
Editor-in-Chief,
AIAA Education Series



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Preface

An unidentified academic once defined aerospace engineering as that branch of engineering most concerned with problems that occur when a fluid meets a solid. Now a quick perusal of an engineering periodical such as AIAA's *Aerospace America* would indicate that the practicing engineer has a much broader range of professional interests than might be suggested by such a definition. Nevertheless, the undergraduate curricula of the 50 or so U.S. aerospace engineering schools, indicate some preoccupation with pressure and thermal loads at the surface of various aerospace vehicles.

At the Naval Surface Weapons Center there is great interest in the pressure and thermal environments of unmanned re-entry vehicles. Perhaps in this intensity of effort there is the remote possibility that fluid dynamicists might have a lack of appreciation or even awareness of the requirements of the flight mechanician. To prevent the development of such a condition, it was decided to offer a short in-house course to provide the nonspecialist some perspective on the methods and concerns of the re-entry vehicle aeromechanician. The present book is an extension of these course notes.

The purpose of the notes and book remains about the same, namely to provide the practicing engineer with a senior-level introduction to the dynamics of re-entry vehicles. No claims of originality are made or intended for the various developments. I have made extensive use of a rich and varied open literature which has developed in the past 25 years around the subject of re-entry vehicle flight mechanics. It would seem appropriate here to acknowledge my debt to such analysts and expositors as H. Ashley, J.J. Martin, K.R. Britting, M.H. Kaplan, H.B. Vaughn, A. Miele, A.D. Whellon, S.H.T. Loh as well as many others.

While the content of this book draws from the works of many aerospace workers, the existence of the book must be attributed to a group of very supportive colleagues. Encouragement for the development of the original notes and course presentation came from the friendly prodding and continual enthusiasm of Dr. A. Morrison of NSWC. Early encouragement to expand the notes into book form came from a variety of sources. I would particularly like to acknowledge the efforts of Professor A. Fuhs of the Naval Postgraduate School and of Lt. Col. W.P. Maiersperger (USAF, ret.). The pivotal decision for publication was made by Dr. J.S. Przemieniecki of the USAF Institute of Technology and Editor-in-Chief of the AIAA Education Series. At this point Dr. Morrison again provided all finan-

cial and clerical support to bring the manuscript into a form acceptable to AIAA's Mr. John Newbauer. I must also extend my thanks to AIAA staff member Ms. Camille S. Koorey for her very competent assistance in moving the project forward.

All technical errors and shortfalls are, of course, my own. Nevertheless, I must acknowledge the invaluable technical review given the manuscript by Drs. Przemieniecki, Morrison, and Schindel (NSWC).

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I. INTRODUCTION

The dynamics of re-entry bodies may be regarded as an application of rational mechanics, the oldest branch of physics. Although the foundations of the classical or nonrelativistic period of rigid body mechanics were virtually completed by the close of the nineteenth century, other areas of mechanics are evolving to the present day. The continual development of aerodynamics/gasdynamics has been motivated largely by the requirements of the design of manned and unmanned vehicles which enter the atmosphere at speeds that are at least a significant fraction of the speed associated with orbital motion. At the much lower speeds typical of the operation of atmospheric flight vehicles, the flowfield primarily imparts a distributed or pressure load to the vehicle. Control then is effected by making local alterations to this pressure field, often by configurational changes such as flap deflections. At the much higher speeds associated with re-entry, the support and control of these distributed loads is an even greater engineering challenge. To complicate matters further, at re-entry speeds the flowfield imparts a distributed thermal load in addition to the pressure load. Coping with this thermal load presents an additional and perhaps more formidable engineering challenge than that encountered in sustaining the pressure loads.

Most of the emphasis in this book will be on the dynamics of the boost and re-entry of suborbital aerospace vehicles. Nevertheless, it might be of some interest to consider for a moment the magnitude of the problem presented by the heat load. For example, a satellite in a circular orbit at 320 km above the Earth's surface has a specific kinetic energy of 3.11×10^7 J/kg (13,162 Btu/lb). An interplanetary vehicle re-entering the Earth's atmosphere from a near-parabolic orbit would have a specific energy approximately twice this circular orbital value. Even though the re-entry vehicles (RV) considered in this book are all suborbital, the specific energy associated with such vehicles can closely approach this value for circular orbit. The significance of a specific energy of 3.11×10^7 J/kg becomes clear when we consider the specific energies of vaporization of water and carbon. Water, for example, will vaporize at a specific energy of 2.32×10^6 J/kg (1000 Btu/lb); carbon, with one of the highest heats of vaporization, will vaporize at a specific energy of 6.03×10^7 J/kg (26,000 Btu/lb).

It obviously becomes a major engineering problem to dissipate this energy or to convert it into a form that does not prove disastrous to the vehicle. Clearly, if all of the body's kinetic energy were to be converted into thermal energy in the body itself, the vehicle would vaporize unless made almost

entirely of carbon. The survival of natural atmospheric entry bodies like meteorites indicates that not all of the thermal energy is absorbed by the body. Frictional drag transfers varying amounts of this thermal energy to the near flowfield. The fractional amount of energy that is absorbed by the vehicle depends upon body shape and the trajectory traversed throughout re-entry. At least three techniques are available for controlling the amount of heat absorbed by the body.

The most straightforward method is to absorb the entire heat load in the vehicle. Using the body as a heat sink puts great demand on material properties. Since the amount of energy available for heat conversion depends upon the difference in vehicle kinetic energies at atmospheric entry and Earth impact, a steep entry trajectory of short duration seems best suited for the heat sink design. It will be shown subsequently how variations in the drag coefficient can vary the altitude at which peak deceleration as well as peak heating takes place. The disadvantage of the heat sink method is the weight penalty of a blunt-nosed, thick-skinned vehicle.

A second engineering response to the heat problem is the radiation-cooled heat shield. If the amount of heat transferred to the vehicle can be bounded so that it is no larger than the heat radiated from the vehicle, the shield can be a thin metal skin of negligible heat capacity. The trajectory characteristic of the radiation-cooled RV will be much more shallow than the trajectory used for the heat sink design. Also, where the heat sink design will be nearly ballistic over its entire trajectory, the radiation-cooled vehicle will have significant lift.

A third method is to use a heat shield constructed of fiberglass or other materials bonded to a heat shield. Under the high stagnation temperatures associated with re-entry, this heat shield sublimates, passing from the solid to the vapor state. This sublimation process, known as ablation, provides a heat capacity in excess of 1.1×10^7 J/kg (5000 Btu/lb). The ablative heat shield provides the lightest solution to the re-entry thermal load problem. Vehicles with ablative heat shields installed have considerable latitude in the selection of the re-entry trajectory. For example, extensive maneuvering might be permitted just prior to impact to make alterations or corrections to an apparent impact point.

We shall not be concerned in this book with the engineering design of either vehicle structures to survive inertial and pressure loads or heat shields to survive thermal loads. It might be sufficient to recognize that the details of heat load distribution and intensity are varied by the vehicle configuration and the re-entry trajectory. Presumably, both vehicle and trajectory shape are under the control of a mission planner. The distributed pressure loads also are influenced by vehicle shape and trajectory, although vehicle shape and trajectory have somewhat different effects on pressure loads than they have on distributed thermal loads. For example, the altitude of peak deceleration is usually somewhat lower than the altitude of maximum heat transfer.

This cause-and-effect relationship (vehicle configuration and trajectory as cause, and thermal and pressure distributions as effect) is a bit of an oversimplification. It is equally true that the existing pressure/thermal distri-

butions will influence subsequent configuration and therefore pressure loads and ultimately the trajectory. For example, the thermal characteristics of the near flowfield will determine the gas constituents and properties in that field, thereby affecting the moment and drag loads on the body. Obviously heat shield shape changes due to unsymmetric ablation can "freeze in" an asymmetry, bringing about a nonzero trim angle, which in turn will generate an aerodynamic load normal to the trajectory. Certainly if such effects are not anticipated the subsequent trajectory will have some surprises, culminating in an embarrassing difference between the predicted and actual impact points.

In spite of the complex interactions among the thermal and pressure distribution, the configuration, and the trajectory, we shall place most of the emphasis in this book on vehicle dynamics. The distributed pressure field will be represented by various "lumped" parameters known as aerodynamic coefficients and stability derivatives. These parameters will vary significantly over the entire trajectory. However, such variations are less important when compared with changes in atmospheric density, which will vary several orders of magnitude from initial entry to impact.

In addition to aerodynamic loads, re-entry bodies also are subjected to forces due to the gravitational attraction of the Earth. The familiar "inverse-square" gravitational model was among the earliest speculations of rational mechanics. However, precise description of a re-entry body's trajectory requires a more detailed model because the Earth's field is only approximately spherical. It is necessary to modify this inverse-square representation to take into account departures of the Earth from sphericity in shape and homogeneity in mass distribution. Thus, we may expect to see "oblateness" and "pear-shaped" terms imposed upon the classical $1/r^2$ form.

If the re-entry body can be regarded as a rigid body immersed and moving through a gas, and subjected to a gravitational field, a vectoral representation of the loads might take the following form:

$$m\mathbf{a}^b = \mathbf{F}_a^b + \mathbf{F}_g^b \quad (1.1)$$

where m is body mass, $\mathbf{a}^b$ is the acceleration of the body, $\mathbf{F}_a^b$ is the aerodynamic load, and $\mathbf{F}_g^b$ is the gravity load. The superscript b indicates that all loads are resolved in some body frame. Equation (1.1) could be regarded as a "generalized force" equation, where m could also represent a moment of inertia, $\mathbf{a}$ an angular acceleration, and the various $\mathbf{F}$ vectors, applied moments. An important part of the subsequent discussion will cover the forms this force equation will take as various frames of reference are considered. For example, vector $\mathbf{F}_g^b$ represents the resolution of the gravity vector in a body frame; however, the gravity vector is defined more fundamentally in an Earth-fixed frame. Thus, considerable attention will be given here to transforming vector and tensor entities (and sometimes their derivatives) from one reference frame to another. In the past, vector algebra was used to formulate the concepts of vector differential calculus. Although use will be made of vector algebra, emphasis will be placed on matrix algebra because of the relative ease with which such manipulations can be described in a digital computer program.

At this point it might be of some interest to summarize the remaining chapters of the text. Since most of the interesting engineering problems associated with the re-entry body occur within the atmosphere, we shall develop an atmospheric model to account for the extensive changes in properties with altitude. Some simplifications to model will be indicated for closed-form trajectory calculations. Throughout its trajectory the re-entry vehicle will be subjected to the Earth's gravitational field; because the RV will never be more than a few tenths of an Earth's radius from the Earth's surface, we shall give attention to departures of the Earth's field from spherical symmetry. We shall then give a fairly general derivation of the force equations describing particle motion and the force and moment equations describing rigid body motion. Because of the fundamental importance of the transfer of vector and tensor quantities (and their derivatives) between rotating reference frames, we shall give lengthy treatment to various ways of describing such transformations. This background material will then be applied to aerospace vehicle motion. Suborbital (or Keplerian) motion will be examined first, followed by closed-form solutions to the point-mass motion of an RV within the atmosphere. Following a brief discussion of flow regimes and the boost trajectory, we shall consider rigid body motion, first exo-atmospheric and then during re-entry. During the endo-atmospheric phase we shall consider body angular motion for both constant and rapidly varying atmospheric density.

II. ATMOSPHERIC MODEL

2.1 Basic Relationships

The atmosphere of the Earth is for the most part a mixture of nitrogen and oxygen. However, merely identifying the major and minor gases in this mix we call the atmosphere would not be sufficient for our purposes. First we must recognize that the atmosphere is not a static medium. It is continuously changing its temperature, pressure, chemical constituents, particulate presence, and electrical characteristics, primarily due to changes in solar radiation. The recognition of some variations has long appeared in the performance analysis of atmospheric vehicles in the "hot day" and "cold day" atmospheres. By way of contrast, for an engineering analysis of RV's (at the level of the analysis in this book), one standard atmosphere may be used. There are at least two reasons for ignoring distinctions between "hot day" and "cold day" atmospheres: First, the thermal environment of the RV is severe, with stagnation temperatures in the neighborhood of 2000 K (3600°R). Consequently, variations in ambient enthalpy are negligible. Secondly, if we convert variations in atmospheric properties (from a standard) at a fixed altitude to variations in altitude, it will be found that such altitude variations are perhaps negligible when compared to the altitude range over which flight takes place. For example, suppose we take 40 K as the *variation* between a "hot" and "cold" atmosphere. If at some altitude the atmosphere has a thermal gradient of 4 K/km, then there is a corresponding atmospheric variation of 10 km. Although 10 km may be greater than the attainable altitude of an atmospheric vehicle, it is only 10% of the altitude range of an RV. We are therefore justified in limiting ourselves to a single atmospheric model in our study of RV dynamics.

Even though we shall establish a single atmospheric model, it will be seen that property variations about this "standard" would be relatively easy to introduce. Again, the argument for using only a simple standard is that variations or uncertainties about this standard, when translated into altitude variations, are assumed negligible when compared with the total range of flight. The reason for establishing an atmospheric model is of course to have available atmospheric properties of interest at any altitude. Our model will cover the altitude range from sea level to about 700 km (2300 kft), although most RV motion of interest occurs at altitudes below 150 km (500 kft). Above 100 km the atmospheric density is so low that the influence of the atmosphere on RV linear and angular motion is usually considered negli-

ble, at least for our purposes. This is fortunate since atmospheric properties above 150 km fluctuate widely with variations in solar radiation.

In establishing an atmospheric model, we first shall establish some simple analytic formulations of the variations of the temperature, pressure, and density with altitude. A simple computer program then will be written, based upon this analysis, to provide us with a fairly compact model of the atmosphere which may be used as a subroutine in dynamic studies. In order for our analytic and later numerical work to be something more than speculation, we must base our work upon measurements and averages of measurements.

Currently there are several measurement-based atmospheric models that may be used. For our purposes, we shall rely upon the U.S. Standard Atmosphere 1962.¹ It might be argued that this 1962 standard atmosphere is not a "model" but merely a tabulation of measurements. However, a perusal of Ref. 1 will indicate a tabulation of average atmospheric properties at close and regularly spaced intervals of altitude. Thus, an analytic model of property variations with altitude must be established to interpolate between measurements. It is realistic to expect that property measurements can be made only at irregular intervals of altitude.

Table 2.1 U.S. Standard Atmosphere (Metric)

Altitude (km)		Molecular Temperature (K)	Pressure Ratio (—)	Lapse Rate (K/km)
Geopotential	Geometric			
0.00	0.000	288.10	1.000	
11.00	11.019	216.65	2.284×10^{-1}	-6.50
20.00	20.063	216.65	5.462×10^{-2}	0.00
32.00	32.162	228.65	8.567×10^{-3}	+1.00
47.00	47.350	270.65	1.095×10^{-5}	+2.80
52.00	52.430	270.65	5.823×10^{-4}	0.00
61.00	61.590	252.65	1.797×10^{-4}	-2.00
79.00	80.000	180.65	1.024×10^{-5}	-4.00
88.70	90.000	180.65	1.622×10^{-6}	0.00
98.45	100.000	210.65	2.980×10^{-7}	+3.00
108.13	110.000	260.65	7.228×10^{-8}	+5.00
117.77	120.000	360.65	2.488×10^{-8}	+10.00
146.54	150.000	960.65	5.000×10^{-9}	+20.00
156.07	160.000	1110.60	3.640×10^{-9}	+15.00
165.57	170.000	1210.65	2.756×10^{-9}	...
184.48	190.000	1350.65	1.660×10^{-9}	+7.00
221.96	230.000	1550.65	6.869×10^{-10}	+5.00
286.47	300.000	1830.65	1.860×10^{-10}	+4.00
367.31	400.000	2160.65	3.977×10^{-11}	+3.30
463.53	500.000	2420.65	1.080×10^{-11}	+2.60
548.23	600.000	2590.65	3.400×10^{-12}	+1.70
630.53	700.000	2700.65	1.176×10^{-12}	+1.10

For our purposes, we are interested in how temperature, pressure, density, and molecular weight vary with altitude. We have depicted variations in temperature T , pressure P , and thermal lapse rate L_z with altitude in Tables 2.1 and 2.2 (metric and English measurements, respectively). The altitude entries have been selected as the "break points" at which the thermal lapse rates (derivative of temperature with altitude) appear to change almost discontinuously. Altitude bands over which the thermal lapse rate is constant will be designated as "layers." In our model we shall set these layers from measurements and then use the analytic model to interpolate for properties within each layer. Precise meaning will be given to terms such as geopotential, geometrical altitude, and molecular temperature. Again we must emphasize that information presented in any atmospheric model is only an estimate, which is based in turn on direct and indirect determinations of atmospheric pressure and molecular temperature. One can expect large variations in any atmospheric property from 90 km to the upper limits of the model.

Referring to Tables 2.1 and 2.2, we see that two models of the atmosphere will be required: one to account for cases where the temperature is not changing with altitude (say from 47 to 52 km) and a second model to account for temperature variations with altitude (say from 20 to 32 km). It will be noted that, in layers where the temperature varies with altitude, such variation is linear; i.e., the lapse rate is constant; note also that this linear lapse rate may be either positive or negative.

In formulating our atmospheric model, we shall make use of four equations from elementary physics. For reference we shall present these equations, deferring their physical justification until later.

Table 2.2 U.S. Standard Atmosphere (English)

Altitude (kft)		Molecular Temperature (°R)	Pressure Ratio (—)	Lapse Rate (°R/kft)
Geopotential	Geometric			
0.0	0.00	518.67	1.000	—3.567
36.1	36.15	389.97	2.284×10^{-1}	0.000
65.6	65.82	389.97	5.462×10^{-2}	+0.549
105.0	105.52	411.57	8.567×10^{-3}	+1.536
154.2	156.35	487.17	1.090×10^{-3}	0.000
170.6	172.01	487.17	5.823×10^{-4}	—1.097
200.1	202.06	454.77	1.797×10^{-4}	—2.195
259.1	262.47	325.17	1.024×10^{-5}	0.000
290.0	295.27	325.17	1.622×10^{-6}	1.646
...	328.10	379.17	2.980×10^{-7}	2.743
...	360.90	469.17	7.228×10^{-8}	5.485
...	393.37	649.17	2.468×10^{-8}	10.973
...	492.12	1729.17	5.000×10^{-9}	8.230
...	524.49	1999.17	3.800×10^{-9}	

(a) The hydrostatic equation relates differential pressure P to differential altitude dz as

$$dP = -\rho g dz \quad (2.1a)$$

where ρ denotes the air density, and g is the gravitational acceleration.

(b) The thermal equation state for an ideal gas relates pressure P , density ρ , and temperature T as

$$P = \rho RT \quad (2.1b)$$

where R is the gas constant.

(c) The polytropic equation of state might be written as an equation relating density ρ and pressure P as

$$P / \rho^n = \text{const} \quad (2.1c)$$

where n is the polytropic exponent.

(d) The inverse law of a spherical gravity field relates the gravitational acceleration at altitude z to sea level conditions as

$$g = g_0 \left[\frac{R_E}{R_E + z} \right]^2 \quad (2.1d)$$

where g_0 is the gravitational acceleration at sea level, and R_E is the Earth's radius.

The hydrostatic equation is derived from the assumption that an element of air must be in a state of equilibrium if its weight is balanced by pressure forces. Figure 2.1 shows an element of air of density ρ subjected to a differential pressure of dP .

Summation of the forces in a vertical direction gives the hydrostatic equation

$$dP = -\rho g dz \quad (2.2)$$

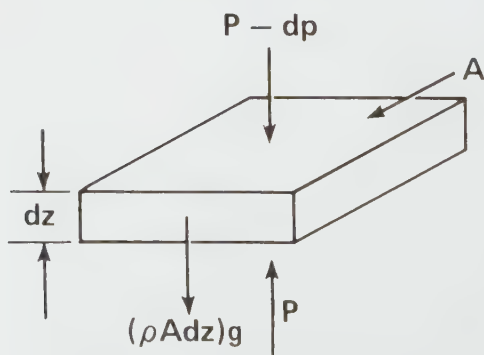


Fig. 2.1 Atmospheric element.

The altitude z is defined here as the measurable altitude, i.e., the geometric altitude. Integration of Eq. (2.2) should provide the functional relationship between pressure and altitude. Since both the density ρ and gravitational constant g will vary with z , it is necessary to provide these functions before the integration can take place. Although it is possible to provide $g = g(z)$ [see Eq. (2.1d)], the problem can be avoided by redefining a new altitude h , called the "geopotential altitude," as

$$g_0 dh = g dz \quad (2.3)$$

Thus, Eq. (2.2) may be written as

$$dP = -\rho g_0 dh \quad (2.4)$$

Whether or not we sidestep the variation of g with altitude by inventing an artificial altitude or provide an analytic form for $g = g(z)$, we must recognize that the density will vary with altitude.

The equation of state of an ideal gas will provide a relationship between pressure P and density ρ . Unfortunately, this relationship introduces temperature T , which Tables 2.1 and 2.2 indicate to be altitude-dependent. Because in the English system the pound is the unit of force whereas in the metric system the kilogram is used as a unit of mass, the state equation takes a different form in each system. In the metric system we write

$$\rho = MP / R^* T \quad (2.5a)$$

where R^* is the universal gas constant, and M is the molecular weight of air.

For the English system, where the molecular weight is in pound-mass moles, conversion must be made to slug-moles if density ρ is to be in units of slugs per cubic foot. Thus, we have for the ideal gas equation in English units

$$\rho = MP / R^* T g_0 \quad (2.5b)$$

At this point, reference may be made to Table 2.3 for the appropriate numerical constants.

In Table 2.3, we indicate that the atmosphere has well-defined layers throughout which the thermal gradient is constant. Thus it would seem that we have replaced the need for a density altitude function with a temperature altitude function. This is true, and we shall represent the atmosphere as having 14 or so layers, each layer being characterized by a distinct thermal gradient. The numerical value of this gradient must come from Tables 2.1 and 2.2 or ultimately from Ref. 1 or a similar source.

It might be of interest to offer some physical justification for the thermal gradient dT/dz . From thermodynamics, we may write the polytropic equation of state as

$$P / \rho^n = \text{const} \quad (2.6)$$

Table 2.3 Useful Constants

Definitions	Symbol	Metric	English
Sea level atmospheric pressure	P_0	$1.01325 \times 10^5 \text{ Pa}$	2116.22 psf
Sea level atmospheric density	ρ_0	1.2250 kg/m^3	$2.377 \times 10^{-3} \text{ slugs/ft}^3$
Sea level atmospheric temperature	T_0	288.15 K	518.6°R
Avogadro's number	N	6.02257×10^{26} /kg-mole	2.73179×10^{26} /(lb-mole)
Universal gas constant	R^*	8.31432×10^3 J/(kg-mole-K)	4.9719×10^4 lb-ft ² /(s ² -lb-mole-°R)
Gas constant for air	R	287 J/(kg-K)	1716.4 s ² /(ft ² -°R)
Radius of Earth	R_E	$6.37 \times 10^6 \text{ m}$	$2.09 \times 10^7 \text{ ft}$
Sea level gravity constant	g_0	9.086 m/s^2	32.174 ft/s^2
Sea level molecular weight	M_0	28.9664	28.9664
Rotational rate of Earth	Ω_E	$7.28 \times 10^{-5} \text{ rad/s}$	$7.28 \times 10^{-5} \text{ rad/s}$

Combining this with Eq. (2.5a), the thermal equation of state may be obtained by first writing the logarithmic differentials of both equations:

$$\frac{dP}{P} = \frac{d\rho}{\rho} + \frac{dT}{T} \quad (2.7a)$$

$$\frac{1}{n} \frac{dP}{P} = \frac{d\rho}{\rho} \quad (2.7b)$$

Eliminating dP/P between Eqs. (2.7) gives

$$\frac{dT}{T} = \frac{n}{n-1} \frac{dT}{T} \quad (2.8a)$$

$$\frac{dT}{dz} = \frac{P}{T} \frac{n}{n-1} \frac{dT}{dz} \quad (2.8b)$$

Using Eq. (2.2), the hydrostatic equation, along with Eq. (2.5) again gives

$$\frac{dT}{dz} = - \frac{(n-1)}{n} \frac{gM}{R^*} \quad (2.9a)$$

$$\frac{dT}{dh} = - \frac{(n-1)}{n} \frac{g_0 M}{R^*} \quad (2.9b)$$

Equation (2.9b) may be integrated to give

$$T = T_i - \left[\frac{(n-1)g_0 M}{nR^*} \right] (h - h_i) \quad (2.10)$$

where the subscript i indicates the values of h and T at the beginning of the i th layer. Equation (2.10) is used to calculate the temperature T throughout the layer as a function of geopotential altitude. Equation (2.10) remains valid until the next layer is encountered, i.e., where dT/dh changes. Since the thermal gradient dT/dh changes discontinuously at various altitudes, it is meaningful to identify, as we have done, altitude ranges or "layers" over which dT/dh is constant.

It might be of some interest to calculate the appropriate value of n , the polytropic constant corresponding to the thermal gradients indicated in Tables 2.1 and 2.2. Taking values of the molecular weight of air M , the universal gas constant R^* , and g_0 from Table 2.3, we obtain Table 2.4.

An interesting comment on the physical significance of the polytropic exponent was made by H. Schlichting²:

"The temperature gradient dT/dh determines the stability of the stratification in the stationary atmosphere. The stratification is more stable when the temperature decrease with increasing height becomes smaller. For $dT/dh = 0$ when $n = 1$, the atmosphere is isothermal and has a very stable stratification. For $n = \gamma = 1.405$, the stratification is adiabatic (isentropic) with $dT/dh = -0.98$ deg K per 100 m. This stratification is indifferent, because an air volume moving upward for a certain distance cools off through expansion at just the same rate as the temperature drops with height. The air volume maintains the temperature of the ambient air and is, therefore, in an indifferent equilibrium at every altitude. Negative temperature gradients of a larger magnitude than 0.98 deg K per 100 m result in unstable stratification."

Table 2.4 Polytropic Exponent at Atmospheric Layer Boundaries

Geopotential Altitude (km)	Thermal Gradient (K/km)	Polytropic Exponent
0.0	-6.5	1.2350
11.0	0.0	1.0000
20.0	1.0	0.9716
32.0	2.8	0.9242
47.0	0.0	1.0000
52.0	-2.0	1.0620
61.0	-4.0	1.1330
79.0	0.0	1.0000
81.7	...	...

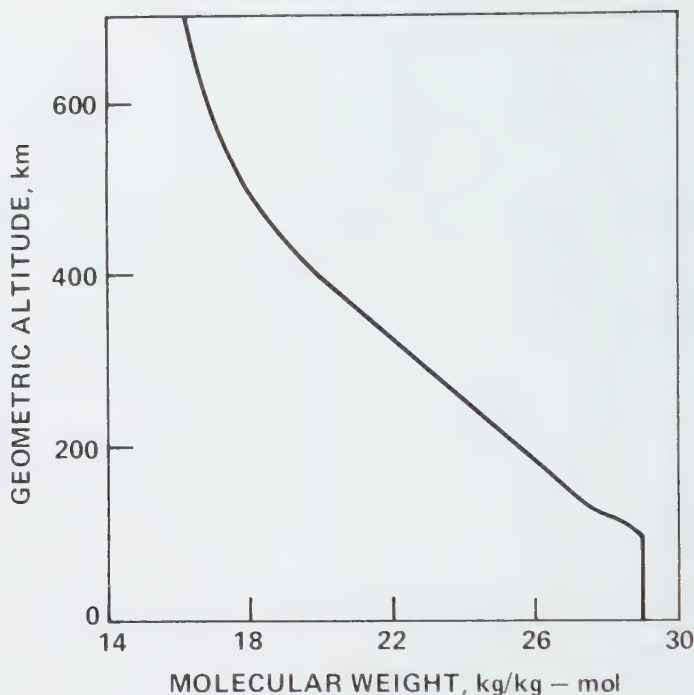


Fig. 2.2 Molecular weight M vs geometric altitude Z .

Returning to the thermal equation of state for an ideal gas, Eqs. (2.5), we again point out that we have replaced the density gradient $d\rho/dh$ with a thermal gradient dT/dh . The convenience seems to lie in the fact that the thermal gradient has only a dozen or so discrete values over the whole altitude range of interest. Unfortunately the molecular weight M also varies with altitude, although it is fairly constant (at 28.964) up to an altitude of 90 km (300 kft). Figure 2.2 indicates the variation of the molecular weight with altitude. Numerical values are given in Table 2.5.

Since both the molecular weight and temperature vary with altitude, we can combine both variations together in a single variable called the "molecular temperature." The molecular temperature T_M is defined as the product of the kinetic temperature T and the ratio of molecular weight at sea level to the molecular weight at altitude as

$$T_M = T[M_0/M] \quad (2.11)$$

Thus, we may replace M/T in Eq. (2.5) by M_0/T_M . It will be recalled that we used a similar type of variable "lumping" in defining the geopotential altitude h in Eq. (2.3). In Tables 2.1 and 2.2, the temperature and temperature gradient are in terms of the molecular temperature. Since the molecular

Table 2.5 Atmospheric Molecular Weight and Thermal Characteristics vs Geometric Altitude

Altitude z (km)	Molecular Scale Temperature T_M (K) ^a	Gradient L_h (K/km)	Molecular Weight M	Kinetic Temperature T (K)
90	180.65		28.9644	180.65
100	210.65	+ 3.0	28.8800	210.02
110	260.65	+ 5.0	28.5600	257.00
120	360.65	+ 10.0	28.0700	349.49
150	960.65	+ 20.0	26.9200	892.79
160	1110.65	+ 15.0	26.6600	1022.20
170	1210.65	+ 10.0	26.4500	1105.50
190	1350.65	+ 7.0	25.8500	1205.50
230	1550.65	+ 5.0	24.6900	1321.70
300	1830.65	+ 4.0	22.6600	1432.10
400	2160.65	+ 3.3	19.9400	1487.40
500	2420.65	+ 2.6	17.9400	1499.20
600	2590.65	+ 1.7	16.8400	1506.10
700	2700.65	+ 1.1	16.1700	1507.60

^aNote that the molecular temperature equals the kinetic temperature for geometric altitudes below 90 km, since the molecular weight is constant at 28.9644 for all altitudes up to 90 km. Refer to Tables 2.1 and 2.2.

Table 2.6 Composition of the Atmosphere to 90 km (Dry Air)

Constituent	Volume Fraction	Mass Fraction
Nitrogen	0.78084	0.75527
Oxygen	0.20947	0.23143
Argon	9.340×10^{-1}	1.282×10^{-2}
Carbon dioxide	3.140×10^{-2}	4.560×10^{-4}
Neon	1.818×10^{-3}	1.250×10^{-5}
Krypton	1.140×10^{-4}	3.300×10^{-6}
Methane	2.000×10^{-4}	7.750×10^{-7}
Nitrous oxide	5.000×10^{-5}	7.600×10^{-7}
Helium	5.240×10^{-4}	7.240×10^{-7}
Xenon	8.700×10^{-6}	3.900×10^{-7}
Hydrogen	5.000×10^{-5}	3.480×10^{-8}

weight is constant at 28.9644 up to 90 km (300 kft). Table 2.5 provides estimates of temperature and molecular weight above 90 km.

In this section our interest in the atmosphere is primarily in those properties which contribute to the aerodynamic loads and to a lesser extent to the thermal loads. Other scientific and engineering disciplines would certainly emphasize other properties of the atmosphere. Reference 1 might partially

satisfy the needs of these specialists. Further information on atmospheric chemistry which might find application in various subspecialties of re-entry body aerodynamics and mechanics is contained in Ref. 1. A succinct coverage of various aspects of atmospheric chemistry, as well as a guide to the literature up to the 1960's, is Martin's book *Atmospheric Reentry*.³ Table 2.6 provides a listing of the constituent gases of the dry atmosphere up to approximately 90 km (300 kft).

2.2 Development of Analytic Atmosphere Model

To review briefly, we have set as our goal the development of an atmospheric model that will provide pressure, density, kinetic temperature, and molecular weight over an altitude band from sea level to about 160 km (525 ft). Above this altitude, diurnal and annual variations in properties are extensive or unknown; further, at the suborbital velocities under consideration we shall postulate that the aerodynamic loads above 160 km are negligible in comparison with gravity.

The first step in the development of an analytic model of the atmosphere began with the derivation of the hydrostatic equation [Eq. (2.3)]. The hydrostatic relationship is a differential equation relating pressure P to geometric altitude z . Integration of this equation requires density ρ and the gravitational constant g as analytic functions of altitude. Density was eliminated through the use of the thermal equation of state of an ideal gas [Eqs. (2.5)]. The gravitational acceleration constant may be expressed as a fairly simple function of altitude [Eq. (2.1d)] or the entire problem avoided at the expense of introducing an artificial altitude called the "geopotential" altitude. This altitude assumes a constant sea level value for g [see Eq. (2.3)].

Introduction of the thermal equation of state replaces a density variation with a thermal and molecular weight variation. Changes of molecular weight M with altitude are bypassed by introducing an artificial temperature T_M , called the "molecular temperature," which bases temperature on the constant sea level value of molecular weight [see Eq. (2.11)]. Thus the main result of the entire exercise is to require a tabulation of thermal gradient, dT_M/dz or dT_M/dh , depending upon whether we choose geometric or geopotential altitude as the independent variable. We further showed that temperature gradients could be related to the exponent n in the polytropic equation of state [Eq. (2.6)]. Tabulated values of either dT_M/dz or n could be used, along with sea level conditions, as empirical inputs into the model. It is arbitrary whether to tabulate n and consider dT_M/dz as a derived quantity [Eqs. (2.9)] or to tabulate dT_M/dz directly. The latter course will be taken.

We now may combine the hydrostatic and thermal state equations to get a relationship between pressure and altitude (geometric or geopotential) as

$$\int_{P_i}^P \frac{dP}{P} = - \int_{h_i}^h \frac{Mg_0 dh}{R^*T(h)} \quad (2.12a)$$

$$\int_{P_i}^P \frac{dP}{P} = \int_{z_i}^z \frac{Mg dz}{R^*T(z)} \quad (2.12b)$$

From Tables 2.1 and 2.2, we have represented the atmosphere as a layered structure, each layer characterized by a distinct value of the thermal gradient. To make the mathematics a bit easier, we shall separate layers with a zero from layers with nonzero gradients. The subscripts i in Eqs. (2.12) refer to the conditions at the beginning of the i th layer. Thus we may identify four cases, zero and nonzero thermal gradients combined with geopotential and geometric altitudes. In other words, the four cases might be identified as

Case 1: $L_h = 0$, $T_M = T_{Mi}$, $g = g_0$

Case 2: $L_z = 0$, $T_M = T_{Mi}$, $g = g(z)$

Case 3: $L_z \neq 0$, $T_M = T_{Mi} + L_h(h - h_i)$, $g = g_0$

Case 4: $L_z \neq 0$, $T_M = T_{Mi} + L_z(z - z_i)$, $g = g(z)$

where the derivatives dT_M/dh and dT_M/dz have been represented by L_h and L_z , respectively.

In Case 1, replacing T_M and g in Eq. (2.12a) and integrating gives

$$\int_{P_i}^P \frac{dP}{P} = - \int_{h_i}^h \frac{M_0 g_0 dh}{R^* T_{Mi}} = - \int_{h_i}^h \frac{g_0 dh}{R T_{Mi}} \quad (2.13a)$$

where $R = R^*/M_0 = 287 \text{ J}/(\text{kg} \cdot \text{mole})$, the gas constant for air. Carrying out the integration indicated in Eq. (2.13a) gives

$$\frac{P}{P_i} = \exp \left\{ - \left[\frac{g_0 (h - h_i)}{R T_{Mi}} \right] \right\} \quad L_h = 0 \quad (2.13b)$$

Since the temperature remains constant, the density and pressure variations with altitude will be identical:

$$\frac{\rho}{\rho_i} = \exp \left\{ - \left[\frac{g_0 (h - h_i)}{R T_{Mi}} \right] \right\} \quad L_h = 0 \quad (2.13c)$$

In Case 2, we shall use the geometric altitude with the isothermal atmosphere. From Newton's inverse-square law of gravity, we have

$$g = g_0 \left[\frac{R_E}{R_E + Z} \right]^2 \simeq g_0 (1 - bz) \quad (2.14)$$

where R_E is the Earth's radius ($6.371 \times 10^6 \text{ m}$ or $2.09 \times 10^7 \text{ ft}$). It will be noted that we have made an approximation to the inverse gravity equation by retaining the first two terms in a binomial expansion. Thus, $b = 2/R_E$ or $3.139 \times 10^{-7} \text{ (1/m)}$ [$9.57 \times 10^{-8} \text{ (1/ft)}$]. For the English system of units, the gas constant for air can be written as $R = R^*g_0/M = 1716 \text{ (lb/slug-ft} \cdot \text{R)}$.

In Case 2, we shall continue to look at an isothermal atmosphere. Therefore, we shall treat the case where $L_z = 0$. Using Eq. (2.12b), we get

$$\int_{P_i}^P \frac{dP}{P} = - \int_{z_i}^z \frac{Mg dz}{R^* T_{M_i}} = - \frac{1}{R T_{M_i}} \int_{z_i}^z g dz \quad (2.15)$$

Since g is a function of altitude z , we get, after replacing g by the approximation of Eq. (2.14),

$$\frac{P}{P_i} = \exp \left\{ - \left\{ \left[\frac{g_0(z - z_i)}{R T_{M_i}} \right] \left[1 - \frac{b}{2}(z + z_i) \right] \right\} \right\} \quad (2.16a)$$

If $b = 0$, the geometric altitude z and the geopotential altitude h are identical, and Eq. (2.16a) reduces to Eq. (2.13b). Again, because the temperature is assumed to be unchanged over the interval, the density ratio takes the same form as the pressure ratio:

$$\frac{\rho}{\rho_i} = \exp \left\{ - \left\{ \left[\frac{g_0(z - z_i)}{R T_{M_i}} \right] \left[1 - \frac{b}{2}(z + z_i) \right] \right\} \right\} \quad (2.16b)$$

In Case 3, we consider a constant thermal lapse rate with geopotential altitude. Letting $L_h = dT/dh$, we have

$$T_M = T_{M_i} + L_h(h - h_i) \quad (2.17)$$

The above linear variation of temperature with altitude has already been noted in Eq. (2.10). Returning to Eq. (2.12a) with T_M replaced by the linear function of Eq. (2.17), we get

$$\int_{P_i}^P \frac{dP}{P} = - \int_{h_i}^h \frac{M_0 g_0 dh}{R^* [(h - h_i)L_h + T_{M_i}]} \quad (2.18)$$

Rearranging terms on the right gives

$$\int_{P_i}^P \frac{dP}{P} = - \frac{g_0}{R L_h} \int_{h_i}^h \frac{dh}{[(h - h_i) + T_{M_i}/L_h]} \quad (2.19)$$

Carrying out the integration gives, for the pressure ratio,

$$\frac{P}{P_i} = \left[\frac{L_h(h - h_i) + T_{M_i}}{T_{M_i}} \right]^{-(g_0/L_h R)} \quad L_h \neq 0 \quad (2.20a)$$

The density ratio follows from the thermal equation of state of an ideal gas, Eqs. (2.5) and (2.20a), as

$$\frac{\rho}{\rho_i} = \left[\frac{L_h(h - h_i) + T_{M_i}}{T_{M_i}} \right]^{-[(g_0/L_h R) + 1]} \quad (2.20b)$$

Finally, for Case 4 we consider the most complex case, where temperature and gravity both vary with altitude. From Eqs. (2.14) and (2.17), we have

$$g = g_0(1 - bz) \quad (2.21a)$$

$$T_M = T_{M_i} + L_z(z - z_i) \quad (2.21b)$$

Returning once again to Eqs. (2.12), we get

$$\int_{P_i}^P \frac{dP}{P} = -\frac{g_0}{RL_z} \int_{z_i}^z \frac{(1 - bz) dz}{[(z - z_i) + T_{M_i}/L_z]} \quad (2.22)$$

After some manipulation, the above expression integrates to

$$\frac{P}{P_i} = \left[\left(\frac{L_z}{T_{M_i}} \right) (z - z_i) + 1 \right] \left\{ \frac{g_0}{RL_z} \left[1 + b \left(\frac{T_{M_i}}{L_z} z_i \right) \right] \right\} \exp \left[\frac{g_0 b}{RL_z} (z - z_i) \right] \quad (2.23a)$$

The density ratio follows from the above equation and the thermal equation of state of an ideal gas [Eq. (2.20a)]:

$$\frac{\rho}{\rho_i} = \left[\frac{L_z}{T_{M_i}} (z - z_i) + 1 \right] \left\{ \frac{g_0}{RL_z} \left[\frac{RL_z}{g_0} + 1 + b \left(\frac{T_{M_i}}{L_z} z_i \right) \right] \right\} \exp \left[\frac{g_0 b}{RL_z} (z - z_i) \right] \quad (2.23b)$$

The preceding analysis provided some fairly analytic expressions from which atmospheric properties such as density, temperature, pressure, and molecular weight may be calculated. We must recognize of course that for some engineering problems other atmospheric properties, such as viscosity, thermal conductivity, collision frequency, mean free path, etc., may be of interest. Because our interest will be primarily in the evaluation of loads and not in EM transmission or heat transfer, no additional properties will be considered. Thus, we shall confine our attention to the variation of pressure,

density, temperature, and molecular weight with geometric altitude. The short computer program given in Table 2.7 will be based upon Eqs. (2.20) for the isothermal layers and upon Eq. (2.23) for the general polytropic layers (where L_z is nonzero). The molecular weight will be a linear interpolation of the values defined at the endpoints of at each layer (see Fig. 2.2). Reference should be made to Tables 2.1, 2.2, and 2.5 for the designations of temperature and molecular weight at the beginning of each layer.

Glancing at Table 2.7, we see that the atmospheric data are input in lines 320 through 530. These data consist of geometric altitude, molecular tem-

Table 2.7 Simple Atmospheric Model

```

00100 REM THIS PROGRAM /SUBROUTINE IS AN ATMOSPHERIC MODEL THAT CALCULATES
00110 REM PRESSURE, DENSITY, TEMPERATURE AND MOLECULAR WEIGHT AT ALTITUDES UP TO
00120 REM 300 KILOMETERS. PROGRAM INPUT IS ALTITUDE Z ( METERS) IN FIRST
00130 REM LINE AFTER REM STATEMENTS. PROGRAM PARAMETERS ARE GAS CONSTANT,
00140 REM R (JOULES/KG-DEG-K), AND THE SEA LEVEL VALUES OF THE GRAVITATIONAL
00150 REM CONSTANT, G0 (M/SEC-SQ), ATMOSPHERIC PRESSURE AT SEA LEVEL, P0
00160 REM (N/M-SQ), THE MOLECULAR WEIGHT, M, AND THE DENSITY, D0 (KG/M-CU).
00170 REM THE FIRST ORDER GRAVITATIONAL CONSTANT IS B (1/M).
00180 REM THE FOLLOWING SUBSCRIPTED VARIABLES ARE USED: T(I) IS THE
00190 REM ALTITUDE (KILO-M), T(I), THE MOLECULAR TEMPERATURE (DEG-K), P(I),
00200 REM THE PRESSURE (N/M-SQ), M(I), THE MOLECULAR WEIGHT (KG-MOLE),
00210 REM D(I), THE DENSITY (KG/M-CU) AND L(I), THE THERMAL LAPSE RATE (DEG-K/KILO-M).
00220 INPUT Z
00230 R = 287
00240 G0 = 9.806
00250 M0 = 28.964
00260 P0 = 1.01325E+5
00270 D0 = 1.225
00280 B = 3.139E-7
00290 DIM Z(25), T(25), P(25), M(25), D(25), L(25)
00300 FOR I = 1 TO 22
00310 READ Z(I), T(I), P(I), M(I)
00320 DATA 0.00, 288.1, 1.000, 28.964
00330 DATA 11.019, 216.65, 2.284E-1, 28.964
00340 DATA 20.063, 216.65, 5.462E-2, 28.964
00350 DATA 32.162, 228.65, 8.567E-3, 28.964
00360 DATA 47.350, 270.65, 1.095E-3, 28.964
00370 DATA 52.43, 270.65, 5.823E-4, 28.964
00380 DATA 61.59, 252.65, 1.797E-4, 28.964
00390 DATA 80.00, 180.65, 1.024E-5, 28.964
00400 DATA 90.00, 180.65, 1.622E-6, 28.964
00410 DATA 100.00, 210.65, 2.98E-7, 28.88
00420 DATA 110.00, 260.65, 7.22E-8, 28.56
00430 DATA 120.00, 360.65, 2.488E-8, 28.07
00440 DATA 150.00, 960.65, 5.0E-9, 26.92
00450 DATA 160.00, 1110.6, 3.64E-9, 26.66
00460 DATA 170.00, 1210.65, 2.756E-9, 26.5
00470 DATA 190.00, 1350.65, 1.66E-9, 25.85
00480 DATA 230.00, 1550.65, 6.869E-10, 24.69
00490 DATA 300.00, 1830.65, 1.86E-10, 22.66
00500 DATA 400.00, 2160.65, 3.977E-11, 19.94
00510 DATA 500.00, 2420.65, 1.08E-11, 17.94
00520 DATA 600.00, 2590.65, 3.40E-12, 16.84
00530 DATA 700.00, 2700.65, 1.176E-12, 16.17
00540 NEXT I
00550 FOR I = 1 TO 22
00560 Z(I) = Z(I)*1000
00570 P(I) = P(I)*P0
00580 D(I) = P(I)/(R*T(I))
00590 NEXT I
00600 FOR I = 1 TO 21
00610 L(I) = (T(I+1)-T(I))/(Z(I+1)-Z(I))
00620 NEXT I
00630 GOSUB 00650
00640 GOTO 00880
00650 FOR I = 1 TO 21
00660 IF Z>Z(I+1) THEN 00860
00670 IF ABS (L(I))<1E-5 THEN 00790
00680 Q1 = 1+B*((T(I)/L(I))-Z(I))
00690 Q2 = (Q1*G0)/(R*L(I))
00700 T = T(I) + L(I)*(Z-Z(I))

```

(Table continued on next page)

Table 2.7(cont.) Simple Atmospheric Model

```

00710 Q3 = T/T(I)
00720 Q4 = Q3*(-Q2)
00730 Q5 = EXP((B*G0*(Z-Z(I)))/(R*L(I)))
00740 Q6 = Q4*Q5
00750 P = P(I)*Q6
00760 Q7 = Q2+1
00770 D = D(I)*(Q3*(-Q7))*Q5
00780 GOTO 00830
00790 T = T(I)
00800 Q8 = (-1)*G0*(Z-Z(I))*(1-(B/2)*(Z+Z(I)))/(R*T(I))
00810 P = EXP(Q8)*P(I)
00820 D = EXP(Q8)*D(I)
00830 M = M(I)*(M(I+1)-M(I))/(Z(I+1)-Z(I))* (Z-Z(I))
00840 T9 = (M*T)/M0
00850 GOTO 00870
00860 NEXT I
00870 RETURN
00880 PRINT #GEO,-ALT#,#PRES,-RATIO#,#DENS,-RATIO#,#MOL-TFMP#,#MOL-WT#
00890 PRINT #METERS#,# - #,# - #,#DEG-K#,# - #
00900 PRINT
00910 PRINT Z,P/P0,D/D0,T,M
00920 PRINT
00930 PRINT
00940 PRINT #GEO-ALT#,#,#PRESSURE#,#DENSITY#,#KIN-TEMP.#
00950 PRINT #METERS#,#N/SQ-M#,#KG/CU-M#,#DEG-K#
00960 PRINT
00970 PRINT Z,P,D,T9
00980 END

```

perature, pressure ratio, and molecular weight. Each entry defines an atmospheric layer by some value of the constant n from the polytropic equation of state [Eq. (2.6)]. If n is unity we have the isothermal layer and use Eqs. (2.20), and if n is some other value we would use, as we have pointed out, Eqs. (2.23). Since molecular temperature is input along with geometric altitude, the thermal gradient is calculated in line 610; in line 670 the decision is made whether or not the layer is isothermal. Depending upon that decision, either Eqs. (2.20) or (2.23) are used for the calculations within that layer.

Table 2.8 Pressure Ratio Comparisons

Geometric Altitude (km)	Pressure Ratio P/P_0 (Computer)	Pressure Ratio P/P_0 (1962 Atmosphere)
0	1.0000	1.0000
2	0.7845	0.7846
5	0.5333	0.5334
8	0.3518	0.3518
10	0.2615	0.2615
20	5.5790×10^{-2}	5.4570×10^{-2}
30	1.1940×10^{-2}	1.1810×10^{-2}
50	7.8760×10^{-4}	7.8730×10^{-2}
60	2.2170×10^{-4}	2.2170×10^{-2}
70	5.4490×10^{-5}	5.4480×10^{-2}
100	2.9800×10^{-7}	2.9680×10^{-7}
200	1.3140×10^{-9}	1.3160×10^{-9}

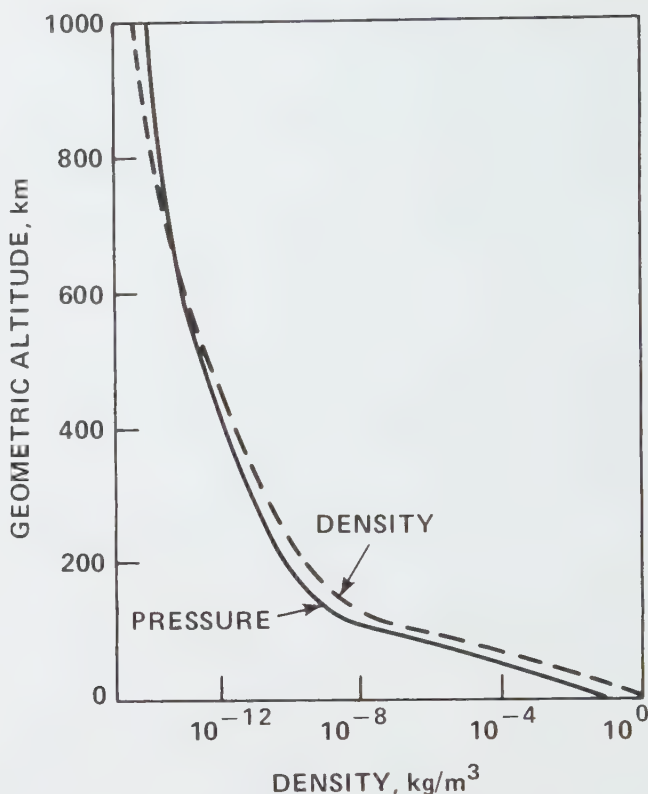


Fig. 2.3 Atmospheric density vs geometric altitude.

The program can be used conveniently as a subroutine by entering in line 230 (with z defined prior to entry) and a RETURN at line 880. Total program length is 66 lines in such a case.

A comparison might be made between atmospheric properties as computed from the preceding program and the same properties from the 1962 atmosphere. Such a comparison is given in Table 2.8.

2.3 Simple Atmospheric Model for Closed-Form Solutions

In the preceding section, we separated the atmosphere into layers labeled according to whether the temperature remained constant or varied linearly within the layer. Based upon this type of model, we developed analytic relationships between the intensive atmospheric properties of density ρ , pressure P , temperature T , and altitude z or h . In some of the analytic work to follow in which we shall seek closed-form solutions to various equations of motion, we shall require a particularly simple analytic form for a relationship between density ρ and altitude h .

In Fig. 2.3, we have plotted density ratio vs altitude. For the atmospheric

portion of the RV flight below about 200 km, atmospheric effects are considered important; above approximately this altitude we usually can ignore such effects. Thus we may fit a straight line to the logarithm of the density ratio for altitudes below 200 km and ignore the fit above this altitude. The result is the particularly simple analytic relationship between density ρ and altitude h :

$$\rho/\rho_0 = \exp(-h/H) \quad (2.24)$$

where $\rho_0 = 1.752 \text{ kg/m}^3$, and $H = 6700 \text{ m}$. Equation (2.24) has the same form as Eq. (2.13c), and it appears that we are assuming that the atmosphere is isothermal over the range of 200 km down to sea level. Of course this assumption is a bit inconsistent with our earlier and more complete model (Tables 2.1 and 2.2). Nevertheless, we shall take Eq. (2.24) as an acceptable approximation for use in simple closed-form solutions that we will encounter in Chapter VI and later.

It was shown in Tables 2.1 and 2.2 that the atmosphere has a thermal lapse rate over the lower layer, and thus Eq. (2.24) is not appropriate over this layer. Even if we accept Eq. (2.24) as a rough fit for density over the altitude range from sea level to 200 km, it might be interesting to compare it with the more complete model at sea level. At $h = 0$, we have $\rho = \rho_0$, or 1.752 kg/m^3 . Also, from Eq. (2.13c),

$$H = RT_0/g = (287)(288)/9.81 = 8547\text{m} \quad (2.25)$$

compared with $H = 6700 \text{ m}$ in our simple model.

Example 2.1: Application of Exponential Atmosphere

We might now consider an example that will illustrate the use of this simple exponential atmosphere model. The two fitting constants, ρ_0 and H , for the terrestrial atmosphere were given in Eq. (2.24). For a Martian atmosphere, the corresponding values of the constants are

$$\rho_{0m} = 0.1420 \text{ kg/m}^3$$

$$H_m = 27,700 \text{ m}$$

It will be noted that the atmospheric density at the Martian surface is only about 8% that of the atmosphere of terrestrial sea level. We might then be led to believe that the Martian atmosphere in comparison with that of the Earth is quite tenuous. As density comparisons at the surface of both planets show, the Martian atmosphere is quite thin. However, a density comparison at increasing altitudes above sea level shows that the densities of both atmospheres become closer and closer. Above a certain altitude the density of the Martian atmosphere is greater than that of the terrestrial atmosphere. The reason is of course that the Martian gravitational attraction at its surface is less than one-half of the Earth at its surface. Consequently, the Martian

atmosphere is more spread out. Our question then is the following: At what altitude above each planet is the atmospheric density equal; i.e., $\rho_m = \rho_e$. From the previously given exponential model we have

$$\frac{\rho_e}{\rho_m} = 1 = \frac{\rho_{0e}}{\rho_{0m}} \exp \left\{ -h \left[\frac{1}{H_e} - \frac{1}{H_m} \right] \right\} \quad (2.26)$$

$$1 = \frac{1.752}{0.142} \exp \left\{ -h \left[\frac{1}{6700} - \frac{1}{27700} \right] \right\}$$

Solving for h , we get 22,200 km (72,855 ft). The values of the atmospheric parameters, especially those of Mars, are only approximate. However, a vehicle entering a Martian atmosphere at an altitude above 25 km, say, will encounter an atmosphere of greater density than it would under similar entry conditions on Earth.

References

¹Dubin, M., et al., *U. S. Standard Atmosphere 1962*, U. S. Government Printing Office, Washington, D. C., 1962.

²Schlichting, H. et al., *Aerodynamics of the Airplane*, McGraw Hill, New York, 1979.

³Martin, J. J., *Atmospheric Reentry*, Prentice-Hall, Englewood, N. J., 1966.

III. EARTH'S FORM AND GRAVITY FIELD

3.1 Fundamental Concepts

One of the foundation stones of rational mechanics, indeed of physics itself, is Newton's Law of Universal Gravitation. This gravitational law does not "explain" gravity in terms of more fundamental speculations, nor does it relate gravity to other field phenomenon. Rather, Newton offered a quantitative description of gravity by relating this pervasive force to certain metrical quantities, namely the masses of the objects producing the field and the distance of their separation. Since gravity is the mutually attractive force between two objects, we might let one of the objects be the Earth of mass M_E and the other an RV of mass m . If R_E is the radius of the Earth and z is the geometric altitude of the RV above the surface of the Earth, Newton's Law of Gravity may be written for this special case as

$$F_g = KM_E m / (R_E + z)^2 \quad (3.1a)$$

The term K is taken as the constant of proportionality between force F_g due to gravity and masses M_E and m causing the gravity. Since $F_g = mg$, where g is the gravitational acceleration of free-fall, we have

$$g = KM_E / (R_E + z)^2 \quad (3.1b)$$

There are several ways of evaluating the universal gravitational constant K and Earth's mass M_E . For heliocentric orbits, the Earth's mass M_E would be expressed in units of solar mass and K in terms of the astronomical unit. (Further discussion of the units of definition of K are contained in texts on celestial mechanics such as Ref. 1.) The astronomical unit is nearly the mean of the Earth's orbital radius. Since we are restricting attention to suborbital flights within the Earth's field, it is more convenient to evaluate the product KM_E in laboratory units (metric, say) at some point in the field. For convenience, this point will be chosen at the Earth's surface, i.e., when $z = 0$. Thus, Eq. (3.1b) becomes

$$g = g_E R_E^2 / (R_E + z)^2 \quad (3.1c)$$

The above relationship also was given in Eq. (2.1d).

There are certain approximations that may be made to Eq. (3.1c). For

example, for atmospheric vehicles, experience seems to indicate that the gravitational field is of constant magnitude. If $z/R_E < 10^{-3}$, an adequate simplification of Eq. (3.1c) is to let $g = g_E$. For suborbital vehicles, where $z/R_E = O(10^{-1})$, a first-order expansion of Eq. (3.1c) gives

$$g = g_E(1 - bz) \quad (3.1d)$$

where we have replaced g_0 by g_E . The coefficient b is easily shown to be $2/R_E$, or twice the reciprocal of the radius of the Earth. The approximation of Eq. (3.1d) was used in the previous chapter to maintain our definition of geometric altitude in the various integrations involving pressure and density.

Even though certain approximations to the inverse-square law are adequate in many applications, it is also true that Eq. (3.1c) may not be entirely satisfactory in its "exact form" for the analysis of highly accurate trajectories. Now raising questions about Eq. (3.1c) does not mean that we regard Newton's gravitational law as being other than fundamental. It is just that Newton's law describes the gravitational attraction between two point masses. In this book we are not particularly concerned with the motion of point masses around each other, but rather with the orbital or suborbital motion of an RV (secondary) around the Earth (primary). We should remember that the RV's altitude above the Earth is never more than 10% of an Earth radius. To apply Newton's gravitational law to the entire Earth without modification would require that the Earth be a sphere with radially symmetric mass distribution. In this chapter we shall discuss an Earth that has neither a spherical shape nor a central gravity field.

If we are going to relax our usual assumptions of sphericity for the Earth's shape and gravity field, we might expect some ambiguities in the definition of some commonly held notions. Let us take a look at the idea of "vertical," for example. We might define "vertical" as the direction parallel to the local gravity vector. The local gravity vector, in turn, might be defined as the direction that a plumb bob would take after appropriate corrections have been made for Earth's rotation. We might also define "vertical" as along the perpendicular to the local tangent plane, although such a plane might be difficult to define due to topographical irregularities. If both definitions of "vertical" were to be compared we would find, in general, that neither would be parallel to the other and that neither, when extended, would pass through the center of the Earth. To be sure, the variations are negligibly small most of the time; however, there is still a class of problems where ambiguities in defining such things as "vertical" must be resolved.

3.2 Geocentric Position Vector

Since the Earth is not exactly spherical in shape, the normal to the surface will not, in general, pass through the center of the Earth. In this chapter, we shall develop the idea of the Earth as an ellipsoid of revolution. Therefore, in Fig. 3.1 the Earth's meridian plane is represented as an elliptical quadrant. This ellipsoid will be taken as a close approximation to a geopotential surface called a "geoid."

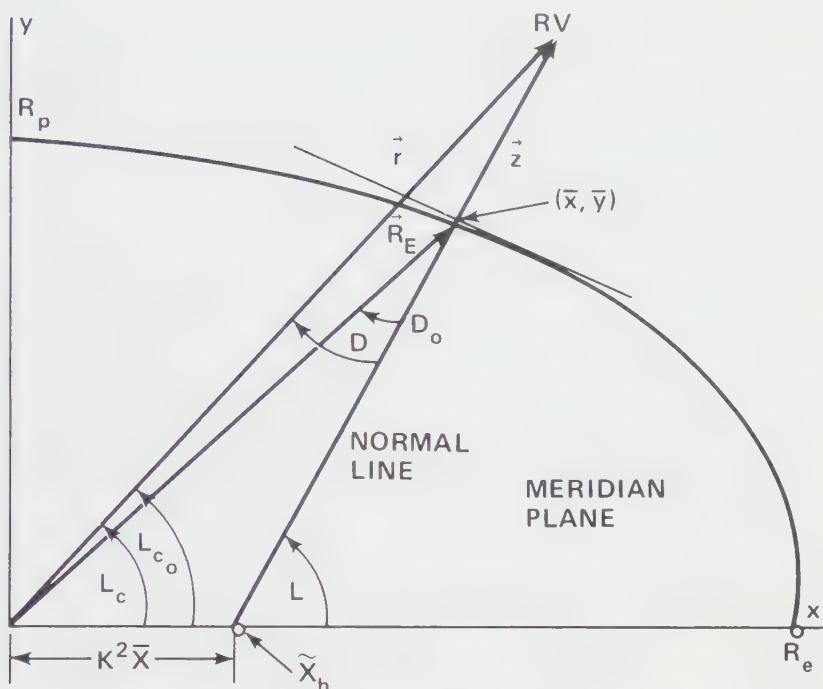


Fig. 3.1 Reference ellipsoid.

The surface of the geoid is mean sea level. In other words, if the Earth were a fluid body but retained its mass distribution and rotation, the resulting figure would define the geoid. Now the geoid, being an equipotential surface, would have a somewhat irregular shape, reflecting the irregularities of the Earth's mass distribution. As it turns out, these variations are relatively minor. Variation in the elevation of the geoid relative to the closest reference ellipsoid is about 1% of the topographic variations in elevation. The Hayford Spheroid of 1909 is the international ellipsoid. It has values of R_e and R_p , the ellipsoidal radii at the equator and poles, respectively, as

$$R_e = 6378.388 \text{ km } (2.09265 \times 10^7 \text{ ft})$$

$$R_p = 6356.912 \text{ km } (2.08560 \times 10^7 \text{ ft})$$

There are two ways of describing the deviation of an ellipse from a circle. One is the "ellipticity" e , defined as

$$e = (R_e - R_p) / R_e = 1 - (R_p / R_e) \quad (3.2a)$$

Using the above values for R_e and R_p , we get an ellipticity $e = 1/297$. A related concept is the "eccentricity" k , defined as

$$k = [1 - (R_p / R_e)^2]^{1/2} \quad (3.2b)$$

“Ellipticity” and “eccentricity” are related as

$$K^2 = 2e[1 - (e/2)] \quad (3.2c)$$

The surface of the geoid is by definition an equipotential surface. Consequently, the gravity gradient is always a maximum along the normal to the geoid. Even though the normal to the geoid does not, in general, pass through the center of the Earth, the vector along the local normal to the geoid defines the direction of “up and down.” The reference ellipsoid illustrated in Fig. 3.1 will be used as a convenient approximation to the geoid. Because the reference ellipsoid is not exactly an equipotential surface, there will be some angular variation between its normal and the vertical as defined in terms of the geoid. This disagreement in vertical between the geoid and reference ellipsoid is called the “deflection of the vertical” (or “station error”) and usually is less than 0.30 mrad; over land masses it seldom exceeds 0.10 mrad. Thus, in this section and for the remainder of the book we shall regard the Earth’s shape to be the reference ellipsoid.

Returning now to Fig. 3.1, we see that we have located a point in the meridian plane by the coordinates (x, y) . Obviously the semimajor and semiminor axes of the ellipse are, respectively, the intersection of the ellipse with the edge-view of the equatorial plane and the normal to this plane, i.e., R_e and R_p . The subscripts refer to equator e and pole p . We previously introduced two shape parameters, the ellipticity e and the eccentricity k , in Eqs. (3.2a) and (3.2b). From elementary geometry, the equation of the ellipse is given as

$$(x^2/R_e^2) + (y^2/R_p^2) = 1 \quad (3.3)$$

The derivative, or slope, of the ellipse at a general point $(\bar{x}, \bar{y})$ is

$$\left. \frac{dy}{dx} \right|_{@ \bar{x}, \bar{y}} = - \left[\frac{\bar{x}}{\bar{y}} \right] \left[\frac{R_p^2}{R_e^2} \right] \quad (3.4)$$

The slope of the normal to the ellipse at the point $(\bar{x}, \bar{y})$ is the negative reciprocal of the slope defined in Eq. (3.4). Now let (x_n, y_n) be arbitrary points on the normal line; using the point-slope form for the equation of a straight line, we have, for the equation of the normal,

$$\frac{x_n - \bar{x}}{y_n - \bar{y}} = \left[\frac{\bar{y}}{\bar{x}} \right] \left[\frac{R^2}{R_p^2} \right] \quad (3.5)$$

The intersection of the normal line with the x axis (as the edge-view of the Earth’s equatorial plane) is found by setting y_n equal to zero in the above equation and solving for x_n , as

$$x_n = \bar{x}_n = \bar{x}[1 - (R_p/R_e)^2] = K^2 \bar{x} \quad (3.6)$$

A glance at Eq. (3.2b) justifies the last entry in Eq. (3.6).

In Fig. 3.1, $\mathbf{r}$ is the vector from the center of the Earth to the RV. The RV has a geometric altitude z along the local normal. Since we are accepting the ellipse as an adequate approximation to the geoid, z is measured along the local vertical. From the law of cosines, we have, for the magnitude of $\mathbf{r}$,

$$r^2 = R_E^2 + z^2 + 2R_E z \cos(D_0) \quad (3.7)$$

The succeeding discussion parallels the more elaborate treatment of Britting.² In Eq. (3.7), D_0 is the angular deviation of the local normal. Also note that R_E is not precisely the radius of the Earth, for with an assumption of an ellipsoidal Earth the term "radius" must be redefined. Here we take R_E as a vector from the Earth's center to the point on the Earth's surface below the RV; "below" is taken along the ellipsoidal normal from the RV to the Earth's surface. R_E will of course be latitude-dependent.

Returning to Eq. (3.7), we complete the square to get

$$r = [(R_E + z)^2 - 2R_E^2 z (1 - \cos D_0)]^{-1/2} \quad (3.8a)$$

$$r = (R_E + z)^2 \left[1 - \frac{2zR_E(1 - \cos D_0)}{(R_E + z)^2} \right]^{-1/2} \quad (3.8b)$$

Now $(1 - \cos D_0)$ is approximately $D_0^2/2$. It will be shown subsequently that D_0 at a latitude of 45 deg is equal to the ellipticity. The ellipticity, as we have noted, is 1/297 rad or about 0.19 deg. Applying the small-angle approximation to D_0 , we can expand Eqs. (3.8) to give

$$r = (R_E + z) - \frac{zR_E D_0^2}{2(R_E + z)} - \frac{z^2 R_E^2 D_0^4}{8(R_E + z)^3} \quad (3.9)$$

We may evaluate the significance of the various terms in Eq. (3.9). Assume that D_0 takes on its value at a latitude of 45 deg, i.e., $D_0 = 1/297$, that the altitude of re-entry is 300 km, and that the Earth's radius is 6370 km. The evaluation of the first two terms in Eq. (3.9) gives

$$r = R_E + z - z(5.413 \times 10^{-6})$$

With $z = 300$ km, the magnitude of the third term is about 1.6 m (5.3 ft). The conclusion, then, is that the following approximation to Eqs. (3.8) is valid to within less than 2 m:

$$r = R_E + z \quad (3.10)$$

3.3 Deviation of the Vertical

When we consider the Earth's geopotential surface to be represented better by an ellipsoid than by a sphere, some commonly held ideas become a bit ambiguous. For example, the idea of vertical or "up-and-down" can have alternate definitions. One definition for vertical might be the vector from the center of the Earth through a point of interest on the Earth's surface. This vertical is called "geocentric vertical." An alternate definition might define vertical as the vector coincident with the gravity vector. Vertical, according to this definition is called "geographic vertical." In a spherically shaped Earth with radially symmetric mass distribution, the geocentric and geographic verticals are identical. For an ellipsoidal Earth, there are two verticals. The angle between the geocentric and geographic vertical is noted in Fig. 3.1 by the symbol D_0 for a point on the Earth's surface and by D for a point at an altitude z above the Earth's surface. From Fig. 3.1, the angles D and D_0 , called the deviation of the normal, are expressible in terms of the geocentric and geographic latitudes, L_c and L , as

$$D = L - L_c \quad D_0 = L - L_{c_0} \quad (3.11)$$

By the law of sines we have

$$\frac{\sin D}{K^2 \bar{x}} = \frac{\sin(\pi - L)}{r} \quad (3.12)$$

We must remember that $\bar{x}$ is the x component of the point on the Earth's surface directly below the RV and that r is the magnitude of the vector from the Earth's center to the RV. We may write $\bar{x}$ as

$$\bar{x} = R_E \cos L_{c_0} \quad (3.13a)$$

but

$$L_{c_0} = L - D_0 \quad (3.13b)$$

Thus, we may write

$$\bar{x} = R_E (\cos L \cos D_0 + \sin L \sin D_0) \quad (3.13c)$$

Substituting Eqs. (3.2c), (3.10), and (3.13c) into Eq. (3.12), we get

$$\begin{aligned} \sin D = e \left[\frac{R_E}{R_E + Z} \right] \left[1 - \frac{e}{2} \right] \sin 2L \cos D_0 \\ + 2e \left[1 - \frac{e}{2} \right] \left[\frac{R_E}{R_E + Z} \right] \sin^2 L \sin D_0 \end{aligned} \quad (3.14)$$

Evaluating D at $z = 0$, i.e., $D = D_0$, we get

$$D_0 = e \sin 2L + \Delta_0 \quad (3.15a)$$

where Δ_0 is given as

$$\Delta_0 = -(e/2) \sin 2L + 2e^2 \sin 2L \sin^2 L \quad (3.15b)$$

At a latitude L of 45 deg and e of $1/297$ rad, Δ_0 is about 5.6×10^{-6} rad or 1.16 arc-s. The first term in Eq. (3.15a) is about 700 arc-s. We might be justified in ignoring this second term in applications.

A similar expansion may be carried out on Eq. (3.14). Following closely the work of Britting,² we get

$$D = e \sin 2L + \Delta \quad (3.16a)$$

where

$$\Delta = -e \sin 2L [(e/2) + (z/R_E)] \quad (3.16b)$$

Again taking the re-entry altitude at 300 km and an Earth's radius of 6370 km, the Δ term is about 1.64×10^{-4} rad or 34 arc-s. The first term on the right in Eq. (3.16a) is about 700 arc-s.

3.4 Earth's Radius

Earlier we introduced the vector R_E as the Earth's radius. Specifically, this vector was defined as a vector from the origin of the reference ellipsoid to the point $(\bar{x}, \bar{y})$ on its surface. From Fig. 3.1, we may write the Cartesian coordinates $(\bar{x}, \bar{y})$ in polar form as

$$\bar{x} = R_E \cos L_c \quad \bar{y} = R_E \sin L_c \quad (3.17)$$

Inserting in Eq. (3.3), we get

$$R_E = \frac{R_p}{[1 - \{1 - (R_p/R_E)^2\} \cos L_c]} \quad (3.18)$$

The quantity in braces in the above equation is the eccentricity squared [see Eq. (3.2b)]. We may expand the denominator in series of K^2 as

$$R_E = R_p \left[1 - \frac{K^2}{2} \cos^2 L_{c_0} + \frac{3K^4}{8} \cos^4 L_{c_0} + \frac{5K^6}{16} \cos^6 L_{c_0} + \cdots \right] \quad (3.19)$$

L_{c_0} is the geocentric latitude; the geographic latitude L can be alternately used instead through Eq. (3.11). The deviation of the vertical, D , has been

written in terms of the latitude L and the ellipticity e in Eqs. (3.14) and (3.15). The eccentricity K also might be written in terms of the ellipticity e after Eq. (3.2c). Thus, an equivalent to Eq. (3.19) might be written in terms of the ellipticity e as

$$R_E = R_p \left[1 - \frac{e}{2}(1 + \cos 2L) + \frac{e^2}{4} \left(\frac{13}{4} + 2 \cos 2L - \frac{5}{4} \cos 4L \right) + \cdots \right] \quad (3.20)$$

If we recognize that

$$R_p = R_e(1 - e)$$

we can get a slightly different form for Eq. (3.19), as

$$R_E = R_e \left[1 - \frac{e}{2}(1 - \cos 2L) + \frac{5}{16}e^2(1 - \cos 4L) - \cdots \right] \quad (3.21)$$

Retaining only the first two terms provides a useful approximation:

$$R_E = R_e(1 - e \sin^2 L) \quad (3.22)$$

The coefficient of the third term in Eq. (3.21), i.e., $R_e(5/16)e^2$, is about 22 m. There seems to be little point in retaining terms of the order of e^2 or higher, since R_e is probably not known more accurately than 20 m. Again Eq. (3.22) provides us with the Earth's radius as a function of geographic latitude L .

3.5 Earth's Gravity Field

In the preceding sections, we have discussed analytic descriptions of the Earth's shape. It was shown that the Earth is not exactly spherical, but may be described more accurately as an ellipsoid of revolution. Since the Earth has neither an exactly spherical shape nor homogeneous mass distribution, we can anticipate small but important departures of the gravity field from the classical inverse-square model.

Accurate knowledge of the Earth's field is important both for RV guidance and navigation and for precise trajectory simulation. If we restrict guidance to the establishment of initial conditions to be set at termination of the boost phase, the influence of gravity must be anticipated over the ballistic trajectory with an accuracy that cannot be met by an inverse-square model. In looking at the navigation problem, we recognize that accelerometer outputs are proportional to the difference between inertially referenced acceleration and the gravitational influence at the system's position. Since acceleration must be inferred from the accelerometer output by vectorially adding the local gravitational force, accuracy of position (integration of acceleration) depends directly upon the accuracy of the gravitational field model. Although it is not applicable to RV guidance and navigation, it

should be pointed out that an aircraft-type navigation requires only two navigational coordinates (latitude and longitude) because a local-level platform is used to maintain the axis of sensitivity of the accelerometers normal to the gravitational vector. In the aircraft case, the gravity vector is resolved in the local frame established by the platform such that the gravity vector is always normal to the platform. However, errors in establishing the gravity vector mean that there will be small components of gravity in the plane of the table.

Even though the gravitational field will be much more complex analytically than the inverse square, the modified representation will still be that of a conservative field. Thus, the gravitational field $\bar{g}$ is a vector field derivable from a scalar function, called the gravity potential Φ , as

$$\bar{g} = \nabla \Phi \quad (3.23)$$

In the equations below, we locate the RV by the vector $\bar{r}$ given in spherical coordinates (r, α, λ) . The potential comes from the distributed mass of the Earth with density D . The mass element dm is located by the vector $\bar{\rho}$, which has spherical coordinates (ρ, β, θ) . Again the vector $\bar{r}$ locates the RV, whereas the vector $\bar{\rho}$ locates the mass element dm . These two vectors might be expressed as column matrices as

$$\bar{r} = \begin{bmatrix} r \sin \alpha \cos \lambda \\ r \sin \alpha \sin \lambda \\ r \cos \alpha \end{bmatrix} \quad \bar{\rho} = \begin{bmatrix} \rho \sin \beta \cos \theta \\ \rho \sin \beta \sin \theta \\ \rho \cos \beta \end{bmatrix} \quad (3.24)$$

The gravitational potential is derived from the following volume integral:

$$\Phi = K \iiint \frac{dm}{|\bar{r} - \bar{\rho}|} \quad (3.25)$$

where K is the universal gravity constant. The denominator in Eq. (3.25) may be written as

$$|\bar{r} - \bar{\rho}| = [r^2 + \rho^2 - 2r\rho \cos \gamma]^{1/2} = r[1 + (\rho/r)^2 - 2(\rho/r) \cos \gamma]^{1/2} \quad (3.26)$$

Evaluating outside the Earth, i.e., $r > \rho$, we can expand the reciprocal of Eq. (3.25) to get

$$\begin{aligned} |\bar{r} - \bar{\rho}|^{-1} = r^{-1} [1 + (\rho/r) \cos \gamma + \frac{1}{2}(\rho/r)^2 (3 \cos^2 \gamma - 1) \\ + (1/r)(\rho/r)^3 \cos \gamma (5 \cos^3 \gamma - 3) + \dots] \end{aligned} \quad (3.27)$$

where γ is the angle between the $\bar{r}$ vector and the $\bar{\rho}$ vector. Since Eq. (3.26) represents the Legendre generating function, we may write the above in

terms of Legendre polynomials as

$$|\vec{r} - \vec{\rho}|^{-1} = r^{-1} \sum_k P_k \cos \gamma \frac{\rho^k}{r^k} \quad (3.28)$$

In Eq. (3.28) we might point out that the Legendre polynomial has the following generating function credited to Rodrigues:

$$P_k(\nu) = \frac{1}{2^k k!} \frac{d^k (\nu^2 - 1)^k}{d\nu^k} \quad (3.29)$$

where $P_k(\cos \gamma)$ can be determined by carrying out the differentiation in the variable ν and then replacing ν by $\cos \gamma$.

Returning now to Eq. (3.25), we get

$$\Phi(r, \alpha, \lambda) = \frac{K}{r} \iiint \sum_k P_k(\cos \gamma) \left(\frac{\rho}{r} \right)^k dm \quad (3.30)$$

In Fig. 3.2 we have indicated a differential element that will be integrated over the volume. The geometry of this differential element is given in Fig. 3.3.

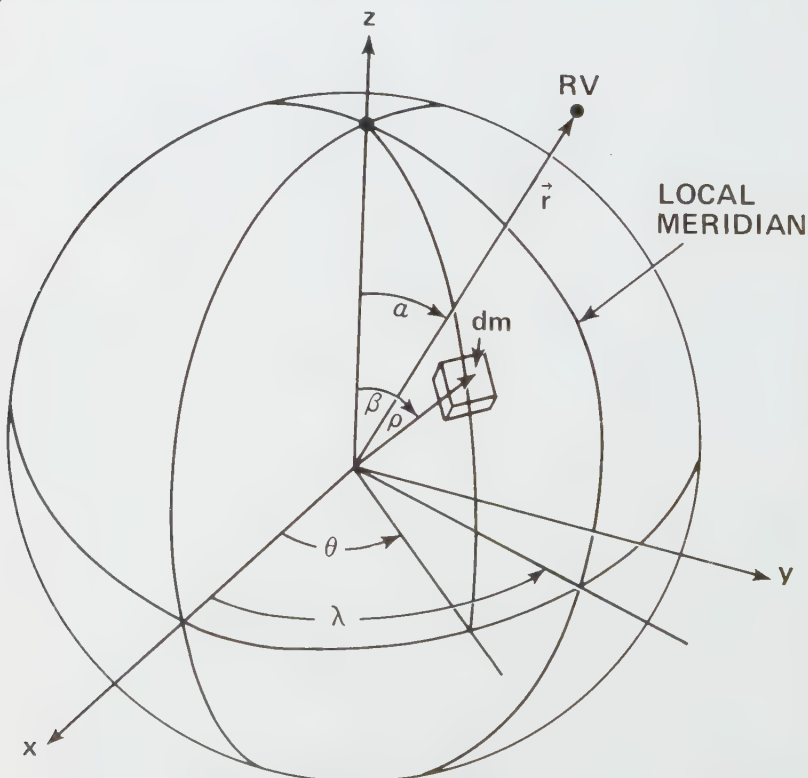


Fig. 3.2 Gravitational geometry.

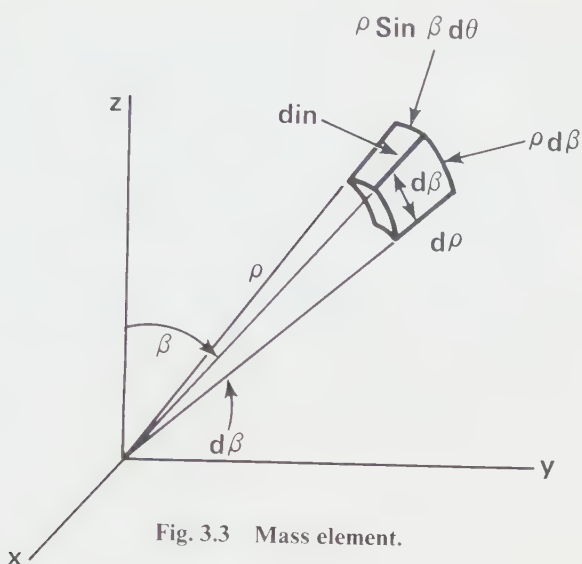


Fig. 3.3 Mass element.

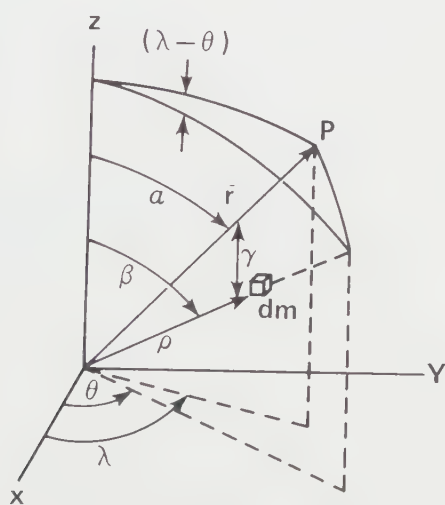


Fig. 3.4 Gravitational potential geometry.

In Eq. (3.30) the fundamental variable is γ , but the volume integral is taken more conveniently over the variables (α, β, θ) . From the law of cosines of spherical trigonometry, we may write

$$\cos \gamma = \cos \beta \cos \alpha + \sin \beta \sin \alpha \cos (\lambda - \theta) \quad (3.31)$$

Some of the geometry is shown in Fig. 3.4.

Equation (3.31) then is used to convert variables from γ to the more convenient set (α, β, θ) . Through the addition formula of spherical harmonics, we get

$$P_k(\cos\gamma) = P_k(\cos\alpha) P_k(\cos\beta) + 2 \sum_{j=1}^k \frac{(k-j)!}{(k+j)!} \cos j(\lambda - \theta) P_k^j(\cos\alpha) P_k^j(\cos\beta) \quad (3.32)$$

where $P_k^j(\nu)$ are the associated Legendre functions of the first kind, degree k , order j , and related to the Legendre polynomials through the following:

$$P_k^j(\nu) = (1-\nu^2)^{j/2} \frac{d^j}{d\nu^j} P_k(\nu) \quad (3.33)$$

Using Eq. (3.32), we may now write for Eq. (3.30), after it has been transformed by using Eq. (3.32),

$$\Phi(r, \alpha, \lambda) = \sum_{k=0}^{\infty} \frac{A_k}{r^{k+1}} P_k(\cos\alpha) + \text{periodic functions in } \theta \quad (3.34)$$

where

$$A_k = \iiint \rho^k P_k(\cos\beta) dm \quad (3.35)$$

We have not put the rather complicated relationships, which we have designated as "periodic functions in θ ," into Eq. (3.32). For the case of the Earth, this term will not make any contribution because the reference ellipsoid has symmetry about the polar axis z ; i.e., the density D depends only on ρ and β as

$$D(\rho, \beta, \theta) = D(\rho, \beta)$$

Equation (3.32) may be written in three terms by letting $k=0$, $k=1$, $k \geq 2$ to give

$$\begin{aligned} \Phi(r, \alpha) = & \frac{k}{r} \iiint dm + \frac{k}{r^2} \cos\alpha \iiint \rho \cos\beta dm \\ & + \sum_{k=2}^{\infty} \frac{A_k}{r^{k+1}} P_k(\cos\alpha) \end{aligned} \quad (3.36)$$

The second term may be dropped when it is recognized that $\rho \cos\beta$ is the distance from the equatorial plane to the elementary mass dm . In our reference ellipsoid, the center of the ellipsoid is at the center of mass, and so the integral of $\rho \cos\beta$ over the body must vanish. Finally we have for the gravita-

tional potential

$$\Phi(r, \alpha) = \frac{\mu}{r} + \sum_{k=2}^{\infty} \frac{A_k}{r^{k+1}} P_k(\cos \alpha) \quad (3.37)$$

where $\mu = kM_e$ or the product of the Earth's mass with the gravitational constant k . Letting

$$\frac{R_e^k}{R_e^k r^{k+1}} = \left[\frac{R_e}{r} \right]^k \frac{1}{R_e^k r} \quad (3.38)$$

we may rewrite Eq. (3.37) as

$$\Phi(r, \alpha) = \frac{\mu}{r} \left[1 - \sum_{k=2}^{\infty} \left[\frac{R_e}{r} \right]^k J_k P_k(\cos \alpha) \right] \quad (3.39)$$

where

$$J_k = -(A_k / R_e^k \mu)$$

The coefficients J_k are determined experimentally. Table 3.1 is a listing of the numerical values of quantities that appear in Eq. (3.39). The Legendre polynomials with indices $k = 2, 3, 4$ are given as

$$P_2(\cos \alpha) = \frac{1}{2}(3\cos^2 \alpha - 1) = \frac{1}{4}(3\cos 2\alpha + 1)$$

$$P_3(\cos \alpha) = \frac{1}{2}(5\cos^3 \alpha - 3\cos \alpha) = \frac{1}{8}(5\cos 3\alpha + 3\cos \alpha)$$

$$P_4(\cos \alpha) = \frac{1}{8}(35\cos^4 \alpha - 3\cos^2 \alpha + 3) = (1/64)(35\cos 4\alpha + 20\cos 2\alpha + 9)$$

Table 3.1 Earth Constants

Symbol	Definition	Value	Uncertainty	Units
μ	$kM_e, g_e R_e^2$	3.9853×10^{14}	0.1 (%)	m^3/s^2
R_e	Equatorial radius	6379.388	22 (m)	km
J_2	...	1.0827×10^{-3}	0.00020×10^{-3}	...
J_3	...	-2.3×10^{-6}	0.1×10^{-6}	...
J_4	...	1.8×10^{-6}	0.2×10^{-6}	...
e	Ellipticity	1/297	...	...

Finally we get, for the Earth's potential through J_4 ,

$$\begin{aligned}\Phi(r, \alpha) = & \mu \frac{1}{r} - \frac{J_2}{2} \frac{R_e^2}{r^3} (3\cos^2\alpha - 1) - \frac{J_3}{2} \frac{R_e^3}{r^4} (5\cos^3\alpha - 3\cos\alpha) \\ & - \frac{J_4}{8} \frac{R_e^4}{r^5} (35\cos^4\alpha - 30\cos^2\alpha + 3)\end{aligned}\quad (3.40)$$

The even harmonics are seen to give rise to the oblate terms, which are symmetric about the equatorial radius; the odd harmonics are antisymmetric about the equatorial radius and are the so-called "pear-shape" terms. It should be re-emphasized that the angle α is the co-latitude.

In Eq. (3.23) we indicated that the gravity vector is the gradient of the potential Φ . Thus we may write

$$G_r \mathbf{i}_r + G_\alpha \mathbf{i}_\alpha = \left[\frac{\partial}{\partial r} (\) \mathbf{i}_r + \frac{1}{r} \frac{\partial}{\partial \alpha} (\) \mathbf{i}_\alpha \right] \Phi \quad (3.41)$$

Inserting Eq. (3.40) into (3.41) gives

$$\begin{aligned}G_r = & -\frac{\mu}{r^2} \left[1 - \frac{3}{2} J_2 \left[\frac{R_e}{r} \right]^2 (3\cos^2\alpha - 1) - 2J_3 \left[\frac{R_e}{r} \right]^3 \cos\alpha (5\cos^2\alpha - 3) \right. \\ & \left. - \frac{5}{8} J_4 \left[\frac{R_e}{r} \right]^4 (35\cos^4\alpha - 30\cos^2\alpha + 3) \right]\end{aligned}\quad (3.42a)$$

$$\begin{aligned}G_\alpha = & 3 \frac{\mu}{r^2} \left[\frac{R_e}{r} \right]^2 \sin\alpha \cos\alpha \left[J_2 + \frac{1}{2} J_3 \left[\frac{R_e}{r} \right] \sec\alpha (5\cos^2\alpha - 1) \right. \\ & \left. + \frac{5}{6} J_4 \left[\frac{R_e}{r} \right]^2 (7\cos^2\alpha - 3) \right]\end{aligned}\quad (3.42b)$$

The gravity vector may be written in column form as

$$\bar{\mathbf{g}} = \begin{bmatrix} -G_r \\ 0 \\ -G_\alpha \end{bmatrix}$$

where, because of the assumptions of the Earth's rotational symmetry, there is no component of the gravity vector out of the meridian plane.

A comment might be in order concerning Eq. (3.42a). It would appear that the gravitational acceleration is a minimum at the Earth's poles, since the term containing J_2 is a maximum at zero co-latitude. However, if the

variable r is taken as the distance from the Earth's center, then we have shown in Eq. (3.22) that, if $r = R_E$, then r is latitude-dependent. Replacing L by $90 - \alpha$, we get for r

$$r = R_E = R_e(1 - e \cos^2 \alpha) \quad (3.43)$$

It can be shown that the gravitational acceleration diminishes monotonically from the poles to the equator (as α goes from 0 to 90 deg).

A computer program is given in Table 3.2 for calculating the gravitational

Table 3.2 Computer Program for Calculating the Gravitational Acceleration

```

00100 REM THIS PROGRAM CALCULATES THE MAGNITUDE OF THE GRAVITY
00110 REM VECTOR COMPONENTS G1 AND G2 IN THE RADIAL, R, AND COLATITUDINAL, A
00120 REM DIRECTIONS. M1 IS THE GRAVITY CONSTANT TIMES THE EARTH'S
00130 REM MASS. R1 IS THE EQUATORIAL RADIUS OF THE EARTH. J2 AND J4
00140 REM ARE THE OBLATENESS CONTRIBUTIONS AND J3 IS THE PEAR-SHAPE (PS)
00150 REM CONTRIBUTION. THE RADIAL COMPONENT OF THE GRAVITY VECTOR, G1,
00160 REM OBLATENESS TERM, AND K3, THE PEAR SHAPE TERM. THE
00170 REM CO-LATITUDINAL COMPONENT, G2, IS SEPARATED INTO THE H2,
00180 REM OBLATENESS TERM, AND H3, THE PEAR SHAPE TERM. THE
00190 REM COORDINATES OF A POINT IN THE FIELD ARE R, THE RADIUS ( METERS)
00200 REM AND A, THE CO-LATITUDE (DEG). IF USED TO CALCULATE GRAVITY AT
00210 REM THE EARTH'S SURFACE E IS THE EARTH ELLIPTICITY.
00220 R1 = 6.378388E6
00230 M1 = 3.9853E14
00240 J2 = 1.0827E-3
00250 J3 = (-1)*2.3E-6
00260 J4 = (-1)*1.8E-6
00270 E = 1/297
00280 Z9 = 3.14159/(180)
00290 FOR J = 1 TO 2
00300 FOR I = 0 TO 90 STEP 5
00310 A = I*Z9
00320 R = R1*(1-E*((COS(A))^2))
00330 R2 = R1/R
00340 S = SIN(A)
00350 C = COS(A)
00360 K1 = (-1)*M1*(1/(R^2))
00370 K2 = (3/2)*J2*(R2^2)*(3*C*C-1)
00380 K2 = K2+(5/8)*J4*(R2^4)*(35*(C^4)-30*C*C+3)
00390 K2 = (-1)*K1*K2
00400 K3 = (-1)*K1*2*J3*(R2^3)*C*(5*C*C-3)
00410 G1 = K1+K2+K3
00420 H = 3*M1*(1/(R^2))*(R2^2)*S*C
00430 H2 = H*(J2+(5/6)*J4*(R2^2)*(7*C*C-3))
00440 H3 = 3*M1*(1/(R^2))*(K2^3)*S*(.5)*J3*(5*C*C-1)
00450 G2 = H2 + H3
00460 IF J =>2 THEN 00530
00470 IF I > 0 THEN 00510
00480 PRINT #CO-LAT.,#,INV. SQ.,#TOT. RAD. AC.,#TOT. CO-L AC.,#RADIUS#
00490 PRINT #DEGREES#,M/SEC-SQ.,#M/SEC-SQ.,#M/SEC-SQ.,#K1,K2,K3
00500 PRINT
00510 PRINT A/Z9,K1,G1,G2,R/1000
00520 IF J = < 1 THEN 00590
00530 IF I > 0 THEN 00580
00540 PRINT
00550 PRINT
00560 PRINT #CO-LAT.,#,RAD-OBL.,#,RAD PS.,#CO-LAT OBL.,#CO-LAT PS#
00570 PRINT
00580 PRINT A/Z9,K2,K3,H2,H3
00590 NEXT I
00600 NEXT J
00610 END

```


acceleration at the Earth's surface as a function of co-latitude. This program could be used as a subroutine when it is necessary to calculate the gravitational acceleration at a general point above the Earth's surface. Adaptation would involve eliminating lines 100 to 210 and 480 to 610 inclusive. In line 320, R would have to be replaced by $R_E + z$, where R_E is given in Eq. (3.43) and z is the distance above the Earth along the local normal to the reference ellipsoid (see Fig. 3.1). Program output at 5-deg increments in co-latitude is given in Table 3.3. It will be noted that the gravitational acceleration decreases somewhat with increasing co-latitude.

The column indicated as "INV.-SP" provides the simple spherical gravity field introduced at the beginning of this chapter [see Eq. (3.1c)]. The reason that this quantity is not constant with co-latitude is that the surface of the Earth, as represented by the ellipsoid, is not at a constant distance from the Earth's mass-center. The remaining columns are the total radial and total

Table 3.3 Gravitational Acceleration Components

CO-LAT. DEGREES	INV. SQ. M/SEC-SQ	TOT. RAD. AC M/SEC-SQ	TOT. CO-L AC M/SEC-SQ	RADIUS KM
0	-9.86208	-9.83001	0	6356.91
5.	-9.86158	-9.82987	2.77249E-3	6357.08
10.	-9.86007	-9.82945	5.46071E-3	6357.56
15.	-9.85762	-9.82877	7.98294E-3	6358.35
20.	-9.85429	-9.82785	1.02626E-2	6359.42
25.	-9.85019	-9.82671	1.22304E-2	6360.75
30.	-9.84544	-9.82539	1.38268E-2	6362.28
35.	-9.8402	-9.82393	1.50035E-2	6363.98
40.	-9.83461	-9.82238	1.57248E-2	6365.79
45.	-9.82885	-9.82079	1.59691E-2	6367.65
50.	-9.8231	-9.8192	1.57292E-2	6369.51
55.	-9.81752	-9.81766	1.50124E-2	6371.32
60.	-9.8123	-9.81622	1.38406E-2	6373.02
65.	-9.80758	-9.81492	1.22492E-2	6374.55
70.	-9.8035	-9.81381	1.02863E-2	6375.88
75.	-9.8002	-9.81291	8.01130E-3	6376.95
80.	-9.79777	-9.81225	5.49270E-3	6377.74
85.	-9.79628	-9.81185	2.80654E-3	6378.22
90.	-9.79578	-9.81172	3.38378E-5	6378.39

CO-LAT.	RAD-UBL.	RAD PS	CO-LAT OBL	CO-LAT PS
0	3.21599E-2	-9.16538E-5	0	0
5.	3.17925E-2	-8.95597E-5	2.78436E-3	-1.18669E-5
10.	3.07019E-2	-8.34146E-5	5.48367E-3	-2.29617E-5
15.	2.89216E-2	-7.36214E-5	8.01550E-3	-3.25663E-5
20.	2.65065E-2	-6.08190E-5	1.03026E-2	-4.00664E-5
25.	2.35308E-2	-4.58377E-5	1.22754E-2	-4.49943E-5
30.	2.00861E-2	-2.96402E-5	1.38739E-2	-4.70599E-5
35.	.016278	-1.32546E-5	1.50497E-2	-4.61702E-5
40.	1.22228E-2	2.29656E-6	1.57673E-2	-4.2431E-5
45.	8.04442E-3	1.60660E-5	1.60053E-2	-3.61488E-5
50.	3.86993E-3	2.72451E-5	.015757	-2.77872E-5
55.	-1.73995E-4	3.52168E-5	1.50304E-2	-1.79538E-5
60.	-3.96498E-3	3.95944E-5	1.38479E-2	-7.34791E-6
65.	-7.38861E-3	4.02448E-5	1.22459E-2	3.28612E-6
70.	-1.03418E-2	3.72942E-5	1.02731E-2	1.32087E-5
75.	-1.27359E-2	3.11166E-5	7.98956E-3	2.17347E-5
80.	-.014499	2.23058E-5	5.46443E-3	2.82784E-5
85.	-1.55785E-2	1.16344E-5	2.77414E-3	3.23923E-5
90.	-1.59419E-2	1.79359E-10	4.23910E-8	3.37955E-5

co-lateral components of gravitational acceleration as given by Eq. (3.39). The oblate and pear-shape components are given in the second series of columns.

Further descriptions of the Earth's gravity field would include a discussion of the deflection of the gravity vector due to topological irregularities such as mountain ranges, coastal shelves, and oceanic trenches. Such information is available from gravimetric surveys. Gravity deflections due to such effects have components both in and out of the meridian plane. Out of the meridian plane, the deflection can be as high as 10 arc-s, which is equivalent to about 5×10^{-4} m/s². Comparison shows that local out-of-meridian deflection can be an order of magnitude greater than the pear-shape effect. Further information on gravity vector deviations may be found in Heiskanen and Venig-Meinesz's book.³ This chapter is a close copy of parts of Brittings' excellent treatment.² Another readable short treatment of the Earth's field is that of Kaplan.⁴

References

¹Koelle, H.H. (ed.), *Handbook of Astronautical Engineering*, McGraw-Hill, New York, 1961, Chap. 4.

²Britting, K. R., *Inertial Navigation Systems Analysis*, Wiley, and Sons, Inc., New York, 1971.

³Heiskanen, W. A. and Venig-Meinesz, F.A., *The Earth and Its Gravity Field*, McGraw Hill, New York, 1958.

⁴Kaplan, M., *Modern Space Craft Dynamics and Control*, Wiley, New York, 1976.

IV. FORCE EQUATIONS

4.1 Newton's Second Law

In addition to his speculations concerning gravity, Newton also laid the foundations for mechanics in what has become known as Newton's "three laws of motion." The most significant of these laws for our purposes is the second, which relates force, the agent that causes motion, to the kinematic entity called acceleration. The constant of proportionality between the applied force and the resulting acceleration is the mass. A re-entry body or any endo- or exo-atmospheric body is subjected to a variety of forces, some of which may be aerodynamic, gravitational, solar, magnetic, or others. Some of these forces may be negligibly small over certain portions of the trajectory, only to dominate the physics at a later segment. For example, for an RV moving along an exo-atmospheric trajectory, the dominant force would be gravitational. Atmospheric, i.e., aerodynamic, forces would be negligibly small or, by definition, zero. On the other hand, during a later endo-atmospheric portion of the trajectory, when aerodynamic drag can cause decelerations of the order of 100 g, we might be justified in ignoring gravitational effects. Before we attempt to write the force equations, it is important to review some concepts in vector algebra.

4.2 Vector Differentiation

Before exploring the rudiments of vector calculus, a few words should be said about vectors and their physical significance. Frequently, there seems to be confusion between the meaning or definition of a vector and the definition of the frame into which the vector is resolved. One example may clarify this point. Suppose that we track an RV by measuring the range to the RV and the angles that this range vector makes with some convenient orthogonal axis system or station at the Earth's surface. The availability of the magnitude and directional angles of this range vector is equivalent to saying that we have determined its components in the station frame. Determination of the components of a vector in a given frame is defined as resolving the vector in that frame. However, the components of a vector will vary depending upon the frame orientation. Furthermore, the vector that represents the rate of change of a vector will depend, in addition, upon the angular velocity of the frame. Thus, the derivative of a vector cannot be uniquely defined in terms of its components in a particular frame.

In the present example of an RV moving above an Earth-based station, the range vector is defined as the vector from the origin of this Earth frame to the center of mass of the RV. This definition says nothing about the frame into which the vector may be resolved. The range vector can be resolved into the Earth frame; it also can be resolved into a frame fixed to the RV. Resolving the vector into a frame fixed to the RV often causes some confusion. It might appear that we are describing the position of the RV's center of mass from the RV's center of mass. However, the vector that we are resolving is not the null vector (RV center of mass to RV center of mass) but still is the vector from the Earth station to the center of mass of the RV. Such a resolution may have to be carried out conceptually; there might well be no instruments in existence that will measure the components of a given vector directly in a given frame: this condition of practical measurement of components is irrelevant in carrying out vector manipulations, although it might well set engineering requirements. The main point here is that the definition of a vector and the definition of the frame into which it may be resolved are two distinct concepts.

In this section we also shall consider the rate of change of a vector. This vector derivative is also a vector. Returning to our example of the RV being observed from a surface station, we can resolve the range vector in the Earth station frame at successive time intervals. It might appear that we then can define the vector derivative, i.e., the vector representing the rate of change of the range vector, by numerically differentiating components of the range vector. However, the Earth frame into which we are resolving our vector has the angular rotation that it shares with the Earth. Thus, part of the observed rate of change of components is due to range vector directional change, and part is due to the angular rotation of the Earth. In this section we shall consider how to separate the two contributions to obtain the rate of rotation of the range vector relative to some Earth frame that is nonrotating (inertial frame). The important distinction here is that a vector representing the derivative of a position vector relative to a moving frame is different from the vector representing the derivative of the same position vector relative to an inertial or nonrotating frame. Even though the two vector derivatives are distinctly different, each can be resolved in any frame of convenience. For example, the vector representing the rate of change of the range vector relative to an inertial frame can be resolved in the moving Earth frame.

A vector is an entity that is useful in describing a physical phenomenon having both magnitude and direction. Our immediate physical application is to define the position of some object from a selected point called the origin. In the nontrivial case, the separation distance (magnitude) and direction are represented by the nonzero vector $\mathbf{R}$. Let R be the magnitude of $\mathbf{R}$ and $\mathbf{1}_R$ the unit vector indicating the direction of the object from the origin, as

$$\mathbf{R} = R \mathbf{1}_R \quad (4.1)$$

Since both magnitude and direction may change with time, we get, for the derivative of Eq. (4.1),

$$\frac{d\mathbf{R}}{dt} = \frac{dR}{dt} \mathbf{1}_R + R \frac{d\mathbf{1}_R}{dt} \quad (4.2)$$

Clearly the first term is the rate of change of the magnitude and the second the rate of change of direction. The vectoral sum of these two contributions provides the vector describing the rate of change of the vector $\mathbf{R}$ (see Fig. 4.1).

We note that the vector $\mathbf{R}$ at time t has changed both direction and magnitude at time $t + \Delta t$. The rotational vector ω_R is normal to the plane defined by the vectors $\mathbf{R}$ and $\mathbf{R} + \Delta\mathbf{R}$ or, equivalently, $\mathbf{R}(t)$ and $\mathbf{R}(t + \Delta t)$. Let us look down (along the negative of the ω_R vector) at the plane defined by $\mathbf{R}(t)$ and $\mathbf{R}(t + \Delta t)$ (see Fig. 4.2).

We make a distinction between the direction of the unit vectors $\mathbf{i}_R$ and $\mathbf{i}'_R$ before (without prime) and after (with prime) rotation. However, we shall ignore this distinction, since $\Delta\theta$ is considered to be an infinitesimal quantity, or will be when we apply limits in defining derivatives.

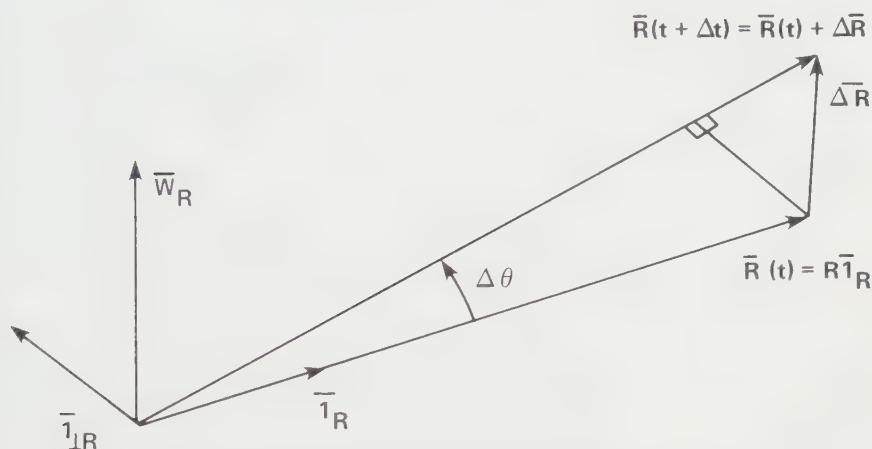


Fig. 4.1 Geometry of vectoral change in $\mathbf{R}$ vector.

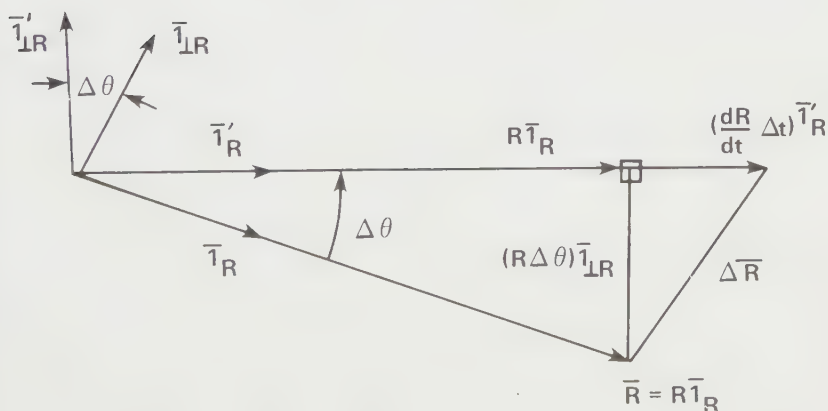


Fig. 4.2 Planar view of vectoral change in $\mathbf{R}$ vector.

The total vectoral change, $\Delta \mathbf{R} = (d\mathbf{R}/dt)\Delta t$, has two components:

$$\mathbf{l}_R \left\{ \frac{dR}{dT} \right\} \Delta t \quad (R \Delta \theta) \mathbf{l}_{\perp R} \quad (4.3)$$

as shown above. If the vector ω_R is the rotational rate of $\mathbf{R}$, then

$$\mathbf{l}_{\perp R} = \frac{\omega_R \times \mathbf{R}}{|\omega_R| R \sin(\omega_R, R)} = \frac{\omega_R \times \mathbf{R}}{\omega_R R} \quad (4.4)$$

$$\omega_R R \mathbf{l}_{\perp R} = \omega_R \times \mathbf{R}$$

where we have defined ω_R as always perpendicular to $\mathbf{R}$. From Eq. (4.2), we may rewrite the second term on the right using Eq. (4.4):

$$R \left\{ \frac{d\mathbf{l}_R}{dt} \right\} \Delta t = R \Delta \theta \mathbf{l}_{\perp R} = R \omega_R \Delta t \mathbf{l}_{\perp R} = (\omega_R \times \mathbf{R}) \Delta t \quad (4.5)$$

or

$$\frac{d\mathbf{l}_R}{dt} = \omega_R \times \mathbf{l}_R \quad (4.6)$$

Thus Eq. (4.2) becomes

$$\frac{d\mathbf{R}}{dt} = \frac{dR}{dt} \mathbf{l}_R + \omega_R \times \mathbf{R} \quad (4.7)$$

The first term on the right gives the velocity magnitude along the line of sight from the origin (observer) to the moving object. The second term gives the velocity due to motion of the object at a right angle to the line of sight.

The acceleration of the object follows from a differentiation of Eq. (4.7). In carrying out this differentiation, we shall encounter derivatives of $\mathbf{l}_R$, $\mathbf{R}$, and ω_R . The derivatives of $\mathbf{l}_R$ and $\mathbf{R}$ follow from Eqs. (4.6) and (4.7); the derivative of ω_R will lie along ω_R and perpendicular to $\mathbf{R}$, since ω_R by definition is the instantaneous rotation of $\mathbf{R}$ and must perforce be normal to the plane defined by $\mathbf{l}_R$ and $\mathbf{l}_{\perp R}$. Thus we get

$$\frac{d^2 \mathbf{R}}{dt^2} = \frac{d^2 R}{dt^2} \mathbf{l}_R + 2(\omega_R \times \mathbf{l}_R) \frac{dR}{dt} + \frac{d\omega_R}{dt} \times \mathbf{R} + \omega_R \times (\omega_R \times \mathbf{R}) \quad (4.8)$$

The four terms on the right may be defined as follows:

$$\frac{d^2 R}{dt^2} \mathbf{l}_R : \quad \text{due to acceleration in length of } \mathbf{R} \text{ vector}$$

$2(\omega_R \times \mathbf{l}_R) \frac{dR}{dt}$: due to changing length of the $\mathbf{R}$ vector while rotating at constant ω_R (strictly speaking, *not* the Coriolis acceleration)

$\frac{d\omega_R}{dt} \times \mathbf{R}$: tangential acceleration due to rate of change of angular velocity at constant position vector $\mathbf{R}$

$\omega_R \times (\omega_R \times \mathbf{R})$: centripetal acceleration

4.3 Equation of Coriolis

The equation of Coriolis arises when there are two axis systems rotating relative to each other. The rate of change of the vector $\mathbf{R}$ is different when viewed from each frame. Often in applications one frame is taken as inertial (nonrotating, nonaccelerating), whereas the second frame is taken as rotating relative to the first. Let us take one frame as O' (inertial) and a second as O (rotating).

We have (Fig. 4.3)

$$\mathbf{R} = \mathbf{R}_0 + \mathbf{r} \quad (4.9)$$

as the vector $\mathbf{R}$ in inertial space and

$$\dot{\mathbf{R}} = \dot{\mathbf{R}}_0 + \dot{\mathbf{r}} \quad (4.10)$$

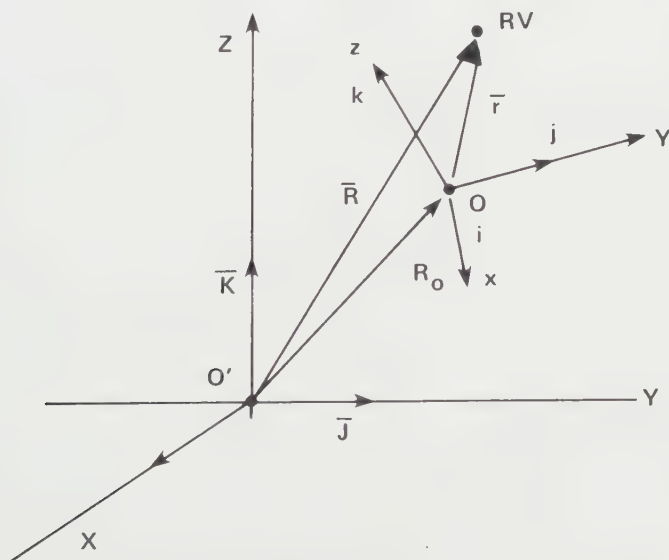


Fig. 4.3 Observation of $\mathbf{RV}$ from rotating and inertial frames.

as the derivative with respect to inertial space. The vectors $\mathbf{R}$, $\mathbf{R}_0$, $\mathbf{r}$ can be expressed in component form ($\mathbf{R}$ and $\mathbf{R}_0$ in the inertial frame and $\mathbf{r}$ in the moving frame):

$$\mathbf{R} = X\mathbf{I} + Y\mathbf{J} + Z\mathbf{K} \quad (4.11)$$

$$\mathbf{R}_0 = X_0\mathbf{I} + Y_0\mathbf{J} + Z_0\mathbf{K} \quad (4.12)$$

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad (4.13)$$

and, differentiating with respect to inertial space,

$$\dot{\mathbf{R}} = \dot{X}\mathbf{I} + \dot{Y}\mathbf{J} + \dot{Z}\mathbf{K} \quad (4.14)$$

$$\dot{\mathbf{R}}_0 = \dot{X}_0\mathbf{I} + \dot{Y}_0\mathbf{J} + \dot{Z}_0\mathbf{K} \quad (4.15)$$

$$\dot{\mathbf{r}} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k} + x\dot{\mathbf{i}} + y\dot{\mathbf{j}} + z\dot{\mathbf{k}} \quad (4.16)$$

Since the unit vectors $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ can change only at right angles to themselves and in a plane normal to the rotation vector $\boldsymbol{\omega}$, we have

$$\dot{\mathbf{i}} = \boldsymbol{\omega} \times \mathbf{i} \quad \dot{\mathbf{j}} = \boldsymbol{\omega} \times \mathbf{j} \quad \dot{\mathbf{k}} = \boldsymbol{\omega} \times \mathbf{k} \quad (4.17)$$

It then follows that

$$\left. \frac{d\mathbf{r}}{dt} \right|_I = \left. \frac{d\mathbf{r}}{dt} \right|_m + \boldsymbol{\omega}_{lm} \times \mathbf{r} \quad (4.18)$$

Equation (4.18) has some similarity to Eq. (4.7). However, Eq. (4.18) is a relationship between the velocity measured in a moving frame and the velocity measured in an inertial frame, i.e.,

$$\left. \frac{d\mathbf{r}}{dt} \right|_m \quad \left. \frac{d\mathbf{r}}{dt} \right|_I$$

respectively. The vector $\boldsymbol{\omega}_{lm} \times \mathbf{r}$ accounts for the fact that one frame is rotating relative to the other and thus that measurement of the "same thing," i.e., velocity, will differ by the vector $\boldsymbol{\omega}_{lm} \times \mathbf{r}$.

The second derivative of $\mathbf{r}$ or the acceleration may be written as follows:

$$\begin{aligned} \left. \frac{d^2\mathbf{r}}{dt^2} \right|_I &= \frac{d}{dt} \left[\left. \frac{d\mathbf{r}}{dt} \right|_m \right]_I + \frac{d}{dt} [\boldsymbol{\omega}_{lm} \times \mathbf{r}]_I = \frac{d}{dt} [\dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k}]_I + \frac{d\boldsymbol{\omega}_{lm}}{dt} \\ &\quad \times \mathbf{r} + \boldsymbol{\omega}_{lm} \times \left. \frac{d\mathbf{r}}{dt} \right|_I = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \ddot{z}\mathbf{k} + \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \ddot{z}\mathbf{k} + \frac{d\boldsymbol{\omega}_{lm}}{dt} \end{aligned}$$

$$\begin{aligned}
 \times \mathbf{r} + \boldsymbol{\omega}_{lm} \times \left. \frac{d\mathbf{r}}{dt} \right|_m + \boldsymbol{\omega}_{lm} \times \mathbf{r} &= \left. \frac{d^2\mathbf{r}}{dt^2} \right|_m + \boldsymbol{\omega}_{lm} \times \left. \frac{d\mathbf{r}}{dt} \right|_m + \frac{d\boldsymbol{\omega}_{lm}}{dt} \\
 \times \mathbf{r} + \boldsymbol{\omega}_{lm} \times \left. \frac{d\mathbf{r}}{dt} \right|_m + \boldsymbol{\omega}_{lm} \times (\boldsymbol{\omega}_{lm} \times \mathbf{r}) & \quad (4.19)
 \end{aligned}$$

Collecting terms, we may write

$$\begin{aligned}
 \left. \frac{d^2\mathbf{R}}{dt^2} \right|_I &= \left. \frac{d^2\mathbf{R}_0}{dt^2} \right|_I + \left. \frac{d^2\mathbf{r}}{dt^2} \right|_I \\
 \left. \frac{d^2\mathbf{R}}{dt^2} \right|_I &= \left. \frac{d^2\mathbf{R}_0}{dt^2} \right|_I + \left. \frac{d^2\mathbf{r}}{dt^2} \right|_m + 2\boldsymbol{\omega}_{lm} \times \left. \frac{d\mathbf{r}}{dt} \right|_m + \frac{d\boldsymbol{\omega}_{lm}}{dt} \times \mathbf{r} + \boldsymbol{\omega}_{lm} \times (\boldsymbol{\omega}_{lm} \times \mathbf{r}) \quad (4.20)
 \end{aligned}$$

Equations (4.18) and (4.20) are statements of Coriolis' theorem, i.e., relating the measurement of the same entity (velocity and acceleration) in two frames, one of which is rotating relative to the other. It is important that all five terms on the right of Eq. (4.20) be defined:

$\left. \frac{d^2\mathbf{R}_0}{dt^2} \right|_I$: linear acceleration of the moving frame relative to inertial space

$\left. \frac{d^2\mathbf{r}}{dt^2} \right|_m$: apparent acceleration of p in the moving frame

$2\boldsymbol{\omega}_{lm} \times \left. \frac{d\mathbf{r}}{dt} \right|_m$: Coriolis acceleration due to motion of p in moving frame

$\frac{d\boldsymbol{\omega}_{lm}}{dt} \times \mathbf{r}$: apparent acceleration of p due to angular acceleration of moving frame

$\boldsymbol{\omega}_{lm} \times (\boldsymbol{\omega}_{lm} \times \mathbf{r})$: apparent acceleration of p due to angular velocity of moving frame

The word "moving" has been applied to one of the frames. The first term above indicates that "moving" includes linear acceleration of the frame, i.e.,

$$\left. \frac{d^2\mathbf{R}_0}{dt^2} \right|_I$$

4.4 Motion Over the Earth's Surface

We pose the question concerning what is the apparent (or observed) acceleration of a body to an observer on the Earth's surface. The observer may select a natural frame such as "south, east, and up." Since the Earth is rotating relative to inertial space, such a frame is noninertial. Figure 4.4 illustrates the situation we are discussing.

First let us apply Eq. (4.20) to the point P on the Earth's surface. We have a moving frame and an inertial frame, both with origins coincident at the Earth's center at O :

$$\mathbf{R}_0 = 0 \quad (4.21)$$

$$\mathbf{r} = \mathbf{R}_E \quad \left. \frac{d\mathbf{r}}{dt} \right|_m = \dot{\mathbf{R}}_E = 0 \quad (4.22)$$

$$\boldsymbol{\omega}_{Im} = \boldsymbol{\Omega}_E \quad \dot{\boldsymbol{\omega}}_{Im} = \dot{\boldsymbol{\Omega}}_E = 0 \quad (4.23)$$

Thus, Eq. (4.20) gives for the acceleration at the Earth's surface at point P :

$$\mathbf{R}_E \equiv \left. \frac{d^2 \mathbf{R}_E}{dt^2} \right|_I = \boldsymbol{\Omega}_E \times (\boldsymbol{\Omega}_E \times \mathbf{R}_E) \quad (4.24)$$

All other terms in Eq. (4.20) vanish. Now let us put an inertial frame at O (the Earth's center) and a moving frame at the Earth's surface at P and look at the motion of object RV . Let

$$\left. \frac{d^2 \mathbf{r}}{dt^2} \right|_m = \mathbf{a}_B \quad \left. \frac{d\mathbf{r}}{dt} \right|_m = \mathbf{v}_B \quad \boldsymbol{\omega}_{Im} = \boldsymbol{\Omega}_E \quad (4.25)$$

$$\left. \frac{d^2 \mathbf{R}_0}{dt^2} \right|_I = \left. \frac{d^2 \mathbf{R}_E}{dt^2} \right|_I = \boldsymbol{\Omega}_E \times (\boldsymbol{\Omega}_E \times \mathbf{R}_E) \quad (4.26)$$

Now we use Eq. (4.20) again to get

$$\left. \frac{d^2 \mathbf{R}_B}{dt^2} \right|_I = \boldsymbol{\Omega}_E \times (\boldsymbol{\Omega}_E \times \mathbf{R}_E) + \mathbf{a}_B + \boldsymbol{\Omega}_E \times (\boldsymbol{\Omega}_E \times \mathbf{r}) + 2\boldsymbol{\Omega}_E \times \mathbf{v}_B \quad (4.27)$$

Again we must emphasize that $\mathbf{a}_B$ and $\mathbf{v}_B$ are the acceleration and velocity as observed in the moving frame, that is, the frame fixed to the Earth.

Now from Newton's second law,

$$\left. \frac{d^2 \mathbf{R}_B}{dt^2} \right|_I = \frac{\mathbf{F}}{m} = -g \frac{\mathbf{R}_E}{R_E} \quad (4.28)$$

where we have assumed that only gravity is acting upon the point RV. The gravitational direction is taken as along the negative of the unit vector from the Earth's center. Inserting Eq. (4.28) into (4.27) and solving for $\mathbf{a}_B$, the apparent acceleration, we get

$$\mathbf{a}_B = -g[\mathbf{R}_E/R_E] - 2\boldsymbol{\Omega}_E \times \mathbf{v}_B \quad (4.29)$$

We have neglected the two centrifugal terms in Eq. (4.27) as being negligible in comparison to $\boldsymbol{\Omega}_E \times \mathbf{v}_B$, provided we confine our attention to typical re-entry speeds, i.e., 3 km/s. For speeds less than about 3 km/s, the centrifugal term(s) might properly be included. Equation (4.29) indicates that the acceleration $\mathbf{a}_B$ measured by ground station at point P will not satisfy Newton's second law because of an "error" term $2\mathbf{R}_e \times \mathbf{v}_B$, which appears because the tracking or observation station at P is not an inertial frame. Sometimes a fictitious force called the "Coriolis force" is postulated to "explain" the observed acceleration. Such a "force" should not be introduced, since the perceived acceleration does not originate with the object under examination but is instead a consequence of observing an object that has a linear velocity from a frame that has angular velocity. The apparent vectoral change or directional change in the linear velocity vector is not due to an imposed force but is merely a consequence of making an observation of a kinematic quantity, i.e., velocity, from a frame that is rotating. The term Coriolis "effect" is equally inappropriate, since this so-called effect is merely an illusion.

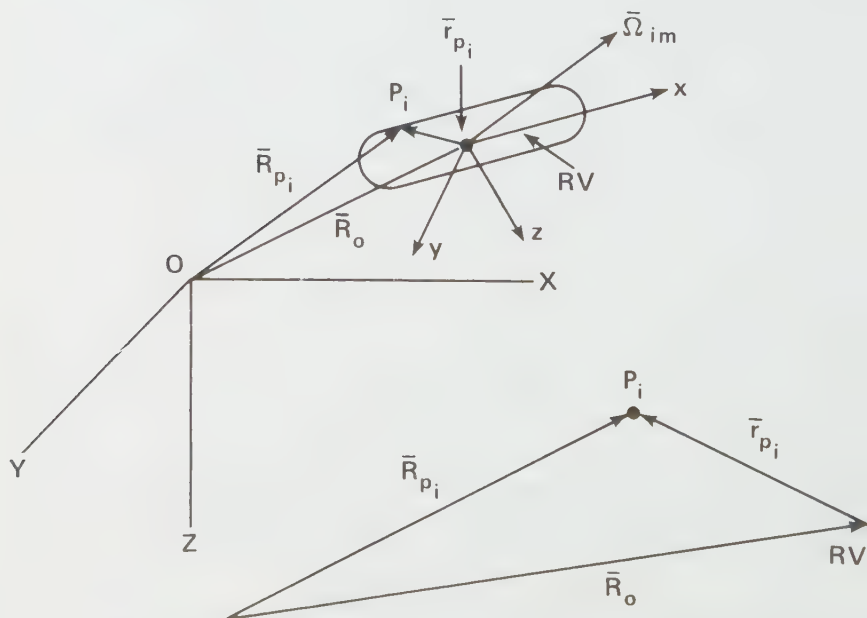


Fig. 4.5 Body generic element referred to inertial and body frames.

4.5 Force Equations

In Fig. 4.5, we have established an inertial frame (X, Y, Z) and a body-fixed moving frame (x, y, z). We note that the moving frame is located by vector $\mathbf{R}_0$ from the inertial frame O , the origin of the moving frame. The moving frame has a rotation rate $\boldsymbol{\Omega}_{\text{Im}}$. Thus we have

$$\mathbf{R}_{p_i} = \mathbf{R}_0 + \mathbf{r}_{p_i} \quad (4.30)$$

The meaning of each of the above three symbols is obvious from the vector diagram. The first derivative of the above equation becomes

$$\left. \frac{d\mathbf{R}_{p_i}}{dt} \right|_I = \left. \frac{d\mathbf{R}_0}{dt} \right|_I + \left. \frac{d\mathbf{r}_{p_i}}{dt} \right|_I$$

Letting

$$\mathbf{V}_0 = \left. \frac{d\mathbf{R}_0}{dt} \right|_I$$

we get

$$\left. \frac{d\mathbf{R}_{p_i}}{dt} \right|_I = \mathbf{V}_0 + \left. \frac{d\mathbf{r}_{p_i}}{dt} \right|_I = \mathbf{V}_0 + \left. \frac{d\mathbf{r}_{p_i}}{dt} \right|_m + \boldsymbol{\Omega}_{\text{Im}} \times \mathbf{r}_{p_i} \quad (4.31)$$

The second derivative becomes

$$\begin{aligned} \left. \frac{d^2\mathbf{R}_{p_i}}{dt^2} \right|_I &= \frac{d}{dt} \left[\left. \mathbf{V}_0 + \frac{d\mathbf{r}_{p_i}}{dt} \right|_m + \boldsymbol{\Omega}_{\text{Im}} \times \mathbf{r}_{p_i} \right]_m + \boldsymbol{\Omega}_{\text{Im}} \\ &\times \left[\left. \mathbf{V}_0 + \frac{d\mathbf{r}_{p_i}}{dt} \right|_m + \boldsymbol{\Omega}_{\text{Im}} \times \mathbf{r}_{p_i} \right] \end{aligned}$$

Carrying out the indicated manipulations, we get

$$\begin{aligned} \left. \frac{d^2\mathbf{R}_{p_i}}{dt^2} \right|_I &\equiv \mathbf{a}_{p_i} = \left. \frac{d\mathbf{V}_0}{dt} \right|_m + \left. \frac{d^2\mathbf{r}_{p_i}}{dt^2} \right|_m + 2\boldsymbol{\Omega}_{\text{Im}} \times \left. \frac{d\mathbf{r}_{p_i}}{dt} \right|_m + \frac{d\boldsymbol{\Omega}_{\text{Im}}}{dt} \times \mathbf{r}_{p_i} \\ &+ \boldsymbol{\Omega}_{\text{Im}} \times \mathbf{V}_0 + \boldsymbol{\Omega}_{\text{Im}} \times \left[\boldsymbol{\Omega}_{\text{Im}} \times \mathbf{r}_{p_i} \right] \end{aligned} \quad (4.32)$$

Note in the above equation that we are letting $\mathbf{a}_{p_i}$ be the acceleration of the particle p_i with respect to *inertial* space. Now if we multiply both sides by the mass m_i of the particle p_i , we get

$$\begin{aligned} \mathbf{a}_{p_i} m_i = \mathbf{F}_i + \sum_j \mathbf{F}_{ij} = m_i \left. \frac{d\mathbf{V}_0}{dt} \right|_m + \left. \frac{d^2 \mathbf{r}_{p_i}}{dt^2} \right|_m + 2\boldsymbol{\Omega}_{\text{Im}} \times \left. \frac{d\mathbf{r}_{p_i}}{dt} \right|_m \\ + \frac{d\boldsymbol{\Omega}_{\text{Im}}}{dt} \times \mathbf{r}_{p_i} + \boldsymbol{\Omega}_{\text{Im}} \times \mathbf{V}_0 + \boldsymbol{\Omega}_{\text{Im}} \times \left(\boldsymbol{\Omega}_{\text{Im}} \times \mathbf{r}_{p_i} \right) \end{aligned} \quad (4.33)$$

where $\Sigma \mathbf{F}_{ij}$ is the sum of the forces of all of the j particles on the i particle. Now sum the preceding over all of the i particles:

$$\sum_i \mathbf{F}_i + \sum_i \sum_j \mathbf{F}_{ij} = \sum_i \mathbf{F}_i = \mathbf{F} \quad i \neq j \quad (4.34)$$

as

$$\sum_i \sum_j \mathbf{F}_{ij} = 0 \quad i \neq j$$

because $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$ due to Newton's third law, that is, the force of the j th particle on the i th particle, is equal and opposite to the force of the i th particle on the j th particle. Let us take the following definitions:

$$\sum_i m_i = M \quad (4.35a)$$

$$\mathbf{r}_p = \frac{1}{M} \sum_i \mathbf{r}_{p_i} m_i \quad (4.35b)$$

Equation (4.35a) indicates that a summation of all of the mass particles is the total mass. Equation (4.35b) defines the center of mass or, equivalently, the center of gravity, i.e., the c.g. Using Eqs. (4.34) and (4.35) in (4.33), we get

$$\begin{aligned} \mathbf{F} = M \left[\left. \frac{d\mathbf{V}_0}{dt} \right|_m + \left. \frac{d^2 \mathbf{r}_p}{dt^2} \right|_m + 2\boldsymbol{\Omega}_{\text{Im}} \times \left. \frac{d\mathbf{r}_p}{dt} \right|_m + \frac{d\boldsymbol{\Omega}_{\text{Im}}}{dt} \times \mathbf{r}_p \right. \\ \left. + \boldsymbol{\Omega}_{\text{Im}} \times \mathbf{V}_0 + \boldsymbol{\Omega}_{\text{Im}} \times (\boldsymbol{\Omega}_{\text{Im}} \times \mathbf{r}_p) \right] \end{aligned} \quad (4.36)$$

We now can consider special cases of Eq. (4.36):

(a) c.g. does not move relative to a fixed body axis system, i.e.,

$$\left. \frac{d\mathbf{r}_p}{dt} \right|_m = 0$$

$$\mathbf{F} = M \left[\left. \frac{d\mathbf{V}_0}{dt} \right|_m + \frac{d\boldsymbol{\Omega}_{lm}}{dt} \times \mathbf{r}_p + \boldsymbol{\Omega}_{lm} \times \mathbf{V}_0 + \boldsymbol{\Omega}_{lm} \times (\boldsymbol{\Omega}_{lm} \times \mathbf{r}_p) \right] \quad (4.37)$$

(b) Axis system origin is located and remains located at the c.g., i.e. ($\mathbf{r}_p \equiv 0$),

$$\mathbf{F} = M \left[\left. \frac{d\mathbf{V}_0}{dt} \right|_m + \boldsymbol{\Omega}_{lm} \times \mathbf{V}_0 \right] \quad (4.38)$$

The derivation of Eq. (4.36) may have seemed lengthy and overdone. However, it is necessary from time to time to return to fundamentals when treating certain engineering problems. Equation (4.36) might be of interest when a moving mass control system is in use. In such a system, the center of gravity is moved in order to develop a rolling moment or perhaps to alter static stability by varying the distance between the center of gravity and the center of pressure.

If we were studying an aerodynamic control system like a split windward flap, then we might locate a frame with origin at the hinge axis which might not be the center of gravity of the flap. Thus, Eq. (4.37) would have application. In most of what follows, we shall be looking at a re-entry body with a body-fixed axis system whose origin is at the body's center of gravity. Thus, our interest would center around Eq. (4.38) for the most part.

Example 4.1: Application of Force Equations

In this section, we shall apply the force equation to a simple problem to illustrate the utility of the preceding development. Figure 4.6 shows a portion of the Earth's surface. At a point P on the surface of a spherical Earth, we have erected a coordinate system such that x points due east, y points due north, and z points up. Assume that we are launching a small rocket in the xz plane at an initial velocity magnitude of V_0 m/s and at an angle to the local horizontal of γ_E .

In the two auxiliary sketches, we see the meridian planes and the east-west plane (the xz plane). The question we pose is this: what are the coordinates of the impact point, i.e., $(x, y, 0)$? For a spherical Earth the z coordinate at impact would not be zero, but we shall assume that the initial speed is so small compared with the escape velocity that the region over which flight takes place is to a good approximation represented by a flat Earth.

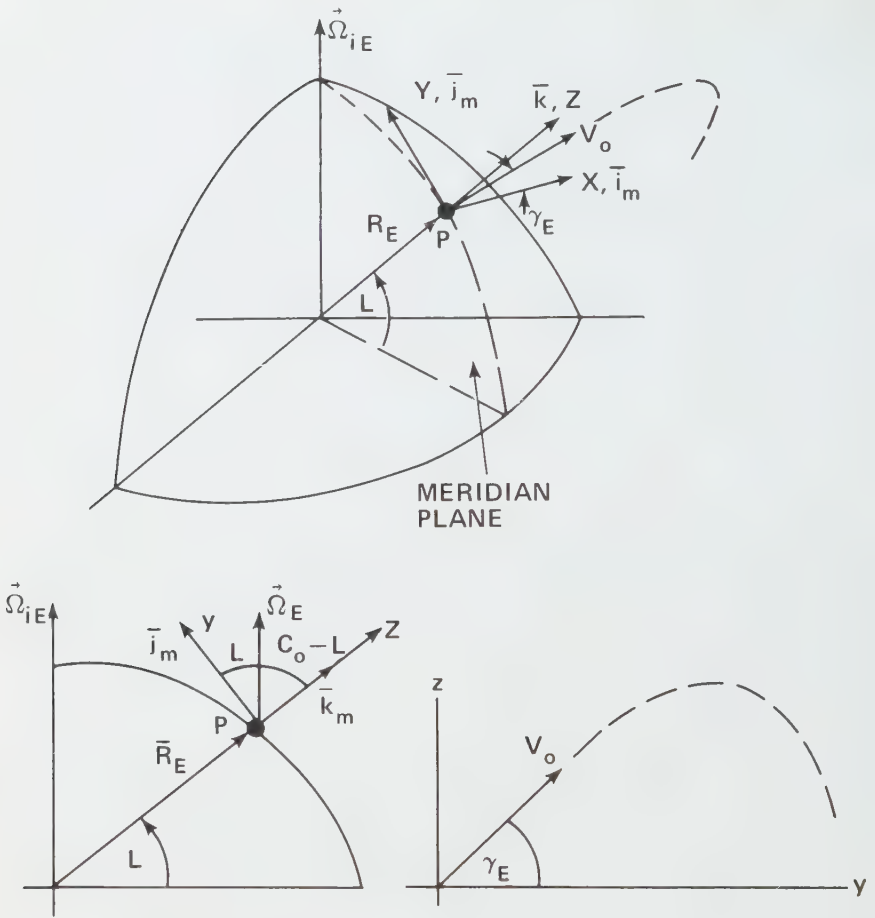


Fig. 4.6 Rocket trajectory observed from local (curved) Earth frame.

We already have developed a vector equation for motion over the Earth's surface. This equation is (4.27) and is repeated below:

$$\left. \frac{d^2 \mathbf{R}_B}{dt^2} \right|_I = \boldsymbol{\Omega}_E \times (\boldsymbol{\Omega}_E \times \mathbf{R}_E) + \boldsymbol{\Omega}_E \times (\boldsymbol{\Omega}_E \times \mathbf{r}) + 2\boldsymbol{\Omega}_E \times \mathbf{V}_B + \mathbf{a}_B$$

We recognize immediately from Fig. 4.6 that the Earth's rotation vector may be written in vector form, as

$$\boldsymbol{\omega}_{Im} = \boldsymbol{\Omega}_E = (\Omega_E \cos L) \mathbf{j}_m + (\Omega_E \sin L) \mathbf{k}_m \quad (4.39)$$

Now if we wanted a numerical solution we could write the position vector $\mathbf{r}$ and velocity vector $\mathbf{V}_B$ as

$$\mathbf{r} = x\mathbf{i}_m + y\mathbf{j}_m + z\mathbf{k}_m \quad \mathbf{V}_B = \dot{x}\mathbf{i}_m + \dot{y}\mathbf{j}_m + \dot{z}\mathbf{k}_m \quad (4.40)$$

The specific force (force per unit mass) vector and the gravitational vector would be written using Newton's second law as

$$\left. \frac{d^2\mathbf{R}_B}{dt^2} \right|_I = -g\mathbf{k}_m - \frac{\mathbf{F}_D}{M} \quad (4.41a)$$

The gravity vector follows from Eq. (4.28), and the force vector $\mathbf{F}_D$, made up entirely of aerodynamic drag, is

$$\frac{\mathbf{F}_D}{M} = -\frac{\rho V_B^2 S C_D}{2M} \frac{\mathbf{V}_B}{V_B} = \left[\frac{\rho V_B S C_D}{2M} \right] \mathbf{V}_B \quad (4.41b)$$

where C_D is the drag coefficient, S is the reference area upon which the drag coefficient is based, and M is the body mass.

Now Eqs. (4.39), (4.40), and (4.41) could be inserted into Eq. (4.27) to get three second-order coupled differential equations written in the variables (x, y, z) . These equations then could be integrated numerically. This would be an appropriate skill-building exercise in computer programming, but of little value in understanding vector mechanics.

Let us return to our original problem: find in some approximate way the impact point of the rocket relative to its launch point. Since the trajectory equations can be integrated in closed form for a flat, nonrotating Earth with no atmosphere, we can use such results to find the effect of Earth rotation on impact point. Thus, from Eq. (4.41a) we have

$$\left. \frac{d^2\mathbf{R}_B}{dt^2} \right|_I = -g \left[\frac{\mathbf{R}_E}{R_E} \right] = -g\mathbf{k} \quad (4.42)$$

The above result follows from Newton's second law relating the applied force per unit mass to the acceleration of the unit mass relative to inertial space. The velocity vector relative to the moving launch frame follows from elementary physics for motion over a flat Earth without atmosphere:

$$\mathbf{V}_B = (V_0 \cos \gamma_E) \mathbf{i}_m + [(V_0 \sin \gamma_E) - gt] \mathbf{k}_m \quad (4.43)$$

From Fig. 4.6, we may write the Earth's radius vector as

$$\mathbf{R}_E = R_E \mathbf{k}_m \quad (4.44)$$

If we assume throughout the flight that the vector $\mathbf{r}$ from the launch site to

the rocket is at all times insignificant compared in magnitude with the Earth's radius vector, i.e., $|\mathbf{r}| \ll |\mathbf{R}_E|$, then

$$\Omega_E \times (\Omega_E \times \mathbf{r}) \approx 0 \quad (4.45)$$

Thus, Eq. (4.27) may be written as

$$-g\mathbf{k} = \mathbf{a}_b + 2\Omega_E \times \mathbf{V}_B + \Omega_E (\Omega_E \cdot \mathbf{R}_E) - \mathbf{R}_E (\Omega_E \cdot \Omega_E) \quad (4.46)$$

where we have used the vector identity

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \equiv \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$$

$$-g\mathbf{k}_m = \ddot{x}\mathbf{i}_m + \ddot{y}\mathbf{j}_m + \ddot{z}\mathbf{k}_m + 2 \begin{vmatrix} \mathbf{i}_m & \mathbf{j}_m & \mathbf{k}_m \\ 0 & \Omega_E \cos L & \Omega_E \sin L \\ V_0 \cos \gamma_E & 0 & (V_0 \sin \gamma_E - gt) \end{vmatrix} + [\Omega_E \cos L] \mathbf{j}_m + (\Omega_E \sin L) \mathbf{k}_m [R_E \Omega_E \sin L] - R_E \Omega_E^2 \mathbf{k}_m \quad (4.47)$$

Since we are interested only in the (x, y) coordinates at impact, we can ignore the z equation. Thus, we can collect the $\mathbf{i}_m, \mathbf{j}_m$ components to get

$$x + 2\Omega_E \cos L (V_0 \sin \gamma_E - gt) = 0 \quad (4.48a)$$

$$y + 2\Omega_E V_0 \sin L \cos \gamma_E + \Omega_E^2 R_E \sin L \cos L = 0 \quad (4.48b)$$

In order to obtain the coordinates (x, y) at impact, we must integrate Eqs. (4.48) twice with respect to time. The first integration gives

Table 4.1 Program for Calculating Impact Point of Trajectory

```

00100 REM THIS PROGRAM CALCULATES THE IMPACT POINT ON THE EARTH'S SURFACE
00110 REM OF AN OBJECT LAUNCHED AT LATITUDE (DEG) I AT AN INITIAL ANGLE G1 (DEG),
00120 REM AND AN INITIAL VELOCITY MAGNITUDE, V0 (M/SEC). IMPACT POINT IS
00130 REM DESIGNATED (EAST +) AND NORTH +).
00140 V0 = 250
00150 G1 = 60
00160 L = 45
00170 W = 7.28E-5
00180 R = 6.370E6
00190 Z9 = 3.14159/180
00200 G = 9.81
00210 G1 = G1*Z9
00220 L = L*Z9
00230 T = 2*V0*SIN(G1)/G
00240 X = V0*COS(G1)*T
00250 X = X - (2*W*COS(L))*((V0*SIN(G1))*(T*T/2) - G*(T*T/2))
00260 Y = (-1)*(2*W*V0*SIN(L)*COS(G1))*(T*T/2)
00270 Y = Y - (W*W*R*COS(L)*SIN(L))*(T*T/2)
00280 PRINT #EAST#, #NORTH#, #LAT#, #V0#, #G1#
00290 PRINT #METERS#, #METERS#, #DEG.#, #M/SEC.#, #DEG#
00300 PRINT
00310 PRINT X, Y, L/Z9, V0, G1/Z9
00320 END

```

$$\dot{x} = V_0 \cos \gamma_E - 2\Omega_E \cos L [(V_0 \sin \gamma_E)t - (gt^2/2)] \quad (4.49a)$$

$$\dot{y} = -(2\Omega_E V_0 \sin L \cos \gamma_E)t - (\Omega_E^2 R_E \sin L \cos L)t \quad (4.49b)$$

where the initial conditions on $(\dot{x}, \dot{y})$ are $(V_0 \cos \gamma_E, 0)$. There is of course an initial condition on $\dot{z}$, but at this point we are tracking only the (x, y) equations. We shall make use of the z equation to obtain a guess on the time of flight, i.e., time elapsed between launch and impact. A second integration gives (x, y) , the displacement of the rocket at impact relative to the launch point:

$$x = (V_0 \cos \gamma_E)t - 2\Omega_E \cos L [(V_0 \sin \gamma_E)(t^2/2) - (gt^3/6)] \quad (4.50a)$$

$$y = -[2\Omega_E V_0 \sin L \cos \gamma_E](t^2/2) - [\Omega_E^2 R_E \sin L \cos L](t^2/2) \quad (4.50b)$$

An approximation to the time of flight follows from the z equation, whose two integrals follow from Eq. (4.47) as

$$z = -gt \quad \dot{z} = V_0 \sin \gamma_E - gt \quad (4.51a)$$

$$z = (V_0 \sin \gamma_E)t - (gt^2/2) \quad (4.51b)$$

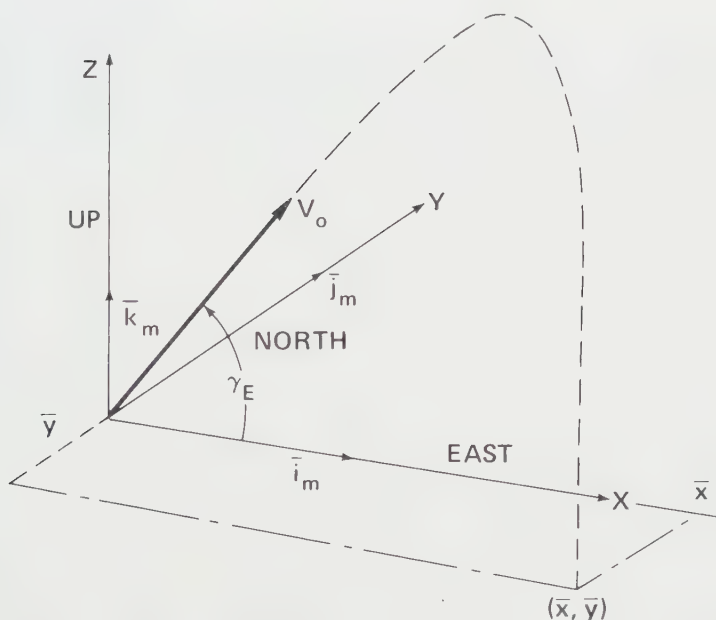


Fig. 4.7 Rocket trajectory observed from local (flat) Earth frame.

Solving for the time of flight from Eq. (4.51b), we get, when $z = 0$,

$$t = t_f = 2V_0 \sin \gamma_E / g \quad (4.52)$$

A short program has been written for solving Eqs. (4.50). The program is listed in Table 4.1.

Some typical output might be given. Suppose the rocket has a launch velocity magnitude of 150 m/s and an angle of launch of 60 deg. The launch vector relative to the unit vectors ($\mathbf{i}_m, \mathbf{j}_m, \mathbf{k}_m$) is shown in Fig. 4.7.

If the rocket is launched at an angle of 60 deg to the horizontal, i.e., $\gamma_E = +60$, the output is

EAST METERS	NORTH METERS	LAT DEG.	VU M/SEC.	GL DEG
5510.26	-28.9808	45.	250	60.

To remove the contribution of the Earth's rotation, the Earth's rate is set equal to zero (line 170) to give

EAST METERS	NORTH METERS	LAT DEG.	VU M/SEC.	GL DEG
5517.5	0	45.	250	60.

It will be noted that, in the absence of Earth rotation, the rocket does not drift north or south. However, Earth's rotation rate causes the eastward-launched rocket to impact about 30 m to the south of the launch point. Earth rotation rate has a smaller effect on the east-west coordinate, causing a 7 m shortfall of the rocket at impact.

A vertical launch causes the trajectory to impact about 22 m south and 11 m west of the launch point:

EAST METERS	NORTH METERS	LAT DEG.	VU M/SEC.	GL DEG
-11.127	-21.9253	45.	250	90.

Without Earth's rotation rate, the vertically launched rocket passes through the launch point:

EAST METERS	NORTH METERS	LAT DEG.	VU M/SEC.	GL DEG
1.69062E-2	0	45.	250	90.

It should be emphasized that this example is limited to an east-west

launching at point P . It would be easy to generalize this exercise for a launch in any direction. V_B in Eq. (4.43) would have to be rewritten to include a component along the y axis. This would be quite simple. Equation (4.43) would have to account for the rocket's azimuth at launch:

$$\mathbf{V}_B = (V_0 \cos \gamma_E \sin \Psi) \mathbf{i}_m + (V_0 \cos \gamma_E \cos \Psi) \mathbf{j}_m + (V_0 \sin \gamma_E - gt) \mathbf{k}_m \quad (4.53)$$

The centrifugal term $\Omega_E \times (\Omega_E \times \Omega_E)$ would not be affected by the redefinition of $\mathbf{V}_B$. However, the term $2\Omega_E \times \mathbf{V}_B$ would now become

$$\begin{aligned} 2\Omega_E \times \mathbf{V}_B &= \begin{vmatrix} \mathbf{i}_m & \mathbf{j}_m & \mathbf{k}_m \\ 0 & \Omega_E \cos L & \Omega_E \sin L \\ V_0 \cos \gamma_E \sin \Psi & V_0 \cos \gamma_E \cos \Psi & V_0 \sin \gamma_E - gt \end{vmatrix} \\ &= 2\mathbf{i}_m \{[(\Omega_E \cos L)(V_0 \sin \gamma_E - gt)] - [\Omega_E V_0 \sin L \cos \gamma_E \cos \Psi]\} \\ &\quad + 2\mathbf{j}_m [\Omega_E V_0 \sin L \cos \gamma_E \sin \Psi] + \mathbf{k}_m [-\Omega_E V_0 \cos L \cos \gamma_E \sin \Psi] \quad (4.54) \end{aligned}$$

Equations (4.48) then become:

$$x + 2\Omega_E V_0 [(\cos L)(V_0 \sin \gamma_E - gt) - \sin L \cos \gamma_E \cos \Psi] = 0 \quad (4.55a)$$

$$y + 2\Omega_E V_0 \sin L \cos \gamma_E \sin \Psi + \Omega_E^2 R_E \sin L \cos L = 0 \quad (4.55b)$$

In the example just given, the launch was in the xz plane, so that the azimuth angle was $\Psi = 90$ deg. Inserting $\Psi = 90$ deg into Eq. (4.55) gives Eqs. (4.48). Two integrations of Eq. (4.55) provide the impact points $(x, y, 0)$. Completion of this problem to find the impact point with rocket azimuth included will be left as an exercise for the reader.

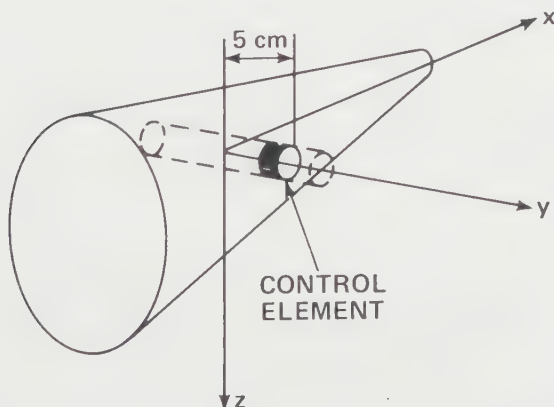


Fig. 4.8 Moveable control element.

Example 4.2: Application of Force Equation

Figure 4.8 depicts an RV moving-element control system. By moving the element along the y axis, a couple is created between the center of pressure, which presumably lies in the xz plane, and the RV's center of gravity, which is moved slightly to the right. The magnitude of this couple and hence the rolling moment on the vehicle are controlled by varying the position of the control element. The null position of the moving element is at the origin of the axis system, which is the plane of vehicle symmetry. The vehicle is decelerating at $3g$ and has a linear velocity of 3000 m/s . Both the velocity and the acceleration vectors are along the x body axis. The RV is rotating at 20 rad/s about the x axis. The element has a mass of 0.1 kg . When the RV is at rest in a laboratory, it is noted that to move the element at 3 m/s along the y axis requires a force of 2 N . Find the additional forces that must be applied to the moving mass if the element is moving at a speed relative to the RV of 3 m/s when at a position of 5 cm from null along the positive y axis.

Let us use Eq. (4.36) in this problem. According to the previous information, we may write

$$\frac{d\mathbf{V}_0}{dt} = -(3)(9.81)\mathbf{i}_m = -29.43\mathbf{i}_m$$

$$\mathbf{V}_0 = 3000\mathbf{i}_m$$

$$\Omega_{lm} = 20\mathbf{i}_m$$

$$\mathbf{r}_p = (0.05)\mathbf{j}_m$$

$$\left. \frac{d\mathbf{r}_p}{dt} \right|_m = 3\mathbf{j}_m$$

$$\left. \frac{d^2\mathbf{r}_p}{dt^2} \right|_m = 0 \quad \frac{d^2\Omega_{lm}}{dt^2} = 0$$

$$\begin{aligned} \mathbf{F}_p = & 0.1 [(-29.43)\mathbf{i}_m + 20\mathbf{i}_m \times (3000)\mathbf{i}_m + 0 \\ & + 2(20\mathbf{i}_m) \times 3\mathbf{j}_m + 0 + (20\mathbf{i}_m) \times (20\mathbf{i}_m \times 0.05 \times \mathbf{j}_m)] \end{aligned}$$

which becomes

$$\mathbf{F}_p = -2.94\mathbf{i}_m + 12\mathbf{k}_m - 2\mathbf{j}_m \quad (4.56)$$

Since the control element can move only along the $\mathbf{j}_m$ or y body axis, the forces -2.94 N along the x axis and 12 N along the z axis are forces of constraint. Depending upon the frictional characteristics of the control system, these forces of constraint may require additional forces on the element along the $\mathbf{j}_m$ direction and hence require additional force to move the element. If we were designing such a control system, the effect of these forces of con-

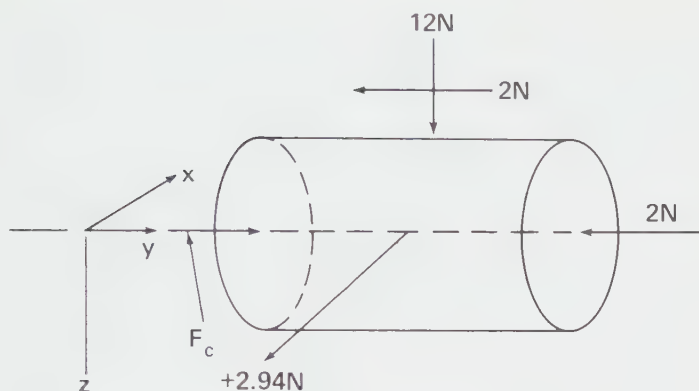


Fig. 4.9 Control element.

straint in creating friction forces would be given serious consideration. We do note in Eq. (4.56) that there is a force directly impeding motion along the tube. The freebody diagram of the element is shown in Fig. 4.9. It shows that an additional force of 2 N will be required to move the mass, because the RV has linear acceleration and linear and angular velocity in flight, but has none of these (presumably) under laboratory conditions.

In the laboratory, a measurement was made of 2 N to move the element. To this frictional force, we must add the force of 2 N caused by vehicle dynamics. From Eq. (4.56), it will be noted that this "dynamic force" is caused by the centrifugal acceleration term:

$$\begin{aligned}\Omega_{lm} \times (\Omega_{lm} \times \mathbf{r}_p) &= \dot{\phi} \mathbf{i}_m \times (\dot{\phi} \mathbf{i}_m \times r_p \mathbf{j}_m) \\ &= (\dot{\phi})^2 r_p\end{aligned}\quad (4.57)$$

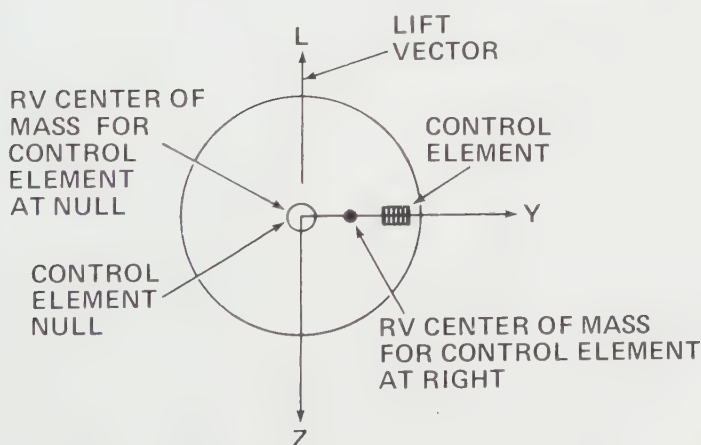


Fig. 4.10 Base view of RV.

where ϕ is the roll angle and $\dot{\phi}$ is the roll rate. Note that this force will *not* change sign if the rotation rate changes sign. However, the sign of this force *will* change if the element is located to the left (negative y axis) at the same distance (5 cm in this case). Thus, vehicle dynamics will create an apparent force which will always oppose a control motion that is tending to increase the distance of the element from the xz plane. Of course as we move the element away from the xz plane (either right or left) we are tending to increase the roll acceleration and its integral, roll rate. By moving the element to the right (say), we are moving the RV center of mass to the right, causing a positive torque and hence a positive angular acceleration about the x axis (see Fig. 4.10).

This angular acceleration integrates to give the angular rate Ω_{lm} , also positive about the x axis. In Chapter VII, we shall discuss the moment equation in detail. For the present, if we accept that the axial moment inertial I_x times the roll acceleration $\ddot{\phi}$ is equal to the applied roll torque, which we assume proportional to the position of the control element, we have

$$I_x \ddot{\phi} = k_1 r_p$$

or

$$\Omega_{lm} \triangleq \dot{\phi} = k_1 \int_0^t \frac{r_p(\tau)}{I_x} d\tau \quad (4.58)$$

Now we have shown that the force on the control element due to dynamic (rolling) motion is

$$F_{dy} = k_2 (\dot{\phi})^2 r_p = k_1 k_2 \left[\int_0^t \frac{r_p(\tau)}{I_x} d\tau \right]^2 r_p \quad (4.59)$$

The force on the control element then must be equal to the frictional force F_f , which is measured under static laboratory conditions plus the force due to dynamics, or

$$\begin{aligned} F_c &= F_f + F_{dy} \\ &= F_f + k_1 k_2 \left[\int_0^t \frac{r_p(\tau)}{I} d\tau \right]^2 r_p \end{aligned} \quad (4.60)$$

We observe the presence of a nonlinear stiffness force with the stiffness "constant" proportional to the square of the integral of the control element position.

V. KEPLERIAN MOTION: FREE-FLIGHT/COAST PHASE

5.1 Re-Entry Vehicle Trajectory

In Chapter IV we developed the force equations in a fairly general form. These equations will be referred to again, particularly after we have derived the moment equations. However, we can develop considerable insight into RV motion by regarding the RV as a point-mass, subjected primarily to aerodynamic and gravitational forces. Within this point-mass assumption, the force equations may be applied directly to the various phases of RV flight. We have just considered a simple case of motion of a low-velocity rocket above a rotating but atmosphere-free Earth. Of course the Earth does have an atmosphere, and the presence of this atmosphere cannot be ignored when specifying vehicle performance or mission requirements. However, although an RV begins and ends its flight within the atmosphere, a considerable portion of its trajectory takes place outside the tenable atmosphere. During this portion of the flight, all atmospheric effects are negligible or by definition zero. In honor of J. Kepler, whose speculations concerning planetary motion led to his three empirical laws, this portion of the RV's flight is known as "Keplerian motion." The term "Keplerian" is more than merely honorary: it will be shown that during this phase the three laws of Kepler, linked by Newton's Law of Universal Gravitation, serve to describe RV motion quite accurately.

Before proceeding with a discussion of Keplerian motion, let us identify the other phases of RV motion. It has more or less been established that the trajectory of a ballistic missile may be separated into three phases: powered flight, free flight, and re-entry flight. The first is sometimes called the boost phase. Our only concerns with the boost phase at this point are the position and velocity vectors of the RV at the terminus of this phase. The possibly unwarranted assumption will be made that the initial conditions in $|\mathbf{r}|$ and $|\mathbf{V}|$ at RV off-loading (the end of boost) can be selected arbitrarily from a range of values. As we shall show, the subsequent trajectory-free flight and re-entry phases will be influenced strongly by this selection of r and V .

If the ballistic missile is regarded as an extension of conventional artillery, the boost phase corresponds to the interior ballistics of the cannon-launched projectile. Just as the muzzle velocity and gun barrel azimuth and elevation may be selected to shape the trajectory of exterior ballistics, so too may the boost phase be regarded only at its terminus, where the initial conditions are set for the "exterior ballistics" of the free-fall and re-entry trajectories.

Table 5.1 Brief Description of Phases

Phase	Passive Forces	Active Forces	Medium
Launch (<i>AB</i>)	Gravitational hydrodynamic	Controlled rocket thrust	Water
Boost (<i>BC</i>)	Gravitational aerodynamic	Controlled rocket thrust	Air
Keplerian (<i>CD</i>)	Gravitational	Controlled jet thrust	None
Re-entry (<i>DE</i>)	Gravitational aerodynamic	Aerodynamic	Air

In this book, we shall consider each of the three phases separately and in some detail. To the three phases of boost (powered), Keplerian (coast), and re-entry (maneuvering) we might add a fourth: launch phase (Fig. 5.1 and Table 5.1). Now the launch phase might be included in the boost phase, but in some cases the physics influencing motion might be sufficiently different to warrant making the distinction. An example might be the submarine-launched RV. During the launch phase, the vehicle is subjected to potential and frictional forces of an incompressible medium; cavitation may be present just prior to broach. Also, portions of the vehicles may be covered with a gas bubble or sheath. In the transition between the launch and boost phases, i.e., at broach, the vehicle experiences a discontinuous change in medium density, with the medium density during boost $1/850$ th that of launch.

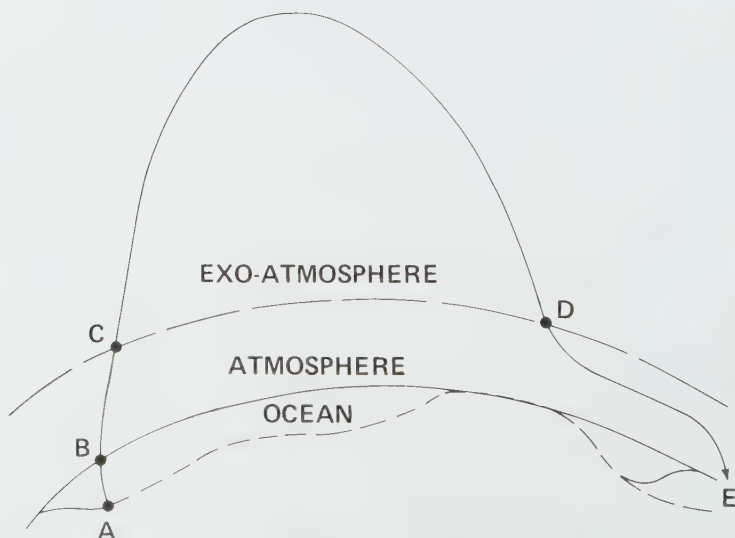


Fig. 5.1 Re-entry vehicle trajectory phases.

5.2 Equation of Motion

Free flight will be defined as that portion of the total RV trajectory between RV off-loading and atmospheric re-entry. Consistent with this definition is the requirement that the free-flight phase be entirely exo-atmospheric and that the only force acting upon the RV is the gravitational force due to the Earth. We shall leave others to consider the effects of solar radiation, residual atmosphere, and nonterrestrial gravity fields.

The description that we shall use interchangeably with the free-flight phase is "Keplerian motion." During this phase, the RV's motion may be described as orbital (even though the orbit intersects the Earth). Kepler's

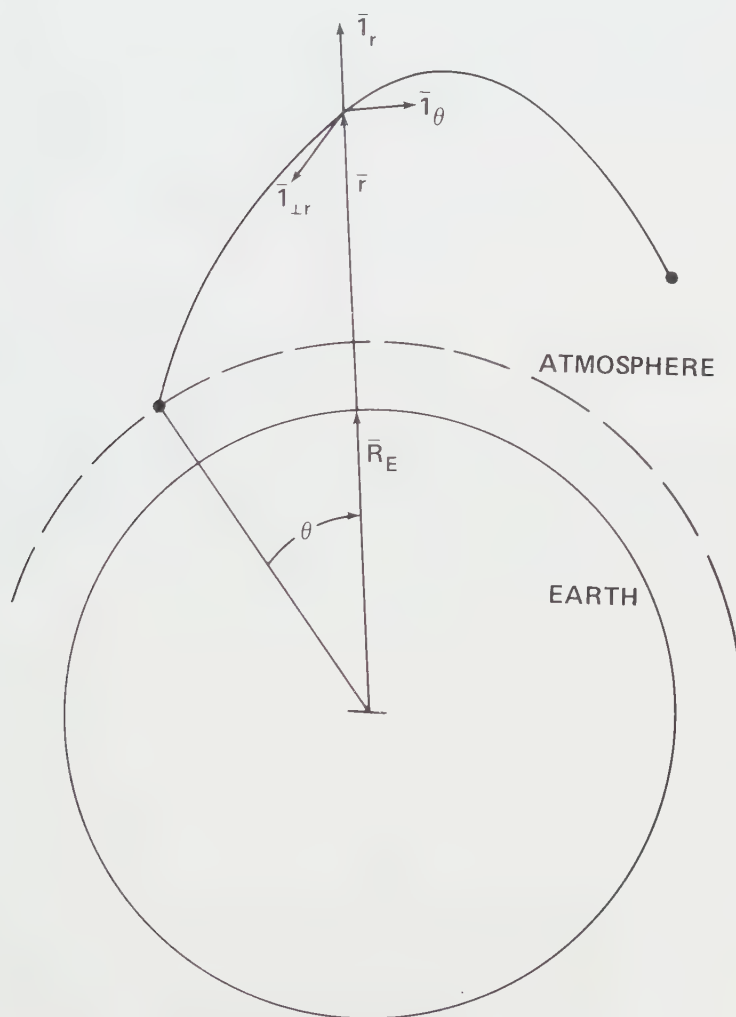


Fig. 5.2 Variables in Keplerian phase.

empirical observations of planetary motion were distilled by him into three laws. Newton showed that all three laws could be derived analytically once the assumption was made that gravity was a central-inverse-square field.

From Eq. (4.20) we can derive two differential equations in the polar coordinate variables r and θ (see Fig. 5.2). Integration (in principal) will lead to specification of the trajectory. Unfortunately it does not turn out that the radius vector varies harmonically with time but rather that the reciprocal of the radius vector varies harmonically with the central angle θ . The time variable is embedded in the problem. It seems impossible to develop an analytic formulation of r and θ as time functions. The best that can be done is to express time as a function of θ and then imply r as a function of time through the relationship of $r = r(\theta)$. However, we are getting ahead of the development. We have, from Eq. (4.20),

$$\begin{aligned} \left. \frac{d^2 \mathbf{R}}{dt^2} \right|_I &= \left. \frac{d^2 \mathbf{r}}{dt^2} \right|_I \\ &= \left. \frac{d^2 \mathbf{r}}{dt^2} \right|_m + 2\boldsymbol{\omega}_{Im} \times \left. \frac{d\mathbf{r}}{dt} \right|_m + \frac{d\boldsymbol{\omega}_{Im}}{dt} \times \mathbf{r} + \boldsymbol{\omega}_{Im} \times (\boldsymbol{\omega}_{Im} \times \mathbf{r}) \end{aligned}$$

where

$$\left. \frac{d^2 \mathbf{r}}{dt^2} \right|_I = \mathbf{g} \quad (5.1)$$

Because all forces other than those due to the Earth's field are ignored, Eq. (5.1) indicates that during Keplerian motion the RV is indeed in free fall.

We select three unit vectors, one along the position vector $\mathbf{r}$, a second along the angular velocity vector $\boldsymbol{\omega}_{Im}$, and the third normal to the first two. These unit vectors are, respectively, $\mathbf{l}_r$, $\mathbf{l}_\theta$, and $\mathbf{l}_{\perp r}$; and $\mathbf{l}_\theta \times \mathbf{l}_{\perp r}$ (see Fig. 5.3). Note that in Eq. (5.1) the system origin is at the Earth's center; i.e., R_0 of Eq. (4.20) is zero. In terms of the unit vectors, we have

$$\mathbf{r} = r \mathbf{l}_r \quad \boldsymbol{\omega}_{Im} = \dot{\theta} \mathbf{l}_\theta \quad \mathbf{g} = -(\mu/r^2) \mathbf{l}_r \quad (5.2)$$

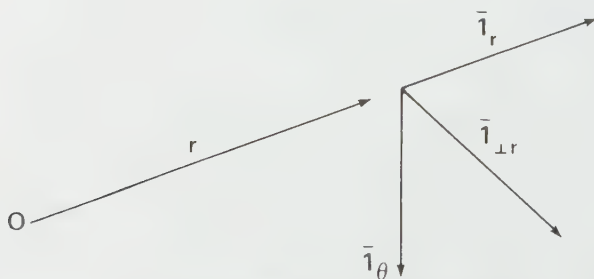


Fig. 5.3 Position vector components.

Inserting Eqs. (5.2) into Eq. (5.1) gives two differential equations in $\{r, \theta\}$:

$$\ddot{r} - r(\dot{\theta})^2 + (\mu/r^2) = 0 \quad (5.3a)$$

$$r\ddot{\theta} + 2\dot{\theta}\dot{r} = 0 \quad (5.3b)$$

If both sides of Eq. (5.3b) are multiplied by $\dot{r}$, they may be combined in a single derivative:

$$\frac{d}{dt}(r^2\dot{\theta}) = 0 \quad (5.4)$$

or, equivalently, $r^2\dot{\theta} = \text{const.}$ Thus, a characteristic of Keplerian motion is that trajectory arcs subtend equal areas in equal time, i.e., Kepler's area law. Equation (5.4) also indicates that angular momentum of the RV about the Earth's center is conserved. Thus, the angular momentum throughout the trajectory must remain equal to its values at RV off-loading, i.e., the terminus of the boost phase,

$$r^2 \frac{d\theta}{dt} = r_0 V \cos \gamma = P \quad (5.5)$$

$$r_0 = R_E + z$$

where r_0 , V , and γ are the magnitude of the radius vector, magnitude of the velocity, and the flight-path angle at boost termination or RV off-loading (see Fig. 5.4); z is the altitude at boost termination, and P is the angular momentum per unit mass.

The conservation of angular momentum of the RV relative to the Earth's center follows because the only force acting is Earth's gravity, and because of the assumed centrality of this field (along negative $\mathbf{r}$) there is no force normal to $\mathbf{r}$. Of course this conservation is no longer valid during re-entry when at times the drag load, roughly normal to $\mathbf{r}$, may be more than an order of

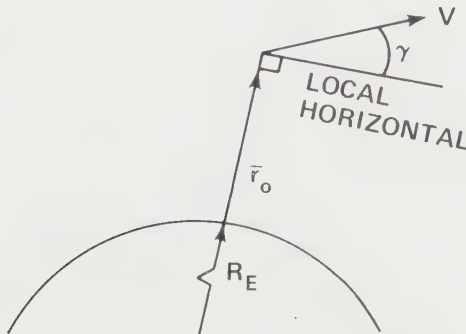


Fig. 5.4 Definition of local horizontal.

magnitude greater than the gravitational force. We may make further use of Eq. (5.5) in transforming the independent variable from time t to central angle θ and from radius r to $u = 1/r$:

$$\frac{d}{dt}(\) = \frac{d\theta}{dt} \frac{d}{d\theta}(\) = Pu^2 \frac{d}{d\theta}(\) \quad (5.6)$$

where P is again the angular momentum (per unit mass) and was defined in Eq. (5.5). Clearly

$$\frac{dr}{dt} = -P \frac{du}{d\theta}$$

and

$$\frac{d^2r}{dt^2} = -P^2 u^2 \frac{d^2u}{d\theta^2}$$

and so Eq. (5.3) may be written as

$$\begin{aligned} \frac{d^2u}{d\theta^2} + u &= \frac{\mu}{P^2} = \frac{\mu}{r_0^2 V^2 \cos^2 \gamma} \\ &= \frac{1}{\lambda r_0 \cos^2 \gamma} \end{aligned} \quad (5.7a)$$

where γ has been shown to be the angle between the velocity vector and the local horizontal at boost terminus. Local horizontal is of course normal to r_0 . It will be noted that for later convenience we have introduced a parameter λ , which is the ratio of twice the linear kinetic energy per unit mass to the potential energy per unit mass, or

$$\lambda = \frac{2(\frac{1}{2}V^2)}{\mu/r_0} = \frac{r_0 V^2}{\mu} = \frac{V^2}{\mu/r_0} \quad (5.7b)$$

Subsequently it will be shown that μ/r_0 is the square of the linear velocity in a circular orbit of radius r_0 . Thus an equivalent definition of γ is the square of the ratio of the velocity magnitude at boost termination to the circular orbital velocity. Since we are restricting attention to suborbital trajectories, λ will always be less than unity.

Our next interest is in obtaining solutions to Eq. (5.7a). We should expect the solution of the homogeneous equation to be that of a harmonic oscillator with constant bias. The particular solution of Eq. (5.7a) is

$$u_p = 1/\lambda r_0 \cos^2 \gamma \quad (5.8)$$

and the homogeneous solution is

$$u_h = A \sin \theta + B \cos \theta \quad (5.9)$$

Obviously $u = u_p + u_h$ satisfies Eq. (5.5). To find the constants A and B we rely upon the following initial conditions on u and $du/d\theta$:

$$u_0 \equiv u_{\theta=0} = 1/r_0$$

$$\left. \frac{du}{d\theta} \right|_0 \equiv \left. \frac{du}{d\theta} \right|_{\theta=0} = -\frac{1}{r_0} \tan \gamma \quad (5.10)$$

It is easy then to show that the solution of the equation of motion [Eq. (5.7a)] is

$$r_0 u(\theta) = \frac{r_0}{r(\theta)} = \frac{1 - \cos \theta}{\lambda \cos^2 \gamma} + \frac{\cos(\gamma + \theta)}{\cos \gamma} \quad (5.11)$$

Free-fall motion in a central force field requires that the reciprocal of the radius vector vary harmonically with the central angle θ . We might have preferred to have the radius vector vary harmonically with time, but the physics of an inverse-square, central force model do not work out that way.

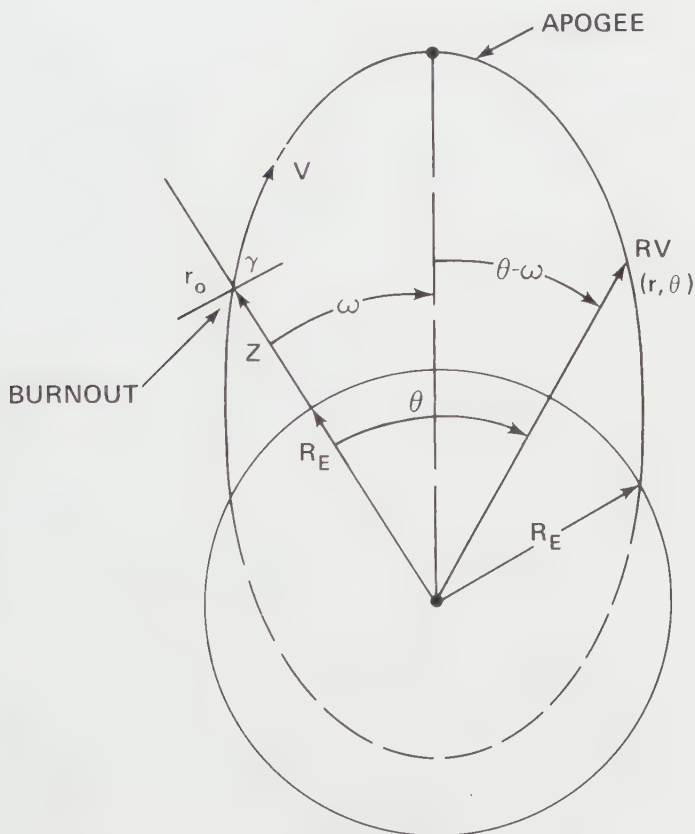


Fig. 5.5 Keplerian trajectory.

In Eq. (5.11), r_0 and λ are set at burnout. It is fairly straightforward to manipulate Eq. (5.11) to give $r = r(\theta)$ in the familiar conic section in polar coordinates. This will be done in the next few pages. It will be noted in Eq. (5.11) that we have three parameters: radius vector $r_0 = R_E + z$ or, equivalently, altitude z at burnout, flight-path angle γ at burnout, and the velocity parameter λ , also evaluated at burnout. We may recall that λ is the ratio of square of the burnout velocity magnitude to the square of the velocity magnitude in circular orbit (at radius r_0). If we were to use the equation for conic sections, then we should also expect three parameters, the eccentricity e , the semilatus rectum l , or maybe preferably the semimajor axis a , and the argument of the apogee ω .

In Fig. 5.5 we see a typical Keplerian orbit with the RV at some point along the trajectory having the polar coordinates (r, θ) . Clearly the Earth's center is at the further focus of the ellipse; consequently, when the RV is at its zenith, it is also at orbital apogee.

In Fig. 5.5, we have depicted the elliptical orbit of the coast or Keplerian phase. We recognize that, in general, burnout will not occur at orbital apogee. Thus, we have designated the location of apogee from burnout by the angle ω . Note that because of the finite size of the Earth only a portion of the orbit will exist; the dotted lines indicate the continuation of the orbit if the Earth were a mass point.

In any standard text on analytic geometry, it is shown that the polar equation for an ellipse may be written as

$$r = \frac{a(1-e^2)}{1-e\cos(\theta-\omega)} \quad (5.12)$$

where a is the semi-axis, e the eccentricity, and ω the argument of the apogee. The term $a(1-e^2)$ is called the "semilatus rectum." We may manipulate Eq. (5.11) to give the form of (5.12):

$$r = \frac{r_0 \lambda \cos^2 \gamma}{1 - \cos \theta + \lambda \cos \gamma \cos(\theta + \gamma)}$$

or

$$r = \frac{r_0 \lambda \cos^2 \gamma}{1 - e \{ \cos \theta [(1/e)(1 - \lambda \cos^2 \gamma)] + \sin \theta [(1/e)(\lambda \cos \gamma \sin \gamma)] \}} \quad (5.13)$$

Since in Eq. (5.12)

$$\cos(\theta - \omega) = \cos \theta \cos \omega + \sin \theta \sin \omega$$

we may make a comparison with the terms within the braces of Eq. (5.13):

$$e \sin \omega = \lambda \cos \gamma \sin \gamma \quad (5.14a)$$

$$e \cos \omega = 1 - \lambda \cos^2 \gamma \quad (5.14b)$$

Squaring both sides and adding, we arrive at an expression for the eccentricity:

$$e = [\lambda(\lambda - 2)\cos^2\gamma + 1]^{1/2} \quad (5.15)$$

Next, taking the ratio of Eqs. (5.14), we get, for the argument of the apogee,

$$\omega = \tan^{-1} \left[\frac{\lambda \cos\gamma \sin\gamma}{1 - \lambda \cos^2\gamma} \right] \quad (5.16)$$

Recognizing that

$$r_0 \lambda \cos^2\gamma = a(1 - e^2)$$

we get, for the semimajor axis, using Eq. (5.15),

$$a = r_0 / (2 - \lambda) \quad (5.17)$$

We know that in a circular orbit the eccentricity must be zero and the velocity must always be perpendicular to the radius vector. Clearly if $\lambda = 1$ and $\gamma = 0$, we conclude from Eq. (5.15) that the eccentricity $e = 0$. We could reverse the process and set $e = 0$ in Eq. (5.15) and solve for λ . The only real solution would require that $\gamma = 0$, which in turn would require $\lambda = 1$ for both roots. Because of the definition of λ [see Eq. (5.7b)], the velocity in circular orbit must be

$$V_0 = (\mu/r_0)^{1/2} = [(R_E + z_0)g(h)]^{1/2} \cong (R_E g_0)^{1/2} \quad (5.18)$$

The second relationship follows from the definition of μ and the third from the approximation that $z \ll R_E$. Inserting the values of R_E as 6.378×10^6 and $g = 9.81 \text{ m/s}^2$, we get, for the circular orbital velocity, $V_0 = 7910 \text{ m/s}$. The equivalent in English units is $R_E = 2.09 \times 10^7 \text{ ft}$, $g = 32.174 \text{ ft/s}^2$, and $V_0 = 25,931 \text{ ft/s}$.

At escape velocity the RV would be on a parabolic trajectory. The eccentricity e of a parabola is unity. From Eq. (5.15) there are three ways to achieve a parabolic trajectory: $\lambda = 0$, $\lambda = 2$, and $\gamma = \pi/2$. For $\lambda = 0$, we have the flat Earth parabolic trajectory from elementary physics, since the vehicle velocity is negligible in comparison to the orbital velocity. This particular trajectory is of no interest in the study of RV trajectories. For $\gamma = \pi/2$, we have a sort of degenerate parabola, since the RV would move only in a vertical line in an Earth-centered inertial frame. Obviously, such a trajectory would have no practical value. For $\lambda = 2$, we have a parabolic escape trajectory. From the definition of λ given in Eq. (5.7b), we have

$$\lambda = \frac{V^2}{\mu / (R_E + z)} \quad (5.19)$$

Now replacing λ by 2 and solving for V , we get

$$V = V_{\text{escape}} = [2\mu/(R_E + Z)]^{1/2} = [2(R_E + Z)g(z)]^{1/2} \quad (5.20)$$

$$V_{\text{escape}} \simeq [2(R_E g_0)]^{1/2}$$

The idea of escape velocity being associated with $\lambda = 2$ is also consistent with Eq. (5.17), since we should expect the semimajor axis to become unbounded as an ellipse becomes parabolic: clearly $a \rightarrow \infty$ when $\lambda \rightarrow 2$.

We return now to Eq. (5.11), which we shall use in the remainder of this section. Suppose that burnout conditions are such that the velocity vector is at right angles to the radius vector; i.e., $\gamma = 0$. Equation (5.11) then would become

$$\frac{r_0}{r(\theta)} = \frac{1 - \cos\theta + \lambda \cos\theta}{\lambda} \quad (5.21)$$

Obviously, the initial conditions are satisfied if $\theta = 0$ when $r = r_0$. Also, if λ is taken as unity, $r(\theta) = r_0$ for all values of θ , and the orbit is circular. Still keeping $\gamma = 0$, we may solve Eq. (5.15) for λ to get

$$\lambda = 1 \pm e$$

or

$$e = 1 - \lambda$$

and

$$e = \lambda - 1 \quad (5.22)$$

Clearly for an elliptical orbit λ may take on any value over a range of 0 to 2, with $\lambda = 1$ corresponding to the special case of a circular orbit and $\lambda = 0, 2$ the parabolic orbits; thus elliptical orbits are defined for λ over the two non-intersecting domains: $0 < \lambda < 1$ and $1 < \lambda < 2$. How can this be? If V , the velocity burnout, is less than circular, the trajectory is elliptical; if it is greater than circular but less than escape (parabolic), it is elliptical again. In other words, assume for a simple argument that the RV is launched normal to the radius vector ($\gamma = 0$) at a velocity magnitude well below circular. The trajectory is characterized by an elliptical orbit that intersects the Earth. If the velocity at burnout is increased, then the ellipse becomes less eccentric; i.e., $e \rightarrow 0$. If the burnout velocity magnitude is equal to $[\mu(Re)]^{1/2}$, then the orbit is circular. A further increase in the velocity at burnout means that the orbit becomes elliptic, the eccentricity now increasing with increasing velocity magnitude at burnout. Finally, at a velocity magnitude of $(2\mu Re)^{1/2}$ the orbit becomes parabolic. For $0 \leq \lambda < 1$, the burnout point is the apogee of the orbit; for $1 < \lambda < 2$, the burnout point is the perigee. Figure 5.6 illustrates orbits corresponding to various values of λ . Since we are considering suborbital motion from burnout to re-entry we shall confine our attention to orbits for which $\lambda < 1$.

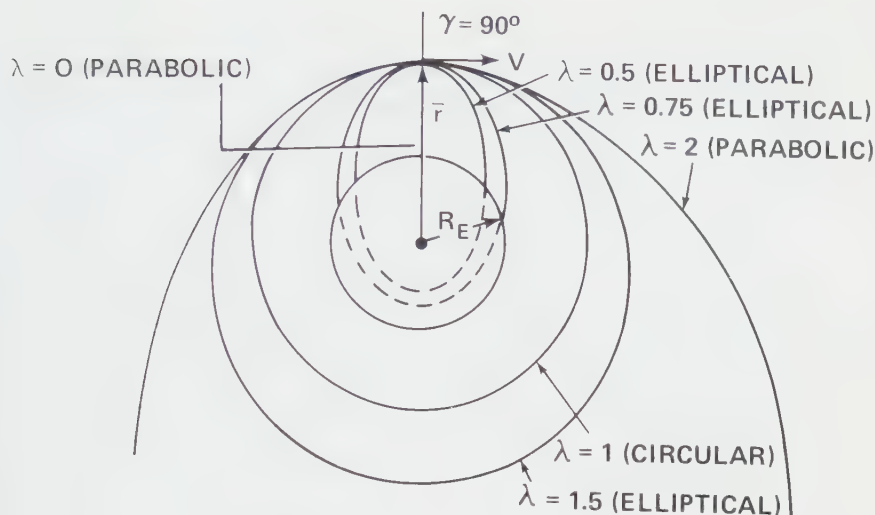


Fig. 5.6 Trajectory geometry.

Table 5.2 Program to Calculate Burnout Speed
as a Function of Range Angle and Flight-Path Angle

```

00100 REM THIS PROGRAM CALCULATES THE BURNOUT SPEED AS A FUNCTION
00110 REM OF BURNOUT ANGLE WITH RANGE ANGLE AND ALTITUDE AS PARAMETERS.
00120 REM V IS THE BURNOUT VELOCITY MAGNITUDE (M/SEC.) Z IS THE BURNOUT
00130 REM ALTITUDE ( METERS). G IS THE FLIGHT PATH ANGLE (DEG.) AND
00140 REM R9 IS THE RANGE ANGLE (DEG.). R IS THE EARTH'S RADIUS (METERS).
00150 REM M1 IS THE GRAVITATIONAL CONSTANT TIMES THE EARTH'S MASS.
00160 REM THE PRINT OUT INCREMENT IN RANGE ANGLE IS T1.
00170 REM THE PRINT OUT INCREMENT IN FLIGHT PATH ANGLE IS G1.
00180 PRINT #VELOCITY#, #FLT PATH ANG#, #RANGE ANG#, #ALTITUDE#
00190 PRINT #M/SEC.#, #DEGREES#, #DEGREES#, #FEET#
00200 Z = 0
00210 T1 = 15
00220 G1 = 10
00230 M1 = 3.9853E14
00240 R = 6.378E6
00250 R9 = 3.14159/180
00260 T1 = T1*R9
00270 G1 = G1*R9
00280 R1 = R*Z
00290 FOR I = 1 TO 10
00300 FOR J = 1 TO 8
00310 T = T1*I
00320 G = G1*J
00330 V = (1-COS(T))/((R1/R)*((COS(G))^2)-COS(T*G)*COS(G))
00340 V = SQRT((M1/R1)*V)
00350 PRINT V,G/R9,T/R9,Z
00360 NEXT J
00370 PRINT
00380 NEXT I
00390 END

```


5.3 Impact Equation

Let us define Earth impact to occur when $r = R_E$ and the corresponding central angle, $\theta = \theta_i$, to be the range angle. We can define the burnout velocity magnitude V for a given λ and θ_i by solving Eq. (5.11) for V :

$$V = \left\{ \frac{\mu}{r_0} \left[\frac{(1 - \cos \theta_i)}{(r_0/R_E) \cos^2 \gamma - \cos(\theta_i + \gamma) \cos \gamma} \right] \right\}^{1/2} \quad (5.23a)$$

Equation (5.23a) is quite useful for developing a physical sense of how the burnout parameters control range. A program to calculate $V = f(\gamma; \theta_i)$ based upon Eq. (5.23a) is given in Table 5.2.

Equation (5.23a) provides the magnitude of the burnout velocity as a function of range angle θ_i , flight-path angle γ , and altitude z (in $r_0 = R_E + z$). A program to calculate $V = f(\gamma, \theta_i)$ based upon Eq. (5.23a) is given in Table 5.2. A sample output from this program follows:

VELOCITY M/SEC.	FLIGHT PATH DEGREES	RANGE AND DEGREES	ALTITUDE FEET
5247.96	10.	15.	0
4335.52	20.	15.	0
3933.22	30.	15.	0
3800.1	40.	15.	0
3878.72	50.	15.	0
4201.88	60.	15.	0
4942.19	70.	15.	0
6856.57	80.	15.	0

The radius vector at burnout of course also follows from Eq. (5.11), with $r(\theta) = R_E$ and $\theta = \theta_i$:

$$\frac{r_0}{R_E} = \frac{1 - \cos \theta_i}{\gamma \cos^2 \gamma} + \frac{\cos(\gamma + \theta_i)}{\cos \gamma} \quad (5.23b)$$

Equations (5.23) are not independent but are alternate expressions of Eq. (5.11)

Equations (5.23) can be programmed easily on a desk calculator. Figure 5.7 is a plot of velocity magnitude at burnout vs flight-path angle at burnout, with range angle taken as a parameter. We noticed immediately that for a given range angle, say 30 deg, there are two burnout angles γ at the same velocity magnitude that will give this range. For example, at a burnout velocity magnitude of 6000 m/s, flight-path angles of about 7 and 63 deg will give this range of 30 deg. The trajectory corresponding to the low flight-path angle is called "under-lofted." The trajectory corresponding to the highest value of flight-path angle is "over-lofted." Because of the multivalued nature of the inverse of $V = V(\gamma)$, there are usually two trajectories corresponding to each burnout velocity for a fixed range.

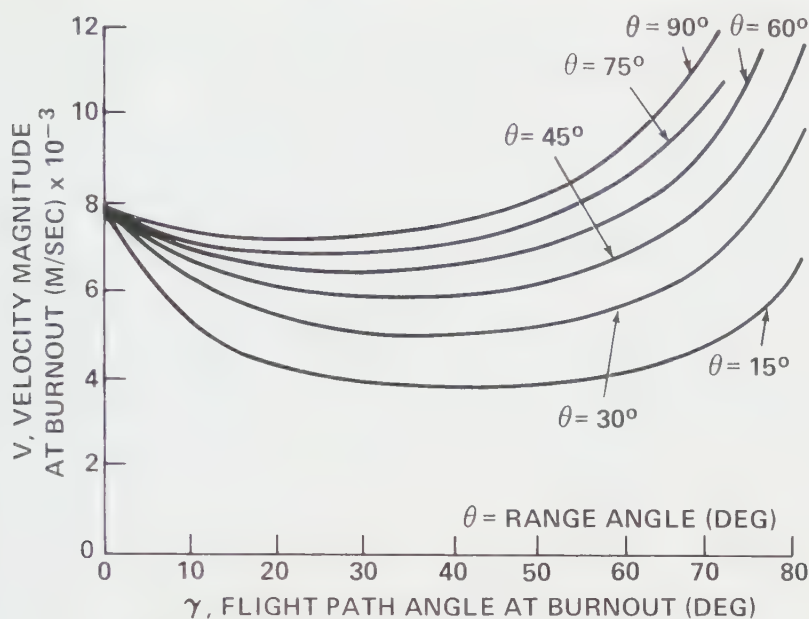


Fig. 5.7 Velocity magnitude at burnout vs flight-path angle at burnout for several range angles.

As might be expected, the familiar flat Earth approximation falls out of Eq. (5.23a) by letting $\theta_i \ll 1$:

$$V^2 = \left[\frac{\mu}{2R_E} \right] \left[\frac{\theta_i}{\sin \gamma \cos \gamma} \right] = \frac{gR_E}{\sin 2\gamma} \quad (5.24)$$

An important conclusion which follows from Eq. (5.23a) should be emphasized: there is an infinite number of pairs of (V, γ) corresponding to a fixed range angle θ_i .

Of at least academic interest is the so-called "optimum" burnout angle γ^* . This optimum will be taken to mean the angle γ which corresponds to the smallest velocity required to subtend a given range angle θ_i . It should be pointed out that more than likely there will be considerations which will dictate some trajectory other than this "optimum." Clearly γ^* goes from 0 for a range of 180 deg to 45 for a range of 0 deg (or flat Earth trajectory). One can compute γ^* by differentiating Eq. (5.23a), setting the result equal to zero, and solving for γ^* . We get

$$\tan 2\gamma^* = \frac{\sin \theta_i}{(1 + z/R_E) - \cos \theta_i} \quad (5.25a)$$

In the event that the altitude z is considered negligible compared to the

Earth's radius, the above expression may be simply written as

$$\gamma^* = (\pi/4) - (\theta_i/4) \quad (5.25b)$$

The familiar flat Earth optimum range angle γ^* follows as $\pi/4$. This result also could have been obtained from Eq. (5.24) by differentiating both sides with respect to γ , setting the derivative to zero, and solving for $\gamma = \gamma^*$.

5.4 Time of Flight

As pointed out earlier, the analytic relationship containing time is expressed as $t = t(\theta)$ rather than the preferred form, $\theta = \theta(t)$. The reason for accepting $t = t(\theta)$ is that this is apparently the only form in which t can be expressed using elementary functions. The procedure for obtaining $t = t(\theta)$ is fairly straightforward. Using Eq. (5.5), we may write

$$\int_0^{\theta(t)} r^2 dx = \int_0^t r_0 V \cos \gamma dt = (r_0 V \cos \gamma) t \quad (5.26)$$

where x is, of course, a dummy variable of integration; the quantity $r_0 V \cos \gamma$ may be removed from the integral on the right-hand side because of Eq. (5.5), i.e., constancy of angular momentum in a central force field. We may replace r in the integral on the left-hand side by $r = r(\theta)$ from Eq. (5.11). Thus,

$$t = \frac{r_0}{V \cos \gamma} \int_0^{\theta_i(T)} \left[\frac{1 - \cos x}{\lambda \cos^2 \gamma} + \frac{\cos(\gamma + x)}{\cos \gamma} \right]^{-2} dx \quad (5.27a)$$

If the integration is carried out, we get

$$t = \frac{r_0}{V \cos \gamma} \left\{ \frac{\tan \gamma (1 - \cos \theta) + (1 - \lambda) \sin \theta}{(2 - \lambda) \{ [(1 - \cos \theta) \gamma \cos \gamma^2] + [\cos(\gamma + \theta) / \cos \gamma] \}} \right. \\ \left. + \frac{2 \cos \gamma}{\lambda [(2/\lambda) - 1]^{3/2}} \tan^{-1} \left[\frac{2/\lambda - 1}{\cos \gamma \cot(\theta/2) - \sin \gamma} \right] \right\} \quad (5.27b)$$

If we define total time of flight as the elapsed time from burnout (or RV off-loading at boost termination) to impact with the Earth or the beginning of atmospheric re-entry, then we may replace t and θ in Eq. (5.27b) by T and θ_i , respectively:

$$T = \frac{r_0}{V \cos \gamma} \left\{ \frac{\tan \gamma (1 - \cos \theta_i) + (1 - \lambda) \sin \theta_i}{(2 - \lambda) \{ [(1 - \cos \theta_i) / \gamma \cos \gamma^2] + [\cos(\gamma + \theta_i) / \cos \gamma] \}} \right. \\ \left. + \frac{2 \cos \gamma}{\lambda [(2/\lambda) - 1]^{3/2}} \tan^{-1} \left[\frac{[(2/\lambda) - 1]^{1/2}}{\cos \gamma \cot(\theta/2) - \sin \gamma} \right] \right\} \quad (5.28)$$

Equation (5.28) expresses the time of flight T in the Keplerian phase to cover the range angle θ_i . Thus, we have

$$T = T(\theta_i; r_0, \lambda, \gamma, V) \quad (5.29)$$

However, it has been noted in Eq. (5.7b) that λ is a nondimensional velocity parameter; i.e., $\lambda = V^2/(\mu/r_0)$, and so we may write Eq. (5.29) in terms of V , recognizing that, if V and r_0 are known, then so is λ . Thus,

$$T = T(\theta_i; r_0, \gamma, V) \quad (5.30)$$

The time of flight T is shown to be a function of one terminal variable θ_i , and three initial variables, r_0 , γ , and V . Since Eq. (5.28) provides time of flight, or time elapsed from boost termination to trajectory termination, we must use the impact equation [Eq. (5.23)] to establish V (and λ). Thus V , the velocity magnitude at boost termination, is not an independent variable but depends upon θ_i , r_0 , γ . To calculate the time of flight T , we should use the following equations:

$$T = T(\theta_i; r_0, \gamma, V) \quad (5.31)$$

$$V = V(\theta_i; r_0, \gamma) \quad (5.23a)$$

$$\lambda = V^2/(\mu/r_0) \quad (5.7b)$$

There is no single procedure that we must follow in using the above equations. However, one path we might take is to regard the range angle θ_i to be a given quantity. Now Eq. (5.23a) and Fig 5.7 show that there is a bounded and continuous set of pairs of (V, γ) which will allow us to subtend this range angle. It should be clear at this point that setting a range angle θ_i does not uniquely define a set (V, γ) . If $\gamma < \gamma^*$, then we shall have trajectories that we shall define as "under-lofted" (more or less), and in like manner, if $\gamma > \gamma^*$, then we shall have trajectories that we shall define as "over-lofted" (more or less). Thus, we first might specify a range angle θ_i , then a flight-path angle γ , and finally use Eq. (5.23a) to obtain the magnitude of the velocity at burnout, V . All of this would assume that $r_0 = R_E$; if $r_0 = R_E + z$, then sometime before obtaining V we first would have to specify the altitude z . To simplify things in the remainder of the discussion, we shall assume that $z = 0$, although one might specify a typical burnout altitude of 300 km.

An alternate argument might require that the speed at burnout, V , be specified before flight-path angle γ , at burnout. We might reason that V is related to the energy requirements that must be met during the boost phase. If energy is critical, then we probably should select the minimum burnout speed for a given range angle. Minimum burnout speed would require that the flight-path angle at burnout γ , be equal to γ^* . As we shall see when we consider the boost phase, there is usually surplus energy that must be "wasted" if the same launch vehicle is to be used to boost an RV with a set of range options. Since all of the boost propellant must be consumed, vari-

ous boost trajectories must be followed in order to achieve various velocity magnitudes at burnout. Nevertheless, it might seem more "logical" to select velocity magnitude at burnout, and then compute the appropriate flight-path angle. Assuming that the range angle θ_i and the speed at burnout V are fixed, we have a minor difficulty on our hands. We must obtain the inverse of Eq. (5.23a), that is,

$$\gamma = \gamma(\theta_i; V, z) \quad (5.32)$$

If we consider the transcendental nature of Eq. (5.23a), we might anticipate considerable difficulty in obtaining the above inverse. The above inverse is also double-valued, since there are two flight-path angles for every velocity magnitude at burnout, except of course for $\gamma = \gamma^*$.

Without belaboring the point any longer, let us assume that we shall follow the procedure outlined below:

(a) Select a range angle θ_i .

(b) From Fig 5.7 select a velocity magnitude and two flight-path angles, one corresponding to an under-lofted trajectory and the other to an over-lofted trajectory.

Table 5.3 Calculation of Time of Flight

```

00100 REM THIS PROGRAM CALCULATES THE TIME OF FLIGHT OF A SUBORBITAL
00110 REM REENTRY VEHICLE. V IS THE VELOCITY MAGNITUDE AT BURNOUT IN
00120 REM METERS PER SECOND. G IS THE FLIGHT PATH ANGLE AT BURNOUT
00130 REM IN DEGREES. A IS THE RANGE ANGLE IN DEGREES. Z IS THE ALTITUDE
00140 REM AT BURN OUT IN METERS. R1 IS THE RADIUS OF THE EARTH IN METERS.
00150 REM M1 IS THE GRAVITATIONAL CONSTANT TIMES THE EARTH'S MASS.
00160 G = 60
00170 Z = 0
00180 R1 = 6.378E6
00190 M1 = 3.9853E14
00200 A = 90
00210 R9 = 3.14159/180
00220 R = R1 + Z
00230 G = G/R9
00240 A = A/R9
00250 V = (1-COS(A))*M1/R
00260 V = V/((R/R1)*(COS(G)^2)-COS(A+G)*COS(G))
00270 L = R*V/M1
00280 V = SQR(V)
00290 T8 = TAN(G)*(1-COS(A))+(1-L)*SIN(A)
00300 T9 = (1-COS(A))/(L*(COS(G)^2))
00310 T9 = (T9*(COS(G+A)/COS(G)))*(2-L)
00320 T7 = T8/T9
00330 T6 = (COS(G)/TAN(A/2))-SIN(G)
00340 IF ABS(T6)> 0.001 THEN 00370
00350 T9 = 3.14159/2
00360 GOTO 00420
00370 T8 = (SQR((2/L)-1))/T6
00380 T9 = ATN(T8)
00390 IF T8>0 THEN 00420
00400 T9 = T9+3.14159
00410 GOTO 00420
00420 T9 = T9*2*COS(G)/(L*((2/L)-1)^(3/2))
00430 T8 = T7*T9
00440 T = T8*(R/(V*COS(G)))
00450 PRINT #FLT-TIME#,#BO SPEED#,#BO FLT ANG#,#RANGE ANG#
00460 PRINT #SECONDS#,#M/SEC#,#DEGREES#,#DEGREES#
00470 PRINT
00480 PRINT T,V,G/R9,A/R9
00490 END

```

(c) Using the range angle θ_i and the flight-path angle at burnout, γ , find the velocity magnitude at burnout from Eq. (5.23a).

(d) With V available, calculate the velocity parameter λ from Eq. (5.7b).

(e) Now with V , θ_i , γ , λ available, use Eq. (5.31) to calculate the time of flight, T .

(f) Repeat step (e) using the flight-path angle at burnout, γ , corresponding to the over-lofted trajectory.

A short computer program has been written to allow us to follow the above procedure systematically. This is essentially a programming of Eqs. (5.23a) and (5.31). A listing is given in Table 5.3.

Let us assume that we wish to compare the time of flight of an under-lofted and over-lofted trajectory with a range angle of 75 deg. We might take a flight-path angle of 10 deg for the under-lofted trajectory. Then, going to Fig 5.7, it may be found that, for the same velocity magnitude, the over-lofted trajectory would have a flight-path angle at burnout of 42.5 deg. The output from Table 5.3 gives the time of flight for both trajectories:

FLT-TIME SECONDS	BO SPEED M/SEC	BO FLT ANG DEGREES	RANGE ANG DEGREES
1271.18	7238.03	10.	75.
2577.69	7238.03	42.5	75.

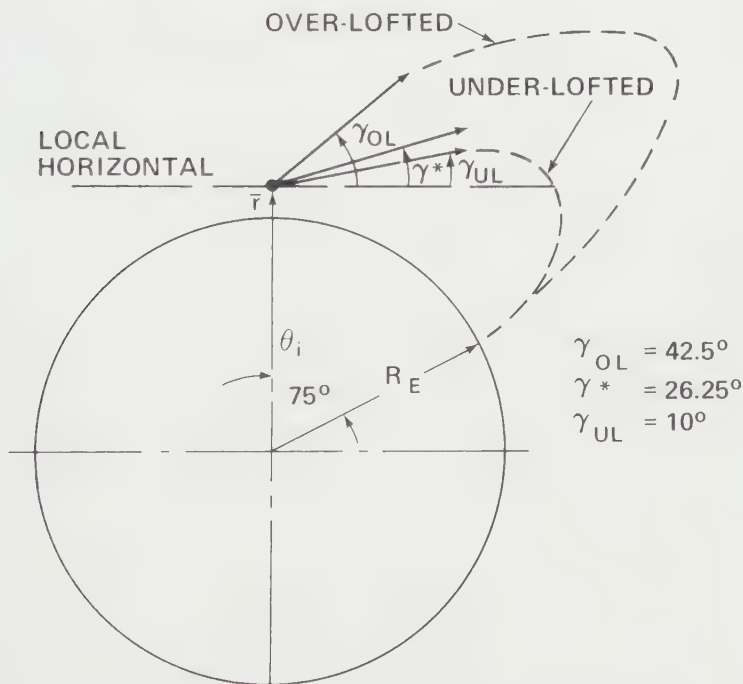


Fig. 5.8 Geometric relationships among γ_{OL} , γ_{UL} , γ^* , V , and θ_i .

It will be noted that the time of flight for the over-lofted trajectory is about twice that for the under-lofted trajectory. Some of the appropriate geometry is illustrated in Fig 5.8. The value for γ^* follows from Eq. (5.25b) as

$$\gamma^* = 45^\circ - 75^\circ/4 = 26.25^\circ$$

5.5 Error Analysis

The analysis and estimation of errors in a ballistic missile delivery system is an extremely complex subject involving the formulation of an error differential equation from the system differential equations. Such analyses are a direct outgrowth of the classical methods of error analysis which involved the insertion of a nominal variable plus a perturbative variable into the system differential equations. The resulting differential equation in the perturbative variable then must be solved. We shall use a somewhat more direct approach here. Rather than introduce a perturbative variable into the system's differential equation [Eq. (5.7)], we shall introduce the perturbative variable into the solution of this equation [Eq. (5.23)]. The easiest way to do this is to take the differential of Eq. (5.23a) or (5.23b). The differential variables are V , h , γ , and θ_i (the speed, altitude, and flight-path angle at burn-out, and the range angle). After some manipulation, we get

$$\begin{aligned} & \frac{2\delta V}{V} \left[\frac{1 - \cos\theta_i}{\lambda \cos^2\gamma} \right] + \frac{\delta h}{E^R} \left[1 + \frac{1 - \cos\theta_i}{\lambda \cos^2\gamma} \right] \\ &= \delta\gamma \left[\frac{\sin(\gamma + \theta_i)}{\cos\gamma} - \left[\frac{\sin\gamma}{\cos^3\gamma} \right] \cos(\gamma + \theta_i) - \tan\gamma \left[\frac{1 - \cos\theta_i}{\gamma \cos^2\gamma} \right] \right] \\ & \quad + \delta\theta_i \left[\frac{\sin\theta_i}{\lambda \cos^2\gamma} - \frac{\sin(\gamma + \theta_i)}{\cos\gamma} \right] \end{aligned} \quad (5.33)$$

It will be noted in Eqs. (5.23) that we used the variable r_0 as the position of the RV at burnout, i.e., distance of RV from Earth's center. In Eq. (5.33), we have replaced r_0 by $R_E + z$, where z is the altitude at burnout. Also note that, in deriving Eq. (5.33), we have made the approximation $1/(R_E + z) = 1/R_E$. If we recognize that the range error δR may be expressed approximately as

$$\delta R = R_E \delta\theta_i \quad (5.34)$$

then we should get

$$\frac{\delta R}{\delta V} = \frac{2R_E}{V} \left[\frac{1 - \cos\theta_i}{\sin\theta_i - \lambda \sin(\gamma + \theta_i) \cos\gamma} \right] \quad (5.35)$$

If we make the approximation that $r_0 \simeq R_E$ (thereby eliminating z as a variable), we can solve the impact equation [Eq. (5.23b)] for λ , the energy or velocity parameter, and, using this value, eliminate λ from Eq. (5.35) to get

$$\frac{\delta R}{\delta V} = \frac{2R_E}{V} [\sin\theta_i + \cot\gamma(1 - \cos\theta_i)] \quad (5.36)$$

The speed at burnout, V , still remains; this is not an independent variable, since it also must satisfy the impact equation [Eq. (5.23a)], which means that V varies directly with θ_i and parametrically with γ . This being understood, Eq. (5.36) indicates that the range error at impact per unit velocity magnitude error at burnout depends upon the range θ_i and the flight-path angle γ at burnout. Obviously, Eqs. (5.23a) and (5.36) could be combined to give $\delta R / \delta V$ vs γ plots with θ_i as a parameter, or vs θ_i with γ as a parameter, as suggested above. Let us assume that the range angle $\theta_i = 90$ deg. Referring to Eq. (5.25), the optimum burnout angle γ^* at this range is 22.5 deg. According to Eq. (5.23a), the required velocity magnitude will be 7195 m/s. Inserting the values as needed in Eq. (5.36), we get

$$\frac{\delta R}{\delta V} \simeq 6 \frac{\text{km}}{(\text{m/s})} \quad (5.37)$$

An error of 1 m/s out of 7195 m/s will result in a range error of 6 km. From Eq. (5.34), we note that a lofted trajectory (high γ) is less sensitive to speed errors than the "optimum," i.e., $\gamma = \gamma^*$, or the under-lofted trajectories. In comparing the over- and under-lofted trajectories with the optimum, there is a performance penalty since velocity magnitude, V , at burnout is higher for both nonoptimum trajectories. However, larger V means smaller error coefficient, $\delta R / \delta V$. Thus, from the point of view of minimum range error due to velocity, it would appear that the over-lofted is better than either the minimum velocity ($\gamma = \gamma^*$) or the under-lofted ($\gamma < \gamma^*$).

Next we can consider range errors due to errors in controlling the flight-path angle at burnout, γ . We return to Eq. (5.33), making use of Eq. (5.34) to get

$$\frac{\delta R}{\delta \gamma} = \frac{\lambda \sin(\gamma + \theta_i) \cos^2 \gamma - \lambda \sin \gamma \cos(\gamma + \theta_i) - \tan \gamma (1 - \cos \theta_i) \cos \gamma}{\sin \theta_i \cos \gamma - \lambda \cos^2 \gamma \sin(\gamma + \theta_i)} \quad (5.38)$$

Again solving Eq. (5.23b) for λ , the energy parameter, and inserting in Eq. (5.38) we get

$$\frac{\delta R}{\delta \gamma} = 2R_E \left[1 - \frac{\sin(\theta_i + 2\gamma)}{\sin 2\gamma} \right] \quad (5.39)$$

It should be noted that, if $\gamma = \gamma^*$, or the flight-path angle at burnout corresponding to the minimum velocity magnitude at burnout, is inserted

into Eq. (5.39), then the error coefficient $\delta R / \delta \gamma = 0$, regardless of the range angle θ_i . For $\gamma < \gamma^*$, $\delta R / \delta \gamma$ is negative, and for $\gamma > \gamma^*$, $\delta R / \delta \gamma$ is positive.

Let us assume a range angle of 75 deg and a flight-path angle at burnout of 15 deg. Inserting these quantities in Eq. (5.39), we get

$$\frac{\delta R}{\delta \gamma} = -5.28 \frac{\text{km}}{\text{mrad}} \quad (5.40)$$

Thus, for the conditions just quoted, if the flight-path angle at burnout is in error by 1 mrad, then there will be a corresponding range error of 5.28 km (short).

By a procedure similar to the development of Eqs. (5.36) and (5.39), we can find the variation in range at impact with errors in burnout height:

$$\frac{\delta R}{\delta h} = 2 \cot \gamma - \left[\frac{\cos(\gamma + \theta_i)}{\cos \gamma} \right] \quad (5.41)$$

Let us assume that $\gamma = 22.5$ deg and $\theta_i = 90$ deg; from Eq. (5.41), we get

$$\frac{\delta R}{\delta h} = 5.24 \frac{\text{km}}{\text{km}} \quad (5.42)$$

Another error source that is important for ballistic RV's is that due to lateral velocity error. If there is a small lateral velocity error δV_n , the actual trajectory will be rotated slightly relative to the intended or nominal trajectory through the angle $\delta V_n / V \cos \gamma$. Figure 5.9 illustrates the basic geometry. Clearly the angle between the nominal and actual trajectories is $\delta V_n / V \cos \gamma$. Although the orbits may be elliptical, the track on the Earth will follow the great circle arcs shown. The arc length that measures the difference between

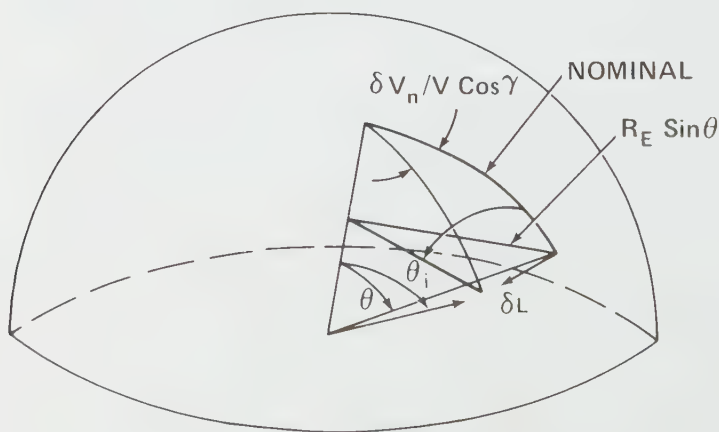


Fig. 5.9 Lateral velocity error.

the two trajectories at impact is δL or

$$\delta L = (R_E \sin \theta_i) \delta V_n / (V \cos \gamma) \quad (5.43)$$

Rearranging, we get

$$\frac{\delta L}{\delta V_n} = \frac{R_E \sin \theta_i}{V \cos \gamma} \quad (5.44)$$

By way of example, we assume that boost termination takes place at apogee ($\gamma = 0$) and that the trajectory has a range angle θ_i of 60 deg. Equation (5.23a) gives a burnout speed V of 7908 m/s (25,947 ft/s), resulting in an error coefficient of

$$\frac{\delta L}{\delta V_n} = 0.697 \frac{\text{km}}{\text{m/s}} \quad (5.45)$$

A velocity error of 10 m/s, normal to the intended trajectory, can result in an error at impact of 6.97 km. The lateral velocity error is quite important in practice, particularly for ballistic vehicles where the range angle θ_i is about 90 deg. Notice in Eq. (5.44) that for flights halfway around the world, $\theta_i = 180$ deg, this lateral error vanishes, since all great circle planes between antipodes are equivalent.

Another source of error at impact has its origin in displacement error at burnout. In other words, if end of boost, i.e., burnout, occurs at a point displaced from its intended position in the nominal trajectory, what is the corresponding error at impact? Now since position error at burnout is a vector, we might think of separating this vector into two mutually perpendicular components; the first component will be taken within the nominal trajectory, defining it as the downrange position error at burnout, the second component will be taken normal to the nominal trajectory and designated as crossrange error at burnout. The first component, that within the nominal trajectory, we shall designate as D , and the second, normal to the nominal trajectory, as N . In Fig. 5.10 the downrange error may be designated either by a central angle error $\delta\theta$ or by the downrange displacement δD . The two are related as

$$\delta D = R_E \delta\theta \quad (5.46)$$

The range error at impact, δR , is given in terms of the central angle at impact, $\delta\theta_i$, as

$$\delta R = R_E \delta\theta_i \quad (5.47)$$

Since $\delta\theta = \delta\theta_i$, we have

$$\frac{\delta D}{\delta R} = 1 \quad (5.48)$$

The reasoning is, of course, that a downrange error serves only to rotate the actual trajectory (or orbital segment) relative to the nominal trajectory, as seen in Fig. 5.10.

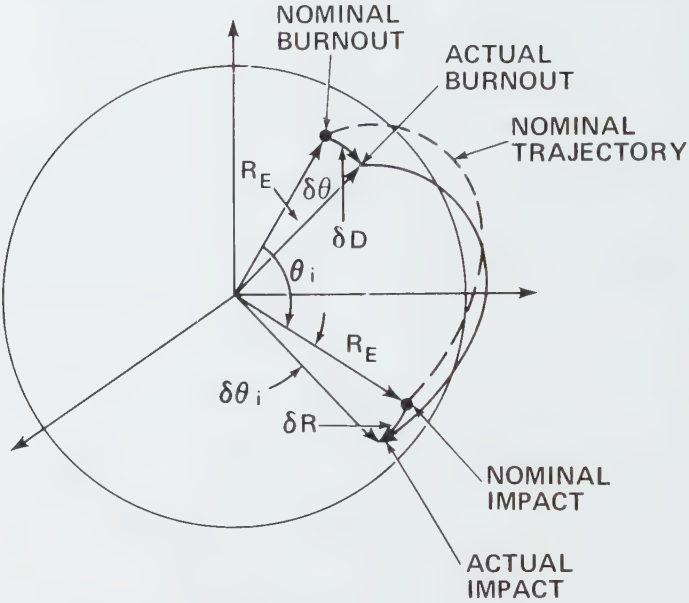


Fig. 5.10 Downrange error.

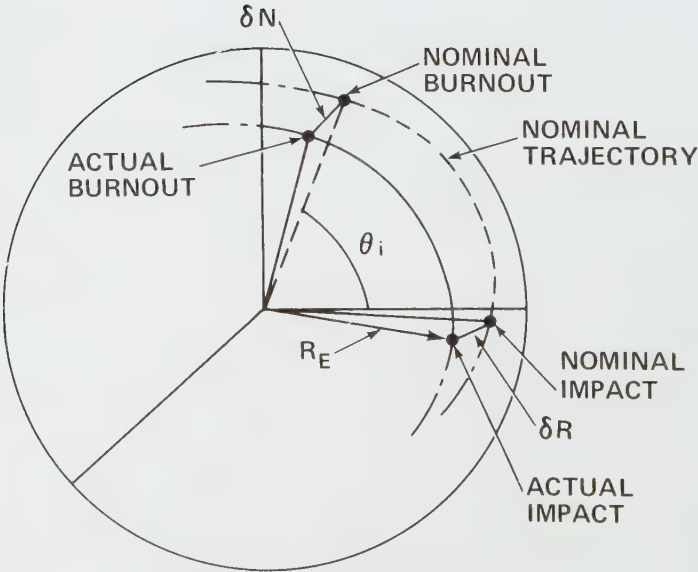


Fig. 5.11 Lateral displacement error.

We have indicated the second displacement error source as being normal to the nominal trajectory. This situation is shown in Fig. 5.11. If δN is the displacement of the burnout point normal to its intended position in the nominal trajectory, then the actual trajectory is rotated relative to the nominal trajectory. It can easily be shown that

$$\frac{\delta R}{\delta N} = \cos \theta_i \quad (5.49)$$

It will be noted that the error at impact due to the lateral *displacement* is zero for θ_i , equal to 90 deg, which is the range angle for a trajectory spanning one quarter of the Earth. The error is a maximum at 180 deg. This relationship with central angle may be compared with the error due to the lateral *velocity* [Eq. (5.44)]. For the lateral velocity error, the impact error is a maximum for a central angle of 90 deg and zero for a central angle of 180 deg.

5.6 Launch Site/Target Motion

Up to now we have assumed that the Earth is nonrotating. Although such an assumption might be acceptable for the error analysis carried out in the last section, it is inadequate in defining an impact point. It is easy to show that the target rotates through a distance d (measured along a minor circle of longitude) during flight time T , as

$$d = \Omega_E R_E (\cos L_i) T \quad (5.50)$$

where Ω_E is the Earth's rotational rate (7.29×10^{-5} rad/s), R_E the Earth's radius, and L_i the target's latitude. If L_i is taken as 45 deg, R_E as 6370 km, and a typical time of flight T taken as 2000 s, then Eq. (5.50) indicates that d will be in excess of 650 km.

In the rest of this section, we shall outline the equations involved in accounting for the Earth's rotation in setting both the velocity vector at burnout and the time of flight at the beginning of Keplerian motion until target impact. There is no hope of effecting a closed-form solution. However, the equations will be set up so that the interrelationship among the variables may be seen.

In this analysis, we shall assume that the RV is subjected to a central-inverse-square gravity field, i.e., Keplerian motion. In Fig. 5.12, the RV is launched at point B , i.e., burnout, at a latitude of L_B and intended to strike or impact the target at latitude L_T . Let Ψ_0 be the angular arc subtended in the equatorial plane by the polar great circles passing through the burnout point and the target at burnout; Ψ_0 is equivalent to the longitudinal difference between the burnout point and the target. The angular arc subtended in the equatorial plane by the polar great circles through the burnout point and the target at impact is given by

$$\Psi = \Psi_0 + \Omega_E T \quad (5.51)$$

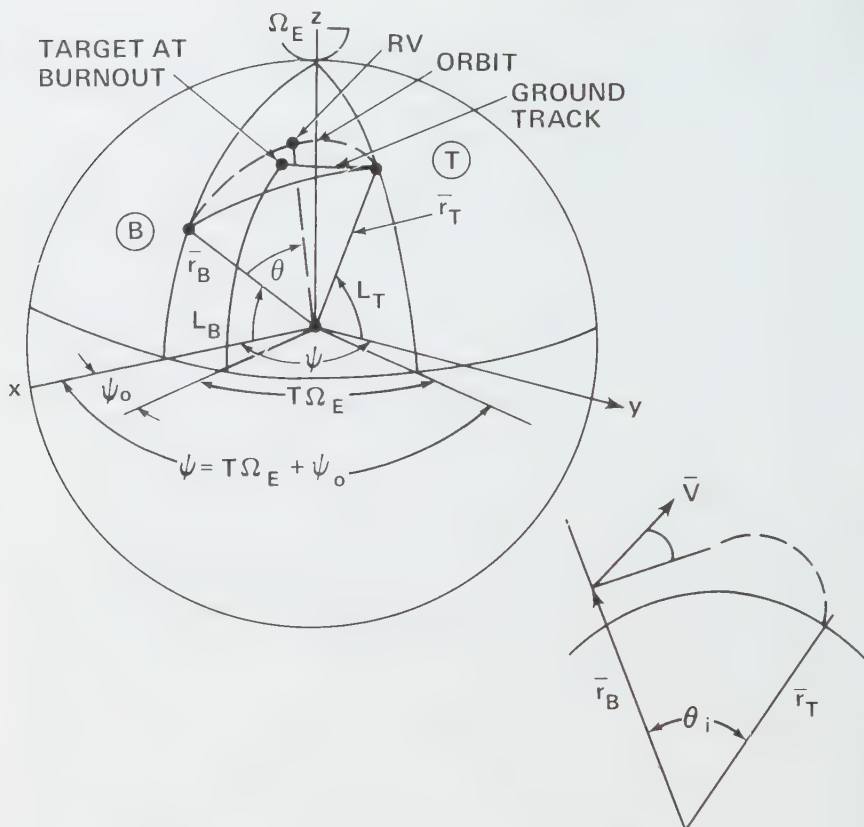


Fig. 5.12 Target motion due to Earth rotation.

It should be pointed out that Ψ is *not* the longitude in an Earth-fixed frame but the angular displacement between two Earth meridian planes, one plane containing the RV at burnout, and the second containing the RV at impact. The angle Ψ measures the angle about the Earth's polar axis which the RV must subtend in passing from burnout to target. Equation (5.51) shows that part of the angle is due to the longitudinal separation of the burnout point and target point at launch, and part of this angle is due to the rotation of the Earth during the time of flight T .

The burnout and target vectors, $\mathbf{r}_B$, $\mathbf{r}_T$, may conveniently be expressed in inertial (x, y, z) frame of Fig. 5.12. Note that the RV is, by definition, in the $x-z$ plane at burnout or boost termination:

$$\mathbf{r}_B = (r_0 \cos L_B) \mathbf{i} + (r_0 \sin L_B) \mathbf{k} \quad (5.52a)$$

$$\mathbf{r}_T = (R_E \cos L_T \cos \Psi) \mathbf{i} + (R_E \cos L_T \sin \Psi) \mathbf{j} + (R_E \sin L_T) \mathbf{k} \quad (5.52b)$$

where $r_0 = R_E + z$ as before [Eq. (5.5)]. The central angle still will be desig-

nated by θ , with θ_i the range angle. From the definition of the inner product, we have

$$\cos\theta_i = (\mathbf{r}_B \cdot \mathbf{r}_T) / r_0 R_E = \cos L_B \cos L_T \cos \Psi + \sin L_B \sin L_T \quad (5.53)$$

If we take a glance at Eq. (5.28), we recognize that we must have the range angle θ_i in order to calculate the time of flight, but, since Ψ depends upon the time of flight T , Eq. (5.53) indicates that θ_i cannot be calculated until we know the time of flight. Obviously, an iterative procedure must be employed. Apparently, we first must guess T , calculate Ψ from Eq. (5.51), calculate θ_i from Eq. (5.53), and finally use Eq. (5.28) to calculate a new T . The procedure should rapidly converge. If the RV is unguided, there may be an onboard provision for continual measurement of time as sort of an open-loop indication of position. It should be emphasized at this point that the velocity magnitude at burnout relative to inertial space (nonrotating Earth) should be inserted for V in Eq. (5.28). For a rotating Earth, we will find it necessary to differentiate between velocity magnitudes relative to inertial space and relative to a rotating Earth. This distinction is drawn in Eq. (5.55) below. To avoid confusion, U will be taken as the velocity magnitude relative to inertial space, and V will now be the velocity magnitude relative to the rotating Earth. For a nonrotating Earth, U and V are identical.

The guidance relationships for a ballistic trajectory on a rotating Earth are a three-dimensional version of the impact equation [Eq. (5.23)]. The goal is to develop the equations of the velocity vector at burnout. It would seem that there are four quantities that must be set at burnout to meet the impact conditions: velocity magnitude and three-directional cosines. However, there are only three conditions that can be set independently:

- (a) The velocity vector at burnout must lie in the trajectory plane, i.e., a plane determined by $\mathbf{r}_B$ and $\mathbf{r}_T$.
- (b) The vector $\mathbf{r}$ to the RV must coincide with the target vector $\mathbf{r}_T$ at the end of T seconds.
- (c) The directional cosines satisfy the normalization rule.

Since we have four quantities and only three equations, we shall assume that the velocity magnitude V has been determined from other considerations, and we shall express the directional cosines (l, m, n) as functions of V . This is equivalent in the two-dimensional case discussed earlier to expressing flight-path angle γ at burnout as a function of velocity magnitude V , i.e., the inverse of Eq. (5.23a) (see Fig. 5.7).

In the inertial coordinates, the missile velocity is the vectoral sum of the velocity relative to the Earth, $\mathbf{V}$, and the linear velocity due to the Earth's rotational rate, i.e., $\Omega_E r_0 \cos(L_B)$. If $\mathbf{U}$ is the velocity vector of the RV relative to inertial space, then we may write

$$\mathbf{U} = \mathbf{V} + [\Omega_E r_0 \cos(L_B)] \mathbf{j} \quad (5.54)$$

Let us express the velocity vector of the RV relative to the Earth in terms of

directional cosines:

$$\mathbf{V} = (lV)\mathbf{i} + (mV)\mathbf{j} + (nV)\mathbf{k} \quad (5.55a)$$

We may now write, for Eq. (5.54)

$$\mathbf{U} = (lV)\mathbf{i} + [mV + \Omega_E r_0 \cos(L_B)]\mathbf{j} + (nV)\mathbf{k} \quad (5.55b)$$

The first guidance condition noted in condition (a) above may be written as

$$\mathbf{U} \cdot (\mathbf{r}_B \times \mathbf{r}_T) = 0 \quad (5.56a)$$

which becomes, upon using Eq. (5.52) and (5.55),

$$\begin{aligned} [m + (\Omega_E r_0 / V) \cos L_B] (\sin L_B \cos L_T \cos \Psi - \sin L_T \cos L_B) \\ + n \cos L_B \cos L_T \sin \Psi - l \sin L_B \cos L_T \sin \Psi = 0 \end{aligned} \quad (5.56b)$$

The second guidance condition follows from the planar impact equation given earlier [see Eq. (5.23b)]:

$$\frac{r_0}{R_E} = \frac{\mu}{r_0 U^2} \frac{1 - \cos \theta_i}{\cos^2 \gamma} + \frac{\cos(\gamma + \theta_i)}{\cos \gamma} \quad (5.57)$$

The central angle θ_i follows Eqs. (5.53). The magnitude of the velocity vector $\mathbf{U}$ is

$$U = V \left[l^2 + \left[m + \frac{\Omega_E r_0 \cos L_B}{V} \right]^2 + n^2 \right]^{1/2} \quad (5.58)$$

and the flight-path angle, defined as the angle between the velocity vector $\mathbf{U}$ and the local horizontal, follows from Eqs. (5.52a) and (5.55) as

$$\begin{aligned} \sin \gamma &= \frac{\mathbf{U} \cdot \mathbf{r}_b}{|\mathbf{U}| |\mathbf{r}_B|} \\ &= \frac{l \cos L_B + n \sin L_B}{\left\{ l^2 + \left[m + (\Omega_E r_0 \cos L_B) / V \right]^2 + n^2 \right\}^{1/2}} \end{aligned} \quad (5.59)$$

Finally the third guidance condition regarding the orthogonality of the inertial axes gives

$$l^2 + m^2 + n^2 = 1 \quad (5.60)$$

As pointed out above, we shall regard V , the magnitude of the burnout

velocity relative to the Earth, to be given. Equations (5.56), (5.57), and (5.60) will then, in principle, allow us to express the direction cosines (at burnout) (l, m, n) as functions of V .

Unfortunately the flight time T is embedded in the three equations, since Ψ is a function of the time of flight [see Eq. (5.51)]. The flight time T is a function of the range angle θ_i , which according to Eq. (5.53) is a function of Ψ and hence T . We can get time of flight into the problem by using Eqs. (5.27) as

$$T = \frac{r_0}{|U| \cos \gamma} \int_0^{\theta_i(T)} \left[\left(\frac{\mu}{r_0 U^2} \right) \left(\frac{1 - \cos x}{\cos^2 \gamma} \right) + \frac{\cos(\gamma + x)}{\cos \gamma} \right]^{-1/2} dx \quad (5.61)$$

As shown by Eqs. (5.27), we can integrate Eq. (5.61) analytically.

It might be of some interest to bring this section to completion by listing the various equations that are of use in dealing with the three-dimensional rotating-Earth Keplerian trajectory. Before doing this, however, let us review briefly the development of the two-dimensional nonrotating Earth trajectory. In that effort, we first specified the central angle θ_i subtended by the trajectory from end of boost (burnout) to impact. For the nonrotating-Earth problem, knowledge of the latitude of both the burnout and impact point is sufficient for the determination of θ_i . This can be done either by spherical trigonometry or by vectoral procedures, such as that which led to Eq. (5.53). It will be noted in Eq. (5.53) that, if the Earth does not rotate, then $\Omega_E = 0$ and $\Psi = \Psi_0$. Thus θ_i , the range angle depends upon the *difference* in longitude of launch and target points and the latitudes of these points. It was shown that, with θ_i available, a set of values of flight-path angle γ , inserted into Eq. (5.23a), leads to a corresponding set of values of the magnitude of the burnout velocity V . Plotting results of this exercise led to Fig. 5.7. The graphical presentation of V vs γ means that we have essentially inverted Eq. (5.23a) and have obtained γ as a "function" of V . [The quotation marks are used because the inverse, $\gamma = f(V)$, is multivalued and not, therefore, a function.] An important point here is that nowhere did time of flight T enter the problem. Time of flight was evaluated as something of an afterthought; this was done in Eq. (5.31), and a program was given to compute T for values of γ , V , θ_i which were already calculated. The important point is that time of flight is not necessary to set the conditions at end of boost, i.e., V and γ , to meet the range requirements, or to set the central angle at impact θ_i for the nonrotating Earth.

For the equivalent development of the Keplerian trajectory over a rotating Earth, the problem becomes much more difficult. We shall not try to solve this problem, but an appreciation for the complexity may be obtained by listing the equations we have developed so far.

Assume that we have available the latitude of the burnout point and target, i.e., L_B and L_T . Also, assume that we have the difference between the longitude angles of target and burnout point, i.e., Ψ_0 . (If the Earth had no atmosphere or we were willing to neglect the trajectory of the boost phase, then Ψ_0 would be equal to the difference in longitude of target and launch

points.) The goal we have then set is to find the velocity vector V (magnitude and direction) at burnout so that the RV impacts the target. As we saw in the two-dimensional nonrotating Earth case, setting a range requirement, i.e., setting θ_i , does not determine a unique pair (V, γ) . Thus we sought $V = V(\gamma, \theta_i)$ or, equivalently (in principle), the function $\gamma = \gamma(V, \theta_i)$. To specify uniquely the direction of the velocity at burnout, i.e., γ , we must specify *both* θ_i , the range angle, and V , the magnitude of the velocity relative to the Earth at burnout. Any criterion that might be applied in selecting a value for V is beyond our concern here but might involve questions concerning the magnitude of the maximum heat or inertial loads or boost rocket motor requirements.

For the rotating Earth our goal remains the same: we want to find the directional cosine (l, m, n) of the velocity at burnout as functions of the velocity magnitude V and the range angle θ_i . Since the target or impact point is fixed to a rotating Earth, merely locating burnout and target longitude and latitude is insufficient. The time of flight is intimately involved in setting the range angle θ_i , since θ_i depends upon Ψ [Eq. (5.53)], and Ψ depends upon time of flight [Eq. (5.51)]. First let us list the relevant equations:

$$T = \frac{r_0}{U \cos \gamma} \left\{ \frac{\tan \gamma (1 - \cos \theta_i) + (1 - \lambda) \sin \theta_i}{(2 - \lambda) \{ [(1 - \cos \theta_i) / \gamma \cos \gamma^2] + [\cos(\gamma + \theta_i) / \cos \gamma] \}} + \frac{2 \cos \gamma}{\lambda [(2/\lambda) - 1]^{3/2}} \tan^{-1} \left[\frac{[(2/\lambda) - 1]^{1/2}}{\cos \gamma \cot(\theta/2) - \sin \gamma} \right] \right\} \quad (5.28)$$

$$\Psi = \Psi_0 + \Omega_E T \quad (5.51)$$

$$U = V \left[l^2 + \left[m + \frac{\Omega_E r_0 \cos L_B}{V} \right]^2 + n^2 \right]^{1/2} \quad (5.58)$$

$$\lambda = r_0 |U|^2 / \mu \quad (5.7b)$$

$$\cos \theta_i = \frac{\mathbf{r}_B \cdot \mathbf{r}_T}{r_0 R_E} = \cos L_B \cos L_T \cos \Psi + \sin L_B \sin L_T \quad (5.53)$$

$$|U| = U = \left[\frac{\mu}{r_0} \left[\frac{(1 - \cos \theta_i)}{(r_0 / R_E) \cos^2 \gamma - \cos(\theta_i + \gamma) \cos \gamma} \right] \right]^{1/2} \quad (5.23a)$$

$$\sin \gamma = \frac{U \cdot \mathbf{r}_b}{|U| |\mathbf{r}_B|} = \frac{l \cos L_B + n \sin L_B}{\left\{ l^2 + \left[m + (\Omega_E r_0 \cos L_B) / V \right]^2 + n^2 \right\}^{1/2}} \quad (5.59)$$

$$[m + (\Omega_E r_0 / V) \cos L_B] (\sin L_B \cos L_T \cos \Psi - \sin L_T \cos L) + n \cos L_B \cos L_T \sin \Psi - l \sin L_B \cos L_T \sin \Psi = 0 \quad (5.56b)$$

$$\frac{r_0}{R_E} = \left[\frac{\mu}{r_0 |\mathbf{U}|^2} \right] \left[\frac{1 - \cos \theta_i}{\cos^2 \gamma} \right] + \frac{\cos(\gamma + \theta_i)}{\cos \gamma} \quad (5.57)$$

$$l^2 + m^2 + n^2 = 1 \quad (5.60)$$

$$r_0 = R_E + z$$

It would seem that we have 11 equations and 12 variables: $|\mathbf{V}|$, γ , λ , θ_i , T , $|\mathbf{U}|$, r_0 , Ψ , l , m , n . However, Eqs. (5.23a) and (5.57) are not independent. Remember that Eqs. (5.23a) and (5.23b) are the impact equations (with $\mathbf{V}$ being replaced by $\mathbf{U}$ and with $|\mathbf{V}|$ now taking on the meaning of the velocity magnitude at burnout relative to a rotating Earth). Since Eqs. (5.23a) and (5.57) are not independent, we might drop Eq. (5.39) and require that r_0 be specified by some criteria; r_0 can be specified by first choosing a value for burnout altitude z , and then using the last equation. In the simple analysis developed earlier, z was set equal to zero, and then $r_0 = R_E$. Now we have 10 equations and 11 variables. However, the speed at burnout, $\mathbf{U}$, can be given a value (as was done in the planar, nonrotating Earth case) and the above set of equations solved for l , m , n , the directional cosines of the velocity vector $\mathbf{U}$.

5.7 Oblateness Effects

We might ask if the error coefficients developed under the assumption of a nonrotating Earth have any utility for the rotating Earth model. Since the error coefficients represent a linear measure of departure from a nominal trajectory, they should have validity whether the nominal trajectory is over a rotating or a nonrotating Earth.

The material used in this section was taken almost entirely from a paper by A. D. Wheelon,¹ which was printed in 1959 by the American Rocket Society (now the American Institute of Aeronautics and Astronautics). Almost a quarter of a century has elapsed since Wheelon's landmark paper was published. Certainly the technology for implementing the ideas expressed by Wheelon has developed enormously over this period of time. Nevertheless, Wheelon's paper remains a clear and valuable expression of the fundamental physics of ballistic RV's. The interested reader will find many other interesting topics covered by Wheelon. One such topic is a development from first principles of the equations for estimating the effect of Earth oblateness on the downrange and crossrange impact point. In other words, if we compute an impact point based upon a Keplerian trajectory in a central gravitational field, we ask the question as to where the impact point would "move" if we included the oblateness term J_2 in the gravitational potential.

It has never been the author's intention to prepare a handbook; rather, all ideas presented have been developed from first principles. Some departure from this ideal seems necessary at this point in that the analysis of the effect

of the Earth's oblateness on the Keplerian trajectory is lengthy. Since some discussion was given in Chapter III of corrections to the simple inverse-square gravity field, it seems appropriate to include at least the results of Wheelon's analysis of oblateness effects.

Figure 5.13 defines some of the variables that we shall use. It will be noted that L_B and L_T are the latitudes of the burnout point and target or impact point, respectively; θ_i is the range angle, and Ψ is the difference in longitude between the target and burnout points. It should be recalled from Eq. (5.53) that the range angle is deduced from L_B and L_T as well as from the longitude difference Ψ if we assume a nonrotating Earth and $\Psi = \Psi_0$.

From Eq. (5.55) we may express the gravitational potential per unit mass up to and including the oblateness J_2 term as

$$\Phi(r, \gamma) = \mu \left[\frac{1}{r} + \frac{3J_2}{2} \left(\frac{1}{3} - \cos^2 \alpha \right) \right] \quad (5.62)$$

Wheelon defines the following two constants:

$$A = \sin(L_B) \quad (5.63a)$$

$$B = \cos L_B \left[1 - \frac{\cos^2 L_T \sin^2 \Psi}{\sin^2 \theta_i} \right] \quad (5.63b)$$

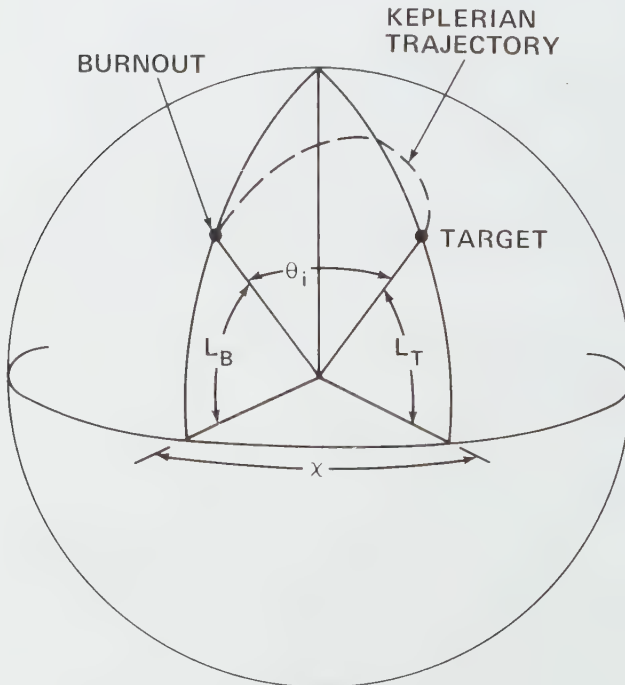


Fig. 5.13 Burnout point/target geometry.

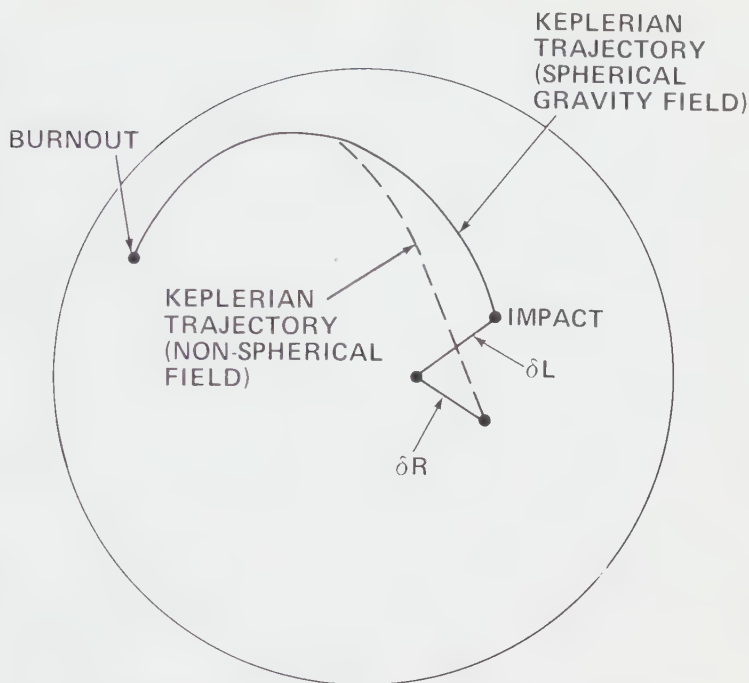


Fig. 5.14 Effective Earth oblateness on impact.

The Lagrangian applicable to the Keplerian trajectory is easily written from Eq. (5.62) as

$$L = \frac{1}{2} \left[\left(\frac{dr}{dt} \right)^2 + r^2 \left(\frac{d\theta}{dt} \right)^2 \right] + \phi(r, \alpha) \quad (5.64)$$

The equation of motion can now be developed along the lines of Eqs. (5.4) and (5.7), which were written earlier for the simple inverse-square central field; now we have a course included, a term to account for the oblateness of the Earth and hence a (slight) departure from the simple central field. After some manipulation, Wheelon obtains the following:

$$\begin{aligned} \delta R = & \frac{-(3/2)JR_E}{\lambda^2 \cos^3 \gamma \sin \gamma} \left\{ 2 \left[\frac{1}{3} - A^2 \right] (1 - \cos \theta_i) \right. \\ & + \left[\lambda \cos^2 \gamma - \frac{2}{3} \right] [(1 - \cos \theta_i) + A^2 (\cos^2 \theta_i + \cos \theta_i - 2) \\ & \left. - B^2 (1 - \cos \theta_i)^2 - 2AB \sin \theta_i (1 - \cos \theta_i)] \right\} \quad (5.65a) \end{aligned}$$

$$\delta L = \frac{-(3JR_E)}{\lambda \cos^2 \gamma} \left[\frac{\sin \Psi \cos L_B \cos L_T}{\sin \theta_i} \right] [A(1 - \cos \theta_i) + B(\theta_i - \sin \theta_i)] \quad (5.65b)$$

where δR and δL are the downrange and crossrange "errors" in impact point which are caused by neglecting the Earth's oblateness. In other words, if we compare the impact point using a central force field with the impact point in which the Earth's oblateness is included in the gravity model, we shall find differences in the downrange and crossrange location of the impact point, i.e., of δR and δL . Again the impact point of a trajectory in which oblateness effects are included in the gravity model will be δR km (say) downrange of the impact point of a Keplerian trajectory in which the gravity field was entirely spherical. δR is measured downrange within the Keplerian trajectory. The impact point also will, in general, lie at a distance δL km (say) crossrange or normal to the Keplerian trajectory. The δR and δL variables are defined in Fig. 5.14.

Equations (5.65) have been programmed; a listing is given in Table 5.4. The remarks section of the program indicates variable definition. It will be noted in line 310 that the range angle, $R = \theta_i$, is calculated from the latitude of the burnout and impact points ($L_B = L_1$ and $L_T = L_2$) and the longitude difference, $\Psi_0 = P$.

For illustrative purposes, we have run a single case where we have set burnout condition as $\gamma = 20$ deg and $V = 6850$ m/s. The longitude difference between target and burnout is $\Psi = 75$ deg, and the latitude of both burnout and impact points is 45 deg. Under these circumstances, the output given below shows that including the oblateness term J_2 moves the impact point about 8 km downrange and about 8 km crossrange (to the left). These changes in the impact point are certainly a strong argument for representing the gravitational field as accurately as possible:

LAT BO DEG.	LAT TARG DEG.	RANGE ANG DEG.	DWN RANGE KM	CROS RANGE KM
45.	45.	50.9935	7.93011	-3.8837

In this section we have developed the concept of the Keplerian trajectory as a segment of an elliptical orbit between the burnout point and the impact point. We have shown that, by including terms in the gravitational potential to account for noncentral gravity effects, due primarily to Earth's oblateness, a significant difference can be made in the location of the impact point. We have also shown that there are great demands on the accuracy of the flight-path angle and velocity magnitude at burnout in order to control the impact point to within 1 km.

In the next chapter we shall consider the final portion of the trajectory: the portion of the flight entirely within the atmosphere. Although the duration of the atmospheric trajectory is two orders of magnitude less than that

Table 5.4 Calculation of Oblation Effects

```

00100 REM THIS PROGRAM CALCULATES THE DOWNRANGE AND CROSRANGE
00110 REM ERRORS DUE TO EARTH'S OBLATION. L1 IS THE LATITUDE
00120 REM OF THE BURNOUT POINT; L2 IS THE LATITUDE OF THE TARGET.
00130 REM P IS THE DIFFERENCE BETWEEN THE LONGITUDE OF THE
00140 REM TARGET AND THE BURNOUT POINT. V IS THE MAGNITUDE OF THE
00150 REM VELOCITY AT BURNOUT AND G IS THE FLIGHT PATH ANGLE AT
00160 REM AT BURNOUT.
00170 L1 = 45
00180 L2 = 45
00190 G = 20
00200 P = 75
00210 V = 6850
00220 J = 1.0823E-3
00230 M1 = 3.9853E14
00240 R1 = 6.378E6
00250 L = R1*V*V/M1
00260 R9 = 3.14159/180
00270 L1 = L1*R9
00280 L2 = L2*R9
00290 P = P*R9
00300 G = G*R9
00310 R = COS(L1)*COS(L2)*COS(P)*SIN(L1)*SIN(L2)
00320 R = SQR(1-R*R)/R
00330 R = ATN(R)
00340 A = SIN(L1)
00350 B = COS(L1)*SQR(1-((COS(L2)*SIN(P)/SIN(R))2))
00360 Z9 = 2*((1/3)-(A*A))*(1-COS(R))
00370 Z8 = (B*B)*((1-COS(R))2)
00380 Z8 = Z8+2*A*B*SIN(R)*(1-COS(R))
00390 Z8 = (1-COS(R))+A*A*((COS(R)2+COS(R)-2)-Z8
00400 Z8 = (L*(COS(G)2)-(2/3))*Z8+Z9
00410 D1 = (-1)*(J*(3/2)*R1/(L*L*(COS(G)3*SIN(G)))*Z8
00420 Z9 = A*(1-COS(R))+B*(R-SIN(R))
00430 Z8 = (SIN(P)*COS(L1)*COS(L2)/SIN(R))*Z9
00440 D2 = (-1)*2*J*R1*Z8/(L*(COS(G)2))
00450 PRINT #LAT BU#, #LAT TARG#, #RANGE ANG#, #DOWN RANGE#, #CROS RANGE#
00460 PRINT #DEG., #, #DEG., #, #DEG., #, #KM#, #KM#
00470 PRINT
00480 PRINT L1/R9, L2/R9, R/R9, D1/1000, D2/1000
00490 END

```

of the Keplerian trajectory, it is during this last phase that the RV must survive extreme environmental conditions. Most of our concern in this phase will be with inertial loads, although some mention will be made of thermal loads as well.

Reference

¹ Wheelon, A. D., "Free Flight of a Ballistic Missile," *ARS Journal*, Vol. 29, Dec. 1959, pp. 915-926.

VI. RE-ENTRY BODY PARTICLE MECHANICS

6.1 Re-Entry Physics

In the previous section we considered Keplerian motion, in which the assumption was made that the only force acting upon the RV was the classical inverse-square central gravity field. Aerodynamic loads were ignored because the atmosphere was considered to have negligible density. At an altitude somewhere below 150 km, atmospheric forces become appreciable. It should be remembered that the sensible (for our purposes) atmosphere is no greater than 2% of the Earth's radius; peak deceleration occurs at an atmospheric height which is 0.2-0.5% of the Earth's radius, less than the proportional thickness of an apple's peel to its radius. In short, the Earth's atmosphere appears to be almost a discontinuity to an exo-atmospheric vehicle encountering the atmosphere at near (circular) orbital speeds. During atmospheric re-entry the gravitational force is often a secondary effect. Drag deceleration may be an order of magnitude or more greater than the gravitational force; maneuvering loads (normal to the velocity vector) can be more than two orders of magnitude greater than the gravitational force.

Most considerations in this section will treat the RV as a mass point. Consequently an important criterion to be applied to any trajectory will be the magnitude of the aerodynamic loads. Obviously the magnitude of such loads, their duration, and direction are important engineering considerations. In addition to protecting components against such loads, even more difficult is the problem of protecting the RV against its thermal environment. An RV entering the atmosphere at 6000 m/s (about 20,000 f/s) ($\lambda = 0.57$) possesses a specific enthalpy of 1.8×10^7 J/kg (7752 Btu/lb). As discussed earlier this is a considerable energy content, because as we have pointed out in the Introduction water vaporizes at a specific enthalpy of 2.32×10^6 J/kg (1000 Btu/lb), and carbon vaporizes at a specific enthalpy of 6×10^7 J/kg (2500 Btu/lb). Obviously a considerable amount of material would be lost if all the heat were to be dissipated by vaporization of the RV. The survival of manned and unmanned RV's as well as meteors (which may enter the atmosphere on high-energy parabolic orbits) indicates that some of the potential thermal energy must be dissipated by radiation and in the near flowfield by complex irreversible flow processes.

An important consideration at this point is the altitudes at which peak loading and peak heating occur. The drag force might be written in

coefficient form as

$$D = \frac{1}{2} \rho V^2 S C_D \quad (6.1)$$

where S is the reference area, ρ the atmospheric density, V the speed, and C_D the shape-dependent drag coefficient. Obviously peak drag loading occurs when ρV^2 is a maximum (assuming C_D remains constant). On the other hand, heat rate per unit area (W/m^2) can be written as a fraction f of kinetic energy per unit time per unit area, or power per unit area of the flow

$$q = f C_D (\rho V^3 / 2)$$

Thus peak heating will occur when ρV^3 is a maximum. It will turn out that peak heating rate takes place at a somewhat higher altitude than does peak deceleration in most cases. The situation may be seen qualitatively in Fig. 6.1.

6.2 Equations of Motion

For the present we shall study motion in the atmosphere by assuming that the velocity vector remains entirely in a plane. To carry out the analysis we

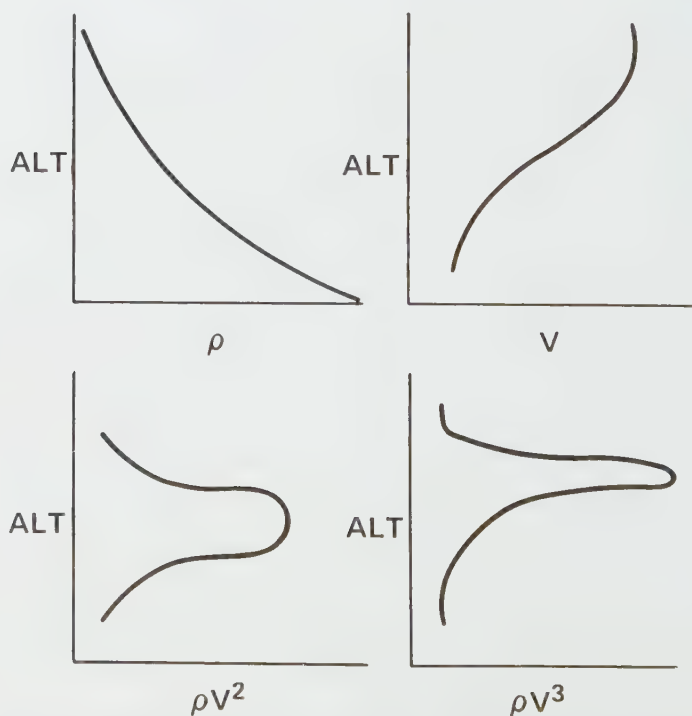


Fig. 6.1 Density ρ , velocity V , ρV^2 and ρV^3 variation with altitude.

shall postulate three axis systems:

(a) (X^i, Y^i, Z^i) : an inertial frame Earth-centered such that the $Z^i Y^i$ plane contains the velocity vector.

(b) (x^l, y^l, z^l) : a moving frame centered at the RV's c.g. such that the velocity vector is always in the $x^l z^l$ plane and the z^l axis is along the gravity vector (positive toward the Earth).

(c) (x^m, y^m, z^m) : a moving frame centered at the RV's c.g. such that the x_m axis is along the velocity vector and the $x^m z^m$ plane contains the velocity vector. Note that we are taking the flight-path angle γ as positive when the velocity vector is *below* the horizon. The orientation of the inertial frame is not important, i.e., we are interested in θ , not θ . However, the Z^i axis can be assumed to pass through either the burnout point or the re-entry point, whatever is convenient. These axes systems can be seen in Fig. 6.2.

Let us write Eq. (4.36). The acceleration in the i and l frames are related as

$$\left. \frac{d\mathbf{V}}{dt} \right|_i = \left. \frac{d\mathbf{V}}{dt} \right|_l + \boldsymbol{\Omega}_{il} \times \mathbf{V} \quad (6.2a)$$

and the acceleration in the m and l frames are related as

$$\left. \frac{d\mathbf{V}}{dt} \right|_m = \left. \frac{d\mathbf{V}}{dt} \right|_l + \boldsymbol{\Omega}_{lm} \times \mathbf{V} \quad (6.2b)$$

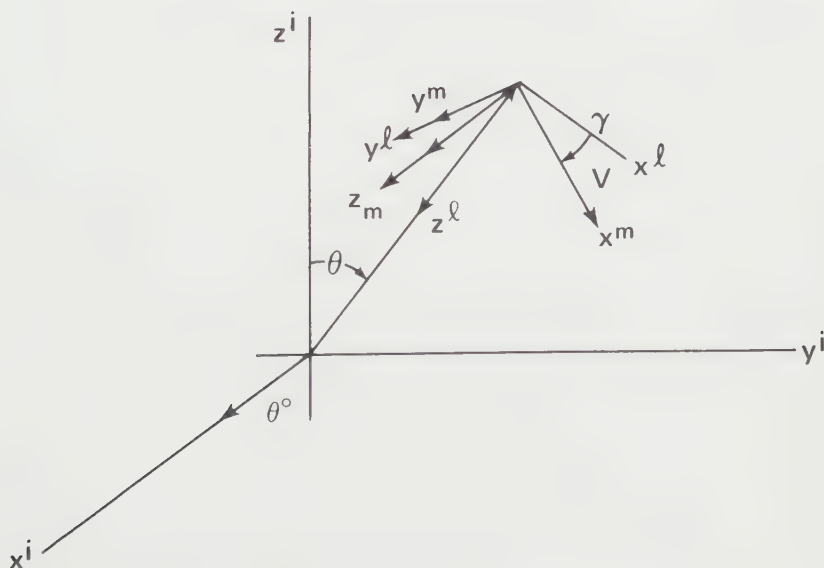


Fig. 6.2 Inertial and moving reference frames.

Combining Eqs. (6.2a) and (6.2b) we get

$$\left. \frac{d\mathbf{V}}{dt} \right|_I = \left. \frac{d\mathbf{V}}{dt} \right|_I + (\boldsymbol{\Omega}_{II} + \boldsymbol{\Omega}_{Im}) \times \mathbf{V} \quad (6.3)$$

According to our definition of axes we have

$$\mathbf{V} = V\mathbf{i}_m \quad \boldsymbol{\Omega}_{II} = -\dot{\theta}\mathbf{j}_m \quad \boldsymbol{\Omega}_{Im} = -\dot{\gamma}\mathbf{j}_m \quad (6.4)$$

Also from Newton's second law we have

$$\left. \frac{d\mathbf{V}}{dt} \right|_I = \frac{\mathbf{F}_a}{M} + \frac{\mathbf{F}_g}{M} \quad (6.5)$$

where the aerodynamic force $\mathbf{F}_a$ will consist of lift (normal to $\mathbf{V}$) and drag (along negative $\mathbf{V}$)

$$\frac{\mathbf{F}_g}{M} = \frac{-\rho V^2 SC_L}{2M} \mathbf{k}_m + \frac{-\rho V^2 SC_D}{2M} \mathbf{i}_m \quad (6.6a)$$

and the gravitational force, $\mathbf{F}_g$ equals

$$\mathbf{F}_g/M = (g \cos \gamma) \mathbf{k}_m + (g \sin \gamma) \mathbf{i}_m \quad (6.6b)$$

Again remember that the x^m axis is along the velocity vector and the z^m axis is normal to the velocity vector.

Inserting Eqs. (6.4-6.6) into Eq. (6.3) we get

$$\begin{aligned} & \left[\frac{-\rho S V^2 C_D}{2m} \right] \mathbf{i}_m + \left[\frac{-\rho S V^2}{2m} \right] \mathbf{k}_m \\ &= \frac{dV}{dt} \mathbf{i}_m + (\dot{\theta} + \dot{\gamma}) V \mathbf{k}_m - (g \sin \gamma) \mathbf{i}_m + (-g \cos \gamma) \mathbf{k}_m \end{aligned} \quad (6.7)$$

We shall now introduce the "ballistic factor" β . This parameter, while dimensional, is a measure of the relative effect of the inertial force mg to the aerodynamic retarding force (or drag) $C_D S$, where C_D is the drag coefficient and S is the reference area. In this report the reference area S will always be the maximum cross-sectional area. For consistency with the so-called English units, β will be taken equal to $mg/C_D S$, although some papers written in metric units would define β as $m/C_D S$. In the English system β has the units of lb/ft^2 , while in the metric system β has the units (at least here) of N/m^2 (sometimes called Pascals). In any event β will have the units of pressure, whether measured in the English or metric system.

$$\frac{dV}{dt} = - \left[\frac{\rho g V^2}{2\beta} \right] + (g \sin \gamma) \quad (6.8a)$$

$$V \left[\frac{d\gamma}{dt} + \frac{d\theta}{dt} \right] = g \cos \gamma - \frac{\rho g V^2}{2\beta} \left[\frac{C_L}{C_D} \right] \quad (6.8b)$$

Equations (6.8) are two simultaneous equations in (V, γ, θ) . For much of the endo-atmospheric motion we shall regard changes in the central angle θ to be of little consequence. Of course in general we cannot discount $d\theta/dt$ so we must add another equation. We know that in the local frame the velocity components might be written

$$V \cos \gamma = \frac{dx'}{dt} \quad V \sin \gamma = \frac{dz'}{dt} = -\frac{dh}{dt}$$

where h is the altitude. Since

$$\frac{dx'}{dt} = (R_E + h) \frac{d\theta}{dt}$$

we may write

$$(R_E + h) \frac{d\theta}{dt} = V \cos \gamma \quad (6.8c)$$

$$\frac{dh}{dt} = -V \sin \gamma \quad (6.8d)$$

Equations (6.8) are a determinate system in the variables: θ is the central angle, V the velocity magnitude, h the altitude, and γ the flight-path angle. If we regard $h \ll R_E$ we may write Eqs. (6.8c) and (6.8d) as:

$$R_E \frac{d\theta}{dt} = V \cos \gamma = \frac{dX_I}{dt} \quad (6.9)$$

$$\frac{dh}{dt} = -V \sin \gamma \quad (6.10)$$

Equations (6.8c) and (6.8d) or (6.9) and (6.10) allow us to solve the navigation problem as was done in Chapter V during Keplerian motion, i.e., find (r, θ) or (x', h) . However, we are more interested in this chapter in obtaining closed-form approximate solutions which will allow us to calculate the velocity magnitude as a function of altitude (for estimating the magnitude of the maximum inertial and thermal loads and the altitude at which such maxima occur).

We may confine our attention at most to Eqs. (6.8a) and (6.9). Note that θ , the central angle, occurs only in derivative form. It should be appreciated that the altitude h is implicit in both the density ρ and gravitational constant g in Eqs. (6.8). We might allow g to remain constant with small error, but in Chapter II we saw that the pressure, and hence density, can vary six orders of magnitude in 200 km (see Table 2.8). We shall use the exponential form for a density variation with altitude given in Eq. (2.24) and then use Eq. (6.11b) to transform the independent variable from time t to altitude h .

We shall now examine a number of cases in which we shall obtain closed-form solutions of Eqs. (6.8a) and (6.8b) using Eqs. (6.11a) and (6.11b) as required (to eliminate θ and to transform from time t to altitude h). Note that we are designating altitude by the symbol h . Earlier we restricted this symbol to the geopotential altitude; however, in the present usage, h may be designated as either a geopotential or geometric altitude.

6.3 Case I: Horizontal Flight-Drag/No Lift

We shall analyze the horizontal flight case twice, once in the time domain and a second time in the arc length domain. To do both we require Eq. (6.8a); we write, since $\gamma \simeq 0$,

$$\frac{dV}{dt} = -\frac{\rho g V^2}{2\beta} \quad (6.11)$$

A simple integration gives

$$V = (2\beta V_i)/2\beta + V_i \rho g t \quad (6.12)$$

where V_i is the initial velocity. If we define arc length as s , we have

$$\frac{ds}{dt} = V$$

Equation (6.11) then becomes

$$\frac{dV}{dt} = \frac{dV}{ds} \frac{ds}{dt} = V \frac{dV}{ds} = -\frac{\rho g V^2}{2\beta} \quad (6.13)$$

The above equation may be integrated after dividing through by V

$$V = V_i \exp - \left[\frac{\rho g}{2\beta} s \right] \quad (6.14)$$

The above equation is very useful for appreciating the relationship among the variables V , ρ , β , and s . Unfortunately in the integration of Eq. (6.14) the density was held constant. Since RV re-entry is characterized by density changes of several orders of magnitude, Eq. (6.14) could be applied only over trajectory segments where some kind of constant average density could be reasonably used. Equation (6.14) might be called the “ballistic range equation.” In ballistic ranges the projectile’s trajectory is virtually at constant altitude, i.e., constant ambient density. We can now see at least qualitatively the relationship between the ballistic factor β , velocity decrement $\ln(V/V_i)$, and arc length s . A configuration having a high-drag coefficient, i.e., small β , will clearly cause a larger velocity decrement over the same arc length than will a body of low-drag coefficient.

Example 6.1: The Use of the Ballistic Equation

We may now use the ballistic equation to solve a rather simple problem. When the British Monarch Henry VIII decided to establish a permanent artillery corps, a classification was established for the various pieces of ordnance. This classification has continued down to the present day at least within the British Army. (In keeping with tradition it seems only appropriate to use British units in expressing our solution to the problem.) Guns were classified according to the weight of the spherical iron projectile that

loosely fit the bore rather than the bore diameter itself.* A case in point is the "10-pounder" which had a bore of 4.5 in. and, with a tolerance of 0.25 in., fired a projectile with a diameter of 4.25 in. Henry's gunners also called the 10-pounder a "Culvering," meaning in French/Latin a serpent or adder. Colorful as this name is, the term 10-pounder is sufficient. Assume we have obtained the following data relative to this projectile:

$$C_D = 0.2$$

$$W = 10 \text{ lb}$$

$$S = \pi \left[\frac{d}{2} \right]^2 = \pi \left[\frac{4.25}{(2)(12)} \right]^2 = 0.0985 \text{ ft}^2$$

$$\beta = \frac{W}{C_D S} = \frac{10}{(0.2)(0.0985)} = 507.5 \text{ lb/ft}^2$$

Now we ask the question: What is the projectile velocity magnitude at 800 yd if the muzzle velocity is 1000 ft/s?

From Eq. (6.14) we have

$$\begin{aligned} V_f &= V_i \exp \left[-\frac{\rho g}{2\beta} s \right] \\ &= (1000) \exp - \left[\frac{(0.00238)(32.174)(3)(800)}{(2)(507.5)} \right] \\ &= 843 \text{ ft/s} \end{aligned} \tag{6.15}$$

Thus we see that the speed of the projectile has lost about 20% of its initial velocity magnitude by the time the projectile reaches 800 yd.

We can gain an appreciation of the ballistic factor as a performance parameter. Suppose for example that we now consider a 40-pounder (ball diameter of 6.75 in.). We have then for the cross-sectional area and the ballistic factor:

$$S = \pi \left[\frac{d}{4} \right]^2 = \pi \left[\frac{6.75}{(2)(12)} \right]^2 = 0.2485 \text{ ft}^2 \tag{6.16}$$

so

$$\beta = \frac{W}{C_D S} = \frac{40}{(0.2)(0.2485)} = 804.8 \tag{6.17}$$

* As an aside in this discussion of artillery classification, according to the amount of weight per ball, is the shotgun gage classification based upon the number of balls per weight. In other words, gage indicated gun-bore diameter by the number of balls (each of which just fits the bore) that can be cast from 1 lb of lead.

As before the velocity magnitude at the target is

$$V_f = (1000) \exp - \left[\frac{(0.00238)(32.174)(3)(800)}{(2)(804.8)} \right]$$

$$V_f = 892 \text{ ft/s}$$

We have the interesting result that a 40-pounder projectile will have 4.94×10^5 ft-lb of energy at the target while the 10-pound projectile will have 1.08×10^5 ft-lb at the target. By increasing the weight by a factor of 4 we have increased the kinetic energy at the target by a factor of 5. Also if we compare the ratio of initial to final (at 800 yd) kinetic energy for each projectile we get 0.695 for the 10 lb ball and 0.795 for the 40 lb ball.

6.4 Case II: Curved Flight Path over Flat Earth

From Eq. (6.11) we may write

$$\frac{d\theta}{dt} = \left[\frac{1}{R_E} \right] V \cos \gamma \quad (6.18)$$

If we neglect the Earth's curvature, i.e., $R_E \rightarrow \infty$, then $d\theta/dt = 0$ and Eqs. (6.8a) and (6.8b) may be written assuming $\gamma \ll 1$ as

$$V \frac{d\gamma}{dt} = g - \left[\frac{\rho g V^2}{2\beta} \right] \frac{C_L}{C_D} \quad (6.19)$$

$$\frac{dV}{dt} = \frac{-\rho g V^2}{2\beta} + g \sin \gamma \simeq \frac{-\rho g V^2}{2\beta} \quad (6.20)$$

Converting from time t to arc length s as the independent variable, Eqs. (6.19) and (6.20) become:

$$\frac{dV}{ds} = -\frac{\rho g V}{2\beta} \quad (6.21a)$$

$$\frac{d\gamma}{ds} = \frac{g}{V^2} - \left[\frac{\rho g}{2\beta} \right] \frac{C_L}{C_D} \quad (6.21b)$$

Equation (6.21a) has been integrated with the results presented in Eq. (6.14). If this expression is inserted in Eq. (6.21b) we get

$$\gamma = \gamma_i + \left[\frac{\beta}{\rho V_i^2} \right] \left[\exp \left[\frac{\rho g}{\beta} s \right] - 1 \right] - \left[\frac{\rho g}{2\beta} \right] \left[\frac{C_L}{C_D} \right] s \quad (6.22)$$

where the subscript i indicates initial conditions.

Example 6.2: Free Flight over a Flat Earth

Ballistic ranges are useful for simulating some re-entry conditions in a ground-based facility. Ballistic ranges are deceptively simple in operation. A model is launched from a gun into a long corridor or range where various optical recording devices are located. At precise time intervals the position and altitude of the model is relaunched. Appearances of simplicity are deceiving: obtaining aerodynamic data in the ballistic range, particularly at launch velocities approaching circular orbital speeds, presents awesome engineering problems. A relatively minor operational problem forms the basis of our example.

Suppose we want to launch a body with a lift-to-drag ratio of about unity. An example of a controlled lift or maneuvering RV is shown in Fig. 6.3. It will be noted that the model has a flap at the base. While the model is conical in shape there is also a cutout near the base. This cutout is a configurational asymmetry which effects the body pressure field in an asymmetric manner. The result is that the body tends to trim nose up, generating lift in the same direction. The relative position of the flap alters the effect of the base cutout; therefore the flap, by being set at various angles, can alter the lift on the body. The flap is called a windward flap because in normal usage it is always on the windward side of the body. If such a model is launched horizontally in the range, the generated lift will cause the model to rise and possibly strike the ceiling or recording equipment. Thus our question is the following: By what angle must the model be launched downward in order that the distance through which the model climbs will be less than the ceiling of the range? The ballistic range is illustrated in Fig. 6.4. The available downrange distance is 200 m and the possible height is 2 m. How-

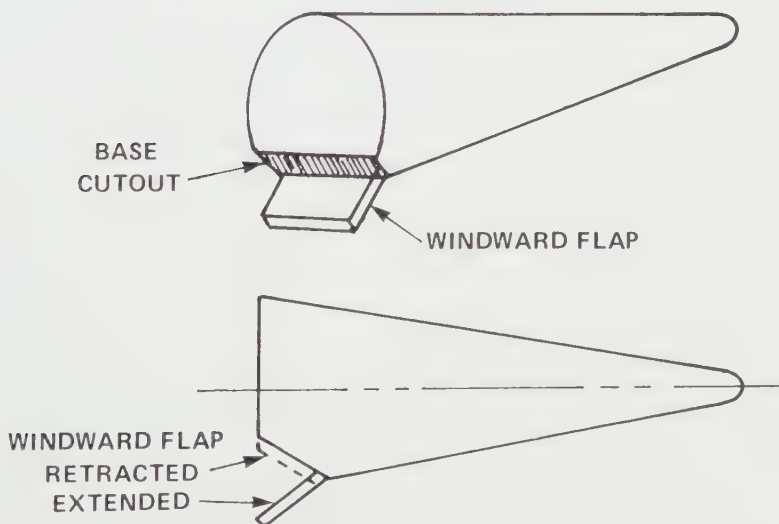


Fig. 6.3 Controllable lift re-entry body using windward flap.

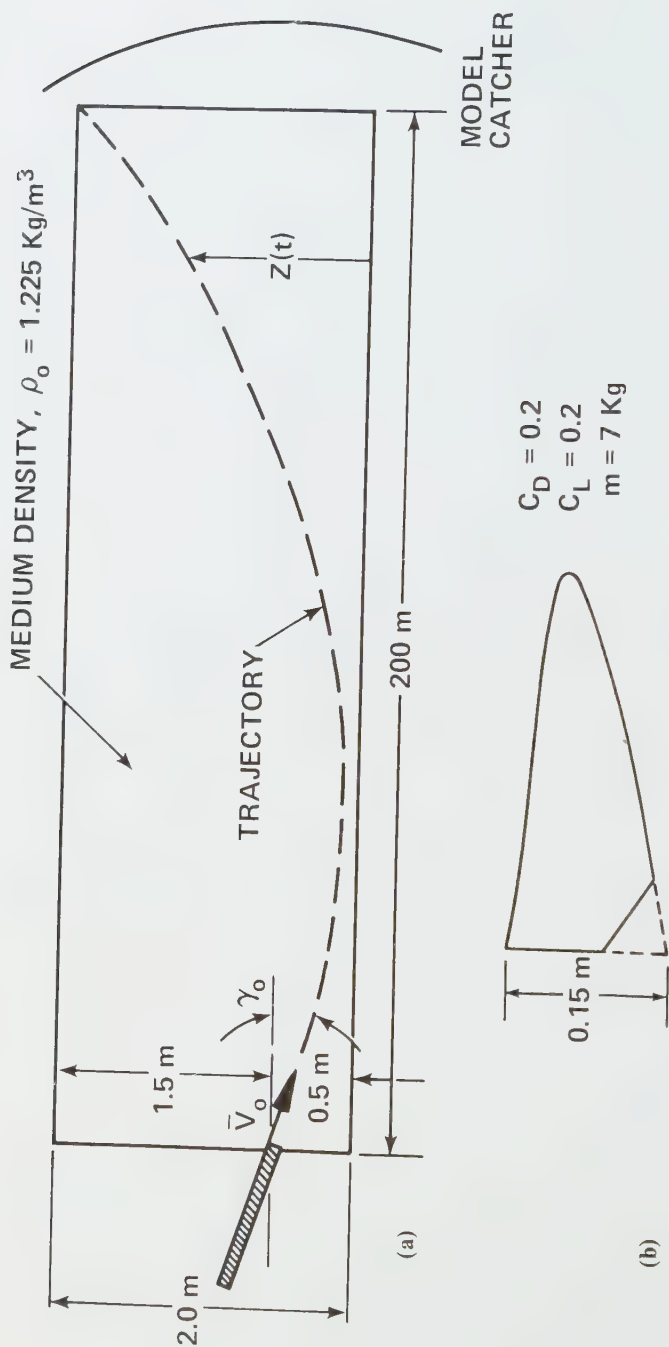


Fig. 6.4 (a) Model trajectory in the ballistic range.
 (b) Representative re-entry vehicle model.

ever, the gun position is 0.5 m above the floor so the available rise is 1.5 m. Obviously a precise answer would require a numerical integration of Eqs. (6.19) and (6.20). However, it is possible to obtain a perfectly adequate answer by making a few approximations. Remember we do not care what the trajectory is along the range. All we want to know is the height of the model above the base or floor when it is 200 m downrange. Stated slightly differently, we want the smallest angle γ_0 such that model's height at the end of the range is ≤ 2 m. γ_0 is the flight-path angle at launch or, equivalently, the angle downward that the launcher must have to meet the 2 m altitude requirement. If the launch velocity magnitude, V_0 is set at 750 m/s. Figure 6.5 illustrates the model weight and dimensions.

The ballistic factor β is given as

$$\beta = \frac{mg}{C_D S} = \left[\frac{(7.0)(9.81)(4)}{(0.2)(\pi)(0.15)^2} \right] = 1.943 \times 10^4 \text{ Pa} \quad (6.23)$$

Now the transverse acceleration is $V d\gamma/dt$ positive downward. Taking z positive upward as the model height above the floor we have

$$-V \frac{d\gamma}{dt} = \frac{d^2 Z}{dt^2} = \left[\frac{\rho g V^2}{2\beta} \right] \left[\frac{C_L}{C_D} \right] - g \quad (6.24)$$

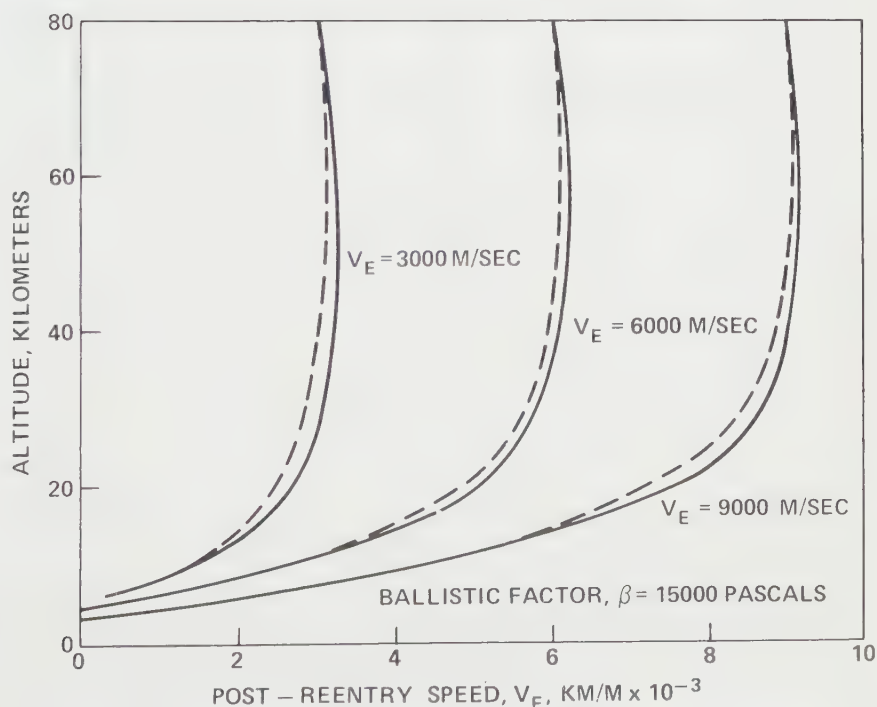


Fig. 6.5 Altitude vs post-reentry speed for a ballistic factor of 15,000 Pa.

If V is regarded as a constant in order to simplify the problem we might take the “average” velocity $(V_0 + V_f)/2$. Now V_f , the velocity magnitude at the end of the range, i.e., 200 m from the launcher, may be calculated from Eq. (6.15)

$$V_f = V_0 \exp - \left[\frac{\rho g}{2\beta} s \right]$$

$$V_f = (750) \exp - \frac{(1.225)(9.81)(200)}{(2)(19,430)}$$

$$V_f = 705 \text{ m/s}$$

An “average” speed $\tilde{V}$ might be taken as

$$\tilde{V} = (V_0 + V_f)/2 = 727 \text{ m/s} \quad (6.25)$$

We may now integrate Eq. (6.24) once to get

$$\frac{dZ}{dt} = \frac{dZ}{dt} \bigg|_0 + \left[\frac{\rho g \tilde{V}^2}{2\beta} \right] \left[\frac{C_L}{C_D} \right] t - gt \quad (6.26)$$

Letting $(dZ/dt)_0 = -V_0 \sin \gamma_0$ we can integrate Eq. (6.26) again to get for $z = Z_f$ at $t = t_f$, where t_f is the time of flight or the time it takes the model to cover 200 m,

$$Z_f = (-V_0 \sin \gamma_0)t + \left[\frac{\rho \tilde{V}^2}{2\beta} \left[\frac{C_L}{C_D} \right] - 1 \right] \frac{gt^2}{2} + Z_0 \quad (6.27)$$

Now with:

$$Z_0 = 0.5 \text{ m}$$

$$Z_f = 2.0 \text{ m}$$

$$C_L / C_D = 1$$

$$\rho = 1.225 \text{ kg/m}^3$$

$$\beta = 1.943 \times 10^{+4} \text{ Pa}$$

$$V_0 = 750 \text{ m/s}$$

we should be able to solve Eq. (6.27) for the initial flight-path angle γ_0 , or the angle of the launcher relative to the horizontal. First, we must obtain the time of flight.

The time of flight can be obtained either from Eq. (6.13) as

$$t_f = \frac{2\beta [(V_0/V_f) - 1]}{\rho g V_0}$$

or more approximately using the average velocity $\tilde{V}$:

$$t_f = 200/\tilde{V}$$

In the first, more exact case, $t_f = 0.275$. To three significant figures this is the same value we get from the second and more approximate relationship. Inserting into our previous relationship for Z_f we get

$$2.0 = 0.5 - (750)(0.275)\sin\gamma_0 + \left[\frac{(1.225)(727)^2(1)}{(2)(1.943 \times 10^4)} - 1 \right] (9.81) \frac{(0.275)^2}{2} \quad (6.28)$$

Solving for γ_0 we get

$$\gamma_0 = 1.94 \text{ deg}$$

Thus, the model must be launched 1.94 deg downward in order that the climb, due to lift, will not cause the model to strike the ceiling of the range. It is easy to show that if γ_0 equals zero, i.e., the model was launched horizontally, the height of the model at the end of the range would be about 6.3 m.

We might also ask: What is the axial and transverse acceleration of the model just after leaving the gun? (Express answer in g .) From Eq. (6.20) we have for axial and transverse accelerations $n_a, n_\perp$:

$$n_a \simeq -\frac{\rho V^2}{2\beta} = -\frac{(1.225)(750)^2}{(2)(1.943 \times 10^4)} = 17.7 g$$

$$n_\perp \simeq \left[1 - \frac{\rho V^2}{2\beta} \left[\frac{C_L}{C_D} \right] \right] = \left[1 - \frac{(1.225)(750)^2}{(2)(1.943 \times 10^4)} \left[\frac{1}{1} \right] \right]$$

$$n_\perp = -16.7 g$$

Obviously if the lift-to-drag ratio C_L/C_D were zero, then $n_\perp$ would be 1 g due to gravity.

Equation (6.22) can be called the "skip-glider" equation. Remember that in this section we are defining γ , the flight-path angle, to be positive when the velocity vector is below the local horizon. Obviously there is some interplay among the variables. We might identify β as a design condition. The ballistic factor depends upon size, shape, and weight. The entrance or initial angle γ_i might be considered as an operational variable since the flight-path angle at re-entry depends upon the preceding Keplerian trajectory which in turn depends upon initial conditions, i.e., (V, γ) at burnout. The lift-to-drag

ratio may or may not be a control variable, depending upon whether the vehicle is a fixed-trim or a variable-trim vehicle. If C_L/C_D is exactly zero, γ will always be positive and the RV will enter the atmosphere. However, above a certain level of C_L/C_D , the RV will "skip" out of the atmosphere. Equation (6.22) was programmed and the results are shown for $C_L/C_D = 0, 0.5, 1.0$. A listing of the program is given in Table 6.1.

We have run three cases (Tables 6.2, 6.3, and 6.4) using the program listed in Table 6.1. Now if C_L/C_D (line 320) is 0, the RV enters the atmosphere and the flight-path angle γ gets increasingly positive. For all three cases the initial flight-path angle is 10 deg and entry velocity magnitude is 7000 m/s (line 190). In the readout column FPA-1 is the second term on the right in Eq. (6.22) and the column FPA-2 is the third term. The first term is of course the initial flight-path angle. It will be noted that the third term tends to make the flight-path angle more negative, i.e., cause the velocity vector to point above the horizon. For the no-lift case the third term remains zero and the RV "digs deeper" into the atmosphere.

If the lift-to-drag ratio is set at 0.5, then we note some skipping. However,

Table 6.1 Skip Glider

```

00100 REM THIS PROGRAM CALCULATES THE COMPONENTS AND SUM OF THE
00110 REM FLIGHT PATH ANGLE FOR A SHALLOW ATMOSPHERIC ENTRY.
00120 REM D IS THE DRAG COEFFICIENT; L IS THE LIFT COEFFICIENT.
00130 REM D1/H ARE THE DENSITY/SLTITUDE PARAMETERS OF THE
00140 REM EXPONENTIAL ATMOSPHERE. D2 IS THE DENSITY AT ALTITUDE Z.
00150 REM D3 IS THE BODY DIAMETER (METERS); M IS THE BODY MASS (KG.)
00160 REM X IS THE RANGE (METERS). C0 IS THE ENTRY SPEED (M/SEC.).
00170 REM G IS THE FLIGHT PATH ANGLE (DEG)/G1 IS THE FLIGHT PATH
00180 REM ANGLE AT ENTRY (DEG); B IS THE BALLISTIC FACTOR ( PASCALS)
00190 V0 = 7000
00200 G1 = 10
00210 D3 = 0.4
00220 M = 40
00230 D = 0.2
00240 L = 0
00250 Z = 30000
00260 X9 = 2000
00270 D1 = 1.752
00280 H = 6700
00290 R9 = 3.14159/180
00300 X = 0
00310 G1 = G1*R9
00320 R = L/D
00330 D2 = D1*EXP((-1)*(Z/H))
00340 B = M*9.81/(D*3.14159*((D3/2)*2))
00350 PRINT #BAL FAC.#,#LIFT/DRAG#,#ALT.#,#DENSITY#
00360 PRINT #PASCALS#,# - #,#METERS#,#KG/CU-M#
00370 PRINT
00380 PRINT B,R,Z,D2
00390 PRINT
00400 PRINT #FPA-1#,#FPA-2#,#FPA-TOT#,#VEL.#,#RANGE#
00410 PRINT #DEG.#,#DEG.#,#DEG.#,#M/SEC.#,#METERS#
00420 FOR I = 0 TO 25
00430 X = 1*X9
00435 V = V0*EXP((-1)*D2*9.81*X/(2*B))
00440 G8 = (B/(D2*V*V))*(EXP((D2*9.81/B)*X)-1)
00450 G9 = (-1)*(D2*9.81/(2*B))*R*X
00460 G = G1*G8*G9
00470 PRINT G8/R9,G9/R9,G/R9,V,X
00480 NEXT I
00490 END

```

Table 6.2 Flight-Path Angle vs Range $C_L/C_D = 0$

BAL FAC. PASCALS	LIFT/DRAG -	ALT. METERS	DENSITY KG/CU-M	
15613.1	0	30000	1.99036E-2	
FPA-1 DEG.	FPA-2 DEG.	FPA-TOT DEG.	VEL. M/SEC.	RANGE METERS
0	0	10.	7000	0
2.38194E-2	0	10.0238	6913.	2000
4.94639E-2	0	10.0495	6827.09	4000
7.70424E-2	0	10.077	6742.24	6000
.10667	0	10.1067	6658.45	8000
.138467	0	10.1385	6575.7	10000
.172562	0	10.1726	6493.98	12000
.20909	0	10.2091	6413.27	14000
.248193	0	10.2482	6333.57	16000
.29002	0	10.29	6254.85	18000
.334729	0	10.3347	6177.12	20000
.382487	0	10.3825	6100.35	22000
.43347	0	10.4335	6024.54	24000
.487863	0	10.4879	5949.66	26000
.545859	0	10.5459	5875.72	28000
.607666	0	10.6077	5802.7	30000
.6735	0	10.6735	5730.58	32000
.743588	0	10.7436	5659.36	34000
.818172	0	10.8182	5589.03	36000
.897505	0	10.8975	5519.57	38000
.981853	0	10.9819	5450.97	40000
1.0715	0	11.0715	5383.23	42000
1.16674	0	11.1667	5316.33	44000
1.26788	0	11.2679	5250.26	46000
1.37526	0	11.3753	5185.01	48000
1.48922	0	11.4892	5120.57	50000

Table 6.3 Flight-Path Angle vs Range $C_L/C_D = 0.5$

BAL FAC. PASCALS	LIFT/DRAG -	ALT. METERS	DENSITY KG/CU-M	
15613.1	.5	30000	1.99036E-2	
FPA-1 DEG.	FPA-2 DEG.	FPA-TOT DEG.	VEL. M/SEC.	RANGE METERS
0	0	10.	7000	0
2.38194E-2	-1.358265	9.66555	6913.	2000
4.94639E-2	-1.71053	9.33293	6827.09	4000
7.70424E-2	-1.1048	9.00225	6742.24	6000
.10667	-1.43306	8.67361	6658.45	8000
.138467	-1.79133	8.34714	6575.7	10000
.172562	-2.14959	8.02297	6493.98	12000
.20909	-2.50786	7.70123	6413.27	14000
.248193	-2.86612	7.38207	6333.57	16000
.29002	-3.22439	7.06563	6254.85	18000
.334729	-3.58265	6.75208	6177.12	20000
.382487	-3.94092	6.44157	6100.35	22000
.43347	-4.29918	6.13429	6024.54	24000
.487863	-4.65745	5.83042	5949.66	26000
.545859	-5.01571	5.53015	5875.72	28000
.607666	-5.37396	5.23369	5802.7	30000
.6735	-5.73224	4.94126	5730.58	32000
.743588	-6.09051	4.65308	5659.36	34000
.818172	-6.44877	4.3694	5589.03	36000
.897505	-6.80704	4.09047	5519.57	38000
.981853	-7.1653	3.81655	5450.97	40000
1.0715	-7.52357	3.54793	5383.23	42000
1.16674	-7.88183	3.28491	5316.33	44000
1.26788	-8.2401	3.02770	5250.26	46000
1.37526	-8.59836	2.7769	5185.01	48000
1.48922	-8.95663	2.53259	5120.57	50000

Table 6.4 Flight-Path Angle vs Range $C_L/C_D = 1.0$

BAL FAC. PASCALS	LIFT/DRAG -	ALT. METERS	DENSITY KG/CM ³	
15613.1	1	30000	1.99036E-2	
FPA-1 DEG.	FPA-2 DEG.	FPA-TOT DEG.	VEL. M/SEC.	RANGE METERS
0	0	10.	1000	0
2.38194E-2	-7.1653	9.30723	5713.	2000
4.94639E-2	-1.43306	8.6164	6621.09	4000
1.0424E-2	-2.14939	7.92145	6742.24	6000
.10661	-2.06612	7.24033	6658.43	8000
.138461	-3.36665	6.55502	6573.1	10000
.172562	-4.29918	5.87338	6493.92	12000
.20909	-5.01571	5.19338	6413.21	14000
.248193	-5.73224	4.51595	6333.37	16000
.29002	-6.44877	3.84123	6254.63	18000
.334729	-7.1653	3.16943	6177.12	20000
.382481	-7.88183	2.50066	6100.33	22000
.433347	-8.59836	1.83511	6024.54	24000
.487863	-9.31489	1.17297	5949.66	26000
.545859	-10.0314	.514438	5875.72	28000
.607666	-10.748	-.140255	5802.71	30000
.6735	-11.4645	-.790922	5730.38	32000
.743588	-12.181	-1.43742	5659.36	34000
.818172	-12.8975	-2.07937	5589.03	36000
.897505	-13.6141	-2.71657	5519.37	38000
.981853	-14.3306	-3.34875	5450.97	40000
1.0715	-15.0471	-3.97563	5383.23	42000
1.16674	-15.7637	-4.59693	5316.33	44000
1.26788	-16.4802	-5.21231	5250.26	46000
1.37526	-17.1967	-5.82146	5185.01	48000
1.48922	-17.9133	-6.42404	5120.37	50000

the velocity magnitude has diminished sufficiently so that the second term in Eq. (6.22) dominates. By 90 km downrange the velocity vector is again pointing below the local horizontal.

Finally if the lift-to-drag ratio is increased to unity the RV must be 115 km downrange before the skip peaks. Even though the RV enters the atmosphere at a flight-path angle of 10 deg, the skip brings the velocity vector almost 20 deg above the local horizon.

The importance of Tables 6.2, 6.3, and 6.4 is as illustrations pointing out that the lift-to-drag ratio is critical when re-entering the Earth's atmosphere at velocities approaching that of a circular orbit, i.e., $\lambda = 0(1)$. A lift-to-drag ratio that seems modest by the standards of atmospheric flying machines is sufficient to cause skipping. An interesting engineering question concerns the normal acceleration $n_\perp$ encountered in the skipping maneuver. The normal acceleration (normal to the velocity vector) follows from Eq. (6.8b) (with $d\theta/dt = 0$) or from Eq. (6.19) as

$$n_\perp = 1 - \left[\frac{\rho V^2}{2\beta} \right] \frac{C_L}{C_D} \quad (6.29)$$

where $n_\perp$ is the acceleration in g . This equation had been used earlier in the ballistic range safety problem. Clearly $f_\perp = n_\perp - 1$ is the specific force (in g) so

$$f_{\perp} = \left[\frac{\rho V^2}{2\beta} \right] \frac{C_L}{C_D} \quad (6.30)$$

It is the specific force that is "felt" by crew or instruments within the RV. Let us consider the normal acceleration experienced by the RV of the last table with a lift-to-drag ratio of 1.0. At peak skip $V = 3518$ m/s, or the density ρ and ballistic factor β are:

$$\rho = 1.99 \times 10^{-2} \text{ kg/m}^3$$

$$\beta = 15,613 \text{ Pa}$$

Thus from Eq. (6.30)

$$f = \frac{\rho V^2}{2\beta} \frac{C_L}{C_D} = 7.89 g$$

At entrance the transverse specific force is almost four times as large or 28.6 g. Thus at an altitude of 30 km (100 kft) the RV can experience large transverse inertial loads, even with a lift-to-drag ratio miniscule by the standards of atmospheric vehicles, and at an altitude far above that reachable by such vehicles.

6.5 Case III: Vertical Re-Entry without Lift

Again we return to Eqs. (6.8) for our basic equations for planar motion after re-entry. Since we shall take $\gamma = \pi/2$, we shall also neglect centrifugal effects, i.e., $V d\theta/dt \equiv 0$. Thus we have from Eq. (6.8)

$$\frac{dV}{dt} = - \left[\frac{\rho g V^2}{2\beta} \right] + g \quad (6.31)$$

Unlike Cases I and II, where the flight was essentially horizontal, steep re-entry will require that variations in density be accounted for prior to carrying out integration. We shall refer to Eq. (2.24) for the atmospheric model that we shall use in the integrations.

$$\rho = \rho_0 e^{-Z/H} \quad (6.32)$$

We shall let $H = 6700$ m (22 kft) and $\rho_0 = 1.752$ kg/m³ (0.0036 slugs/ft³).

Inserting Eq. (6.32) into Eq. (6.31) we get

$$\frac{dV}{dt} = \frac{dZ}{dt} \frac{dV}{dZ} = (-V \sin \gamma) \frac{dV}{dZ} \simeq -V \frac{dV}{dZ} = - \frac{\rho_0 g e^{-Z/H} V^2}{2\beta} + g \quad (6.33)$$

If we let $\xi = V^2$ we get from the above equation

$$\frac{d\xi}{dZ} = + \frac{\rho_0 g \xi e^{-Z/H}}{\beta} - 2g \quad (6.34a)$$

Again we can review the assumptions made in arriving at Eq. (6.34): exponential atmosphere with constants ρ_0 and H , $\gamma \equiv \pi/2$, g , constants, and drag coefficient constant, i.e., β constant. Now we shall refer to Appendix A which provides a fairly general method for solving a differential equation like Eq. (6.34). Comparing Eqs. (6.34) and (A.1) we make the following association

$$P(Z) = -\rho_0 g e^{-Z/H} / \beta \quad Q(Z) = -2g \quad (6.34b)$$

Let us look at the homogeneous equation first, i.e., $Q(z) = -2g = 0$, where gravity is neglected. The solution to Eq. (6.34b) is, under such circumstances,

$$V = K \exp - \left[\frac{\rho_0 g H e^{-Z/H}}{2\beta} \right] \quad (6.35)$$

where ξ has been replaced by V^2 and K is a constant of integration. To evaluate K we might let

$$\beta > 500 \text{ Pa} \quad (6.36)$$

and define re-entry as occurring at $Z = 100 \text{ km}$ (300 kft) and at $V = V_E$. Inserting into Eq. (6.35) we have

$$V = V_E = K(0.99) \quad (6.37)$$

So for the case of steep descent we may replace the constant K with the velocity magnitude at re-entry. Thus the velocity magnitude equation for Case IIIa, steep re-entry with no lift and negligible gravity, is

$$V = V_E \exp - \left[\frac{\rho_0 g H}{2\beta} e^{-Z/H} \right] \quad (6.38)$$

The deceleration along the flight path is always of engineering interest. The above equation might be differentiated; however, an easier approach is to insert Eq. (6.38) into Eq. (6.31) to give

$$n_a = \frac{\dot{V}}{g} = - \left[\frac{\rho_0}{2\beta} \right] e^{-Z/H} \left[V_E^2 \exp \left[- \frac{\rho_0 g H}{2\beta} e^{-Z/H} \right] \right] \quad (6.39)$$

We shall plot Eqs. (6.38) and (6.39) subsequently when we consider Case IIIb, steep re-entry without lift but including gravity.

We return now to the nonhomogeneous equation, Eq. (6.34). Letting $P(Z)$ and $Q(Z)$ of Eq. (A.1) take on the values given in Eq. (6.34b) we get

$$\xi = \exp \left[\int \frac{\rho_0 g e^{-Z/H}}{\beta} dZ \right] \left[-2g \int \exp \left[- \int \left[\frac{\rho_0 g e^{-Z/H}}{\beta} \right] dZ \right] dZ + C \right] \quad (6.40)$$

First we can carry out the integrals in the exponents to get:

$$\xi = \exp - \left[\frac{\rho_0 g H}{\beta} e^{-Z/H} \right] \left[2gH \int \exp \left[\frac{\rho_0 g e^{-Z/H}}{\beta} \right] d(-Z/H) + C \right] \quad (6.41)$$

The easiest way to handle the remaining integral is to expand the exponential in a series and then integrate term by term. Note that all integrals are indefinite. Thus C is the (as yet) unspecified constant of integration. Since $\xi = V^2$, we may replace ξ by V by taking the square root of the result.

$$V = \exp \left\{ - \left[\frac{\rho_0 g H e^{-Z/H}}{2\beta} \right] \right\} \times \left[2gh \sum_{n=1}^{n=\infty} \frac{(\rho_0 g H e^{-Z/H} / \beta)^n}{n n!} - 2gZ + C \right]^{1/2} \quad (6.42)$$

Equation (6.42) may be compared with Eq. (6.38). It will be recalled that Eq. (6.38) is the solution of Eq. (6.34a) with $g = 0$.

From Eq. (6.31) we may solve for the counterpart to Eq. (6.39):

$$n_a = - \frac{dV/dt}{g} = \left[\frac{\rho_0 V^2}{2\beta} \right] e^{-Z/H} - 1 \quad (6.43)$$

Inserting Eq. (6.42) we get

$$n_a = \frac{\rho_0 e^{-Z/H}}{2\beta} \left\{ \exp - \left[\frac{\rho_0 g H e^{-Z/H}}{\beta} \right] \left[2gh \sum_{n=1}^{n=\infty} \frac{(\rho_0 g H e^{-Z/H} / \beta)}{nn!} - 2gZ + C \right] \right\} - 1 \quad (6.44)$$

The constant of integration C can be evaluated at re-entry conditions. A computer program has been written to evaluate Eqs. (6.42) and (6.44). This program is given in Table 6.5.

The input quantities are defined in the REM statements. It will be noted that the constant of integration C is evaluated in line 350. A set of runs was made for an RV with a ballistic factor of 15,000 Pa. A typical output is given in Table 6.6, followed by a graphical presentation of velocity magnitude and axial deceleration vs altitude in Fig. 6.5.

It will be noted that the acceleration is negative and of magnitude of about $1g$ at re-entry. The reason is of course that the aerodynamic loads are negligible at this altitude and the RV is essentially in free-fall. At about 65

km (213 kft) the situation begins to change and the aerodynamic load begins to be felt. Aerodynamic loads begin to dominate at about 40 km (130 kft). At 20 km the deceleration exceeds 100 g. The solid lines use Eqs. (6.42) and (6.44); the dotted lines use Eqs. (6.38) and (6.39). The ballistic factor β is 15,000 P.

From Fig. 6.5 we see that the speed increases slightly until the aerodynamic drag begins to have some influence. Then as the drag increases in proportion to ρV^2 , the speed diminishes. It appears that ignoring gravity during most of the re-entry trajectory leads to an acceptable approximation. In other words, Eqs. (6.38) and (6.39) may be completely adequate as a replacement for Eqs. (6.42) and (6.44).

We note in Fig. 6.6 that the peak deceleration occurs at the same altitude regardless of the entry velocity magnitude. An analytic justification for this observation will be given in Eq. (6.61). It will be shown in Eq. (6.61) that the altitude at which maximum deceleration occurs is not dependent upon

Table 6.5 Velocity Magnitude/Deceleration vs Altitude Steep Descent—No Lift

```

00100 REM THIS PROGRAM CALCULATES THE MAGNITUDE OF THE VELOCITY, V.
00110 REM AND THE AXIAL DECELERATION, N, AS A FUNCTION OF ALTITUDE, Z.
00120 REM INPUTS ARE INITIAL ALTITUDE, Z0, INITIAL VELOCITY MAGNITUDE, V0,
00130 REM ATMOSPHERIC CONSTANTS, D0 (DENSITY), AND H (GAS CONSTANT TIMES THE
00140 REM TEMPERATURE), GRAVITY, G, BALLISTIC FACTOR, B.
00150 REM AND THE ALTITUDE INCREMENT, Z9.
00160 Z0 = 320000
00170 V0 = 21250
00180 Z9 = 10000
00190 H = 22000
00200 D0 = 0.0034
00210 G = 32.174
00220 B = 750
00230 FOR J = 1 TO 50
00240 Z = Z0 - Z9*(J-1)
00250 IF Z < (-1) THEN 00550
00260 LET Z1 = (D0*G*H/B)*EXP((-1)*Z0/H)
00270 P = 1
00280 Z3 = 0
00290 T = (V0*2)*EXP(Z1)
00300 FOR I = 1 TO 10
00310 Z2 = (Z1*I)/(I*(I*P))
00320 Z3 = Z3 + Z2
00330 P = P*I
00340 NEXT I
00350 C = T - 2*G*H*Z3 + 2*G*Z0
00360 Z4 = (D0*G*H/B)*EXP((-1)*Z/H)
00370 P = 1
00380 Z3 = 0
00390 FOR I = 1 TO 10
00400 Z5 = (Z4*I)/(I*(I*P))
00410 Z3 = Z3 + Z5
00420 P = P*I
00430 NEXT I
00440 V = SQR(EXP((-1)*Z4)*(2*G*H*Z3 - 2*G*Z*C))
00450 N = ((D0*(V*2)/(2*B))*EXP((-1)*Z/H)) - 1
00460 IF J > 1 THEN 00530
00470 PRINT #BAL. FACT#, INITIAL VEL., #, INITIAL ALT., #
00480 PRINT B, V0, Z0
00490 PRINT
00500 PRINT #VELOCITY#, #DECEL., #, ALTITUDE#
00510 PRINT #FT/SEC#, #FT/SEC-SQ., #, KILO-FT., #
00520 PRINT
00530 PRINT V, N, Z/1000
00540 NEXT J
00550 END

```

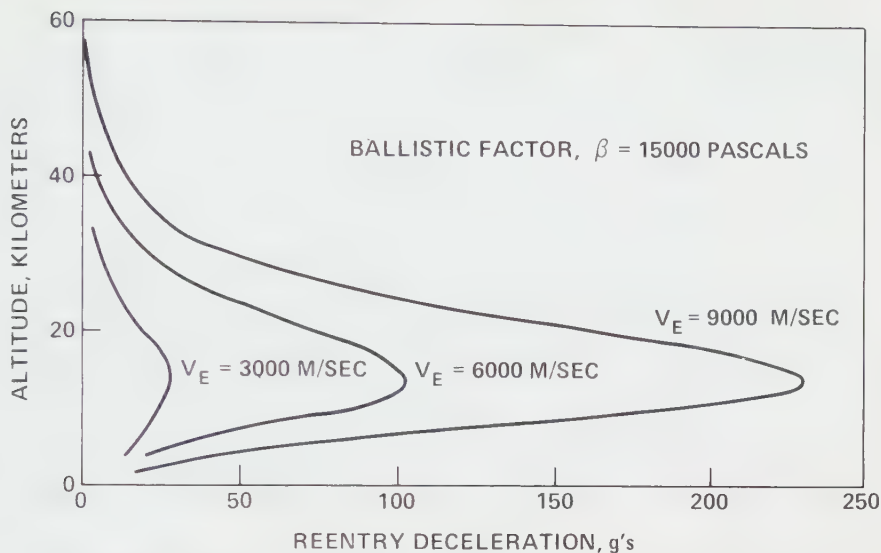


Fig. 6.6 Altitude vs re-entry deceleration for a ballistic factor of 15,000 Pa.

Table 6.6 Typical Output

BAL. FACT 150	INITIAL VEL. 21000	INITIAL ALT. 32000
VEL. CLTY FT/SEC	DECEL. FT/SEC-SQ.	ALTITUDE KILO-FT.
21000.	-.444507	320
21005.1	-.444222	310
21010.2	-.443772	300
21015.3	-.443063	290
21020.4	-.442044	280
21025.4	-.440710	270
21030.4	-.439033	260
21035.3	-.437040	250
21040.2	-.434760	240
21044.9	-.432129	230
21049.3	-.429176	220
21053.5	-.425866	210
21057.1	-.422131	200
21060.9	-.418020	190
21064.3	-.413572	180
21067.5	-.408794	170
21070.1	-.403651	160
21072.3	-.398111	150
21074.0	-.392151	140
21075.5	-.385792	130
21076.3	-.379049	120
21076.5	-.371915	110
21076.2	-.364317	100
21075.4	-.356273	90
21073.0	-.347760	80
21069.1	-.338795	70
21064.1	-.329314	60
21058.5	-.319349	50
21051.4	-.308813	40
21043.0	-.297734	30
21033.0	-.286099	20
21020.43	-.273055	10
4430.73	43.4477	6

entry speed, but does depend upon the ballistic factor β . Thus, an observation of the altitude of peak deceleration may be used as an indicator of the ballistic factor of an RV under observation. It will be shown in Eq. (6.62) that the magnitude of the peak deceleration does depend upon the velocity magnitude at entry, but is independent of the ballistic factor.

6.6 Case IV: Steep, Nonvertical Re-Entry without Lift

In the previous section we analyzed vertical re-entry. It was shown that the decelerative loads due to aerodynamic drag are so large that gravity may be neglected. Now our basic equations for all planar motion have been given in Eqs. (6.8a) and (6.8b). If g may be neglected in Eq. (6.8a), the $g \sin \gamma$ which is less than g might also be neglected in nonvertical trajectories. However, in a nonvertical trajectory the centrifugal force, opposed by gravity, may not be negligible. From Eq. (6.8b) the difference between centrifugal force and the gravity component normal to the trajectory is

$$V \frac{d\theta}{dt} - g \cos \gamma = \frac{\Delta F}{m} \quad (6.45)$$

From Eq. (6.8c) we may write for $d\theta/dt$

$$\frac{d\theta}{dt} = \frac{V \cos \gamma}{R_E + Z} \quad (6.46)$$

where we have replaced h by Z (geometric altitude). Thus Eq. (6.45) may be written as

$$\Delta F = m \left[\frac{V^2}{(R_E + Z)} - g \right] \cos \gamma \quad (6.47)$$

When the gravitational force just balances the centrifugal force we should get circular orbital motion. So if $\Delta F = 0$, the following must occur

$$V_{\odot} = \left[(R_E + Z)g \right]^{1/2} \quad (6.48)$$

Clearly $V_{\odot}$ is called the velocity in circular orbit, sometimes known as the "first cosmic velocity." (The second cosmic velocity is the parabolic or "escape velocity" which is $\sqrt{2}$ times $V_{\odot}$.) The circular orbital velocity was used in the definition of the orbital parameter λ in Eq. (5.7b).

For ballistic RV or meteoric flight ΔF will not be zero. However, the entry speed V_E is large enough to make ΔF , or the difference term in the brackets of Eq. (6.47) is quite small relative to $(-V/g)_{\max}$. For example, if $V = V_E = 6200$ m/s and $Z = Z_E = 90$ km, then

$$\Delta F = \frac{V^2}{R_E + Z} - g = g \frac{(6200^2)/g}{6.37 \times 10^6 + 9 \times 10^4} - 1 = (-0.393 g) \quad (6.49)$$

Since $0.393 g$ may be considered negligible across the trajectory and we are

assuming that C_L , or the lift is zero, it follows from Eq. (6.8b) that $d\gamma/dt = 0$. In Eq. (6.8a) we shall neglect gravity since, as we have pointed out, the gravitational load is negligible when compared to the aerodynamic decelerative load. Equation (6.8a) now becomes

$$\frac{dV}{dt} = \left[\frac{-\rho_0 g V^2}{2\beta} \right] e^{-Z/H} \quad (6.50)$$

In Case I we transformed from time t to arc length s [see Eq. (6.13)]. For a vertical trajectory, s and geometric altitude Z are interchangeable. In the present nonvertical steep re-entry ($\gamma < \pi/2$) we shall use geometric altitude Z . The transformation then becomes

$$\frac{dV}{dt} = \frac{dV}{dZ} \frac{dZ}{dt} = -\frac{dV}{dZ} V \sin \gamma \simeq -\frac{dV}{dZ} V \sin \gamma_E \quad (6.51)$$

Using the above transformation, Eq. (6.50) becomes

$$\begin{aligned} \int_{V_E}^V \left[\frac{dV}{V} \right] &= \int_{Z_E}^Z \left[\frac{\rho_0 g e^{-Z/H}}{2\beta \sin \gamma_E} \right] dZ \\ &= - \int_{Z_E}^{Z/H} \left[\frac{\rho_0 g H e^{-Z/H}}{2\beta \sin \gamma_E} \right] d(-Z/H) \end{aligned} \quad (6.52)$$

Carrying out the indicated integration we get

$$V = V_E \exp \left[\frac{\rho_0 g H e^{-Z_E/H}}{2\beta \sin \gamma_E} \right] \left\{ \exp - \left[\frac{\rho_0 g H e^{-Z/H}}{2\beta \sin \gamma_E} \right] \right\} \quad (6.53)$$

The first exponential can be shown equal to unity for typical re-entry conditions. For example, if

$$\rho_0 = 1.752 \text{ kg/m}^3$$

$$H = 6700 \text{ m}$$

$$Z_E = 90 \text{ km}$$

$$\beta = 1500 \text{ Pa}$$

Inserting into the first exponential gives

$$\exp (7.95 \times 10^{-5}) \simeq 1 \quad (6.54)$$

The result is that the above expression becomes

$$V = V_E \exp - \left[\frac{\rho_0 g H e^{-Z/H}}{2\beta \sin \gamma_E} \right] \quad (6.55)$$

If we take a_E as the "trajectory parameter," that is

$$a_E = \left[\frac{\rho_0 g H}{2\beta \sin \gamma_E} \right] \quad (6.56)$$

then Eq. (6.55) becomes

$$V = V_E e^{-a_E e^{-Z/H}} \quad (6.57)$$

Equation (6.57) certainly resembles Eq. (6.38). The only difference is of course the presence of $\sin \gamma_E$ in the trajectory parameter a_E [Eq. (6.56)]. Equation (6.57) may be inserted in Eq. (6.8a) to give:

$$-(\dot{V}/g) = n_a = (\rho_0 V_E^2 e^{-[2a_E e^{-Z/H}]} e^{-Z/H})/2\beta \quad (6.58)$$

where n_a is the axial deceleration in g .

An interesting question concerns finding the altitude $\tilde{Z}$ at which the axial deceleration is a maximum. Using

$$\frac{dn_a}{dZ} = 0 \quad (6.59)$$

we insert Eq. (6.58) to get

$$a_E e^{-\tilde{Z}/H} = 1/2 \quad (6.60)$$

Solving for $\tilde{Z}$ we get

$$\tilde{Z} = H \ln \left[\frac{\rho_0 g H}{\beta \sin \gamma_E} \right] \quad (6.61)$$

Note that if $\beta \sin \gamma_E > (\rho_0 g H)$, then there will be no extremum to Eq. (6.58) for positive Z . The maximum deceleration will occur at impact. The value of the maximum deceleration $\tilde{n}_a$ is found by inserting Eq. (6.61) into Eq. (6.58):

$$\tilde{n}_a = V_E^2 \sin \gamma_E 2geH \quad (6.62)$$

where e is the base of the natural logarithms. To find the velocity at which maximum deceleration occurs we insert Eq. (6.61) into Eq. (6.57) to get

$$V = V_E e^{-1/2} = (0.6065)V_E \quad (6.63)$$

Figures 6.7 and 6.8 indicate the variation of velocity and deceleration with altitude. Note that in keeping with Eq. (6.61) the altitude at which maximum deceleration occurs decreases with increasing ballistic factor β . In keeping with this comment we note, as we have done earlier from Eq. (6.62), that the magnitude of the maximum deceleration is independent of the ballistic factor, but depends only upon (V_E, γ_E) , the velocity and flight-path angle at re-entry.

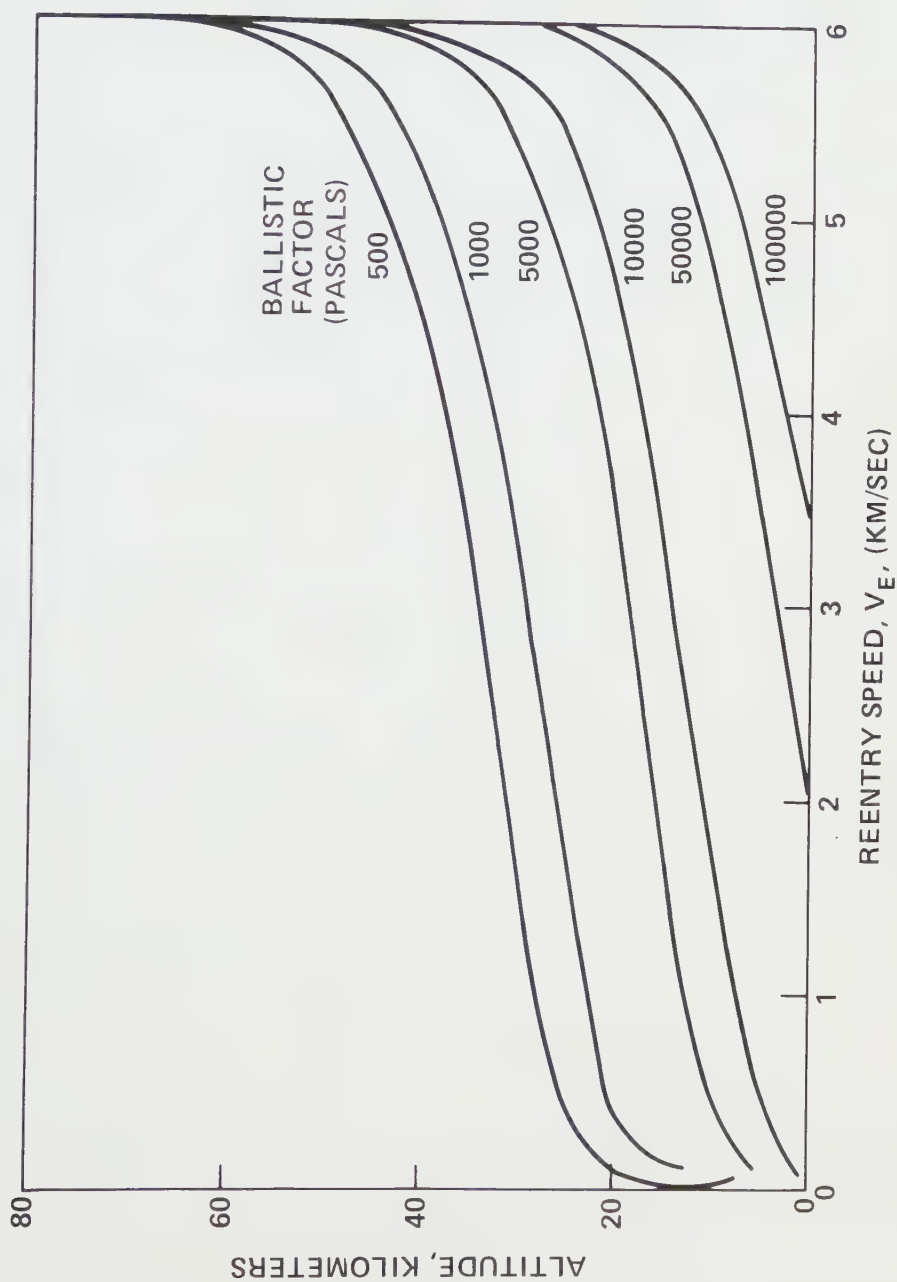


Fig. 6.7 Altitude vs re-entry speed for several ballistic factors.

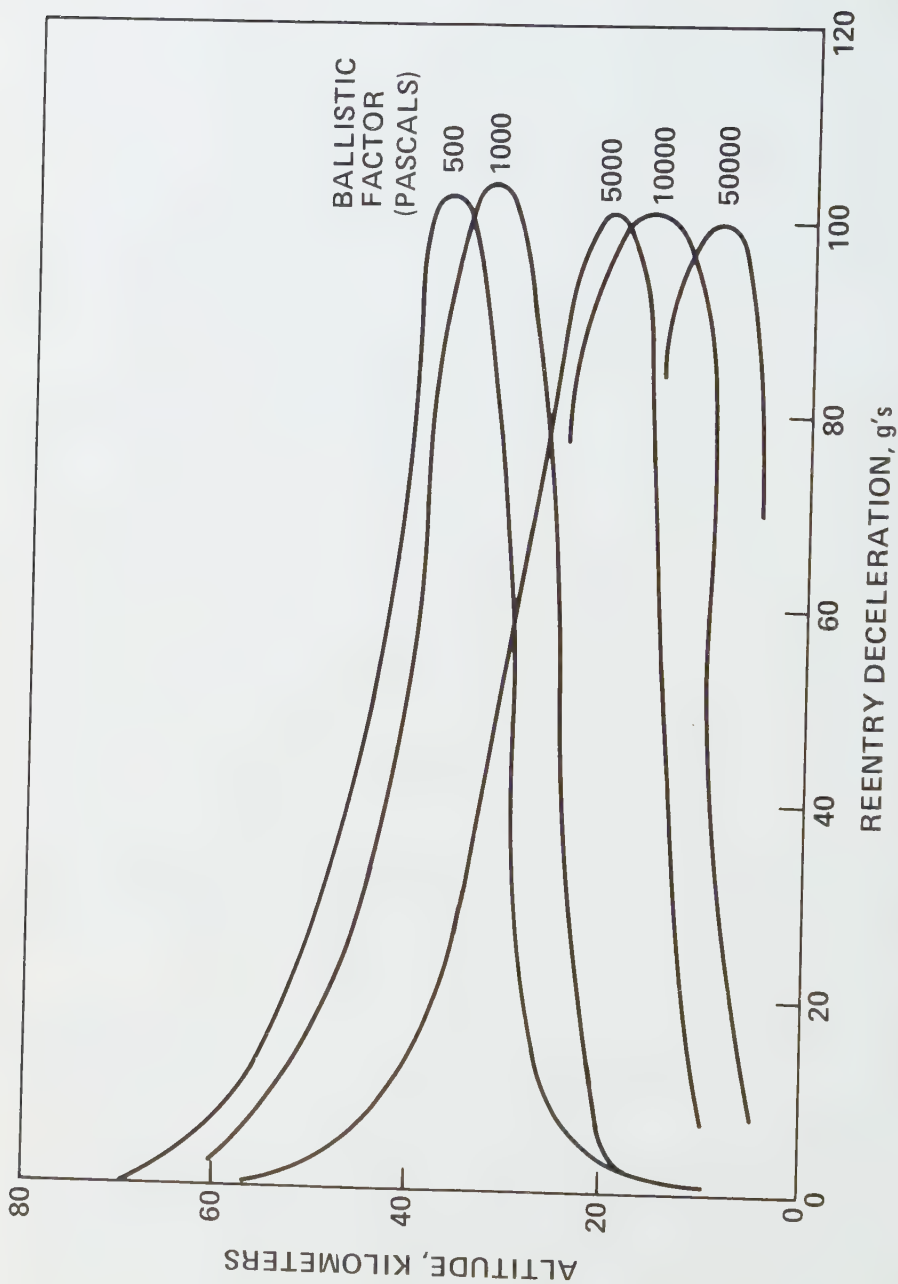


Fig. 6.8 Altitude vs re-entry deceleration for several ballistic factors.

6.7 Case V: Lifting Re-Entry

There are many special integrations of Eqs. (6.8a) and (6.8b) that can be made. We shall consider here one final closed-form solution—lifting re-entry. We shall continue to assume that gravity and centrifugal forces are approximately balanced. Consequently Eqs. (6.8a) and (6.8b) become

$$-V \frac{d\gamma}{dt} = \left[\frac{\rho_{\infty} g V^2}{2\beta} \right] \left[\frac{C_L}{C_D} \right] \quad (6.64a)$$

$$-\frac{dV}{dt} = \left[\frac{\rho_{\infty} g V^2}{2\beta} \right] \quad (6.64b)$$

Dividing Eq. (6.64a) by Eq. (6.64b) we get

$$\begin{aligned} \frac{dV/dt}{V} &= \left[\frac{C_L}{C_D} \right]^{-1} \frac{d\gamma}{dt} \\ \frac{dV}{V} &= \left[\frac{C_L}{C_D} \right]^{-1} d\gamma \end{aligned} \quad (6.65)$$

Equation (6.65) integrates to

$$V = V_E \exp - \left[\frac{\gamma_E - \gamma}{C_L / C_D} \right] \quad (6.66)$$

By itself Eq. (6.66) is not of much use as the independent variable is Z , the geometric altitude, and V varies only implicitly with Z through $\gamma = \gamma(Z)$. Therefore we must find a function to represent the variation of γ with Z .

We have shown that

$$\frac{dZ}{dt} = -V \sin \gamma \quad (6.67)$$

We may rewrite Eq. (6.64a) to eliminate velocity magnitude V :

$$\sin \gamma d\gamma = \left[\frac{\rho_0 g H}{2\beta} \right] \left[\frac{C_L}{C_D} \right] e^{-Z/H} d \left[\frac{Z}{H} \right] \quad (6.68)$$

Integrating from γ_E to γ and Z_E to Z (entry conditions to local conditions)

$$\cos \gamma = \left[\frac{\rho_0 g H}{2\beta} \right] \left[\frac{C_L}{C_D} \right] \left[e^{-Z/H} - e^{-Z_E/H} \right] + \cos \gamma_E \quad (6.69)$$

Since Z_E is about 100 km and H is 6.7 km, we might assume that

$$e^{-Z_E/H} \simeq 0 \quad (6.70)$$

We may now solve for γ to get

$$\gamma = \cos^{-1} \left[\left(\frac{\rho_0 g H}{2\beta} \right) \left(\frac{C_L}{C_D} \right) e^{-Z/H} + \cos \gamma_E \right] \quad (6.71)$$

Equation (6.71) gives γ , the flight-path angle, as a function of geometric altitude Z . If Eq. (6.71) is inserted in Eq. (6.66) we get the velocity magnitude V as a function of altitude, ballistic factor β , and lift-to-drag ratio C_L/C_D :

$$V = V_E \times \exp - \left\{ \frac{1}{(C_L/C_D)} \left[\gamma_E - \cos^{-1} \left[\left(\frac{\rho_0 g H}{2\beta} \right) \left(\frac{C_L}{C_D} \right) e^{-Z/H} + \cos \gamma_E \right] \right] \right\} \quad (6.72)$$

If $\mathbf{a}$ is the acceleration vector we may write after Eqs. (6.8a and 6.8b) the deceleration components along the trajectory $\mathbf{n}_v$ and perpendicular to the trajectory $\mathbf{n}_{\perp v}$ as

$$-\frac{\mathbf{a}}{g} = \mathbf{n} = - \left(\frac{\dot{V}}{g} \right) \mathbf{i}_m + \left(\frac{V\dot{\gamma}}{g} \right) \mathbf{k}_m = n_v \mathbf{i}_m + n_{\perp v} \mathbf{k}_m \quad (6.73)$$

From Eqs. (6.8a) and (6.8b) we have neglecting gravity,

$$n_v = -\frac{\dot{V}}{g} = \left(\frac{\rho_0 V^2}{2\beta} \right) e^{-Z/H} \quad (6.74a)$$

$$n_{\perp v} = -\frac{\dot{\gamma}V}{g} = \frac{\rho_0 V^2}{2\beta} \left(\frac{C_L}{C_D} \right) e^{-Z/H} \quad (6.74b)$$

The magnitude of the deceleration vector $\mathbf{n}$ is

$$n = n_v^2 + n_{\perp v}^2 \quad \left| n_v^2 + n_{\perp v}^2 \right|^{1/2} = \left(\frac{\rho_0 V^2}{2\beta} \right) e^{-Z/H} \left[1 + \left(\frac{C_L}{C_D} \right)^2 \right]^{1/2} \quad (6.75a)$$

The vector $\mathbf{n}$ makes an angle with respect to the flight path, i.e., with respect to the velocity vector $\mathbf{V}$ of

$$\arg \mathbf{n} = \sin(C_L/C_D) \tan^{-1}(C_L/C_D) \quad (6.75b)$$

Let us now find the maximum deceleration load which we shall designate $n_{\max} = n_m$. The corresponding altitude, flight-path angle, and velocity magnitudes at which this maximum load occurs are also designated as Z_m , γ_m , and V_m , respectively. From Eqs. (6.75a) and (6.72) the load magnitude n is

$$n = \frac{\rho_0 V_E^2}{2\beta} e^{-Z/H} \left[1 + \left(\frac{C_L}{C_D} \right)^2 \right]^{1/2} \times \exp - \left\{ \frac{2}{(C_L/C_D)} \left[\gamma_E - \cos^{-1} \left[\left(\frac{\rho_0 g H}{2\beta} \right) \left(\frac{C_L}{C_D} \right) e^{-Z/H} + \cos \gamma_E \right] \right] \right\} \quad (6.76)$$

If the derivative of the above expression is taken with respect to Z and equated to zero we get

$$\left[\frac{\rho_0 g H}{2\beta \sin \gamma_m} \right] e^{-Z_m/H} = 1/2 \quad (6.77)$$

Solving for $\sin \gamma_m$ and making use of Eq. (6.56) we have

$$\sin \gamma_m = (2a_E \sin \gamma_E) e^{-Z_m/H} \quad (6.78)$$

A second relationship between Z_m and γ_m may be found in Eq. (6.71). If we eliminate γ_m between these equations we get

$$Z_m = H \ln \left\{ a_E \left[\left(4 + \left(\frac{C_L}{C_D} \right)^2 \csc^2 \gamma_E \right)^{1/2} + \left(\frac{C_L}{C_D} \right) \cot \gamma_E \right] \right\} \quad (6.79)$$

Equation (6.79) gives the altitude at which the maximum deceleration takes place. Inserting this expression into Eq. (6.71) we get the flight-path angle γ_m at which maximum deceleration takes place:

$$\gamma_m = \cos^{-1} \left\{ \frac{C_L/C_D \sin \gamma_E}{[4 + C_L^2/C_D^2 \csc^2 \gamma_E]^{1/2} + (C_L/C_D) \cot \gamma_E} + \cos \gamma_E \right\} \quad (6.80)$$

The velocity magnitude at which the maximum deceleration takes place is from Eq. (6.66):

$$V_m = V_E \exp - \left[\frac{\gamma_E - \gamma_m}{C_L/C_D} \right] \quad (6.81)$$

and finally the value of the maximum deceleration n_m is from Eq. (6.76):

$$n_m = \frac{\rho_0 V_E^2}{2\beta} \left[1 + \left(\frac{C_L}{C_D} \right)^2 \right]^{1/2} V_m^2 \quad (6.82)$$

From Eq. (6.79) it is clear that the altitude at which maximum deceleration takes place depends upon the mass and lift/drag characteristics as well as the entry angle γ_E through the presence of a_E [see Eq. (6.56)]. However, the flight-path angle γ_m and the velocity magnitude V_m , at which maximum deceleration n_m takes place, depend only on the entry angle and the lift-to-drag ratio C_L/C_D . Remembering that C_L/C_D can be either positive or negative, it is apparent that the deceleration profile (with altitude) as well as the maximum value of deceleration and the altitude Z_m at which this maximum occurs can be modified by controlling C_L/C_D .

The preceding discussion covered five special cases of planetary entry. For each of these cases fairly simple analytic relationships were developed containing variables of interest as functions of time, arc length, or altitude. We can regard these analytic expressions as approximate solutions to the four differential equations given in Eqs. (6.8). If these equations were to be numerically integrated, a graphical or tabular presentation of the various dependent/independent variable pairs would be called the "exact" solutions to the same set of differential equations. The sometimes self-imposed burden on the engineering analyst is to compare his approximate solutions with the numerical or exact solutions. The graphical comparison seems to be even more impressive if sprinkled liberally with data symbols from appropriate experiments. Mutual corroboration among the three methodologies: analytic (approximate), numerical (exact), and experimental is the engineering equivalent of the navigators three-point-fix. An acceptable fix gives the analyst, like the navigator, great confidence in the essential correctness of his labors. If the analyst is an engineer, then he draws pride in his ability to discern the essential physics in the problem before committing his analytic skills; if the analyst feels that his forte is analytic virtuosity, then he will emphasize his repertoire of mathematical methods or manipulative skills. (It would be a bit embarrassing to struggle with a series solution only to find after going public that his great analytic insight was an established part of the tool kit of his nineteenth century counterpart.)

If there is an unacceptable deviation between the numerical and/or experimental solutions, then the analyst usually retreats and reconsiders his thought process. Of course the "exact" solution may also be dead wrong for a variety of reasons such as the use of inappropriate or incomplete differential equations in the first place or round-off errors during computation. Experimental studies could also be unreliable, perhaps resting on the shifting sands of instrument error or spurious or unaccounted for inputs. The analyst usually sees only graveyard humor in the shortfall of the computer modeler or the experimentalist. The analyst seeks a benchmark, if not a gold standard, in the work of his collaborators in the laboratory or at the computer terminal.

If we could be assured of the correctness of experiment and computer modeling, why, we may ask, do we place any emphasis on closed-form solutions? During the pre-Cambrian epoch when the electronic computer was only a gleam in the eye of Charles Babbage, approximate analytic solutions were probably the foundations of any design decisions. Today with the computer increasingly available and versatile, we need only call up the

appropriate subroutine to tame some formidable nonlinear differential equations. Still there remains an essential place for the closed-form solution. Such solutions have the advantage of clearly indicating the relative importance of the various components of the physical model. We may ask, what are the design parameters that are of primary significance in meeting mission requirements? It may be seen, for example, in Eq. (6.61) the relative importance of vehicle drag, vehicle weight, and atmospheric properties in setting the altitude for maximum deceleration.

All of the analysis set forth in this chapter was drawn from a more extensive discussion by Loh.¹ Loh provides an almost encyclopedic coverage of closed-form analysis or re-entry body physics. The propriety of the various concepts may be found in his extensive bibliography. In addition, there is a discussion of re-entry body particle dynamics in Ashley's work² which is based upon Loh's earlier compilation; however, Ashley's treatment is more lucid, with an improvement in symbolism. In addition, Ashley also includes some interesting insights into re-entry body dynamics.

It might be of some interest to trace the development of Loh's effort in order to develop some insight of our own into re-entry body mechanics. Let us first restate Eqs. (6.8):

$$\frac{dV}{dt} = - \left[\frac{\rho g V^2}{2\beta} \right] + (g \sin \gamma) \quad (6.83a)$$

$$V \frac{d\gamma}{dt} = - V \frac{d\theta}{dt} + g \cos \gamma - \left[\frac{\rho g V^2}{2\beta} \right] \left[\frac{C_L}{C_D} \right] \quad (6.83b)$$

$$(R_E + Z) \frac{d\theta}{dt} \simeq R_E \frac{d\theta}{dt} = V \cos \gamma \quad (6.83c)$$

$$\frac{dZ}{dt} = - V \sin \gamma \quad (6.83d)$$

In atmospheric vehicle analysis the flight-path angle γ is usually taken positive when the velocity vector is *above* the local horizontal. However, for RV's, the velocity vector is usually *below* the local horizontal so we shall define γ , the flight-path angle, as positive under such circumstances. We can rewrite Eq. (6.83a) by changing the independent variable from time t to altitude Z as:

$$\frac{d(\quad)}{dt} = \frac{dZ}{dt} \frac{d(\quad)}{dZ} \quad (6.84)$$

Using Eq. (6.83d), the above chain rule becomes

$$\frac{d(\quad)}{dt} = - V \sin \gamma \frac{d(\quad)}{dZ} \quad (6.85)$$

Equation (6.83a) then becomes

$$-V \sin \gamma \frac{dV}{dZ} = - \left[\frac{\rho g V^2}{2\beta} \right] + g \sin \gamma \quad (6.86)$$

After some manipulation we get

$$\frac{1}{2} \frac{d(V^2)}{dZ} - \left[\frac{\rho g V^2}{2\beta \sin \gamma} \right] = -g \quad (6.87)$$

Nondimensionalizing the speed V by the orbital speed (at sea level) and the altitude by the atmospheric constant H gives

$$\frac{d(V^2/gR_E)}{d(Z/H)} - \frac{\rho g (V^2/R_E g) H}{\beta \sin \gamma} = - \frac{2H}{R_E} \quad (6.88a)$$

We may also transform Eq. (6.83b) in the same manner. In doing this we recognize that the central angle θ appears only as derivative. Thus we can use Eq. (6.83c) to eliminate $d\theta/dt$ in Eq. (6.83b). After some manipulation we get

$$\frac{d(\cos \gamma)}{d(Z/H)} - \frac{H \cos \gamma}{R_E} \left[\frac{g R_E}{V^2} \right] - 1 = - \left[\frac{C_L}{C_D} \right] \left[\frac{\rho g H}{2\beta} \right] \quad (6.88b)$$

We could recognize that $\lambda = V^2/gR_E$, as introduced in Eq. (5.7b), and rewrite Eqs. (6.88) as

$$\frac{d\lambda}{d(Z/H)} - \left[\frac{\rho g H}{\beta \sin \gamma} \right] \lambda = - \frac{2H}{R_E} \quad (6.89a)$$

$$\frac{d(\cos \gamma)}{d(Z/H)} - \frac{H \cos \gamma}{R_E} \left[\frac{1}{\lambda} - 1 \right] = - \frac{C_L}{C_D} \left[\frac{\rho g H}{2\beta} \right] \quad (6.89b)$$

Equations (6.89) can be integrated numerically giving the result shown in Figs. 6.9 and 6.10 which were taken from Ashley's book (Ref. 2), which was in turn an adaptation of figures in Loh's book (Ref. 1).

The presentation in Figs. 6.10 was developed for $\gamma_E = +0.05$ deg. For Earth the parameter R_E/H takes on a value of 950; for Mars R_E/H is something like 130.

Each line in Figs. 6.9 and 6.10 corresponds to a fixed lift-to-drag ratio. In carrying out these integrations we should note the presence of $\sin(\gamma)$ in the denominator of Eqs. (6.88a) and (6.89a). This term can be small at least at the beginning of re-entry; in a skipping trajectory γ can pass through zero (see Tables 6.3 and 6.4).

Loh used the exponential atmosphere which we introduced in Eq. (6.32). From this expression it is easily shown that

$$d\rho = -\rho d(Z/H) \quad (6.90)$$

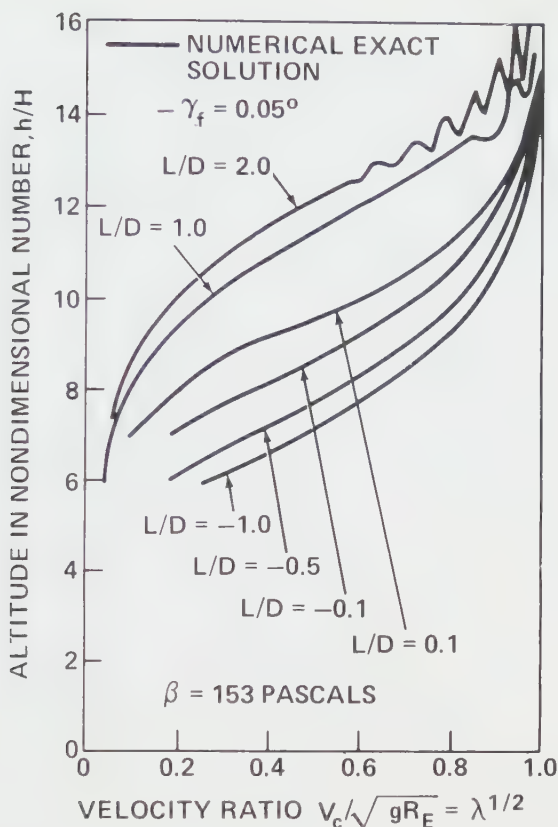


Fig. 6.9 Dimensionless altitude Z/H vs velocity ratio, λ .
(Reprinted courtesy of Addison-Wesley, Reading, Mass.²)

With the above replacement for the differential $d(Z/H)$ we have

$$\frac{d\lambda}{d\rho} + \left[\frac{gH}{\beta \sin \gamma} \right] \lambda = \frac{2H}{\rho R_E} \quad (6.91a)$$

$$\frac{d(\cos \gamma)}{d\rho} + \frac{H \cos \gamma}{\rho R_E} \left[\frac{1}{\lambda} - 1 \right] = \left[\frac{C_L}{C_D} \right] \left[\frac{gH}{2\beta} \right] \quad (6.91b)$$

In examining numerical solutions to Eqs. (6.91) Loh came to the conclusion that the second term in Eq. (6.91b) is virtually a constant. Thus, even though ρ , λ , and γ vary throughout the trajectory, for any practical trajectory, the term

$$\frac{H \cos \gamma}{\rho R_E} \left[\frac{1}{\lambda} - 1 \right]$$

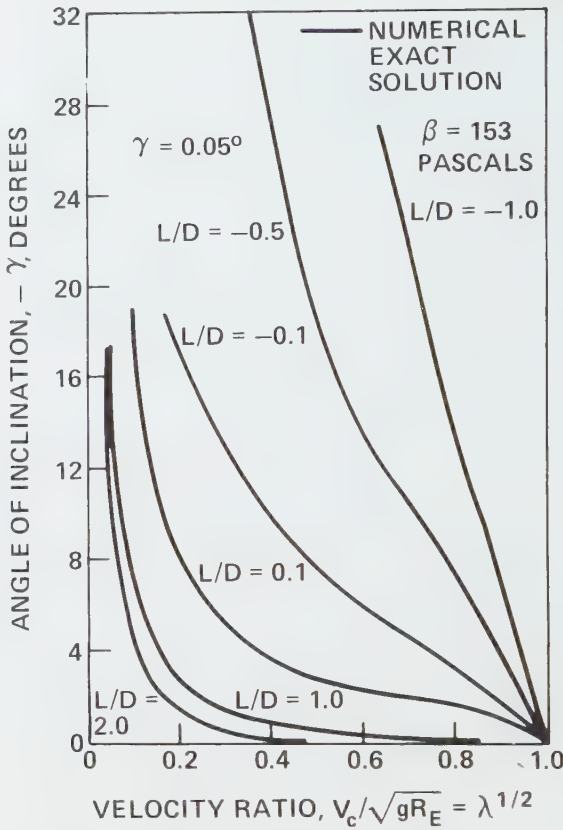


Fig. 6.10 Flight-path angle γ vs velocity ratio λ .
(Reprinted courtesy of Addison-Wesley, Reading, Mass.²)

is nearly constant. In keeping with this assumption we may write for the integral of Eq. (6.91b)

$$\cos\gamma = \cos\gamma_E + \left\{ \frac{C_L}{C_D} \left[\frac{gH}{2\beta} \right] + \left[\frac{\cos\gamma}{\rho} \right] \frac{H}{R_E} \left[1 - \frac{1}{\lambda} \right] \right\} (\rho - \rho_E) \quad (6.92a)$$

or solving explicitly for $\cos(\gamma)$ we get

$$\cos\gamma = \left\{ \cos\gamma_E + \left[\frac{C_L}{C_D} \right] \left[\frac{gH}{2\beta} \right] (\rho - \rho_E) / 1 + \frac{H}{R_E} \left[\frac{1}{\lambda} - 1 \right] \left[1 - \frac{\rho_E}{\rho} \right] \right\} \quad (6.92b)$$

In the development of Eq. (6.71) we neglected centrifugal effects, i.e., we assumed that R_E was infinite. In Eq. (6.92) we may remove centrifugal effects letting $R_E \rightarrow \infty$ and, in addition, ignore the density at entry, i.e., $\rho_E \approx 0$. If we then replace density with its exponential variation with altitude [Eq. (6.32)], Eqs. (6.92) and (6.71) are identical.

Ashley obtains an approximate solution to Eq. (6.91a) by setting the term on the right equal to zero. While we have noted that R_E/H is about 950, such a simplification may not always be appropriate. If we neglect the term on the right of Eq. (6.91a), we would be in effect saying that the following inequality must hold

$$\left[\frac{gH}{\beta \sin \gamma} \right] \lambda \gg \frac{2H}{\rho R_E} = \frac{2e^{Z/H}}{\rho_0(R_E/H)} \quad (6.93)$$

Obviously the inequality can be made to hold under all conditions if we restrict the flight-path angle to arbitrarily small values. Let us however take the following as typical values of RV's and re-entry conditions of interest:

$$\lambda = 0.5$$

$$\gamma_E = 10 \text{ deg}$$

$$\beta = 1000 \text{ Pa}$$

$$Z_E = 50 \text{ km}$$

$$\rho_0 = 1.752 \text{ kg/m}^3$$

$$H = 6700 \text{ m}$$

We get for the left-hand term of the inequality 3.78×10^2 as compared with 2.1 for the term on the right. The assumption seems justified, but it must be remembered that a ballistic factor of 1000 Pa is small. Values of 10^4 and even 10^5 are possible. In such cases the assumption would be invalid. However, we shall continue with this simplification, keeping in mind the restrictions on the size of the ballistic factor β . Equation (6.91a) now takes on the particularly simple form:

$$\sin \gamma \frac{d\lambda}{d\rho} = - \left[\frac{gH}{\beta} \right] \lambda \quad (6.94a)$$

Next we can invoke Loh's approximation to write Eq. (6.91b) as

$$\sin \gamma \frac{d\gamma}{d\rho} = \frac{H \cos \gamma}{\rho R_E} \left[\frac{1}{\lambda} - 1 \right] - \left[\frac{C_L}{C_D} \right] \left[\frac{gH}{2\beta} \right] \quad (6.94b)$$

Dividing Eq. (6.94a) by Eq. (6.94b) we get

$$\frac{d\lambda}{d\gamma} = \left[\frac{-gH}{\beta} \right] \lambda \left/ \frac{H \cos \gamma}{\rho R_E} \left[\frac{1}{\lambda} - 1 \right] - \frac{C_L}{C_D} \left[\frac{gH}{2\beta} \right] \right. \quad (6.95)$$

The above expression may be easily integrated since the denominator is a constant (Loh's assumption). The result is

$$\lambda = \lambda_E \exp \frac{gH}{\beta} (\gamma_E - \gamma) \bigg/ \frac{H \cos \gamma}{\rho R_E} \left[\frac{1}{\lambda} - 1 \right] - \frac{C_L}{C_D} \left[\frac{gH}{2\beta} \right] \quad (6.96)$$

With a flat Earth assumption, i.e., $R_E \rightarrow \infty$ (equivalent to neglecting centrifugal effects), Eq. (6.96) reduces to Eq. (6.66). Backing up slightly, Eq. (6.91a), when restricted to a flat Earth and steep trajectory, i.e., $\gamma \simeq \pi/2$, integrates to give Eq. (6.66).

Equations (6.92) and (6.96) are approximate solutions to Eqs. (6.91). Equations (6.91) have three unknowns: ρ [or altitude z if we use Eq. (6.32)], velocity magnitude ratio λ , and flight-path angle γ . Equations (6.92) and (6.96) are two equations in the three unknowns. Therefore if one variable is prescribed, then the other two may be computed in a fairly straightforward manner using a desk calculator.

As Ashley points out (Ref. 2) the solution to Eqs. (6.89) or (6.91) depends upon the following three parameters:

(a) C_L/C_D , the lift-to-drag ratio, which is essentially a function of the configuration. If we assume a control-configuration vehicle (CCV) such as the windward flap discussed earlier, then C_L/C_D is a control variable.

(b) R_E/H , a constant which characterizes the planet size and its atmosphere.

(c) H/β , an atmosphere and RV interaction parameter.

Let us return briefly to Eqs. (6.92) and (6.96). We pointed out earlier that if we ignore centrifugal effects, that is letting $R_E \rightarrow \infty$, Eq. (6.92) is identical to Eq. (6.71). Let us now consider a ballistic re-entry trajectory. In other words, we shall assume a finite value for R_E , but shall set C_L/C_D identically equal to zero. With the density at re-entry ρ_E also set equal to zero, we have from Eqs. (6.92) and (6.96):

$$\cos \gamma = (\cos \gamma_E) \bigg/ 1 + \frac{H}{R_E} \left[\frac{1}{\lambda} - 1 \right] \quad (6.97a)$$

$$\lambda = \lambda_E \exp \left\{ \left[\frac{gH}{\beta} (\gamma_E - \gamma) \right] \bigg/ \frac{H \cos \gamma}{\rho R_E} \left[\frac{1}{\lambda} - 1 \right] \right\} \quad (6.97b)$$

First we note in Eq. (6.97a) that if we use the terrestrial value of R_E/H of 950 and consider suborbital motion only (where λ might be adequately covered by the range $0.2 < \lambda < 1.0$), then the second term in the denominator of Eq. (6.97a) may be neglected. The obvious conclusion then is that the trajectory of the ballistic RV is essentially one of constant flight-path angle. This means that the RV retains its entry angle, i.e., $\gamma = \gamma_E$. As Ashley points out this observation is a good representation for flight-path angles greater than 5 deg (Ref. 2). Even for $\gamma > 5$ deg a constant flight-path angle throughout the trajectory is only an approximation. If the flight-path angle did not change from its value at entry then the velocity magnitude would remain constant over the trajectory. This is easily seen in Eq. (6.97b). If we

do not make the assumption that the parameter H/R_E is negligibly small, then Eq. (6.97a) could be solved for λ or the velocity ratio to provide a relationship between velocity magnitude and flight-path angle.

Example 6.3: Relationship between Velocity Magnitude and Flight-Path Angle

To complete this discussion of RV particle dynamics, suppose we have an RV which has two restrictions. First, it has a limited lift-to-drag ratio; second, it has an upper limit to the magnitude of the transverse acceleration which can be imposed on the vehicle. Both of these limitations can be set as parameters.

As the RV enters the atmosphere it is to be expected that full use of the available lift-to-drag ratio will have virtually no effect on the transverse acceleration since the atmospheric density is negligible. However, as the vehicle descends into the lower and denser atmosphere, the lift-to-drag ratio must be diminished in order to keep the transverse load within the required bounds.

The equations that we shall use are Eqs. (6.66), (6.71), and (6.74b). For convenience we have listed them below:

$$V = V_E \exp - \left[\frac{\gamma_E - \gamma}{C_L / C_D} \right] \quad (6.66)$$

$$\gamma = \cos^{-1} \left[\left[\frac{\rho_0 g H}{2\beta} \right] \left[\frac{C_L}{C_D} \right] e^{-Z/H} + \cos \gamma_E \right] \quad (6.71)$$

$$n_{\perp v} = -\frac{\dot{\gamma} V}{g} = \frac{\rho_0 V^2}{2\beta} \left[\frac{C_L}{C_D} \right] e^{-Z/H} \quad (6.74b)$$

Table 6.7 is a listing of a program that calculates the velocity magnitude V , the flight-path angle γ , and the transverse acceleration $n_{\perp}$ in g . In line 360 a check is made on the acceleration level using the maximum value for the lift-to-drag ratio. Obviously we have two possibilities: first, the calculated transverse acceleration is less than or equal to the tolerance level; and second, the calculated value is above the tolerance level. In the first case, where the calculated value of transverse acceleration is below the tolerance level, this value of the transverse acceleration is retained and the lift-to-drag ratio is set at the maximum level. In the second case, where the transverse acceleration is above the tolerance level, the transverse acceleration is set at the tolerance level and the appropriate lift-to-drag ratio is calculated in line 450. This lift-to-drag ratio will be less than the vehicle is capable, but must be maintained in order to keep the transverse loads within tolerance.

A sample output from the above program is given in Table 6.8. It will be noted that initially the lift-to-drag ratio is at the maximum and that the transverse acceleration is well below the tolerance level. However, at about 45 km, the lift-to-drag ratio must be diminished in order to keep the transverse acceleration within tolerance.

Table 6.7 Simple Maneuvering Trajectory

```

00100 REM THIS PROGRAM CALCULATES THE VELOCITY MAGNITUDE ,V1,
00110 REM THE FLIGHT PATH ANGLE, G,THE LIFT-TO-DRAG RATIO,L2,
00120 REM WITH A FIXED L-D RATIO AND A RESTRICTED TRANSVERSE
00130 REM ACCELERATION, N. B IS THE BALLISTIC FACTOR, V0, IS THE
00140 REM INITIAL VELOCITY MAGNITUDE, Z0 THE REENTRY ALTITUDE,
00150 REM AND G0 THE INITIAL FLIGHT PATH ANGLE. D0/H ARE
00160 REM THE ATMOSPHERIC FITTING CONSTANTS.
00170 N = 4.0
00180 L2 = (-1)*1.5
00190 B = 10000
00200 V0 = 6000
00210 Z0 = 100000
00220 G0 = 10
00230 Z9 = 5000
00240 D0 = 1.752
00250 H = 6700
00260 R9 = 3.14159/180
00270 G0 = G0*R9
00280 G = G0
00290 V1 = V0
00300 Z = Z0
00310 PRINT #ALTITUDE#,#SPEED#,#L/D#,#TRANS AC#,#FLT ANG#
00320 PRINT #KM#,#M/SEC#,# - #,#G'S#,#DEG#
00330 PRINT
00340 FOR I = 1 TO 25
00350 Z = Z0-Z9*(I-1)
00360 N1 = ((L2*D0*(V1*2))/(2*B))*EXP((-1)*Z/H)
00370 IF ABS(N1)>4 THEN 00410
00380 N = N1
00390 L1 = L2
00400 GOTO 00460
00410 N = (-1)*4
00420 L1 = (N*2*B/(D0*(V1*2)))*EXP(Z/H)
00430 IF ABS(L1)<=ABS(L2) THEN 00460
00440 L1 = L2
00450 N = ((L1*D0*(V1*2))/(2*B))*EXP((-1)*Z/H)
00460 P8 = ((D0*9.81*H)/(2*B))*L1*EXP((-1)*Z/H)+COS(G0)
00470 G = ATN(SQR((1-(P8*2))/P8))
00480 V = V0*EXP((-1)*(G0-G)/L1)
00490 V1 = V
00500 IF Z<= 1000 THEN 00530
00510 PRINT Z/1000,V,L1,N,G/R9
00520 NEXT I
00530 END

```

Table 6.8 Sample Output from Maneuvering Program

ALTITUDE KM	SPEED M/SEC	L/D -	TRANS AC G'S	FLT ANG DEG
100	6005.10	-1.5	-1.55916E-3	9.92619
95	6005.08	-1.5	-3.29410E-3	9.92721
90	6004.93	-1.5	-6.94748E-3	9.92935
85	6004.62	-1.5	-1.46523E-2	9.93388
80	6003.95	-1.5	-3.09001E-2	9.94343
75	6002.55	-1.5	-6.51575E-2	9.96352
70	5999.6	-1.5	-.13736	10.0058
65	5993.42	-1.5	-.289424	10.0443
60	5980.6	-1.5	-.609173	10.2783
55	5954.41	-1.5	-1.27933	10.6555
50	5902.5	-1.5	-2.67467	11.408
45	5799.67	-1.0824	-4	12.106
40	5586.22	-.531559	-4	12.1763
35	5164.7	-.271658	-4	12.3334
30	4391.52	-.150684	-4	12.6944
25	3178.37	-9.88159E-2	-4	13.5974
20	1795.84	-8.94436E-2	-4	16.1819
15	917.148	-.132838	-4	24.2955
10	590.452	-.241478	-4	42.0198
5	201.322	-.276241	-4	63.728

6.8 Heat Transfer and Dynamics

The emphasis of this chapter has been on the particle dynamics of RV's. Nevertheless it would be evident that vehicular motion alters the near flowfield and that this flowfield effects subsequent motion. Clearly there is a cause and effect relationship between the properties of the flowfield and the linear and angular motion of the RV. In addition to the inertial loads due to linear and angular accelerations, the near flowfield transfers heat to the surface of the RV. Designing the RV to survive the heat loads often proves more difficult than providing protection against the inertial loads. As pointed out earlier an RV entering the atmosphere at 6000 m/s (about 20,000 ft/s) has a specific enthalpy of 1.87×10^7 J/kg (about 7752 Btu/lb). The significance of such a level of energy may be appreciated when it is recalled that water vaporizes a specific enthalpy of 2.32×10^6 J/kg (1000 Btu/lb), and that carbon vaporizes at a specific enthalpy of 6×10^7 J/kg (26,000 Btu/lb).

No attempt shall be made in this section to consider the engineering aspects of heat transfer. However, we shall show briefly the interaction that exists between the heating loads and vehicle dynamics. The sequel follows closely the dynamic heat-transfer interaction as presented by A. Miele in Ref. 3.

Let us define the heating rate per unit area $\dot{q}$ (J/s/m²) as

$$\dot{q} \equiv \frac{dq}{dt} = k(T_r - T_w) \quad (6.98)$$

where k is the convective heat-transfer coefficient (J/m²-s-K), T_r the recovery temperature, and T_w the wall temperature (both temperatures are in degrees Kelvin). If the Prandtl number is taken as unity, the recovery temperature may be written as

$$T_r = T_\infty \{1 + [(\gamma - 1)/2]M^2\} \quad (6.99)$$

where r is the recovery factor which we shall take as unity. Subtracting local wall temperature T_{wl} from both sides we may write

$$(T_r - T_w)_l = (T_\infty - T_{wl}) + M^2 T_\infty (\gamma - 1)/2 \quad (6.100)$$

Now for large Mach numbers the second term on the right dominates leading to the approximation

$$(T_r - T_w)_l = T_r - T_{wl} = M^2 T_\infty (\gamma - 1)/2 \quad (6.101)$$

where the subscript l refers to local conditions. By definition of the Mach number we may write:

$$M^2 = V^2/a^2 = V^2/\gamma RT \quad (6.102)$$

$$M^2 T = \frac{V^2}{\gamma R} = \frac{V^2}{(c_p/c_v)(c_p - c_v)} = \frac{V^2}{c_p(\gamma - 1)} \quad (6.103)$$

Equation (6.101) now becomes

$$T_r - T_{w_l} = V^2 / 2c_p \quad (6.104)$$

where γ , R , and c_p are thermodynamic constants: c_p is the specific heat at constant pressure, γ the ratio c_p / c_v , the denominator being the specific heat at constant volume, and R the gas constant for air.

Reynold's analogy relates the convective heat-transfer coefficient k to the local skin-friction coefficient:

$$c_{f_l} = \tau_w [(\rho_\infty / 2) V^2] \quad (6.105)$$

where τ_w is the local wall stress (Pascals).

According to Reynold's analogy we may write:

$$k = \frac{1}{2} c_{f_l} c_{p_l} \rho_l V_l \quad (6.106)$$

where c_{f_l} is the friction coefficient at local conditions, c_{p_l} the specific heat at constant pressure at local conditions, ρ_l and V_l are density and velocity also at local conditions. The total heat-transfer rate $\dot{Q}$ may be written as the surface integral $\dot{q}$ over the wetted areas S_w

$$\dot{Q} = \int_{S_w} \dot{q} ds = \int_{S_w} k (T_r - T_w) ds \quad (6.107)$$

Inserting Eqs. (6.104) and (6.106) into Eq. (6.107) gives

$$\begin{aligned} \dot{Q} &= \int_{S_w} \frac{c_{f_l} c_{p_l} \rho_l V_l V^2}{4c_p} ds \\ &= \frac{V^2}{4c_p} \int_{S_w} c_{f_l} c_{p_l} \rho_l V_l ds \end{aligned} \quad (6.108)$$

where the approximation can be made that

$$c_{p_l} \simeq c_p \quad V_l \simeq V$$

and

$$\int_{S_w} c_{f_l} ds = C_F S_w$$

to give

$$\dot{Q} = C_F \rho S_w V^3 / 4 \quad (6.109)$$

It appears that if we define $\dot{q}$ as the average rate of change of heat transfer per unit area, we may observe from Eq. (6.109) that

$$\dot{q} = C_F \rho V^3 / 4 \quad (6.110)$$

meaning that $\dot{q}$ is a maximum when ρV^3 is a maximum. The altitude and velocity magnitude at which this maximum occurs are easily found. Assuming steep descent with no lift we may replace ρ with the atmospheric model of Eq. (6.32) and V with Eq. (6.57) to get

$$\dot{q} = \frac{1}{4} C_F \rho_0 V_E^3 \left[e^{-3a_E e^{-Z/H}} \right] e^{-Z/H} \quad (6.111)$$

Differentiating $\dot{q}$ with respect to altitude Z and setting this derivative to zero gives

$$Z_m = H \ln(3a_E) = H \ln \left[\frac{3\rho_0 g H}{2\beta \sin \gamma_E} \right] \quad (6.112)$$

as the altitude at which the maximum average heat rate takes place. A comparison of Z_m with $\bar{Z}$ of Eq. (6.61) indicates that the altitude of maximum heating is about 1.10 times that of maximum axial load. Inserting Eq. (6.112) into Eq. (6.57) gives the velocity at maximum heating rate of

$$V_m \simeq 0.72 V_E \quad (6.113)$$

which may be contrasted with the velocity for maximum load of Eq. (6.63).

Rewriting Eqs. (6.82) and (6.74a) we have

$$\dot{Q} = \frac{1}{4} C_F \rho S_w V^3 \quad (6.82)$$

$$\dot{V} = -\frac{\rho_0 g V^2}{2\beta} e^{-Z/H} = -\frac{\rho g V^2}{2\beta} \quad (6.74b)$$

Dividing the second equation into the first we have

$$\frac{dQ}{dV} = -\frac{W}{2g} \left[\frac{C_F S_w}{C_D S} \right] v \quad (6.114)$$

where we have replaced β by $W/C_D S$.

Letting the initial and final values of heat transferred be

$$Q_i = 0 \quad Q_f = Q$$

and assuming C_F and C_D to be some kind of average values we get, upon integration of Eq. (6.114),

$$Q = \frac{W}{4g} \left[\frac{C_F S_w}{C_D S} \right] \left[V_E^2 - V_f^2 \right] \quad (6.115)$$

If we are to take the final velocity magnitude V_f at impact, i.e., $Z = 0$, then from Eq. (6.57)

$$V_f = V_E e^{-a_E} \quad (6.116)$$

The meaning of Eq. (6.115) becomes clear. If the law of conservation of

energy is applied to the path and assuming the variation in potential energy to be negligible, then the initial kinetic energy $(W/2g)V_E^2$ must equal the final kinetic energy plus the energy dissipated through drag. Part of the energy dissipated in drag is contained in the wake, the remainder is transferred to the body in the form of heat. This heat fraction is given in Eq. (6.115).

Let us now define the heat transferred to the RV as a fraction of the total kinetic energy

$$Q = Q / (W/2g)V_E^2 \quad (6.117)$$

Using Eq. (6.116) we may recast Eq. (6.115) as

$$Q = \frac{1}{2} \left[\frac{C_F S_w}{C_D S} \right] \left[1 - e^{-2a_E} \right] \quad (6.118)$$

where from Eq. (6.56),

$$a_E = \frac{\rho_0 g H}{2\beta \sin \gamma_E} = \frac{\rho_0 C_D S H}{2m \sin \gamma_E} \quad (6.119)$$

We shall close this section by looking at two extreme cases of Eq. (6.118). Assume first we have an extremely light RV such that a_E is large. Thus Eq. (6.118) becomes

$$Q = \frac{1}{2} \left[\frac{C_F S_w}{C_D S} \right] \quad (6.120)$$

For a light RV, the fraction of the initial kinetic energy, transferred to the body by convective heating, is one half of the ratio of the friction drag to total drag. Therefore, in order to minimize the heat convected to the RV the frictional drag must be made small relative to the pressure drag. The engineering implications are that the light RV must have large pressure drag, i.e., a blunt shape. The graph in Fig. 6.11 shows how the drag coefficient varies with nose blunting.

Figure 6.11 provides some quantitative information to support the intuitive feeling that increasing nose bluntness increases the pressure drag. It might be interesting to see how large a β an RV may have and still meet the approximation of Eq. (6.120). If we set as an upper bound

$$e^{-2a_E} \leq 0.1$$

then

$$\beta \leq - \frac{\rho_0 g H}{\ln(0.1) \sin \gamma_E} \quad (6.121)$$

Letting $\gamma_E = 75$ deg, $\rho_0 = 1.752$ kg/m³, and $H = 6700$ m, we get a $\beta \leq 5.18 \times 10^4$ Pa. Many RV's have a ballistic factor well below this value so the approximation of Eq. (6.120) is of more than academic interest. For a

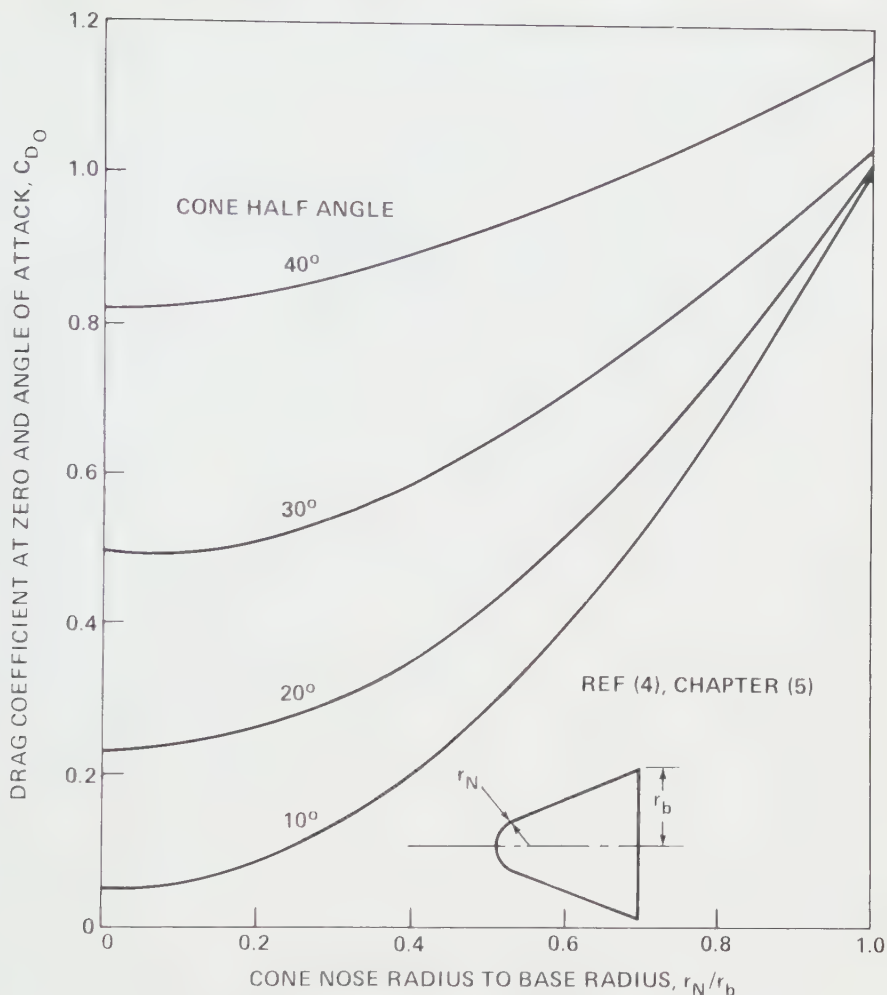


Fig. 6.11 Cone drag coefficient vs cone bluntness.

typical practical design $C_F S_h / C_D S$ is about 1/10. Inserting this value in Eq. (6.120) indicates that the heat energy transferred to the RV is about 1/20 of the total kinetic energy present at re-entry. As pointed out earlier, this is significant in that the amount of energy present at re-entry is of the same order of magnitude as that required to vaporize the RV.

For the relatively heavy RV we may approximate Eq. (6.118) as

$$Q = \frac{1}{2} \frac{C_F S_w}{C_D S} 1 - (1 - 2a_E)$$

$$Q = \frac{\rho_0 g H C_F S_w}{2W \sin \gamma_E} = \frac{\rho_0 H C_F S_w}{2m \sin \gamma_E} \quad (6.122)$$

Thus for a heavy RV the fraction of the initial kinetic energy transferred to the RV as heat is proportional to the friction coefficient; thus a shape of low-friction drag must be employed.

References

¹ Loh, W.H.T., *Reentry and Planetary Entry Physics and Technology Dynamics, Physics, Radiation, Heat Transfer and Ablation*, Springer-Verlag, New York, 1968.

² Ashley, H., *Engineering Analysis of Flight Vehicles*, Addison-Wesley, Reading, Mass., 1974.

³ Miele, A., *Flight Mechanics*, Addison-Wesley, Reading, Mass., 1962.

VII. MOMENT EQUATIONS

It might seem appropriate as the next topic to discuss the boost trajectory since we have already discussed the Keplerian and re-entry portion of the trajectory. However, in analyzing the boost trajectory we shall require the use of directional cosine matrices (DCM). The DCM's will be derived in Chapter VIII, so we shall defer discussion of the boost phase until Chapter XI.

In Chapter IV we derived the force equations. Equation (4.36) was developed as the most general form of the force equation. Equation (4.36) was then subjected to various assumptions; from these assumptions we developed simpler forms of the force equation. One of these forms was then applied in Chapter V to Keplerian motion and in Chapter VI to planar motion in the atmosphere. In this chapter, we shall develop the moment equations in a general form. Then as we did in Chapter IV, we shall subject the most general form to various simplifications to develop some useful but more tractable equations.

7.1 Derivation of Moment Equations

We shall now derive the equations for the angular momentum of a rigid body shown in Fig. 7.1 where we designated an inertial (nonrotating, nonaccelerating) axis system (X_I, Y_I, Z_I) . A second axis has been affixed to the RV. This axis system has been designated as (x^b, y^b, z^b) . In the dynamics of aerospace vehicles it is common practice to have an axis system whose origin, although affixed to the body, does not share all body angular rates. A frame which we shall use in Chapter X, called the aeroballistic axes, does not share body roll. In much of our studies of RV dynamics, we shall assume that the (x^b, y^b, z^b) axes are fixed to the body and share all body angular motion; however, the origin of this system will not necessarily be at the center of mass of the RV.

The vector $\mathbf{R}_0$ locates the origin of the (x^b, y^b, z^b) system, $\mathbf{R}_i$ locates the generic mass particle m_i from the inertial system (X_I, Y_I, Z_I) , and $\mathbf{r}_i$ locates the same particle from the body system (x^b, y^b, z^b) . Clearly then

$$\mathbf{R}_i = \mathbf{R}_0 + \mathbf{r}_i \quad (7.1)$$

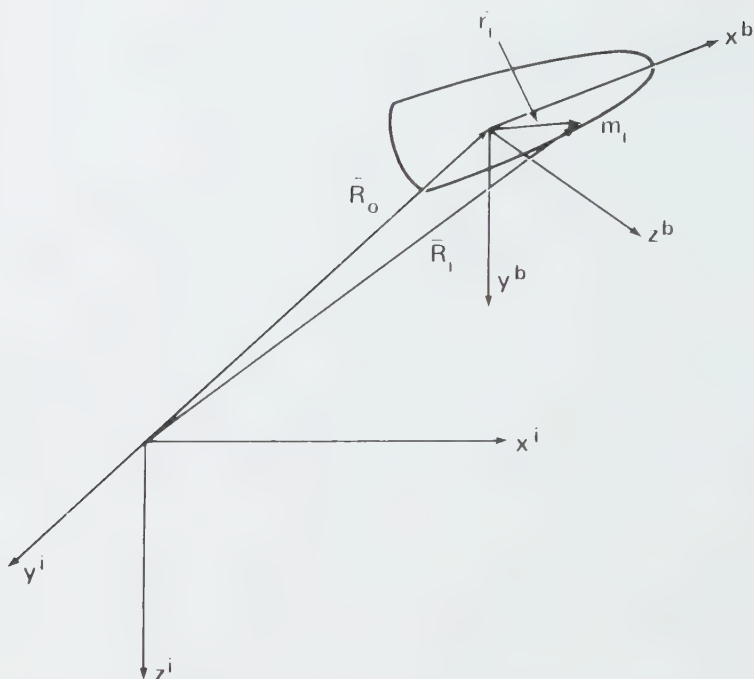


Fig. 7.1 Location of generic mass element in re-entry vehicle.

The linear momentum of the particle m_i relative to the inertial system (X_I, Y_I, Z_I) is

$$m_i \left. \frac{d\mathbf{R}_i}{dt} \right|_I \quad (7.2)$$

The angular momentum $\mathbf{H}_i$ is the moment of linear momentum, so

$$\mathbf{H}_i = \mathbf{R}_i \times m_i \left. \frac{d\mathbf{R}_i}{dt} \right|_I \quad (7.3a)$$

Using Eq. (7.1) and the theorem of Coriolis (see Chapter IV) gives

$$\begin{aligned} \mathbf{H}_i = (\mathbf{R}_0 + \mathbf{r}_i) \times m_i \left. \frac{d\mathbf{r}_i}{dt} \right|_m &+ (\mathbf{R}_0 + \mathbf{r}_i) \times m_i (\boldsymbol{\omega}_{Im} \times \mathbf{r}_i) \\ &+ (\mathbf{R}_0 + \mathbf{r}_i) \times m_i \left. \frac{d\mathbf{R}_0}{dt} \right|_I \end{aligned} \quad (7.3b)$$

Now letting

$$\left. \frac{d\mathbf{R}_0}{dt} \right|_I = \mathbf{V}_0$$

Eq. (7.3b) becomes

$$\begin{aligned} \mathbf{H}_i = m_i \mathbf{R}_0 \times \left. \frac{d\mathbf{r}_i}{dt} \right|_m + m_i \mathbf{r}_i \times \left. \frac{d\mathbf{r}_i}{dt} \right|_m + \mathbf{R}_0 \times (\boldsymbol{\omega}_{lm} \times \mathbf{r}_i m_i) \\ + m_i \mathbf{r}_i \times (\boldsymbol{\omega}_{lm} \times \mathbf{r}_i) + \mathbf{R}_0 \times m_i \mathbf{V}_0 + m_i (\mathbf{r}_i \times \mathbf{V}_0) \end{aligned} \quad (7.4)$$

We may sum over all the particles of the rigid body to get

$$\begin{aligned} \mathbf{H} = \sum_i \mathbf{H}_i = \mathbf{R}_0 \times \frac{d}{dt} \left[\sum_i m_i \mathbf{r}_i \right] \Big|_m + \sum_i \left[(\mathbf{r}_i m_i) \times \left. \frac{d\mathbf{r}_i}{dt} \right|_m \right] \\ \mathbf{R}_0 \times \left[\boldsymbol{\omega}_{lm} \times \sum_i \mathbf{r}_i m_i \right] + \sum_i \mathbf{r}_i \times (\boldsymbol{\omega}_{lm} \times \mathbf{r}_i m_i) \\ + (\mathbf{R}_0 \times \mathbf{V}_0) M + \left[\sum_i m_i \mathbf{r}_i \right] \times \mathbf{V}_0 \end{aligned} \quad (7.5)$$

where we have taken $\sum m_i = M$, or the total vehicle mass. Note also that since m_i is invariant with time, it is permissible to place it inside the derivative operator. Of course in some devices of engineering interest m_i would vary with time. One example is the rocket powered RV booster which during engine operation has a continuous mass loss. In such a case the time rate of change of this term in the momentum (angular) can contribute to a damping type moment.¹

For our purpose the origin of the inertial axis system (X_I, Y_I, Z_I) is arbitrary since we are considering the case of an isolated rigid body. Let us therefore put the origin of the inertial axes at the origin of the body axes. The result is that $\mathbf{R}_0 \equiv 0$. Equation (7.5) then becomes

$$\mathbf{H} = \sum_i (m_i \mathbf{r}_i) \times \left. \frac{d\mathbf{r}_i}{dt} \right|_m + \sum_i [\mathbf{r}_i \times (\boldsymbol{\omega}_{lm} \times \mathbf{r}_i) m_i] + \left[\sum_i m_i \mathbf{r}_i \right] \times \mathbf{V}_0 \quad (7.6)$$

By making $\mathbf{R}_0$ identically zero we are ignoring the contribution to the total angular momentum arising from curvilinear motion of the body about some point external to the body. Consider orbital motion, for example, where $\mathbf{R}_0$ is the point from the Earth's center to the RV in orbit. Even if the origin of the body frame (x^b, y^b, z^b) is at the RV's center of mass, i.e., $\sum_i m_i \mathbf{r}_i \equiv 0$, we

should retain the term in Eq. (7.5), $(\mathbf{R}_0 \times \mathbf{V}_0) M$. If we take $r \equiv |\mathbf{R}_0|$ and $r\theta = |\mathbf{V}_0|$, then the term $(\mathbf{R}_0 \times \mathbf{V})$ is identical to that in the parentheses in Eq. (5.4).

Returning to Eq. (7.6) we can make further simplifications. First, if we are limiting ourselves to a rigid body, then the first term is identically zero since $d\mathbf{r}_i/dt|_m = 0$. In other words, there is no motion of mass particles within the body. Expressed in derivative form

$$\left. \frac{d\mathbf{r}_i}{dt} \right|_m \equiv 0 \quad (7.7)$$

Before abandoning this term it must be pointed out that there are control systems that vary vehicle trim by moving a small mass within the RV. In such a case while most of the RV's constituent mass would obey Eq. (7.7), the moving mass would not. In more advanced treatments it might be necessary to return to Eq. (7.6) and retain the first term. Further, we shall define the center of mass $\bar{\mathbf{r}}$ as

$$M\bar{\mathbf{r}} = \sum_i \mathbf{r}_i m_i \quad (7.8)$$

Using Eqs. (7.7) and (7.8) in Eq. (7.6) we get

$$\mathbf{H} = M\bar{\mathbf{r}} \times \mathbf{V}_0 + \sum_i [\mathbf{r}_i \times (\boldsymbol{\omega}_{lm} \times \mathbf{r}_i) m_i] \quad (7.9)$$

The first term on the right will vanish if any one of the following three conditions is met:

- (a) (x^b, y^b, z^b) system has its origin at the center of mass, i.e., $\bar{\mathbf{r}} \equiv 0$.
- (b) The origin of the (x^b, y^b, z^b) system is fixed, i.e., $\mathbf{V}_0 \equiv 0$, even though the origin may *not* be at the center of mass.
- (c) If the velocity of the origin $\mathbf{V}_0$ is in a direction parallel to the vector from the origin to the center of mass, i.e., $\mathbf{V}_0$ and $\bar{\mathbf{r}}$ are colinear.

For our purposes we shall assume (a) above, recognizing again that a moving mass control system will invalidate this assumption. In any event, if the origin of (x^b, y^b, z^b) is taken at some point other than the center of mass, then there will be an angular momentum contribution, $M\bar{\mathbf{r}} \times \mathbf{V}_0$. Now we shall use the vector identity

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \equiv \mathbf{b}(\mathbf{c} \cdot \mathbf{a}) - \mathbf{c}(\mathbf{b} \cdot \mathbf{a})$$

Equation (7.9) (with $\bar{\mathbf{r}} \equiv 0$) now becomes

$$\mathbf{H} = \sum_i m_i [\boldsymbol{\omega}_{lm}(\mathbf{r}_i \cdot \mathbf{r}_i) - \mathbf{r}_i(\boldsymbol{\omega}_{lm} \cdot \mathbf{r}_i)] \quad (7.10)$$

Now with

$$\mathbf{r}_i = x_i \mathbf{i}^b + y_i \mathbf{j}^b + z_i \mathbf{k}^b \quad \boldsymbol{\omega}_{lm} = p \mathbf{i}^b + q \mathbf{j}^b + r \mathbf{k}^b$$

where $(\mathbf{i}^b, \mathbf{j}^b, \mathbf{k}^b)$ are the unit vectors along the moving (x^b, y^b, z^b) frame, Eq. (7.10) becomes

$$\begin{aligned}
\mathbf{H} = & \left[p \sum_i (y_i^2 + z_i^2) m_i - q \sum_i (x_i y_i) m_i - r \sum_i (x_i z_i) m_i \right] \mathbf{i}^b \\
& + \left[-p \sum_i (y_i x_i) m_i + q \sum_i (x_i^2 + z_i^2) m_i - r \sum_i (y_i z_i) m_i \right] \mathbf{j}^b \\
& + \left[-p \sum_i (z_i x_i) m_i - q \sum_i (y_i z_i) m_i + r \sum_i (x_i^2 + y_i^2) m_i \right] \mathbf{k}^b \quad (7.11)
\end{aligned}$$

In the above equation r is the component of angular velocity along the z^b axis; a distinction must be made with the use of r as a radius vector. Since all mass elements m_i of the RV have the same rotational rates, we may remove the angular velocity components (p, q, r) outside of the summation signs. The body's mass distribution may be characterized by the following set of constants

$$\begin{aligned}
I_x &= \sum_i m_i (y_i^2 + z_i^2) \\
I_y &= \sum_i m_i (z_i^2 + x_i^2) \\
I_z &= \sum_i m_i (x_i^2 + y_i^2) \\
I_{xz} &= I_{zx} = \sum_i m_i (x_i z_i) \\
I_{xy} &= I_{yx} = \sum_i m_i (x_i y_i) \\
I_{yz} &= I_{zy} = \sum_i m_i (y_i z_i) \quad (7.12)
\end{aligned}$$

The first three quantities are called the "moments of inertia" about the indicated (x^b, y^b, z^b) axes and the second three are the "products of inertia" with respect to the indicated planes, e.g., xy, xz, yz . The symmetry in the products of inertia has been noted. Also we note that the moments of inertia may only be positive, but the products of inertia may be either positive or negative. Using the definitions of Eqs. (7.12), Eq. (7.10) becomes

$$\mathbf{H} = H_x \mathbf{i}^b + H_y \mathbf{j}^b + H_z \mathbf{k}^b \quad (7.13)$$

where

$$\begin{aligned}
H_x &= I_x p - I_{xy} q - I_{xz} r \\
H_y &= -I_{xy} p + I_y q - I_{yz} r \\
H_z &= -I_{xz} p - I_{yz} q + I_z r \quad (7.14)
\end{aligned}$$

Clearly Eq. (7.14) may be written in matrix form as

$$H = \begin{bmatrix} H_x \\ H_y \\ H_z \end{bmatrix} = \begin{bmatrix} I_x & -I_{xy} & -I_{xz} \\ -I_{xy} & I_y & -I_{yz} \\ -I_{xz} & -I_{yz} & I_z \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (7.15)$$

The 3×3 matrix on the right is called the inertia tensor or dyadic. Equation (7.15) may be written more compactly as the dyadic dot product with angular rate ω .

$$\mathbf{H} = \mathbf{I} \cdot \boldsymbol{\omega} \quad (7.16)$$

We may now show that the rate of change of angular momentum relative to inertial space is equal to the moment applied to the RV. From Eq. (7.3a) we have

$$\left. \frac{d\mathbf{H}}{dt} \right|_I = \frac{d}{dt} \left[\sum_i \mathbf{H}_i \right] = \sum_i \mathbf{R}_i \times m_i \left. \frac{d^2 \mathbf{R}_i}{dt^2} \right|_I + \sum_i \left. \frac{d\mathbf{R}_i}{dt} \right|_I \times m_i \left. \frac{d\mathbf{R}_i}{dt} \right|_I \quad (7.17)$$

The second term obviously vanishes and the first term may be rewritten using Newton's second law:

$$\mathbf{F}_i = m_i \left. \frac{d^2 \mathbf{R}_i}{dt^2} \right|_I \quad (7.18)$$

to become

$$\sum_i \left. \frac{d\mathbf{H}_i}{dt} \right|_I = \sum_i \mathbf{R}_i \times m_i \left. \frac{d^2 \mathbf{R}_i}{dt^2} \right|_I = \sum_i \mathbf{R}_i \times \mathbf{F}_i = \sum_i \mathbf{M}_i \quad (7.19)$$

Carrying out the summation we get

$$\mathbf{M} = \sum_i \mathbf{M}_i = \sum_i \left. \frac{d\mathbf{H}_i}{dt} \right|_I = \left. \frac{d\mathbf{H}}{dt} \right|_I = \left. \frac{d\mathbf{H}}{dt} \right|_m + \boldsymbol{\omega}_{lm} \times \mathbf{H} \quad (7.20)$$

where we have again used the theorem of Coriolis.

For reference the moment equation may be written in vector form as

$$\mathbf{M} = \left. \frac{d\mathbf{H}}{dt} \right|_m + \boldsymbol{\omega}_{lm} \times \mathbf{H} \quad (7.21)$$

where the angular momentum $\mathbf{H}$ is given in Eqs. (7.13-7.15).

7.2 Calculation of Moment and Products of Inertia

In this section we shall discuss how to calculate the moments of inertia in one axis system given measurements in another axis system. In Fig 7.2 we

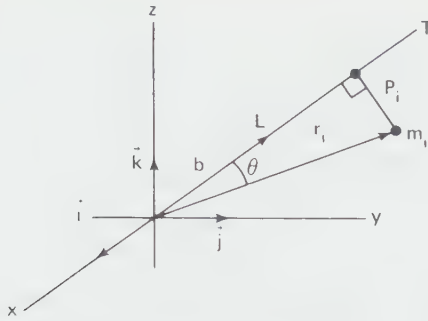


Fig. 7.2 Location of generic mass element relative to an arbitrary line.

have depicted a mass particle m_i and a line L with unit direction vector $\mathbf{b}$. The moment of inertia of a particle of mass m_i about a line L is $p_i^2 m_i$, where p_i is the perpendicular distance from the line L to the particle m_i .

The direction of vector of line L is established by the unit vector $\mathbf{b}$ given in terms of the direction cosines (a, b, c) as

$$\mathbf{b} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k} \quad (7.22a)$$

The vector $\mathbf{r}_i$ to the mass element m_i is

$$\mathbf{r}_i = x_i\mathbf{i} + y_i\mathbf{j} + z_i\mathbf{k} \quad (7.22b)$$

and by definition $|\mathbf{p}_i|$ is

$$p_i = |\mathbf{r}_i \times \mathbf{b}| = r_i \sin(\theta) = r_i \sin\theta \quad (7.22c)$$

Thus we have

$$I = \sum_i p_i^2 m_i = \sum_i |\mathbf{r}_i \times \mathbf{b}|^2 m_i \quad (7.23)$$

Inserting Eqs. (7.22a and b) in Eq. (7.23) we get

$$I = \sum_i m_i [(a\mathbf{i} + b\mathbf{j} + c\mathbf{k}) \times (x_i\mathbf{i} + y_i\mathbf{j} + z_i\mathbf{k})]^2 \quad (7.24)$$

Using the definitions given in Eqs. (7.12) we have

$$I = I_x a^2 + I_y b^2 + I_z c^2 - 2I_{xy} ab - 2I_{xz} ac - 2I_{yz} bc \quad (7.25)$$

It should be clear that the six inertia parameters $I_x, I_y, I_z, I_{xy}, I_{xz}, I_{yz}$ will vary as the (x, y, z) axis system rotates relative to the line L . As the axis system rotates the six inertial parameters and the directional cosines (a, b, c) will vary. However, since the line L maintains a fixed direction relative to the body, the moment of inertia I will not change. We shall assume at this point that it is possible to rotate the (x, y, z) frame to a position such that all

the products of inertia vanish, i.e., Eq. (7.25) might be written as

$$I = I_x a^2 + I_y b^2 + I_z c^2 \quad (7.26)$$

where I_x , I_y , I_z are the principal moments of inertia and the axis system in which they are quantized is called the "principal axes." Principal axes, then, are the axis system for which the products of inertia vanish. In such an axis system, the matrix of Eq. (7.15) will become diagonal. The method of eigenvalues can be used to "diagonalize" any inertia tensor which means that once the six inertial parameters are measured or "known" in any arbitrary frame, a purely mathematical technique (not requiring further measurements) can be used to find the principal axes and the values of the principal moments of inertia.² A discussion of the method of eigenvalues pertinent to determining the principal moments of inertia will be given in Chapter X.

The approach taken here will be to let the geometry suggest principal axes. Let us assume that an RV has a plane of symmetry. It should be remembered that even though an RV has geometric symmetry of some kind, say a plane of mirror symmetry, the distribution of mass internal to the RV will not necessarily be symmetric with respect to this geometric plane. Thus, while the principal axes exist, the plane of mass symmetry will not lie necessarily in this plane of geometric symmetry. Continuing with our assumption that the plane of geometric symmetry is also a plane of mass symmetry, we get from Eq. (7.25)

$$I = I_x a^2 + I_y b^2 + I_z c^2 - 2I_{xz} ac \quad (7.27)$$

Equation (7.27) applies to Fig. 7.3 which depicts an initially conical RV which has been subjected to a base cutout. The cutouts and flaps destroy the RV's rotational symmetry but the RV still retains a plane of symmetry.

Assume that we have measured I_x , I_z , and I_{xz} in the xyz frame. Since the x - z plane is a plane of symmetry we would expect the principal axes x , z to lie in this plane, and that $y = y$ because the y axis is normal to a plane of symmetry. Thus,

$$I_y = I_y$$

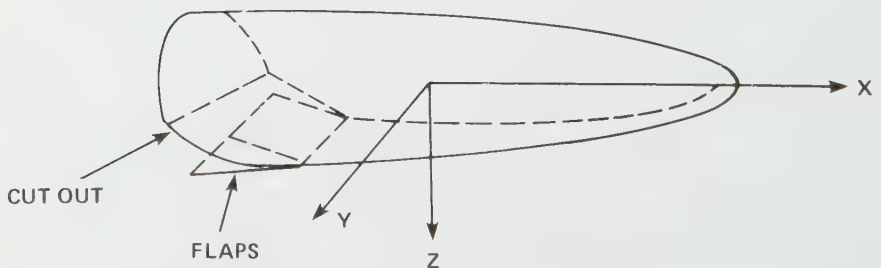


Fig. 7.3 Re-entry vehicle with plane of symmetry.

where bold print indicates principal values. The task then is to locate the principal axes ($\bar{x}, \bar{z}$) from the general axes (x, z) in which the moment of inertia measurements were made. We note again that from our assumption of a plane of symmetry, $y = y$. We shall use Eq. (7.27) to find $I_{\bar{x}}$, $I_{\bar{z}}$, the principal moments of inertia.

For the principal moment of inertia about the $\bar{x}$ axis, we have from Fig. 7.4

$$a = \cos\theta$$

$$b = 0$$

$$c = \cos(\pi/2 + \theta) = -\sin\theta$$

So from Eq. (7.27) we have

$$I_{\bar{x}} = I_x \cos^2\theta + I_z \sin^2\theta + 2I_{xz} \sin\theta \cos\theta \quad (7.28a)$$

For $I_{\bar{z}}$ we have

$$a = \cos(\pi/2 - \theta) = \sin\theta$$

$$b = 0$$

$$c = \cos\theta$$

and again applying Eq. (7.27) we have

$$I_{\bar{z}} = I_x \sin^2\theta + I_z \cos^2\theta - 2I_{xz} \sin\theta \cos\theta \quad (7.28b)$$

From Fig. 7.4 it follows that

$$X_i = X_i \cos\theta - Z_i \sin\theta$$

$$Y_i = Y_i$$

$$Z_i = X_i \sin\theta + Z_i \cos\theta \quad (7.29a)$$

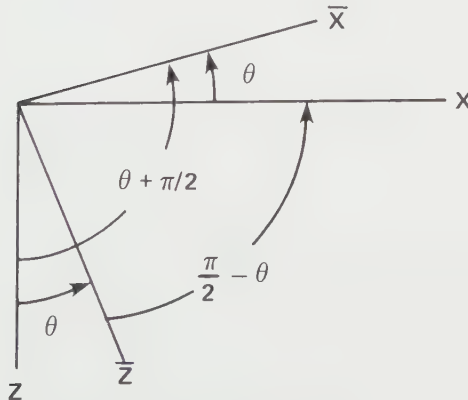


Fig. 7.4 Rotation of general axes to principal axes.

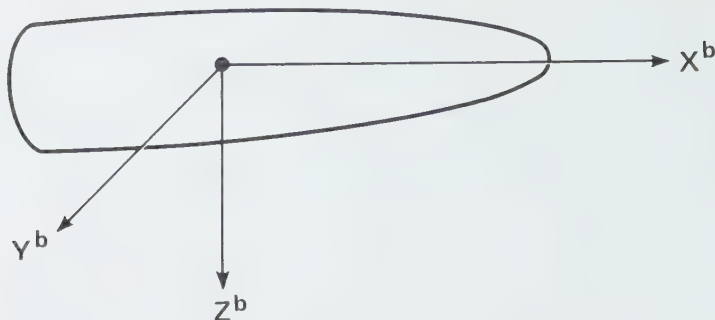


Fig. 7.5 Re-entry vehicle with mass and configurational plane of symmetry.

or in matrix form

$$\begin{bmatrix} X_i \\ Y_i \\ Z_i \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \end{bmatrix} \quad (7.29b)$$

where the subscript i indicates that the coordinates (X_i, Y_i, Z_i) and (X_i, Y_i, Z_i) locate the generic mass element m_i . Now by definition the product of inertia I_{xz} vanishes in the principal axes. The absence of I_{xz} may be used to locate the principal axes (X_i, Y_i, Z_i) from the general system (X_i, Y_i, Z_i) :

$$\begin{aligned} I_{xz} &\equiv 0 = \sum_i m_i X_i Z_i \\ 0 &= \sum_i m_i (X_i \cos\theta - Z_i \sin\theta)(X_i \sin\theta + Z_i \cos\theta) \\ 0 &= \frac{1}{2}(I_z - I_x)\sin 2\theta + I_{xz} \cos 2\theta \end{aligned}$$

on solving for θ

$$\theta = \frac{1}{2} \tan^{-1} [-2I_{xz} / (I_z - I_x)] \quad (7.30)$$

Equation (7.30) must be solved first to get θ . Then it is possible to obtain I_x and I_z from Eqs. (7.28).

In Fig. 7.5 we have depicted an RV with mass distribution such that the x - z plane is both a plane of mass and configurational symmetry. Let us assume that the origin of the coordinate system is at the center of mass. Let the mass distribution be represented by dumbbells as shown in Fig. 7.6. We assume that dumbbells A and B are of the same mass and lie along the x axis at an equal distance from the origin. Masses C and D are also an equal distance from the origin and lie along the y axis. Masses E and F are equal to each other and are placed equidistant from the origin. However, while masses E and F are equal and confined to the x - z principal plane, they have x - z coordinates of $(-\xi, -\zeta)$ and (ξ, ζ) , respectively. The product of

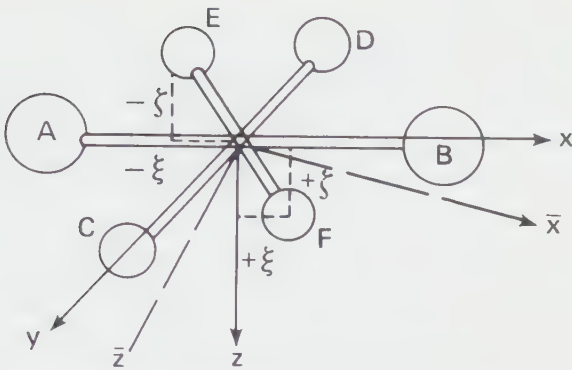


Fig. 7.6 Lumped approximation to continuous mass distribution.

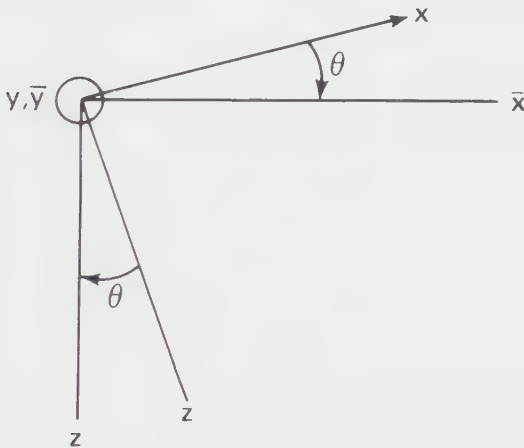


Fig. 7.7 Rotation of principal axes to general axes.

inertia then follows according to its definition in Eqs. (7.12) as

$$I_{xz} = \sum_{i=1}^{i=2} m_i X_i Z_i = M_{EF}(-\xi)(-\zeta) + M_{EF}(\xi)(\zeta) \quad (7.31)$$

$$I_{xz} = 2M_{EF}\xi\zeta \quad (7.32)$$

The reader should be able to answer the question as to why mass pairs $A-B$ and $C-D$ do not contribute to a product of inertia. Since I_{xz} is positive and $I_z > I_y$, Eq. (7.30) indicates that the x, z axes must be rotated about the y axis in a negative sense to establish the principal axes $(\bar{x}, \bar{y}, \bar{z})$. The situation looks something like that illustrated in Fig. 7.7.

The important idea is that if I_x, I_z, I_{xz}, I_y are measured in one frame, i.e., x, y, z ; we can find the amount the frame must be rotated to establish the

principal axes from Eq. (7.30) and the corresponding principal moments of inertia from Eqs. (7.28). The process can be reversed of course. Given the principal moments of inertia, I_x, I_z (I_y remains unchanged if there is a plane of symmetry) and the rotation angle to the new frame, we can derive an expression for the general moment of inertia. First using the definition for I_x as,

$$I_x = \sum_i m_i (y_i^2 + z_i^2) \quad (7.33)$$

Inverting our earlier relationship between (X_i, Y_i, Z_i) and (X, Y, Z) we have

$$\begin{bmatrix} X_i \\ Y_i \\ Z_i \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad (7.34)$$

Inserting in Eq. (7.33) where appropriate we have

$$\begin{aligned} I_x &= \sum_i m_i [(Y_i^2 + Z_i^2 \cos^2\theta + X_i^2 \sin^2\theta - 2X_i Z_i \sin\theta \cos\theta)] \\ &= \sum_i m_i [Y_i^2 + Z_i^2] \cos^2\theta + (Y_i^2 + X_i^2) \sin^2\theta - 2X_i Z_i \sin\theta \cos\theta \\ &= I_x \cos^2\theta + I_z \sin^2\theta - 2I_{xz} \sin\theta \cos\theta \end{aligned} \quad (7.35)$$

but $I_{xz} = 0$ by definition of principal axes, so

$$I_x = I_x \cos^2\theta + I_z \sin^2\theta \quad (7.36a)$$

A similar type of development provides I_z

$$I_z = I_x \sin^2\theta + I_z \cos^2\theta \quad (7.36b)$$

To review briefly, there are simple relationships among the general and principal moments of inertia when there is a plane of symmetry. If we know I_x, I_z, I_{xz} we can use Eqs. (7.28) and (7.30) to get I_x, I_z , and θ . The angle θ is of course the angle through which the xz axes must be rotated about the y axis to establish the principal frame. On the other hand if we are given the principal moments of inertia I_x, I_y, I_z and the angle to a new frame, we can use Eq. (7.36) to obtain I_x and I_z and Eq. (7.30) to obtain I_{xz} . Note we must replace θ by $(-\theta)$ if we are to define θ as a positive rotation about the y axis to go from the general frame to the principal frame. Thus,

$$I_{xz} = [(I_z - I_x)/2] \tan 2\theta \quad (7.37)$$

Suppose we want to go from one general frame to another, i.e., frame (1) to (2) by a positive angular rotation θ about the y axis (see Fig. 7.8). From the same development in deriving Eqs. (7.28a), (7.28b), and (7.30) we obtain

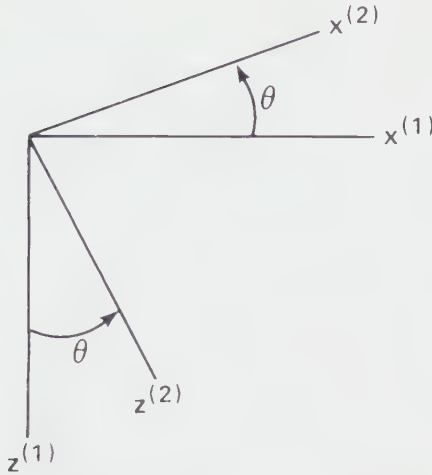


Fig. 7.8 Rotation between two general axes.

$$I_x^{(2)} = I_x^{(1)} \cos^2 \theta + I_z^{(1)} \sin^2 \theta + I_{xz}^{(1)} \sin 2\theta$$

$$I_z^{(2)} = I_x^{(1)} \sin^2 \theta + I_z^{(1)} \cos^2 \theta - I_{xz}^{(1)} \sin 2\theta$$

$$I_{xz}^{(2)} = \frac{1}{2}(I_z^{(1)} - I_x^{(1)}) \sin 2\theta + I_{xz}^{(1)} \cos 2\theta$$

$$I_y^{(2)} = I_y^{(1)} \quad (7.38)$$

We might pose an interesting generalization of Eqs. (7.38) as follows: assume that the moments and products of inertia have been quantized in a fixed body frame, i.e., we can assign numerical values to the nine elements of the inertia tensor. If we define a second axis system as an arbitrary rotation from the first system, what are the new values of the inertia tensor? We might also pose the following related question: What systematic methods are available for establishing a second coordinate frame which has the property of diagonalizing the inertia tensor? By definition the moments of inertia in such a frame would have no products of inertia and hence such a frame would be by definition a principal frame. This problem has already been solved for the case when the RV has a plane of symmetry [Eqs. (7.28a), (7.28b), and (7.30)]. In Chapter VIII we shall discuss this problem when there is no plane of symmetry. Both of the above questions will be answered after we have discussed the Euler angles and directional cosine matrices.

References

- ¹Thomson, W. T., *Introduction to Space Mechanics*, Wiley, New York, 1960.
- ²Kaplan, M. H., *Modern Spacecraft Dynamics and Control*, Wiley, New York, 1960.

VIII. AXIS TRANSFORMATIONS

8.1 Angular Rotations

It should be clear by now that vectors are important entities in the description and solution of engineering problems in RV dynamics. The computation or measurement of vectors means the assignment of a numerical value to the vector's components in some axis system. Certain vectors may be measurable, in a practical sense, only in a specific coordinate frame. For example, body velocity measurements in a strapdown inertial measurement unit must, of necessity, be *measured* in a frame attached to the body, yet Newton's second law relates force and acceleration only when acceleration is measured relative to an inertial (nonaccelerating) frame. Navigational information, on the other hand, might be *expressed* most meaningfully in an Earth-based frame. This chapter considers ways of relating measurements or computations made in one frame with their equivalent numerical components in a second frame.

There are several methods for describing frame rotation. In this book we shall limit attention to Euler angles, directional cosines matrices (DCM's), and quaternions (or the closely related Euler four-parameter method). In principle all such methods are equivalent; however, one method may have certain advantages in particular applications such as ease of computation, maintenance of matrix orthogonality, and freedom from singularities.

It might seem that too much emphasis is being placed upon such a variety of techniques for describing frame rotation when our theme is the dynamics of re-entry bodies. It is true that we could proceed with our development in the most frugal manner by selecting one method, say DCM's, and denying attention to all other methods. However, if we, for example, wish to simulate the navigation problem using strapdown systems, the inadequacies of the DCM method would become readily apparent, even though the DCM is the most straightforward of all methods.

8.1.1 Euler Angles: Scheme 1

In this and the next section we shall confine attention to Euler angles which relate measurements in two frames by a series of three orthogonal rotations between the two frames. The analytic form of the transformation matrix between the two frames will depend upon the order in which the

rotations are taken. In this chapter we shall follow two schemes; the first universally used in the study of aircraft dynamics is adapted to the case of an RV with a plane of mass symmetry; the second scheme, long used in the study of top dynamics and aeroballistics, is particularly suited to RV's with rotational symmetry. Let us consider first the aircraft sequence.

The three rotations that we shall use are:

(1) Rotate positively about the Z axis through the finite rotation angle Ψ (see Fig. 8.1). The matrix relating vector components (x', y', z') to vector components (x_I, y_I, z_I) is

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos\Psi & \sin\Psi & 0 \\ -\sin\Psi & \cos\Psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_I \\ y_I \\ z_I \end{bmatrix} = C \begin{bmatrix} x_I \\ y_I \\ z_I \end{bmatrix} \quad (8.1)$$

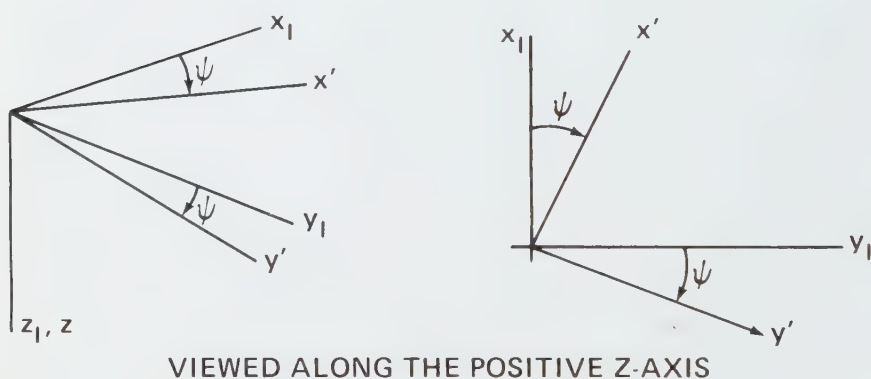


Fig. 8.1 First rotation: Euler angle scheme 1.

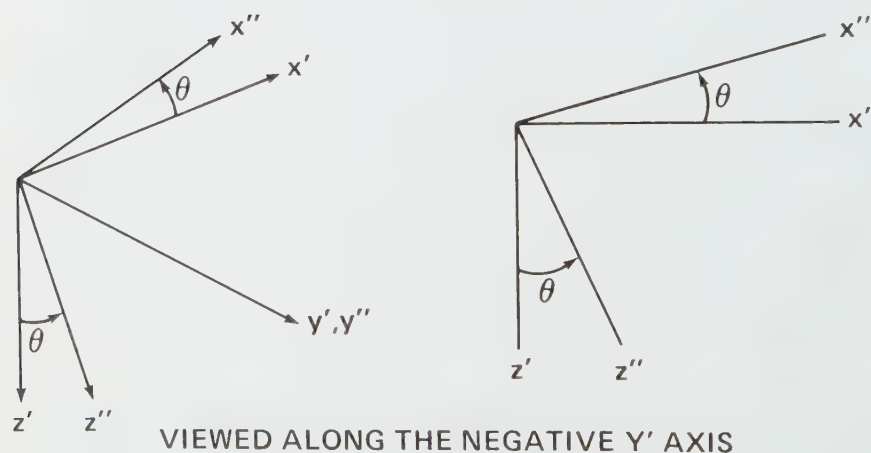


Fig. 8.2 Second rotation: Euler angle scheme 1.

(2) Next rotate about the y' axis through the finite rotational angle θ to give the (x'',y'',z'') frame (see Fig. 8.2). The matrix relating the (x'',y'',z'') axes and the (x',y',z') axes is

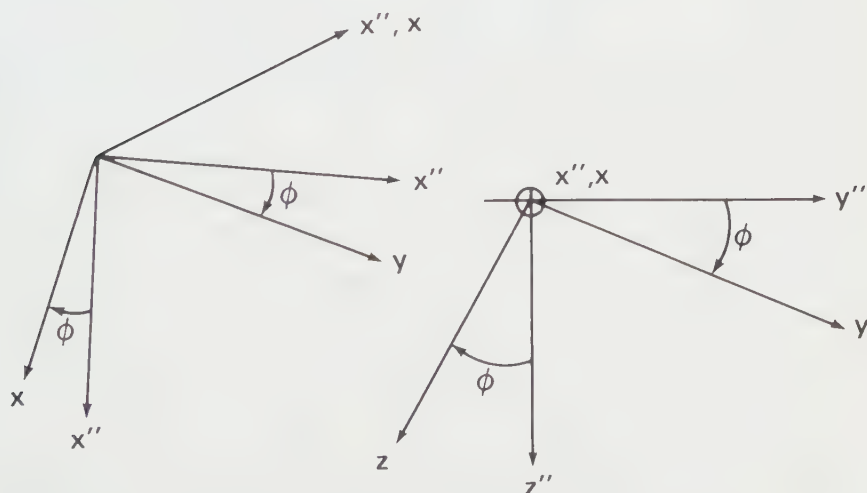
$$\begin{bmatrix} x'' \\ y'' \\ z'' \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = B \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = BC \begin{bmatrix} x_I \\ y_I \\ z_I \end{bmatrix} \quad (8.2)$$

(3) Finally, rotate positively about the x'' axis, through the positive rotation angle ϕ to give the (x,y,z) frame (see Fig. 8.3). The matrix relating the vector components (x,y,z) to the vector components (x'',y'',z'') is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} x'' \\ y'' \\ z'' \end{bmatrix} = A \begin{bmatrix} x'' \\ y'' \\ z'' \end{bmatrix} = ABC \begin{bmatrix} x_I \\ y_I \\ z_I \end{bmatrix} \quad (8.3)$$

Clearly the relationship between vector components (x,y,z) in the body frame and the original inertial reference frame, (x_I,y_I,z_I) frame is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = ABC \begin{bmatrix} x_I \\ y_I \\ z_I \end{bmatrix} = D \begin{bmatrix} x_I \\ y_I \\ z_I \end{bmatrix} \quad (8.4)$$



VIEWED ALONG THE POSITIVE X AXIS

Fig. 8.3 Third rotation: Euler angle scheme 1.

By matrix multiplication the matrix product, $ABC = D$ is

$$D = \begin{bmatrix} \cos\theta\cos\Psi & \cos\theta\sin\Psi & -\sin\theta \\ \sin\theta\sin\theta\cos\phi & \sin\Psi\sin\theta\sin\phi & \sin\phi\cos\theta \\ -\sin\Psi\cos\phi & +\cos\Psi\cos\phi & \\ \cos\Psi\cos\phi\sin\theta & \sin\Psi\cos\phi\sin & \cos\phi\cos\theta \\ +\sin\Psi\sin\phi & -\cos\Psi\sin\theta & \end{bmatrix} \quad (8.5)$$

Since the matrix D was formed from the multiplication of orthogonal matrices, it too is orthogonal and its inverse is simply its transpose. The inverse is used, of course, to relate the components of (x_I, y_I, z_I) to those of (x, y, z) . From Eq. (8.5), the inverse is

$$D^{-1} = \begin{bmatrix} \cos\theta\cos\Psi & \sin\phi\sin\theta\cos\Psi & \cos\Psi\cos\phi\sin\theta \\ & -\sin\Psi\cos\phi & \sin\Psi\sin\phi \\ \cos\theta\sin\Psi & \sin\Psi\sin\theta\sin\phi & \sin\Psi\cos\phi\sin\theta \\ & +\cos\Psi\cos\phi & -\cos\Psi\sin\phi \\ -\sin\theta & \sin\phi\cos\theta & \cos\phi\cos\theta \end{bmatrix} \quad (8.6)$$

Subsequently we shall use a somewhat more descriptive notation for the matrices D and D^{-1} . It will be noted that the matrix D is used to transform vector components in the (x_I, y_I, z_I) or inertial frame to the (x, y, z) body frame. The reverse is true for the D^{-1} matrix. Consequently, we shall use the symbol C for the matrix with sub/superscripts with frame "from" as subscript and frame "to" as superscript. Thus, $C_i^b = D$ and $C_b^i = D^{-1}$.

One vector that we shall encounter extensively is the angular velocity vector. The components of the angular velocity vector in the (x, y, z) frame are (p, q, r) . These components can be related to the corresponding components in the (x_I, y_I, z_I) frame by the following:

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = AB \begin{bmatrix} 0 \\ 0 \\ \frac{d\Psi}{dt} \end{bmatrix} + A \begin{bmatrix} 0 \\ \frac{d\theta}{dt} \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{d\phi}{dt} \\ 0 \\ 0 \end{bmatrix} \quad (8.7)$$

where A and B are from Eqs. (8.2) and (8.3)

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\theta & \cos\phi \end{bmatrix} \quad B = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \quad (8.8)$$

and the product AB clearly is by the rules of matrix multiplication.

$$AB = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ \sin\phi\sin\theta & \cos\phi & \sin\phi\cos\theta \\ \cos\phi\sin\theta & -\sin\phi & \cos\phi\cos\theta \end{bmatrix} \quad (8.9)$$

Then the vector (p, q, r) is

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ \sin\phi\sin\theta & \cos\phi & \sin\phi\cos\theta \\ \cos\phi\sin\theta & -\sin\phi & \cos\phi\cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \frac{d\Psi}{dt} \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} 0 \\ \frac{d\theta}{dt} \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{d\phi}{dt} \\ 0 \\ 0 \end{bmatrix} \quad (8.10)$$

or in component form

$$\begin{aligned} p &= -\frac{d\Psi}{dt}(\sin\theta) + \frac{d\phi}{dt} \\ q &= \frac{d\Psi}{dt}(\sin\phi\cos\theta) + \frac{d\theta}{dt}\cos\phi \\ r &= \frac{d\phi}{dt}(\cos\phi\cos\theta) - \sin\theta\frac{d\theta}{dt} \end{aligned} \quad (8.11)$$

and in matrix form

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi\cos\theta \\ 0 & -\sin\phi & \cos\phi\cos\theta \end{bmatrix} \begin{bmatrix} \frac{d\phi}{dt} \\ \frac{d\theta}{dt} \\ \frac{d\Psi}{dt} \end{bmatrix} = M \begin{bmatrix} \frac{d\phi}{dt} \\ \frac{d\theta}{dt} \\ \frac{d\Psi}{dt} \end{bmatrix} \quad (8.12)$$

To find the components of

$$\frac{d\phi}{dt} \quad \frac{d\theta}{dt} \quad \frac{d\Psi}{dt}$$

we could proceed from a geometric argument or we could use the formalism of taking an inverse of the above matrix M . Following the latter course we have as the determinant of M ,

$$\det |M| = \cos\theta \quad (8.13)$$

The adjoint of M is then

$$\text{adj } |M| = \frac{1}{\cos\theta} \begin{bmatrix} \cos\theta & 0 & 0 \\ \sin\phi\sin\theta & \cos\phi\cos\theta & \sin\phi \\ \cos\phi\sin\theta & -\sin\phi\cos\theta & \cos\phi \end{bmatrix} \quad (8.14)$$

Finally the inverse of M is

$$M^{-1} = \frac{1}{\cos\theta} \begin{bmatrix} \cos\theta & \sin\phi\sin\theta & \cos\phi\sin\theta \\ 0 & \cos\phi\cos\theta & -\sin\phi\cos\theta \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \quad (8.15)$$

Now the vector components of the angular velocity vector in the (x_I, y_I, z_I) frame may be written as

$$\begin{aligned} \frac{d\phi}{dt} &= p + q\sin\phi\tan\theta + r\cos\phi\tan\theta \\ \frac{d\theta}{dt} &= q\cos\phi - r\sin\phi \\ \frac{d\Psi}{dt} &= q\sin\phi\sec\theta + r\cos\phi\sec\theta \end{aligned} \quad (8.16)$$

The argument often leveled against the use of Euler angles is that there are singularities in two of the rates of the Euler angles. In the scheme we have used, the guilty angles are the roll angle ϕ and yaw angle Ψ . These rates may be written as

$$\frac{d\phi}{dt} = p + \frac{q\sin\phi\sin\theta + r\cos\phi\sin\theta}{\cos\theta} \quad (8.17a)$$

$$\frac{d\Psi}{dt} = \frac{q\sin\phi + r\cos\phi}{\cos\theta} \quad (8.17b)$$

The singularities occur in both roll and yaw when $\theta = \pi/2$. We might have suspected a problem when we noticed that $\det |M|$ [of Eq. (8.13)] is zero at $\theta = \pi/2$ [consequently the matrix M [Eq. (8.12)] will not invert when $\theta = \pi/2$]. However, the singularities of Eqs. (8.17) are removable using L'Hospital's Rule and we are able to obtain expressions that are finite when θ is 90 deg. The practical implication of all this is that as θ approaches 90 deg, the numerators and denominators of Eqs. (8.17) become smaller and smaller. Roundoff errors ultimately make Eqs. (8.17) useless well before $\theta = \pi/2$. Equations (8.17) must be replaced by other expressions from which the singularity has been removed.* The programmer must then decide at some point to switch from Eqs. (8.17) to the new expressions. Because this point is seldom included in textbooks it will now be outlined.

*The author is grateful to Dr. G. T. Schmidt of Draper Laboratory for providing the outline of this method.

First, set θ equal to $\pi/2$ in Eq. (8.11); we then get

$$p = -\frac{d\Psi}{dt} + \frac{d\phi}{dt} \quad q = \frac{d\theta}{dt}(\cos\theta) \quad r = -\frac{d\theta}{dt}(\sin\theta) \quad (8.18)$$

Now let us look at $d\Psi/dt$:

$$\frac{d\Psi}{dt} = \frac{q \sin\phi + r \cos\phi}{\cos\theta} \quad (8.19)$$

or using the above expressions in Eqs. (8.17):

$$\frac{d\Psi}{dt} = \frac{(d\theta/dt)\cos\phi\sin\theta - (d\theta/dt)\sin\theta\cos\phi}{\cos\theta} \bigg|_{\theta=\pi/2} = \frac{0}{0} \quad (8.20)$$

Similarly using Eqs. (8.17) and (8.18) $d\theta/dt$ becomes

$$\frac{d\theta}{dt} = p + \frac{(d\theta/dt)\sin\phi\cos\phi\sin\theta - \cos\phi\sin\theta\sin\phi(d\theta/dt)}{\cos\theta} \bigg|_{\theta=\pi/2} = p + \frac{0}{0} \quad (8.21)$$

Clearly we have indeterminate forms in the expressions for $d\Psi/dt$ and $d\theta/dt$. Let us look first at $d\Psi/dt$:

$$\frac{d\Psi}{dt} = \frac{q \sin\phi + r \cos\phi}{\cos\theta} \quad (8.22)$$

Using L'Hospital's rule we get {Note that we have made use of the derivative substitution: $d/d\theta(\) = (1/\theta)[d(\)/dt]$.

$$\begin{aligned} \frac{d\Psi}{dt} &= \lim_{\theta \rightarrow \pi/2} \left\{ \frac{(d/d\theta)(q \sin\theta + r \cos\theta)}{(d/d\theta)(\cos\theta)} \right\} \\ &= \lim_{\theta \rightarrow \pi/2} \left\{ \frac{1}{\theta} \left[\frac{(d/dt)(q \sin\phi + r \cos\phi)}{-\sin\phi} \right] \right\} \\ \frac{d\Psi}{dt} &= -\frac{1}{\theta} [\dot{q} \sin\phi + q(\cos\phi)\dot{\phi} + \dot{r} \cos\phi - r(\sin\phi)\dot{\phi}] \bigg|_{\theta=\pi/2} \quad (8.23) \end{aligned}$$

Now using Eq. (8.18) Eqs. (8.23) become,

$$\begin{aligned} \frac{d\Psi}{dt} &= \lim_{\theta \rightarrow \pi/2} - \left\{ \frac{1}{\theta} (\dot{q} \sin\phi + \theta \dot{\phi} \cos^2\phi + \dot{r} \cos\phi + \theta \dot{\phi} \sin^2\phi) \right\} \\ \frac{d\Psi}{dt} &= \lim_{\theta \rightarrow \pi/2} - \left\{ \frac{1}{\theta} [\dot{q} \sin\phi + \theta \dot{\phi} \cos\phi + \dot{r} \cos\phi] \right\} \quad (8.24) \end{aligned}$$

Again from Eqs. (8.18) we may write

$$\dot{\phi} = \dot{\Psi} + p \quad (8.25)$$

Equation (8.24) used in the above expression gives

$$\begin{aligned}\dot{\phi} &= p - \frac{1}{\theta} \{ \dot{q} \sin \phi + \dot{\theta} \dot{\phi} + \dot{r} \cos \phi \} \\ 2\dot{\phi} &= p - \frac{1}{\theta} \{ \dot{q} \sin \phi + \dot{r} \cos \phi \} \\ \dot{\phi} &= \frac{p}{2} - \frac{1}{2\theta} \{ \dot{q} \sin \phi + \dot{r} \cos \phi \}\end{aligned}\quad (8.26a)$$

Now from the middle equation of Eq. (8.16)

$$\dot{\phi} = \frac{p}{2} - \frac{1}{2} \left\{ \frac{\dot{q} \sin \phi + \dot{r} \cos \phi}{q \cos \phi - r \sin \phi} \right\} \quad (8.26b)$$

Rewriting Eq. (8.18) or Eq. (8.25) slightly we get

$$\dot{\Psi} = \dot{\phi} - p \quad (8.27)$$

Equation (8.24) then becomes

$$\dot{\Psi} = -\frac{p}{2} - \frac{1}{2} \left\{ \frac{\dot{q} \sin \phi + \dot{r} \cos \phi}{q \cos \phi + r \sin \phi} \right\} \quad (8.28)$$

Now the second and third equation of Eqs. (8.18) gives us

$$\begin{aligned}\tan \phi &= -(r/q) \\ \sin \phi &= \pm r / \sqrt{r^2 + q^2} \quad \cos \phi = \pm q / \sqrt{r^2 + q^2}\end{aligned}\quad (8.29)$$

Now using Eqs. (8.16) and the above expressions we get

$$\begin{aligned}\dot{\theta} &= q \cos \phi - r \sin \phi \\ \dot{\theta} &= (q^2 / \sqrt{q^2 + r^2}) - (-r^2 \sqrt{q^2 + r^2}) = \pm \sqrt{r^2 + q^2} \frac{1}{2}\end{aligned}\quad (8.30)$$

We must emphasize that the above expression is valid at (or near) $\theta = \pi/2$ since we used Eq. (8.18) in its formulation; Eq. (8.18) is of course subjected to the restriction of being valid only at (or near) $\theta = \pi/2$. It follows from Eq. (8.28) that

$$\begin{aligned}\dot{\Psi} &= -\frac{p}{2} - \frac{1}{2} \left[\frac{+1}{(q^2 + r^2)^{1/2}} \right] \left[\frac{\dot{r}q}{(q^2 + r^2)^{1/2}} - \frac{\dot{q}r}{(q^2 + r^2)^{1/2}} \right] \\ \dot{\Psi} &= -\frac{p}{2} - \frac{1}{2} \frac{\dot{r}q - \dot{q}r}{q^2 + r^2}\end{aligned}\quad (8.31)$$

The corresponding expression for $d\phi/dt$ quickly follows:

$$\dot{\phi} = \frac{1}{2} \left[p - \frac{\dot{r}q - q\dot{r}}{q^2 + r^2} \right] \quad (8.32)$$

The expressions in Eqs. (8.31) and (8.32) are valid at (or near) 90 deg only. When away from the 90 deg region then we should use Eqs. (8.16). Actually this is not the whole story; in the expression for $d\theta/dt$ in Eq. (8.30) we must use the $\pm$ sign depending on whether θ is increasing (+) or decreasing (−) on its passage through 90 deg.

8.1.2 Euler Angles: Scheme 2

We can take a brief look at a second scheme of Euler angles. In the past this second scheme has been applied extensively to top dynamics.¹ If an RV is a body of revolution and we wish to describe the interaction of spin rate and coning motion, we shall find this Euler representation to be quite useful.

As in the previously described scheme, the Euler matrix [Eq. (8.5)] was formed as the product of three matrices, each representing a successive rotation about one of three axes. In Fig. 8.4 rotation is about the x axis through the angle Ψ , about the y axis through the angle θ , and a final rotation about the x axis (again) through the angle ϕ . The composite rotation is given in Fig. 8.4.

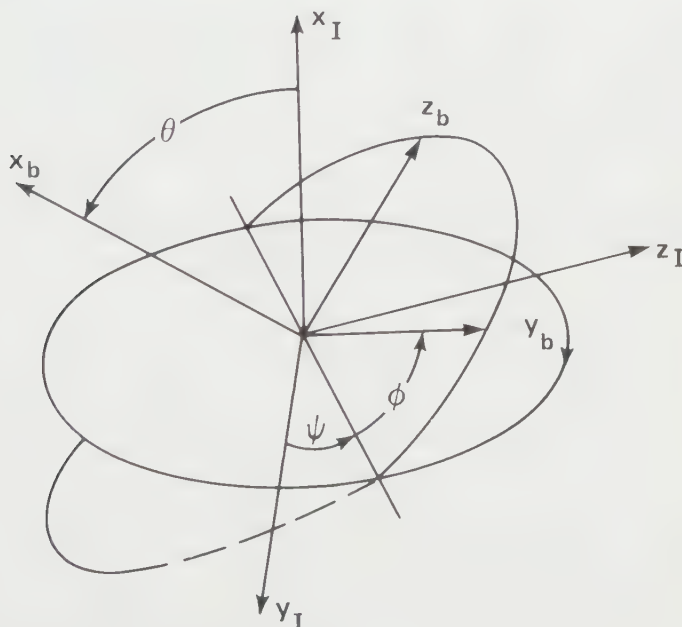
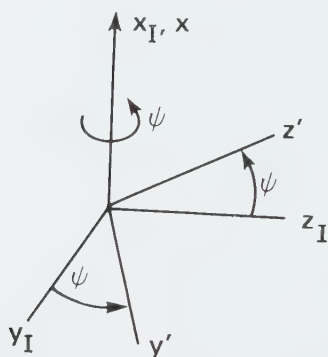
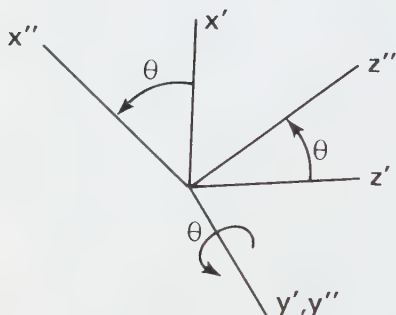


Fig. 8.4 Euler angles: scheme 2.

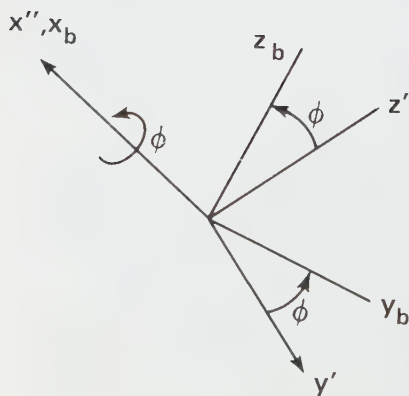
In Fig. 8.5 we have illustrated the three successive rotations; i.e., $C \rightarrow B \rightarrow A$, accompanying each sketch by the corresponding matrix



$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\Psi & \sin\Psi \\ 0 & -\sin\Psi & \cos\Psi \end{bmatrix} \quad (8.33a)$$



$$B = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \quad (8.33b)$$



$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \quad (8.33c)$$

Fig. 8.5 Rotation sequence for second Euler scheme.

The composite matrix D , representing the transformation from the inertial frame to the body frame, is the product of ABC or

$$D = C_i^b$$

$$= \begin{bmatrix} \cos\theta & \sin\theta\sin\Psi & -\sin\theta\cos\Psi \\ \sin\phi\sin\theta & \cos\phi\cos\Psi - \sin\phi\cos\theta\sin\Psi & \cos\phi\sin\Psi + \sin\phi\cos\theta\cos\Psi \\ \cos\phi\sin\theta & -\sin\phi\cos\Psi - \cos\theta\sin\Psi\cos\phi & -\sin\phi\sin\Psi + \cos\phi\cos\theta\cos\Psi \end{bmatrix} \quad (8.34)$$

Equation (8.34) corresponds to the matrix D given in Eq. (8.5). The inverse is easily written since it is the transpose of D .

$$D^{-1} = C_b^i$$

$$= \begin{bmatrix} \cos\theta & \sin\phi\sin\theta & \cos\phi\sin\theta \\ \sin\theta\sin\Psi & \cos\phi\cos\Psi - \sin\phi\cos\theta\sin\Psi & -\sin\phi\cos\Psi - \cos\theta\sin\Psi\cos\theta \\ -\sin\theta\cos\Psi & \cos\phi\sin\Psi + \sin\phi\cos\theta\cos\Psi & -\sin\phi\sin\Psi + \cos\phi\cos\theta\cos\Psi \end{bmatrix} \quad (8.35)$$

Next we must obtain expressions relating the angular rates in body axes; i.e., (p, q, r) to the derivatives of the Euler angles (Ψ, θ, ϕ) . Following a development along the lines of Eqs. (8.7), we may write

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = AB \begin{bmatrix} \frac{d\Psi}{dt} \\ 0 \\ 0 \end{bmatrix} + A \begin{bmatrix} 0 \\ \frac{d\theta}{dt} \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{d\phi}{dt} \\ 0 \\ 0 \end{bmatrix} \quad (8.36)$$

Inserting the matrix product AB and the matrix A as appropriate we get

$$p = \frac{d\Psi}{dt} \cos\theta + \frac{d\phi}{dt}$$

$$q = \frac{d\Psi}{dt} \sin\phi\sin\theta + \frac{d\theta}{dt} \cos\phi$$

$$r = \frac{d\phi}{dt} \cos\phi\sin\theta - \frac{d\theta}{dt} \sin\phi \quad (8.37a)$$

This Euler angle scheme is useful for analyzing the motion of an RV model in a wind-tunnel dynamic test.² In such a test the model has three degrees of freedom (roll, pitch, yaw). The x_I axis can be regarded as the direction of the velocity vector, θ measures the model pitch in the wind plane (plane defined by the x -body axis and the x -inertial or wind axis), and Ψ is the measure of the azimuth of this wind plane.

For reference we can invert Eq. (8.37a) to obtain expressions for the derivatives of the Euler angles (Ψ, θ, ϕ) in terms of (p, q, r) :

$$\begin{aligned}\frac{d\Psi}{dt} &= \frac{(q \sin\phi + r \cos\phi)}{\sin\theta} \\ \frac{d\theta}{dt} &= q \cos\phi - r \sin\phi \\ \frac{d\phi}{dt} &= p - q \sin\phi \cot\theta - r \cos\phi \cot\theta\end{aligned}\quad (8.37b)$$

8.2 Directional Cosine Matrices

Consider a vector $\mathbf{r}$ which has components (x, y, z) in an unprimed coordinate frame and components (x', y', z') in a primed coordinate frame. Assume both frames have a common origin although in general none of the axes of the one frame is coincident with the axes of the other frame. The vector $\mathbf{r}$ in the unprimed frame may be written as

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad (8.38a)$$

and in the primed frame

$$\mathbf{r} = x'\mathbf{i}' + y'\mathbf{j}' + z'\mathbf{k}' \quad (8.38b)$$

We now pose the problem of expressing the components (x', y', z') in terms of the components (x, y, z) (Fig. 8.6). We might expect such relationships to be transcendental algebraic equations. However, since one frame may not only be rotated but rotating relative to the other, we can anticipate differential equations as well. Let us assume we are looking at the instantaneous rotation of one frame with respect to the other.

It is possible to express the unit vectors in the primed system in terms of those in the unprimed system as,

$$\begin{aligned}\mathbf{i}' &= (\mathbf{i}' \cdot \mathbf{i})\mathbf{i} + (\mathbf{i}' \cdot \mathbf{j})\mathbf{j} + (\mathbf{i}' \cdot \mathbf{k})\mathbf{k} \\ \mathbf{j}' &= (\mathbf{j}' \cdot \mathbf{i})\mathbf{i} + (\mathbf{j}' \cdot \mathbf{j})\mathbf{j} + (\mathbf{j}' \cdot \mathbf{k})\mathbf{k} \\ \mathbf{k}' &= (\mathbf{k}' \cdot \mathbf{i})\mathbf{i} + (\mathbf{k}' \cdot \mathbf{j})\mathbf{j} + (\mathbf{k}' \cdot \mathbf{k})\mathbf{k}\end{aligned}\quad (8.39)$$

Since the unit vectors have unit magnitude, the dot (inner) product of any two is simply the cosine of the angle between them

$$\begin{aligned}\mathbf{i}' \cdot \mathbf{i} &= \cos(\mathbf{i}', \mathbf{i}) \\ \mathbf{i}' \cdot \mathbf{j} &= \cos(\mathbf{i}', \mathbf{j}) \\ \mathbf{i}' \cdot \mathbf{k} &= \cos(\mathbf{i}', \mathbf{k})\end{aligned}\quad (8.40)$$

The same process can be applied to unit vectors $\mathbf{j}'$ and $\mathbf{k}'$. The result would be nine cosines, each one being the cosine of the angle between one axis in

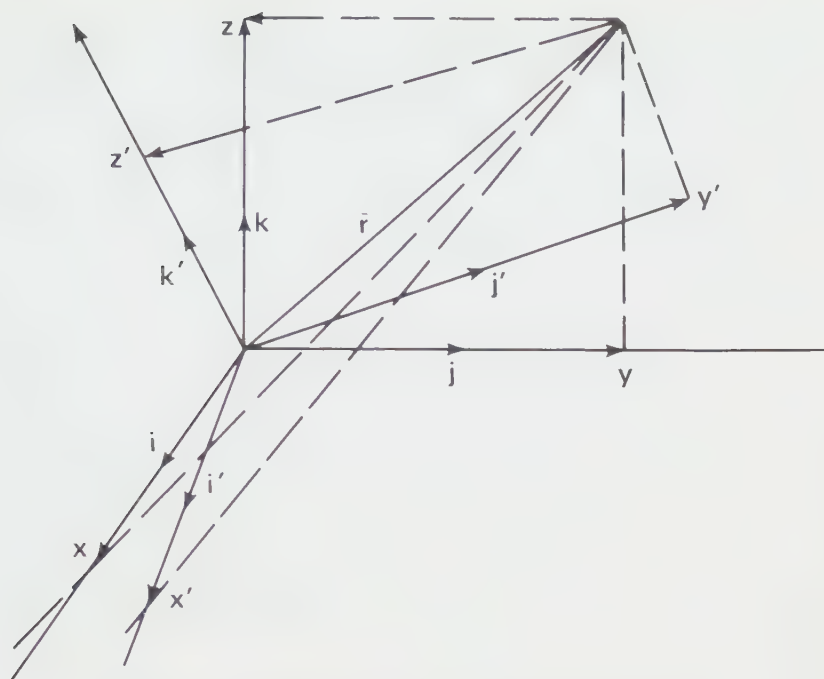


Fig. 8.6 Components of a vector in two Cartesian frames.

one system and one axis in the other system. A shorthand that might be used is to assign numbers 1, 2, 3 to the axes x, y, z . A double subscripted symbol can be used, with the first subscript referring to the primed system and the second to the unprimed system, as for example,

$$C_{23} = \cos(\mathbf{j}', \mathbf{k}) \quad (8.41)$$

Equation (8.39) then becomes

$$\begin{aligned} \mathbf{i}' &= C_{11}\mathbf{i} + C_{12}\mathbf{j} + C_{13}\mathbf{k} \\ \mathbf{j}' &= C_{21}\mathbf{i} + C_{22}\mathbf{j} + C_{23}\mathbf{k} \\ \mathbf{k}' &= C_{31}\mathbf{i} + C_{32}\mathbf{j} + C_{33}\mathbf{k} \end{aligned} \quad (8.42)$$

With the symbol C_{ij} retaining the same meaning; i.e., first subscript referring to the primed system and the second to the unprimed system, the inverse of Eq. (8.37) might be written as

$$\begin{aligned} \mathbf{i} &= C_{11}\mathbf{i}' + C_{21}\mathbf{j}' + C_{31}\mathbf{k}' \\ \mathbf{j} &= C_{12}\mathbf{i}' + C_{22}\mathbf{j}' + C_{32}\mathbf{k}' \\ \mathbf{k} &= C_{13}\mathbf{i}' + C_{23}\mathbf{j}' + C_{33}\mathbf{k}' \end{aligned} \quad (8.43)$$

It is now possible to express the components of $\mathbf{r}$ in the (x', y', z') system in terms of the components in the (x, y, z) system using Eqs. (8.39) and (8.42)

$$\begin{aligned}x' &= \mathbf{r} \cdot \mathbf{i}' = C_{11}(\mathbf{r} \cdot \mathbf{i}) + C_{12}(\mathbf{r} \cdot \mathbf{j}) + C_{13}(\mathbf{r} \cdot \mathbf{k}) \\y' &= \mathbf{r} \cdot \mathbf{j}' = C_{21}(\mathbf{r} \cdot \mathbf{i}) + C_{22}(\mathbf{r} \cdot \mathbf{j}) + C_{23}(\mathbf{r} \cdot \mathbf{k}) \\z' &= \mathbf{r} \cdot \mathbf{k}' = C_{31}(\mathbf{r} \cdot \mathbf{i}) + C_{32}(\mathbf{r} \cdot \mathbf{j}) + C_{33}(\mathbf{r} \cdot \mathbf{k})\end{aligned}\quad (8.44)$$

Since $\mathbf{r} \cdot \mathbf{i} = x$, $\mathbf{r} \cdot \mathbf{j} = y$, $\mathbf{r} \cdot \mathbf{k} = z$, the above set becomes

$$\begin{aligned}x' &= C_{11}x + C_{12}y + C_{13}z \\y' &= C_{21}x + C_{22}y + C_{23}z \\z' &= C_{31}x + C_{32}y + C_{33}z\end{aligned}\quad (8.45a)$$

In matrix form Eq. (8.45a) becomes

$$(\mathbf{r})' = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = C \begin{bmatrix} x \\ y \\ z \end{bmatrix} = [C_{ij}] \begin{bmatrix} x \\ y \\ z \end{bmatrix}\quad (8.45b)$$

The nine elements in matrix C , $C_{11} \dots C_{33}$ are called direction cosines and the matrix C is called the "direction cosine matrix" or DCM. According to Eq. (8.43) the matrix C has an inverse

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = C^{-1} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}\quad (8.46a)$$

where

$$C^{-1} = \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix}\quad (8.46b)$$

The DCM provides a systematic means of transferring the components of a vector from one frame to a second frame which is rotated relative to the first frame. However, this statement requires some clarification. A vector is changed only by changing its magnitude or direction. Strictly speaking, a vector remains unchanged by changes in coordinate systems. However, the numerical value of the components will change as the frame in which the vector is being resolved rotates or if the vector is being resolved in two distinct frames. Thus Eqs. (8.45) and (8.46) relate components resolved in one frame to components resolved in a second frame. We have used primed and unprimed frames. We might just as well have used a and b to label the two frames.

$$\mathbf{r}^b = C_a^b \mathbf{r}^a\quad (8.47)$$

The symbolism of Eq. (8.47) was introduced briefly in Eqs. (8.5) and (8.34); it will be used extensively throughout the sequel. The symbol C_a^b means the operation of transferring the components from frame a (subscript) to frame b (superscript). The matrix for carrying out this operation is a DCM. For operations in three-space, the DCM is a three-by-three matrix, and, as will be seen, there are certain implied algebraic relationships among the elements of the DCM. Notice also that the vectors that are being transformed are represented as column matrices. Equation (8.47) is shorthand for

$$\mathbf{r}^b = \begin{bmatrix} x^b \\ y^b \\ z^b \end{bmatrix} = \begin{bmatrix} C_{11}^a & C_{12}^a & C_{13}^a \\ C_{21}^a & C_{22}^a & C_{23}^a \\ C_{31}^a & C_{32}^a & C_{33}^a \end{bmatrix} \begin{bmatrix} x^a \\ y^a \\ z^a \end{bmatrix} = C_a^b \mathbf{r}^a \quad (8.48)$$

C_{32} , for example, is the cosine of the angle between the z axis in the b frame and the y axis in the a frame, i.e., $\cos(\mathbf{k}^b, \mathbf{j}^a) = \mathbf{k}^b \cdot \mathbf{j}^a$.

Let us return to Eq. (8.47). We noted in Eq. (8.43) that if the components in frame a , say, could be expressed in terms of components in frame b , we could expect an inverse relationship, i.e., expressions of components in frame b in terms of components in frame a . From Eq. (8.47) we might expect

$$\mathbf{r}^a = C_b^a \mathbf{r}^b = C_b^a C_a^b \mathbf{r}^a \quad (8.49a)$$

Clearly then

$$C_b^a C_a^b = [1] \quad (8.49b)$$

or that C_a^b is the inverse of C_b^a . Replacing the matrices by their elements we would get

$$\begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (8.50)$$

Thus we obtain six redundancy equations among the nine elements of the DCM.

$$C_{11}^2 + C_{21}^2 + C_{31}^2 = 1$$

$$C_{12}^2 + C_{22}^2 + C_{32}^2 = 1$$

$$C_{13}^2 + C_{23}^2 + C_{33}^2 = 1$$

$$C_{11}C_{12} + C_{21}C_{22} + C_{31}C_{32} = 0$$

$$C_{11}C_{13} + C_{21}C_{23} + C_{31}C_{33} = 0$$

$$C_{12}C_{13} + C_{22}C_{23} + C_{32}C_{33} = 0 \quad (8.51)$$

The existence of relationships among the elements (or “redundancy”) is not surprising. It means that only three elements are independent. In the Euler angle development of the preceding section, three independent parameters were required: $\{\theta, \phi, \Psi\}$. Thus it takes only three separate or independent quantities to establish the orientation of an axis system. If four or more parameters are used, then there must be one or more interrelationships among the parameters. If there are n parameters then there must be $n-3$ redundancy relationships. However, there are alternate forms to Eqs. (8.51). Another set is based on the orthogonality of the frames, i.e., $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are all perpendicular to each other. It then follows that

$$\begin{aligned} \mathbf{i}' \times \mathbf{j}' &= \mathbf{k}' & \mathbf{j}' \times \mathbf{k}' &= \mathbf{i}' & \mathbf{k}' \times \mathbf{i}' &= \mathbf{j}' \\ \mathbf{i} \times \mathbf{j} &= \mathbf{k} & \mathbf{j} \times \mathbf{k} &= \mathbf{i} & \mathbf{k} \times \mathbf{i} &= \mathbf{j} \end{aligned} \quad (8.52)$$

First we shall give the redundancy relationships and then justify some of them.

$$\begin{aligned} C_{11} &= C_{22}C_{33} - C_{23}C_{32} \\ C_{21} &= C_{13}C_{32} - C_{12}C_{33} \\ C_{31} &= C_{12}C_{23} - C_{13}C_{22} \\ C_{12} &= C_{23}C_{31} - C_{21}C_{33} \\ C_{22} &= C_{11}C_{33} - C_{13}C_{31} \\ C_{32} &= C_{13}C_{21} - C_{11}C_{23} \\ C_{13} &= C_{21}C_{32} - C_{22}C_{31} \\ C_{23} &= C_{12}C_{31} - C_{11}C_{32} \\ C_{33} &= C_{11}C_{22} - C_{12}C_{21} \end{aligned} \quad (8.53)$$

To justify C_{31}, C_{32}, C_{33} , let us write

$$\mathbf{k}' = \mathbf{i}' \times \mathbf{j}'$$

From Eqs. (8.40) we have

$$(C_{31}\mathbf{i} + C_{32}\mathbf{j} + C_{33}\mathbf{k}) = (C_{11}\mathbf{i} + C_{12}\mathbf{j} + C_{13}\mathbf{k}) \times (C_{21}\mathbf{i} + C_{22}\mathbf{j} + C_{23}\mathbf{k}) \quad (8.54)$$

Carrying out the cross product operation on the right and then equating coefficients we justify three redundancy equations of Eqs. (8.53).

Example 8.1: Use of Redundancy Relationships

We can now use a fairly simple problem to illustrate the use of the redundancy equations. Suppose we have the DCM given below. Our task is to fill in the unknown elements indicated by the x 's.

$$\begin{bmatrix} 0.3473 & 0.9224 & -0.1692 \\ x & 0.2411 & -0.4504 \\ x & x & 0.8767 \end{bmatrix} = C$$

To compute the missing elements we make use of the redundancy relationships of Eqs. (8.53)

$$C_{33} = C_{11}C_{22} - C_{12}C_{21}$$

or

$$C_{21} = \frac{C_{33} - C_{11}C_{22}}{-C_{12}} = \frac{(0.8767) - (0.3473)(0.2411)}{-0.9224}$$

$$C_{21} = -0.8597$$

$$\begin{aligned} C_{32} &= C_{11}C_{21} - C_{12}C_{23} \\ &= (-0.1692)(-0.8597) - (0.3473)(-0.4504) \\ &= 0.3019 \end{aligned}$$

$$\begin{aligned} C_{31} &= C_{12}C_{23} - C_{13}C_{22} \\ &= (0.9224)(-0.4504) - (-0.1692)(0.2411) \\ &= -0.3747 \end{aligned}$$

Example 8.2: Relationship between Euler Angles and the Directional Cosine Matrix

Equation (8.48) shows that the DCM is a nine-element square matrix. Of course, so is the Euler angle matrix developed in Eq. (8.5). If both matrices (the DCM and the Euler) are used to relate vector components in two distinct frames, the application of either matrix will give identical results. The distinction between the Euler and DCM formalism is one of element definition: in the Euler angle matrix, the elements are expressed as sums and products of sine/cosine functions of the Euler angles; in the DCM the elements are defined as the cosine of the angle between each of the axes in one frame and each of the axes in the other frame, taken two at a time and according to some orderly sequence. In order to relate the Euler angle matrix to the DCM, let us assume that a microprocessor dedicated to the inertial navigation requirements of the Space Shuttle can provide us with a continuously updated DCM between a body-axis fixed to the Shuttle's fuselage $(x^b, y^b, z^b)^T = r^b$ and a tangent Earth-frame (up, east, north). We shall designate this tangent Earth-frame as (N) or the navigation frame. A display of the numerical values of the DCM to the pilot would be meaningless; however, if these elements could be converted into Euler angles, $\Psi = H$, heading, $\theta = E$, elevation or pitch, and $\phi = B$, bank, then such a display can be quite meaningful to the pilot.

Letting S = sine and C = cosine we may write for the D^{-1} matrix of Eq. (8.6):

$$C_b^N = \begin{bmatrix} CH\ CE & CH\ SE\ SB - SH\ SB & CH\ SE\ CB + SH\ SB \\ SH\ CE & SH\ SE\ SB + CH\ CB & SH\ SE\ CB - CH\ SB \\ -SE & CE\ SB & CE\ CB \end{bmatrix} \quad (8.55)$$

or as a DCM

$$C_b^N = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \quad (8.56)$$

If the Shuttle is using a strapdown navigator, then the utility of the above matrix is to transform specific force components measured primarily in a body-frame into the equivalent components in some other convenient frame, in this case a local Earth or navigation frame as

$$V^N = C_b^N V^b$$

Subsequently we shall have a few more remarks to make concerning inertial navigation. We shall assume for the present that the DCM, C_b^N is continuously available in some accessible on-board memory unit. Our goal is to obtain from these nine elements, the three Euler angles, (H, E, B) . From the above matrix we may write

$$\begin{aligned} E &= -\sin^{-1}(C_{31}) \\ H &= \sin^{-1} \left[\frac{C_{21}}{\cos(E)} \right] = \sin^{-1} \left[\frac{C_{21}}{[1 - (C_{31})^2]^{1/2}} \right] \\ B &= \sin^{-1} \left[\frac{C_{32}}{\cos(E)} \right] = \sin^{-1} \left[\frac{C_{32}}{[1 - (C_{31})^2]^{1/2}} \right] \end{aligned} \quad (8.57)$$

It was pointed out in the first part of this chapter and in Eq. (8.15) that the transformation matrix between H, E, B and (p, q, r) is undefined at $E = \pm 90$ deg. We also showed that this singularity is removable. However, we can sidestep this added complication by restricting the domain of E as

$$-90 < E < 90$$

The above restriction leaves $\cos(E)$ always positive so we do not have a quadrant problem with the elevation angle E . However, we do have a quadrant angle problem with both the heading angle H and the bank angle B . Let

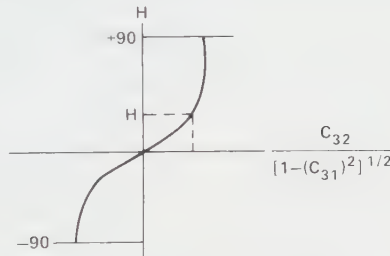
$$C_b^N = \begin{bmatrix} CH & CE_1 & x & x \\ SH & CE_1 & x & x \\ -SE_1 & CE & SB & CE & CB \end{bmatrix} = \begin{bmatrix} C_{11} & x & x \\ C_{21} & x & x \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \quad (8.58)$$

We have shown that E, H, B are all inverse-sine functions of certain elements of the matrix C_b^N . Thus, once the argument to the inverse sine is known we need only a first and fourth quadrant routine for $\sin^{-1}()$.

To account for the quadrants we note that since $\cos(E)$ is always positive, the sign of C_{33} is determined by $\cos(B)$. Since the cosine is positive in the first and fourth quadrants, the $\sin^{-1}()$ routine can be used directly if C_{33} is

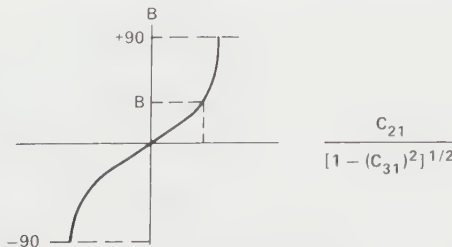
positive; thus $C_{33} > 0$ means go directly to the $\sin^{-1}()$ routine. In other words, since E must be either in the first or fourth quadrant, the correct quadrant will always be found from the sign of C_{31} . If C_{31} is negative, E is in the first quadrant and if C_{31} is positive E will be in the fourth quadrant. If C_{33} is positive, the angle of bank B must be either in the first or fourth quadrant; if the sign of C_{32} is positive, then B must be in the first quadrant; if C_{32} is negative, then B must be in the fourth quadrant. Similar remarks apply to the heading angle H .

Now let us consider the situation when C_{33} is negative. If $C_{33} < 0$, then $\cos(B)$ must be negative and so B must be in either the second or third quadrant. C_{32} must then be examined. If C_{32} is negative, then $\sin(B)$ must be negative; and if C_{32} is positive, then $\sin(B)$ must be positive. Positive $\sin(B)$ means that B must be in the second quadrant (positive sine, negative cosine). If C_{32} is negative, then B must be in the third quadrant by the same line of reasoning. The algorithms for both the bank angle B and heading angle H can then be summarized below in Figs. 8.7 and 8.8.



- (1) $C_{33} > 0$; go directly to subroutine and exit.
- (2) $C_{33} < 0, C_{32} > 0$; go to subroutine, evaluate B , and replace B as $B = 180 - B$.
- (3) $C_{33} < 0, C_{32} < 0$; go to subroutine, evaluate B , and replace B as $B = -180 - B$.

Fig. 8.7 Inverse sine function to obtain the bank angle.



- (1) $C_{11} > 0$; go directly to subroutine and exit.
- (2) $C_{11} < 0, C_{21} > 0$; go to subroutine, evaluate H , and replace H as $H = 180 - H$.
- (3) $C_{11} < 0, C_{21} < 0$; go to subroutine, evaluate H , and replace H as $H = -180 - H$.

Fig. 8.8 Inverse sine function to obtain the heading angle.

A computer program has been written to check out the above algorithms. This program is listed in Table 8.1. In lines 170, 180, 190, the heading, elevation, and bank angles H , E , B are entered. The DCM C_b^N is calculated in lines 250 through 330. From lines 340 through 640 the preceding algorithms are used to extract the Euler angles. The extracted angles are designated by $H1$, $E1$, and $B1$. Comparisons are made between the input angles and the output angles in the PRINT statements.

For a sample output let us concentrate on the bank angle B . Assume that the bank angle is set successively at 45, 135, and 225 deg. It will be noted that in the first two cases the algorithms extract the input; i.e., $B1 = B$. However, for a bank angle of 225 deg, the display would indicate -135 deg, the equivalent angle, but indicating rotation in a negative direction. Thus, a bank angle B of 180 deg indicates inverted flight, obtained by rotating clockwise 180 deg about the x^b axis; a bank angle of -180 deg also indicates inverted flight, but achieved by a counterclockwise rotation. Similarly, the heading angle H is positive for rotations in which the vehicle's nose moves right and negative when vehicle's nose moves left. An additional statement could be added in which negative values of H are added to 360 deg to make the heading angle compatible with the compass rose.

$H = 45.$	$H = 45.$
$E = 22.$	$E = 22.$
$B = 45.$	$B = 225.$
$H = 45.$	$H1 = 45.$
$E = 22.$	$E1 = 22.$
$B = 135.$	$B1 = 135.$
$H1 = 45.$	$H1 = 45.$
$E1 = 22.$	$E1 = 22.$
$B1 = 45.$	$B1 = -135.$

8.3 Derivative of Direction Cosine Matrix

In Eq. (8.42) we related unit vectors in a primed and unprimed frame through the direction cosines $C_{11} \dots C_{33}$. Instead of primed and unprimed frames we shall use a b or a body frame and an i inertial frame. According to the nomenclature developed in the last section, the unit vectors are related as

$$\begin{bmatrix} \mathbf{i}^i \\ \mathbf{j}^i \\ \mathbf{k}^i \end{bmatrix} = C_b^i \begin{bmatrix} \mathbf{i}^b \\ \mathbf{j}^b \\ \mathbf{k}^b \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \begin{bmatrix} \mathbf{i}^b \\ \mathbf{j}^b \\ \mathbf{k}^b \end{bmatrix} \quad (8.59)$$

A unit vector can change only in direction, never in magnitude, so, for example

$$\frac{d\mathbf{i}^b}{dt} = \boldsymbol{\omega}_{ib} \times \mathbf{i}^b \quad (8.60)$$

Table 8.1 Computation of Euler Angle Directional Cosine Matrix

```

00100 REM THIS PROGRAM COMPUTES THE ELEMENTS OF THE DIRECTIONAL
00110 REM COSINE MATRIX FROM A BODY FRAME TO LOCAL EARTH OR
00120 REM NAVIGATIONAL FRAME. ALGORITHMS ARE AVAILABLE TO RECONSTRUCT
00130 REM THE EULER ANGLES FROM THE DCM. COMPARISONS ARE MADE
00140 REM BETWEEN THE INPUT EULER ANGLES AND THE RECONSTRUCTED
00150 REM EULER ANGLES. THE DCM IS PRINTED.
00160 BASE 1
00170 H = 45
00180 E = 22
00190 B = 225
00200 DIM D(3,3)
00210 R = 3.14159/180
00220 H = H*R
00230 E = E*R
00240 B = B*R
00250 D(1,1) = COS(H)*COS(E)
00260 D(1,2) = COS(H)*SIN(E)*SIN(B)-SIN(H)*COS(B)
00270 D(1,3) = COS(H)*SIN(E)*COS(B)+SIN(H)*SIN(B)
00280 D(2,1) = SIN(H)*COS(E)
00290 D(2,2) = SIN(H)*SIN(E)*SIN(B)+COS(H)*COS(B)
00300 D(2,3) = SIN(H)*SIN(E)*COS(B)-COS(H)*SIN(B)
00310 D(3,1) = (-1)*SIN(E)
00320 D(3,2) = COS(E)*SIN(B)
00330 D(3,3) = COS(E)*COS(B)
00340 E1 = ATN((-1)*D(3,1)/(SQR(1-(D(3,1))^2)))
00350 H2 = D(2,1)/SQR(1-((D(3,1))^2))
00360 B2 = D(3,2)/SQR(1-((D(3,1))^2))
00370 GOTO 00430
00380 Y1 = X1/SQR(1-((X1)^2))
00390 Z1 = ATN(Y1)
00400 RETURN
00410 IF D(3,3) > 0 THEN 00470
00420 IF D(3,2) >= 0 THEN 00450
00430 IF D(3,3) >= 0 THEN 00510
00440 X1 = B2
00450 GOSUB 00380
00460 IF D(3,2) < 0 THEN 00490
00470 B1 = 180*R-Z1
00480 GOTO 00540
00490 B1 = (-1)*180*R-Z1
00500 GOTO 00540
00510 X1 = B2
00520 GOSUB 00380
00530 B1 = Z1
00540 IF D(1,1) > 0 THEN 00620
00550 X1 = H2
00560 GOSUB 00380
00570 IF D(2,1) < 0 THEN 00600
00580 H1 = 180*R-Z1
00590 GOTO 00650
00600 H1 = (-1)*180*R-Z1
00610 GOTO 00650
00620 X1 = H2
00630 GOSUB 00380
00640 H1 = Z1
00650 H = H/R
00660 E = E/R
00670 B = B/R
00680 H1 = H1/R
00690 E1 = E1/R
00700 B1 = B1/R
00710 PRINT #H = #,H,#H1 = #,H1
00720 PRINT #E = #,E,#E1 = #,E1
00730 PRINT #B = #,B,#B1 = #,B1
00740 PRINT
00750 PRINT #DIRECTIONAL COSINE MATRIX-BODY TO NAV. FRAME#
00760 PRINT
00770 MAT PRINT D
00780 END
    
```


Equation (8.60) will be recognized as a restatement of Eq. (4.4). In the above expression ω_{ib} is the rotational velocity vector of the b body frame relative to inertial space which we assume to have coordinates in the body frame. Equation (8.60) indicates that the only way the unit vector $\mathbf{i}^b$ can change is normal to itself and the rotational velocity vector. We now take the derivative of the unit vector $\mathbf{i}^i$ as in Eq. (8.59) to get

$$\frac{d\mathbf{i}^i}{dt} = 0 = \dot{C}_{11}\mathbf{i}^b + \dot{C}_{12}\mathbf{j}^b + \dot{C}_{13}\mathbf{k}^b + C_{11}\frac{d\mathbf{i}^b}{dt} + C_{12}\frac{d\mathbf{j}^b}{dt} + C_{13}\frac{d\mathbf{k}^b}{dt} \quad (8.61a)$$

with similar expressions for $\mathbf{j}^i$ and $\mathbf{k}^i$. Using Eq. (8.60) the last three terms in Eq. (8.61a) may be written as

$$\begin{aligned} C_{11}\frac{d\mathbf{i}^b}{dt} + C_{12}\frac{d\mathbf{j}^b}{dt} + C_{13}\frac{d\mathbf{k}^b}{dt} &= C_{11}\omega_{ib} \times \mathbf{i}^b + C_{12}\omega_{ib} \times \mathbf{j}^b + C_{13}\omega_{ib} \times \mathbf{k}^b \\ &= \omega_{ib} \times \mathbf{i}^i \end{aligned} \quad (8.61b)$$

where we have used Eq. (8.42) and replaced the “prime” by i and the “unprimed” by b . Inserting (8.61b) into Eq. (8.61a) we get

$$\begin{aligned} \frac{d\mathbf{i}^i}{dt} &= \dot{C}_{11}\mathbf{i}^b + \dot{C}_{12}\mathbf{j}^b + \dot{C}_{13}\mathbf{k}^b + \omega_{ib} \times \mathbf{i}^i = 0 \\ \frac{d\mathbf{j}^i}{dt} &= \dot{C}_{21}\mathbf{i}^b + \dot{C}_{22}\mathbf{j}^b + \dot{C}_{23}\mathbf{k}^b + \omega_{ib} \times \mathbf{j}^i = 0 \\ \frac{d\mathbf{k}^i}{dt} &= \dot{C}_{31}\mathbf{i}^b + \dot{C}_{32}\mathbf{j}^b + \dot{C}_{33}\mathbf{k}^b + \omega_{ib} \times \mathbf{k}^i = 0 \end{aligned} \quad (8.62)$$

For example, we might write for the one of the cross products:

$$\omega_{ib} \times \mathbf{i}^i = \begin{bmatrix} \mathbf{i}^b & \mathbf{j}^b & \mathbf{k}^b \\ p & q & r \\ C_{11}^1 & C_{12} & C_{13} \end{bmatrix} \quad (8.63)$$

where we have made use of the first of Eqs. (8.42). For the other two cross products the last row will change according to Eq. (8.42). If the cross products are evaluated (by expanding the appropriate determinant) we get three vector equations or nine scalar equations. The first vector equation follows from inserting Eq. (8.63) into Eq. (8.62)

$$0 = [\dot{C}_{11} + qC_{13} - rC_{12}]\mathbf{i}^b + [\dot{C}_{12} + C_{11}r - C_{13}p]\mathbf{j}^b + [\dot{C}_{13} + C_{12}p - qC_{11}]\mathbf{k}^b \quad (8.64)$$

with two other vector equations following from Eq. (8.62).

The nine scalar equations are

$$\begin{aligned}\dot{C}_{11} &= C_{12}r - C_{13}q & \dot{C}_{12} &= C_{13}p - C_{11}r & \dot{C}_{13} &= C_{11}q - C_{12}r \\ \dot{C}_{21} &= C_{22}r - C_{23}q & \dot{C}_{22} &= C_{23}p - C_{21}r & \dot{C}_{23} &= C_{21}q - C_{22}p \\ \dot{C}_{31} &= C_{32}r - C_{33}q & \dot{C}_{32} &= C_{33}p - C_{31}r & \dot{C}_{33} &= C_{31}q - C_{32}p\end{aligned}\quad (8.65)$$

If we were engaged in DCM-updating in a strapdown inertial navigation problem, we would have to calculate C_b^i (body-to-inertial). The elements in this DCM would be time dependent (since the body is presumed rotating relative to inertial space). We would need to solve between four and nine differential equations to get the nine elements of the C_b^i DCM. The reason that we would not necessarily have to solve nine differential equations is that we could use the redundancy equations developed in the last section [Eq. (8.53)], replacing some of the differential equations with algebraic equations. The angular rates $\{p, q, r\}$ in the body frame would come from the strapdown gyros.

Equations (8.65) can be put in a more compact matrix form as

$$\frac{d}{dt}[C_b^i] \equiv \dot{C}_b^i = \begin{bmatrix} \dot{C}_{11} & \dot{C}_{12} & \dot{C}_{13} \\ \dot{C}_{21} & \dot{C}_{22} & \dot{C}_{23} \\ \dot{C}_{31} & \dot{C}_{32} & \dot{C}_{33} \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix}\quad (8.66)$$

The 3×3 angular velocity matrix [second matrix on the far right in Eq. (8.66)] is called the "skew-symmetric" form of the angular velocity vector. Premultiplying any vector (in column form) by the skew-symmetric form of a second vector is the matrix equivalent of the vector operation called the "cross product" of the second vector times the first, e.g., Eq. (8.63). The angular velocity may thus be written in either column or skew-symmetric form. The symbols we shall use for these forms are

$$\omega_{ib}^b = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad \Omega_{ib}^b = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix}\quad (8.67)$$

Equation (8.66) may now be written as

$$\dot{C}_b^i = C_b^i \Omega_{ib}^b\quad (8.68)$$

The symbols used for the elements in Eq. (8.67) are $\{p, q, r\}$. This is aircraft usage and in some documents concerned with RV navigation/dynamics symbols $\{\omega_x, \omega_y, \omega_z\}$ or $\{\omega_1, \omega_2, \omega_3\}$ are frequently appearing alternates.

Let us write as we have done earlier in Eq. (8.49a)

$$C_b^i C_i^b = [I]$$

Taking the derivative of the above product

$$\dot{C}_b^i C_l^b + C_b^i \dot{C}_i^b = 0$$

or using Eq. (8.68)

$$C_b^i \Omega_{ib}^b C_i^b = -C_b^i C_l^b \Omega_{bi}^i = C_b^i C_i^b \Omega_{ib}^i = \Omega_{ib}^i$$

since $\Omega_{ib}^i = -\Omega_{bi}^i$ because b relative to i is of opposite sign from i relative to b . Thus we have

$$\Omega_{ib}^i = C_b^i \Omega_{ib}^b C_i^b \quad (8.69)$$

The above equation indicates that if we have an angular rate of the b frame relative to the i inertial frame (i must be an inertial frame) and this rate vector is resolved in the b frame, we must premultiply by the b to i DCM and postmultiply by the i to b DCM in order to resolve the angular rate of the b frame relative to the i frame in the i frame.

At this point a brief aside might lend some illumination to ideas of rotating frames and the above matrix symbolism. Whether using the column or skew-symmetric form of the angular velocity vector, we note that the superscript identifies the frame in which the vector's coordinates are expressed. The subscripts might be a source of some confusion. Let us consider two noncoincident body frames, a and b . Since both are by definition affixed to the body (which is undergoing angular rotation), we can identify a and b with a general moving frame m . Hence the following symbols indicate exactly the same concept, i.e., the numerical values of the elements in the matrices would be the same in all three cases:

$$\Omega_{ib}^i \quad \Omega_{ia}^i \quad \Omega_{im}^i \quad (8.70)$$

The subscripts indicate that the entity being quantized is the rotational rate of the body, represented by an a , b , or m frame, and the superscript i indicates that this rate is quantized in the i or inertial frame. If a and b are noncoincident body frames, then

$$\Omega_{ib}^b = \Omega_{ia}^b \quad \Omega_{ab}^b = \Omega_{ab}^a = 0 \quad \Omega_{ib}^b \neq \Omega_{ib}^a \quad (8.71)$$

8.3.1 Force and Moment Equations Expressed in Matrix Notation

We can now show that the force and moment equations derived in Secs. 4.4 and 7.1, respectively, may be written in matrix form. First we shall derive the matrix equivalent to the vector Eqs. (4.8) and (4.9).

Let $\mathbf{r}$ be a vector to the center of mass of a body from some origin. The vector $\mathbf{r}$ can be resolved in either the inertial frame as r^i or the body frame r^b . Such vectors are column matrices. The relationship between r^i and r^b is

$$r^i = C_b^i r^b \quad (8.72)$$

Differentiating and using Eq. (8.68) we get the matrix equivalent to Eq. (4.8):

$$\begin{aligned}\dot{\mathbf{r}}^i &= C_b^i \dot{\mathbf{r}}^b + \dot{C}_b^i \mathbf{r}^b = C_b^i \dot{\mathbf{r}}^b + C_b^i \Omega_{ib}^b \mathbf{r}^b \\ \dot{\mathbf{r}}^i &= C_b^i [\dot{\mathbf{r}}^b + \Omega_{ib}^b \mathbf{r}^b]\end{aligned}\quad (8.73)$$

The first term in the brackets is the rate of change of the vector $\mathbf{r}$, with respect to the b frame; the second term accounts for the rotation of the frame. Thus the entire term in the brackets is the rate of change of $\mathbf{r}$ with respect to inertial space quantized in the body frame. Premultiplying by the DCM C_b^i converts to the inertial frame. Differentiating Eq. (8.73) gives the acceleration in matrix form:

$$\ddot{\mathbf{r}}^i = C_b^i [\ddot{\mathbf{r}}^b + 2\Omega_{ib}^b \dot{\mathbf{r}}^b + \dot{\Omega}_{ib}^b \mathbf{r}^b + \Omega_{ib}^b \Omega_{ib}^b \mathbf{r}^b] \quad (8.74)$$

Note that Eq. (8.74) is the matrix equivalent of Eq. (4.9); it is not the equivalent of Eq. (4.6). The four terms in the brackets may be identified as the apparent acceleration or acceleration relative to the moving b frame, the Coriolis acceleration, the tangential acceleration, and the centrifugal acceleration.

Returning to Eq. (8.73) we may let V_0^b be the quantity in the brackets or

$$V_0^b = \dot{\mathbf{r}}^b + \Omega_{ib}^b \mathbf{r}^b$$

and with V_0^i identified with $\dot{\mathbf{r}}^i$, Eq. (8.73) becomes:

$$V_0^i = C_b^i V_0^b \quad (8.75)$$

The vector $\mathbf{V}_0$ is the linear velocity of RV's c.g. relative to inertial space; the superscripts b or i indicate the frames in which the components are expressed or quantized. From Newton's second law we have [paralleling the development of Eq. (4.20)]:

$$\begin{aligned}F^i &= \frac{d}{dt} [M V_0^i] = M \frac{d}{dt} [C_b^i V_0^b] \\ F^i &= M C_b^i [\dot{V}_0^b + \Omega_{ib}^b V_0^b]\end{aligned}\quad (8.76)$$

where M is the mass of the body. The forces F^i are divided into aerodynamic and gravitational contributions, as

$$F^i = F_a^i + F_g^i$$

Equation (8.76) becomes upon premultiplication by $(C_b^i)^T$:

$$(C_b^i)^T (F_a^i + F_g^i) = C_b^i F_a^i + C_b^i F_g^i = M [\dot{V}_0^b + \Omega_{ib}^b V_0^b] \quad (8.77)$$

Since the aerodynamic forces are measured in the b body frame and the gravitational forces are measured in an e - or Earth frame, we have:

$$F_a^b + C_e^b C_i^e F_g^i = F_a^b + C_e^b F_g^e = M [\dot{V}_0^b + \Omega_{ib}^b V_0^b] \quad (8.78)$$

In Sec. 8.1 we identified the matrix D [Eq. (8.5)] as C_i^b . In some aircraft work it is acceptable to consider an Earth-based frame as indistinguishable from an inertial frame. In the present application to RV's we assume that a local Earth's frame is noninertial since an Earth frame would continually rotate with the Earth as the RV moved above the Earth's surface. We shall assume that $D = C_e^b$. The gravitational force F_g^e in the Earth's frame is

$$F_g^e = \begin{bmatrix} 0 \\ 0 \\ Mg \end{bmatrix} \quad (8.79)$$

Using Eq. (8.5) we have

$$C_e^b F_g^e = \begin{bmatrix} -Mg \sin \theta \\ Mg \sin \phi \cos \theta \\ Mg \cos \phi \cos \theta \end{bmatrix} \quad (8.80)$$

The aerodynamic force F_a^b for now will be expressed in column form as

$$F_a^b = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad (8.81)$$

The linear and angular velocities V_0^b , ω_{ib}^b , Ω_{ib}^b may be expressed as

$$V_0^b = \begin{bmatrix} u \\ v \\ w \end{bmatrix} \quad \omega_{ib}^b = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad \Omega_{ib}^b = \begin{bmatrix} 0-r & q \\ r & 0-p \\ -q & p & 0 \end{bmatrix} \quad (8.82)$$

Clearly the insertion of Eqs. (8.80-8.82) into Eq. (8.78) would provide three scalar equations.

Let us now consider the moment equation in matrix form. It was shown leading up to Eq. (7.21) that the applied moment M_i is related to the angular momentum H^i as,

$$M^i = \frac{d}{dt}[H^i] = \frac{d}{dt}[C_b^i H^b]$$

From Eqs. (7.16) we have the matrix equivalent to the dyadic inner product:

$$H^b = I^b \omega_{ib}^b \quad (8.83a)$$

The moment equation then becomes

$$M^i = \dot{C}_b^i I^b \omega_{ib}^b + C_b^i I^b \dot{\omega}_{ib}^b \quad M^i = C_b^i \Omega_{ib}^b I^b \omega_{ib}^b + C_b^i I^b \dot{\omega}_{ib}^b$$

Finally, multiplying both sides by $(C_b^i)^T$ we get

$$M^b = I^b \dot{\omega}_{ib}^b + \Omega_{ib}^b I^b \omega_{ib}^b \quad (8.83b)$$

It will be assumed that all moments are of aerodynamic origin and are most readily expressed in a body frame as

$$M^b = \begin{bmatrix} L \\ M \\ N \end{bmatrix} \quad (8.84)$$

The inertia tensor I^b has been expressed in Eq. (7.15) as a 3×3 matrix; Ω_{ib}^b and ω_{ib}^b have been given in Eq. (8.82) as skew-symmetric and column (vector) forms of the angular velocity. We shall show subsequently that by a similarity transformation we can express the inertia tensor in a second frame provided it is quantized in a first frame and the DCM between the frames is available.

8.4 Practical Application of Direction Cosine Matrices: Strapdown Inertial Navigational System

We can now give a practical application of the use of DCM's. The example we shall choose is a strapdown inertial navigation system. Before getting into the technical details that are unique to the strapdown inertial system, let us point out some requirements that are fairly generally applied to all navigation systems. The purpose of any navigator is to have available and on demand an unambiguous and quantitative description of vehicle position. Our primary interest in this book is with an RV in suborbital flight. Thus, we might accept as a description of vehicle position the Cartesian coordinates of the vehicle's center of mass in some inertial frame. Alternately, vehicle position might be expressed in terms of longitude, latitude, and altitude above the Earth as measured along the local normal to the geoid. For an inertial navigator there are various constraints that do not necessarily apply to other navigators, namely, that this positional information be obtained only by operations performed on measurements made by on-board instruments. In our application, these instruments will be sensitive to linear accelerations and angular velocities relative to inertial space. The "strap-down" modifier applied earlier indicates that the instruments are fixed or "strapped-down" to the vehicle structure.

The inertial navigator's sensing devices are the gyro and accelerometer. A brief discussion of the ring laser gyro (RLG) and the torque-to-balance pendulous accelerometer (TBPA) are discussed briefly in Appendix D. Other instruments are available to measure angular velocity and linear acceleration; however, the RLG and the TBPA represent the best in current technology.

In Fig. 8.9, taken from Ref. 3, we have presented the essential concepts of a strapdown inertial navigator computing in inertial coordinates. We shall see how the descriptive material developed in this chapter concerning DCM's and their derivatives can make the discussion of strapdown systems quite systematic. In Fig. 8.9 we have designated three distinct computational processes. First, we must generate the body-to-inertial frame matrix C_b^i . The reason is, of course, that the specific force is measured in a frame

affixed to the RV while the computations are to be carried out in an inertial frame. One scheme, often applied to aircraft, is to perform the computations in a frame at the Earth's surface, directly beneath the vehicle. This frame moves with the vehicle and is called variously a geographic, local level, or navigation frame. While we shall use this frame in carrying out some transformations, as pointed out above, computations will be performed on acceleration components in an inertial frame. For unmanned vehicles, where it is not necessary to provide displays of heading, elevation, and bank angles, computations are more straightforward if carried out in an inertial frame. Thus, we must compute the C_b^i matrix which transforms body-fixed instrument readings to an inertial frame.

Since the RV is continuously changing attitude, the elements of C_b^i are time varying. The transformation computer must continually update C_b^i in order that a current value is always available to transform f^b into f^i , as

$$f^i = C_b^i f^b \quad (8.85)$$

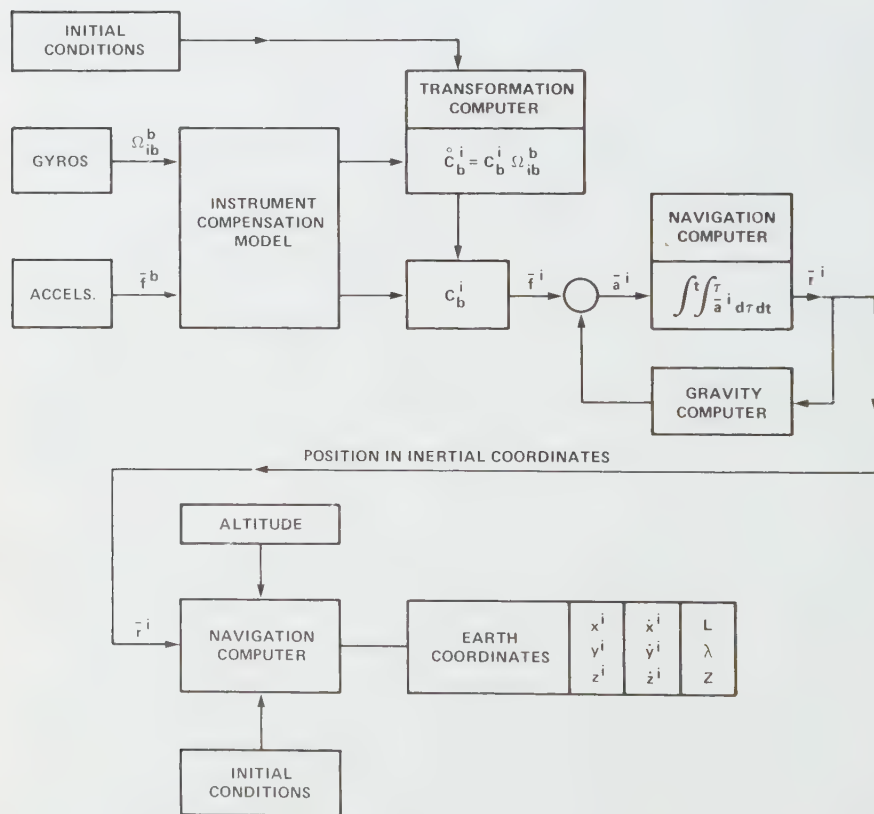


Fig. 8.9 Schematic of a strapdown inertial navigator.

where f^i and f^b are the specific force vector with coordinates in the i frame and b frame, respectively. The second computation indicated in Fig. 8.9 transforms the components of the gravity vector from a "natural" Earth frame to the inertial frame where computation is being carried out. As noted from Eq. (D.9) addition of G^i to f^i provides the acceleration of the body relative to inertial space with the components resolved in the selected inertial frame or,

$$\ddot{r}^i \equiv a^i = f^i + G^i = C_b^i f^b + C_e^i G^e \quad (8.86)$$

The final set of computations take the positional coordinates of the RV relative to the inertial frame and convert them into Earth coordinates such as longitude, latitude, and altitude along the local normal to the geoid. These operations are centered in the navigational computer. Various other computations must also be carried out such as the double integration of a^i to r^i .

We must now specify some frames of reference. In order to keep this discussion within bounds, we shall introduce only a minimum number of frames; in a practical problem (simulation of a working system or perhaps an error analysis) we would require or at least find it convenient to introduce several additional frames. For example, the only inertial frame we shall introduce will be Earth-centered with one axis along the Earth's rotation vector and a second pointing at some fixed star. While this system is useful, it would be difficult to establish at the end of boost. Consequently another

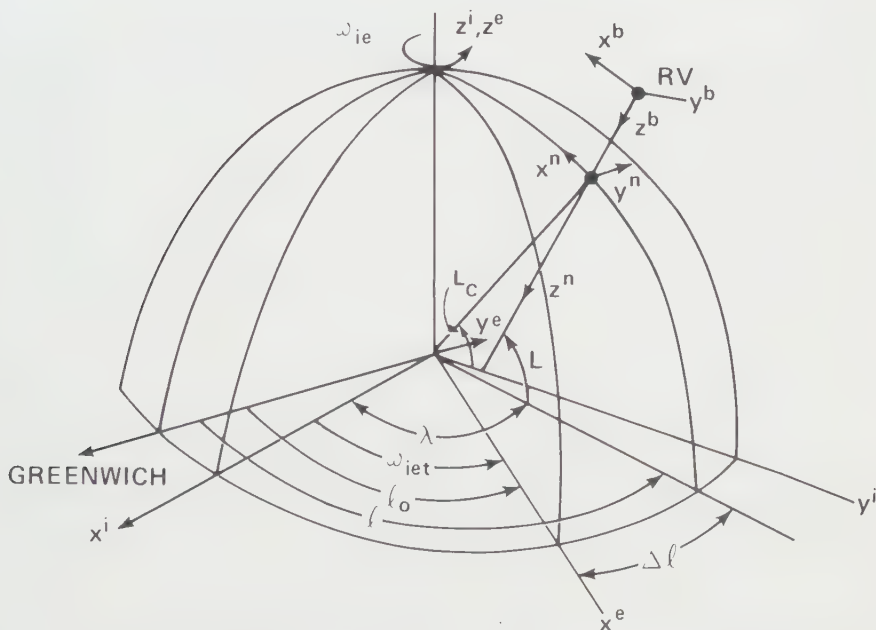


Fig. 8.10 Frame definitions for strapdown navigation.

inertial frame would have to be established from star sightings that are convenient to the RV's position at the end of boost. However, since the DCM's between all inertial frames must be available prior to any computations, all inertial frames are essentially equivalent.

In Fig. 8.10 we have introduced four axis systems; (1) an inertial frame centered at the Earth with the z^i axis along the Earth's rotation vector and the x^i axis pointing to some fixed star; (2) an Earth-fixed frame (x^e, y^e, z^e) with the z^e axis colinear with the z^i axis and the x^e - y^e plane coincident with the x^i - y^i plane at the beginning of navigation; (3) a geographic frame (x^n, y^n, z^n) located at the Earth's surface beneath the RV and oriented such that the z^n axis is along the negative to the local normal to the geoid. x^n is north and y^n is east. The Earth's angular velocity vector is in the x^n - z^n plane. We have used the superscript n since the geographic frame is also a navigation frame as will become apparent; (4) a body or b frame fixed to the RV.

Let us consider briefly the greatest computational demands on the strap-down system: updating the C_b^i DCM. We have already pointed out that C_b^i is continuously changing. From Eq. (8.68) we can express the derivative of C_b^i in terms of the matrix product of C_b^i with the skew-symmetric form of the angular velocity matrix Ω_{ib}^b as

$$\dot{C}_b^i = C_b^i \Omega_{ib}^b \quad (8.87a)$$

In principle at least we can get a new value of C_b^i by integrating Eq. (8.87a).

The integral of Eq. (8.87) can be given as

$$C_b^i(t) = C_b^i(0) e^{(\Omega_{ib}^b t)} \quad (8.87b)$$

The matrix exponential can be written as an infinite series (which converges for any matrix) as

$$\begin{aligned} C_b^i(t) &= C_b^i(0) \sum_{k=0}^{\infty} \frac{(\Omega_{ib}^b)^k}{k!} \\ &= C_b^i(0) \left[I + \Omega_{ib}^b t + \Omega_{ib}^b \Omega_{ib}^b \frac{t^2}{2} + \cdots \right] \end{aligned} \quad (8.88)$$

Unfortunately the above representation is of limited use since Ω_{ib}^b is not a constant, but is continuously changing during the vehicles's trajectory. However, we can regard Ω_{ib}^b to be a constant over a small interval of time, say, Δt . Thus to update C_b^i over a small interval Δt or equivalently from time $t = t_j$ to time $t = t_{j+1}$, where $t_{j+1} = t_j + \Delta t$, we might truncate the above series to the first two terms to give

$$C_b^i(t_{j+1}) = C_b^i(t_j) [I + \Omega_{ib}^b \Delta t] \quad (8.89)$$

We could write Eq. (8.89) in terms of the elements of the various matrices we have used to define frame rotation. With nine elements we shall of

course get nine algebraic equations; let us look at just two, C_{11} and C_{23} .

$$C_{11}(t_{j+1}) = C_{11}(t_j) + C_{12}(t_j)r(t_j)\Delta t - C_{13}(t_j)q(t_j)\Delta t \quad (8.90a)$$

$$C_{23}(t_{j+1}) = C_{23}(t_j) + C_{21}(t_j)q(t_j)\Delta t - C_{12}(t_j)p(t_j)\Delta t \quad (8.90b)$$

Thus, having available the elements of C_b^i and Ω_{ib}^b at time t_j we can in principle update the elements over the interval Δt to obtain the new elements of C_b^i at the new time $t_{j+1} = t_j + \Delta t$. It should be noted that this Euler integration follows directly from Eq. (8.87a) since

$$\dot{C}_{11} = C_{12}r - C_{13}q \quad (8.90c)$$

$$\dot{C}_{23} = C_{21}q - C_{22}p \quad (8.90d)$$

Assuming that the right sides of Eqs. (8.90c) and (8.90d) are constant, integration of Eqs. (8.90c) and (8.90d) gives Eqs. (8.90a) and (8.90b). Much more sophisticated update algorithms are required for a practical strapdown system with flight times of about 2×10^3 s, particularly if body roll rates during Keplerian motion are of the order of 10 rad/s.

It should be clear at this point that the purpose of the gyros in the strapdown inertial navigator is to provide the angular velocity components on the right-hand side of Eqs. (8.90a) and (8.90b). The gyros are not used as a directional indicator, as a mechanical gyro might be in an artificial horizon for aircraft. Nevertheless, the strapdown navigator is capable of providing the H, E, B (heading, elevation, bank) angles discussed in the second example of Sec. 8.3. With C_b^i available it is possible to transform the body measurements of specific force into the inertial frame where the gravity vector is added and the integration of the acceleration vector to the position vector is carried out.

Once we have position available in the inertial frame, we would possibly find it more convenient to locate the RV in an Earth coordinate frame. The navigation frame, as defined previously, has its origin on the Earth's surface directly beneath the RV. We therefore would like to transform position in the inertial frame to the navigational frame. This will require the DCM C_i^n .

Referring to Fig. 8.10 we can show in a fairly straightforward manner that C_i^n is given as follows:

$$C_i^n = \begin{bmatrix} -\sin L \cos \lambda & -\sin L \sin \lambda & \cos L \\ -\sin \lambda & \cos \lambda & 0 \\ -\cos L \cos \lambda & -\cos L \sin \lambda & -\sin L \end{bmatrix} \quad (8.91a)$$

The angular rate of the n frame relative to the i frame may also be arrived at from a consideration of the frame geometries. This angular rate may be presented as a column vector with coordinates in either the n frame or the i frame.

$$\begin{aligned} \omega_{in}^n &= [\dot{\lambda} \cos L, -\dot{L}, -\dot{\lambda} \sin L]^T \\ \omega_{in}^b &= [\dot{L} \sin \lambda, -\dot{L} \cos \lambda, \dot{\lambda}]^T \end{aligned} \quad (8.91b)$$

The skew-symmetric forms of the column vectors follow from Eq. (8.67). It also follows from Fig. 8.10 that

$$l = l_0 + \lambda - \omega_{ie} t \quad (8.92)$$

where l is the terrestrial longitude from Greenwich. l_0 is the initial value of l . λ is the angle from the x^i axis to the meridian containing the RV (sometimes called "celestial longitude"), ω_{ie} is the Earth's rotational rate, and t is the time. Also note that we have been using the geographic latitude L . Geocentric latitude L_c and geographic latitude L are related in Eq. (3.11) where the deviation of the normal D can be expressed in terms of the geographic latitude L and the Earth's ellipticity e [see Eq. (3.15a)].

Since l_0 is known from initial conditions and $\omega_{ie} t$ is available from a clock, we must find λ in order to calculate the longitude l . Since r^i is available (as processed data from the accelerometers), we can use the DCM C_i^n to find L and λ (or L_c and λ) to finally provide use with L and l . The position vector of the RV from the Earth's center in the n frame is $r^n = [0, 0, -(R_E + h)]^T$. The known vector r^i is related to r^n as

$$r^i = C_i^n r^n = (C_i^n)^T r^n \quad (8.93a)$$

or

$$r^i = (R_E + h)[\cos L \cos \lambda, \cos L \sin \lambda, \sin L]^T \quad (8.93b)$$

Since $r^i = (x^i, y^i, z^i)^T$ is known, it appears that we can solve Eqs. (8.93) for the three unknowns: h , L , λ . Mathematically this is acceptable and it will be left as an exercise for the reader to obtain h , L , and λ from x^i , y^i , and z^i . Unfortunately, there is one flaw in this procedure. In inertial navigation, if the altitude computation is based upon inertial measurements (as we seem to be suggesting), the computation loop is mathematically divergent. Thus, we shall assume that RV altitude h is available from some outside source such as an altimeter and that consequently Eq. (8.86b) will be used to calculate only L and λ . The upshot of all this is that not only do we know where we are, but that the DCM C_i^n is available for further computations.

In Eq. (8.86) we pointed out that we must have the gravity vector $\mathbf{G}$ available in an inertial frame to add to the specific force vector $\mathbf{f}$ (also in an inertial frame) before we can obtain the kinematic acceleration $\dot{r}^i$. The gravity vector $\mathbf{G}$ was presented earlier in Eqs. (3.39). In this equation, $\mathbf{G}$ was written in spherical coordinates; the following matrix gives the $\mathbf{G}$ vector in the Earth frame

$$G^e = \begin{bmatrix} G_r \cos L_c - G_\alpha \sin L_c \\ 0 \\ G_r \sin L_c + G_\alpha \cos L_c \end{bmatrix} \quad (8.94)$$

where α is the co-latitude and L_c is the geocentric latitude, i.e., $L_c + \alpha = \pi/2$. Since we are assuming that the latitude L_c is available, numerical values can be assigned to the elements of G^e . Of course we are

also assuming that some type of gravity model is available for on-board computations. Equations (3.39) are an analytic form of a possible model. The vector G^e is easily transformed into inertial space as:

$$G^i = C_e^i G^e \quad (8.95a)$$

where the DCM C_e^i follows from Fig. 8.10 as

$$C_e^i = \begin{bmatrix} \cos\omega_{ie} t & -\sin\omega_{ie} t & 0 \\ \sin\omega_{ie} t & \cos\omega_{ie} t & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (8.95b)$$

If a display of vehicle attitude angles relative to a local Earth frame, i.e., H, E, B is desired, then we would have to generate the matrix C_n^b . This can be accomplished as a single matrix product

$$C_n^b = C_i^b C_n^i = (C_b^i)^T (C_i^n)^T \quad (8.96)$$

Matrix C_b^i is developed from the updating procedure and C_i^b becomes useful when h, L , and λ are available. We have already discussed in the second example in Sec. 8.3 one way in which the angles (H, E, B) might be deduced from the elements of C_n^b .

Another navigational quantity that might be of some interest is the velocity of the vehicle over the Earth's surface. In other words, we wish to find $v^e \equiv \dot{r}^e$, or the rate of change of the position vector in Earth coordinates. Since C_i^n and C_e^i are available we shall be able to find C_e^n . We have

$$v^n = C_e^n \dot{r}^e$$

but

$$r^e = C_i^e r^i$$

taking the derivative of both sides gives,

$$\dot{r}^e = \dot{C}_i^e r^i + C_i^e \dot{r}^i \quad (8.97)$$

Now from Eq. (8.68) we have

$$\dot{C}_e^i = C_e^i \Omega_{ie}^e \quad (8.98)$$

taking the transpose of both sides gives

$$\dot{C}_i^e = [C_e^i \Omega_{ie}^e]^T = (\Omega_{ie}^e)^T (C_e^i)^T \quad (8.99)$$

Now from Eq. (8.67)

$$(\Omega_{ie}^e)^T = -\Omega_{ie}^e$$

so

$$\dot{C}_i^e = -\Omega_{ie}^e C_i^e$$

which leads to

$$\begin{aligned}\dot{r}^e &= C_i^e \dot{r}^i - \Omega_{ie}^e C_i^e r^i \\ &= C_i^e [\dot{r}^i - \Omega_{ie}^e r^i]\end{aligned}\quad (8.100)$$

From Eq. (identity) (8.69) the above becomes

$$\dot{r}^e = C_i^e [\dot{r}^i - \Omega_{ie}^i r^i] \quad (8.101)$$

where $\omega_{ie}^i = (0, 0, \omega_{ie})$ and $(\dot{r}^i, r^i)$ are available from integrations of $\dot{r}^i$. Finally,

$$\begin{aligned}v^n &= C_e^n \dot{r}^e = C_e^n C_i^e [\dot{r}^i - \Omega_{ie}^i r^i] \\ v^n &= C_i^n [r^i - \Omega_{ie}^i r^i]\end{aligned}\quad (8.102)$$

The matrix C_i^n has been given in Eqs. (8.91). The components of v^n are (v_N, v_E, v_D) or speed north, speed east, and speed down (rate of climb/descent).

8.5 Transformation of the Inertia Tensor

Another application of DCM's is to transform the inertia tensor from an axis system in which it is quantized into a second system. In Chapter VII we showed how to change the moments and products of inertia if the RV has a plane of symmetry and if axis rotation is about the normal to this plane. We can now indicate how the numerical values in the inertia tensor change for an arbitrary change in axis system orientation. Suppose we have quantized the inertia tensor in the a body frame. Now what are the numerical values in another body frame (designated here as the b frame)? Assume the DCM from a to b is C_a^b .

From Eq. (7.16) the angular momentum $\mathbf{H}$ in body frames a and b are

$$H^a = I^a \omega_{im}^a = I^a \omega_{ia}^a \quad (8.103a)$$

$$H^b = I^b \omega_{im}^b = I^b \omega_{ib}^b = I^b \omega_{ia}^b \quad (8.103b)$$

where ω_{im}^b is the angular velocity of the body (moving) frame m with respect to the inertial frame i with the coordinates expressed in the b frame. The frame m could be a or b or any body frame. The numerical value of ω_{im}^b will of course depend upon the frame selected, hence the superscripts a and b . We have

$$\begin{aligned}\omega_{im}^a &= C_b^a \omega_{im}^b \\ H^a &= C_b^a H^b\end{aligned}\quad (8.104)$$

Inserting Eq. (8.104) into Eq. (8.103a) we have

$$C_b^a H^b = I^a C_b^a \omega_{im}^b \quad (8.105)$$

Multiplying both sides by the inverse of C_b^a , i.e., C_a^b , we have

$$H^b = (C_a^b I^a C_b^a) \omega_{im}^b \quad (8.106)$$

Equating the term in parentheses with its equivalent in Eq. (8.103b) we get

$$I^b = C_a^b I^a C_b^a \quad (8.107)$$

Clearly if we have quantized the moment of inertia I in one frame and have the DCM of a second frame relative to the first, we can carry out the multiplication indicated in Eq. (8.107) to quantize the tensor I in the second frame.

Example 8.3: Application of Moment of Inertia Coordinate Transformation

We can give an illustration of the application of Eq. (8.107) to transforming the moment of inertia from one frame to another. Suppose an RV has a body axis system defined relative to some identifying marks on the structure. We can identify this body frame as reference or r frame. As pointed out in Sec. 8.4, we indicated that the accelerometers instrument an orthogonal axis system. This axis system or triad has been defined as the input axes. Let us assume that this triad or t -axis system is not co-linear with the reference axes (see Fig. 8.11). The t and r axes are related by the following

$$g^t = C_r^t g^r$$

where the vector g represents an accelerometer triad output vector when the accelerometers are subjected to a gravitational field only. If we wish to transform the moment of inertia from the r axes to the instrumented t axes we must first obtain the elements of the C_r^t DCM. One way of obtaining the

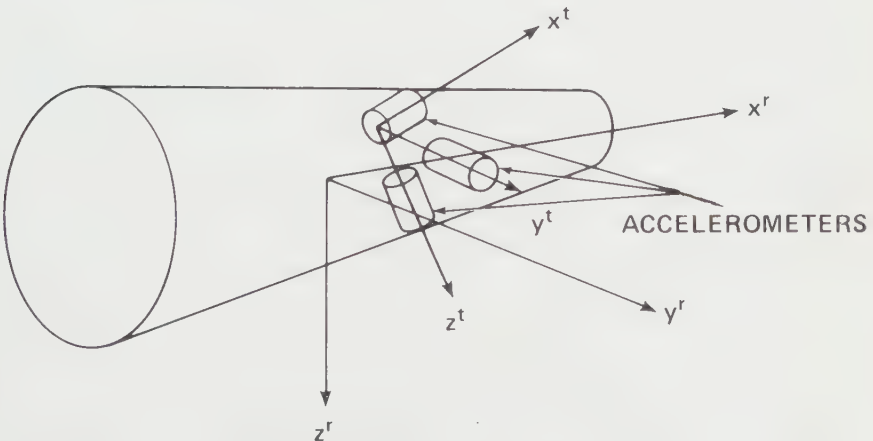


Fig. 8.11 Reference axes and input axes.

elements of C_r^t is to rotate the RV so that the Z^r axis is aligned with the vertical (and the X^r Y^r is plane horizontal). Reading the X and Y accelerometers will give g_x^t and g_y^t as

$$g_x^t = g \cos(X^t, Z^r) = gA_{13} \quad g_y^t = g \cos(Y^t, Z^r) = gA_{23}$$

or

$$A_{13} = g_x^t / g \quad A_{23} = g_y^t / g$$

where g is the local gravitational constant, say, 9.81 m/s^2 .

Let us assume that $g_x^t = 0.372 \text{ m/s}^2$ and $g_y^t = 0.605 \text{ m/s}^2$. If the RV is then rotated so that the X axis is now along the local vertical, we can read the x accelerometer to obtain

$$A_{12} = g_x^t / g$$

Assuming that we measure $g_x^t = -0.211 \text{ m/s}^2$; we can now numerically identify three elements of the C_r^t DCM. A program has been written to calculate the remaining elements of C_r^t . This program given in Table 8.2 uses the redundancy relationships of Eqs. (8.51) and (8.53). In lines 220 the moment of inertia I is entered. This matrix is quantized in the reference body frame; the C_r^t DCM is then used to transform the moment of inertia into the new or triad t frame. Program output presents the two moments of inertia as well as the transform matrix.

8.6 Rotational Energy

In Sec. 8.2 on DCM's we found that there were several redundancy relationships which are useful for completing or "filling in" the elements of a matrix describing an axis transformation. These redundancy relationships are based upon the fact that the scalar or length of a vector remains invariant under a coordinate transformation. We shall now show that the kinetic energy of rotation of a body is also invariant with transformation of the body axis system.

As we have done earlier, we shall develop this idea from first principles. In Fig. 8.12 we have established two frames, one an inertial frame (nonrotating, nonaccelerating) and a second body-fixed frame. In this body frame we shall track all the mass particles which are needed to describe the rigid body.

As before we have

$$\mathbf{R}_i = \mathbf{R}_0 + \mathbf{r}_i \quad (8.108)$$

taking the derivative relative to inertial space of both sides of Eq. (8.108) we get

$$\begin{aligned} \left. \frac{d\mathbf{R}_i}{dt} \right|_I &= \left. \frac{d\mathbf{R}_0}{dt} \right|_I + \left. \frac{d\mathbf{r}_i}{dt} \right|_b \\ \left. \frac{d\mathbf{R}_i}{dt} \right|_I &= \mathbf{V}_0 + \left. \frac{d\mathbf{r}_i}{dt} \right|_b + \boldsymbol{\omega}_{Ib} \times \mathbf{r}_i \end{aligned}$$

$$\frac{d\mathbf{R}_i}{dt} = \mathbf{V}_0 + \boldsymbol{\omega}_{ib} \times \mathbf{r}_i \quad (8.109)$$

Since we are considering a rigid body only, we set to zero the derivative $d\mathbf{r}_i/dt|_b$. The kinetic energy T_i of the mass particle m_i is

$$T_i = \frac{1}{2} M_i \left. \frac{d\mathbf{R}_i}{dt} \right|_I \cdot \left. \frac{d\mathbf{R}_i}{dt} \right|_I \quad (8.110)$$

Table 8.2 Transformation of the Moment of Inertia

```

00100 BASE 1
00110 DIM A(3,3),B(3,3),D(3,3),I2(3,3),I1(3,3)
00120 G = 9.8125
00130 A(1,3) = 0.372/G
00140 A(2,3) = 0.605/G
00150 A(1,2) = (-1)*.211/G
00160 A(3,3) = SQR(1-(A(1,3)**2+A(2,3)**2))
00170 A(2,2) = SQR(1-(A(1,2)**2+A(3,2)**2))
00180 A(3,1) = A(1,2)*A(2,3)-A(1,3)*A(2,2)
00190 A(1,1) = SQR(1-(A(1,2)**2+A(1,3)**2))
00200 A(3,2) = (A(2,2)*A(3,3)-A(1,1))/A(2,3)
00210 A(2,1) = A(1,3)*A(3,2)-A(1,2)*A(3,3)
00220 MAT READ I1
00230 DATA 1.5,0,-0.15,0,.32,0,-0.15,0,10.15
00240 MAT B = INV(A)
00250 MAT D = I1*B
00260 MAT I2 = A*D
00270 PRINT #FIRST MOMENT OF INERTIA#
00280 PRINT
00290 MAT PRINT I1
00300 PRINT #NEW MOMENT OF INERTIA#
00310 PRINT
00320 MATPRINT I2
00330 PRINT #TRANSFORM MATRIX#
00340 PRINT
00350 MAT PRINT A
00360 PRINT
00370 END
    
```

FIRST MOMENT OF INERTIA

1.5	0	-.15
0	9.32	0
-.15	0	10.15

NEW MOMENT OF INERTIA

1.505	-.16227	.188321
-.165786	9.31812	5.46773E-2
.189172	.029598	10.1469

TRANSFORM MATRIX

.77705	-7.15032E-2	3.79108E-2
2.02756E-2	.977759	6.16551E-2
-3.72279E-2	-3.08671E-2	.997377

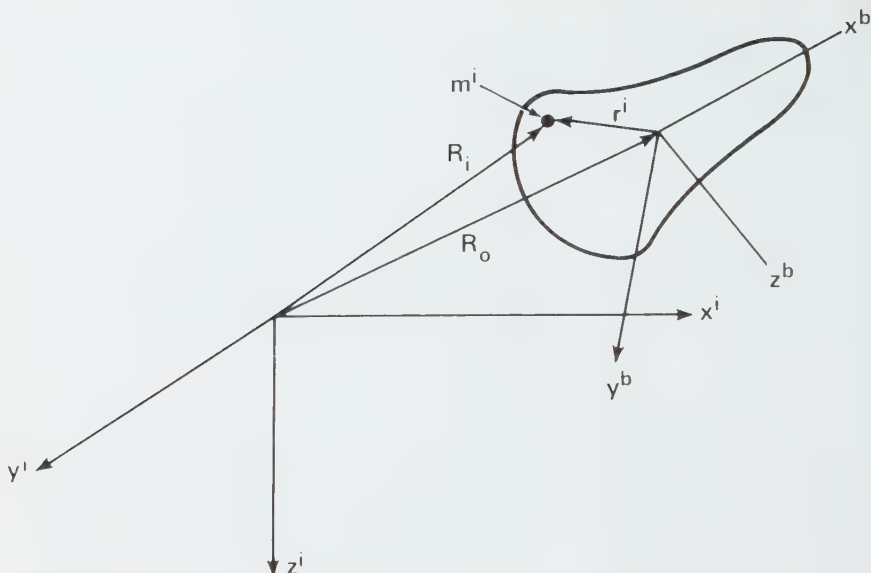


Fig. 8.12 Location of generic mass element from body-fixed and inertial frames.

The kinetic energy of the complete RV is the sum of the kinetic energy of each particle as

$$T = \sum_i T_i = \frac{1}{2} \sum_i m_i \left. \frac{d\mathbf{R}_i}{dt} \right|_I \cdot \left. \frac{d\mathbf{R}_i}{dt} \right|_I \quad (8.111)$$

Inserting $d\mathbf{R}_i/dt|_I$ from Eq. (8.109) we get

$$T = \frac{1}{2} \sum M_i [\mathbf{V}_0 \cdot \mathbf{V}_0 + 2\mathbf{V}_0 \cdot (\boldsymbol{\omega}_{Ib} \times \mathbf{r}_i) + (\boldsymbol{\omega}_{Ib} \times \mathbf{r}_i) \cdot (\boldsymbol{\omega}_{Ib} \times \mathbf{r}_i)] \quad (8.112)$$

The total mass M of the RV may be set equal to the sum of the mass particles m_i as

$$M = \sum_i m_i$$

Equation (8.112) now may be written as the sum of three terms

$$T = \left[\frac{1}{2} m V_0^2 \right] + \left[\mathbf{V}_0 \cdot (\boldsymbol{\omega}_{Ib} \times \sum_i \mathbf{r}_i m_i) \right] + \left[\frac{1}{2} \sum m_i (\boldsymbol{\omega}_{Ib} \times \mathbf{r}_i) \cdot (\boldsymbol{\omega}_{Ib} \times \mathbf{r}_i) \right] \quad (8.113)$$

The first term on the right may be ignored since it represents the kinetic energy due to translation; we are now interested only in the kinetic energy due to rotation. Also, if we assume that the origin of the body axis system is

at the center of gravity, we may ignore the second term as well. The kinetic energy due to rotation may now be written in vector form as

$$T = \frac{1}{2} \sum_i m_i [(\boldsymbol{\omega}_{Ib} \times \mathbf{r}_i) \cdot (\boldsymbol{\omega}_{Ib} \times \mathbf{r}_i)] \quad (8.114)$$

In component form we may write the equivalent to Eq. (8.114) as

$$T = \frac{1}{2} \sum_i C_i [(\omega_y z - \omega_z y)^2 + (\omega_z x - \omega_x z)^2 + (\omega_x y - \omega_y x)^2] \quad (8.115)$$

Expanding the three products in Eq. (8.115) and making use of Eqs. (7.12) we get

$$\begin{aligned} 2T &= \sum_i m_i [\omega_x^2 (y_i^2 + z_i^2) + \omega_y^2 (x_i^2 + z_i^2) + \omega_z^2 (y_i^2 + x_i^2) \\ &\quad - 2\omega_x \omega_y x_i y_i - 2\omega_x \omega_z x_i z_i - 2\omega_y \omega_z y_i z_i] \\ 2T &= \omega_x^2 I_x + \omega_y^2 I_y + \omega_z^2 I_z - 2\omega_x \omega_y I_{xy} - 2\omega_x \omega_z I_{xz} - 2\omega_y \omega_z I_{yz} \\ 2T &= \omega_x [\omega_x I_x - \omega_y I_{xy} - \omega_z I_{xz}] + \omega_y [\omega_y I_y - \omega_x I_{xz} - \omega_z I_{yz}] \\ &\quad + \omega_z [\omega_z I_z - \omega_x I_{xz} - \omega_y I_{yz}] \end{aligned} \quad (8.116)$$

The terms in the brackets are the components of the angular momentum $\mathbf{H}$ [see Eq. (7.14)]. Thus a convenient shorthand for Eq. (8.116) is

$$2T = \boldsymbol{\omega} \cdot \mathbf{H} = \boldsymbol{\omega}^T \mathbf{H} \quad (8.117a)$$

Using Eq. (7.16) we get

$$2T = \boldsymbol{\omega} \cdot \mathbf{I} \cdot \boldsymbol{\omega} \quad (8.117b)$$

The matrix equivalent of the above vector-dyadic triple product is

$$2T = (\boldsymbol{\omega}_{Ib}^b)^T \mathbf{I}^b \boldsymbol{\omega}_{Ib}^b \quad (8.117c)$$

Regardless of how the body-frame b is oriented relative to the inertial frame I , the above triple product will always equal twice the magnitude of the rotational kinetic energy.

Let us now return briefly to Sec. 7.2. There it was shown that the moment of inertia of a body about an axis having the direction cosines (a, b, c) relative to a body frame could be written in terms of the moments/products of inertia quantized in this body frame. For reference, Eq. (7.25) will be rewritten as

$$I = I_{xx} a^2 + I_{yy} b^2 + I_{zz} c^2 - 2ab I_{xy} - 2ac I_{xz} - 2bc I_{yz} \quad (8.118)$$

The above equation permits us to calculate the moment of inertia of the body about an arbitrary axis having the directional cosines (a, b, c) . Let us assume that this arbitrary axis locates the instantaneous direction of the angular velocity vector in the body frame where $a = \omega_x / \omega_{Ib}$, $b = \omega_y / \omega_{Ib}$, and $c = \omega_z / \omega_{Ib}$ (see Fig. 8.13). Thus, if I_ξ is the moment of inertia of the

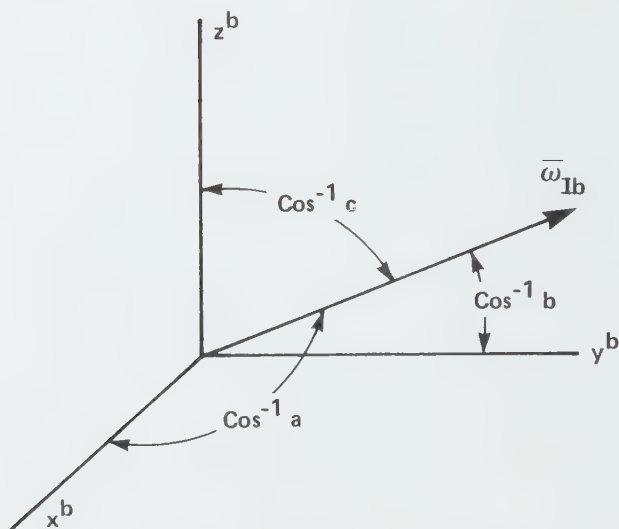


Fig. 8.13 Angular velocity vector.

body about the ω_{Ib} axis, then from Eq. (7.25) we have, with $\omega \equiv |\omega_{Ib}|$,

$$I_{\xi} = \left(\frac{\omega_x}{\omega} \right)^2 I_{xx} + \left(\frac{\omega_y}{\omega} \right)^2 I_{yy} + \left(\frac{\omega_z}{\omega} \right)^2 I_{zz} - 2 \left(\frac{\omega_x}{\omega} \right) \left(\frac{\omega_y}{\omega} \right) I_{xy} - 2 \left(\frac{\omega_x}{\omega} \right) \left(\frac{\omega_z}{\omega} \right) I_{xz} - 2 \left(\frac{\omega_y}{\omega} \right) \left(\frac{\omega_z}{\omega} \right) I_{yz} \quad (8.119)$$

Equation (8.116) may be written as

$$2T = \omega^2 \left[\left(\frac{\omega_x}{\omega} \right)^2 I_{xx} + \left(\frac{\omega_y}{\omega} \right)^2 I_{yy} + \left(\frac{\omega_z}{\omega} \right)^2 I_{zz} - 2 \left(\frac{\omega_x}{\omega} \right) \left(\frac{\omega_y}{\omega} \right) I_{xy} - 2 \left(\frac{\omega_x}{\omega} \right) \left(\frac{\omega_z}{\omega} \right) I_{xz} - 2 \left(\frac{\omega_y}{\omega} \right) \left(\frac{\omega_z}{\omega} \right) I_{yz} \right] \quad (8.120)$$

or from Eq. (8.119)

$$T = \frac{1}{2} I_{\xi} \omega^2 \quad (8.121)$$

In words, Eq. (8.121) says that the kinetic energy T is equal to one-half the product of the angular velocity (magnitude) squared times the moment of inertia about the axis co-linear with the velocity vector.

Equation (8.121) is quite simple; however, there are certain disadvantages. First, assuming that we are given a body axis system, we must find the directional cosines of ω_{Ib} in this frame. We then must compute the moment of inertia I_ξ about this axis. A second and related disadvantage is that every change in direction of ω_{Ib} in the body frame requires a recomputation of I_ξ . If the direction of the angular velocity vector is continually changing then the value of I_ξ must also change. It would seem easier to eschew Eq. (8.121) in favor of Eqs. (8.117b) and (8.117c). In these latter equations a body fixed axis is used and hence the elements of the moment of inertia tensor remain constant.

If we are going to maintain a body axis system, we might still ask the question as to whether the moment of inertia might be simplified by choosing certain body axes. First let us write Eq. (7.25) as follows:

$$I_{xx}x^2 + I_{yy}y^2 + I_{zz}z^2 - 2I_{xy}xy - 2I_{xz}xz - 2I_{yz}yz = 1 \quad (8.122)$$

where $x = a\rho$, $y = b\rho$, $z = c\rho$ and $\rho = 1/\sqrt{I_\xi}$.

Equation (8.122) depicts a quadratic form known as the ellipsoid of inertia. Any body, regardless of how arbitrary its shape or mass distribution, must have its inertia characteristics defined by this quadratic form. Since an ellipsoid is a figure having three axes of symmetry, if we choose our body frame along these principal axes then the ellipsoid of inertia takes the particular simple form:

$$\tilde{I}_{xx}\tilde{x}^2 + \tilde{I}_{yy}\tilde{y}^2 + \tilde{I}_{zz}\tilde{z}^2 = 1$$

or

$$\left[\frac{\tilde{x}}{\sqrt{1/I_{xx}}} \right]^2 + \left[\frac{\tilde{y}}{\sqrt{1/I_{yy}}} \right]^2 + \left[\frac{\tilde{z}}{\sqrt{1/I_{zz}}} \right]^2 = 1 \quad (8.123)$$

In Fig. 8.14 we have depicted an RV surrounded by its ellipsoid of inertia. The magnitude of the radius vector from the origin of the axis system to the ellipsoid equals the reciprocal of the square root of the moment of inertia of the RV about this radius vector. If a body axis system is chosen to align with the principal axes, then the products of inertia (off-diagonal terms) vanish from the inertia tensor. Let us assume that we have the DCM between an arbitrary body frame b and the principal axes p also a body frame. Let this DCM be C_p^b . We may write the components of the angular velocity in the b frame in terms of the angular velocity in the p frame as

$$\omega_{Ib}^b = C_p^b \omega_{Ip}^p \quad (8.124)$$

Equation (8.117c) then may be written as

$$2T = (C_p^b \omega_{Ip}^p)^T I^b (C_p^b \omega_{Ip}^p)$$

or

$$2T = (\omega_{Ib}^b)^T [C_p^{bT} I^b C_p^b] \omega_{Ib}^b \quad (8.125a)$$

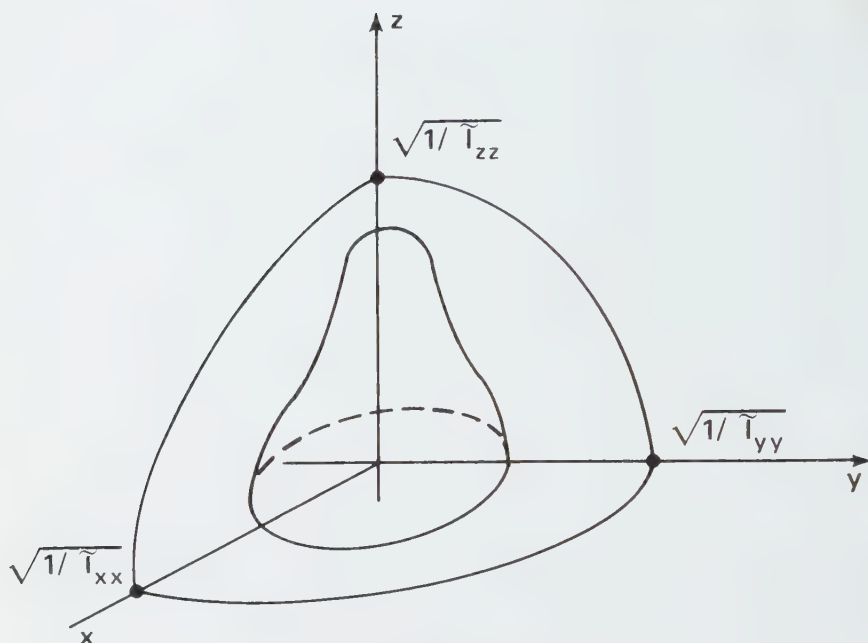


Fig. 8.14 Ellipsoid of inertia.

Using Eq. (8.107) allows us to write Eq. (8.125a) as

$$2T = (\omega_{lb}^b)^T I^b \omega_{lb}^b \quad (8.125b)$$

where

$$I^p = C_b^p I^b C_p^b \quad (8.125c)$$

Equation (8.125c) indicates that we can diagonalize the inertia tensor if we have the appropriate DCM C_b^p , which represents a coordinate frame rotation from the arbitrary b body frame to the principal axis frame p . Equation (8.125c) is more or less a definition of the C_b^p and DCM; however, it provides us with no information as to how to formulate C_b^p . In the next section we shall show that the computation of the Eigen values and Eigen vectors of I^b will lead to algorithms for constructing C_b^p .

8.7 Eigen Values and Eigen Vectors

In this section we shall briefly consider a procedure for computing the principal moments of inertia and the DCM between a general body-frame b and the principal frame p . According to Eq. (8.125c) the availability of this DCM, i.e., C_b^p , will permit computation of the principal moments of inertia as well as being a means of locating the principal axes.

It was shown in Eq. (7.16) that the angular momentum $\mathbf{H}$ equals the inner product of the inertial dyadic with the angular velocity vector ω_{Ib} . In matrix notation the equivalent expression may be written as

$$H^b = I^b \omega_{Ib}^b \quad (8.126)$$

From the above expression we may conclude that in general the angular momentum and angular velocity vectors are not co-linear. Figure 8.15 indicates this general condition.

In Eq. (8.126) the inertia tensor I^b might be thought of as a transformation which both rotates and stretches the angular velocity vector ω_{Ib} to provide the angular momentum vector $\mathbf{H}$. Although it is rather obvious we should emphasize the fact that even if the body axis system we have chosen is a principal axis (inertial tensor diagonal), the angular momentum and angular velocity vectors, i.e., $\mathbf{H}$ and ω_{Ib} are not generally co-linear: axis systems are descriptive only and do not influence the physics of the problem. However, it is true that if the rotation does take place about any of the principal axes, then the $\mathbf{H}$ and ω_{Ib} vectors are co-linear. (In Chapter XII we shall consider the stability of rotations about principal axes.) This fact provides us with a method of locating the principal axes relative to the arbitrary body axes, i.e., defining a DCM between the b and p frames. In other words, if rotation takes place about a principal axis, H^b is simply a multiplicative constant times ω_{Ib} or

$$H^b = \lambda [1] \omega_{Ib}^b \quad (8.127)$$

where $[1]$ is the 3×3 unit matrix.

Since the general case of vehicle rotation about any body axis must include the special case of rotation about each of the principal axes, then

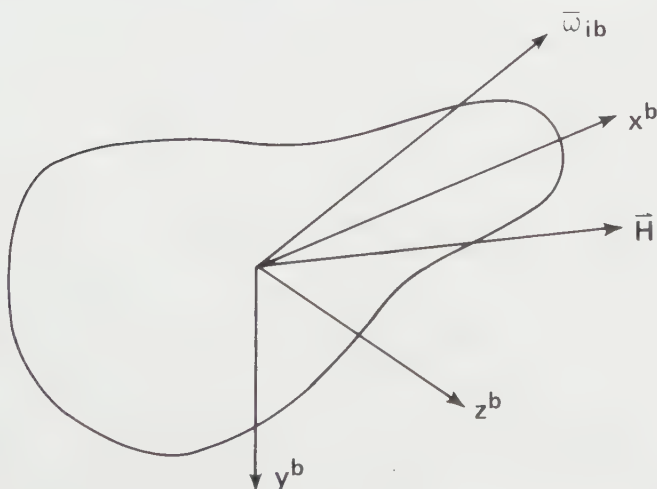


Fig. 8.15 Angular velocity and angular momentum vectors.

Eqs. (8.126) and (8.127) must be equal for rotation about the principal axes. Thus,

$$H^b = I^b \omega_{Ib}^b = \lambda [1] \omega_{Ib}^b$$

We must then have

$$(I^b - \lambda[1])\omega_{Ib}^b = 0 \quad (8.128)$$

or in the form of linear equations we have

$$\begin{aligned} (I_{xx} - \lambda)\omega_x - I_{xy}\omega_y - I_{xz}\omega_z &= 0 \\ I_{xy}\omega_x + (I_{yy} - \lambda)\omega_y - I_{yz}\omega_z &= 0 \\ -I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz} - \lambda)\omega_z &= 0 \end{aligned} \quad (8.129)$$

The characteristic determinant must be zero in order that ω_{Ib} be nonzero (nontrivial case). Expansion of the characteristic determinant of Eqs. (8.129) provides us with the roots or "Eigen values" of the inertia tensor. These roots will be designated a λ_1 , λ_2 , and λ_3 . These roots are the principal moments of inertia. We seem to have found the principal moments of inertia without recourse to Eq. (8.125c). Figure 8.16, indicating rotation about one of the principal axes, reviews our work so far.

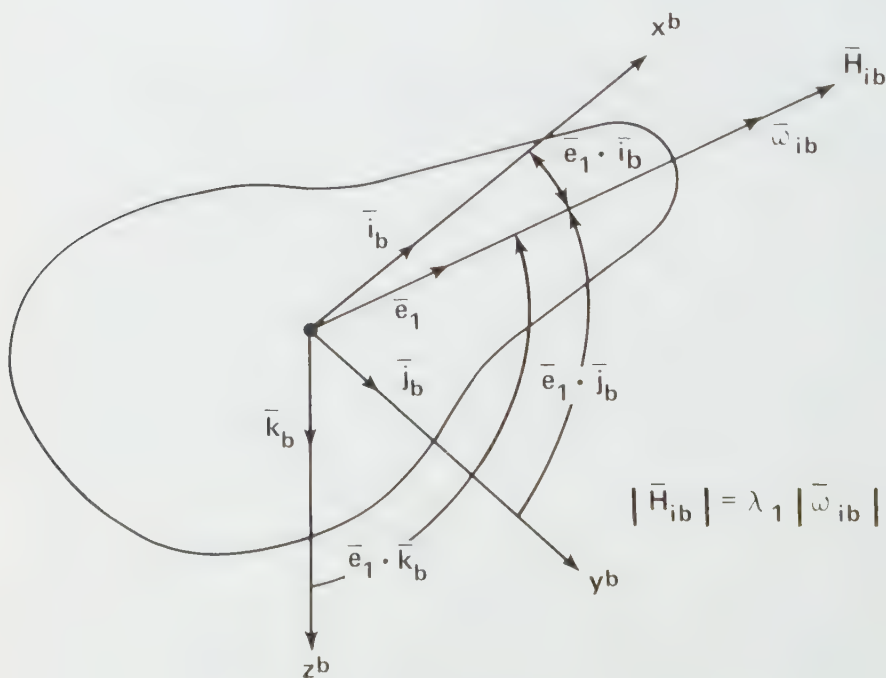


Fig. 8.16 Rotation about a principal body axis.

The RV is seen in Fig. 8.16 to be rotating about the principal axis we have designated as 1. The direction of this principal axis is indicated by the unit vector $\mathbf{e}_1$. The angular location of $\mathbf{e}_1$ relative to the body-frame b is provided by the inner or dot product of $\mathbf{e}_1$ with each of the b frame unit vectors ($\mathbf{i}, \mathbf{j}, \mathbf{k}$) in succession. This inner product is of course the cosine of the directional angles. In Fig. 8.16 we have set the angular momentum and angular velocity as co-linear with the angular momentum equal in magnitude to a constant λ times the magnitude of the angular velocity. Since we are assuming knowledge of the components of ω_{Ib} in the body frame, i.e., $\omega_{Ib}^b = (\omega_x, \omega_y, \omega_z)^T$, we can identify the directional cosines between the $\mathbf{e}_1$ vector along H^b and ω_{Ib} and the unit vectors along the body axes. For example, paralleling Eq. (8.39) we may write

$$\mathbf{e}_1 = (\mathbf{e}_1 \cdot \mathbf{i}_b) \mathbf{i}_b + (\mathbf{e}_1 \cdot \mathbf{j}_b) \mathbf{j}_b + (\mathbf{e}_1 \cdot \mathbf{k}_b) \mathbf{k}_b \quad (8.130a)$$

where the inner products of $\mathbf{e}_1$ with ($\mathbf{i}, \mathbf{j}, \mathbf{k}$) successively provides the directional angles of the unit vector $\mathbf{e}_1$ in the body frame. Thus,

$$\mathbf{e}_1 \cdot \mathbf{i}_b = \frac{\omega_x}{\omega_{Ib}} \quad \mathbf{e}_1 \cdot \mathbf{j}_b = \frac{\omega_y}{\omega_{Ib}} \quad \mathbf{e}_1 \cdot \mathbf{k}_b = \frac{\omega_z}{\omega_{Ib}} \quad (8.130b)$$

where ω_{Ib} is the magnitude of ω_{Ib}^b , the angular velocity vector of the body relative to inertial space.

The procedure to find $\mathbf{e}_1$ (and later $\mathbf{e}_2$ and $\mathbf{e}_3$) is to assume a constant value for one component of ω_{Ib}^b , say, $\omega_x = 1$, and solve two of the three equations of (8.129) for ω_y and ω_z . The magnitude ω_{Ib} is calculated and the ratios provide the directional cosines of $\mathbf{e}_1$ in the body frame as the fractions indicated in Eq. (8.130b). The procedure is repeated for the Eigen values λ_2 and λ_3 . We now have the Eigen vectors $\mathbf{e}_1, \mathbf{e}_2$, and $\mathbf{e}_3$, which allow us to locate the principal frame relative to the body frame, i.e., to establish C_b^p .

Following the procedure of Eqs. (8.39) and (8.40) we have now defined the DCM C_b^p as

$$\left[C_b^p \right]^T = \begin{bmatrix} \left[\mathbf{e}_1 \cdot \mathbf{i}_b \right] & \left[\mathbf{e}_2 \cdot \mathbf{i}_b \right] & \left[\mathbf{e}_3 \cdot \mathbf{i}_b \right] \\ \left[\mathbf{e}_1 \cdot \mathbf{j}_b \right] & \left[\mathbf{e}_2 \cdot \mathbf{j}_b \right] & \left[\mathbf{e}_3 \cdot \mathbf{j}_b \right] \\ \left[\mathbf{e}_1 \cdot \mathbf{k}_b \right] & \left[\mathbf{e}_2 \cdot \mathbf{k}_b \right] & \left[\mathbf{e}_3 \cdot \mathbf{k}_b \right] \end{bmatrix} = C_p^b \quad (8.131)$$

Because of the assumed orthogonality of the b and p frames, $(C_p^b)^{-1} = (C_p^b)^T$. The symbolism $\mathbf{e}_1 \cdot \mathbf{k}_b$ of course means the cosine of the angle between the z -body axis and the principal axis we have designated as 1. For our purposes we shall designate the principal axes in order of increasing Eigen values, i.e., increasing moments of inertia.

A computer program is given in Table 8.3 for calculating the principal moments of inertia and establishing the required DCM C_p^b . The moments of inertia are entered in line 200 where $I(1,1) = I_{xx}$, $I(2,2) = I_{yy}$, $I(3,3) = I_{zz}$, $I(1,2) = I_{xz}$ and $I(2,3) = I_{yz}$. Lines 240-260 accounts for the necessary diag-

Table 8.3 Program for Calculating Eigen Values and Eigen Vectors
of Moment of Inertia Matrix

```

00100 REM THIS PROGRAM CALCULATES THE EIGENVALUES AND THE
00110 REM EIGEN VECTORS OF A GENERAL INERTIA TENSOR. THE MOMENTS
00120 REM OF INERTIA ABOUT THE DESIGNATED X,Y,Z AXES AND THE
00130 REM PRODUCTS OF INERTIA XY,XZ,YZ ARE ENTERED. EIGENVALUES
00140 REM ( VALUES OF THE PRINCIPAL MOMENTS OF INERTIA) ARE
00150 REM PRINTED OUT IN ORDER OF INCREASING MAGNITUDE. CORRESPONDING
00160 REM EIGENVECTORS ARE PRINTED OUT AS COLUMNS IN THE DCM
00170 REM (PRINCIPAL-TO-GENERAL AXES). PRINCIPAL MOMENT OF
00180 REM INERTIA MATRIX IS PRINTED.
00190 BASE 1
00200 READ I(1,1),I(2,2),I(3,3),I(1,2),I(1,3),I(2,3)
00210 DATA 2.1,3.05,1.55,-1.0,-1.5,0
00220 DIM I(3,3),I1(3,3),A(3,3),A1(3,3),Z1(3,3)
00230 DIM P(3,3),P8(3,3),P9(3,3),Z(3)
00240 I(2,1) = I(1,2)
00250 I(3,1) = I(1,3)
00260 I(3,2) = I(2,3)
00270 A0 = I(1,1)*(I(2,3)^2)+I(2,2)*(I(1,3)^2)+I(3,3)*(I(1,2)^2)
00280 A0 = A0-I(1,1)*I(2,2)*I(3,3)-2*I(1,2)*I(1,3)*I(2,3)
00290 A1 = I(1,1)*I(2,2)+I(1,1)*I(3,3)+I(2,2)*I(3,3)
00300 A1 = A1-(I(1,2)^2)+I(1,3)^2+I(2,3)^2)
00310 A2 = (-1)*(I(1,1)+I(2,2)+I(3,3))
00320 A3 = 1
00330 DEF FNY(X) = A0+A1*X+A2*(X^2)+A3*(X^3)
00340 DEF FND(X)=A1+2*A2*X+3*A3*(X^2)
00350 J = 1
00360 X1 = (-1)*I(2,2)*10
00370 X2 = I(2,2)*10
00380 S = .01
00390 FOR X = X1 TO X2 STEP S
00400 Y1 = FNY(X)
00410 Y2 = FNY(X+S)
00420 IF SGN(Y1) = SGN(Y2) THEN 00530
00430 X0 = X
00440 Y1 = FNY(X0)
00450 IF ABS(Y1)< 1E-3 THEN 00510
00460 X1 = FND(X0)
00470 IF X1 = 0 THEN 00530
00480 D = (-1)*Y1/X1
00490 X0 = X0+D
00500 GOTO 00440
00510 Z(J) = X0
00520 J = J+1
00530 NEXT X
00540 FOR L = 1 TO 3
00550 MAT P9 = IDN(3,3)
00560 MAT P8 = (Z(L))*P9
00570 MAT P = I-P8
00580 W1 = 1
00590 W2 = P(2,2)*P(1,3)-P(1,2)*P(2,3)
00600 W3 = (P(2,1)*P(1,2)-P(1,1)*P(2,2))/W2
00610 W2 = (P(1,1)*P(2,3)-P(2,1)*P(1,3))/W2
00620 W = SQR(W2^2+W1^2+W3^2)
00630 A(1,L) = W1/W
00640 A(2,L) = W2/W
00650 A(3,L) = W3/W
00660 NEXT L
00670 MAT A1 = TRN(A)
00680 MAT Z1 = I*A
00690 MAT I1 = A1*Z1
00700 PRINT #EIGEN VALUES#
00710 MAT PRINT Z
00720 PRINT #EIGEN VECTORS#
00730 MAT PRINT A
00740 PRINT #PRINCIPAL MOMENT OF INERTIA MATRIX#
00750 MAT PRINT I1
00760 PRINT #ORIGINAL MOMENT OF INERTIA MATRIX#
00770 MAT PRINT I
00780 END

```

(Table continued on next page)

Table 8.3 (cont.) Program for Calculating Eigen Values and Eigen Vectors of Moment of Inertial Matrix

EIGEN VALUES		
.149074		
2.52184		
4.02903		
EIGEN VECTORS		
.664389	.378856	.64424
.229027	.71731	-.657988
.711431	-.584751	-.389881
PRINCIPAL MOMENT OF INERTIA MATRIX		
.14913	-4.89376E-5	-3.72595E-5
-4.89376E-5	2.52184	1.92560E-4
-3.72595E-5	1.92560E-4	4.02903
ORIGINAL MOMENT OF INERTIA MATRIX		
2.1	-1	-1.5
-1	3.05	0
-1.5	0	1.55

onal symmetry for the inertia tensor. The portion of the program up to and including line 540 computes the Eigen values of the roots of the characteristic equation. This portion of the program is a direct application of the program for finding the real roots of a polynomial given in Appendix D. Lines 550-700 compute the Eigen vectors $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$. The DCM C_p^h is printed under the Eigen vector heading. According to Eq. (8.131) these Eigen vectors are the columns of the C_p^h matrix. For the sample printout, the Eigen vector $\mathbf{e}_1$ is given as

$$\mathbf{e}_1 = 0.664\mathbf{i}_b + 0.229\mathbf{j}_b + 0.711\mathbf{k}_b \quad (8.132)$$

$\mathbf{e}_1$ locates the axis of the smallest principal moment of inertia relative to the body frame.

The procedure for identifying the components of $\mathbf{e}_2$ and $\mathbf{e}_3$ follows an identical process.

References

- ¹Goldstein, H., *Classical Mechanics*, Addison-Wesley, Reading, Mass., 1950, Chap. 5.
- ²Regan, F. T., "Dynamic Testing with the NSWC Three Degree of Freedom Gas Bearing," Naval Surface Weapons Center, Silver Springs, Md., NSWC TR 81-491, 1982.
- ³Schmidt, G. T., "Strapdown Inertial Systems—Theory and Application—Instruction and Overview," AGARD Lecture Series Strap-down Inertial Systems LS-95, May 1978.

IX. FLOWFIELD DESCRIPTION

The subject of aerodynamics is of course crucial to an understanding of re-entry body dynamics. The pressure distribution imparted by the near flowfield determines the forces and moments acting on the vehicle. Unfortunately it is not possible to give proper weight to the subject of gasdynamics in a book devoted to vehicle dynamics. Any attempt to cover the various aspects of gasdynamics would completely overwhelm the discussion of dynamics. In this chapter we shall only allude to the various flight regimes encountered by the RV, and then only in a brief manner. We shall, however, calculate the forces and moments on a conical vehicle using a fairly simple flow representation known as the Newtonian impact method.

9.1 Flow Regimes

Figure 9.1 indicates five regimes which have distinguishing flow characteristics. Tagged within the regions are significant RV aerodynamic and dynamic events. We shall refer to Fig. 9.1 frequently as we justify or explain these occurrences.

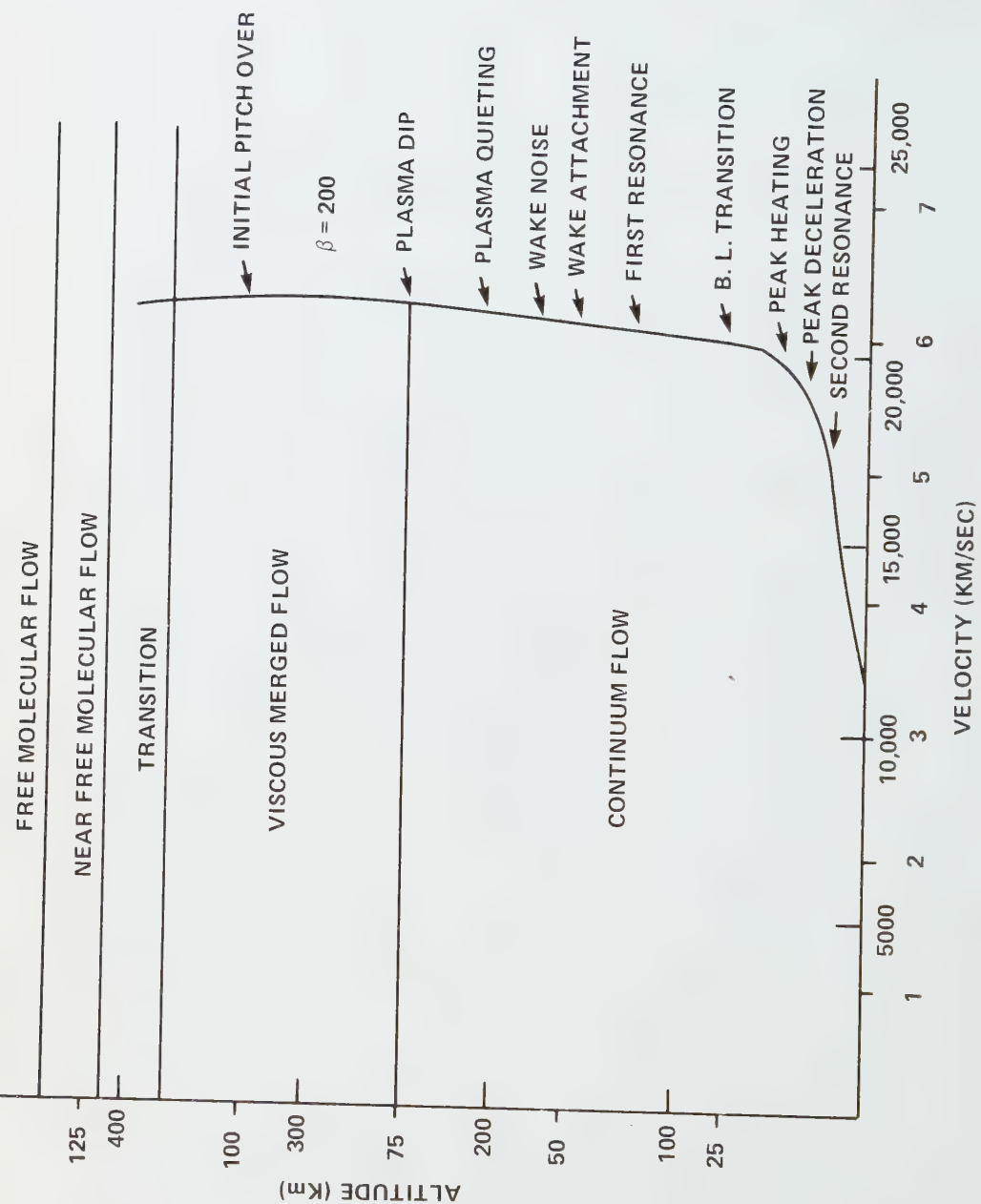
The five regimes are described as follows:

(a) *Free Molecular Regime*: In this regime the molecular mean free path λ is large compared to a characteristic body dimension such as nose radius. Molecules which collide with a boundary attain the state of that boundary after a single collision.

(b) *Near Free Molecular Flow*: This regime is sometimes called the slip region because the gas molecule acquires the momentum of the moving boundary only after there have been several collisions. If on the average a molecule does not acquire the energy corresponding to the energy of the boundary after one collision one says that there is a lack of "accommodation." In the case of slip, molecules near the wall will have a mean velocity different from that of the wall. The lack of accommodation means for example that the temperature will be a nearly discontinuous function of distance from the surface.

(c) *Transition Regime*: The transitional regime is very difficult to treat analytically so that little is known about aerodynamic quantities such as lift, drag, or heat transfer. For engineering purposes these quantities are usually estimated by a judicious faring between regimes.

(d) *Viscous Merged Layer Regime*: The viscous effects in the forming shock wave and boundary layer must be treated in a unified manner. Even if the



shock wave and boundary layer are treated separately neither can be considered as discontinuities. The presence of a boundary layer on the wall alters the boundary conditions for the shock wave while at the same time the large pressure gradients due to the shock wave strongly alter the boundary-layer flow. Essentially we have in the viscous merged layer regions shock-wave-boundary-layer interaction.

(e) *Continuous Regime*: In this region the classical fluid mechanics of high Reynolds number applies and the shock waves and boundary layer are treated as discontinuities. This region is often subdivided into categories such as incompressible, transonic, supersonic, and hypersonic, with lines of demarcation set by the Mach number. However, for most flow conditions encountered by re-entry vehicles it turns out that Mach number at impact is often greater than five. Thus if we define hypersonic flows as flows having Mach numbers in excess of five, it is the hypersonic regime that is of overwhelming interest in RV engineering.

The boundaries of the five flow regimes just noted may be given quantitative definition. The parameter useful for this purpose is the Knudsen number which we shall designate by the symbol K . The Knudsen number is the ratio of the molecular mean free path to a characteristic body dimension, as

$$K = \lambda / R_N \quad (9.1)$$

where λ is the mean free path and R_N the nose radius. The question arises as to the flow location of the mean free path. In rarefied flow where the molecular collisions are important the mean free path λ is taken in the flow emitted from the surface *after* impact. The relationship between the mean free path after collision λ_c and the mean free path in the oncoming flow λ_∞ is

$$\lambda_c = \left[\frac{4}{\sqrt{\pi \gamma}} \right] \left[\frac{T_w}{T_\infty} \right]^{1/2} \left[\frac{\lambda_\infty}{M_\infty} \right] \quad (9.2)$$

where T_w and T_∞ are the temperature at the wall and the temperature in the freestream ahead of flow disturbance caused by the RV. If the wall temperature is nearly the same as the freestream temperature, then

$$\lambda_c = 1.9 \lambda_\infty / M_\infty \quad (9.3)$$

In the continuum flow regime we shall use the mean free path of the molecules immediately behind the shock wave. We shall designate this mean free path by λ_1 and then relate λ_1 to the mean free path of the oncoming flow as

$$\lambda_1 = \frac{\lambda_\infty}{\rho_s / \rho_\infty} \quad (9.4)$$

where ρ is the flow density and the subscript s and ∞ refer, respectively, to conditions downstream and upstream of the bow shock.

Now we are able to define the five flow regimes in terms of the Knudsen number K_c [based on λ_c of Eq. (9.2)] or K_∞ (based on λ_∞).

- (a) *Free Molecular Flow*: $K_c \geq 10$, or $K_\infty > 5.24M_\infty$
- (b) *Near Free Molecular Flow*: $10 \leq K_c \leq 1/3$, or $5.4M_\infty \leq K_\infty \leq 0.175M_\infty$
- (c) *Transition Flow*: $0.175M_\infty \leq K_\infty \leq 1$
- (d) *Viscous Merged Layer*: $1 \leq K_\infty \leq 0.1/(\rho_s/\rho_\infty)$
- (e) *Continuum Flow*: $K_\infty > 0.1/(\rho_s/\rho_\infty)$

In the continuum flow regime we shall expect to encounter hypersonic speeds for most RV's. Earlier we somewhat arbitrarily defined a hypersonic flow as one for which the Mach number was in excess of 5.0. However no universal definition of hypersonics exists. We should point out that hypersonic flow is not merely an extension of supersonic flows. Hypersonic flows have the following three characteristics not found at lower Mach numbers:

(a) The shock waves in hypersonic flows lie close to the body causing strong interaction between shock wave and boundary layer. The boundary layer in hypersonic flows is 10-100 times thicker than that encountered at lower speeds. The change in the effective body shape brings a change in the effective body bluntness which in turn brings about a change in the shape of the nose shock.

(b) High-temperature gas effects become significant. These effects include excitation of additional degrees of freedom in the gas molecule and dissociation of the gas into ions.

(c) An important characteristic of hypersonic flows is the nonlinearity of loads (forces and moments) with angle of attack.

Figure 9.2 identifies some of the significant regions in the flowfield around a body in hypersonic flow in the continuum region. One prominent feature of such flow is the strong forward bow shock wave. A packet of air at ambient conditions ($\rho_\infty, T_\infty, p_\infty, V_\infty$) approaches the body and passes

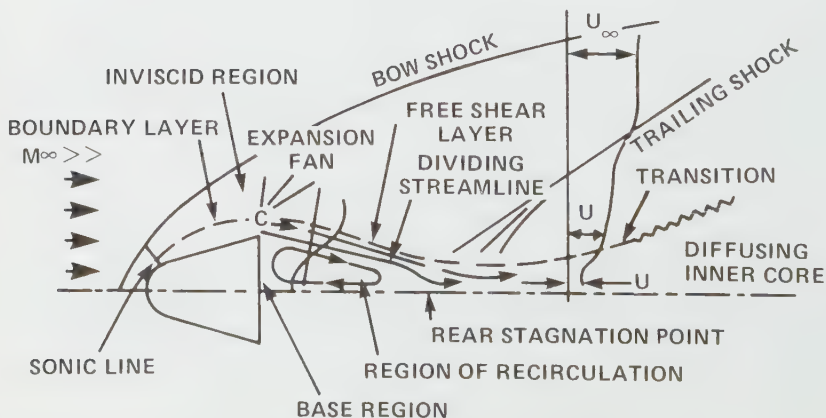


Fig. 9.2 Hypersonic flow characteristics.

through this strong near-normal bow shock wave. Changing thermodynamic conditions cause pressure, density, and temperature p, ρ, T to increase and V to decrease. Also total temperature does not change but total pressure decreases resulting in an entropy rise approximately proportional to this decrement in total head. The flow now accelerates around the body reaching the speed of sound at the sonic line. Some of the flow becomes entrained in the boundary layer, the remainder in the essentially inviscid shock layer. Somewhere beyond the maximum body area there is a Prandtl-Meyer expansion fan through which the flow expands to a lower pressure that is still somewhat greater than ambient. At the base of the body there is shear flow (free shear layer) between the "dead water" or base recirculation region. This shear layer is made up primarily of the boundary layer detached from the body. The shear layer converges on an axis which is a rearward extension of the body's axis of symmetry. An oblique trailing shock wave develops at the neck or point of convergences of this shear layer. Downstream of this point of convergence the shear layer develops into the portion of the wake called the viscous core. Flow which remained outside of the boundary layer and the developed free shear layer crosses the oblique shock formed at the shear layer neck. That region between the trailing oblique shock and the viscous core is called the inviscid wake. As this fluid crosses the oblique shock the pressure temperature and density increase and in continuing downstream this outer (inviscid) wake combines with the diffusing inner (viscous) wake.

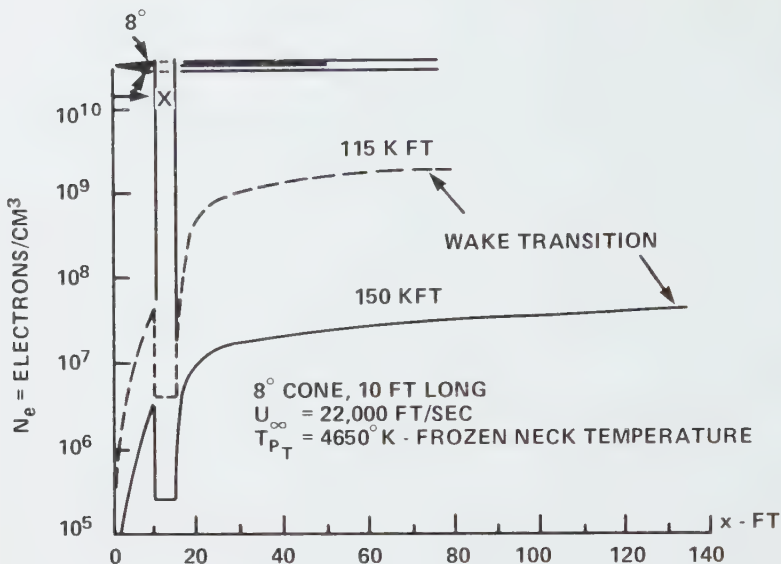
At hypersonic flow conditions the gas may be ionized by three processes: (a) thermal ionization caused by compressive shock waves; (b) thermal ionization generated by viscous effects; and (c) chemical ionization which may include evaporated material from the ablative heat shield. All three processes will contribute electrons and ions to the plasma sheath. Because the mass of electrons is negligible when compared with that of the ions, the electrons have a much greater effect on radar reflection or electromagnetic waves propagated from the RV. Radar amplitude (reflected signal) drops off very quickly for radar frequencies above the plasma frequency which is the natural oscillating frequency of the electrons in the plasma. This frequency f_p is given as

$$f_p = \sqrt{(e^2/\pi M_e) N_e} = 8970 N_e^{1/2} \text{ Hz} \quad (9.5)$$

where e is the electron charge $= 4.8 \times 10^{-10}$ esu, M_e the electron mass $= 9.1 \times 10^{-28}$ g, and N_e the number of free electrons/cm³.

The hardest quantity to come by is N_e . Figure 9.3 taken from the paper of Fernandez and Levinsky¹ presents estimates for N_e , where X is the distance behind the vehicle's vertex. Using N_e of 10^9 electrons/cm³ we have $f_p = 2.8 \times 10^8$ or 0.28 g-Hz. All radar bands X (10 g-Hz), C (5 g-Hz), S (2.5 g-Hz), and L (1 g-Hz) would have severe amplitude attenuation.

In Chapter VI we considered particle motion of the RV and as a consequence produced velocity-altitude histories. If re-entry is taken at 120 km, it may be seen in the following sketch that the velocity remains nearly constant until the RV has reached an altitude of 30 km. (Actually the velocity mag-

Fig. 9.3 Estimates for N_e .

nitude will, in some circumstances, increase slightly down to 30 km because of gravity.) In Eq. (6.25) we calculated peak axial loading as a function of altitude z , entry velocity magnitude V_E , and ballistic coefficient β . Also note from Eq. (6.19) that peak axial deceleration load varies with dynamic pressure $\rho V_\infty^2/2$, and inversely with ballistic coefficient β . Peak axial loading occurs at altitudes between 3 and 15 km. Peak heating occurs at a somewhat higher altitude.

In Eq. (6.17) we indicated that transverse acceleration depends upon the lift-to-drag ratio. Both drag C_D and lift C_L coefficients will increase with angle of attack, although C_L will increase more rapidly with the result that C_L/C_D will increase with angle of attack. In Figs. (9.4) and (9.5) we have plotted representative total angle of attack and normal acceleration variations with altitude (as well as dynamic pressure, velocity magnitude, and normal acceleration). Note that initially the angle of attack does not oscillate and that it rises during early re-entry (indicated by the α band or envelope). The plot given above might not be typical of all RV's under all initial conditions. For example in many cases the oscillations may not continue to impact but dissipate completely above 30 km. Obviously the closer to zero angle of attack the RV is at re-entry, the smaller the oscillatory amplitude. (Oscillatory motion will be discussed in Chapter XIII.) One way to assure re-entry at small angles of attack is to stabilize the body at the proper orientation at off-loading from the equipment section of the boost vehicle in order that the RV centerline has the proper inertial orientation for re-entry (see Chapter XI). This orientation is stabilized by spinning the RV at something like 700-1000 deg/s. The spinning RV acts like a free gyro. In Chapter XII we shall discuss the angular motion of the RV prior to re-entry,

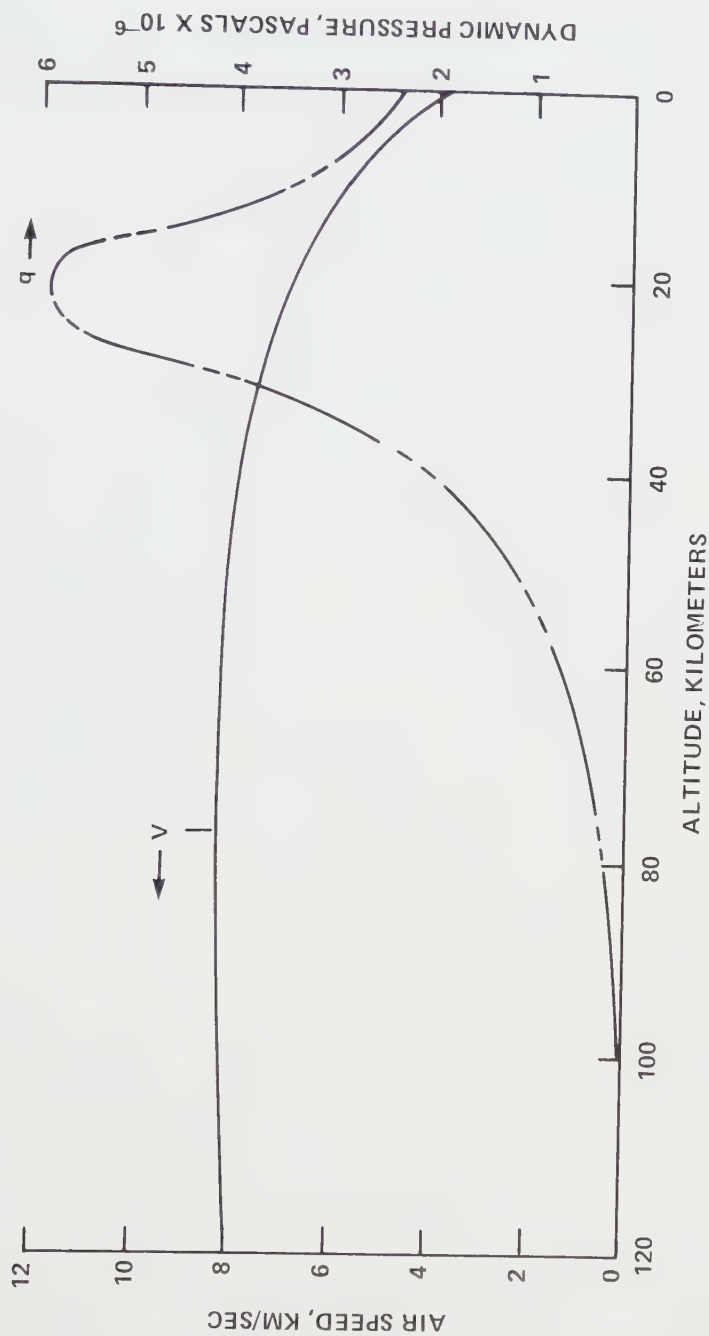


Fig. 9.4 Air speed and dynamic pressure vs altitude.

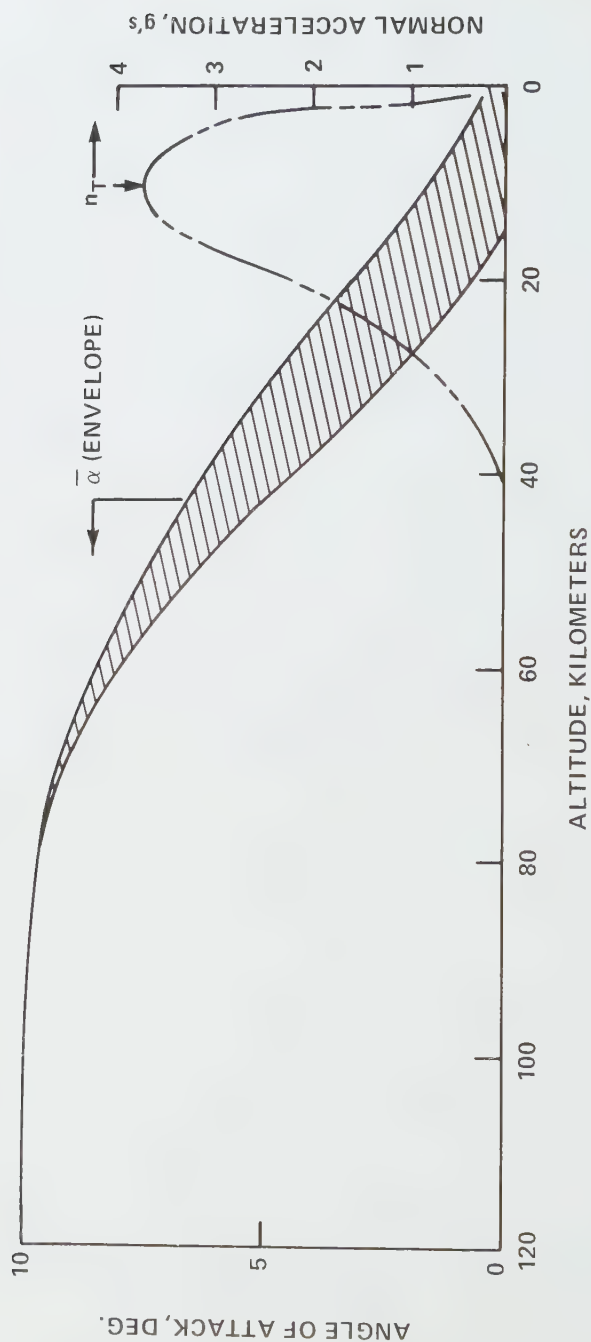


Fig. 9.5 Normal acceleration and angle of attack vs altitude.

i.e., during the Keplerian phase. It shall be shown that if the RV at off-loading is spun about an axis close to but not coincident with the minor principal axis or axis of mass symmetry, then the RV's axis of symmetry will "cone" or precess about some axis fixed in inertial space. In Chapter XII we shall show that the direction of this fixed axis is that of the angular momentum vector.

A small error in angle of attack should cause no significant error at impact; however, large angle-of-attack errors and/or lateral or yaw angle errors cause coning motion which integrates into a large coning radius of many miles. Even though the angular motion dissipates to near zero, the displacement error causes a corresponding large impact error as indicated in Fig. 9.6.

In Fig. (9.1) we indicated the occurrence of the events "first resonance" and "second resonance." It shall be shown in Chapters X and XIII that resonance in angle-of-attack magnitude occurs when the spin or roll rate of the RV equals (or nearly equals) the pitch/yaw frequency.

Another important event in the RV trajectory has been noted as "boundary-layer transition." In the continuum flow regime viscous effects are essentially confined to a small layer called the "boundary layer." Within this layer the details of fluid motion—whether such motion is orderly or random—is crucial in setting the level of skin friction and heat transfer from the flow. When the shear stresses in the fluid are comparable to the pressures responsible for fluid motion, the flow within the boundary layer is orderly and streamlines are parallel to each other. As the shear stresses becomes smaller relative to the pressure causing fluid flow, the flow within the boundary layer becomes more random. The concept of a boundary layer remains valid whether the fluid is orderly (laminar) or random (turbulent). However, the velocity profile through the boundary layer and the

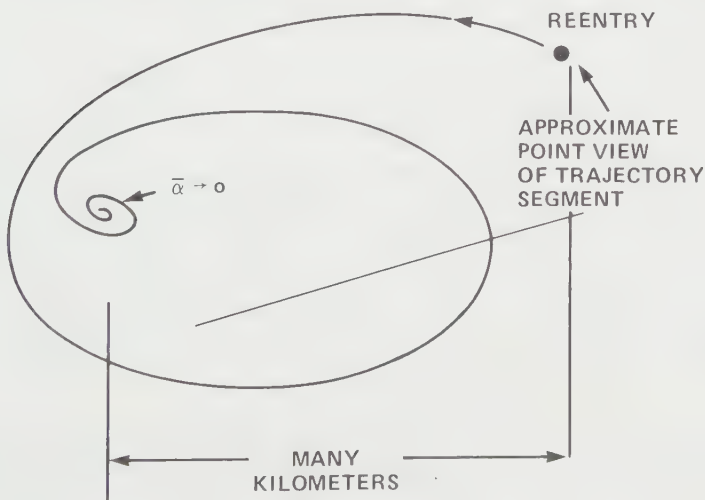


Fig. 9.6 Lateral displacement due to angle of attack.

corresponding boundary-layer thickness are markedly different between the two types of flow. There is not enough space to do justice to even an introduction of boundary-layer transition. However, a commonly used parameter for characterizing the boundary layer is the Reynolds number. The Reynolds number is the ratio of the inertial forces to the viscous forces, as

$$\frac{\text{inertial forces}}{\text{viscous forces}} = \frac{\Delta(\text{momentum})/(\text{unit time})}{\text{shear stress} \times \text{unit area}} = R_e$$

For a fluid to support a shear stress there must be relative motion (velocity differential) between adjacent layers, i.e., $\tau = \mu(dV/dy)$ where τ is the stress, μ the dynamic viscosity, and y a coordinate normal to the direction of motion. Using the above definition for the Reynolds number R_e

$$R_e = \frac{d(mV)/dt}{\mu(dV/dy)A} = \frac{[\rho L^3 V/(L/V)]}{\mu(V/L)L^2} = \frac{\rho VL}{\mu} = \frac{VL}{\nu}$$

where ν is the kinematic viscosity or μ/ρ .

The Reynolds number is a useful parameter for characterizing whether the boundary layer is turbulent or laminar. A laminar boundary layer exists when the viscous forces are large enough to damp out the oscillations caused by the dynamic forces. Thus laminar flow is characterized by low Reynolds number. Conversely, when the dynamic forces overcome the viscous forces the boundary layer becomes turbulent. In laminar flow the fluid particles move in parallel streamlines with no mixing except at the molecular level. In turbulent flow there is random mixing resulting in a large momentum exchange among fluid particles. Flow over a solid object is laminar until a critical Reynolds number is reached after which the flow is turbulent. If x is the distance from the leading edge of a flat plate and δ is the boundary-layer thickness, then the thickness of the laminar and turbulent boundary layers are

$$\frac{\delta_L}{x} = \frac{5.0}{\sqrt{R_e}} \quad \frac{\delta_T}{x} = \frac{0.3747}{R_e^{0.2}} \quad (9.6)$$

When the Reynolds number reaches the critical value, transition begins. In this transition region the flow is neither entirely turbulent nor entirely laminar. It is mixed, with the upper part turbulent and the lower part laminar. A region of partial separation occurs near the surface in this transition region but disappears with the onset of fully turbulent flow. The laminar boundary layer continues to exist as a small sublayer near the surface where the velocity profile is nearly linear and very steep. The turbulent boundary layer grows roughly at ten times the rate of the laminar layer as seen in Fig. 9.7.

The velocity profiles in the laminar and turbulent boundary layers show significant differences. The magnitude of the velocity in the turbulent boundary layer is much greater, particularly near the surface. In other words the turbulent boundary layer has much more energy near the object's surface

than does the corresponding laminar layer. The velocity profile determines the skin friction on the surface and we can expect greater heat transfer to the surface in the presence of the turbulent boundary layer as shown in Fig. 9.8.

The boundary layer on a body of revolution is more complex than that on a flat plate. Nevertheless the critical Reynolds number must be identified. We might do this by noting the altitude over which transition from a fully laminar to a nearly fully turbulent boundary layer takes place as shown in Fig 9.9.

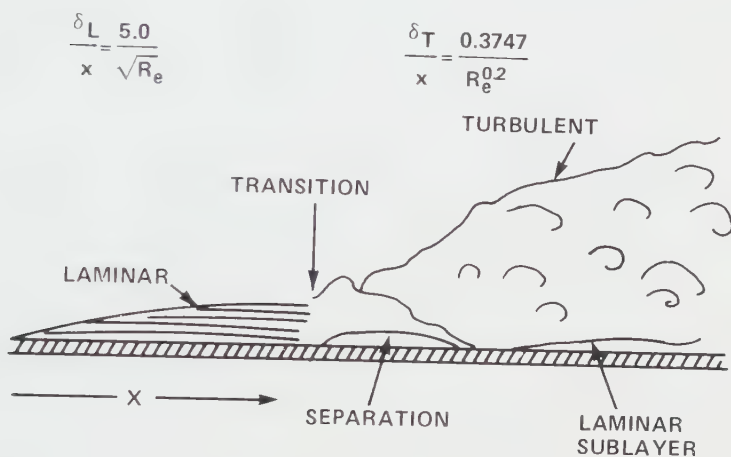


Fig. 9.7 Boundary-layer types.

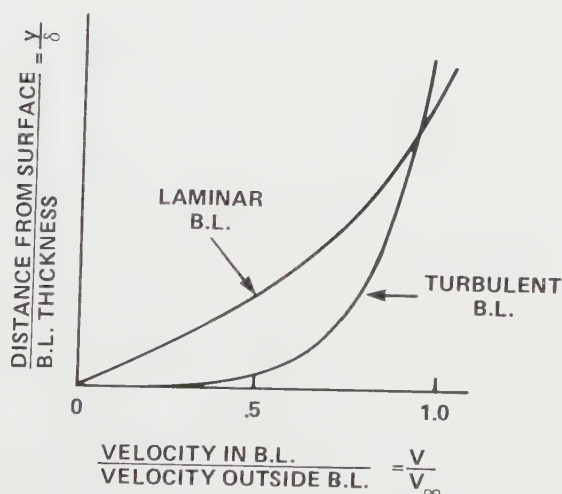


Fig. 9.8 Boundary-layer profiles.

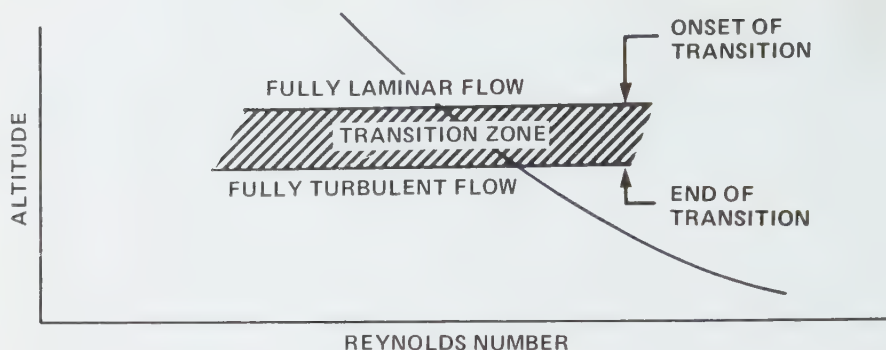


Fig. 9.9 Altitude vs Reynolds number.

9.2 Aerodynamic Coefficients and Stability Derivatives

The concept of aerodynamic force coefficients was introduced in Chapter VI, with the drag coefficient C_D defined in Eq. (6.1). Essentially the aerodynamic force derivatives are a normalization of the dimensional aerodynamic force by a reference area (here we use base area) and dynamic pressure; the moment derivatives are the normalization of the aerodynamic moment by a reference area, reference length (taken in this book as body diameter), and dynamic pressure as

$$\begin{aligned} -C_A = C_x &= \frac{X}{q} S & C_D &= \frac{F_D}{qS} = \frac{D}{qS} & C_y &= \frac{Y}{qS} \\ -C_N = C_z &= \frac{Z}{qS} & C_L &= \frac{F_L}{qS} = \frac{L}{qS} \end{aligned} \quad (9.7)$$

for the force coefficients and

$$C_l = \frac{L}{qSd} \quad C_m = \frac{M}{qSd} \quad C_n = \frac{N}{qSd} \quad (9.8)$$

for the moment coefficients. As pointed out the dynamic pressure q equals $\frac{1}{2}\rho V^2$ and the reference area S equals $\pi d^2/4$, where d is the maximum body diameter. C_A and C_N , the axial and normal force coefficients, are alternate terminology to C_x and C_z since C_x is always negative and C_z is usually negative for positive angles of attack. C_D and C_L are the drag and lift coefficients. Whether (C_D, C_L) or (C_A, C_N) are used depends upon the choice of axis systems. If a wind axis system is used then (C_D, C_L) would be appropriate. A body axis would require (C_x, C_z) or their equivalent (C_A, C_N) . The body axis system has been used in most of the analysis. A comparison between both axis systems is shown in Fig. 9.10.

Clearly whether the wind axis system (C_D, C_L) or the body axis system

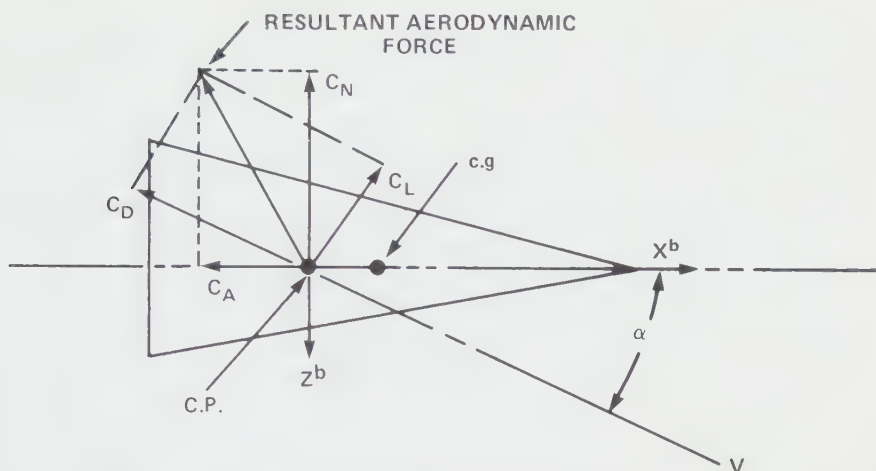


Fig. 9.10 Components of aerodynamic force (coefficient form).

(C_A, C_N) is used, the vectorial sum must be equal the total load on the body. The utility of aerodynamic coefficients is that they permit generalization of aerodynamic data/measurements. The aerodynamic coefficients are functions of body shape and for other than inviscid and incompressible flows, they are also functions of the various similarity parameters. The three such parameters introduced so far, the Knudsen number, the Mach number, and the Reynolds number are the only ones which we shall assume to be significant.

The aerodynamic flowfield in encountering an RV sets up a pressure distribution about the RV. The surface integral of this pressure distribution produces the force. The first moment of this distribution gives the aerodynamic moment. Appropriate normalization of these factors and moments by dynamic pressure and reference area and length defines the aerodynamic coefficients as we have done in Eqs. (9.7) and (9.8).

In the succeeding chapters, especially Chapters X, XI, and XIII we shall come to grips with aerodynamic forces and moments which may vary with the state variables in a fashion that is complex and at times imperfectly understood. For example the pitching moment C_m may be highly nonlinear in the vicinity of a particular trim-angle of attack. Nevertheless we shall endeavor to bring the applied aerodynamics into some tractable form, recognizing that at times such simplifications may not be appropriate. Our approach will be to express the force and moment as the first (linear) terms of a Taylor series expansion in various state variables. The variables chosen will be those which experience has shown to have a significant influence on a given force or moment. A typical expansion shall be given in Eq. (13.4). At this point we shall merely define some of the derivatives and leave their significance to later chapters.

Let us take as an example the pitching moment coefficient C_m . If we assume that the RV is undergoing perturbational motion from trim condi-

tion, then state variables of interest might be (u, v, w, p, q , and r). We can nondimensionalize velocity magnitudes by dividing by the steady-state velocity magnitude V_∞ as

$$u/V_\infty = \mathbf{u} \quad v/V_\infty = \tan\beta \cong \beta \quad w/V_\infty = \tan\alpha \cong \alpha$$

The variables α and β are recognized as the angle of attack and the angle of sideslip; $\mathbf{u}$ is the nondimensional velocity component along the x axis. The angular velocity components are nondimensionalized by multiplying by an entity which has the dimensions of time. Such an entity may be taken as d/V_∞ , where d is the maximum body diameter. Thus

$$pd/V_\infty \quad qd/V_\infty \quad rd/V_\infty$$

are appropriate variables representing nondimensional components of the angular velocity. Returning now to the pitching moment coefficients C_m we have as the linear expansion of this coefficient,

$$C_m = C_{m_0} + \left[\frac{\partial C_m}{\partial \mathbf{u}} \right] \mathbf{u} + \left[\frac{\partial C_m}{\partial \beta} \right] \beta + \left[\frac{\partial C_m}{\partial \alpha} \right] \alpha \\ + \frac{\partial C_m}{\partial (pd/V_\infty)} \left[\frac{pd}{V_\infty} \right] + \frac{\partial C_m}{\partial (qd/V_\infty)} \left[\frac{qd}{V_\infty} \right] + \frac{\partial C_m}{\partial (rd/V_\infty)} \left[\frac{rd}{V_\infty} \right] \quad (9.9)$$

It turns out that often the effects represented by the following derivatives are negligible.

$$\frac{\partial C_m}{\partial u} \quad \frac{\partial C_m}{\partial \beta} \quad \frac{\partial C_m}{\partial (pd/V_\infty)} \quad \frac{\partial C_m}{\partial (rd/V_\infty)}$$

On the other hand, there may be significant effects due to the time derivative of angle of attack (essentially the effect in question is due to an acceleration along the z axis). A moment due to $\dot{w}$ or $\dot{\alpha}$ may be represented as

$$\left[\frac{\partial C_m}{\partial \dot{\alpha}} \right] \dot{\alpha} = \frac{\partial C_m}{\partial (\dot{\alpha}d/V_\infty)} \left[\frac{\dot{\alpha}d}{V_\infty} \right] = C_{m_{\dot{\alpha}}} \left[\frac{\dot{\alpha}d}{V_\infty} \right]$$

We have treated $\dot{\alpha}$ as an angular rate and nondimensionalized it appropriately. It also turns out that for rapidly rolling configurations, we might need to retain some coupling terms from the second derivatives in the Taylor series. The more important of these coupling terms is called the "Magnus effect," giving a pitching moment in the combined presence of angle of sideslip and roll rate. The appropriate derivative is

$$\frac{\partial^2 C_m}{\partial(pd/V_\infty) \partial \beta} \left[\frac{pd}{V_\infty} \right] \beta = C_{m_{p\beta}} \left[\frac{pd}{V_\infty} \right] \beta$$

It might be pointed out for most problems in RV dynamics neither the angle-of-attack rate derivative C_{m_α} nor the Magnus derivative $C_{m_{p\beta}}$ are particularly significant. C_{m_α} represents the inability of a flow to adjust instantaneously to an angular change of the configuration or equivalently, the inability of the pressure field to adjust instantaneously to a change in flow direction. However, in hypersonic flows, the transverse velocity imposed on a fluid particle to match the linear velocity of a surface element of the body due to body acceleration is negligibly small in comparison to the velocity of the particle due to motion in the undisturbed flowfield. For example in Fig. 9.11 the RV is shown with a linear acceleration along the z axis (nearly perpendicular to the flow velocity vector V_∞). Because of the definition of the angle of attack α , as $\alpha = w/V_\infty$, an acceleration $\dot{w}$ would kinematically appear as an angular rate $\dot{\alpha}$ of the RV relative to the velocity vector. As a result of this acceleration $\dot{w}$, the apparent rotation of a point on the surface at a distance $|\mathbf{r}|$ from the origin of the body frame would appear to have a velocity $|\mathbf{r}| \dot{\alpha}$ nearly at right angles to V_∞ . In hypersonic flows $|\mathbf{r}| \dot{\alpha}$ would be negligible in comparison to V_∞ .

The Magnus derivative $C_{m_{p\beta}}$ requires both a roll rate p and an angle of sideslip β . For most RV's the roll rate is sufficiently small that the Magnus effect may be neglected. Again, remember the roll variable is the nondimensional roll-rate pd/V_∞ and not the roll-rate p alone. Since V_∞ is large in hypersonic flows, the nondimensional roll rate is often negligible.

We may write the series expansion of Eq. (9.9) in subscript form as

$$C_m = C_{m_0} + C_{m_\alpha} \alpha + C_{m_q} (qd/V_\infty) \quad (9.10)$$

where we have used the subscript rotation to indicate differentiation as

$$\frac{\partial C_m}{\partial \alpha} \triangleq C_{m_\alpha} \quad \frac{\partial C_m}{\partial(qd/V_\infty)} \triangleq C_{m_q}$$

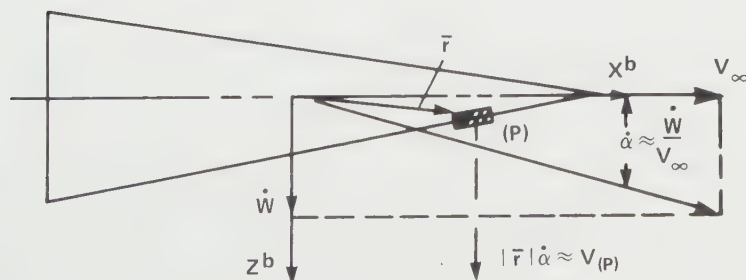


Fig. 9.11 Linear acceleration and angular rate.

The term C_{m_0} is the moment that exists at $\alpha = 0$. If trim is defined as the condition where the applied moment and all rates of change of angular variables are zero, then we may define the trim angle-of-attack α_t by setting C_m and q to zero in Eq. (9.10).

$$0 = C_{m_0} + C_{m_\alpha} \alpha_t$$

or

$$C_{m_0} = -C_{m_\alpha} \alpha_t$$

The series of Eq. (9.9) may be rewritten as

$$C_m = C_{m_\alpha}(\alpha - \alpha_t) + C_{m_q}(qd/V_\infty) \quad (9.11)$$

Equation (9.11) is nearly identical to Eq. (13.4e). The other five coefficients may be expanded in a similar fashion. The various terms in such series will have to be assessed for their physical significance. The only point to be made here is to emphasize the convention used in nondimensionalizing the various derivatives. If ξ is an angle and m is a moment, then the rate of change of the moment coefficient C_m with respect to ξ is

$$\frac{\partial C_m}{\partial \xi} \xi \triangleq C_{m_\xi} \xi$$

If $\dot{\xi}$ is an angular rate, then we have

$$\frac{\partial C_m}{\partial \dot{\xi}} \dot{\xi} = \frac{\partial C_m}{\partial (\dot{\xi} d / V_\infty)} \left(\frac{\dot{\xi} d}{V_\infty} \right) \triangleq C_{m_{\dot{\xi}}} \left(\frac{\dot{\xi} d}{V_\infty} \right)$$

The final term consists of the nondimensionalized derivative multiplied by the nondimensionalized variable. For what interest it may have if ξ is an angular acceleration (dimensions of $1/s^2$) then the nondimensionalized derivative and variable become:

$$\frac{\partial C_m}{\partial \ddot{\xi}} \ddot{\xi} = \frac{\partial C_m}{\partial (\ddot{\xi} d^2 / V_\infty^2)} \left(\frac{\ddot{\xi} d^2}{V_\infty^2} \right) \triangleq C_{m_{\ddot{\xi}}} \left(\frac{\ddot{\xi} d^2}{V_\infty^2} \right)$$

Regardless of how we obtain aerodynamic coefficients and derivatives—measurement, calculation, or even inspired guess—they represent applied aerodynamics' contribution to aeronautical dynamics. Space does not permit an adequate general treatment of load measurement and calculation. However, it is possible to give a sample of the calculation of the axial and normal force moment coefficients in hypersonic continuum flow.

9.3 Newtonian Impact Theory

Newtonian impact theory is a convenient method for calculating loads on fairly blunt bodies at hypersonic speeds. The Newtonian impact theory assumes that the impact of fluid particles at the vehicle's surface completely destroys the component of fluid momentum normal to the surface. Thus the normal component of momentum is converted to a pressure on the body while the tangential component remains unchanged: normal momentum of the fluid is lost while the tangential component is conserved.

In Fig. 9.12 we have depicted a cone of length L and base radius R . We also have indicated the elemental area dA as a band around the conical model. The band has an axial thickness of dx . Let us examine the pressure acting at this elemental ring.

In Fig. 9.13(a), at left is an isometric view of the elemental area dA which takes the form of the surface of a truncated cone. At the right in Fig. 9.13(a), we have presented a small strip, taken from the element dA . The width of this strip is $y d\phi$, and the length is $d\ell$. We shall identify the area of this strip as da which of course equals $y d\phi d\ell$. We shall assume that the width $y d\phi$ is so narrow that curvature may be neglected. Since the elementary strip da is assumed to be planar, we can depict an edge view of this plane.

At the left in Fig. 9.13(b) we see that because of the impermeable nature of the RV's surface, the freestream is turned tangent to the surface or the velocity vector $\mathbf{V}_\infty$ changes direction to become the tangential velocity vector $\mathbf{V}_T$.

The tangential velocity $\mathbf{V}_T$, the freestream velocity $\mathbf{V}_\infty$, and the normal velocity $\mathbf{V}_N$, obey the following vector relationship,

$$\mathbf{V}_T = \mathbf{V}_\infty + \mathbf{V}_N \quad (9.12)$$

At the right in Fig. 9.13(b) we have shown a control volume of cross-

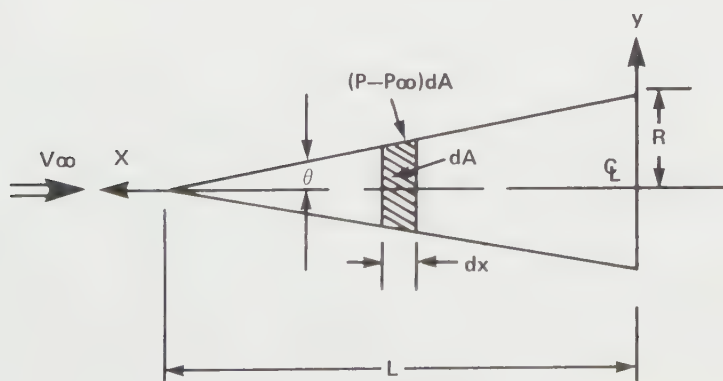


Fig. 9.12 Incremental pressure on cone.

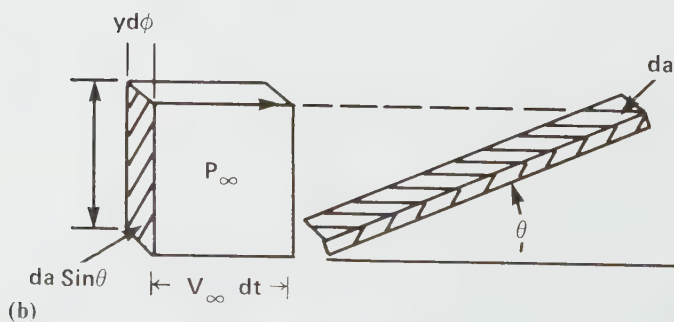
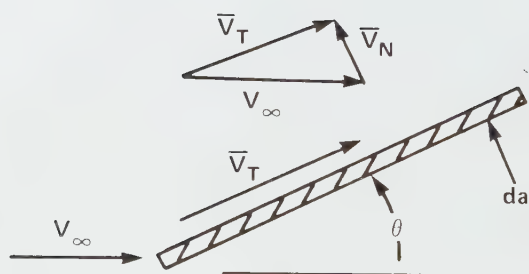
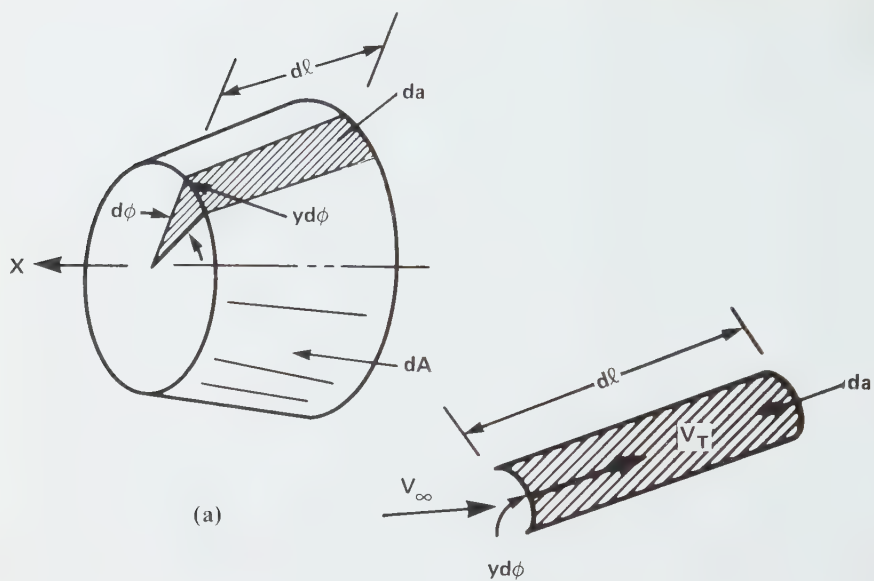


Fig. 9.13 Elemental pressure.

sectional area $(\sin\theta)da$ and length $V_\infty dt$. Consequently the volume $d\forall$ of the control volume is

$$d\forall = (V_\infty \sin\theta)da dt \quad (9.13)$$

If ρ_∞ is the density of the air in this control volume, then an expression for the mass flow follows from Eq. (9.13) as

$$\dot{m} \triangleq \frac{dm}{dt} = (V_\infty \rho_\infty \sin\theta)da \quad (9.14)$$

Newton's second law of motion as applied to fluids requires that the normal force exerted by the fluid on the element area, da equals the product of the mass flow $-\dot{m}$ times the vectorial change in the velocity,

$$d\mathbf{F} = -\dot{m} \Delta \mathbf{V} = \dot{m} (\mathbf{V}_\infty - \mathbf{V}_T) = \dot{m} \mathbf{V}_N \quad (9.15a)$$

Now the elementary force $d\mathbf{F}$ must equal the pressure acting on the area da . The direction of this force is along the local normal $\mathbf{n}$. Thus, we have from the previous equation

$$\begin{aligned} d\mathbf{F} &= \dot{m} \mathbf{V}_N = (\rho_\infty V_\infty da \sin\theta) \mathbf{V}_N = (p - p_\infty) da \mathbf{n} \\ |d\mathbf{F}| &= \rho_\infty V_\infty (\sin\theta) V_N da \end{aligned} \quad (9.15b)$$

Solving for the pressure difference $p - p_\infty$ we get:

$$(p - p_\infty) = (\rho_\infty V_\infty \sin\theta) V_N$$

but since $V_N = V_\infty \sin\theta$, we have

$$(p - p_\infty) = \rho_\infty V_N^2 \quad (9.16)$$

It is fairly straightforward procedure to use Eq. (9.16) to calculate the loads on simple shapes such as cones and hemispheres. This we shall do subsequently. Let us first consider how we might, at least in principle, compute the forces and moments on a fairly arbitrary configuration. In Fig. 9.14, we have depicted such an arbitrary shape. For convenience in carrying out subsequent integrations we have established a coordinate system at the vertex such that the x axis points rearward. The y and z axes are orthogonal to each other and to the x axis, but otherwise quite arbitrary in direction. On the surface of the configuration we have located an elemental area da and a local surface normal $\mathbf{n}$. The freestream velocity vector $\mathbf{V}_\infty$ makes an angle α with respect to the x axis.

The elemental area is located by a vector $\mathbf{r}$ from the origin of the axis system as by the magnitude r_b the distance normal to the x axis. If the body's

shape function $g(x, y, z) = 0$ is known, then the vector $\mathbf{n}$ is easily found from the familiar vector operations:

$$\mathbf{n} = \left\{ \left[\frac{\partial g}{\partial x} \right] \mathbf{i} + \left[\frac{\partial g}{\partial y} \right] \mathbf{j} + \left[\frac{\partial g}{\partial z} \right] \mathbf{k} \right\} / \left[\left[\frac{\partial g}{\partial x} \right]^2 + \left[\frac{\partial g}{\partial y} \right]^2 + \left[\frac{\partial g}{\partial z} \right]^2 \right]^{1/2}$$

$$= \frac{\Delta g}{|\Delta g|} \quad (9.17)$$

Assuming that the velocity vector $\mathbf{V}_\infty$ lies in the x - y plane, we have

$$\mathbf{V}_\infty = (V_\infty \cos \alpha) \mathbf{i} + (V_\infty \sin \alpha) \mathbf{j} \quad (9.18)$$

The force differential $d\mathbf{F}$ acting *on* the area element da is

$$d\mathbf{F} = -(p - p_\infty) \mathbf{n} da \quad (9.19a)$$

Now it will be noted in Fig. 9.14 that we have also constructed at the vertex, the conventional body axes (X^b, Y^b, Z^b). Thus, we may write $d\mathbf{F}$ in coefficient form as

$$d\mathbf{F} = (\rho_\infty V_\infty^2 \pi R^2 / 2) [d(C_A) \mathbf{i} + d(C_N) \mathbf{j} - d(C_Y) \mathbf{k}] \quad (9.19b)$$

where C_A , C_N , and C_Y are respectively the axial force coefficient, the normal force coefficient, and the side force coefficient. The term πR^2 is the reference area, and if R is the base radius, then the reference area is the base area. In the sequel the reference area will be the base areas; however, if we do not confine our attention to bodies of revolution then we might replace πR^2 by base area itself since the base radius R would not have meaning.

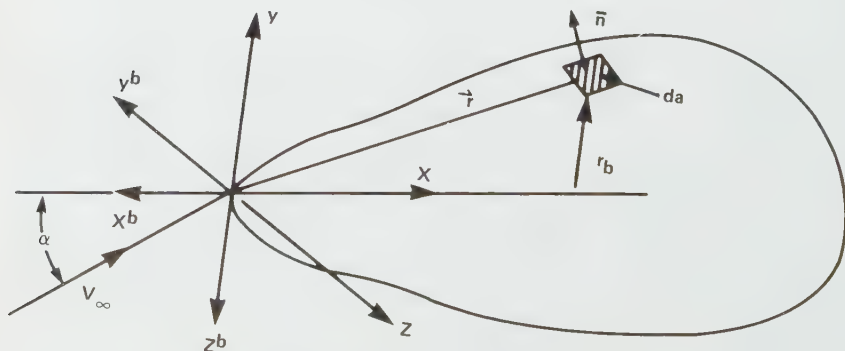


Fig. 9.14 Axis systems.

We can now equate Eqs. (9.19a) and (9.19b) to provide formal expressions for the coefficients. If we confine attention to C_A and C_N we have

$$C_A = - \left[\frac{2}{\rho_\infty V_\infty^2 \pi R^2} \right] \iint_S (\mathbf{n} \cdot \mathbf{i})(p - p_\infty) da \quad (9.20a)$$

$$C_N = - \left[\frac{2}{\rho_\infty V_\infty^2 \pi R^2} \right] \iint_S (\mathbf{n} \cdot \mathbf{j})(p - p_\infty) da \quad (9.20b)$$

Obviously if we can express the local normal $\mathbf{n}$, the pressure difference $(p - p_\infty)$ and the area element da in terms of the same coordinate variables (i.e., x, y, z) then in principle we can obtain the above force coefficients. However unless we confine our attention to simple shapes, such as cones and hemispheres, it is almost impossible to carry out closed-form integrations of Eqs. (9.20). Before we accept this limitation let us find similar expressions for the moment coefficient. The elemental moment vector $d\mathbf{M}$ may be written as

$$d\mathbf{M} = \mathbf{r} \times d\mathbf{F} = -(p - p_\infty)(\mathbf{r} \times \mathbf{n}) da \quad (9.21a)$$

Next let us write the elemental moment vector in coefficient form. In order for these coefficients to be nondimensional, we must divide the moment by a reference length as well as a reference pressure $\rho_\infty V_\infty^2 / 2$ and reference area πR^2 . Anticipating our interest in bodies of revolution we shall use the base diameter $2R$ as this reference length. Thus

$$d\mathbf{M} = (\rho_\infty V_\infty^2 \pi R^3) [-d(C_l)\mathbf{i} - d(C_m)\mathbf{k} - d(C_n)\mathbf{j}] \quad (9.21b)$$

where C_l , C_m , and C_n are, respectively, the rolling moment, pitching moment, and yawing moment coefficients about the conventional body axis. Let us confine our attention to the pitching moment coefficient C_m . Equating Eqs. (9.21a) and (9.21b) we get

$$C_m = + \left[\frac{1}{\rho_\infty V_\infty^2 \pi R^3} \right] \iint_S (\mathbf{r} \times \mathbf{n}) \cdot \mathbf{k} (p - p_\infty) da \quad (9.21c)$$

Again to restate the obvious, the vectors $\mathbf{r}$, $\mathbf{n}$, $\mathbf{k}$, the pressure difference, $(p - p_\infty)$, and the area element da must be expressed in terms of the coordinates (x, y, z) ; also if we are entertaining any idea of a closed-form solution, we must restrict ourselves to simple shapes.

The procedure we must now follow has two steps: (a) select a body shape, which is equivalent to establishing analytic expressions for the area element da , the radius vector $\mathbf{r}$, the area element and the configuration normal $\mathbf{n}$; (b) establish a model for the flow physics which will provide us with the pressure difference $p - p_\infty$ defined at least piecewise continuously over the surface of the configuration.

To illustrate the procedure of using Eqs. (9.20) and (9.21) to obtain the

force and moment coefficients we shall restrict our attention to a right circular cone which is subjected to a Newtonian pressure field. Figure 9.15 shows a right circular cone of half angle θ of length L and base radius R .

We can first find the normal to a generic surface element from Eq. (9.17). The required configuration function for a cone is easily shown to be

$$g(x, y, z) = (y^2 + z^2) - (R/L)^2 x^2 = 0 \quad (9.22)$$

The derivatives in the numerator of Eq. (9.17) are easily shown to be

$$\frac{\partial g}{\partial x} = -2x \left[\frac{R}{L} \right]^2 = -2x \tan^2 \theta \quad (9.23a)$$

$$\frac{\partial g}{\partial y} = 2y \quad (9.23b)$$

$$\frac{\partial g}{\partial z} = 2z \quad (9.23c)$$

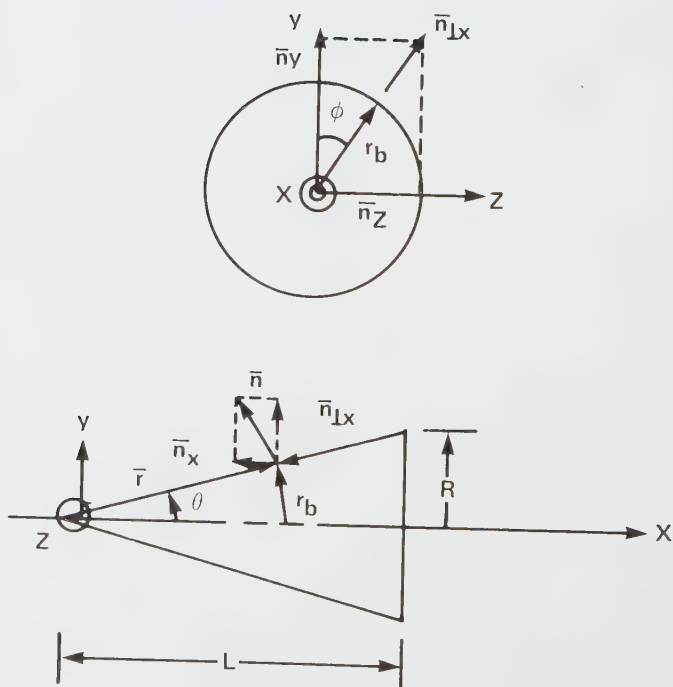


Fig. 9.15 Conical model.

Since a cone is a curved plane, we would expect that only two coordinates are needed to describe the location of the surface element da . From Fig. (9.15) we note that

$$y = r_b \cos\phi = x \tan\theta \cos\phi \quad (9.24a)$$

$$z = r_b \sin\phi = x \tan\theta \sin\phi \quad (9.24b)$$

Thus, for a cone (of fixed vertex angle θ) we have two coordinates to locate the surface element da : x and ϕ , an axial and meridional coordinate. Using Eqs. (9.23) and (9.24) we obtain for the denominator of Eq. (9.17),

$$|\nabla g| = 2x \tan\theta \sec\theta \quad (9.25)$$

Inserting Eqs. (9.23–9.25) into Eq. (9.17) we get

$$\mathbf{n} = (-\sin\theta)\mathbf{i} + (\cos\theta \cos\phi)\mathbf{j} + (\cos\theta \sin\phi)\mathbf{k} \quad (9.26)$$

The freestream velocity vector $\mathbf{V}_\infty$ stated in Eq. (9.18) will be given again as

$$\mathbf{V}_\infty = (V_\infty \cos\alpha)\mathbf{i} + (V_\infty \sin\alpha)\mathbf{j} \quad (9.27)$$

The radius vector from the origin of the axis system to the area element da is given as

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

or using Eqs. (9.24),

$$\mathbf{r} = x\mathbf{i} + (x \tan\theta \cos\phi)\mathbf{j} + (x \tan\theta \sin\phi)\mathbf{k} \quad (9.28)$$

In Fig. 9.16 we indicated the elemental area da on the surface of the cone.

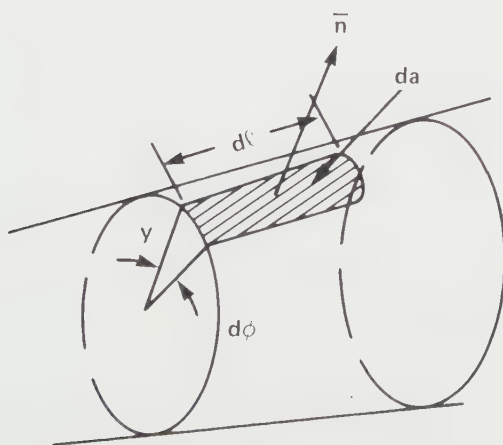


Fig. 9.16 Area element.

We shall now proceed to integrate the surface pressure over surface of the cone, taking the component of the force normal to the x axis as the normal force and the component along the x axis as the axial force.

Using the Newtonian hypothesis, the pressure differential $p - p_\infty$ may be written, from Eq. (9.16) as

$$(p - p_\infty) = \rho_\infty V_N^2 = \rho_\infty (\mathbf{V}_\infty \cdot \mathbf{n})^2 \quad (9.29)$$

From Fig. 9.3 we may express the element da in terms of the coordinates (x, ϕ) as

$$da = r_b d\phi dl = x \tan\theta \sec\theta dx d\phi \quad (9.30)$$

where we have made use of the relationships

$$r_b = x \tan\theta \quad dl = \sec\theta dx$$

Since

$$\mathbf{n} \cdot \mathbf{i} = -\sin\theta$$

$$\mathbf{n} \cdot \mathbf{j} = \cos\theta \cos\phi$$

$$\mathbf{V}_\infty \cdot \mathbf{n} = [-\sin\theta \cos\alpha + \cos\theta \sin\alpha \cos\phi] V_\infty$$

we may write Eqs. (9.20) as

$$C_A = \frac{2}{\pi R^2} \left[\int_0^L \int_0^{2\pi} (\sin^2\theta \cos^2\alpha - 2\sin\theta \cos\theta \sin\alpha \cos\alpha \cos\phi + \cos^2\theta \sin^2\alpha \cos^2\phi) \tan^2\theta dx d\phi \right] \quad (9.31a)$$

$$C_N = \frac{-2}{\pi R^2} \left[\int_0^L \int_0^{2\pi} (\sin^2\theta \cos^2\alpha - 2\sin\theta \cos\theta \sin\alpha \cos\alpha \cos\phi + \cos^2\theta \sin^2\alpha \cos^2\phi) \cos\phi \tan\theta dx d\phi \right] \quad (9.31b)$$

Let us consider the integration of Eq. (9.31a). Integrating with respect to ϕ first, we get

$$\begin{aligned} & \int_0^{2\pi} (\sin^2\theta \cos^2\alpha + 2\sin\theta \cos\theta \sin\alpha \cos\phi + \cos^2\theta \sin^2\alpha \cos^2\phi) d\phi \\ &= 2\pi(\sin^2\theta \cos^2\alpha) + \pi(\cos^2\theta \sin^2\alpha) \end{aligned}$$

The second integration with respect to x then becomes

$$\begin{aligned} C_A &= \frac{2}{R^2} \left[\int_0^L (2\sin^2\theta \cos^2\alpha + \cos^2\theta \sin^2\alpha) \tan^2\theta x \, dx \right] \\ &= \frac{L^2}{R^2} (2\sin^2\theta \cos\alpha + \cos^2\theta \sin^2\alpha) \tan^2\theta \end{aligned}$$

but $R/L = \tan\theta$, so

$$C_A = 2\sin^2\theta + \sin^2\alpha(1 - 3\sin^2\theta) \quad (9.32)$$

We may now return to Eq. (9.31b) and carry out the indicated integration, first with respect to ϕ .

$$\begin{aligned} \int_0^{2\pi} [\sin^2\theta \cos^2\alpha \cos\phi - 2\sin\alpha \cos\alpha \sin\theta \cos\theta \cos^2\phi \\ + \cos^2\theta \sin^2\alpha \cos^3\phi] d\phi \\ = 2\pi \sin\alpha \cos\alpha \sin\theta \cos\theta \\ = -\pi \cos\theta \sin\theta \sin 2\alpha \end{aligned}$$

Integrating with respect to x we now get

$$\begin{aligned} C_N &= \frac{2}{R^2} \left[\int_0^L (\cos\theta \sin\theta \sin^2\alpha) \tan\theta x \, dx \right] \\ &= \frac{2}{R^2} \left[\int_0^L \sin^2\theta \sin^2\alpha x \, dx \right] = \left[\frac{L}{R} \right]^2 \sin^2\theta \sin^2\alpha \end{aligned} \quad (9.33)$$

Again noting that $L/R = \cot\theta$, we get

$$C_N = \cos^2\theta \sin 2\alpha \quad (9.34)$$

We might point out that when the angle-of-attack α is zero, the zero-lift drag coefficient follows from Eq. (9.32) as,

$$C_{A_{\alpha=0}} = C_{D_0} = 2\sin^2\theta \quad (9.35)$$

Before abandoning our force equations let us calculate the drag coefficient for a hemisphere. Many conical RV's have blunted noses and this bluntness often takes the form of a hemisphere. It is a fairly straightforward exercise to show that the drag coefficient of a hemisphere is unity if the base area is used for reference.

Figure 9.17 depicts a hemisphere with a Cartesian coordinate system at the center of the semicircular meridian. It will be noted that the x axis is in a direction opposite to that shown in Fig. 9.15. Consequently

$$\mathbf{V}_\infty = -V_\infty \mathbf{i} \quad (9.36)$$

and

$$C_A \big|_{\alpha=0} = C_{D_0} \left[\frac{2}{\pi r_b^2 \rho_\infty V_\infty^2} \right] \iint_S (\mathbf{n} \cdot \mathbf{i}) \rho_\infty (\mathbf{V}_\infty \cdot \mathbf{n})^2 da \quad (9.37)$$

The minus sign in Eq. (9.20a) has been avoided because of the reversal in the direction of the x axis.

The area element da may be written in terms of r_b , ϕ , and θ as follows

$$da = r_b^2 \sin\theta d\theta d\phi \quad (9.38)$$

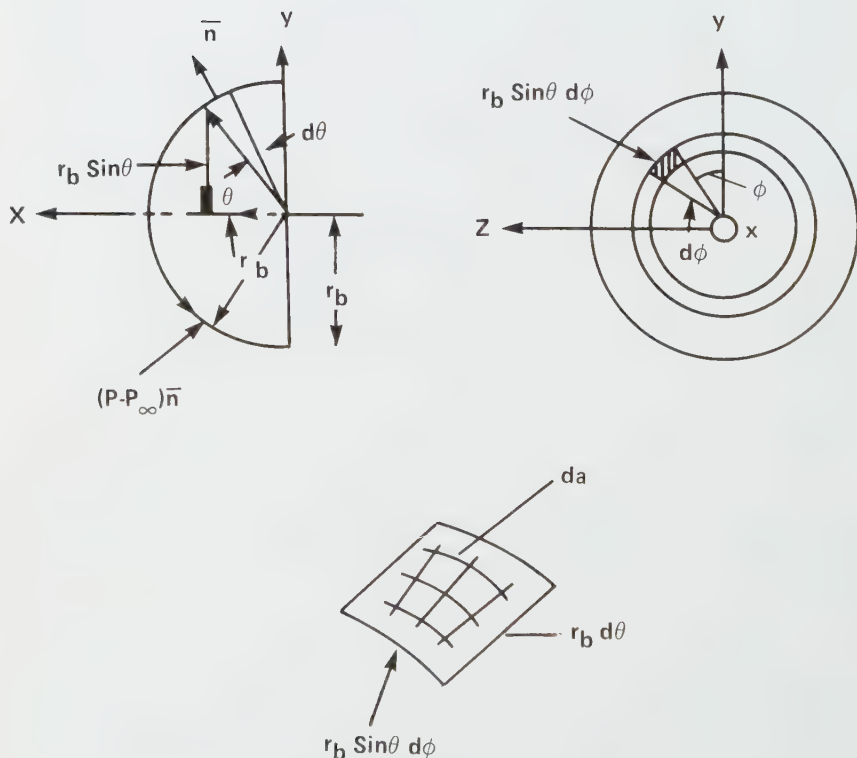


Fig. 9.17 Hemisphere coordinates.

The shape function $g(x, y, z)$ for a hemisphere is

$$g(x, y, z) = (x^2 + y^2 + z^2 - r_b^2) = 0 \quad (9.39a)$$

where

$$x = r_b \cos\theta \quad y = r_b \sin\theta \cos\phi \quad z = r_b \sin\theta \sin\phi \quad (9.39b)$$

From Eq. (9.17) we get:

$$\mathbf{n} = (\cos\theta)\mathbf{i} + (\sin\theta \cos\phi)\mathbf{j} + (\sin\theta \sin\phi)\mathbf{k} \quad (9.40)$$

Clearly

$$(\mathbf{V}_\infty \cdot \mathbf{n})^2 = (-\cos\theta)^2 = \cos^2\theta \quad \mathbf{n} \cdot \mathbf{i} = \cos\theta$$

Thus, Eq. (9.37) becomes

$$C_D = \left[\frac{2}{\rho_\infty V_\infty^2 \pi r_b^2} \right] \left[\int_0^\pi \int_0^{2\pi} (\rho_\infty V_\infty^2 r_b^2) \cos^3\theta \sin\theta d\phi d\theta \right]$$

Integrating with respect to ϕ gives

$$C_D = 4 \int_0^{\pi/2} \cos^3\theta \sin\theta d\theta = -4 \int_0^{\pi/2} \cos^3\theta d(\cos\theta)$$

and integrating again with respect to $\cos\theta$ gives

$$C_D = 1$$

Example 9.1: Calculation of Aerodynamic Coefficients and Center of Pressure

In the integrations leading to expressions for the axial and normal force coefficients for a cone, R is the base radius of the cone; in the present discussion of the drag coefficient of a hemisphere, r_b is the radius of the base. Suppose we require the drag coefficient of a blunted cone (see Fig. 9.18) where r_b is the radius of the hemispherical nose and R is the base radius of the cone. Let us define “nose bluntness” (or simply “bluntness”) as $B = r_b/R$.

First we must recognize that we only require that portion of the hemispherical nose that is tangent to the cone. If the cone has a half-angle of measure θ_c , then we must modify the second integration (i.e., over the variable θ) of the hemispherical nose. Thus,

$$C_D = C_{D_h} = \left[\frac{2}{\rho_\infty V_\infty^2 r_b^2} \right] \left[\int_0^{\pi/2 - \theta_c} \int_0^{2\pi} (\rho_\infty V_\infty^2 r_b^2) \cos^3\theta \sin\theta d\phi d\theta \right]$$

where the upper limit on θ must be diminished by the cone half-angle θ_c to meet tangency requirements. The above expression is easily shown to be:

$$C_{D_h} = (1 - \sin^4 \theta_c) \quad (9.41)$$

Now if D_c is the drag of the cone and D_{c_1} is the drag of the nose section, which will be replaced by the hemisphere, then the drag D_{c_2} of the truncated cone is

$$D_{c_2} = D_c - D_{c_1}$$

From Eq. (9.35) we may rewrite the above expression as

$$D_{c_2} = 2q\pi \sin^2 \theta_c (R^2 - r_b^2)$$

where q is the dynamic pressure, θ_c is the cone half-angle, and R and r_b are the base radius of the cone and conical nose tip, respectively. From Eq. (9.41) the drag D_h of the hemispherical nose is

$$D_h = C_{D_h} q \pi r_b^2 = q \pi r_b (1 - \sin^4 \theta_c)$$

The drag D_{ch} of the truncated cone to which has been added the (tangential) hemispherical nose tip may be written as:

$$D_{ch} = 2q\pi \sin^2 \theta_c (R^2 - r_b^2) + \pi r_b^2 q (1 - \sin^4 \theta_c)$$

Collecting terms and defining bluntness B as r_b / R , we get:

$$C_{d_{ch}} = 2\sin^2 \theta_c + B^2 (1 - 2\sin^2 \theta_c - \sin^4 \theta_c)$$

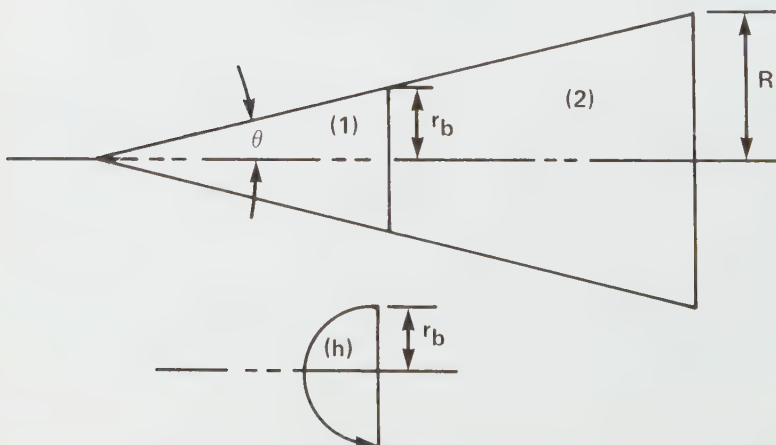


Fig. 9.18 Conical and hemispherical noses.

The above formula for $C_{d_{ch}}$, the zero-lift drag coefficient of a hemispherically blunted cone, has been presented graphically in Fig. 6.8. Note that if we have a hemisphere, i.e., B equals unity, and $\theta_c = 0$. The drag coefficient is unity in agreement with Eq. (9.41); if we have a sharp cone, i.e., B equals zero, we are in agreement with Eq. (9.35).

We can make comparisons between the pressure drag coefficient using the above equation and that obtained from a finite difference calculation based upon a three-dimensional inviscid compressible flow model.² Taking as typical values

$$\theta = 20 \text{ deg} \quad B = 0.2293$$

the above formula provides

$$C_{D_{ch}} = 0.2735$$

while the value from the numerical calculations is

$$C_{D_{ch}} = 0.2584$$

at a Mach number of 20.

Let us now calculate the pitching moment coefficient C_m using the Newtonian hypothesis. The pitching moment was given as an integral expression in Eq. (9.21c). From Eqs. (9.26) and (9.28) it is easy to show that

$$(\mathbf{r} \times \mathbf{n}) \cdot \mathbf{k} = (x \cos \phi \cos \theta + x \tan \theta \cos \phi \sin \theta) \quad (9.42a)$$

since

$$\begin{aligned} p - p_\infty &= \rho_\infty (V_N^2) = \rho_\infty (\mathbf{V}_\infty \cdot \mathbf{n})^2 \\ p - p_\infty &= \rho_\infty V_\infty^2 (\sin^2 \theta \cos^2 \alpha - \sin 2\alpha \sin \theta \cos \theta \cos \phi \\ &\quad + \cos^2 \theta \sin^2 \alpha \cos^2 \phi) \end{aligned} \quad (9.42b)$$

and taking da , the differential area, from Eq. (9.30), we have for the moment equation

$$\begin{aligned} C_m &= \frac{1}{\pi R^3} \left[\int_0^L \int_0^{2\pi} [\sin^2 \theta \cos^2 \alpha - \sin 2\alpha \sin \theta \cos \theta \cos \phi \right. \\ &\quad \left. + \cos^2 \theta \sin^2 \alpha \cos^2 \phi] [x \cos \phi \cos \theta + x \tan \theta \cos \phi \sin \theta] \right. \\ &\quad \left. [x \tan \theta \sec \theta dx d\phi] \right] \end{aligned} \quad (9.43a)$$

While the above expression seems formidable, we must remember only integrals containing $\cos^2 \phi$ will be nonzero over the domain 0 to 2π . These two integrals are

$$\frac{1}{\pi R^3} \left[\int_0^L \int_0^{2\pi} [-x^2 \sin 2\alpha \sin^2 \theta \cos^2 \phi d\phi dx] \right] = I \quad (9.43b)$$

and

$$\frac{1}{\pi R^3} \left[\int_0^L \int_0^{2\pi} [-x^2 \sin 2\alpha \sin^2 \theta \tan^2 \theta \cos^2 \phi d\phi dx] \right] = II \quad (9.43c)$$

Equations (9.43) give the following for I and II

$$I = \frac{1}{3} \cot^3 \theta \sin^2 \theta \sin 2\alpha \quad II = -\frac{1}{3} \cot \theta \sin^2 \theta \sin 2\alpha$$

Since $C_m = I + II$ we have

$$\begin{aligned} C_m &= -\frac{1}{3} \cot^3 \theta \sin^2 \theta \sin^2 \alpha - \frac{1}{3} \cot \theta \sin^2 \theta \sin 2\alpha \\ C_m &= -\frac{1}{3} \cot \theta [\cot^2 \theta + 1] \sin^2 \theta \sin 2\alpha \\ C_m &= C_{m_0} = -\frac{1}{3} \cot \theta \sin 2\alpha \end{aligned} \quad (9.44)$$

where the subscript 0 indicates the moment reference at the vertex.

Equations (9.34) and (9.44) allow us to compute the stability derivatives C_{N_α} and C_{m_α} . Thus

$$C_{N_\alpha} \equiv \left. \frac{\partial C_N}{\partial \alpha} \right|_{\alpha=0} = 2 \cos^2 \theta \quad (9.45a)$$

$$C_{N_{\alpha_0}} \equiv \left. \frac{\partial C_M}{\partial \alpha} \right|_{\alpha=0} = -(\frac{2}{3}) \cot \theta \quad (9.45b)$$

If we define

$$C_{N_q} \equiv \frac{\partial C_N}{\partial (q d / V_\infty)} \quad C_{m_q} \equiv \frac{\partial C_M}{\partial (q d / V_\infty)}$$

Then we could also show that the Newtonian hypothesis leads to

$$C_{N_q} = +\frac{2}{3} \cot \theta \quad (9.45c)$$

$$C_{m_q} = \frac{1}{3} \csc^2 \theta \quad (9.45d)$$

Equations (9.34) and (9.44) allow us to locate the center of pressure from the vertex. Although we have used the body diameter $2R$ as our reference length in defining the moment coefficient C_m it might be more convenient to

express the center of pressure location in terms of the body length L , i.e., $\mathbf{x} = x/L$. Thus the center of pressure may be written as

$$\mathbf{x} = \left[\frac{x}{L} \right] = \left[\frac{2R}{L} \right] \left[\frac{\frac{2}{3}\cot\theta}{2\cos^2\theta} \right]$$

$$\mathbf{x} = \frac{2}{3}\sec^2\theta \quad (9.46)$$

which may be easily found by proceeding $2/3$ of the way along the conical generator and then intersecting the center line perpendicular from this generator as shown in Fig. 9.19.

Equations (9.45) provide us with four stability derivatives C_{N_α} , C_{m_α} , C_{N_q} , and C_{m_q} evaluated at the body vertex. If the derivatives associated with the vertex are designated by the subscript 0, we can obtain the equivalent derivative at a distance x from the vertex. As in Eq. (9.46) we shall express x in fractions of bodylength L or $\mathbf{x} = x/L$. Since our derivatives are based upon body diameter $D = 2R$ as reference length, rather than body length, the ratio L/D will appear. This ratio can be designated as the fineness ratio f_R . Thus the axis transfer relationships among the four derivatives are:

$$C_{N_{\alpha_0}} = C_{N_\alpha} \quad (9.47a)$$

$$C_{N_q} = C_{N_{q_0}} - \mathbf{x} f_R C_{N_\alpha} \quad (9.47b)$$

$$C_{m_\alpha} = C_{m_{\alpha_0}} + \mathbf{x} f_R C_{N_\alpha} \quad (9.47c)$$

$$C_{m_q} = C_{m_{q_0}} + \mathbf{x} f_R C_{N_{q_0}} - \mathbf{x} f_R C_{m_{\alpha_0}} - \mathbf{x}^2 f_R C_{N_\alpha} \quad (9.47d)$$

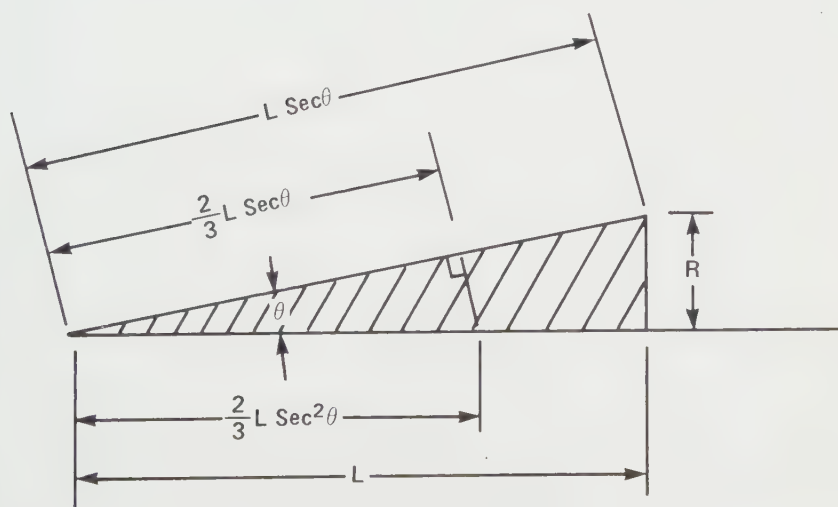


Fig. 9.19 Locating the center of pressure.

The subscript 0 refers to derivatives evaluated at the vertex. These derivatives are available from Eq. (9.45). The x variable locates the moment reference center aft of the vertex in fractions of body length L .

In this chapter we have calculated the aerodynamic coefficients, C_A , C_N , and C_M for blunt and sharp cones using the Newtonian hypothesis to define the surface pressure field. Other pressure models and other, more general shapes, could in principle be used in similar calculations to meet other engineering requirements. However, we should appreciate at this point that the analytic difficulties associated with closed-form integration are likely to be insurmountable as the analytic forms describing the pressure field and the configurational geometry become more complicated. Even though the ultimate fate of Eqs. (9.20) and (9.21c) must be numerical integration, the role of analytic virtuosity should not be overlooked.

References

¹Fernandez, F. I. and Levinsky, E. S., "Air Ionization in the Hypersonic Laminar Wake of Sharp Cones," *AIAA Journal*, Vol. 2, Oct. 1964, pp. 1892-1832.

²Morrison, A. F. et al., *Handbook of Inviscid Sphere-Cone Flow Fields and Pressure Distributions Volume II*, NSWC/WOL/TR-75-45, 1975.

X. MOMENT EQUATIONS IN CONSTANT DENSITY ATMOSPHERE

Earlier we had shown in Eq. (8.65) that the moment applied to the RV may be written as a column matrix (vector) and equated to the rate of change of angular momentum as

$$\mathbf{M}^i = \frac{d}{dt}(\mathbf{H}^i) \quad (10.1)$$

For convenience we can regard the applied moment $\mathbf{M}^i$ as the vector sum of two constituents: a passive moment and an active or control moment. We shall regard the passive moment as having an aerodynamic origin. More specifically the passive moment is the first moment of the pressure distribution over that part of the entire configuration which is not deliberately changed during flight. The control moment may or may not be entirely aerodynamic. Aerodynamic controls are effected by making changes to the configuration. For example the windward flap discussed earlier alters the pressure distribution over the rearward portion of the RV. Differential deflection of a split flap can provide roll control as well. A partial aerodynamic control might be a moving mass control system where the center of pressure/center of gravity offset is controlled by moving a mass within the RV. Mass ejection in the boundary layer can develop a control force that is partially due to the reaction of the effluent and partially due to alterations in the pressure field.

Even though we might draw a distinction between the active and passive moments, there are some interesting interactions. First, control moments, if they are to deserve the name, must be able to overcome the passive moments. Thus, in discussing the adequacy of any control system we must have in hand a detailed understanding of the passive moments; we must know, for example, how such passive moments vary with RV angular attitude, and nondimensional flow parameters such as Mach number and Reynolds number. There can also be an interaction between the control system aerodynamics and passive aerodynamics that may be incidental or deliberate.

In this chapter we shall ignore any complex interactions between the passive and active aerodynamics. We shall simply expand the total moment in a Taylor series in several variables, to represent both the angular excursions of the RV and the angular deflections of the controls.

10.1 Equations of Motion: Slight Configuration Asymmetries

In Eq. (10.1) we showed that the total moment applied to the RV causes a rate of change in angular momentum relative to inertial space. The angular momentum relative to inertial coordinates can be referred to body coordinates by application of the body-frame b to inertial frame i DCM. In this chapter we shall use a modified frame of reference called an "aeroballistic axes." The aeroballistic axes are an axis system which does *not* share body roll rate but *does* share all other body angular rates.* Figure 10.1 shows the three frames: i inertial, a aeroballistic, and b body sharing a common origin at the RV's center of gravity. Equation (10.1) may be written as

$$\mathbf{M}^i = \frac{d}{dt}(\mathbf{H}^i) = \frac{d}{dt}[\mathbf{C}_b^i \mathbf{H}^b]$$

or

$$\mathbf{M}^i = \frac{d}{dt}[\mathbf{C}_a^i \mathbf{C}_b^a \mathbf{H}^b] \quad \mathbf{M}^i = \frac{d}{dt}[\mathbf{C}_a^i \mathbf{C}_b^a \mathbf{I}^b \boldsymbol{\omega}_{ib}^b]$$

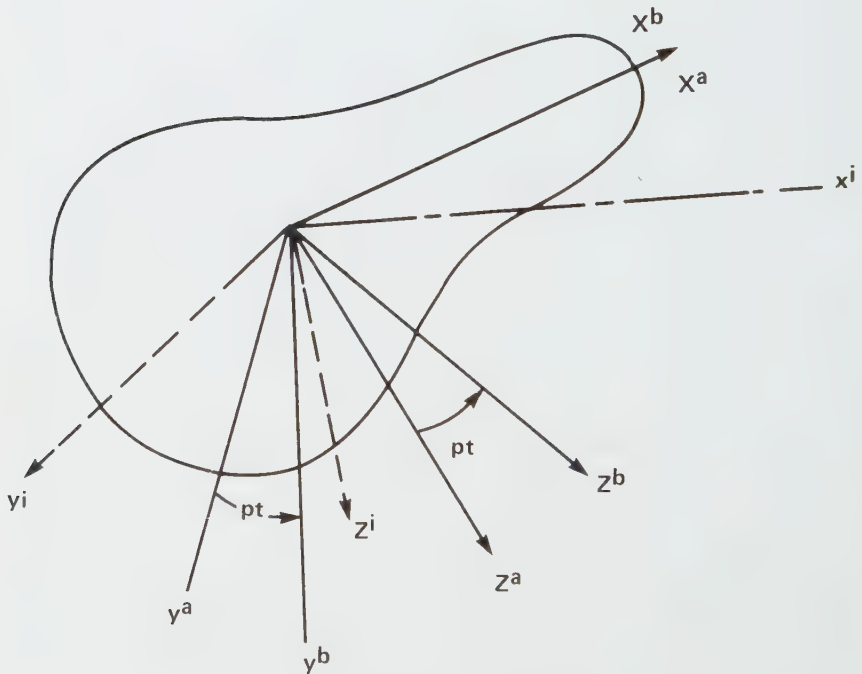


Fig. 10.1 Aeroballistic and body axes.

*An interesting anecdote concerning frame selection is given by D. H. Platus in his excellent paper, "Ballistic Re-entry Vehicle Dynamics," *Journal of Guidance, Control, and Dynamics*, Vol. 5, Jan.-Feb. 1982, pp. 4-16.

where we have replaced the angular momentum by the product of the inertia tensor with the body's angular velocity vector, both of which are measured in the body frame. The components of the angular velocity may be taken in the a frame to give:

$$\mathbf{M}^i = \frac{d}{dt} [\mathbf{C}_a^i \mathbf{C}_b^a \mathbf{I}^b \mathbf{C}_a^b \boldsymbol{\omega}_{ib}^a]$$

Using Eq. (8.81) we have:

$$\mathbf{M}^i = \frac{d}{dt} [\mathbf{C}_a^i \mathbf{I}^a \boldsymbol{\omega}_{ib}^a]$$

Carrying out the differentiation and then premultiplying both sides by $(\mathbf{C}_a^i)^T$, we get:

$$\mathbf{M}^a = \mathbf{I}^a \dot{\boldsymbol{\omega}}_{ib}^a + \boldsymbol{\Omega}_{ia}^a \mathbf{I}^a \boldsymbol{\omega}_{ib}^a \quad (10.2)$$

A few comments concerning the utility of Eq. (10.2) might be in order.

First we note that we have two symbolic forms for angular rates: $\boldsymbol{\Omega}_{ib}^a$ and $\boldsymbol{\omega}_{ib}^a$. The former is the skew-symmetric form of the angular rate of the a frame with respect to inertial space and with coordinates in the a frame; the latter is the column form of the angular velocity of the body or b frame with respect to inertial space and with components taken in the a frame. These vectors may be expressed as

$$\boldsymbol{\Omega}_{ia}^a = \begin{bmatrix} 0 & -r & q \\ r & 0 & 0 \\ -q & 0 & 0 \end{bmatrix} \quad \boldsymbol{\omega}_{ib}^a = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (10.3)$$

where p is the roll rate, q the pitch rate, and r the yaw rate of the RV. Note that, according to the definition of the a frame, the body roll-rate p is not included in the $\boldsymbol{\Omega}_{ia}^a$ matrix.

A second comment on Eq. (10.2) concerns the term $\mathbf{I}^a$. $\mathbf{I}^a$ indicates the inertia tensor quantized in the a frame which is not body fixed. Since we are considering the dynamics of an RV with "slight" configurational asymmetries, we shall let

$$\mathbf{I}^a = \mathbf{I}^b \quad (10.4)$$

The above statement is bound up in the definition of the word "slight." A configuration asymmetry is considered as "slight" when its presence does not appreciably affect the mass distribution, but its presence is sufficient to be felt as the generator of significant asymmetric aerodynamic moments. Such asymmetries may be controlling moments or simply alterations in the symmetric passive aerodynamics, possible due to unsymmetric ablation.

Equation (10.2) may be written in component form using Eqs. (10.3) and (10.4) to give:

$$\mathbf{M}^a = \begin{bmatrix} L \\ M \\ N \end{bmatrix} = \begin{bmatrix} I_x \dot{p} \\ I \dot{q} \\ I \dot{r} \end{bmatrix} + \begin{bmatrix} 0 & -r & q \\ r & 0 & 0 \\ -q & 0 & 0 \end{bmatrix} \begin{bmatrix} I_x & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

or

$$L = I_x \dot{p} \quad (10.5a)$$

$$M = I \dot{q} + r p I_x \quad (10.5b)$$

$$N = I \dot{r} - q p I_x \quad (10.5c)$$

Note that we have taken the body axis as principal and have dropped some subscripts in the moment of inertia as no longer essential. By letting $I_{xx} = I_{yy} = I_{zz} = I$ we are assuming the RV to be a body of revolution. This restriction could be avoided, but it allows subsequent analysis to proceed more easily while still retaining the essential physics.

The roll rate can be uncoupled by assuming that its rate of change is zero, i.e., $\dot{p} = 0$. This assumption leaves Eqs. (10.5b) and (10.5c) to be solved. A more complete treatment would include the force equations as well. However, the moment equations will enable us to uncover the basic physics of inertially and aerodynamically coupled angular motion. The next step is to specify the aerodynamic moment components M and N . It should be noted that the components M and N could also contain some constituents that have nonaerodynamic or only partial aerodynamic origins such as surface ablation or perhaps a jet-type control. For our purposes we shall regard M and N entirely of aerodynamic origin. The following linear expansion in a Taylor series will be used:

$$M = M_\alpha \alpha + M_{\dot{\alpha}} \dot{\alpha} + M_q q + M_{p\beta} p \beta \quad (10.6a)$$

$$N = N_\beta \beta + N_{\dot{\beta}} \dot{\beta} + N_r r + M_{p\alpha} p \alpha \quad (10.6b)$$

The above series expansion has been in the variables α , β , $\dot{\alpha}$, $\dot{\beta}$, q and r . In addition we shall also keep the aerodynamic or Magnus coupling term. As will be seen the Magnus term aerodynamically couples the yaw and pitch equations through the products $p\alpha$ and $p\beta$. The derivatives on the right-hand side can be nondimensionalized according to Eqs. (9.9).

$$\begin{aligned}
\frac{\partial M}{\partial \alpha} &\equiv M_\alpha = C_{m_\alpha} q S d = \frac{\partial C_m}{\partial \alpha} q S d \\
\frac{\partial M}{\partial \dot{\alpha}} &\equiv M_{\dot{\alpha}} = C_{m_{\dot{\alpha}}} \frac{q S d^2}{2u_0} = \frac{\partial C_m}{\partial (\dot{\alpha} d / 2u_0)} \frac{q S d^2}{2u_0} \\
\frac{\partial M}{\partial q} &\equiv M_q = C_{m_q} \frac{q S d^2}{2u_0} = \frac{\partial C_m}{\partial (q d / 2u_0)} \frac{q S d^2}{2u_0} \\
\frac{\partial^2 M}{\partial p \partial \beta} &\equiv M_{p\beta} = C_{m_{p\beta}} \frac{q S d^2}{2u_0} = \frac{\partial^2 C_m}{\partial (\dot{\alpha} d / 2u_0) \partial (p d / 2u_0)} \frac{q S d^2}{2u_0} \\
\frac{\partial N}{\partial \beta} &\equiv N_\beta = C_{n_\beta} q S d = \frac{\partial C_n}{\partial \beta} (q S d) \\
\frac{\partial N}{\partial \dot{\beta}} &\equiv N_{\dot{\beta}} = C_{n_{\dot{\beta}}} \frac{q S d^2}{2u_0} = \frac{\partial C_n}{\partial (\dot{\beta} d / 2u_0)} \frac{q S d^2}{2u_0} \\
\frac{\partial N}{\partial r} &\equiv N_r = C_{n_r} \frac{q S d^2}{2u_0} = \frac{\partial C_n}{\partial (r d / 2u_0)} \frac{q S d^2}{2u_0} \\
\frac{\partial^2 N}{\partial p \partial \alpha} &\equiv N_{p\alpha} = C_{n_{p\alpha}} \frac{q S d^2}{2u_0} = \frac{\partial^2 C_n}{\partial \alpha \partial (p d / 2u_0)} \frac{q S d^2}{2u_0} \quad (10.7)
\end{aligned}$$

It will be shown subsequently that the necessary and sufficient criterion for static stability is that $C_{m_\alpha} < 0$. It will also be shown that the Magnus effect, the aerodynamic coupling of the yaw and pitch equations through roll rate, will be a key in the stabilization of rapidly spinning vehicle. Most RV's do not have a rapid spin rate (in comparison to a projectile) so it may turn out that the Magnus effect may not have great significance.

Most rapidly spinning bodies do not have control surfaces; however, even for rapidly spinning bodies like projectiles, there may be some body cutouts which slightly destroy configurational and hence aerodynamic symmetry. Thus we would have a trim condition (sum of moments are zero) where the body's attitude toward the flow is no longer symmetric and hence side-loads are developed on the vehicle. We shall define a "trim moment" as a moment existing under no-lift conditions which causes angular motion of the body to a condition of trim. We can represent the trim moment caused by this body-fixed asymmetry as $M_\delta \delta$. Since we are taking our moment components in the a frame, this trim asymmetry commutes as follows in Fig. 10.2. It should be noted that the a frame does not rotate; the b frame does rotate, however, and it carries the asymmetry or trimming moment $M_\delta \delta$ with it.

When all the moment contributions expressed as terms in Eq. (10.6) are entered in the moment Eqs. (10.5) we obtain:

$$I\dot{q} + rpI_x - M_\alpha\alpha - M_\alpha\dot{\alpha} - M_qq - M_{p\beta}p_\beta = M_\delta\delta\cos pt \quad (10.8a)$$

$$I\dot{r} - pqI_x - N_\beta\beta - N_\beta\dot{\beta} - N_rr - N_{p\alpha}P_\alpha = M_\delta\delta\cos pt \quad (10.8b)$$

We can solve the above set of equations as two second-order coupled differential equations. However, because of the assumed rotational symmetry we may replace Eqs. (10.8) by a single second-order equation by introducing the complex variable ξ as

$$\xi = i\alpha + \beta \quad (10.9)$$

The angle ξ is called the "complex angle of attack;" the justification for such usage is that the imaginary variable i indicates that angular motions described by α and β are at right angles to each other. The RV's x axis is measured relative to the velocity vector which is assumed inertially fixed by the angles α and β . As noted earlier the rate of rotation of the a frame relative to inertial space has a component q along with the y axis and component r along the z axis. These rates can also be expressed in terms of a complex variable Ω as

$$\Omega = q + ir \quad (10.10)$$

A consequence of our assumption that the force equations are identically zero is that the velocity vector is nonrotating relative to inertial space. Thus, we may make the following simplifying assumptions:

$$q = \dot{\alpha} \quad r = -\dot{\beta} \quad \dot{q} = \ddot{\alpha} \quad \dot{r} = -\ddot{\beta} \quad (10.11)$$

We may now combine Eqs. (10.9) and (10.10) as

$$\Omega = -i\dot{\xi} = \dot{\alpha} - i\dot{\beta} \quad (10.12)$$

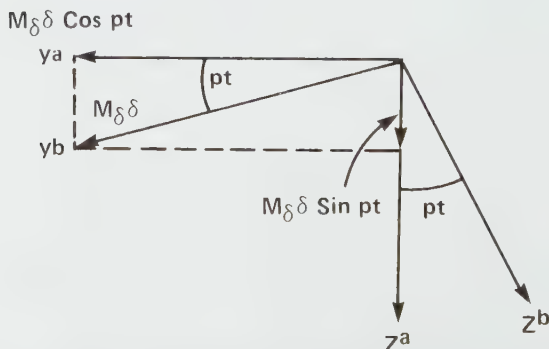


Fig. 10.2 Asymmetric moment resolved in the aeroballistic frame.

Let us return to Eqs. (10.8), multiplying the first by i and adding it to the second:

$$I(\dot{q} + i\dot{r}) - I_x(iq - r)p - M_\alpha\alpha - iN_\beta\beta - M_qq - iN_rr - M_\alpha\dot{\alpha} - iN_\beta\dot{\beta} - M_\delta\delta[\cos pt + i\sin pt] - M_{p\beta}p\beta - iN_{p\alpha}p\alpha = 0 \quad (10.13)$$

As a consequence of rotational symmetry we may assume the equivalence of the following aerodynamic derivatives:

$$M_\alpha = -N_\beta \quad M_q = N_r \quad M_\alpha = -N_\beta \quad M_{p\beta} = N_{p\alpha} \quad (10.14)$$

Equation (10.13) may be rewritten as

$$I(\dot{q} + i\dot{r}) - I_x p(iq - r) - M_\alpha(\alpha - i\beta) - M_q(q + ir) - M_\alpha(\dot{\alpha} - i\dot{\beta}) - M_\delta\delta(\cos pt + i\sin pt) - M_{p\beta}(i\alpha + \beta)p = 0 \quad (10.15)$$

Using Eqs. (10.9), (10.10), and (10.12) we get:

$$I(-\ddot{\xi}i) - I_x(\dot{\xi})p - M_\alpha(i\xi) - M_\alpha(-i\dot{\xi}) - M_\delta\delta e^{ipt} - M_{p\beta}\xi p = 0 \quad (10.16)$$

Further manipulation gives

$$\ddot{\xi} + \left[-ip\frac{I_x}{I} - \frac{M_q + M_\alpha}{I} \right] \dot{\xi} + \left[-ip\frac{M_{p\beta}}{I} - \frac{M_\alpha}{I} \right] \xi = i \left[\frac{M_\delta\delta}{I} \right] e^{ipt} \quad (10.17)$$

Obviously if we have no control deflections (or equivalently no aerodynamic asymmetry) $\delta = 0$, and Eq. (10.17) is homogeneous. We may replace the coefficients in Eq. (10.17) if we first define N_1 , N_2 , and N_3 as

$$N_1 = -ip\frac{I_x}{I} - \frac{M_q + M_\alpha}{I} \quad (10.18a)$$

$$N_2 = ip\left[\frac{M_{p\beta}}{I} \right] - \frac{M_\alpha}{I} \quad (10.18b)$$

$$N_3 = i \left[\frac{M_\delta\delta}{I} \right] \quad (10.18c)$$

Equation (10.17) then becomes

$$\ddot{\xi} + N_1\dot{\xi} + N_2\xi = N_3e^{ipt} \quad (10.19)$$

10.2 Solutions of Equations of Motion

Clearly the homogeneous solution of Eq. (10.19) has two roots which we shall designate as m_1 and m_2 . Thus, we may express the homogeneous solution as

$$\xi_h = K_1 e^{m_1 t} + K_2 e^{m_2 t} = \xi_1(t) + \xi_2(t) \quad (10.20)$$

where

$$m_{1,2} = -\frac{N_1}{2} \pm \sqrt{\left[\frac{N_1}{2}\right]^2 - N_2} \quad (10.21a)$$

In order for Eq. (10.19) to have an oscillatory solution, it is necessary that m_1 and m_2 be complex. Anticipating the complex nature of the exponential roots, we may write Eq. (10.21a) as

$$m_{1,2} = \lambda_{1,2} + \omega_{1,2}i \quad (10.21b)$$

where $\lambda_{1,2}$ is the magnitude of the real part and $\omega_{1,2}$ is the magnitude of the imaginary part.

The particular solution to Eq. (10.19) may be found by using the method of variation of parameters. An outline of the development of this method is given in Appendix A. The method of variation of parameters requires that we have available the solutions to the homogeneous equation, i.e., ξ_1 and ξ_2 . Adapting Eq. (A.14) we have:

$$\xi_p = \int^t \left[\frac{\xi_1(\tau)\xi_2(t) - \xi_1(t)\xi_2(\tau)}{\xi_1(\tau)\xi_2'(\tau) - \xi_1'(\tau)\xi_2(\tau)} \right] N_3 e^{ip\tau} d\tau \quad (10.22)$$

After replacing $\xi_1(t)$ and $\xi_2(t)$ by the appropriate functions from Eqs. (10.20) we get:

$$\xi_p = K_3 e^{ip t} = \left[\frac{N_3}{(ip - m_1)(ip - m_2)} \right] e^{ip t} \quad (10.23a)$$

Using Eq. (10.24b) we may rewrite Eq. (10.23b) as

$$\xi_p = \left[\frac{N_3}{[i(p - \omega_1) - \lambda_1][i(p - \omega_2) - \lambda_2]} \right] e^{ip t} \quad (10.23b)$$

Now if the asymmetric or "trim" moment $M_{\delta\delta}$ is nonzero, N_3 will also be nonzero. It then looks as if the complex angle-of-attack ξ might become large if $p \cong \omega_1$, if λ_1 is small.[†] We will consider the practical implications

[†] It might appear that p , the roll rate, could also equal ω_2 ; however, it will be seen presently that ω_2 always rotates in a direction opposite to p , i.e., ω_2 is negative for positive p so $|p| = |\omega_2|$ will not cause any difficulties.

of this observation subsequently. In any event, the solution to Eq. (10.17) may be written as

$$\xi = \xi_h + \xi_p, \quad \dot{\xi} = K_1 e^{m_1 t} + K_2 e^{m_2 t} + K_3 e^{ip t} \quad (10.24)$$

In Eq. (10.24) we have three constants: K_1 , K_2 , and K_3 . Looking at Eqs. (10.18) and (10.24) we note that K_3 is a complex number influenced through N_3 by the aerodynamic asymmetric moment $M_\delta \delta$, the transverse moment of inertia I , the roll rate p , and the exponential solutions to the homogeneous equation, i.e., m_1 and m_2 . As we shall see, these exponents m_1 and m_2 are functions of the moment of inertia I , the roll rate p , and various aerodynamic coefficients. The point to be emphasized here is that K_3 is *not* dependent upon initial conditions, i.e., ξ and $\dot{\xi}$ at $t = 0$. However, the constants K_1 and K_2 which appear in the solution to the homogeneous solution *are* dependent upon initial conditions. Let us examine the solution to the moment equation at $t = 0$:

$$\xi \big|_{t=0} \equiv \xi_0 = K_1 + K_2 + K_3 \quad (10.25a)$$

and its derivative also at time $t = 0$:

$$\left. \frac{d\xi}{dt} \right|_{t=0} = \dot{\xi}_0 = K_1 m_1 + K_2 m_2 + K_3 (ip) \quad (10.25b)$$

Solution of Eqs. (10.25) for K_1 and K_2 are:

$$K_1 = \frac{\dot{\xi}_0 - (\lambda_2 + i\omega_2)\xi_0 - [(ip - \lambda_2) - i\omega_2]K_3}{(\lambda_1 - i\omega_1) - (\lambda_2 - i\omega_2)} \quad (10.26a)$$

$$K_2 = \frac{\dot{\xi}_0 - (\lambda_1 + i\omega_1)\xi_0 - [(ip - \lambda_1) - i\omega_1]K_3}{(\lambda_2 - i\omega_2) - (\lambda_1 - i\omega_1)} \quad (10.26b)$$

Not only are Eqs. (10.26) fairly complicated, but they are dependent upon the exponential terms m_1 and m_2 . These exponents will be evaluated in terms of roll-rate p , various dimensional aerodynamic derivatives, and the moments of inertia I and I_x . This is done by combining Eqs. (10.18) and (10.21) to give

$$m_{1,2} = \left[\frac{ipI_x}{2I} + \frac{M_q + M_{\dot{\alpha}}}{2I} \right] \pm \sqrt{\left[-\frac{ipI_x}{2I} \right]^2 + 2 \left[\frac{ipI_x}{2I} \right] \left[\frac{M_q + M_{\dot{\alpha}}}{2I} \right] + \left[\frac{M_q + M_{\dot{\alpha}}}{2I} \right]^2 + ip \left[\frac{M_{p\beta}}{I} \right] + \frac{M_{\dot{\alpha}}}{I}} \quad (10.27)$$

where the subscript (1) applies to the + sign and the subscript (2) applies to the - sign. If we carry out the square operation indicated in the second brackets, we find that the term

$$\left[\frac{M_q + M_{\dot{\alpha}}}{2I} \right]^2$$

may be neglected. Thus Eq. (10.27) becomes:

$$m_{1,2} = \left[\frac{ipI_x}{2I} + \frac{M_q + M_{\dot{\alpha}}}{2I} \right] \pm \sqrt{i^2 \left[\left(\frac{pI_x}{2I} \right)^2 - \frac{M_{\alpha}}{I} \right] + 2 \left[\frac{ipI_x}{2I} \right] \left[\frac{M_q + M_{\dot{\alpha}}}{2I} \right] + ip \frac{M_{p\beta}}{I}} \quad (10.28)$$

Since the first bracketed term under the square root dominates, it is possible to expand the entire term under the square root through the use of the binomial theorem. This expansion somewhat simplifies things to give

$$m_{1,2} = \left[\frac{ipI_x}{2I} + \frac{M_q + M_{\dot{\alpha}}}{2I} \right] \pm \left\{ i \sqrt{\left[\left(\frac{pI_x}{2I} \right)^2 - \frac{M_{\alpha}}{I} \right]} + \left\{ 2 \frac{ipI_x}{2I} \left[\frac{M_q + M_{\dot{\alpha}}}{2I} \right] \right. \right. \\ \left. \left. + ip \frac{M_{p\beta}}{I} \right\} / 2i \sqrt{\left[\left(\frac{pI_x}{2I} \right)^2 - \frac{M_{\alpha}}{I} \right]} \right\} \quad (10.29)$$

which becomes an approximation to Eq. (10.27):

$$m_{1,2} = \lambda_{1,2} \pm i \omega_{1,2} = \left\{ \left[\frac{M_q + M_{\dot{\alpha}}}{2I} \right] \pm \left[\frac{pI_x}{2I} \left[\frac{M_q + M_{\dot{\alpha}}}{2I} \right] \right. \right. \\ \left. \left. + p \frac{M_{p\beta}}{I} \right] / \sqrt{\left[\left(\frac{pI_x}{2I} \right)^2 - \frac{M_{\alpha}}{I} \right]} \right\} + i \left\{ \left[\left(\frac{pI_x}{2I} \right)^2 \pm \sqrt{\left[\left(\frac{pI_x}{2I} \right)^2 - \frac{M_{\alpha}}{I} \right]} \right] \right\} \quad (10.30)$$

We have arranged Eq. (10.30) into real and imaginary parts. The real part we have associated with a damping term λ and the imaginary part we have associated with a frequency term ω . Obviously the arrangement has been based upon experience since for example there is no reason for assuming a priori that

$$\left[\frac{pI_x}{2I} \right]^2 > \left[\frac{M_\alpha}{I} \right]$$

Equation (10.30) may now be written as

$$\begin{aligned} m_{1,2} = \lambda_{1,2} \pm i \omega_{1,2} = & \left[\frac{M_q + M_{\dot{\alpha}}}{2I} \right] \left\{ 1 \pm \left[\left[\frac{pI_x}{2I} \right] / \sqrt{\left[\frac{pI_x}{2I} \right]^2 - \frac{M_\alpha}{I}} \right] \right\} \\ & \pm \left[\left[\frac{pI_x}{2I} \right] \left[\frac{M_{p\beta}}{I_x} \right] / \sqrt{\left[\frac{pI_x}{2I} \right]^2 - \left[\frac{M_\alpha}{I} \right]} \right] + i \left[\frac{pI_x}{2I} \right] \\ & \times \left\{ 1 \pm \left[\sqrt{\left[\frac{pI_x}{2I} \right]^2 - \left[\frac{M_\alpha}{I} \right]} / \left[\frac{pI_x}{2I} \right] \right] \right\} \end{aligned} \quad (10.31)$$

Practice among ballisticians, at least, has been to "lump" the variables according to the following symbols:

$$s = \left[\frac{pI_x}{2I} \right]^2 / \frac{M_\alpha}{I} \quad (10.32a)$$

$$\tau = 1 / \sqrt{1 - \frac{1}{s}} = \frac{pI_x}{2I} / \sqrt{\left[\frac{pI_x}{2I} \right]^2 - \left[\frac{M_\alpha}{I} \right]} \quad (10.32b)$$

The term s is called the "gyroscopic stability factor." As will be seen it is a key performance parameter assessing the relative stability of a spin-stabilized vehicle. A spin-stabilized vehicle is usually one which is statically unstable, i.e., $M_\alpha > 0$, but finds stability if spun or rolled at an appropriate rate. Most RV's are statically stable (although marginally so by the standards of ballisticians) and thus roll is included for reasons other than dynamic stability. In the discussion to follow we shall consider both statically stable and statically unstable (usually spin- or roll-stabilized) vehicles. It appears that s is greater than unity for τ to be real. That s be greater than unity is a necessary but not sufficient condition for the dynamic stability of a

spin-stabilized vehicle. The additional condition needed for sufficiency is that λ_1 and λ_2 of Eq. (10.31) be negative. For a statically stable vehicle, then, s is always less than zero. It might be interesting to plot τ vs s for both statically stable ($s < 0$) and roll-stabilized vehicles ($s < 1$) (see Fig. 10.3).

The stability of a roll-stabilized vehicle has certain similarities to that of a spinning top. In the spinning aeroballistic vehicle the moment $M_\alpha \alpha$ is analogous to the gravity moment applied to a top $M_t g / \sin \theta$, where M_t is the mass of the top, l the distance from the top's point of contact to the top's c.g., and θ the angle of lean. A pictorial comparison between the top and the spin-stabilized vehicle is given in Fig. 10.4.

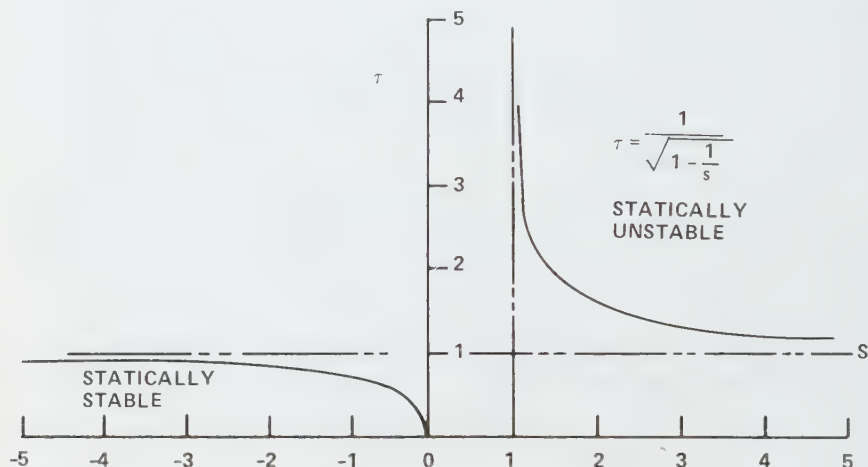


Fig. 10.3 τ vs gyroscopic stability factor S .

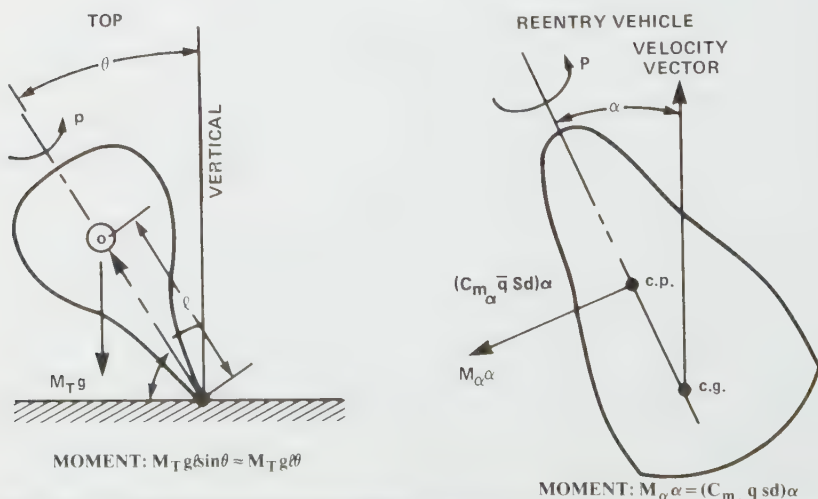


Fig. 10.4 Comparison between a spinning top and a spinning aeroballistic vehicle.

The dynamics of tops were an early application of classical dynamics. One of the results of top theory is the condition for the "sleeping top." The top will "sleep" or spin about the vertical if the spin rate p satisfies the following inequality

$$p^2 \geq (Mgl/I_x^2) \quad (10.33)$$

If p_0 represents the value of p which satisfies the equality in Eq. (10.33), then we may let sp_0^2 be the value of p^2 in general. Obviously s must be greater than unity for the inequality of Eq. (10.33) to be satisfied. Now let us express s in terms of p and p_0 :

$$s = \frac{p^2}{p_0^2} = \frac{p^2 I_x^2}{4M_l g I} = \frac{(p I_x / 2I)^2}{(M_l g / I)} \quad (10.34)$$

I_x is the moment of inertia about the top's axis of symmetry and I is the transverse moment of inertia at the contact point. The above s applies to the "sleeping" top. The analogous situation for the spin-stabilized vehicle is for the x axis or the axis of symmetry to align with the velocity vector. If the sleeping top is disturbed from the vertical it will develop the appropriate moment to bring it back to vertical. Similarly, if the axis of symmetry of the spin-stabilized, statically-unstable vehicle is distributed from the velocity vector, appropriate moments involving a coupling between spin and angular displacement will be generated to bring this axis into coincidence with the velocity vector. The top overturning moment has been shown to be $Mlg\theta$ so its derivative with respect to angular displacement θ is Mlg . Now if we replace this top overturning-moment derivative with the rolling vehicle's overturning-moment derivative M_a , we get:

$$s = \frac{(p I_x / 2I)^2}{(M_a / I)} \quad (10.35)$$

which is of course identical to the s parameter of Eq. (10.32a). Since τ [in Eq. (10.32b)] must be real, and we limit ourselves for the present to statically unstable vehicles, s must be greater than unity. In the case of the top, if s becomes less than unity, then from Eq. (10.33), the top will wobble from the vertical; in the case of spinning re-entry vehicle if s becomes less than unity, the vehicle's axis of symmetry will wobble from the velocity vector.

Using Eqs. (10.32) we may now write Eqs. (10.31) as

$$\lambda_{1,2} \pm i\omega_{1,2} = \left[\left(\frac{M_q + M_{\dot{\alpha}}}{2I} \right) (1 \pm \tau) \pm \frac{M_{p\beta}}{I_x} \tau \right] + i \left[\frac{p I_x}{2I} \left(1 \pm \frac{1}{\tau} \right) \right] \quad (10.36)$$

Concentrating on the frequencies ω_1 and ω_2 first, we get:

$$\omega_1 = \frac{pI_x}{2I} \left[1 + \frac{1}{\tau} \right] \quad (10.37a)$$

$$\omega_2 = \frac{pI_x}{2I} \left[1 - \frac{1}{\tau} \right] \quad (10.37b)$$

since $1/\tau$ is positive but less than unity $\omega_1 > \omega_2$; we can then identify ω_1 as the faster or nutational frequency and ω_2 as the slower or precessional frequency. It follows then that λ_1 is the damping of the nutational mode and λ_2 is the damping of the precessional mode. As noted earlier the roots m_1 and m_2 of Eqs. (10.29) are complex; the magnitude of the imaginary part, i.e., ω_1 and ω_2 are associated with the frequency of motion. The magnitude of the real part, i.e., λ_1 and λ_2 multiplied by time t , appear in the exponential functions that define the envelope of the oscillatory motion. Obviously, if both λ_1 and λ_2 are negative, the envelope of this oscillatory motion will converge, and the vehicle will be considered dynamically stable. On the other hand if either λ_1 or λ_2 (or both) are positive, we shall observe an oscillatory motion that will grow with time. Within the limitations imposed by the linear model of the imposed aerodynamic moments [Eq. (10.6)], diverging oscillatory motion will be characterized as dynamically unstable.

In Eqs. (10.6) we noted the presence of aerodynamic cross-coupling between the yaw and pitch motion. The coefficients describing this coupling are $M_{p\beta}$ and $N_{p\alpha}$; aerodynamic coupling between yaw and pitch which depends upon spin rate and angular displacement is called the "Magnus effect." In the aeroballistic axis system, the Magnus force is nearly always along the negative y axis for positive spin rate (clockwise about the axis of symmetry) and for positive angle of attack (nose above the velocity vector). For nonzero spin rate the presence of angle of attack (in this case essentially pitch angle), a yawing moment is generated, hence the identification of the Magnus effect with aerodynamic cross-coupling. Similarly, for positive spin rate, a positive angle of sideslip, i.e., nose to the left will generate a Magnus force along the positive z axis. Let us concentrate on the angle-of-attack displacement. With the Magnus force always along the negative y axis, the sign of the Magnus moment will depend upon the location of the Magnus center of pressure (c.p.). If the Magnus c.p. is ahead of the moment reference center (m.r.c.), the Magnus moment will be negative; if the Magnus c.p. is behind the m.r.c. then of course the Magnus moment will be positive (see Fig. 10.5).

Thus it may be seen from Eq. (10.36) that the Magnus effect acting ahead of the m.r.c. damps (makes more negative) the nutational mode, but undamps the precessional mode. A Magnus c.p. aft of the m.r.c. has the opposite effect, since an aft c.p. generates a positive moment which undamps the nutational mode and damps the precessional mode. Since

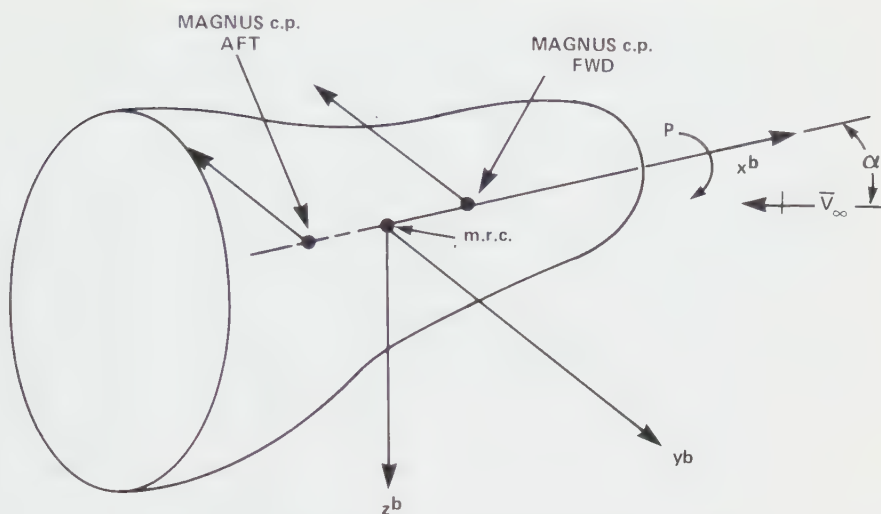


Fig. 10.5 Magnus force location

$M_q + M_u$ is always negative, mode stability is dictated by the Magnus effect, or more precisely the location of the Magnus c.p. relative to the vehicle m.r.c. There is some subtlety here that should be recognized. It turns out that

$$\frac{M_q + M_u}{2I}$$

is of nearly the same size as

$$\frac{M_{p\beta}}{I_x}$$

Further for $s > 1$, τ is always greater than unity (see Fig. 10.3). Thus, in the nutational mode $(1 + \tau)$ is always positive and greater than unity while in the precessional mode $(1 - \tau)$ is always negative. It is therefore quite likely that for either sign of the Magnus moment, the nutational mode will be stable, i.e., $\lambda_1 < 0$. However for the precessional mode,

$$\frac{M_q + M_u}{2I}(1 - \tau)$$

will always be positive. Thus, stability of the precessional mode will nearly always depend upon the sign of the Magnus moment. Magnus c.p. ahead of

the m.r.c., i.e., negative moment, will always result in an unstable precessional (i.e., low-frequency) mode. For the statically stable RV ($M_\alpha < 0$ and hence $s < 0$) the Magnus effect may not be significant since the roll rate may be small. The comments made up to this point are directed toward the statically unstable but spin-stabilized vehicle. We shall discuss the statically stable vehicle subsequently. There is not a great deal of information concerning the location of the Magnus center of pressure in hypersonic flows. However, it is known that for projectile-shaped bodies at supersonic Mach numbers, the Magnus center of pressure is aft of the m.r.c. and usually moves forward with *decreasing* Mach numbers. The result is that somewhere in the transonic region the Magnus moment undamps the precessional mode. As the supersonic vehicle slows down and takes on subsonic speeds, the undamped precessional mode causes the vehicle to wobble. The increase in yaw angle will increase drag, causing a further decrease in speed which exacerbates the wobble.

We have discussed stability in terms of the damping exponentials λ_1 and λ_2 . Earlier we required that s , the gyroscopic stability factor, be greater than unity. We might now pose the question: How does the stability factor change as the speed of the RV changes? From Eq. (10.35) we see that s is proportional to the square of the roll rate, i.e., p^2 and varies inversely with the moment derivative

$$M_\alpha = \frac{1}{2} \rho V^2 S d C_{m_\alpha}$$

Since spin-stabilized vehicles usually have little damping in roll moment, the roll rate p will decrease slowly; however, the inverse relationship of s with the square of the speed, i.e., V^2 means that s will increase as the speed decreases. For a gun-launched projectile, or an RV model set in motion in a ballistic range, s will increase from the value that it has at the muzzle. Figure 6.1 indicates that during re-entry we can usually expect a decrease in velocity magnitude, except for a slight increase during motion in the upper and most tenuous layers of the atmosphere.

The preceding equations, particularly Eq. (10.36), are equally valid for statically stable RV's. Statically stable vehicles have a negative static pitching moment derivative, i.e., $M_\alpha < 0$. From Eq. (10.35) this means that s is also negative and that τ is limited to the range, $0 < \tau < 1$. Since roll rate is not required for stability we can examine the case of zero roll rate first. When the roll-rate p is zero, s is also zero; provided that s approaches zero from the negative side, i.e., $s \rightarrow 0(-)$, τ is also zero. From Eqs. (10.37) we have:

$$\omega_{1,2} = \frac{p I_y}{2I} \left[1 \pm \frac{1}{\tau} \right]$$

For $p = 0$ it would appear that we have an indeterminate form $0/0$. However, the form becomes determinate if $1/\tau$ is replaced by its equivalent form

Eq. (10.32b). Then we have from Eq. (10.37) the following relationship for the frequencies ω_1 and ω_2 :

$$\omega_{1,2} = \pm \sqrt{-\frac{M_\alpha}{I}} = \pm \sqrt{-\frac{C_{m_\alpha} q S d}{I}} \quad (10.38)$$

As pointed out earlier for statically stable vehicles, τ will always be less than unity and consequently $(1 \pm \tau)$ will always be positive. First let us examine the exponential damping terms λ_1 and λ_2 from Eqs. (10.36):

$$\lambda_1 = \left[\frac{M_q + M_{\dot{\alpha}}}{2I} \right] (1 + \tau) + \frac{M_{p\beta}}{I_x} \tau \quad (10.39a)$$

$$\lambda_2 = \left[\frac{M_q + M_{\dot{\alpha}}}{2I} \right] (1 - \tau) - \frac{M_{p\beta}}{I_x} \tau \quad (10.39b)$$

Since we are restricting ourselves to statically stable RV's [i.e., $(1 \pm \tau)$ always positive], we note in the above equation that in the absence of a Magnus effect ($M_{p\beta} \neq 0$) both the precessional and nutational modes will be dynamically stable if $M_q + M_{\dot{\alpha}}$ is negative. Now if the Magnus effect is present, i.e., $M_{p\beta} \neq 0$, this effect will tend to damp one of the modes and undamp the other. If the Magnus center of pressure is ahead of the m.r.c., the Magnus moment derivative will be negative, i.e., $M_{p\beta} < 0$, the Magnus moment will damp the nutational mode and undamp the precessional mode. For the Magnus c.p. aft of the m.r.c. the reverse will be true.

It would appear that there is no dynamic reason for spinning a statically stable vehicle; however, there may be other reasons for deliberately introducing some roll rate such as averaging the effect of aerodynamic asymmetries. Also a certain amount of roll rate may be unavoidable because of the acquisition of configurational asymmetries either prior to or during operation.

Earlier in this discussion we mentioned the presence of a "trimming moment" M_δ , that is a moment causing the RV to trim at some angle of attack relative to the oncoming flow. The result is of course the development of an aerodynamic force normal to the vehicle's axis of symmetry. The source of this moment is a body-fixed asymmetry. This asymmetric moment appears only in the N_3 term of Eqs. (10.18). Consequently, it appears only in the particular solution to Eq. (10.17), i.e., in the K_3 term. From Eqs. (10.18) and (10.23), we may write the K_3 term as

$$K_3 = N_3 / (ip - \lambda_1 - i\omega_1)(ip - \lambda_2 - i\omega_2)$$

The denominator of the above expression may be rewritten to give;

$$K_3 = \frac{N_3}{i[\lambda_2(\omega_1 - p) + \lambda_1(\omega_2 - p)] + [\omega_2(p - \omega_1) + p(\omega_1 - p) + \lambda_1\lambda_2]} \quad (10.40)$$

It appears that K_3 will become large as p approaches ω_1 or ω_2 . To accentuate the problem, let us assume that the exponential damping terms λ_1 and λ_2 are negligible. Thus Eq. (10.40) becomes,

$$K_3 = \frac{-M_\delta \delta / I}{(p - \omega_1)(p - \omega_2)} \quad (10.41)$$

It appears from Eq. (10.40) that we have a singularity in K_3 when the roll-rate p equals either ω_1 or ω_2 . Let us repeat Eq. (10.38) with τ replaced from Eq. (10.32b) as

$$\omega_{1,2} = \left[\frac{pI_x}{2I} \right] \pm \sqrt{\left[\frac{pI_x}{2I} \right]^2 - \frac{M_\alpha}{I}} \quad (10.42)$$

For the roll-stabilized (statically unstable) vehicle,

$$\frac{pI_x}{2I} > \frac{M_\alpha}{I}$$

and therefore it should be clear from Eq. (10.42) that both the nutational and precessional frequencies, i.e., ω_1 and ω_2 are of the same sign. If s is equal to unity, then it is easy to show that

$$\omega_1 = \omega_2 = pI_x / 2I \quad (10.43a)$$

At the other extreme, if $p \rightarrow \infty$ and hence $s \rightarrow \infty$, we have

$$\omega_1 = pI_x / I \quad \omega_2 = 0 \quad (10.43b)$$

Since $I_x < I$ it is clear that neither ω_1 nor ω_2 can ever equal the roll-rate p .

Most RV's are statically stable and such vehicles, if given a roll rate from whatever source, can encounter a resonance condition when the roll rate equals one of the two modal frequencies. For the statically stable vehicle $M_\alpha < 0$ and $s < 0$. When the roll rate is zero, both frequencies are of the same magnitude, but the opposite sign as

$$\omega_{1s} = + \sqrt{-\frac{M_\alpha}{I}} \quad \omega_{2s} = - \sqrt{-\frac{M_\alpha}{I}}$$

where ω_{1s} and ω_{2s} are the static or no-roll rate values of the two modal frequencies.

It is probably not realistic to consider extremely large roll rates, but it may be seen from Eqs. (10.42) that the second (radical) term will always be larger than the first term. Hence the nutation and precessional frequencies will always be of opposite signs. For a positive roll-rate p , the larger frequency (always identified as the nutational frequency) will have the same sign as the roll rate p . For a negative roll rate the nutational frequency will

also be negative. If we define the larger frequency as the nutational frequency, it is clear that regardless of the sign of the roll rate, the nutational frequency and the roll rate will share the same sign. Therefore from Eq. (10.40) it is apparent that if we neglect damping effects altogether a singularity will occur whenever the roll rate and nutational frequency are equal. From Eq. (10.42) we can let this resonance roll rate be $p = p_r$; we can then solve for p_r by writing;

$$\omega_1 = p_r = \left[\frac{p_r I_x}{2I} \right] + \sqrt{\left[\frac{p_r I_x}{2I} \right]^2 - \frac{M_\alpha}{I}}$$

The above expression may be solved for p_r in terms of M_α/I .

$$p_r = \frac{(M_\alpha/I)}{[1 - (I_x/I)]^{1/2}} \quad (10.44)$$

Since

$$M_\alpha = \frac{1}{2} \rho V^2 S d C_{m_\alpha} = \frac{1}{2} \rho V^2 S x C_{N_\alpha} d^2$$

it is clear that the resonant roll-rate p_r will depend upon the speed V atmospheric density ρ , static margin x , shape C_{m_α} , and size S and d as well as mass distribution in the form of I and I_x .

10.3 Further Discussion of Vehicle Dynamics

The details of RV dynamics near resonance are not revealed by the simple expressions given above. However, it is clear that there will be an amplification of any trim angle as $p \rightarrow p_r$. Since I_x/I is of the order of 0.1, the resonance condition occurs when the roll rate is nearly equal to the pitch frequency corresponding to the no roll-rate condition. Since p_r depends upon the product ρV^2 , we might anticipate the possibility of more than one resonance point during the re-entry trajectory. The reason is that the density ρ will increase with diminishing altitude while the velocity magnitude V will diminish. For many RV's we can expect that both a "high" resonance and a "low" resonance will occur during re-entry. We shall have more to say about resonance later in this book when we consider the re-entry vehicle which does not have axial symmetry in either mass distribution or configuration.

Up to now we have expressed the equations of motion of an axial symmetric RV in terms of the complex variable ξ [see Eq. (10.17)]. The solution to this equation was written in terms of this complex variable and five complex constants: K_1, K_2, K_3, m_1 , and m_2 . Subsequently we shall consider the complex constants K_1, K_2 , and K_3 as vectors in the complex plane where the imaginary part is α the angle of attack, and the real part is β the slideslip

angle. Let us express the variables and constants in terms of their corresponding imaginary and real parts:

$$\begin{aligned}
 \xi &= i\alpha + \beta \\
 K_1 &= iA_1 + B_1 \\
 K_2 &= iA_2 + B_2 \\
 K_3 &= iA_3 + B_3 \\
 m_1 &= \lambda_1 + i\omega_1 \\
 m_2 &= \lambda_2 + i\omega_2
 \end{aligned} \tag{10.45}$$

Using the above representations we can now write Eq. (10.20) as

$$\begin{aligned}
 i\alpha + \beta &= (iA_1 + B_1)e^{(\lambda_1 + i\omega_1)t} + (iA_2 + B_2)e^{(\lambda_2 + i\omega_2)t} \\
 &\quad + (iA_3 + B_3)e^{ipt}
 \end{aligned} \tag{10.46}$$

Equating real and imaginary parts gives two equations in the real variables α and β :

$$\begin{aligned}
 \alpha &= A_1(\cos\omega_1 t + B_1\sin\omega_1 t)e^{\lambda_1 t} + (A_2\cos\omega_2 t + B_2\sin\omega_2 t)e^{\lambda_2 t} \\
 &\quad + (A_3\cos pt + B_3\sin pt)
 \end{aligned} \tag{10.47a}$$

$$\begin{aligned}
 \beta &= (B_1\cos\omega_1 t - A_1\sin\omega_1 t)e^{\lambda_1 t} + (B_2\cos\omega_2 t - A_2\sin\omega_2 t)e^{\lambda_2 t} \\
 &\quad + (B_3\cos pt - A_3\sin pt)
 \end{aligned} \tag{10.47b}$$

In Eqs. (10.26) we expressed the constants K_1 and K_2 in terms of the initial conditions on the variable ξ and its first derivative [i.e., $(\xi_0, \dot{\xi}_0)$]. The corresponding equation for K_3 was given in Eq. (10.40). Replacing the K 's by A 's and B 's and ξ_0 by $(i\alpha + \beta)$ and $\dot{\xi}_0$ by $(i\dot{\alpha} + \dot{\beta})$, we can get the following equations for the constants of integration A_1, A_2, A_3, B_1, B_2 , and B_3 .

$$\begin{aligned}
 A_1 &= \left[\frac{[\dot{\alpha}_0 - \lambda_2\alpha_0 - \omega_2\beta_0 + \lambda_2A_3 - (p - \omega_2)B_3](\lambda_1 - \lambda_2)}{(\omega_1 - \omega_2)^2 + (\lambda_1 - \lambda_2)^2} \right] \\
 &\quad - \left[\frac{[\dot{\beta}_0 - \lambda_2\beta_0 + \omega_2\alpha_0 + (p - \omega_2)A_3 + \lambda_2B_3](\omega_1 - \omega_2)}{(\omega_1 - \omega_2)^2 + (\lambda_1 - \lambda_2)^2} \right]
 \end{aligned} \tag{10.48a}$$

$$B_1 = \left[\frac{[\dot{\alpha}_0 - \lambda_2 \alpha_0 - \omega_2 \beta_0 + \lambda_2 A_3 - (p - \omega_2) B_3](\omega_1 - \omega_2)}{(\omega_1 - \omega_2)^2 + (\lambda_1 - \lambda_2)^2} \right] - \left[\frac{[\dot{\beta}_0 - \lambda_2 \beta_0 + \omega_2 \alpha_0 + (p - \omega_2) A_3 + \lambda_2 B_3](\lambda_1 - \lambda_2)}{(\omega_1 - \omega_2)^2 + (\lambda_1 - \lambda_2)^2} \right] \quad (10.48b)$$

$$A_2 = \left[\frac{[\dot{\alpha}_0 - \lambda_1 \alpha_0 - \omega_1 \beta_0 + \lambda_1 A_3 - (p - \omega_1) B_3](\lambda_2 - \lambda_1)}{(\omega_1 - \omega_2)^2 + (\lambda_1 - \lambda_2)^2} \right] - \left[\frac{[\dot{\beta}_0 - \lambda_1 \beta_0 + \omega_1 \alpha_0 + (p - \omega_1) A_3 + \lambda_1 B_3](\omega_2 - \omega_1)}{(\omega_2 - \omega_1)^2 + (\lambda_2 - \lambda_1)^2} \right] \quad (10.48c)$$

$$B_2 = \left[\frac{[\dot{\alpha}_0 - \lambda_1 \alpha_0 - \omega_1 \beta_0 + \lambda_1 A_3 - (p - \omega_1) B_3](\omega_2 - \omega_1)}{(\omega_1 - \omega_2)^2 + (\lambda_1 - \lambda_2)^2} \right] - \left[\frac{[\dot{\beta}_0 - \lambda_1 \beta_0 + \omega_1 \alpha_0 + (p - \omega_1) A_3 + \lambda_1 B_3](\lambda_2 - \lambda_1)}{(\omega_2 - \omega_1)^2 + (\lambda_2 - \lambda_1)^2} \right] \quad (10.48d)$$

$$A_3 = \frac{M_\delta \delta}{I} \left[\frac{\omega_2(p - \omega_1) + p(\omega_1 - p) + \lambda_1 \lambda_2}{[\lambda_2(\omega_1 - p) + \lambda_1(\omega_2 - p)]^2 + [\omega_2(p - \omega_1) + p(\omega_1 - p) + \lambda_1 \lambda_2]^2} \right] \quad (10.48e)$$

$$B_3 = \frac{M_\delta \delta}{I} \left[\frac{\lambda_2(\omega_1 - p) + \lambda_1(\omega_2 - p)}{[\lambda_2(\omega_1 - p) + \lambda_1(\omega_2 - p)]^2 + [\omega_2(p - \omega_1) + p(\omega_1 - p) + \lambda_1 \lambda_2]^2} \right] \quad (10.48f)$$

A computer program to calculate α and β as time functions is given in Appendix E. This program is based upon Eqs. (10.47) with the A 's and B 's given by the expressions in Eqs. (10.48) and the λ 's and ω 's taken from Eqs. (10.31) or (10.36).

Figures 10.4(a) and 10.4(b) presents $\alpha - \beta$ plots of representative RV's having the properties listed in Table (13.1). The appropriate initial conditions are:

$$\alpha_0 = 12\text{deg} \quad \dot{\alpha}_0 = 0 \quad \beta_0 = 0 \quad \dot{\beta}_0 = 0$$

The velocity magnitude is taken at 3 km/s; the roll-rate p is 10 rad/s in Fig. 10.4(a) and 25 rad/s in Fig. 10.4(b). The atmospheric pressure ratio was set at 8.567×10^{-3} which according to Table 2.1 corresponds to an altitude of 32 km.

In Figs. 10.6(a) and 10.6(b) the origin may be interpreted as a point view of either the wind vector (positive direction) or the linear velocity of the

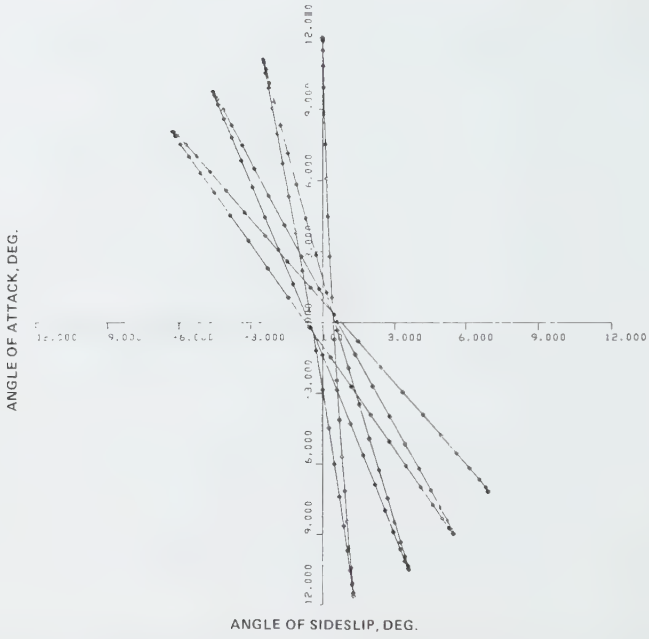


Fig. 10.6(a) Angle of attack vs angle of sideslip for a roll rate of 10 rad/s.



Fig. 10.6(b) Angle of attack vs angle of sideslip for a roll rate of 25 rad/s.

center of mass (negative direction). If the RV has no disturbance and no body-fixed asymmetry, then the vertex of the RV will be at the origin for trim conditions. In these figures the RV was disturbed initially in angle of attack by 12 deg. The statically stable characteristics of the RV together with the inertial coupling through roll rate cause a rotation of the plane of oscillation. As pointed out by Vaughn (Ref. 1) and earlier by Nicolaides (Ref. 2) and Murphy (Ref. 3) $\alpha - \beta$ plots such as those given in the preceding figures may be formed by summing three rotating vectors $\mathbf{K}_1$, $\mathbf{K}_2$, and $\mathbf{K}_3$. These vectors may be written from Eq. (10.24) as

$$\begin{aligned}\mathbf{K}_1 &= K_{1_0} e^{(\lambda_1 + \omega_1 t)t} \\ \mathbf{K}_2 &= K_{2_0} e^{(\lambda_1 + \omega_1 t)t} \\ \mathbf{K}_3 &= K_{3_0} e^{ipt}\end{aligned}\quad (10.49)$$

The damping and frequency terms $\lambda_{1,2}$ and $\omega_{1,2}$ were defined earlier in Eqs. (10.27) and (10.39). The complex constants K_{1_0} and K_{2_0} (or their equivalent in real variables A_1 , B_1 , A_2 , and B_2) represents the initial values of the vectors [see Eqs. (10.26)]; the vector $\mathbf{K}_3$ remains of the same magnitude throughout the angular motion. As pointed out in the discussion centering around Eq. (10.46), the magnitude of $\mathbf{K}_3$ depends upon the size of the body-fixed asymmetry and for statically stable RV's, also upon the relative values of the roll-rate p and the nutational frequency ω_1 . Equation (10.49) indicates that all three vectors rotate in the complex plane, $\mathbf{K}_1$ at the nutational frequency ω_1 , $\mathbf{K}_2$ at the precessional frequency, ω_2 , and $\mathbf{K}_3$ at the roll-rate p . Vectors $\mathbf{K}_1$ and $\mathbf{K}_2$ shrink in magnitude since they are multiplied by the factor $\exp(\lambda_1 t)$ and $\exp(\lambda_2 t)$, providing of course that the RV is dynamically stable.

It was shown in Eq. (10.37) that for a statically *unstable* RV ($s > 1, \tau > 1$) both ω_1 and ω_2 have the same sign. Thus vectors $\mathbf{K}_1$ and $\mathbf{K}_2$ will rotate in the same direction as $\mathbf{K}_3$ (which always rotates at the roll-rate p) although the rates of rotation of all three vectors will be different. For the statically stable RV ($s < 0, \tau < 1$) which is of more practical interest, ω_1 and ω_2 will rotate in opposite directions; the nutational frequency will be of the same sign as that of the roll rate, while the precessional frequency will be of opposite sign. A representation of these three vectors for the statically stable case is given in Fig. 10.7. The angular positions depicted Ψ_1 , Ψ_2 , and Ψ_3 are:

$$\begin{aligned}\Psi_1 &= \omega_1 t + \tan^{-1}(B_1/A_1) \\ \Psi_2 &= \omega_2 t + \tan^{-1}(B_2/A_2) \\ \Psi_3 &= pt + \tan^{-1}(B_3/A_3)\end{aligned}$$

Motion represented by Eq. (10.24) and depicted in Fig. 10.7 is called "tricyclic motion" in recognition of the three vectors as "arms" into which the angular motion is resolved.

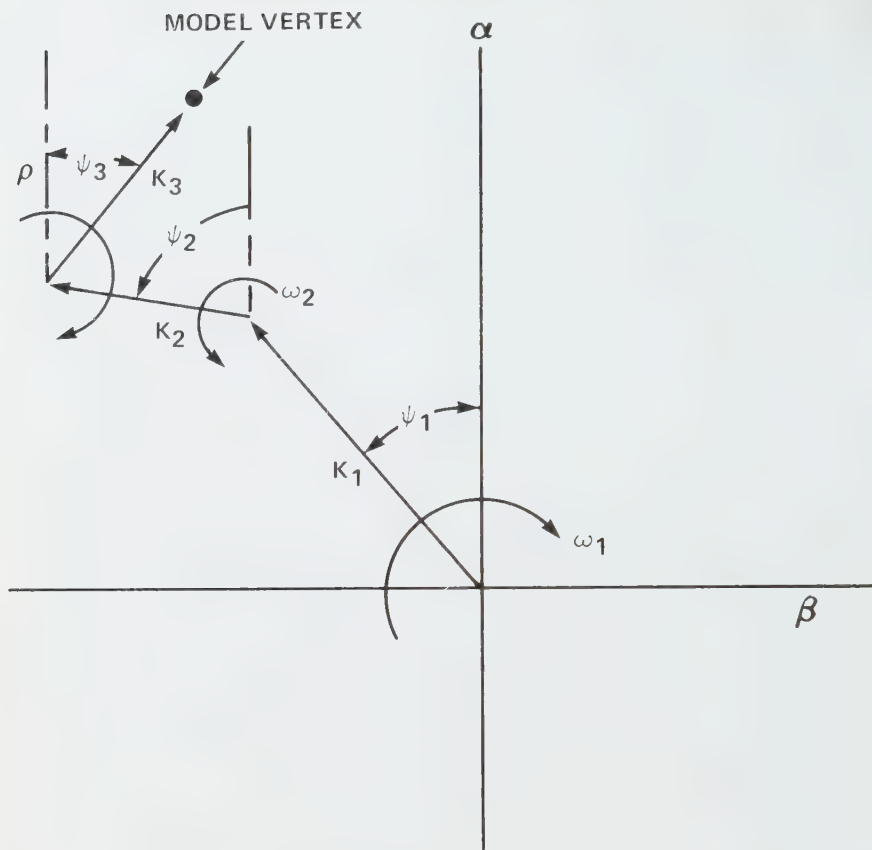


Fig. 10.7 Tricyclic motion.

In Eq. (10.47) it was shown that the motion is initially established by six constants A_1, B_1, A_2, B_2, A_3 , and B_3 . These constants have been expressed in terms of the frequencies, damping rates, and initial conditions $(\alpha_0, \dot{\alpha}_0, \beta_0, \dot{\beta}_0)$ in Eqs. (10.48). Most RV's as well as other statically stable vehicles, have the characteristic that the damping exponents λ_1 and λ_2 are small in comparison to the circular frequencies ω_1 and ω_2 as

$$\lambda_1 \ll \omega_1$$

$$\lambda_2 \ll \omega_2$$

Neglecting λ_1 and λ_2 where they appear in the expressions for the above constants results in an error that is typically about 2%.[‡] With the

[‡] A more exact estimate of this error according to Vaughn (Ref. 1) is: $(2\lambda_1/\omega_1) \times 100, (2\lambda_2/\omega_2) \times 100$.

assumption noted, the six constants may be written as,

$$A_1 = - \left[\frac{\dot{\beta}_0 + \omega_2 \alpha_0 + (p - \omega_2) A_3}{\omega_1 - \omega_2} \right] \cong - \left[\frac{\dot{\beta}_0 + \omega_2 \alpha_0}{\omega_1 - \omega_2} \right]_{\delta=0} \quad (10.50a)$$

$$B_1 = \left[\frac{\dot{\alpha}_0 - \omega_2 \beta_0}{\omega_1' - \omega_2} \right] \quad (10.50b)$$

$$A_2 = - \left[\frac{\dot{\beta}_0 + \omega_1 \alpha_0 + (p - \omega_1) A_3}{\omega_2 - \omega_1} \right] \cong - \left[\frac{\dot{\beta}_0 + \omega_1 \alpha_0}{\omega_2 - \omega_1} \right]_{\delta=0} \quad (10.50c)$$

$$B_2 = \left[\frac{\dot{\alpha}_0 - \omega_1 \beta_0}{\omega_2 - \omega_1} \right] \quad (10.50d)$$

$$A_3 = \frac{M_\delta \delta}{I_y} \left[\frac{1}{\omega_2(p - \omega_1) + p(\omega_1 - p)} \right] \quad p \neq \omega_1 \quad (10.50e)$$

$$B_3 = 0 \quad p = \omega_1 \quad (10.50f)$$

where simplification at the far right assumes that there is no body-fixed asymmetry, i.e., $A_3 \sim \delta = 0$. If A_1 and B_1 are regarded as the components of a polar vector $\mathbf{K}_1$ in the β (horizontal) and α (vertical) directions, respectively, at time $t = 0$ then

$$K_{1t=0} = K_{10} = \sqrt{A_1^2 + B_1^2} \quad (10.51a)$$

A similar relationship may be written for A_2 and B_2 as

$$K_{2t=0} = K_{20} = \sqrt{A_2^2 + B_2^2} \quad (10.51b)$$

and for A_3 and B_3 as

$$K_{3t=0} = K_{30} = \sqrt{A_3^2 + B_3^2} \quad (10.51c)$$

Another interpretation of the A and B constants is as initial values of α and β or

$$\alpha |_{t=0} \equiv \alpha_0 = A_1 + A_2 + A_3 \quad (10.52a)$$

$$\beta |_{t=0} \triangleq \beta_0 = B_1 + B_2 + B_3 \quad (10.52b)$$

We have used the term "tricyclic" to characterize the RV's dynamic motion about the velocity vector and have indicated such motion generally in Fig. 10.7 and in particular in Figs. 10.6(a) and 10.6(b). Because we can vary the four initial conditions over a continuous range of values, we can anticipate an infinite variety of motion patterns. Nevertheless we shall identify some characteristic angular motion and show how the observed motion is dependent upon the initial conditions. For this discussion we can use the simplified relationships for the A and B constants presented in Eqs. (10.50). We shall also assume that there are no configurational asymmetries present, i.e., $M_\delta \delta = 0 = A_3 = B_3$.

Let us first look at the case when the roll-rate p is zero. According to Eqs. (10.38), the nutational and precessional frequencies are of equal magnitude, but of opposite sign, i.e., $\omega_1 = -\omega_2 = \omega$. From Eq. (10.39) it also follows that $\lambda_1 = \lambda_2 = \lambda$. From Eqs. (10.50) the constants of integration can be written as:

$$A_1 = -\frac{\dot{\beta}_0}{2\omega} + \frac{\alpha_0}{2} \quad (10.53a)$$

$$B_1 = -\frac{\dot{\alpha}_0}{2\omega} + \frac{\beta_0}{2} \quad (10.53b)$$

$$A_2 = -\frac{\dot{\beta}_0}{2\omega} + \frac{\alpha_0}{2} \quad (10.53c)$$

$$B_2 = -\frac{\dot{\alpha}_0}{2\omega} + \frac{\beta_0}{2} \quad (10.53d)$$

We can make some simplifications to Eqs. (10.53) to develop some representative trajectories. If we let $\dot{\alpha}_0 = 0$ and $\beta_0 = 0$ then from Eq. (10.48) we have:

$$K_1 = -\frac{\dot{\beta}_0}{2\omega} + \frac{\alpha_0}{2} \quad (10.54a)$$

$$K_2 = -\frac{\dot{\beta}_0}{2\omega} + \frac{\alpha_0}{2} \quad (10.54b)$$

Again using Eq. (10.24) to emphasize the nutational and precessional aspects of the motion we have:

$$\xi = i\alpha + \beta = [K_1 e^{(\omega t)i} + K_2 e^{(-\omega t)i}] e^{\lambda t} \quad (10.55)$$

In Eq. (10.55) we recognize that the K_1 (nutation) arm rotates counterclockwise in the Argand plane and the K_2 (precession) arm rotates clockwise. If $\beta_0 = 0$ then both arms are of equal magnitude, and since they rotate in opposite directions at the same rate, the resulting motion is planar, appear-

ing as a straight line in the complex $\alpha - \beta$ plane. This situation is depicted in Fig. 10.8(a). Now if $\dot{\beta}_0$ is assigned increasingly negative values; then the motion becomes increasingly elliptical since K_1 will increase while K_2 will decrease in magnitude. Both still rotate in opposite directions at the same rate. Figure 10.8(b) and 10.8(c) depict increasingly negative values of $\dot{\beta}_0$. Finally $\dot{\beta}_0 = -\omega\alpha_0$, and the K_2 arm vanishes and the motion becomes circular. This limiting case is shown in Fig. 10.8(d). Thus, the presence of circular motion should not be taken to indicate RV roll rate. A similar sequence could be followed for increasingly positive values being assigned to $\dot{\beta}_0$. Under such circumstances the K_1 arm would eventually vanish and the motion would again be circular, although the rotation rate would be clockwise. Similar reasoning could also be applied if we took $\dot{\alpha}_0$ and β_0 as nonzero and set α_0 and $\dot{\beta}_0$ equal to zero.

If we introduce roll rate into the description of RV dynamics the physics becomes more complicated. First we note from Eqs. (10.50) that the A and B constants will vary with roll rate since the difference between the nutational and precessional frequencies appears in the denominator of expressions for the A 's and B 's. This difference between the nutational and precessional frequencies may be shown from Eq. (10.37) to be;

$$\omega_1 - \omega_2 = 2 \left[\left(\frac{pI_x}{2I} \right)^2 + \omega_s^2 \right]^{1/2} \quad (10.56)$$

where ω_s is the undamped natural frequency of the RV in the absence of roll rate, i.e., $\omega_s = \sqrt{-M_0/I}$. Thus, with $\dot{\alpha}_0 = \dot{\beta}_0 = 0$ and α_0 and β_0 nonzero, we could write the equivalent of Eqs. (10.54). However the resulting expressions are sufficiently complicated to obscure the essential physics we are trying to illuminate in this brief discussion. It is sufficient here to note that the K_1 and K_2 arms will vary according to the relative magnitudes and signs of β_0 and α_0 . For example, to cause the K_2 arm to vanish it would be necessary for $\dot{\beta}_0$ to have the following value:

$$\dot{\beta}_0 = - \left[\left(\frac{pI_x}{2I} \right) + \left\{ \left(\frac{pI_x}{2I} \right)^2 + \omega_s^2 \right\}^{1/2} \right] \alpha_0 \quad (10.57)$$

rather than the simple value developed earlier for the nonrolling case,

$$\dot{\beta}_0 = -\omega_s \alpha_0$$

Also the two characteristic frequencies, although still of opposite sign, no longer have the same magnitude.

Vaughn (Ref. 1) considered four cases in which $\dot{\beta}_0$ becomes increasingly negative. As a consequence the magnitude of the $\mathbf{K}_1$ vector increases relative to the magnitude of the $\mathbf{K}_2$ vector. Vaughn chose the particular case where $\omega_2 = -0.846\omega_1$. Examples of $\alpha - \beta$ petal plots in which $\dot{\beta}_0$ is increasingly negative, are taken from Vaughn (Ref. 1) and presented in Fig. 10.9.

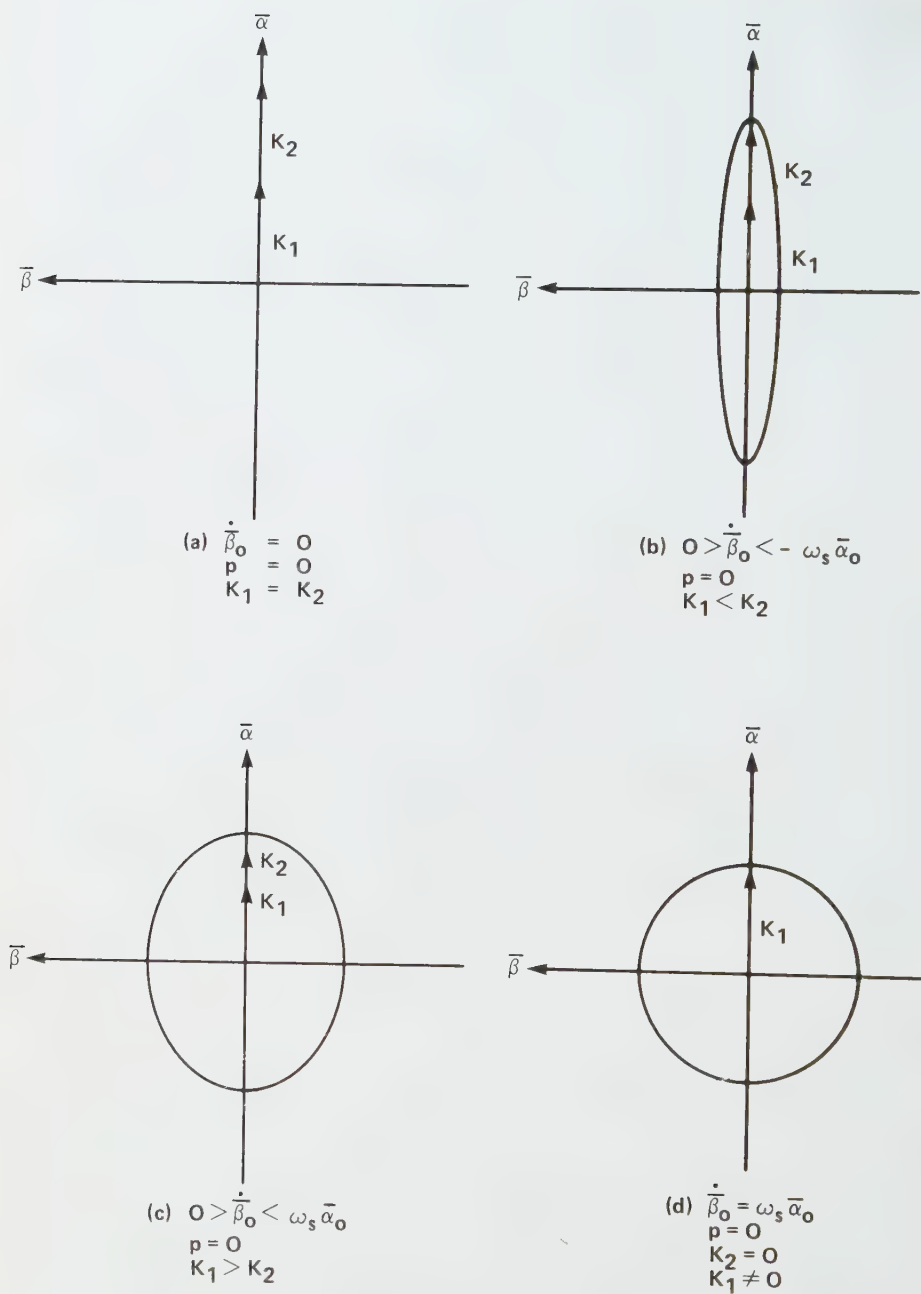
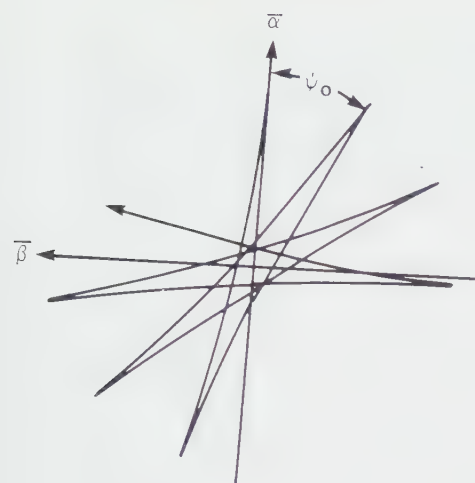


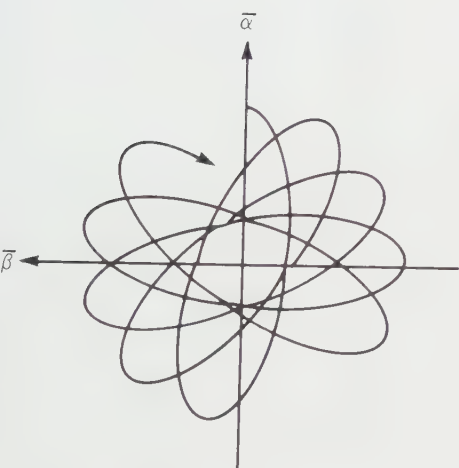
Fig. 10.8 Characteristic angular motion of a nonrolling RV.¹



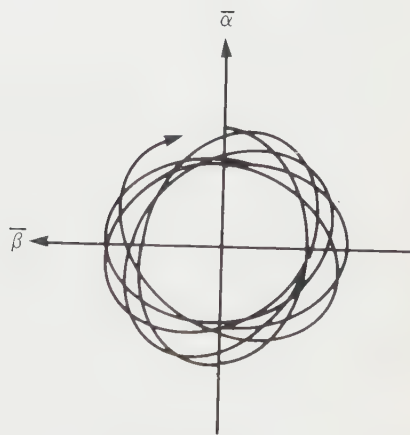
$$\begin{aligned}\dot{\beta}_0 &= 0 \\ \mathbf{p} &\neq 0 \\ \omega_2 &= -.846\omega_1 \\ K_1 &> K_2\end{aligned}$$



$$\begin{aligned}\dot{\beta}_0 &\neq 0 \\ \mathbf{p} &\neq 0 \\ \omega_2 &= -.846\omega_1 \\ K_1 &> K_2\end{aligned}$$



$$\begin{aligned}\dot{\beta}_0 &\neq 0 \\ \mathbf{p} &\neq 0 \\ \omega_2 &= -.846\omega_1 \\ K_1 &> K_2\end{aligned}$$



$$\begin{aligned}\dot{\beta}_0 &\neq 0 \\ \mathbf{p} &\neq 0 \\ \omega_2 &= -.846\omega_1 \\ K_1 &\gg K_2\end{aligned}$$

Fig. 10.9 Characteristic motion of a rolling re-entry vehicle.¹

Note that for $\dot{\beta}_0 = 0$ the petal figure has sharp cusps. As $\dot{\beta}_0$ takes on increasingly negative values the cusps become more rounded until in the final diagram the motion is elliptical, but nearly circular.

An obvious feature of dynamic motion with roll rate; it is the rotation of the "plane of oscillation." This rate, which we shall describe by Ψ (after Vaughn), must equal the sum of the two fundamental frequencies ω_1 and ω_2 as

$$\dot{\Psi} = \omega_1 + \omega_2 = (pI_x / I) \quad (10.58a)$$

Clearly, the rate of rotation of the plane of oscillation is in the same direction as the roll rate. If Ψ is distinguishable in a given petal plot, then the measurement of Ψ may be used as an estimate of the roll rate p . The angle Ψ_0 , defined in Fig. (10.9), is the angle through which the plane of oscillation has rotated during one complete angular oscillation. Vaughn shows that Ψ_0 may be expressed in terms of ω_1 and ω_2 as

$$\Psi_0 = 2\pi \left[\frac{\omega_1}{\omega_2} + 1 \right] \bigg/ \left[\frac{\omega_1}{\omega_2} - 1 \right] \quad (10.58b)$$

In Chapter XIII we shall again discuss the angular motion of the RV within the atmosphere. At that time we shall relax certain assumptions that were made in this chapter such as the restrictions of the configuration to "slight" asymmetries and the atmospheric density to a constant.

Before leaving this chapter we should consider attribution of the many ideas that have been developed and expressed herein. There is some controversy associated with the origins of the equations of aeroballistic motion known as "tricyclic theory." Reference 2 is often considered the seminal paper in the statement of the tricyclic theory. However, Ref. 2 should be recommended as a unifying paper as many of the ideas expressed therein predate this report by many years. Reference 3 is also a broad statement of the tricyclic theory, although it is more terse in development of the elementary theory than either Ref. 1 or 2. It does contain many applications and extensions particularly to problems associated with the ballistic range. Although Ref. 3 has a later date of publication than Ref. 2, it references open literature (both the author of this reference and others) that predate the unifying work of Nicolaides.

Another difference between Nicolaides (Ref. 2) and Vaughn (Ref. 1) on the one hand and Murphy (Ref. 3) on the other is that the former works use time as the independent variable, while the latter uses a nondimensional arc length. Reference 1 by Vaughn, while making no claim to precedence, seems to be one of the most readable efforts to bring the tricyclic theory to the practitioner. It is for this reason that all the material in this chapter follows closely this important exposition by Vaughn.

References

¹Vaughn, H. B., "A Detailed Development of the Tricyclic Theory," Sandia National Laboratory, Albuquerque, N. Mex., Rept. SC-M-67-2933, Feb. 1968.

²Nicholaides, J. D., "On the Free Flight Motion of Missiles Having Slight Configurational Asymmetries," Ballistic Research Laboratories, Rept. No. 858, June 1953, Aberdeen, Md.

³Murphy, C. H., "Free Flight Motion of Symmetric Missiles," Ballistic Research Laboratories, Rept. No. 1216, July 1963, Aberdeen, Md.

XI. BOOST TRAJECTORIES

In this chapter we shall consider the dynamics associated with boosting an RV from the surface of the Earth. Because of the presence of a thruster as well as varying aerodynamic loads, we might anticipate difficulties in finding meaningful closed-form solutions. Since thrusters are an essential part of the boost phase, we shall devote some effort to defining the properties of rocket motors. Then we shall derive the equations of motion of a boosted RV from a nonrotating Earth.

11.1 Rocket Motors

The development of the performance equations of a rocket engine from first principles has been carried out at varying levels of rigor in books on reaction engines. Probably the most readable short account is that due to Ashley (Ref. 1). A more applications-oriented treatment is that by Sutton (Ref. 2). In these as well as many other expositions, the rocket gas, i.e., the products of combustion, is regarded as a thermodynamic fluid with essentially fixed chemical constituents. We shall accept this model for the following development. This thermodynamic fluid will be regarded as a continuum of constant molecular weight that obeys the ideal gas law and the first law of thermodynamics. Since there will not be space to derive the equations of motion of such a fluid, the reader is referred to the excellent and rigorous treatment of these equations of motion by Tsien in Ref. 3. We shall present here a derivation of the performance equations of a rocket that parallels that given by Ashley in Ref. 1.

Assume that a rocket engine is a device which ejects a thermodynamic fluid from the base of a vehicle which is moving through the atmosphere. We shall now surround the vehicle by a cylindrical control volume which is fixed relative to the vehicle; the sides and front face of this cylinder are so remote that the pressure is essentially at the freestream static value, i.e., $p_u = p_\infty = p_s$.

In Fig. 11.1(a) we have depicted the vehicle and the surrounding control volume. The sides of the cylinder, denoted by S_s , are formed from streamlines; hence S_s is a stream surface. The upstream end of the cylinder has a surface S_u and the directed normal to S_u is parallel to the x axis. The downstream surface of the control volume cylinder, designated by S_d , has its

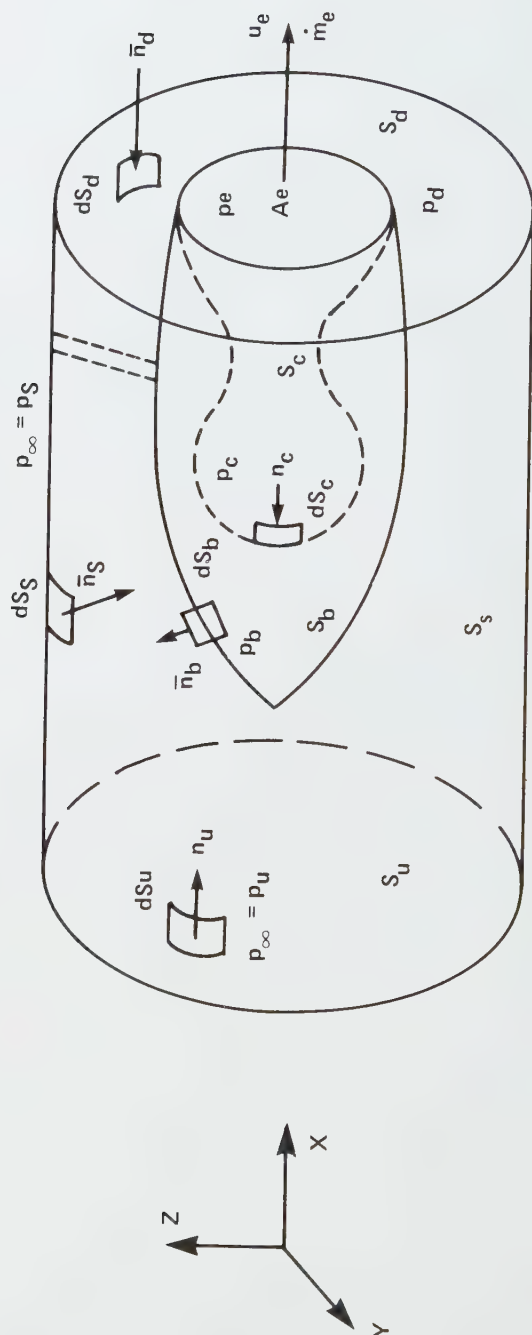


Fig. 11.1(a) Control volume.

normal parallel to the x axis. However, this surface S_d also contains the final cross section of the rocket nozzle. The pressure across S_d surface varies over the surface and is designated here simply as p_d , it will be noted that the area of the exhaust jet or final cross section of the nozzle is designated as A_e .

A_e is a subarea of the surface S_d and for our purposes the pressure p_e over A_e will be regarded as a constant. According to Euler's momentum theorem, the integral of the pressure over any closed surface is related to the momentum flux through the same surface as follows:

$$\iint_S p \mathbf{n} ds = - \iint_S [p \mathbf{u}(\mathbf{u} \cdot \mathbf{n})] ds \quad (11.1)$$

We shall consider the surface S to have the following components: S_s , S_u , and S_d which we have defined above, and S_b and S_c which are, respectively, the body area (which does not include the jet area A_e) and the rocket chamber area which is entirely internal to the body. An integral over the areas S_b and S_c is a closed integral around the vehicle, i.e., externally over S_b and internally over S_c . Let us look at the lefthand side of Eq. (11.1) first and confine our attention to force components in the direction of the x axis. Thus,

$$\begin{aligned} \mathbf{i} \cdot \iint_S p \mathbf{n} ds &= \iint_S p(\mathbf{n} \cdot \mathbf{i}) ds = \int_{S_u} p_\infty(\mathbf{i} \cdot \mathbf{n}_u) ds \\ &+ \int_{S_s} p_\infty(\mathbf{i} \cdot \mathbf{n}_s) ds + \int_{S_d} p_d(\mathbf{i} \cdot \mathbf{n}_d) ds + \int_{S_b} p_b(\mathbf{i} \cdot \mathbf{n}_b) ds + \int_{S_c} p_c(\mathbf{i} \cdot \mathbf{n}_c) ds \end{aligned} \quad (11.2)$$

We first recognize that the second integral is zero since the unit vectors $\mathbf{i}$ and $\mathbf{n}_s$ are perpendicular. We can combine the first and third integrals by rewriting the first integral as

$$\int_{S_u} p_\infty(\mathbf{i} \cdot \mathbf{n}_u) ds = - \int_{S_d - A_e} p_\infty ds - \int_{A_e} p_\infty ds \quad (11.3)$$

Areas S_u and S_d are equal since we have assumed a cylindrical control volume. The third integral may also be written as

$$\int_{S_d} p_d(\mathbf{i} \cdot \mathbf{n}_d) ds = \int_{S_d - A_e} p_d ds + \int_{A_e} p_d ds \quad (11.4)$$

Since the pressure p_d has the constant value p_e over the jet exit area A_e we obtain for the first and third integrals of Eq. (11.2):

$$\int_{S_u} p_\infty(\mathbf{i} \cdot \mathbf{n}_u) ds + \int_{S_d} p_d(\mathbf{i} \cdot \mathbf{n}_d) ds = \int_{S_d - A_e} (p_d - p_\infty) ds + (p_e - p_\infty) A_e \quad (11.5)$$

The fourth and the fifth integrals of Eqs. (11.2) are defined as the drag D and the thrust T , respectively as

$$D = \int_{S_b} p_b(\mathbf{i} \cdot \mathbf{n}_b) ds \quad (11.6)$$

$$-T = \int_{S_c} p_c(\mathbf{i} \cdot \mathbf{n}_c) ds \quad (11.7)$$

In other words, drag D is defined as the integral of all pressure forces over the *exterior* of the body; the thrust T is the integral of the pressure forces over the *interior* of the body. Since we are regarding the x axis as positive downstream, drag is positive and thrust is negative.

It would seem that we are integrating over two closed surface areas, one area which bounds the control volume and a second area which bounds the vehicle. The control volume is bounded by the area components S_u , S_s , and S_d (which includes the jet efflux area A_e); the vehicle is bounded by the area components S_b and S_c , the former the body's surface area and the latter the body's interior or the surface area of the rocket nozzle combustion chamber. It is tacitly assumed that these two closed areas are connected by a vanishingly thin tube shown as a dotted line in Fig. 11.1(a). It is assumed further that a surface integral over the tube's interior and exterior will be of equal magnitude, but of opposite sign and hence will make no contribution. A two-dimensional version of Fig. 11.1(a) is given in Fig. 11.1(b) to clarify these basic ideas.

The surface integral, now represented by a line integral, begins at A and follows the arrows to B around the exterior/interior of the body, then around the control surface, and finally back to A .

Let us return to Eqs. (11.6) and (11.7). Since we are regarding the x axis as positive downstream, the drag force will be positive and the thrust force negative as we mentioned. Glancing at Fig. 11.1(b) we see that the *negative* normal for the external contour of the vehicle is essentially downstream while the *negative* normal in the interior of the vehicle, i.e., the rocket chamber and nozzle, is essentially upstream. Also since the external and

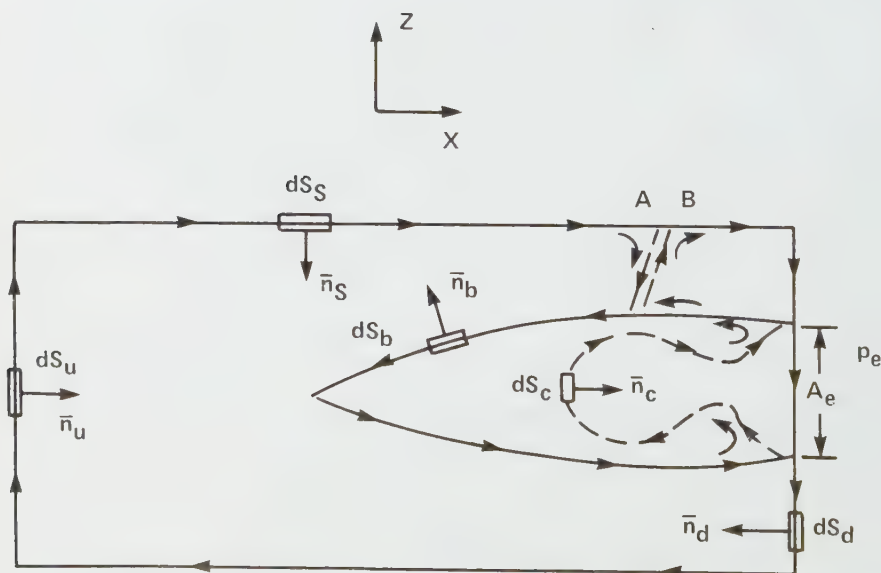


Fig. 11.1(b) Planar projection of control volume.

internal pressure fields have considerably different physical origins, the effects we have called drag and thrust must be considered independently. This does not mean that grey areas do not exist in the definition of the relative contributions of drag and thrust to the pressure integral of Eq. (11.1). Many semantic difficulties in precisely defining thrust and drag arise after an engine has been installed in a vehicle. Ashley (Ref. 1) briefly alludes to the existence of such ambiguities. To summarize, the lefthand side of Eq. (11.1) may now be expressed in terms of the components developed in Eqs. (11.3) and (11.5) as

$$\iint_S p(\mathbf{n} \cdot \mathbf{i}) ds = D - T + \int_{S_d - A_e} (p_d - p_\infty) ds + (p_e - p_\infty) A_e \quad (11.8)$$

We now must return to Eq. (11.1) and examine the righthand side. We could write five integrals paralleling the expressions given in Eq. (11.2). However, we recognize immediately that there is no momentum flux through either the body area S_b or the combustion chamber/nozzle area S_c . There is also no momentum flux through the sides of the cylindrical control volume since S_s is by definition a stream surface and hence the inner product $\mathbf{u} \cdot \mathbf{n}_s$ is identically zero. Thus we may write for the components of the momentum flux integral:

$$\begin{aligned} \iint_S \rho \mathbf{u}(\mathbf{u} \cdot \mathbf{n}) ds &= \int_{S_u} [\rho \mathbf{u}(\mathbf{u} \cdot \mathbf{n}_u)] ds + \int_{S_d} [\rho \mathbf{u}(\mathbf{u} \cdot \mathbf{n}_d)] ds \\ &= -\rho_\infty u_{x\infty}^2 S_u + \int_{S_d - A_e} \rho_d u_{xd}^2 ds + \rho_e u_e^2 A_e \end{aligned}$$

If we recognize that

$$\rho_e u_e^2 A_e = \dot{m}_e u_e \gg \rho_\infty u_{x\infty}^2 A_e$$

we get:

$$\iint_S \rho \mathbf{u}(\mathbf{u} \cdot \mathbf{n}) ds = \int_{S_d - A_e} (\rho_d u_{xd}^2 - \rho_\infty u_{x\infty}^2) ds + \dot{m}_e u_e \quad (11.9)$$

Combining Eqs. (11.8) and (11.9) according to Eq. (11.1) we get:

$$T = \dot{m}_e u_e + (p_e - p_\infty) A_e \quad (11.10a)$$

$$D = \int_{S_d - A_e} (p_\infty - p_d) ds + \int_{S_d - A_e} (\rho_\infty u_{xd}^2 - \rho_d u_{xd}^2) ds \quad (11.10b)$$

It will be noted in Eq. (11.10b) that the presence of drag causes a momentum and pressure defect across the area S_d . Our primary concern is with Eq. (11.10a) which we shall accept as the rocket thrust equation.

A commonly accepted index of performance of a rocket motor is the so-called specific impulse I_{sp} . The specific impulse is simply the ratio of the thrust to mass flow or

$$I_{sp} = T / \dot{m}_e \quad (11.11)$$

Obviously if we choose to ignore the second term on the right in Eq. (11.10a) then the specific impulse is simply the magnitude of the jet velocity u_e . Since the specific impulse term has its origins in engineering practice where at one time there was strong sentiment for measuring both force and mass in units of pounds, I_{sp} has the units of seconds or pounds (thrust)/pounds per second (mass flow). The penalty for insisting that the same unit be used to quantify both force and mass is the appearance in Newton's second law of a constant $1/g$ multiplying the mass. In this book we shall insist the thrust T be expressed in Newtons and that mass flow $\dot{m}_e$ be expressed in kg/s. Consequently, "conventional" values of specific impulse having units of "seconds" must be multiplied by g . For example, a kerosene (RP-1) and oxygen rocket fuel has a specific impulse of 350 s. In mks units the equivalent specific impulse would be 3430 m/s.

It might be of interest to express the specific impulse in terms of the thermodynamic properties of the products of combustion. Let us assume that the combustion of the propellants is essentially complete while such products are still in a condition of stagnation. Under this assumption we may regard the products of combustion to be in chemical equilibrium during passage from combustion chamber to nozzle exit. If we assume the gas to be a continuum, then we can describe the properties of the gas in terms of a gas constant R . We shall also assume that the flow is steady and one-dimensional (properties change only along the thrust axis of the nozzle and are invariant with time at a fixed axial station). Further we shall assume that the flow is essentially frictionless and that once the chemical properties have "frozen" after combustion, no heat is added or removed from the rocket gas. Tsien (Ref. 3) shows that under such circumstances the sum of the specific enthalpy h and the specific kinetic energy $\rho u^2/2$ are conserved. Thus at arbitrary axial stations (1) and (2) we have the relationship:

$$h_1 + \frac{1}{2}\rho_1 u_1^2 = h_2 + \frac{1}{2}\rho_2 u_2^2 \quad (11.12)$$

If (1) is taken in the combustion chamber, after combustion, but still under essentially stagnation conditions and (2) is taken at the end of the nozzle we have:

$$h_1 = h_c \quad u_1 = u_c = 0 \quad h_2 = h_e \quad u_2 = u_e$$

We can express the rocket exhaust velocity magnitude u_e in terms of the enthalpy change from (1) to (2) as

$$u_e = \sqrt{2(h_c - h_e)} \quad (11.13)$$

Since enthalpy h can be shown to a function of temperature only, we may write for the enthalpy difference $h_c - h_e$, or the change in enthalpy between the combustion chamber and nozzle exhaust as

$$h_c - h_e = C_p T_c \left[1 - \frac{T_e}{T_c} \right] = \left[\frac{\gamma}{\gamma - 1} \right] R T_c \left[1 - \left(\frac{P_e}{P_c} \right)^{(\gamma - 1)/\gamma} \right] \quad (11.14)$$

The constant of proportionality between enthalpy and temperature is the so-called specific heat C_p . The above equation is Ashley's Eq. (5.70) of Ref.

10. Equation (11.14) inserted into Eq. (11.13) gives an expression for the exhaust velocity magnitude u_e as

$$u_e = \left[\left[\frac{2\gamma RT_c}{\gamma - 1} \right] \left\{ 1 - \left[\frac{p_e}{p_c} \right]^{(\gamma-1)/\gamma} \right\} \right]^{1/2} \quad (11.15)$$

Since the mass flow must be constant from the combustion chamber to the nozzle exit, we may take conditions at the nozzle throat as reference,

$$\dot{m}_e = \rho_t A_t u_t \quad (11.16)$$

where the subscript t indicates conditions at the throat. Combustion chamber, throat, and exit conditions are shown in Fig. 11.2.

If we continue to assume that the flow is isentropic from station c to e , then the area ratio A_t/A_e is defined in terms of the "back pressure" ratio p_e/p_c . In other words, if a throat-to-exit area ratio A_t/A_e is specified, then the upper limit on the pressure ratio p_e/p_c is also set if isentropic flow is to be assumed between stations t and e . From Sutton's Eq. (3.25) of Ref. 2 we have:

$$\frac{A_t}{A_e} = \left[\frac{\gamma+1}{2} \right]^{1/(\gamma+1)} \left[\frac{p_e}{p_c} \right]^{1/\gamma} \left\{ \left[\frac{\gamma+1}{\gamma-1} \right] \left[1 - \left[\frac{p_e}{p_c} \right]^{(\gamma-1)/\gamma} \right] \right\}^{1/2} \quad (11.17)$$

The mass flow can be expressed in terms of gas properties in the combustion chamber and the throat area. From Ashley (Ref. 1) we have:

$$\dot{m}_e = p_c A_t \left[\frac{\gamma}{RT_c} \left[\frac{2}{\gamma+1} \right]^{(\gamma+1)/(\gamma-1)} \right]^{1/2} \quad (11.18)$$

Finally we may combine Eqs. (11.10a) and (11.15–11.18) to obtain an equation for the thrust T and specific impulse I_{sp} in terms of chamber pressure p_c , exhaust pressure p_e , and throat area A_t .

$$T = p_c A_t \left\{ \left[\left[\frac{2\gamma^2}{\gamma-1} \right] \left[\frac{2}{\gamma+1} \right]^{(\gamma+1)/(\gamma-1)} \left\{ 1 - \left[\frac{p_e}{p_c} \right]^{(\gamma-1)/\gamma} \right\} \right]^{1/2} + \left[\frac{A_e}{A_t} \right] \left[\frac{p_e}{p_c} - \frac{p_\infty}{p_c} \right] \right\} \quad (11.19)$$

$$I_{sp} = \left[\left[\frac{2\gamma RT_c}{\gamma-1} \right] \left\{ 1 - \left[\frac{p_e}{p_c} \right]^{(\gamma-1)/\gamma} \right\} \right]^{1/2} + \left[\frac{A_e}{A_t} \right] \left[\frac{p_e}{p_c} - \frac{p_\infty}{p_c} \right] \left[\frac{RT_c}{\gamma} \left[\frac{\gamma+1}{2} \right]^{(\gamma+1)/(\gamma-1)} \right]^{1/2} \quad (11.20)$$

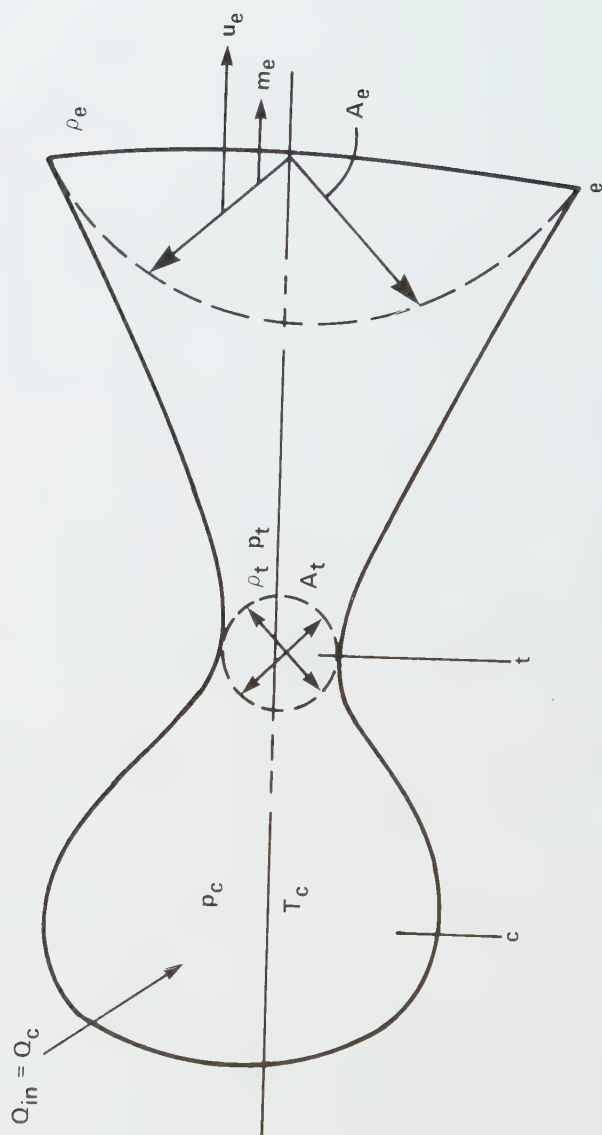


Fig. 11.2 Rocket engine stations.

Equation (11.19) expresses the thrust as a function of the throat area A_t , the combustion chamber pressure p_c , and the "back pressure" ratio p_e/p_c , or the ratio of the exhaust pressure p_e to the combustion chamber pressure p_c . If the exhaust pressure p_e is zero, then the thrust and specific impulse take on the particularly simple forms.

$$T = p_c A_t \left[\frac{2\gamma}{\gamma-1} \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)} \right]^{1/2} \quad (11.21a)$$

$$I_{sp} = \left[\frac{2\gamma R T_c}{\gamma-1} \right]^{1/2} \quad (11.21b)$$

We may note the direct proportionality between thrust and combustion chamber pressure. Also we see that increasing the combustion chamber temperature T_c increases the specific impulse. The specific impulse varies directly with the square root of the gas constant R associated with the products of combustion. Since, $R = R^*/M$, where R^* is the universal gas constant and M is the molecular weight of the products of combustion, we see that the specific impulse varies inversely with the square root of the molecular weight. Thus it would appear desirable for the products of combustion to be of low molecular weight.

In a preliminary design or performance estimate of a rocket nozzle design, the assumption is made that the gas flow is isentropic. An isentropic flow has characteristics which deserve emphasis: if a flow is postulated to be isentropic between two stations, no heat can be added to or removed from the gas between those stations. We must emphasize that heat is an interface property: Heat flow is between a gas and its surroundings, and therefore has no direct meaning in processes occurring in the interior of the gas. The energy represented by the heat, according to the first law of thermodynamics, is absorbed by the gas as internal energy or appears as mechanical work done by the gas on its surroundings. It is to be presumed that in an isentropic flow the energy stored in the gas in the form of internal energy is available for conversion into work at some later time or some later part of the flow process. The concept of an isentropic flow as applied to the flow of a gas in a rocket nozzle means that the energy given to the gas during the combustion process is either converted into mechanical energy or remains in the form of internal energy. This internal energy remains available for conversion into kinetic energy of flow at some subsequent point in the process, but this internal energy is never converted into heat. Thus, the assumption upon which the preceding equations are based is that from the stagnation condition existing just after combustion to flow exit from the nozzle, the internal energy is continually converted into mechanical energy, but at no time is this internal energy increased or decreased by heat flux at the gas boundary (rocket nozzle wall).

As a consequence of the thermodynamic relationships based upon the assumption of an isentropic flow, the mass flow at an arbitrary station (i)

can be expressed in terms of the throat area A_t , the pressure ratio p_t/p_c , and the temperature and pressure in the combustion chamber T_c and p_c , as

$$\dot{m}_e \equiv \dot{m} = \frac{A_t p_c}{R} \left\{ \frac{C_p}{T_c} \left[\left(\frac{p_t}{p_c} \right)^{2/\gamma} - \left(\frac{p_t}{p_c} \right)^{(\gamma+1)/\gamma} \right] \right\}^{1/2} \quad (11.22a)$$

If we postulate that the maximum mass flow per unit area occurs at the nozzle throat, we can differentiate $\dot{m}_e^2/A_t$ with respect to the pressure ratio p_t/p_c to give the following relationship for pressure ratio at the throat:

$$\frac{p_t}{p_c} = \left[\frac{2}{\gamma+1} \right]^{\gamma/(\gamma-1)} \quad (11.22b)$$

It follows from the isentropic relationships that the temperature ratio at the throat and combustion chamber may be written as

$$\frac{T_t}{T_c} = \left[\frac{2}{\gamma+1} \right] \quad (11.22c)$$

The velocity magnitude at the throat may be expressed in terms of either the throat or combustion chamber temperatures as

$$V_t = \left[\left(\frac{2\gamma}{\gamma+1} \right) RT_c \right]^{1/2} \quad (11.22d)$$

$$V_t = [\gamma RT_t]^{1/2} \quad (11.22e)$$

Equation (11.22e) indicates that sonic conditions exist at the throat provided that the pressure ratio is given according to Eq. (11.22b); if p_t/p_c is larger than the ratio given in Eq. (11.22b), then only a subsonic flow may exist at the throat (and everywhere else in the nozzle as well).

Example 11.1: Application of Results

We may now apply some of the preceding relationships to assess the performance of a hypothetical liquid-fueled rocket. Suppose we are given the following information pertaining to a kerosene-oxygen fuel ($RP-1-O_2$):

Molecular weight, $M = 21.6$

Ratio of specific heats, $\gamma = 1.24$

Combustion chamber temperature, $T_c = 3350$ K

Combustion chamber pressure, $p_c = 2 \times 10^6$ N/m²

Exhaust pressure, $p_e = 2 \times 10^4$ N/m²

If we wish the rocket thrust to be 600 N, find the following quantities:

(a) *Ratio of exhaust area to throat area, A_e/A_t .*

From Eq. (11.17) we obtain:

$$\frac{A_t}{A_e} = \left[\frac{1.24+1}{2} \right]^{1/(1.24-1)} (0.01)^{1/1.24} \left\{ \left[\frac{1.24+1}{1.24-1} \right] (1 - (0.01)^{(1.24-1)/1.24}) \right\}^{1/2}$$

$$= 9.174 \times 10^{-2}$$

or

$$A_e / A_t = 10.90$$

(b) Find the temperature, pressure, and gas speed at the nozzle throat.

From Eqs. (11.22) we have

$$T_t = \frac{2}{1.24+1} (3350) = 2291 \text{ K}$$

$$p_t = 1.113 \times 10^6 \text{ N/m}^2$$

To obtain the velocity magnitude at the throat V_t we must first calculate the gas constant:

$$R = R^* / M = 8.3143 \times 10^3 / 21.6 = 385$$

$$V_t = [(1.24)(385)(2291)]^{1/2} = 1195 \text{ m/s}$$

(c) Calculate the exhaust velocity magnitude, V_e .

From Eq. (11.15) with $V_e = u_e$, we obtain:

$$V_e = 2803 \text{ m/s}$$

(d) Find the throat area, A_t .

From Eq. (11.19) with thrust T equal to 600 N, we can solve for A_t to get:

$$A_t = 1.735 \times 10^{-4} \text{ m}^2$$

(e) Find the mass flow, $\dot{m}_e$.

The mass flow follows from Eq. (11.18)

$$\dot{m}_e = (2 \times 10^6)(1.735 \times 10^{-4}) \left[\frac{1.24}{(385)(3350)} \left[\frac{2}{1.24+1} \right]^{(1.24+1)/(1.24-1)} \right]^{1/2}$$

(f) Find the specific impulse, I_{sp} .

From Eq. (11.20) we have:

$$I_{sp} = 2992 \text{ N-s/kg}$$

or

$$I_{sp}/g = 2982/9.81 = 304 \text{ s}$$

It should be noted that if the back pressure ratio p_e/p_c is zero and the rocket exhausts into a vacuum ($p_\infty = 0$), we may calculate the specific impulse using Eq. (11.21b). Under these circumstances

$$I_{sp}/g = 372 \text{ s}$$

The point of view we have taken so far has been to regard the products of combustion as a thermodynamic fluid, i.e., a fluid which obeys the first and second laws of thermodynamics. Since we are considering this fluid to have essentially constant chemical properties, during its passage through the nozzle, it would seem of no consequence whether the rocket fuel was originally in solid or liquid form. While it is true that the preceding relationships among thrust, combustion chamber pressure conditions, mass flow, etc., are valid for the solid propellant rocket there are some additional performance parameters that we should specify for the solid-fueled rocket.

A significant difference between the liquid and solid propellant rocket is that in the latter case, the combustion chamber is continually increasing in volume. Let us consider a radially burning solid propellant. Mass conservation would require that the rate of consumption of solid propellant dm_f/dt equal the sum of the rate of change of mass in the combustion chamber dm_c/dt and the mass flow exiting the nozzle dm_e/dt . Thus

$$\frac{dm_f}{dt} = A_b \rho_b r_b = \frac{dm_c}{dt} + \frac{dm_e}{dt}$$

or

$$A_b \rho_b r_b = \frac{d}{dt}(\rho_c V_c) + \frac{dm_e}{dt} \quad (11.23a)$$

Let us consider the lefthand side of Eq. (11.23a) first. The rate of fuel consumption is equal to the product of the area of the burning surface A_b , the density of the solid fuel ρ_b , and the linear burn or recession rate of the burn surface r_b . On the righthand side, the first term is the rate of change of the mass within the combustion chamber. Since we might assume that the density ρ_b of the combustion products is constant, the first term on the right means that the volume of the combustion chamber V_c is increasing. In a liquid-fuel rocket this combustion chamber mass flow $d(\rho_c V_c)/dt$ would be zero; however, for a solid-fuel rocket the combustion chamber volume is constantly increasing due to the radial recession of the burn surface. The second term on the right is the mass effluent rate which has been identified earlier as $\dot{m}_e$.

For our purposes here we can ignore the first term on the right in Eq. (11.23a) and rewrite the dm_e/dt term using Eq. (11.18) to get:

$$\frac{A_b}{A_t} = \left[\frac{p_c}{r_b \rho_b} \right] \left[\frac{\gamma}{R_c T_c} \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)} \right]^{1/2} \quad (11.23b)$$

The area ratio A_b/A_t of the burning-surface area to the throat area is an important parameter in solid-fuel engine rocket technology. Equation (11.23b) is a relationship between the area ratio A_b/A_t , combustion chamber pressure and temperature p_c, T_c , and propellant properties R_p, ρ_b, r_b , and γ . All propellant properties are available from propellant property tables. Nevertheless, there is a useful empirical relationship between recession rate r_b and chamber pressure p_c . According to Sutton (Ref. 2), we may write

$$r_b = a(p_c^n) \quad (11.23c)$$

From tabular data contained in Ref. 2, (p. 316), we may calculate representative values for a and n . These are:

$$a = 5.88 \times 10^{-5} \quad n = 0.3813$$

if p_c is in Pa and r_b is in m/s. We may calculate a recession rate for a chamber pressure of 1.38×10^7 Pa as

$$\begin{aligned} r_b &= (5.88 \times 10^{-5})(1.38 \times 10^7)^{0.3813} \\ r_b &= 3.10 \times 10^{-2} \text{ m/s} \quad (1.22 \text{ in/s}) \end{aligned}$$

Using Eq. (11.23c) in Eq. (11.23b) we can recognize that for a given set of propellant properties,

$$p_c \sim \left(\frac{A_b}{A_t} \right)^{1/(1-n)} \quad (11.23d)$$

11.2 Boost from a Nonrotating Earth

In this section we shall consider the problem of boost from a nonrotating Earth. Further, we shall restrict the motion of the BV (boost vehicle) to a vertical plane. In spite of these restrictions we should be able to develop a physical sense for the essentials of the boost problem. Since this book is primarily concerned with the mechanics of re-entry, a lengthy treatment of the boost phase does not seem warranted.

Let us consider an axis system (X^i, Y^i, Z^i) at the center of a nonrotating Earth. In Fig. (11.3) we have shown a boost vehicle at an altitude h above the surface on the Earth. In addition to the Earth-centered (X^i, Y^i, Z^i) frame, we shall introduce four other frames. Since we are restricting our attention to planar motion it will be assumed that under all circumstances the Y^i axis is normal to the plane of motion. Consequently we shall describe each axis system in terms of the (X^i, Z^i) axes only. The first frame will be designated as a "local frame," (x^l, z^l) . The z^l axis is along the negative Earth's radius vector; since we shall assume a spherical Earth, the gravity vector is along the positive z^l axis. The second frame is the velocity frame defined such that the x^v axis is always along the velocity vector. In this chapter we shall measure the velocity vector's angle to the local hor-

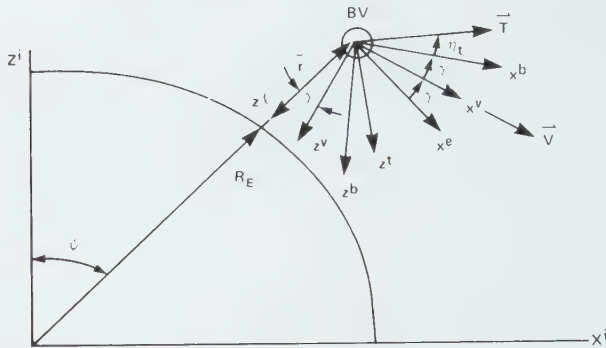


Fig. 11.3 Axis systems.

horizontal by the angle γ taken positive *above* the horizontal. The velocity frame (x^v, z^v) will be used to express the aerodynamic forces of lift (along negative z^v) and drag (along negative x^v). A third frame is the body-frame (x^b, z^b), fixed to the body and aligned with principal inertial axes. The fourth and final axis system is the thrust axes (x^t, z^t) with the thrust vector along the positive x^t axis. Since the y axis of all four frames is normal to the plane of motion at all times, we can take the moment about this normal without being concerned with associating this normal with any one of the four frames just discussed. For this reason the designation of the various y axes is omitted from Fig. (11.3). Nevertheless, when we describe angular motion in the (X^i, Z^i) plane, by taking moments about this normal, we shall associate this moment with the y^b axis since we wish to have constant moments and products of inertia. In this section we shall express forces in the velocity or (v) frame; we might equally have expressed forces in the (b) or body frame. For nonplanar motion a body frame would probably be the more convenient choice.

It will be noted that the vector $\mathbf{r}$ locating the BV from the Earth's center rotates relative to the inertial frame (X^i, Y^i, Z^i). Again because of our planar restriction only two variables are required to describe the location of the BV in the inertial frame: the altitude h and the central angle ψ ; h is measured along the local vertical and ψ is measured from the Z^i as shown.

The components of the velocity vector $\mathbf{V}$ may be expressed in both the inertial (i) frame and the velocity (v) frame; these frames are related by the following DCM:

$$\mathbf{V} = \mathbf{C}_v^i \mathbf{V}^v \quad (11.24a)$$

The first derivative may be expressed as

$$\dot{\mathbf{V}}^i = \mathbf{C}_v^i (\dot{\mathbf{V}}^v + \boldsymbol{\Omega}_{iv}^v \mathbf{V}^v) \quad (11.24b)$$

and from Newton's second law we have

$$\dot{\mathbf{V}}^i = \frac{\mathbf{F}^i}{m} = \frac{\mathbf{F}_a^i}{m} + \frac{\mathbf{F}_g^i}{m} + \frac{\mathbf{F}_t^i}{m} \quad (11.24c)$$

where m is the instantaneous mass of the BV; the forces $\mathbf{F}_a^i$, $\mathbf{F}_g^i$, and $\mathbf{F}_t^i$ are respectively, the aerodynamic, gravitational, and thrust forces. In Eq. (11.24) we have the vector $\mathbf{V}^v$ which is shorthand for the components of the velocity vector in the velocity of (v) frame.

$$\mathbf{V}^v = \begin{bmatrix} V \\ 0 \\ 0 \end{bmatrix} \quad (11.25)$$

Also the skew-symmetric matrix, Ω_{iv}^v , may be written in two parts as

$$\Omega_{iv}^v = \Omega_{il}^v + \Omega_{lv}^v \quad (11.26a)$$

Each of the components may be written as

$$\Omega_{il}^v = \begin{bmatrix} 0 & 0 & -\dot{\Psi} \\ 0 & 0 & 0 \\ \dot{\Psi} & 0 & 0 \end{bmatrix} \quad \Omega_{lv}^v = \begin{bmatrix} 0 & 0 & \dot{\gamma} \\ 0 & 0 & 0 \\ -\dot{\gamma} & 0 & 0 \end{bmatrix}$$

or

$$\Omega_{iv}^v = \begin{bmatrix} 0 & 0 & (\dot{\gamma} - \dot{\Psi}) \\ 0 & 0 & 0 \\ -(\dot{\gamma} - \dot{\Psi}) & 0 & 0 \end{bmatrix} \quad (11.26b)$$

Let us return now to the force expressions $\mathbf{F}^i/m$.

$$\begin{aligned} (\mathbf{C}_v^i)^T \frac{\mathbf{F}^i}{m} &= \dot{\mathbf{V}}^v + \Omega_{iv}^v \mathbf{V}^v \\ (\mathbf{C}_v^i)^T \frac{\mathbf{F}^i}{m} &= \mathbf{C}_i^v \frac{\mathbf{F}_a^i}{m} + \mathbf{C}_i^v \frac{\mathbf{F}_g^i}{m} + \mathbf{C}_i^v \frac{\mathbf{F}_t^i}{m} \\ \frac{\mathbf{F}^v}{m} &= \frac{\mathbf{F}_a^v}{m} + \mathbf{C}_i^v \mathbf{C}_l^i \frac{\mathbf{F}_g^l}{m} + \mathbf{C}_i^v \mathbf{C}_l^i \frac{\mathbf{F}_t^l}{m} \end{aligned}$$

Now from Eq. (11.24c) we have

$$\dot{\mathbf{V}}^v + \Omega_{iv}^v \mathbf{V}^v = \frac{\mathbf{F}^v}{m} = \frac{\mathbf{F}_a^v}{m} + \mathbf{C}_l^v \frac{\mathbf{F}_g^l}{m} + \mathbf{C}_l^v \frac{\mathbf{F}_t^l}{m} \quad (11.27)$$

We now need the DCM's C_l^v and C_t^v . These are written as

$$C_l^v = \begin{bmatrix} \cos\gamma & 0 & -\sin\gamma \\ 0 & 1 & 0 \\ \sin\gamma & 0 & \cos\gamma \end{bmatrix} \quad (11.28a)$$

$$C_t^v = \begin{bmatrix} \cos(\alpha + \eta_t) & 0 & \sin(\alpha + \eta_t) \\ 0 & 0 & 0 \\ -\sin(\alpha + \eta_t) & 0 & \cos(\alpha + \eta_t) \end{bmatrix} \quad (11.28b)$$

The relationship among the (l) and (v) and (t) frames is shown in Fig. 11.4. Let us now consider the force components $\mathbf{F}_a^v$, $\mathbf{F}_g^l$, and $\mathbf{F}_t^l$ in turn.

$$\frac{\mathbf{F}_a^v}{m} = \frac{\mathbf{q}S}{m} \begin{bmatrix} -C_D \\ 0 \\ -C_L \end{bmatrix} \quad (11.29a)$$

The relationship between the (v) frames and (b) frames is shown in Fig. 11.5. It easily follows that,

$$\frac{\mathbf{F}_g^l}{m} = \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} \quad (11.29b)$$

$$\frac{\mathbf{F}_t^l}{m} = \begin{bmatrix} \frac{T}{m} \\ 0 \\ 0 \end{bmatrix} \quad (11.29c)$$

Applying the DCM's as indicated in Eq. (11.27) we get:

$$\frac{\mathbf{F}^v}{m} = \frac{\mathbf{q}S}{m} \begin{bmatrix} -C_D \\ 0 \\ -C_L \end{bmatrix} + \begin{bmatrix} -g \sin\gamma \\ 0 \\ g \cos\gamma \end{bmatrix} + \begin{bmatrix} \frac{T}{m} \cos(\alpha + \eta_t) \\ 0 \\ -\frac{T}{m} \sin(\alpha + \eta_t) \end{bmatrix} \quad (11.30)$$

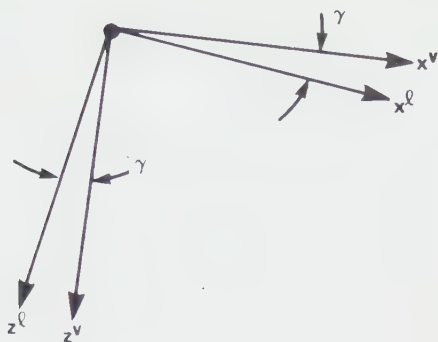
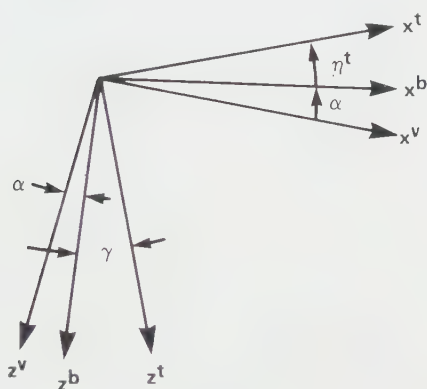
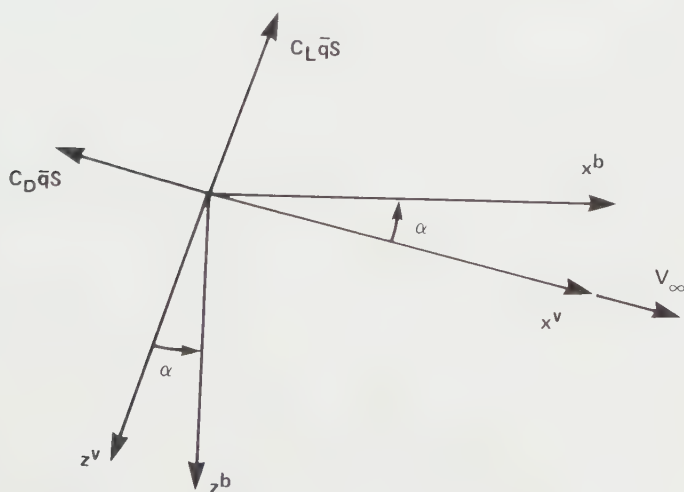

 Fig. 11.4(a) Development of C_l^γ DCM.

 Fig. 11.4(b) Development of the C_l^γ DCM.


Fig. 11.5 Aerodynamic forces in the velocity frame.

Now if we insert Eqs. (11.26b) and (11.30) into Eq. (11.27) we obtain the following two equations for the components of the acceleration vector along and transverse to the velocity vector.

$$\dot{V} = \frac{T}{m} \cos(\alpha + \eta_t) - \frac{C_D \mathbf{qS}}{m} - g \sin \gamma \quad (11.31a)$$

$$V \dot{\gamma} = V \dot{\Psi} + \frac{T}{m} \sin(\alpha + \eta_t) - g \cos \gamma + \frac{\mathbf{qS} C_L}{m} \quad (11.31b)$$

We may next consider the moment applied to the BV. The moment equation has been written in matrix form as

$$\mathbf{M}^i = \frac{d}{dt} \mathbf{H}^i = \frac{d}{dt} \mathbf{C}_b^i \mathbf{H}^b = \frac{d}{dt} \mathbf{C}_b^i \mathbf{I}^b \dot{\boldsymbol{\omega}}_{ib}^b$$

Carrying out the differentiation on the righthand side and then premultiplying both sides by $(\mathbf{C}_b^i)^T$ we get:

$$\mathbf{M}^b = \mathbf{I}^b \dot{\boldsymbol{\omega}}_{ib}^b + \boldsymbol{\Omega}_{ib}^b \mathbf{I}^b \dot{\boldsymbol{\omega}}_{ib}^b \quad (11.32)$$

In our development of the two force equations we resolved all forces in the velocity or (v) frame. In the above moment equations we have resolved the moment in the body frame. If we were to remove our restriction of BV motion to a vertical plane, it might be more convenient to use a body frame for the force equations as well. Under such circumstances the aerodynamic loads, for example, would be expressed in terms of (C_A, C_Y, C_N) rather than $(C_D, 0, C_L)$. However, for planar motion, it is probably slightly easier to sense the physics by considering the stretch and rotation of the velocity vector.

Returning to Eq. (11.32) it is easy to show the following:

$$\boldsymbol{\omega}_{ib}^b = \begin{bmatrix} 0 \\ q \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ (\dot{\alpha} + \dot{\gamma} - \dot{\Psi}) \\ 0 \end{bmatrix} \cong \begin{bmatrix} 0 \\ \dot{\gamma} + \dot{\alpha} \\ 0 \end{bmatrix} \quad (11.33a)$$

$$\boldsymbol{\Omega}_{ib}^b \cong \begin{bmatrix} 0 & 0 & (\dot{\gamma} + \dot{\alpha}) \\ 0 & 0 & 0 \\ -(\dot{\gamma} + \dot{\alpha}) & 0 & 0 \end{bmatrix} \quad (11.33b)$$

$$\mathbf{I}^b = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix}$$

The applied moment $\mathbf{M}^b$ has two constituents, an aerodynamic moment $\mathbf{M}_a^b$ and a thrust moment $\mathbf{M}_t^b$. Let us consider the aerodynamic moment first,

$$\mathbf{M}_a^b \doteq \mathbf{q} S d \begin{bmatrix} 0 \\ C_m \\ 0 \end{bmatrix} \quad (11.34a)$$

Under some circumstances we may justifiably linearize the pitching moment C_m as follows

$$C_m = C_{m_\alpha} (\alpha - \alpha_t) + C_{mq} \frac{qd}{V_\infty} \quad (11.34b)$$

Equation (11.34b) was derived in Eq. (9.11).

In the above expression q is the rate of rotation of the BV relative to inertial space [see Eq. (11.33a)]. Thus $q = \dot{\alpha} + \dot{\gamma} - \dot{\Psi}$; however, since $\dot{\Psi}$ is small in comparison to either $\dot{\alpha}$ or $\dot{\gamma}$, we may let q equal $\dot{\alpha} + \dot{\gamma}$ as a justifiable approximation. Thus

$$C_m = C_{m_\alpha} (\alpha - \alpha_t) + C_{mq} \frac{(\dot{\alpha} + \dot{\gamma})d}{V_\infty} \quad (11.35)$$

where α_t is the trim angle of attack and C_{mq} is the damping in pitch derivative.

The thrust moment is applied to the BV by altering the angle of the thrust vector. The measure of this angle is η_t , taken from the centerline of the BV to the line of action of the thrust vector. The relationships among the thrust vector $\mathbf{T}$, the thrust moment arm, l_t , the thrust angle η_t , the thrust axes, and the body axes are shown in Fig. 11.6.

In general the control moment caused by the thrust vector misalignment is the vector product of $\mathbf{r} \times \mathbf{T}$ or in matrix form,

$$\mathbf{M}_t^b = \mathbf{R}^b \mathbf{T}^b = \mathbf{R}^b \mathbf{C}_t^b \mathbf{T}^t \quad (11.36a)$$

where $\mathbf{R}^b$ is the skew-symmetric form of the vector $\mathbf{r} = (-l_t)\mathbf{i}$. From Fig. 11.6 we may easily write Eq. (11.36) as

$$\mathbf{M}_b^t = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & l_t \\ 0 & -l_t & 0 \end{bmatrix} \begin{bmatrix} \cos \eta_n & 0 & \sin \eta_t \\ 0 & 1 & 0 \\ -\sin \eta_t & 0 & \cos \eta_t \end{bmatrix} \begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{M}_b^t = \begin{bmatrix} 0 \\ -l_t T \sin \eta_t \\ 0 \end{bmatrix} \quad (11.36b)$$

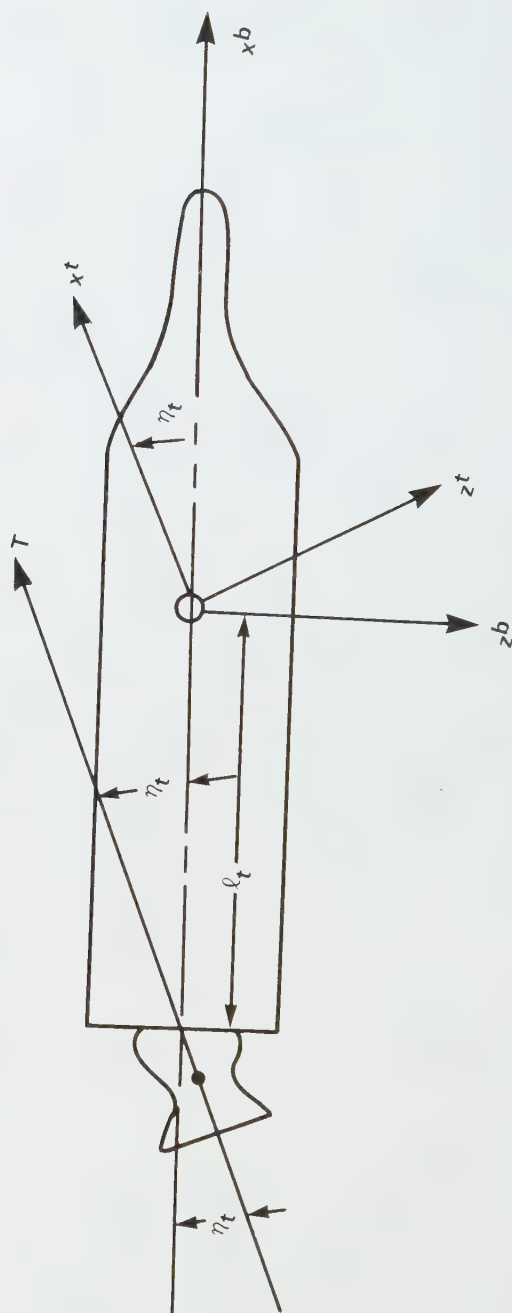


Fig. 11.6 Boost vehicle with thrust vector control.

Combining Eqs. (11.33) (11.34) and (11.36b) and inserting into Eq. (11.32) we get:

$$\ddot{\alpha} = \frac{C_m \alpha q S d}{I} = -\ddot{\gamma} + \ddot{\Psi} = \frac{l_t T \sin \eta_t}{I} \quad (11.37)$$

We can now collect Eqs. (11.31) and (11.37), add some auxiliary equations, and evaluate whether or not we have a determinate system. Before doing so let us recognize that the dynamic pressure q equals $\rho V_\infty^2 / 2$, where ρ is the atmospheric density.

$$\dot{V} = \frac{T}{m} \cos(\alpha + \eta_t) - \left[\frac{C_D(\alpha) \rho V_\infty^2 S}{2m} \right] - \frac{\mu \sin \gamma}{(R_e + h)^2} \quad (11.38a)$$

$$\dot{\gamma} = \dot{\Psi} + \left[\frac{T}{V_\infty m} \right] \sin(\alpha + \eta_t) - \left[\frac{\mu \cos \gamma}{V_\infty (R_e + h)^2} \right] + \frac{\rho S C_L(\alpha) V_\infty}{2m} \quad (11.38b)$$

$$\ddot{\alpha} = \frac{\rho V_\infty^2 S}{2I} C_m(\alpha) - \left[\frac{l_t T}{I} \right] \sin \eta_t + \ddot{\Psi} - \ddot{\gamma} \quad (11.38c)$$

$$m = m_0 - \int \dot{m}_e(t) dt \simeq \dot{m}_0 - \dot{m}_e t \quad (11.38d)$$

$$C_L = C_L(\alpha) \cong C_{L_0} + C_{L_\alpha} \alpha \cong C_{L_\alpha} \alpha \quad (11.38e)$$

$$C_D = C_D(\alpha) \cong C_{D_0} + K_\alpha^2 = C_{D_0} \quad (11.38f)$$

$$C_m = C_m(\alpha) - C_{m_\alpha}(\alpha - \alpha_t) + C_{mq}[(\dot{\alpha} + \dot{\gamma})d / V_\infty] \quad (11.38g)$$

$$\dot{\Psi} = V \cos \gamma / (R_e + h) \quad (11.38h)$$

$$\rho = \rho(h) = \rho_0 e^{-h/H} \quad (11.38i)$$

$$\dot{h} = V \sin \gamma \quad (11.38j)$$

$$\dot{m}_e = T / I_{sp} \quad (11.38k)$$

A few words of justification should be given for each expression in the equation set (11.38). Equations (11.38a) and (11.38b) follow from Eqs. (11.31), or the force equations. Equation (11.38a) accounts for the scalar change in the velocity vector or the component of the linear acceleration along the instantaneous direction of the linear velocity vector; Eq. (11.38b) accounts for the change in direction of the linear velocity vector or the component of linear acceleration normal to the instantaneous direction of the velocity vector. Equation (11.38c) provides us with an equation for the angle-of-attack α which appears as a dependent variable in Eqs. (11.38a-c). The remainder of the equations might be regarded as auxiliary equations among the variables. Equation (11.38d) assumes that the initial mass m_0 is linearly decreased with time. Equations (11.38e-f) represent simple models of the applied aerodynamic force and moment. Equation (11.38h) is a kinematic relationship between the central angle Ψ and the horizontal component of the velocity. Equation (11.38i) originally introduced in Eq. (2.24) is an expression of exponential atmospheric model that has seen much utility. Equation (11.38j), another kinematic relationship, simply recognizes that the vertical component of the velocity vector is equivalent to a time rate of change of altitude. Equation (11.38k) follows from Sec. 11.1, particularly Eq. (11.11). We should also take note that the gravitational acceleration g has been replaced by the familiar relationship,

$$g(h) = \frac{g_E R_E^2}{(R_E + h)^2} = \frac{\mu}{(R_E + h)^2} \quad (11.39)$$

which of course accounts for the altitude variation of gravity according to the hypothesis of a central field.

If we list all the variables appearing in Eqs. (11.38) we obtain 13 in number. These are

$$V, \gamma, \alpha, m, \dot{m}_e, C_D, C_L, C_m, \Psi, \rho, h, \eta_t, T$$

However, there are only 11 equations. Thus, we must specify two variables or provide two additional independent equations containing these variables. If we assume that the thrust vector $\mathbf{T}$ is specified by the two variables (T, η_t) , i.e., the thrust magnitude and directional angle, then we must uncover relationships among these two variables and some (or all) of the other 11 variables. Such relationships are sometimes called a "guidance law." While guidance considerations are beyond the scope of this book, we shall discuss this control moment very briefly.

The particular values that are assigned to the thrust vector (T, η_t) depend upon the trajectory that the BV must follow to meet mission requirements. Mission requirements will be interpreted as the establishment of a velocity magnitude and set of directional cosines of the re-entry vehicle at booster off-loading. The exact values assigned to (T, η_t) will depend upon observations of measurements made onboard the BV during the boost trajectory. These measurements will then be converted, as part of a guidance law, into signals which are then applied to the vector (T, η_t) . Since a solid rocket is used as the booster motor, the thrust magnitude is not under control and

here, at least, will be assumed to be a constant. Thus, the direction of the thrust vector is the only part of the thrust vector under our control. We shall assume that we can measure the thrust angle η_l and within limits control this angle. We should expect some dynamics to be associated with the assignment of thrust angle, but we shall here ignore such considerations and assume the thrust angle η_l to instantaneously accept the assigned value.

A second measurement that we shall assume to have available is the transverse specific force. Specific force $\mathbf{f}$ is the difference between the kinematic acceleration $\mathbf{a}$ and the gravitational acceleration $\mathbf{g}$ as

$$\mathbf{f} = \mathbf{a} - \mathbf{g} \quad (11.40a)$$

Equation (11.40a) may be written in matrix form with components in the body frame as

$$\begin{aligned} \mathbf{f}^b &= \mathbf{a}^b - \mathbf{g}^b \\ &= \mathbf{C}_v^b \mathbf{a}^v - \mathbf{C}_l^b \mathbf{g}^l \end{aligned} \quad (11.40b)$$

where the DCM's $\mathbf{C}_v^b$ and $\mathbf{C}_l^b$ may be written from Figs. 11.4(a) and 11.4(b) as

$$\begin{aligned} \mathbf{C}_v^b &= \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix} \\ \mathbf{C}_l^b &= \begin{bmatrix} \cos(\alpha + \gamma) & 0 & -\sin(\alpha + \gamma) \\ 0 & 1 & 0 \\ \sin(\alpha + \gamma) & 0 & \cos(\alpha + \gamma) \end{bmatrix} \end{aligned}$$

The three vectors of Eq. (11.40a) may now be written in matrix form. First from Eq. (11.27) we see from Eq. (11.26b) that

$$\begin{aligned} \mathbf{a}^v &= \dot{\mathbf{V}}^v + \boldsymbol{\Omega}_{iv}^v \mathbf{V}^v \\ \mathbf{a}^v &= \begin{bmatrix} \dot{V} \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & (\dot{\gamma} - \dot{\Psi}) \\ 0 & 0 & 0 \\ -(\dot{\gamma} - \dot{\Psi}) & 0 & 0 \end{bmatrix} \begin{bmatrix} V \\ 0 \\ 0 \end{bmatrix} \\ \mathbf{a}^v &= \begin{bmatrix} \dot{V} \\ 0 \\ -V(\dot{\gamma} - \dot{\Psi}) \end{bmatrix} \end{aligned} \quad (11.41a)$$

Also

$$\mathbf{f}^b = \begin{bmatrix} f_x \\ 0 \\ f_z \end{bmatrix} \quad (11.41b)$$

$$\mathbf{g}^l = \begin{bmatrix} 0 \\ 0 \\ g_E \end{bmatrix} \quad (11.41c)$$

Now Eq. (11.40b) becomes

$$\begin{bmatrix} f_x \\ 0 \\ f_z \end{bmatrix} = \begin{bmatrix} \cos\alpha & 0 & -\sin\alpha \\ 0 & 1 & 0 \\ \sin\alpha & 0 & \cos\alpha \end{bmatrix} \begin{bmatrix} \dot{V} \\ 0 \\ -V(\dot{\gamma} - \dot{\Psi}) \end{bmatrix}$$

$$\begin{bmatrix} \cos(\alpha + \gamma) & 0 & -\sin(\alpha + \gamma) \\ 0 & 1 & 0 \\ \sin(\alpha + \gamma) & 0 & \cos(\alpha + \gamma) \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ g_E \end{bmatrix}$$

If we confine our attention to the transverse component of specific force, i.e.,

$$f_z = f_{\perp}$$

we have

$$f_{\perp} = \dot{V} \sin(\alpha) - V(\dot{\gamma} - \dot{\Psi}) \cos(\alpha) - \cos(\alpha + \gamma) g_E \quad (11.42)$$

Let us now insert Eqs. (11.38a) and (11.38b) for $\dot{V}$ and $V(\dot{\gamma} - \dot{\Psi})$:

$$f_{\perp} = \frac{T}{m} \cos(\alpha + \eta_t) \sin\alpha - C_D \left[\frac{\rho V^2 S}{2m} \right] \sin\alpha - g_E \sin\alpha \sin\gamma$$

$$- \frac{T}{m} \sin(\alpha + \eta_t) \cos\alpha + g_E \cos\gamma \cos\alpha - \left[\frac{\rho V^2 S}{2m} \right] C_L \cos\alpha - g_E \cos(\alpha + \gamma)$$

which reduces to

$$f_{\perp} = \frac{T}{m} [\cos(\alpha + \eta_t) \sin\alpha - \sin(\alpha + \eta_t) \cos\alpha] + g_E \cos\gamma \cos\alpha$$

$$- \sin\alpha \sin\gamma - \cos(\alpha + \gamma) - \frac{\rho V^2 S C_D}{2M} \left[\sin\alpha + \frac{C_L}{C_D} \cos\alpha \right]$$

to give finally

$$f_{\perp} = -\frac{T}{m}[\sin(\eta_t)] - \left[\frac{\rho V^2 S C_D}{2m} \right] \left[\sin\alpha - \frac{C_L}{C_D} \cos\alpha \right]$$

$$-f_{\perp} \cong \frac{T}{m} \sin(\eta_t) + \left[\frac{\rho V^2 S C_{L\alpha}}{2m} \right] \alpha \quad (11.43)$$

where we have applied a small angle approximation to the angle-of-attack α . We have further assumed that C_D is small in comparison to C_L . This assumption may be substantiated for slender bodies by referring to data given by Ashley in Fig. 10.6 of Ref. 1.

Equation (11.43) provides us with a relationship among three variables: the thrust angle η_t and the transverse specific force $f_{\perp}$, both of which are measurable, and the angle-of-attack α which is not measurable. Thus, we can use Eq. (11.43) to compute the angle-of-attack α as

$$\alpha = - \left[\frac{2m}{\rho V^2 S C_{L\alpha}} \right] \left[\frac{T}{m} \sin(\eta_t) + f_{\perp} \right] \quad (11.44)$$

Now referring to the moment equation [Eq. (11.38c)] we can calculate the required thrust angle, subjected, let us say, to the condition that the angular acceleration of the BV relative to inertial space is zero:

$$\ddot{\alpha} + \ddot{\gamma} - \ddot{\Psi} = 0$$

Solving Eq. (11.38c) for the thrust angle subjected to the above condition, we get:

$$\sin(\eta_t) = \left[\frac{\rho V^2 S d C_{m\alpha}}{2l_t T} \right] \alpha \quad (11.45a)$$

where we have linearized the pitching moment $C_m(\alpha)$. Inserting the angle of attack from Eq. (11.44) we get:

$$\sin(\eta_t) = - \left[\frac{\mathbf{x}}{l_t} \right] \left[\frac{m}{T} \right] \left[f_t + \frac{T}{m} \sin(\eta_t) \right]$$

where the static margin and the thrust moment arm have been nondimensionalized as

$$\mathbf{x} = \frac{C_{m\alpha}}{C_{L\alpha}} \quad l_t = \frac{l_t}{d}$$

Solving for $\sin(\eta_t)$ we get:

$$\begin{aligned} \sin(\eta_t) \left[1 + \frac{\mathbf{x}}{l_t} \right] &= - \left[\frac{\mathbf{x}}{l_t} \right] \left[\frac{m}{T} \right] f_{\perp} \\ \sin(\eta_t) &= - \left\{ \left[\frac{\mathbf{x}}{l_t} \right] \left[\frac{m}{T} \right] f_{\perp} / \left[1 + \frac{\mathbf{x}}{l_t} \right] \right\} \end{aligned} \quad (11.45b)$$

Since $\mathbf{x} \ll l_t$, an approximation to Eq. (11.45b) might be:

$$\eta_t = - \left[\left[\frac{\mathbf{x}m}{l_t T} \right] f_i + \frac{1}{6} \left[\frac{\mathbf{x}m}{l_t T} \right]^3 f_{\perp}^3 \right] \quad (11.45c)$$

Note that the gain term $(\mathbf{x}m / l_t T)$ will change with time since the static margin will increase and the mass will decrease as propellant is expended and the center of mass moves forward on the body. If, for example, a trajectory were specified in terms of the transverse specific force $f_{\perp}$, then Eq. (11.45c) would then specify a thrust-vector angle η_t .

A glance at Eqs. (11.38) indicates a formidable task in modeling the boost trajectory. Again taking as our goal the development of a sense of the physics we may consider a few simple cases. First, let us restrict ourselves to a vertical launch without the presence of any transverse aerodynamic loads. Further we shall approximate the gravitational force by using the sea level value of g , i.e., $g = g_0$. Thus Eq. (11.38a) becomes

$$\frac{dV}{dt} = \frac{T}{m} - \frac{C_{D0}\rho V^2 S}{2m} - g_0 \quad (11.46)$$

The separate integral of each of the terms on the right in Eq. (11.44) will give us a velocity magnitude. Thus

$$\Delta V_t = \int_0^t \frac{T}{m} dt \quad (11.47a)$$

$$\Delta V_d = \int_0^t \frac{C_{D0}\rho V^2 S}{2m} dt \quad (11.47b)$$

$$\Delta V_g = \int_0^t g_0 dt \quad (11.47c)$$

Ashley (Ref. 1) applies the term "characteristic" velocities to each of the contributions listed above. Let us consider the first integral:

$$\Delta V_t = \int_0^t \frac{T}{m} dt = \frac{g_0 \dot{m}_e I_{sp}}{m_0} \int_0^t \frac{dt}{[1 - (\dot{m}_e / m_0)t]}$$

where we have made use of Eqs. (11.38d) and (11.38k). The above equation may be integrated to give:

$$\begin{aligned}\Delta V_t &= g_0 I_{sp} \ln[1 - (\dot{m}_e / m_0)t] \\ \Delta V_t &= g_0 I_{sp} \ln(m_0 / m_f)\end{aligned}\quad (11.48)$$

where m_0 is the initial mass and m_f is the final mass. If we take a mass ratio m_0/m_f to be 3.5 and the specific impulse to be 240 s, we get:

$$\begin{aligned}\Delta V_t &= (9.81)(240)\ln(3.5) \\ &= 2950 \text{ m/s}\end{aligned}$$

A glance at Eq. (11.46) indicates the characteristic velocity we have designated by ΔV_t may be thought of as the inherent velocity capability of the BV in a field-free and atmosphere-free space. The presence of a gravitational field and an atmosphere of course prevents this characteristic velocity ΔV_t from being realized. Let us look at the other characteristic velocities given in Eqs. (11.47b) and (11.47c).

The evaluation of the integral for ΔV_d is difficult to generalize because the density of the atmosphere varies with altitude h [Eq. (11.38i)], the mass m varies with time [Eq. (11.38d)] and, of course, the velocity magnitude depends upon the boost trajectory. A rough approximation might be possible by rewriting Eq. (11.47b) as follows

$$\Delta V_d = \frac{D}{m_0} \int_0^t \frac{dt}{[1 - (\dot{m}_e / m_0)]} = \frac{-D}{m_0} \left[\frac{m_0}{\dot{m}_e} \right] \ln \frac{m_f}{m_0}$$

where m_f is the final mass. From Eq. (11.38k) we may write ΔV_d as

$$\Delta V_d = (D/T)g_0 I_{sp} \ln(m_0/m_f) \quad (11.49)$$

Unfortunately we have only side-stepped the issue by replacing $\rho V^2 SC_{D_0}/2$ by the dimensional drag since it is clear that the density decreases with altitude in a well-defined manner while the velocity-altitude relationship is, as pointed out, mission dependent. Thus we might expect the drag-to-thrust ratio to vary several orders of magnitude during the boost trajectory. If we take D/T to be something like 0.01, I_{sp} to be 240 s, and m_0/m_f to be 3.5, we shall get for ΔV_d ,

$$\begin{aligned}\Delta V_d &= (99.81)(240)(.01)\ln(3.5) \\ &= 29.5 \text{ m/s}\end{aligned}$$

Even with D/T at about 10%, ΔV_d will still be an order-of-magnitude less than the thrust characteristic velocity. Aerodynamic drag, it would seem, has a relatively minor effect on the total velocity capability of the boost vehicle.

We finally can obtain an estimate of ΔV_g , simply by regarding $g = g_0$

$$\Delta V_g = \int_0^{t_f} g \, dt = g_0 t_f \quad (11.50)$$

where t_f is the time of flight. If we assume a time of flight of 100 s the gravitational characteristic velocity ΔV_g is

$$\Delta V_g = 980 \text{ m/s}$$

We can see in comparing the drag and gravitational characteristic velocities that the gravitational field is much more influential in decreasing the velocity potential of the BV than is the drag.

In the brief discussion of the boost trajectory given in this section, we should notice the appearance of the mass ratio m_0/m_f or the initial mass divided by the final mass. From Eq. (11.48) it is obvious that increasing the mass ratio increases the characteristic velocity due to thrust. The advantage of staging in achieving high mass ratios is well known. A readable discussion of staging is given by Ashley in Chapter IX of Ref. 1. An interesting and related discussion of staging is given in Chapter XXIV of Ref. 2.

11.3 Typical Boost Sequence

We may now take a brief look at a typical launch sequence. The case we have selected is the submarine launching of a three stage boost vehicle. The BV has typically a length of about 10 m, a maximum diameter of 1.75 m, and a weight of about 30,000 kg. The range of the BV/RV system might be of the order of 6000 km.

Within the nose fairing is the third stage and what we shall designate as the equipment section. The equipment section contains the RV's and a thruster/control system used in the final deployment of the RV's.

The main events in the boost trajectory are given in Fig. 11.7. During the prelaunch-condition, navigation information is handed-off from the submarine to the navigation equipment aboard the boost vehicle. The BV is ejected from the submarine and the onboard battery is activated. This battery is the sole source of electrical power during the boost trajectory. The first stage of the TVC (thrust vector control) rocket is ignited just after broach. Upon completion of the first stage burn, the second stage ignites and the first stage separates. At a preset level of dynamic pressure the nose fairing is separated from the equipment section and jettisoned. Two jettison motors carry the nose fairing to the side and out of the way of the third stage. The third stage then ignites and at burnout releases the equipment section. Just prior to third stage separation, the PBCS (post boost control system) ignites to provide thrust levels for maneuvering the equipment section during the deployment stage of the trajectory. The PBCS contains a solid chemical propellant which valves a warm gas to a combination of nozzles. Appropriate selection of these nozzles can provide three-axis angular and velocity control of the equipment section. The thrusters of the PBCS

operate in a coarse and vernier mode. In the coarse mode the thrusters move the equipment section away from the jettisoned third state motor; also in the coarse mode the thrusters provide the linear velocity to move the equipment section to the RV release point. Then, in the vernier mode, the thrusters orient the equipment section for RV deployment. After deployment of the RV, the PBCS in the vernier mode backs the equipment section away from the deployed RV.

At this point the RV enters in the Keplerian phase of the re-entry sequence. A discussion of the Keplerian phase was the subject of Chapter V.

In Fig. 11.7 we have depicted three stages of the boost trajectory. All three stage motors are solid propellant rockets. Since no thrust control is

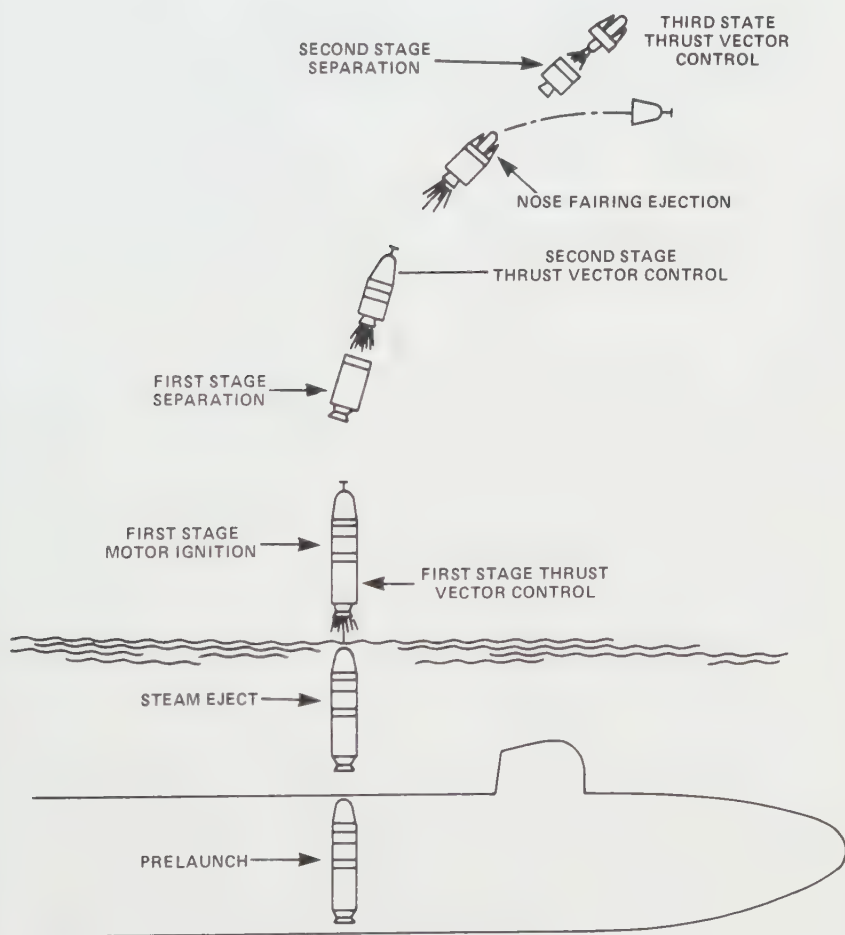


Fig. 11.7 Boost sequence.

available the thrust level cannot be varied during boost to meet varying mission requirements. Let us first recognize that as mission requirements vary so must the RV's position and velocity vary at boost termination. The goal of the boost phase is to place the equipment section at the intended position with the intended velocity together with the simultaneous depletion of the fuel and a null in the steering velocity vector. For mission requirements less than the maximum range, a trajectory must be followed which "wastes" some of the boost energy.

The set of steering commands needed to meet varying mission requirements with a fixed fuel expenditure is called GEM—Generalized Energy Management. A discussion of GEM is well beyond the scope of this book. However, further information may be found in Ref. 4.

It should be pointed out that there are two distinct phases in the generation of steering instructions. First let us define a time $t = t^*$ with the only requirement on t^* being that it occurs prior to third stage separation. For $t < t^*$, the steering instructions are based upon the requirement of keeping the side loads on the BV to a minimum and to do so with a minimum range penalty. For $t > t^*$ the steering instructions are formulated to "waste" sufficient energy to achieve a boost termination condition which meets mission requirements.

Once the RV has separated from the equipment section it is considered to be in free-fall or Keplerian motion until it encounters the tenable atmosphere. In Chapter V we discussed Keplerian motion, but treated the RV as a mass point. In Chapter XII we shall consider angular motion during the Keplerian phase.

References

¹Ashley, H., *Engineering Analysis of Flight Vehicles*, Addison-Wesley, Reading, Mass, 1974.

²Sutton, G. P., *Rocket Propulsion Elements: An Introduction to the Engineering of Rockets*, Wiley, New York, 1966.

³Tsien, H. S., *Fundamentals of Gas Dynamics*, edited by H. W. Emmons, Princeton University Press, Princeton, N. J., 1958, pp. 3-63.

⁴Brand, T. J., "A New Approach to Lambert Guidance," Charles Stark Draper Laboratory, Inc., Cambridge, Mass., R-694, June 1971.

XII. ANGULAR MOTION DURING KEPLERIAN PHASE

In Chapter V we defined Keplerian motion as that portion of the RV's trajectory during which gravitational and only gravitational forces were present. At that time we treated the RV as a mass point. Since the gravitational force is a body force, the RV will experience torque-free motion during the Keplerian trajectory. While torque-free motion implies that the angular momentum vector is fixed in inertial space, a rolling RV will experience an angular velocity vector that is not fixed with respect to either an inertial or a body frame.

12.1 Torque-Free Motion: Small Perturbations

The moment equation may be taken from Eq. (8.65):

$$\mathbf{M}^b = 0 = \mathbf{I}^b \boldsymbol{\omega}_{ib}^b + \boldsymbol{\Omega}_{ib}^b \mathbf{I}^b \boldsymbol{\omega}_{ib}^b \quad (12.1)$$

If we restrict ourselves to principal axes and let $\boldsymbol{\omega}_{ib}^b = \{p, q, r\} = \{\omega_1, \omega_2, \omega_3\}$ we get:

$$\begin{aligned} \mathbf{M}^b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} &= \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\omega}_3 \end{bmatrix} \\ + \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \end{aligned}$$

from which follow the three scalar equations:

$$I_1 \dot{\omega}_1 + \omega_2 \omega_3 (I_3 - I_2) = 0 \quad (12.2a)$$

$$I_2 \dot{\omega}_2 + \omega_1 \omega_3 (I_1 - I_3) = 0 \quad (12.2b)$$

$$I_3 \dot{\omega}_3 + \omega_1 \omega_2 (I_2 - I_1) = 0 \quad (12.2c)$$

A considerable amount of eighteenth and nineteenth century dynamics centers around studies of Eqs. (12.2). Goldstein gives a detailed treatment of obtaining integrals of Eq. (12.1) with nonzero moments when applied to the motion of a symmetric top.¹ Even with the applied moments identically equal to zero, the solution must be expressed in terms of elliptic integrals. The appearance of elliptic integrals seems to greatly impair any physical intuition we might bring to the problem. Thus, what follows is only a nibble at a very heavy offering. In Fig. 12.1 we have depicted an RV with unit vectors $\{\bar{i}_1, \bar{i}_2, \bar{i}_3\}$ along the principal axes.

The angular velocity vector ω will be assumed to be along the $\bar{i}_1$ axis (axis of roll symmetry if appropriate or in the plane of symmetry if appropriate) in undisturbed motion. Thus,

$$\omega = (\omega_0 + \xi)\bar{i}_1 + \omega_2\bar{i}_2 + \omega_3\bar{i}_3 \quad (12.3a)$$

where ξ , ω_2 , and ω_3 are perturbation variables and all of the same order. Since ω_0 is the steady-state value,

$$\dot{\omega} = \dot{\xi}\bar{i}_1 + \dot{\omega}_2\bar{i}_2 + \dot{\omega}_3\bar{i}_3 \quad (12.3b)$$

Since we are considering only second-order effects, the products of the perturbative variables can be neglected when compared to the variables themselves, i.e.,

$$\xi, \omega_2, \omega_3 \gg \xi\omega_2, \xi\omega_3, \omega_2\omega_3$$

Inserting the variables from Eqs. (12.3) into Eqs. (12.2) we get:

$$I_1\dot{\xi} = (I_2 - I_3)\omega_2\omega_3 = 0 \quad (12.4a)$$

$$I_2\dot{\omega}_2 = (I_3 - I_1)(\omega_0 + \xi)\omega_3 = (I_3 - I_1)\omega_0\omega_3 \quad (12.4b)$$

$$I_2\dot{\omega}_3 = (I_1 - I_2)(\omega_0 + \xi)\omega_1 = (I_1 - I_2)\omega_0\omega_2 \quad (12.4c)$$

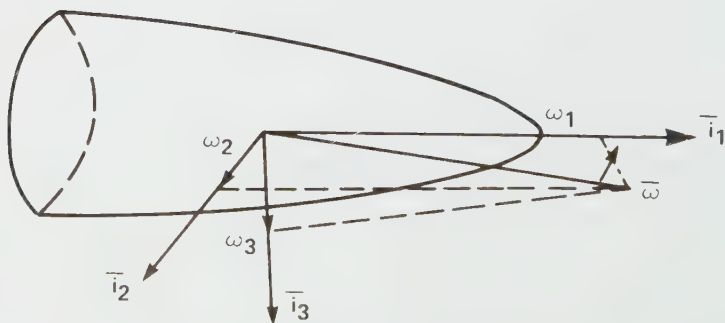


Fig. 12.1 Principal axis system.

Thus

$$\dot{\omega}_2 = \left[\frac{I_3 - I_1}{I_2} \right] \omega_0 \omega_3 \quad \dot{\omega}_3 = \left[\frac{I_1 - I_2}{I_3} \right] \omega_0 \omega_2 \quad (12.5)$$

We may differentiate the first equation and use the second equation to eliminate $\dot{\omega}_3$. The result is:

$$\dot{\omega}_2 + \left\{ \left[\frac{(I_1 - I_2)(I_1 - I_3)}{I_2 I_3} \right] \omega_0^2 \right\} \omega_2 = 0 \quad (12.6a)$$

In like manner we may write:

$$\dot{\omega}_3 + \left\{ \left[\frac{(I_1 - I_2)(I_1 - I_3)}{I_2 I_3} \right] \omega_0^2 \right\} \omega_3 = 0 \quad (12.6b)$$

Letting

$$\lambda = \frac{(I_1 - I_2)(I_1 - I_3)}{I_2 I_3} \omega_0^2 \quad (12.7)$$

we have for Eqs. (12.6)

$$\ddot{\omega}_2 + \lambda^2 \omega_2 = 0 \quad \ddot{\omega}_3 + \lambda^2 \omega_3 = 0 \quad (12.8)$$

Clearly Eqs. (12.8) are of a form associated with an harmonic oscillator; as such their solution is well known:

$$\omega_j = A_{j1} e^{i\lambda t} + A_{j2} e^{-i\lambda t} \quad (12.9a)$$

If λ is real we can use an alternate form to Eq. (12.9a):

$$\omega_j = B_j \sin(\lambda t + \psi_j) \quad (12.9b)$$

If λ is imaginary we shall have one bounded and one unbounded root and the physical motion will be characterized by exponential divergence. For λ that is real we shall have undamped, oscillatory motion. A glance at Eq. (12.7) indicates the conditions for real (stable) or imaginary (unstable) motion: angular motion about the (1) axis is stable when:

$$I_1 > I_2 \quad I_1 > I_3 \quad (12.10a)$$

or when

$$I_1 < I_2 \quad I_1 < I_3 \quad (12.10b)$$

In words, motion is stable for rotation about a major axis or a minor axis; motion about an intermediate axis is unstable. Let us assume that $I_1 > I_2, I_3$. If the RV is released with ω along the $\mathbf{i}_1$ vector, then the RV will behave like a free gyro; if the ω is slightly off the $\mathbf{i}_1$ axis so that there are components of ω along both the $\mathbf{i}_2$ and $\mathbf{i}_3$ axes, then we might expect a wobble to continue throughout the Keplerian phase.

There is one caveat that must be applied to our conclusion that rotation about the minor axis of symmetry is stable. Rotation about the minor axis of symmetry is stable if the body is rigid. However, if the body is not rigid, then any bending motion caused by the rotation can dissipate the energy of rotation. Thus rotational motion associated with maximum kinetic energy cannot be stable. Now we have written several times, the vectoral equation relating the applied moment and the rate of change of angular momentum, viz,

$$\mathbf{M}^i = \frac{d}{dt} \bigg|_i [\mathbf{H}^i] \quad (12.11)$$

The above expression is valid whether the body is rigid or not. More to the point, if the body is not rigid and the angular rotation causes bending motion, the energy associated with this bending must be provided by the rotational energy. Since the bending energy must be dissipated as heat, the energy of rotation must continually decrease. Thus the body will seek a state of minimum kinetic energy.

The equations for kinetic energy were developed in Sec. 8.9. From Eq. (8.89c) we can show that the kinetic energy written in terms of the components of the angular velocity about the principal axis and the principal moments of inertia may be written as

$$T = \frac{1}{2} [I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2] \quad (12.12)$$

The angular momentum also written in a principal body frame is

$$(\mathbf{H}^b)^T = [I_1 \omega_1, I_2 \omega_2, I_3 \omega_3] \quad (12.13)$$

Following a simple argument of Wrigley et al. (Ref. 2), we may use the method of Lagrange to minimize a function T in the presence of another function which remains constant. Now T , the kinetic energy, is a scalar function while the angular momentum $\mathbf{H}^i$, which is to remain constant, is a vector function. If we confine our attention to the magnitude of $\mathbf{H}^i$ only, then we shall have a scalar function. The magnitude of $\mathbf{H}^i$ or $\mathbf{H}^b$ may be found from the inner product as

$$H^2 = (\mathbf{H}^b)^T \mathbf{H}^b = I_1^2 \omega_1^2 + I_1^2 \omega_1^2 + I_1^2 \omega_1^2 \quad (12.14)$$

Therefore we can define the function F as

$$F = T + L H^2 \quad (12.15)$$

where L is the Lagrange multiplier. The angular velocity components which minimize F will also minimize T since both H and L are constants. For a local minimum we have as necessary conditions,

$$\frac{\partial F}{\partial \omega_1} = I_1 \omega_1 + 2L I_1^2 \omega_1 = 0 \quad (12.16a)$$

$$\frac{\partial F}{\partial \omega_2} = I_2 \omega_2 + 2LI_2^2 \omega_2 = 0 \quad (12.16b)$$

$$\frac{\partial F}{\partial \omega_3} = I_3 \omega_3 + 2LI_3^2 \omega_3 = 0 \quad (12.16c)$$

From Eqs. (12.16) we then have from each of the equations taken in turn,

$$L = -\frac{1}{2I_1} \quad L = -\frac{1}{2I_2} \quad L = -\frac{1}{2I_3}$$

If I_1 , I_2 , and I_3 are all different then only one of the preceding equations can be true. Thus we must eliminate two out of the three equations appearing in Eqs. (12.16). If we decide, quite arbitrarily, to omit the first two of these equations, then ω_1 and ω_2 must be identically zero. Physically this means that the angular velocity for a condition of minimum rotational energy must be entirely along one of the principal axes. We have taken this axis to be that labeled "3." No statement has been made as to whether this "3" axis is a major, minor, or intermediate principal axis.

With $\omega_1 = \omega_2 = 0$, we can now write from Eq. (12.14),

$$H^2/I_3^2 = \omega_3^2 \quad (12.17)$$

which from Eq. (12.14) requires that the rotational energy become

$$T = H^2/2I_3 \quad (12.18)$$

It is clear from Eq. (12.18) that the rotational energy will be a maximum if the moment of inertia is a minimum and a minimum if the moment of inertia is a maximum. In other words, rotation about a minimum principal axis for a constant magnitude of angular momentum H^b will be unstable if the vehicle is nonrigid, i.e., is subjected to bending.

In conclusion then we may state that a rigid RV will have stable angular rotation if the rotation is taking place about either a major or minor principal axis; rotation about an intermediate principal axis will be unstable. However, if the vehicle is nonrigid then the vehicle will have stable rotation only when rotating about the major principal axis. Again, these conclusions apply only when the applied moments are zero, i.e., the angular momentum relative to inertial space H^i is constant.

12.2 Torque-Free Motion: Poinot Construction

We noted in the last chapter that a closed-form integration of Eqs. (12.1) is at best difficult. However, there is a geometric construction attributed to Poinot that allows us to visualize torque-free motion of a rigid body.

First, for reference we shall restate Eqs. (8.65a) and (8.89b) which relate angular momentum $\mathbf{H}$ and rotational kinetic energy T to the angular velocity.

$$\mathbf{H}^b = \mathbf{I}^b \boldsymbol{\omega}_{ib}^b \quad \mathbf{H} = \mathbf{I} \cdot \boldsymbol{\omega}_{ib} \quad (12.19a)$$

$$2T = (\boldsymbol{\omega}_{ib}^b)^T \mathbf{I}^b \boldsymbol{\omega}_{ib}^b \quad 2T = \boldsymbol{\omega}_{ib} \cdot \mathbf{I}_{ib} \cdot \boldsymbol{\omega}_{ib} \quad (12.19b)$$

In the above equations we have made use of both the matrix and vector formalisms. As in the previous section we shall continue to take vector components in a principal axis frame; these axes will be designated by the subscripts (1), (2), (3). Equation (12.19a) has been written in principal axis components in Eqs. (12.13); Eq. (12.19b) has been expressed as a quadratic form in Eq. (12.12).

We can now combine Eqs. (12.19a) and (12.19b) as an inner product of $\boldsymbol{\omega}_{ib}$ and $\mathbf{H}$ as

$$2T = (\boldsymbol{\omega}_{ib}^b)^T \mathbf{H}^b \quad (12.20a)$$

or

$$2T = \boldsymbol{\omega}_{ib} \cdot \mathbf{H} \quad (12.20b)$$

For torque-free motion we have:

$$\mathbf{M}^i = \frac{d\mathbf{H}^i}{dt} = 0 \quad (12.21a)$$

or

$$\mathbf{M} = \left. \frac{d\mathbf{H}}{dt} \right|_i = 0 \quad (12.21b)$$

Equations (12.21) state that once the RV is off-loaded from the equipment section the angular momentum vector remains fixed in inertial space. However, since the body frame will be rotating at a rate $\boldsymbol{\omega}_{ib}$ (or $\boldsymbol{\omega}_{ib}^b$), the angular momentum vector will *not* be fixed in the *body frame*. From Eqs. (12.21) we have:

$$\left. \frac{d\mathbf{H}}{dt} \right|_i = \left. \frac{d\mathbf{H}}{dt} \right|_b + \boldsymbol{\omega}_{ib} \times \mathbf{H} = 0$$

or

$$\left. \frac{d\mathbf{H}}{dt} \right|_b = -\boldsymbol{\omega}_{ib} \times \mathbf{H} \quad (12.22a)$$

and in matrix form,

$$\frac{d\mathbf{H}^b}{dt} = -\boldsymbol{\Omega}_{ib}^b \mathbf{H}^b \quad (12.22b)$$

In Eq. (8.65a) we expressed the angular momentum components in a body frame as

$$\mathbf{H}^b = \mathbf{I}^b \boldsymbol{\omega}_{ib}^b \quad (8.65a)$$

It might appear from the above equation that if ω_{ib}^b is fixed in a body frame then $\mathbf{H}^b$ will also be fixed in a body frame. However, Eqs. (12.22) inform us that unless ω_{ib}^b and $\mathbf{H}^b$ are co-linear, $\mathbf{H}^b$ cannot be fixed in a body frame and hence [from Eq. (8.65a)], neither can ω_{ib}^b be fixed in a body frame. The geometrical construction of Poinso is a study of the motion of ω_{ib}^b subjected to the condition of torque-free motion, i.e., constancy of the angular momentum vector in inertial space. The relationship between $\mathbf{H}^b$ and ω_{ib}^b is shown in Fig. 12.2.

From Eq. (12.20) we may write:

$$I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2 = 2T$$

or

$$\begin{aligned} \frac{\omega_1^2}{(\sqrt{2T/I_1})^2} + \frac{\omega_2^2}{(\sqrt{2T/I_2})^2} + \frac{\omega_3^2}{(\sqrt{2T/I_3})^2} &= 1 \\ &= \frac{\omega_1^2}{a^2} + \frac{\omega_2^2}{b^2} + \frac{\omega_3^2}{c^2} \end{aligned} \quad (12.23)$$

where $a = 2T/I_1$, $b = 2T/I_2$, and $c = 2T/I_3$. Equation (12.23) describes a geometrical figure called an ellipsoid. Although not a universal practice, this ellipsoid is often called the ‘‘Poinso ellipsoid’’; it represents the locus of all possible values of the angular velocity vector ω_{ib} which satisfies Eq.

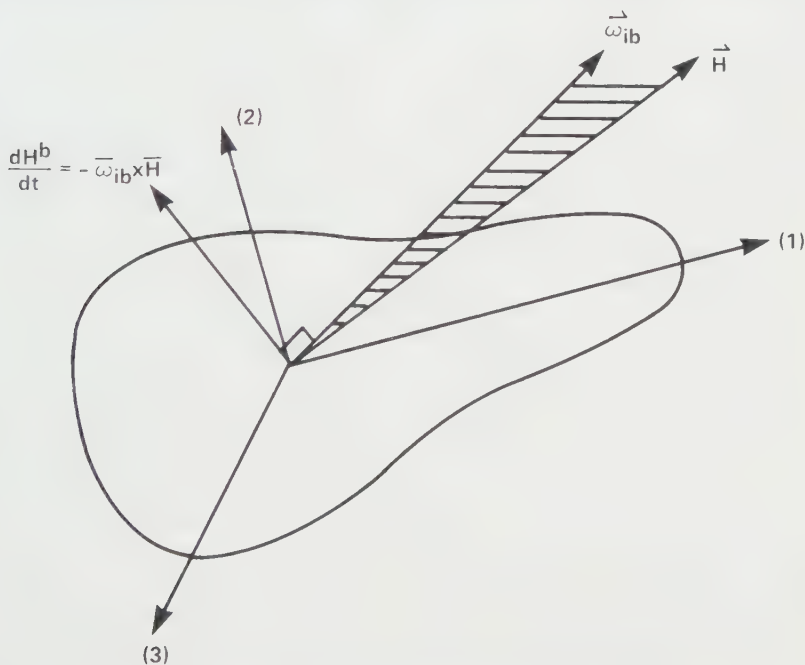


Fig. 12.2 Angular velocity and angular momentum vectors.

(in torque-free motion), first, regardless of how the direction of ω_{ib} changes, its magnitude must always lie along the surface of the Poinsoit ellipsoid; second, the component of ω_{ib} along the angular momentum vector $\mathbf{H}$ must be constant. This latter requirement leads to the corollary that vectoral changes in ω_{ib} must always be at right angles to $\mathbf{H}$. This latter conclusion follows by taking the differential of Eq. (12.24) with respect to inertial space, as

$$d\omega_{ib} \cdot \mathbf{h} + \omega_{ib} \cdot d\mathbf{h} = d(2T/H)$$

Since both $d\mathbf{h}$ and $d(2T/H)$ are zero, we have

$$d\omega_{ib} \cdot \mathbf{h} = 0 \quad (12.25)$$

Two vectors whose inner product is zero must be orthogonal if we ignore the trivial case where one or both vectors are identically zero. The component of ω_{ib} along $\mathbf{H}$ has been identified as $l\mathbf{h}$. Since, according to Eq. (12.25), there can be no vector change in ω_{ib} along $\mathbf{h}$, all changes in ω_{ib} must be confined to its component normal to $\mathbf{h}$ or

$$\omega_{ib} \cdot \mathbf{h} = l = l(\mathbf{h} \cdot \mathbf{h})$$

or

$$(\omega_{ib} - l\mathbf{h}) \cdot \mathbf{h} = 0 \quad (12.26)$$

Thus the vector component of ω_{ib} normal to $\mathbf{h}$ is $(\omega_{ib} - l\mathbf{h})$. This component was shown in Fig. 12.3. A section through the Poinsoit ellipsoid is illustrated in Fig. 12.4. Since $(\omega_{ib} - l\mathbf{h})$ must be normal to $\mathbf{h}$ (or $\mathbf{H}$) and $l\mathbf{h}$ is invariant, the tip of ω_{ib} must lie in a plane that is fixed in inertial space—hence the name “invariable plane.” The tip of ω_{ib} must lie in the invariable plane and, as pointed out earlier, it must at the same time lie on the surface of the Poinsoit ellipsoid. We emphasize that $\mathbf{h}$ is fixed in space, but ω_{ib} is not.

In our development of the force equations (and later the moment equations) we encountered the theorem of Coriolis. We applied this theorem to compare the components of a vector resolved in a (rotating) body frame and in an inertial frame. [see Eq. (4.8)]. The angular velocity vector ω_{ib} provides us with both directional and scalar information. The direction of ω_{ib} sets the axis about which rotation takes place; in other words, if we rotate about an axis aligned with the instantaneous direction of ω_{ib} , we would carry the body frame at time $t = t^*$ into the body frame at time $t = t^* + \Delta t$. We are taking Δt to be of differential magnitude. The second piece of information we obtain from ω_{ib} comes from its magnitude. The angle through which we rotate the body frame from its orientation at $t = t^*$ to its orientation at time $t = t^* + \Delta t$ is $\omega_{ib} \Delta t = \Delta\theta$. Also since ω_{ib} is a vector, we can resolve it in any frame of convenience; frequently we have written ω_{ib} in terms of its components in the instantaneous body frame as $\omega_{ib} = (p, q, r)^T$. Since we are considering Δt to be a infinitesimal quantity, we can regard the angular displacement through which the body frame rotates to be a vector quantity and written in matrix form as $\Delta\theta = (p\Delta t, q\Delta t, r\Delta t)^T$. The important point here is that in our varied work on vector mechanics we regarded the angular velocity vector ω_{ib} to be

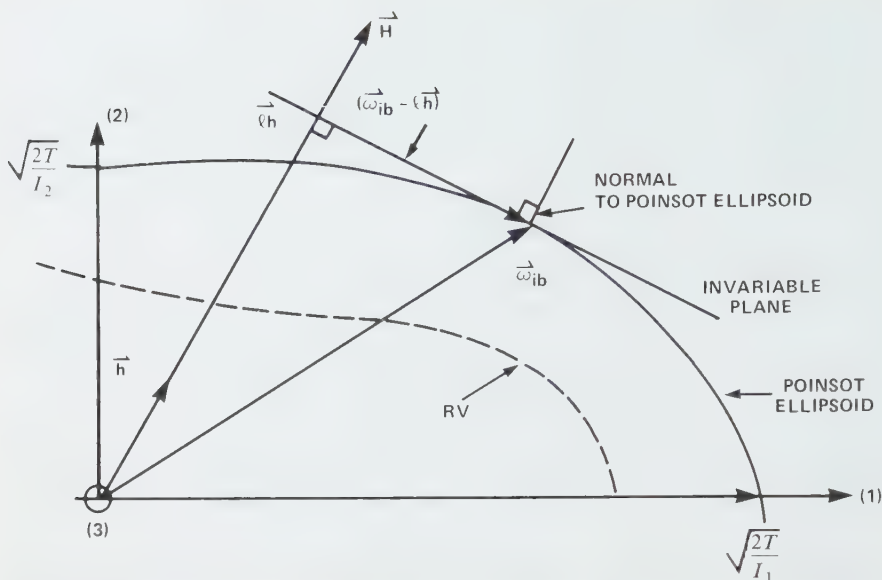


Fig. 12.4 Planar section through the Poincaré ellipsoid and edgeview of the invariable plane.

instantaneously fixed in space: the angular velocity of the angular velocity was given no physical significance. Yet we accepted the fact that if we were to somehow contemplate the angular velocity vector while fixed to either a body or inertial frame we would observe the vector ω_{ib} to change both magnitude and direction. The preceding discussion of the angular velocity's meanderings in an invariable plane while at the same time remaining on the surface of the Poincaré ellipsoid is a geometric description of this vectorial change in ω_{ib} (in torque-free motion).

In Fig. (12.5) we have depicted isometrically the Poincaré ellipsoid and the invariable plane. The Poincaré ellipsoid being fixed to the RV rolls along the invariable plane such that the tip of ω_{ib} is always at the point of tangency. Again we must state that while the Poincaré ellipsoid is fixed to the body and the invariable plane is fixed in inertial space, the angular velocity vector ω_{ib} wanders when viewed from either body-fixed axes or inertial axes (fixed to the invariable plane). For what it is worth the paths traveled by the angular velocity vector ω_{ib} in its meanderings along the invariable plane and along the surface of the Poincaré ellipsoid have been given names. The locus of ω_{ib} on the invariable plane is called the "Herpolhode" and the locus on the Poincaré ellipsoid is called the "Polhode." Details for the construction of these figures is not relevant to our discussion and hence the interested reader is referred to books on dynamics such as Ref. 5 of Chapter III.

It is easy to develop the equation for the invariable plane. In Fig. 12.5 we have depicted a "spin-vector space." Thus, the axes (1), (2), and (3) are measures of the spin rate, although the spin vector for torque-free dynamics must remain on the surface invariable plane and the surface of Poincaré ellip-

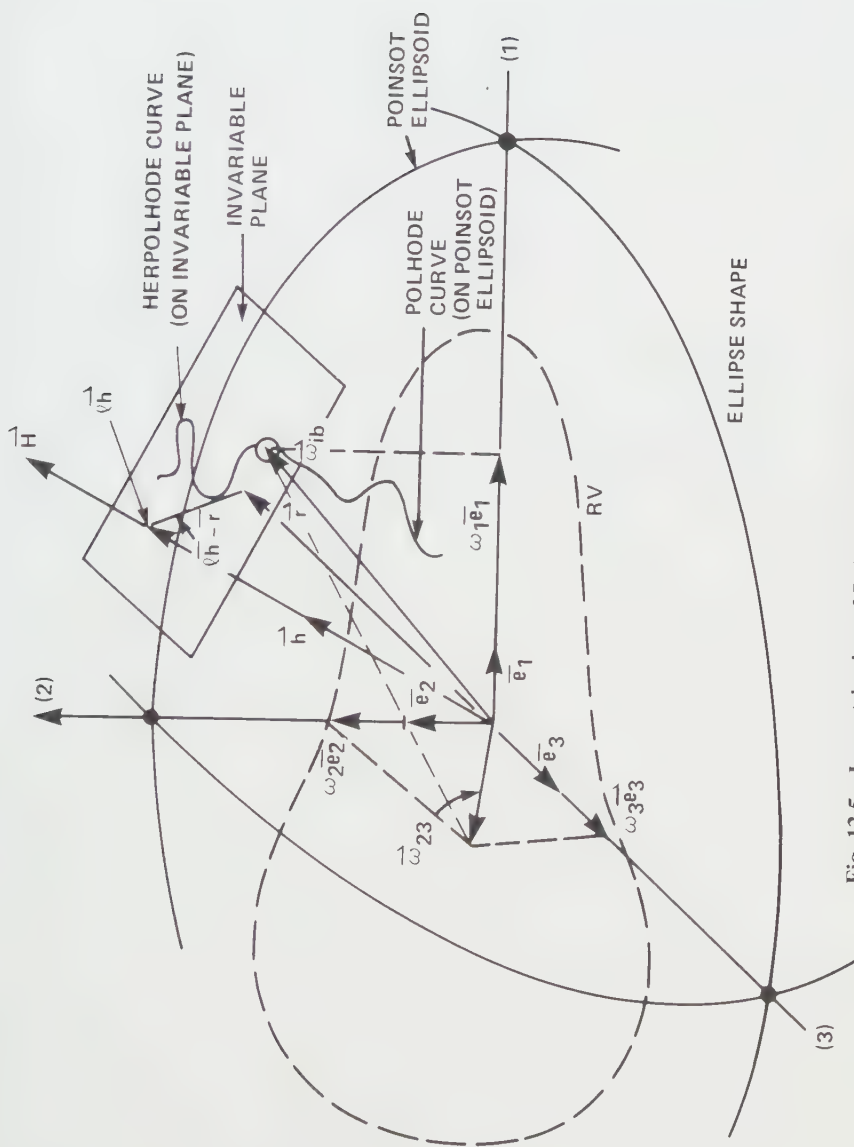


Fig. 12.5 Isometric view of Poinsof ellipsoid and invariable plane.

soid as it rotates on the invariable plane. We can let the vector $\mathbf{r}$ be given as

$$\mathbf{r} = \omega_1 \mathbf{e}_1 + \omega_2 \mathbf{e}_2 + \omega_3 \mathbf{e}_3$$

or

$$\mathbf{r} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + x_3 \mathbf{e}_3 \quad (12.27)$$

where x_1, x_2, x_3 are general points on the invariable plane and used to distinguish between $\omega_1, \omega_2, \omega_3$ which are the body components of the RV's spin vector $\boldsymbol{\omega}_{ib}$ and as such, are confined to the surface of the Poinsoit ellipsoid. In Fig. (12.5) the vector $\mathbf{r}$ locates a general point on the invariable plane. As a consequence $\mathbf{r}$ must satisfy the condition:

$$(\mathbf{r} - l\mathbf{n}) \cdot \mathbf{n} = 0$$

or

$$\mathbf{r} \cdot \mathbf{h} = l \mathbf{h} \cdot \mathbf{h} = l = 2T/H$$

or finally

$$(I_1 \omega_1) x_1 + (I_2 \omega_2) x_2 + (I_3 \omega_3) x_3 = 2T \quad (12.28)$$

where $I_1 \omega_1, I_2 \omega_2, I_3 \omega_3$ are components of the angular momentum vector and the direction numbers of the invariable plane since the angular momentum vector is along the normal. Thus, as we might expect, altering the relative magnitude of the components of the angular momentum vector will change the orientation of the invariable plane; but, of course, for torque-free motion, the direction numbers of the invariable plane are fixed constants.

Before continuing with a more analytic treatment of torque-free motion, let us review the geometrical concepts in terms of an RV off-loading from the equipment section. If the RV is off-loaded with an angular velocity $\boldsymbol{\omega}_{ib}$ oriented arbitrarily with respect to the RV's principal body frame, it is obvious from Eq. (12.19a) that, unless all moments of inertia are equal, the vector $\boldsymbol{\omega}_{ib}$ will not be co-linear with $\mathbf{H}$. We gave special attention to rotation about each of the three (distinct) principal axes. We showed that if rotation took place entirely about the intermediate principal axis, the angular velocity vector would diverge from this axis causing the body-fixed frame to diverge from the fixed direction in space of the angular momentum vector.

If the vehicle is nonrigid, then there is the possibility that rotation can excite bending modes and, if rotation is about the minor principal axis, the angular momentum and angular velocity vectors will also diverge. So far in this section we have set certain restrictions on the magnitude and direction of the angular velocity vector $\boldsymbol{\omega}_{ib}$ as it diverges from the angular momentum vector.

As it turns out the angular velocity vector $\boldsymbol{\omega}_{ib}$ will describe a cone in both the body frame and in the inertial frame. If $I_2 = I_3 = I$ and if $I_1 < I$, the angular motion of the body can be depicted as a rolling of the body-cone on the space cone. The axis of the body-cone is taken along the (1) body-axis and the axis of the space cone is taken along the angular momentum vector $\mathbf{H}$. Figure 12.6 shows the two cones. In Fig. 12.6 $\boldsymbol{\omega}_{ib}$ is the vector describing the instantaneous rotation of the body relative to inertial space. As we have pointed out earlier, this vector is not fixed in space. $\boldsymbol{\omega}_{b\omega}$ is the angular rate

at which ω_{ib} rotates relative to the body frame and $\omega_{i\omega}$ is the rate at which the angular velocity ω_{ib} vector rotates relative to inertial space. Figure 12.6 follows closely Fig. 3.5 of Ref. 1.

The angular velocity of the $\omega_{i\omega}$ vector relative to the body frame follows from a solution of Eqs. (12.2). To make the analysis tractable we can simplify these equations by applying our assumption of mass symmetry about the (1) axis. Thus Eqs. (12.2) become:

$$I_1 \dot{\omega}_1 = 0 \quad (12.29a)$$

$$I \dot{\omega}_3 + \omega_1 \omega_3 (I_1 - I) = 0 \quad (12.29b)$$

$$I \dot{\omega}_2 + \omega_1 \omega_2 (I - I_1) = 0 \quad (12.29c)$$

It immediately follows from Eq. (12.29a) that

$$\omega_1 = \text{const}$$

If we then differentiate Eqs. (12.29b) and (12.29c) we obtain:

$$\ddot{\omega}_2 + \omega_1^2 \left[\frac{I_1 - I_2}{I} \right]^2 \omega_2 = 0 \quad (12.30a)$$

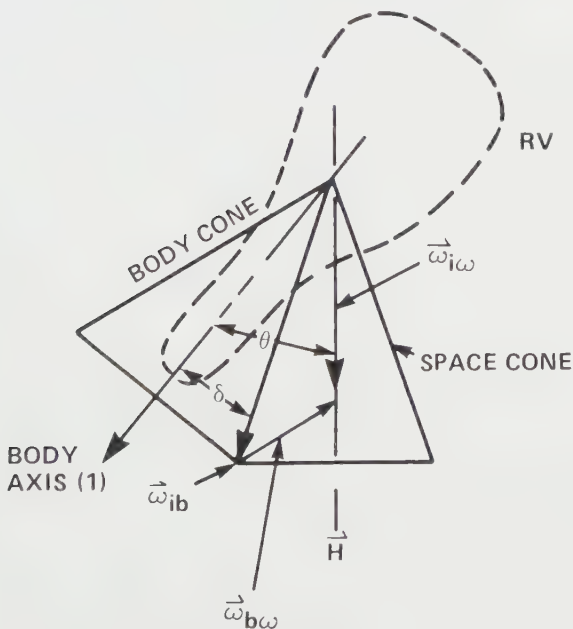


Fig. 12.6 Body and space cones.

$$\ddot{\omega}_3 + \omega_1^2 \left[\frac{I_1 - I_2}{I} \right]^2 \omega_3 = 0 \quad (12.30b)$$

We can recognize that ω_1 is the component of ω_{ib} along the (1) axis. Thus, if $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ are unit vectors along the three principal axes,

$$\omega_1 = \omega_{ib} \cdot \mathbf{e}_1 = \omega_{ib} \cos \delta \quad (12.31a)$$

or if we let

$$|\omega_{b\omega}| \equiv \omega_{b\omega} = \omega_{ib} \cos \delta \left[\frac{I_1 - I}{I} \right] \quad (12.31b)$$

Then Eqs. (12.30) may be rewritten as

$$\ddot{\omega}_2 + \omega_{b\omega}^2 \omega_2 = 0 \quad (12.32a)$$

$$\ddot{\omega}_3 + \omega_{b\omega}^2 \omega_3 = 0 \quad (12.32b)$$

As indicated in Fig. 12.6, δ is the angle between the (1) body axis and the angular velocity vector ω_{ib} . Equations (12.32a) and (12.32b) are recognized as the equations for a harmonic oscillator where $\omega_{b\omega}$ is the oscillatory frequency of ω_2 and ω_3 . If we return to Fig. 12.5 we see that $\omega_2 \omega_3$ are the components of ω_{ib} along the (2) and (3) principal axes.

We can rewrite Eqs. (12.29a-c) using the parameter $\omega_{b\omega}$ as

$$\ddot{\omega}_2 + \omega_{b\omega} \omega_3 = 0 \quad (12.33a)$$

$$\ddot{\omega}_3 - \omega_{b\omega} \omega_2 = 0 \quad (12.33b)$$

Now if we multiply the first equation by ω_2 and the second by ω_3 and add we get:

$$\omega_2 \dot{\omega}_2 + \omega_3 \dot{\omega}_3 = 0$$

$$\frac{d}{dt}(\omega_2^2 + \omega_3^2) = 0$$

or

$$\omega_2^2 + \omega_3^2 \equiv \omega_{23}^2 = \text{const} \quad (12.34)$$

Thus, as a consequence of our assumption of symmetric mass distribution about the (1) axis, the magnitude of the angular velocity of the RV relative to inertial space is constant, as

$$\omega_{ib}^2 \equiv \omega_1^2 + \omega_2^2 + \omega_3^2 = \omega_1^2 + \omega_{23}^2 \quad (12.35)$$

from Eqs. (12.29a) and (12.34).

Now let us introduce the complex variable $\tilde{\omega}_{23}$ as

$$\tilde{\omega}_{23} = i\omega_2 + \omega_3 \quad (12.36)$$

Equations (12.36) are identical to Eq. (10.10), where $\omega_2 = q$ and $\omega_3 = r$. Multiplying Eq. (12.32a) by the imaginary i and adding to Eq. (12.32b) we get:

$$\ddot{\tilde{\omega}}_{23} + \omega_{bw} \tilde{\omega}_{23} = 0 \quad (12.37)$$

We have already shown in Eq. (12.34) that

$$|\tilde{\omega}_{23}| = \text{const} = |\omega_{23}(0)|$$

It is easy to show by substitution that

$$\tilde{\omega}_{23} = |\tilde{\omega}_{23}(0)| e^{i\omega_{bw}t} \quad (12.38)$$

is a solution of Eq. (12.37). The important observation we must make is that the component of ω_{ih} in the (2)-(3) plane rotates at a rate ω_{bw} . This means that the angular velocity vector ω_{ih} rotates in a body frame to form a cone as shown in Fig. 12.6. Again it should be emphasized that all these results rest on the assumption of torque-free motion and mass symmetry about the (1) axis, i.e., $I_2 = I_3 = I$.

In Eq. (12.36) we wrote ω_{23} , the component of ω_{ih} in the (2)-(3) plane, as a complex variable. We can also write ω_{23} as a vector quantity as

$$\omega_{23} = \omega_2 \mathbf{e}_2 + \omega_3 \mathbf{e}_3 \quad (12.39)$$

where $\mathbf{e}_2$ and $\mathbf{e}_3$ are unit vectors along the (2) and (3) principal axes. Now from Eq. (12.19a) we may write the angular momentum as

$$\mathbf{H} = I_1 \omega_1 \mathbf{e}_1 + I_2 \omega_2 \mathbf{e}_2 + I_3 \omega_3 \mathbf{e}_3$$

but since $I_2 = I_3 = I$, we have

$$\mathbf{H} = I_1 \omega_1 \mathbf{e}_1 + I(\omega_2 \mathbf{e}_2 + \omega_3 \mathbf{e}_3)$$

Using Eq. (12.39) the angular momentum vector may also be written as

$$\mathbf{H} = I_1 \omega_1 + I \omega_{23}$$

where

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_{23} \quad \mathbf{H}_{23} = I \omega_{23} \quad (12.40)$$

In Fig. 12.7 we have shown the relationship among the vectors $\mathbf{H}$, ω_{ih} , ($\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$), $\mathbf{H}_{23}$, and ω_{23} .

We can conclude that the vectors $\mathbf{H}$, ω_{ih} , and $\mathbf{e}_1$ (the unit vector along the body's axis of symmetry) all lie in a plane. The reasoning is as follows: the vectors $\mathbf{H}_1 = H_1 \mathbf{e}_1$, $\omega_1 = \omega_1 \mathbf{e}_1$ and $\mathbf{e}_1$ all obviously lie in the same direction as do the vectors $\mathbf{H}_{23}$ and ω_{23} . Because we are limiting ourselves to torque-free motion the angular momentum vector $\mathbf{H}$ must be fixed relative to inertial space. This plane, defined by the vectors $\mathbf{H}$, ω_{ih} , and $\mathbf{e}_1$, cannot be fixed in inertial space since both ω_{ih} and $\mathbf{e}_1$ are not fixed in inertial space. Because all three vectors must be co-planar and $\mathbf{H}$ is fixed, the only conclusion is that

this $(\mathbf{H}, \boldsymbol{\omega}_{ib}, \mathbf{e}_1)$ plane must rotate about the angular momentum vector $\mathbf{H}$. To anticipate our final conclusions, we can provide the following description: an observer nonrotating with respect to inertial space would see the axis of symmetry of the RV rotating (or “precessing”) about some axis in space. The direction of the axis would be the direction of the angular momentum vector. The precession rate would be constant and the axis of symmetry of the RV would describe a cone.

A special case comes to mind: what if $\boldsymbol{\omega}_{ib}$ and $\mathbf{H}$ are both co-linear? This situation is illustrated in Fig. 12.8. For $\boldsymbol{\omega}_{ib}$ and to be co-linear with $\mathbf{H}$ (but not apparently co-linear with $\mathbf{e}_1$, the axis of symmetry of the RV), the following relationship would have to hold true between the transverse and axial moments of inertia I and I_1 :

$$\frac{|\omega_{23}|}{H_{23}} = \frac{(\omega_2^2 + \omega_3^2)^{1/2}}{I(\omega_2^2 + \omega_3^2)^{1/2}} = \frac{\omega_1}{H_1} = \frac{\omega_1}{I_1 \omega_1}$$

which requires that

$$I = I_1 = I_2 = I_3$$

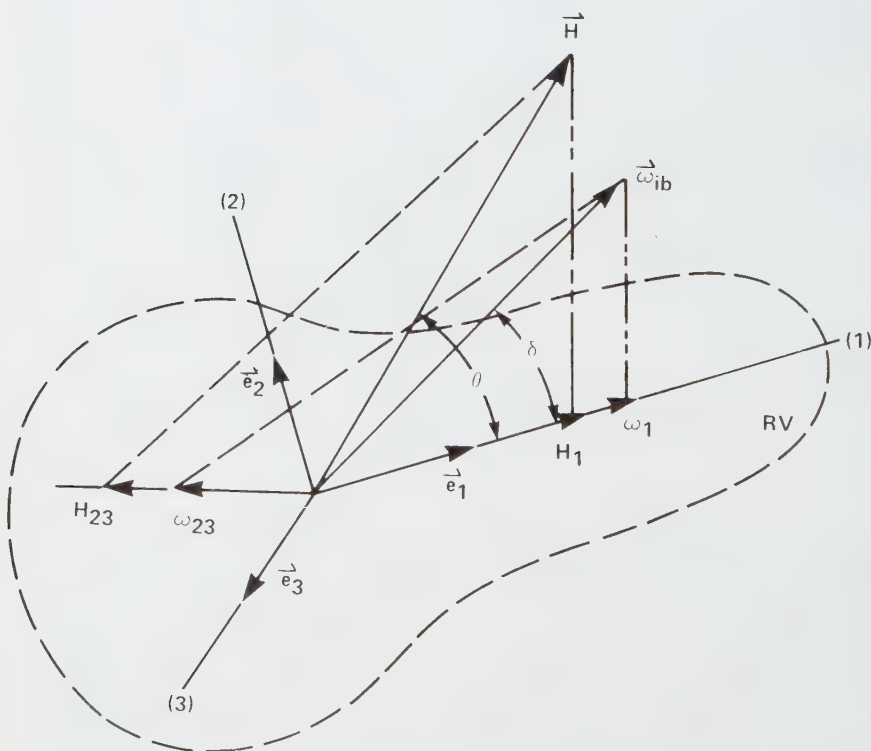


Fig. 12.7 Resolution of the angular momentum and angular velocity vectors in a principal body frame.

Unless $I = I_1$, it is impossible to have coincidence between ω_{ib} and $\mathbf{H}$ without there also being coincidence with $\mathbf{e}_1$. This means of course that if ω_{ib} is coincident with $\mathbf{H}$, then $\omega_{23} = 0$ and $\mathbf{H}_{23} = 0$. Thus, in this special case all three vectors must be co-linear with the principal inertial axis $\bar{\mathbf{e}}_1$, and the ω_{ib} - $\mathbf{H}$ - $\mathbf{e}_1$ plane will be reduced to a single line. The RV would then be spinning about its axis of symmetry. Otherwise the angular momentum vector $\mathbf{H}$ cannot be co-linear with the angular velocity vector ω_{ib} , excepting spherical mass distribution.

Let us return to Fig. 12.7. We note that

$$\tan\theta = \frac{H_{23}}{H_1} = \frac{I(\omega_2^2 + \omega_3^2)^{1/2}}{I_1\omega_1} \quad (12.41a)$$

$$\tan\delta = \frac{\omega_{23}}{\omega_1} = \frac{(\omega_2^2 + \omega_3^2)^{1/2}}{\omega_1} \quad (12.41b)$$

Obviously then

$$\tan\theta = \left[\frac{I}{I_1} \right] \tan\delta \quad (12.42)$$

If we refer to our scheme-2 Euler angle development given in Sec. 8.2, we will identify the angle θ as the orientation of the axis of symmetry from an inertial direction [see Eq. (8.30b)]. The angle θ is often identified as the "nutation angle." Since in our problem $I_1 < I$, according to Eq. (12.40), θ must be larger than δ . Motion may be described as rotation of the RV about the instantaneous direction of the angular velocity vector ω_{ib} at the rate ω_{ib} , while the ω_{ib} vector rotates about the $\mathbf{H}$ vector. Since we have customarily thought of ω_{ib} as being fixed in space while the body frame rotates about the

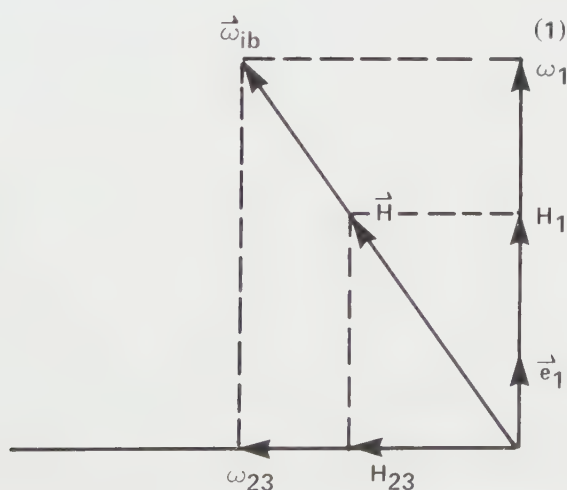


Fig. 12.8 Special case of co-linear angular velocity and angular momentum vectors.

(instantaneously) fixed direction of ω_{ib} , we may find the above description hard to follow. If ω_{ib} remained fixed in space (together with $\mathbf{H}$), the rotation about ω_{ib} would bring the axis of symmetry vector $\mathbf{e}_1$ out of the $\mathbf{H}$ - ω_{ib} - $\mathbf{e}_1$ plane. However, as we have shown, all three vectors must remain coplanar for torque-free motion. Earlier we had designated the rotation rate of the RV's principal axis frame relative to inertial space by the symbol ω_{ib} . The component of ω_{ib} along the axis of symmetry was designated by ω_1 ; yet it seems that this axis is rotating relative to inertial space at some rate other than ω_1 . The confusion may be cleared up if the true meaning of the angular velocity vector ω_{ib} is emphasized. ω_{ib} designates the angular rotation of a frame; it is meaningless to talk of the angular rotation rate of a line, although we have routinely talked about the resolution of ω_{ib} along the three mutually perpendicular lines which form the body axis. Thus, we describe the rotation of the RV relative to inertial space by the vector ω_{ib} , but recognize that the vector, ω_{ib} is itself rotating about $\mathbf{H}$. We must remember that it is the components of ω_{ib} that appear in the inertial terms in the moment equations [see Eqs. (12.27)]. We also note that according to Eqs. (12.41) the nutation angle θ and the angle δ are both constants since $(\omega_2^2 + \omega_3^2)$ is constant and ω_1 is constant.

Since we have noted that the plane containing $(\mathbf{H}, \omega_{ib}, \mathbf{e}_1)$ rotates about the angular momentum vector, we may endeavor to calculate this rate. In passing it seems perfectly consistent with our previous usage of angular velocity vectors to discuss the angular velocity of a plane: an axis system could be associated with the plane as two mutually perpendicular lines would define the plane, and with the directional cosines of these lines known, it is a simple matter to define a perpendicular to these lines which is of course the normal to the plane. In carrying out this work we shall make use of a simple, but elegant development due to Wrigley et al. [Ref. (1)]. If we refer back to Fig. (12.6) we note three vectors: ω_{ib} the rotation rate of the RV relative to inertial space, $\omega_{b\omega}$ the angular rate of the ω_{ib} relative to the RV, and $\omega_{i\omega}$ the angular rate of the ω_{ib} vector relative to inertial space. Since ω_{ib} and $\mathbf{e}_1$ are always in the same plane with $\omega_{i\omega}$ (or $\mathbf{H}$), $\omega_{i\omega}$ is also the rate at which the axis of symmetry appears to rotate about a fixed direction in space. These three vectors obey the following equation:

$$\omega_{i\omega} = \omega_{ib} + \omega_{b\omega} \quad (12.43)$$

Taking the inner product of both sides gives

$$\omega_{i\omega} \cdot \omega_{i\omega} = \omega_{ib} \cdot \omega_{ib} + 2\omega_{ib} \cdot \omega_{b\omega} + \omega_{b\omega} \cdot \omega_{b\omega} \quad (12.44)$$

Using Eq. (12.31) we get

$$\omega_{i\omega}^2 = \omega_{ib}^2 + 2\omega_{ib}^2 \cos^2 \delta \left[\frac{I_1 - I}{I} \right] + \omega_{ib}^2 \left[\frac{I_1 - I}{I} \right] \cos^2 \delta \quad (12.45)$$

If we collect terms, Eq. (12.45) reduces to

$$\omega_{i\omega} = \omega_{ib} [\sin^2 \delta + \cos^2 \delta (I_1 / I^2)]^{1/2} \quad (12.46)$$

In Eq. (12.46) we must remember that δ is the angle between the angular velocity vector ω_{ib} and the body's axis of symmetry. The vector $\omega_{i\omega}$ is called the "precession rate." An observer nonrotating relative to inertial space would observe the axis of symmetry rotating about some fixed direction in space at the nutational rate $\omega_{i\omega}$. Note that if the RV is off-loaded with an angular velocity vector ω_{ib} making an angle δ with respect to the body's axis of symmetry, then the axis of symmetry will rotate about the angular momentum vector at a rate given by Eq. (12.46). Of course the observer will not be able to see the angular momentum vector $\mathbf{H}$, but it is this vector that will define the fixed axis about which the RV seems to be rotating or "precessing."

If we refer to Sec. 8.2 and the development of Euler angles scheme 2 we will remember that ψ is the rate of rotation of the body frame about some fixed direction. We can easily show that ψ is identical to the magnitude of $\omega_{i\omega}$. First we can rewrite Eq. (12.46) as

$$\omega_{i\omega} = \omega_{ib} \cos \delta [\tan^2 \delta + (I_1/I)^2]^{1/2}$$

then using Eq. (12.42) we get

$$\omega_{i\omega} = \cos \delta \left[\frac{I_1}{I} \right] [\tan^2 \theta + 1]^{1/2} \omega_{ib} = \omega_{ib} \left[\frac{I_1}{I} \right] \left[\frac{\cos \delta}{\cos \theta} \right]$$

Finally, from Eq. (12.31a) we have

$$\omega_{i\omega} = \left[\frac{\omega_1}{\cos \theta} \right] \left[\frac{I_1}{I} \right] \quad (12.47)$$

Since all terms on the right are constant, $\omega_{i\omega}$ is again demonstrated to be a constant. Now Kaplan [Ref. (5) of Chapter III] develops ψ in terms of the second scheme of Euler A angles as

$$\dot{\psi} = I_1 \dot{\phi} / (I - I_1) \cos \theta \quad (12.48a)$$

From the first of Eqs. (8.33a) we may replace $\dot{\phi}$ to give

$$\dot{\psi} = \frac{(I_1/I)(\omega_1 - \dot{\psi} \cos \theta)}{[1 - (I_1/I)] \cos \theta}$$

where we have used ω_1 for the component of ω_{ib} along the principal axis (1) rather than the symbol p used in Eq. (8.33a). Collecting terms in the above expression we get:

$$\dot{\psi} = \left[\frac{\omega_1}{\cos \theta} \right] \left[\frac{I_1}{I} \right] \quad (12.48b)$$

which is of course identical to Eq. (12.47). For reference we then have:

$$\dot{\psi} = \omega_{i\omega} = \left[\frac{\omega_1}{\cos\theta} \right] \left[\frac{I_1}{I} \right] = \omega_1 \left[\tan^2\delta + \left[\frac{I_1}{I} \right]^2 \right]^{1/2} \quad (12.49)$$

where

$$\tan\delta = \frac{\omega_{23}}{\omega_1} = \frac{(\omega_2^2 + \omega_3^2)^{1/2}}{\omega_1} \quad (12.50)$$

If, when the RV is off-loaded, we have available the components of the angular velocity along the principal axes, we can then calculate the angle δ from Eq. (12.42). It then is possible to calculate the precessional rate $\dot{\psi}$ from Eq. (12.49). If δ is small, then an approximation to Eq. (12.49) becomes

$$\dot{\psi} = \omega_{i\omega} \cong \omega_1 \left[\frac{I_1}{I} \right] \cong \omega_{ib} \left[\frac{I_1}{I} \right] \quad (12.51)$$

Note from Eq. (12.42) that if δ is zero, then the nutation angle θ is also zero. Under such circumstances the RV's axis of symmetry vector $\mathbf{e}_1$, the angular velocity vector $\boldsymbol{\omega}_{ib}$, and the angular momentum vector $\mathbf{H}$, will all be co-linear.

References

¹Goldstein, H., *Classical Mechanics*, Addison-Wesley, Reading, Mass., 1950, Chap. 5.

²Wrigley, W., Hollister, W. M., and Dehard, W. G., *Gyroscopic Theory: Design and Instruction*, M.I.T. Press, Cambridge, Mass., 1969.

XIII. ANGULAR MOTION DURING RE-ENTRY

In Chapter VI we discussed particle-type motion of the re-entry vehicle (RV) within the atmosphere. Various closed-form solutions were obtained to illuminate the essential physics of atmospheric re-entry. The RV was treated as a point mass and the only aerodynamic forces acting on it were lift and drag. No comment was offered at that time concerning the origin of the lift force although the assumption might have been made that the lifting force was controlled or varied by altering the attitude of the vehicle relative to the velocity vector. The RV attitude is changed by the application of aerodynamic (passive) moments due to RV passage through the atmosphere or control (active) moments due to reactionary, aerodynamic, moving mass, etc., controls. In this chapter we shall confine our attention to moments of aerodynamic origin.

13.1 Force and Moment Equations

In Sec. 8.5 we derived expressions for the force and moment equations in matrix form. These equations will be restated here for convenience:

$$\mathbf{M}[\dot{\mathbf{V}}^b + \boldsymbol{\Omega}_{ib}^b \mathbf{V}^b] = \mathbf{F}_a^b + \mathbf{C}_e^b \mathbf{F}_g^e \quad (13.1a)$$

$$\mathbf{I}^b \dot{\boldsymbol{\omega}}_{ib}^b + \boldsymbol{\Omega}_{ib}^b \mathbf{I}^b \boldsymbol{\omega}_{ib}^b = \mathbf{M}^b \quad (13.1b)$$

If we were to lay the groundwork for a fairly complete simulation of the RV's trajectory, then we would include some additional equations for the dynamics of the control system and for guidance laws (which relate some vehicle observables to control settings) as well as additional coordinate frames. In this chapter we shall be more concerned with obtaining an understanding of the physics of angular dynamics. Toward this end we shall assume that the Earth is flat and nonrotating. Consequently we are essentially assuming that the Earth-frame e is inertial and the DCM $\mathbf{C}_e^b$ is between an inertial frame and a body frame. Thus, $\mathbf{C}_e^b$ is the $\mathbf{D}$ matrix of Eqs. (8.5) and the elements of $\mathbf{C}_e^b$ will be expressed as trigonometric functions of the three Euler angles. The derivatives of these Euler angles are related to the components of the angular velocity vector in the body frame, i.e., $\boldsymbol{\omega}_{ib}^b = [p, q, r]^T$ [Eq. (8.12)], as

$$\omega_{ib}^b = \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \mathbf{M} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \quad (13.1c)$$

If we wish to describe motion of the RV over the flat, nonrotating Earth, we might add the following auxiliary equation which transforms the velocity vector $\mathbf{V}^b$ from a body frame to an Earth frame as

$$\mathbf{V}^e = \mathbf{C}_e^b \mathbf{V}^b = (\mathbf{C}_e^b)^T \mathbf{V}^b \quad (13.1d)$$

The DCM $\mathbf{C}_e^b$ was encountered in Eq. (13.1a) and, as pointed out above, has elements expressible in terms of the Euler angles.

Examination of Eqs. (13.1) indicates that we have nine equations and nine variables provided of course that the forces $\mathbf{F}^b = \mathbf{F}_d^b + \mathbf{C}_e^b \mathbf{F}_s^e$ and moments $\mathbf{M}^b$ may be defined entirely from these nine variables. The nine variables are:

$$\omega_{ib}^b = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad \mathbf{V}^b = \begin{bmatrix} u \\ v \\ w \end{bmatrix} \quad \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}$$

As it turns out an adequate representation of the aerodynamic forces and moments can be made in terms of these variables.

We shall now affix to the RV a conventional body frame (x^b, y^b, z^b) and let the velocity vector, prior to perturbation, be represented as

$$\mathbf{V}_0 = (U_0)\mathbf{i}_b + (V_0)\mathbf{j}_b + (W_0)\mathbf{k}_b \quad (13.2a)$$

If the assumption is made that the velocity vector lies entirely in the $x^b z^b$ plane, prior to perturbation, Eq. (13.2a) becomes

$$\mathbf{V}_0 = (U_0)\mathbf{i}_b + (W_0)\mathbf{k}_b \quad (13.2b)$$

Assume that $\mathbf{v}$, the perturbational velocity vector, may be represented as

$$\mathbf{v} = u\mathbf{i}_b + v\mathbf{j}_b + w\mathbf{k}_b \quad (13.2c)$$

we may write the total velocity vector after perturbation as

$$\mathbf{V} = \mathbf{V}_0 + \mathbf{v} = (U_0 + u)\mathbf{i}_b + v\mathbf{j}_b + (W_0 + w)\mathbf{k}_b \quad (13.2d)$$

or

$$\mathbf{V} = \left[\mathbf{i}_b + \left[\frac{v}{U_0} \right] \mathbf{j}_b + \left[\frac{W}{U_0} + \frac{w}{U_0} \right] \mathbf{k}_b \right] U_0 \quad (13.2e)$$

where we have assumed that $u/U_0 \ll 1$. Also, since v, W_0 , and w are at right angles to U_0 we might recognize the angles between v, W_0, w , and U_0 as new variables to replace the velocities, as

$$w/U_0 = \tan \alpha \cong \alpha \quad W_0/U_0 = \tan \alpha_T \cong \alpha_T \quad v/U_0 = \tan \beta \cong \beta \quad (13.2f)$$

where we have made the small angle assumption consistent with perturbational analysis. Figure 13.1 gives the velocity before perturbation $\mathbf{V}_0^b$, the perturbational velocity $\mathbf{v}^b$, and the total velocity after disturbance $\mathbf{V}^b$ and the angles α, α_T , and β . α_T locates the initial direction of the vector and α locates perturbational changes in the direction of velocity vector $\mathbf{V}^b$. The perturbational velocity $\mathbf{v}^b$ and the initial or steady-state velocity vector $\mathbf{V}_0^b$ are related as

$$\mathbf{V}^b = \mathbf{V}_0^b + \mathbf{v}^b \quad (13.3)$$

The aerodynamic forces and moments can now be expanded in terms of the aerodynamic angles α, β , and α_T and the Euler angles (and their derivatives) ϕ, θ , and ψ . These expansions must be accepted on the basis of experience (wind tunnel, ballistic ranges, flight testing) since many mathematically possible derivatives have been omitted.

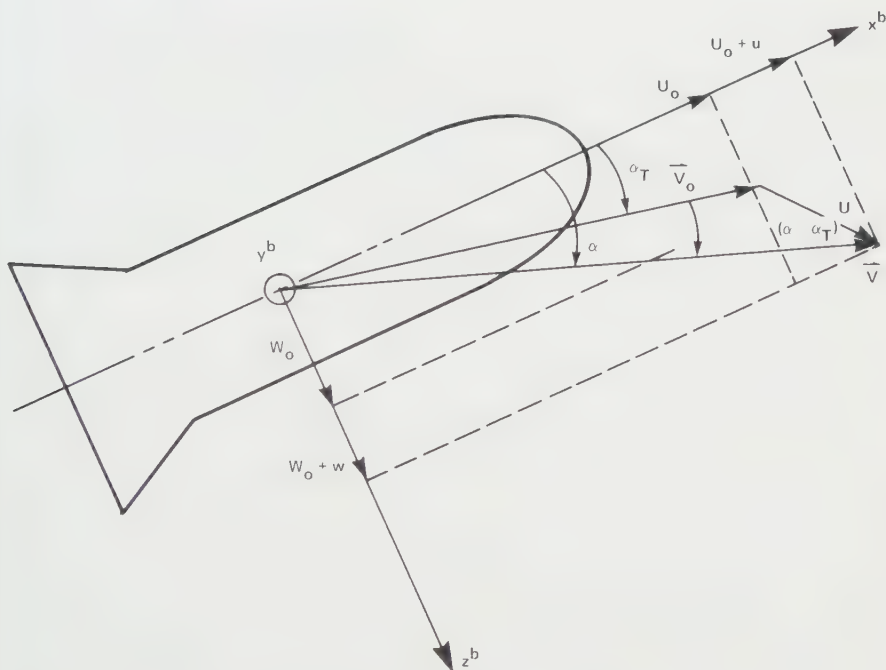


Fig. 13.1 Definition of velocity vector components.

$$X = \frac{1}{2}\rho_{\infty} V_{\infty}^2 S C_x \quad (13.4a)$$

$$Y = \frac{1}{2}\rho_{\infty} V_{\infty}^2 S \left[-\beta C_{y\beta} + \frac{\dot{\beta}d}{V_{\infty}} C_{y\dot{\beta}} \right] \quad (13.4b)$$

$$Z = \frac{1}{2}\rho_{\infty} V_{\infty}^2 S \left[\alpha C_{z\alpha} + \frac{\dot{\alpha}d}{V_{\infty}} C_{z\dot{\alpha}} \right] \quad (13.4c)$$

$$L = \frac{1}{2}\rho_{\infty} V_{\infty}^2 S d \frac{pd}{V_{\infty}} C_{lp} \quad (13.4d)$$

$$M = \frac{1}{2}\rho_{\infty} V_{\infty}^2 S d [(\alpha - \alpha_T) C_{m\alpha} + \left(\frac{pd}{V_{\infty}} \right) \beta C_{m_{p\beta}} + \left(\frac{qd}{V_{\infty}} \right) [C_{m_q} + C_{m\dot{\alpha}}]] \quad (13.4e)$$

$$N = \frac{1}{2}\rho_{\infty} V_{\infty}^2 S d \left[-\beta C_{n\beta} + \frac{pd}{V_{\infty}} \alpha C_{n_{p\alpha}} + \frac{rd}{V_{\infty}} [C_{n_{\dot{\beta}}} - C_{n_r}] \right] \quad (13.4f)$$

The various aerodynamic derivatives were defined earlier in Chapter IX, Eqs. (9.7-9.9).

The force equation was given in matrix form in Eq. (13.1); the components of the force may now be written as

$$\dot{U} + qW - rV = (X/m) - g \sin \theta \quad (13.5a)$$

$$\dot{V} + rU - pW = (Y/m) + g \cos \theta \sin \phi \quad (13.5b)$$

$$\dot{W} + pV - qU = (Z/m) + g \cos \theta \cos \phi \quad (13.5c)$$

where the components of the aerodynamic force vector is given in Eqs. (13.4a-13.4c).

The moment equation, in matrix form, is also given in Eq. (13.1). Scalar equations could of course be written; however, there can be extensive coupling among the three equations if the inertia tensor has off-diagonal terms (products of inertia). For numerical simulation some or all of these inertial couplings may be included; however, for any closed-form solution simplifications must be made. Therefore, the subsequent equations will be predicated upon the assumption that the (x^b, y^b, z^b) axes are principal.

$$[I_x \dot{p} - I_{xy} \dot{q} - I_{xz} \dot{r}] + [(I_z - I_y)rq - I_{xz}pq + I_{xy}pr - I_{yz}q^2 + I_{yz}r^2] = L \quad (13.6a)$$

$$[I_y \dot{q} - I_{yz} \dot{r} - I_{xy} \dot{p}] + [(I_x - I_z)pr - I_{xy}qr + I_{yz}pr - I_{xz}r^2 + I_{xz}p^2] = M \quad (13.6b)$$

$$[I_z \dot{r} - I_{xz} \dot{p} - I_{yz} \dot{q}] + [(I_y - I_x)pq - I_{yz}pr + I_{zx}qr - I_{xy}p^2 + I_{xy}q^2] = N \quad (13.6c)$$

If we now invoke our assumption that the body axes are principal, the products of inertia appearing in Eqs. (13.6) will vanish to give:

$$I_x \dot{p} + (I_z - I_y)rq = L \quad (13.7a)$$

$$I_y \dot{q} + (I_x - I_z)pr = M \quad (13.7b)$$

$$I_z \dot{r} + (I_y - I_x)pq = N \quad (13.7c)$$

13.2 Rolling Moment Equation

In this section we shall consider special cases of RV angular motion. The simplest motion to analyze is pure roll. However, the rolling moment of an RV is not a trivial consideration; it was shown in Chapter X that the presence of roll rate can be a contributor to "resonance" or amplification of the trim angle of attack.

From Eq. (13.6a) we have:

$$I_x \dot{p} + (I_z - I_y)rq = L \quad (13.8a)$$

Assuming rotational symmetry, i.e., $I_z = I_y$, and a rolling moment L separated into a driving moment L_0 , and a damping moment $L_p p \triangleq (\partial L / \partial p)p$, the above equation becomes:

$$I_x \dot{p} = L_0 + L_p p$$

or

$$\dot{p} - (L_p / I_x)p = L_0 / I_x \quad (13.8b)$$

We may represent the driving moment L_0 and damping moment derivative L_p in coefficient form as

$$L_0 = C_{l_0} q S d \quad L_p = C_{l_0} (pd / V_\infty) q S d \quad (13.9)$$

Inserting Eq. (13.9) into Eq. (13.8) we have

$$\frac{d}{dt} \left[p + \frac{C_{l_0} V_\infty}{C_{l_p} d} \right] - \left[\frac{C_{l_p} q S d^2}{I_x V_\infty} \right] \left[p + \frac{C_{l_0} V_\infty}{C_{l_p} d} \right] = 0 \quad (13.10)$$

Integration of the above equation gives

$$p(t) = [p(0) - p_{ss}] e^{\lambda t} + p_{ss} \quad (13.11)$$

where $p(0)$ is the initial spin rate and

$$\lambda = C_{l_p} q S d^2 / I_x V_\infty \quad p_{ss} = -(C_{l_0} V_\infty / C_{l_p} d) \quad (13.12)$$

From Eq. (13.11) we see that at $t = 0$, $p = p(0)$. For roll rate to decay from an initial value, λ must be negative; from Eq. (13.12) this requires that C_{l_p} be negative. If there is a roll driving moment, i.e., $C_{l_0} \neq 0$, then the roll rate will decay to a steady-state value $p = p_{ss}$.

It might be of interest to consider calculations of C_{l_0} and C_{l_p} . There are no general methods available for calculating the roll driving torque since the RV has no fins. The source of such a moment might lie at least partially in the ablative pattern set up in the heat shield.

One contributing source of roll torque can arise in an offset between the center of gravity and the center of pressure. If the RV is a body of revolution the center of pressure will lie on the centerline as shown in Fig. 13.2. If the center of gravity is displaced from the centerline, then there will be a roll torque because of the c.p.-c.g. separation. This contribution is easily seen.

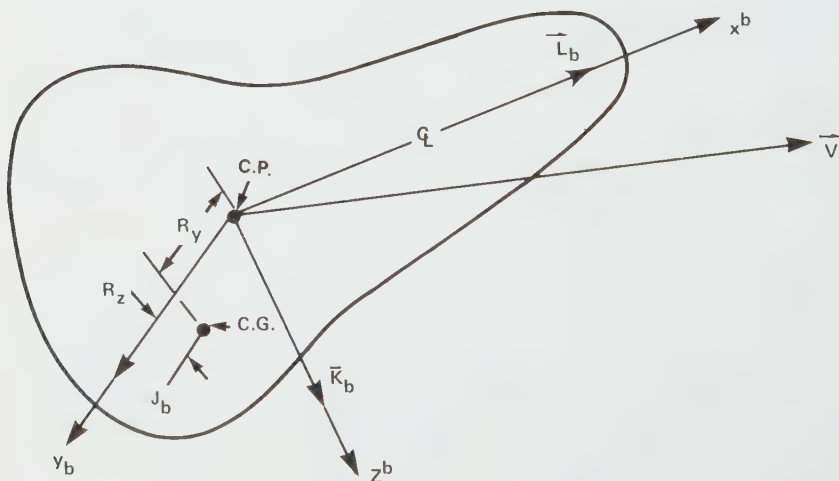


Fig. 13.2 Definition of center of mass offset.

In the analysis subsequent to Eq. (7.5) we ignored the first and third terms, assuming respectively that the center of mass does not move within the vehicle and that the origin of the axis system is at the center of mass. If the axis system remains on the centerline but the c.g. is now removed some distance, then we must retain the third term in Eq. (7.5).

The coordinates of the center of mass are (R_y, R_z) . The body velocity vector is $\mathbf{V}$. Since the three terms in Eq. (7.5) are additive, we may treat them separately. Let $\mathbf{h}_0$ represent the contribution of mass offset to the angular momentum as

$$\mathbf{h}_0 = \left[\sum m_i \mathbf{r}_i \right] \times \mathbf{V} = m [R_y \mathbf{j}_b + R_z \mathbf{k}_b] \times \mathbf{V} \quad (13.13)$$

where m is the body mass.

Simple manipulation gives

$$\mathbf{h}_0 = m \begin{bmatrix} \mathbf{i}_b & \mathbf{j}_b & \mathbf{k}_b \\ 0 & R_y & R_z \\ U & V & W \end{bmatrix} = [\mathbf{i}_b (R_y W - R_z V) + \mathbf{j}_b (R_z U) + \mathbf{k}_b (-R_y U)] m \quad (13.14)$$

The moment $\mathbf{M}$ was shown in Eq. (7.16) to be

$$\mathbf{M} = \left. \frac{d\mathbf{h}_0}{dt} \right|_b + \boldsymbol{\omega}_{ib} \times \mathbf{h}_0 \quad (13.15)$$

Since we are considering only roll rate $\boldsymbol{\omega}_{ib} = \omega_{ib} \mathbf{i}_b$, the moment about the X or $\mathbf{i}_b$ axis follows from Eqs. (13.14) and (13.15) as

$$l_0 = m [R_y \dot{W} - R_z \dot{V}] = m R_y A_z - m R_z A_y \quad (13.16)$$

The total rolling moment must now consist of the driving moment L_0 , the mass offset moment l_0 , and the damping moment $L_p p$ or

$$L = L_0 + L_p p + l_0$$

$$L = C_{l_0} q S d + C_{l_p} \left[\frac{q S d^2 p}{V_\infty} \right] + m R_y A_z - m R_z A_y \quad (13.17)$$

we have omitted any gravity moment due to center of mass offset. The rolling moment equation [Eq. (13.14)] is now being coupled with the Y and Z force equations through the accelerations $\dot{W} = A_z$ and $\dot{V} = A_y$.

It was mentioned earlier that the roll-driving moment C_{l_0} is difficult to isolate; however, there are means for calculating the roll damping moment derivative C_{l_p} .

Consider the case of a nose cone rolling about the x -axis at a rate p . We shall assume that shear stress is developed at the surface with a component τ along the conical generator and a component τ_ϕ perpendicular to τ , where ψ is the angle between the resultant of the stress at any point and the conical generator at that point. ψ can conveniently be defined in terms of τ_ϕ and τ as shown in Fig. 13.3(a):

$$\psi = \tan^{-1}(\tau_\phi / \tau) \quad (13.18a)$$

The elemental area is

$$dA = 2\pi r dx_c$$

The elemental roll torque is

$$dl = -2\pi \tau_\phi r^2 dx_c \quad (13.18b)$$

The roll moment due to shearing stress τ_ϕ is

$$l = \int dl = -2\pi \int_{X_{c_i}}^{X_{c_f}} \tau_\phi r^2 dx_c \quad (13.18c)$$

where x_c is a coordinate along the conical generator of the RV. We can identify the limits in the integral of Eq. (13.18b): X_{c_i} is the point on the generator where the nose radius has its point of tangency with the conical generator; X_{c_f} is the most rearward point on the generator as seen in Fig. 13.3(b).

The roll-damping derivative may now be expressed as

$$C_{l_p} = \frac{\partial C_l}{\partial (pd/V_\infty)} = \frac{l}{qSd(pd/V_\infty)} = \frac{l}{q\pi R_B^2(2R_B)(2pR_B/V_\infty)} \quad (13.19)$$

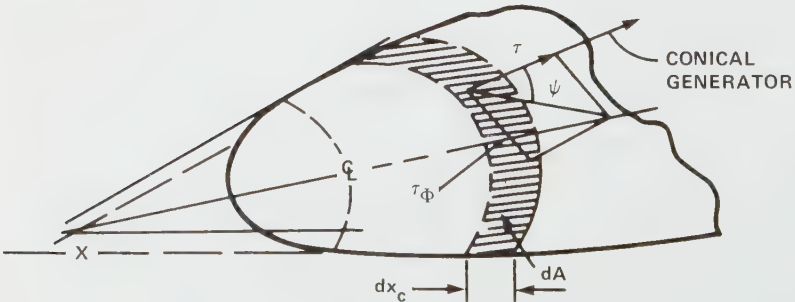


Fig. 13.3(a) Incremental area.

This expression for τ_ϕ may be inserted into Eq. (13.21) to give:

$$C_{l_p} = -\frac{1}{2} \frac{q_c}{q_\infty} \sin^3 \theta \int_{(R_N/R_B)^{\csc \theta}}^{\csc \theta} C_{f_c} \left[\frac{V_\infty}{u_c} \right] \left[\frac{x_c}{R_B} \right]^3 d \left[\frac{x_c}{R_B} \right] \quad (13.22)$$

Obviously once C_{f_c} has been selected as a function of (x_c/R_B) the integration of Eq. (13.22) can be carried out.

It might be of interest to calculate the damping in roll derivative C_{l_p} and to make comparisons with measurements. At a Mach number of 7.70, a Reynolds number of 6.23×10^6 (based on body length), a cone half-angle of 7.25 deg, and a bluntness of 7.5% we have available a measured value of -0.00125 for C_{l_p} .

First we must calculate the skin-friction coefficient C_{f_c} . From Blasius' relationship the skin-friction coefficient for a flat plate with a laminar boundary layer we have

$$C_f = 1.328/(R_e^{1/2}) = 5.32 \times 10^{-4}$$

The skin-friction coefficient C_f will vary with $x_{c_j}^{1/2}$; however, we shall accept a constant value C_f in the interest of simplicity. From Eq. (13.22) we must have q_c/q_∞ and u_c/V_∞ , the ratio of the dynamic pressure and airspeed to their corresponding freestream conditions. Here we shall rely on Linnel's work (see Ref. 1)* and use the following relationships for p_c/p_∞ , T_c/T_∞ (temperature and pressure ratio), and u_c/V_∞ . In the equations to follow, θ_c is the cone half-angle and M_∞ is the freestream Mach number.

$$\frac{T_c}{T_\infty} = [1 + \exp\{-1 - 1.52(M_\infty \sin \theta_c)\}] \left[1 + \frac{(M_\infty \sin \theta_c)^2}{4} \right] \quad (13.23a)$$

$$M_\infty \sin \theta_c > 1$$

$$\frac{T_c}{T_\infty} = 1 + 0.35(M_\infty \sin \theta_c)^{1.5} \quad M_\infty \sin \theta_c < 1 \quad (13.23b)$$

$$\frac{P_c}{P_\infty} = 1 + \frac{0.7 M_\infty^2 (4 \sin^2 \theta_c) [2.5 + 8 \sqrt{M_\infty^2 - 1} \sin \theta_c]}{1 + 16 \sqrt{M_\infty^2 - 1} (\sin \theta_c)} \quad (13.23c)$$

$$u_c/V_\infty = \cos \theta_c \sqrt{1 - (\sin \theta_c / M_\infty)} \quad (13.23d)$$

*The author is grateful to Dr. W. C. Dorsey, NSWC/WOL, for bringing this work to his attention.

Since $M_\infty \sin \theta_c = 0.972$ we shall use the second equation for T_c/T_∞ . It is easily shown from the equation of state of an ideal gas that

$$\frac{q_c}{q_\infty} = \left[\frac{u_c}{V_\infty} \right]^2 \left[\frac{p_c}{p_\infty} \right] \left[\frac{T_c}{T_\infty} \right]^{-1} \quad (13.24)$$

Using $\theta_c = 7.25$ deg and $M_\infty = 7.7$ we can easily calculate

$$u_c/V_\infty = 0.984 \quad p_c/p_\infty = 2.644 \quad T_c/T_\infty = 1.335$$

Thus

$$q_c/q_\infty = 1.917$$

Now carrying out the integration indicated in Eq. (13.22) we have:

$$C_{l_p} = -\frac{1}{8} \left[\frac{q_c}{q_\infty} \right] \csc \theta_c C_{f_c} \left[\frac{V_\infty}{u_c} \right] \left[1 - \left[\frac{R_N}{R_B} \right]^4 \right]$$

$$C_{l_p} = 0.00103$$

which we see is in fair agreement with the measured value of C_{l_p} of -0.00125 .

We might investigate the effect of a roll-damping coefficient of the calculated level on the roll rate. From Eq. (13.12) we can calculate the roll constant λ at an altitude of 15 k.

$$V_\infty = 4500 \text{ m/s}$$

$$q_\infty = \frac{1}{2} \rho_\infty V_\infty^2 = \frac{1}{2} \rho_0 e^{-h/H} V_\infty^2$$

$$= (\frac{1}{2})(1.752)(e^{-15/6.7})(4500)^2$$

$$= 1.89 \times 10^6 \text{ ns/m}^2 (\text{Pa})$$

$$d = 0.44 \text{ m}$$

$$S = \pi d^2/4 = 0.152 \text{ m}^2$$

$$I_x = 0.972 \text{ kg/m}^2$$

Thus from Eq. (13.12)

$$\lambda = \frac{C_{l_p} q S d^2}{I_x V_\infty} = -0.0131$$

Assuming $C_{l_0} = 0$, i.e., $p_{ss} = 0$, we obtain

$$p/p_0 = \exp(-0.0131t)$$

The time to half-amplitude, $t = t_{1/2}$ is

$$t_{1/2} = 52.9 \text{ s}$$

Thus, it is clear that in a horizontal run at low altitude the roll damping, even on an apparently smooth body can cause significant roll decay. Glover has shown that it is possible to relate the damping-in-roll derivative C_{l_p} to the axial force coefficient C_A .² The axial force coefficient is simply the component of the frictional force coefficient acting in a downstream direction as shown in Fig. 13.3(c)

$$C_A = \frac{\cos\theta}{q_\infty \pi R_B^2} \int_{X_{c_i}}^{X_{c_f}} C_{f_c} q_c (2\pi r) dX_c$$

or

$$C_A = \frac{q_c}{q_\infty} \cos\theta \int_{(R_N/R_B)\csc\theta}^{\csc\theta} 2C_{f_c} \left[\frac{r}{R_B} \right] d \left[\frac{X_c}{R_B} \right]$$

but since $r/R_B = \sin\theta$ we have

$$C_A = \frac{q_c}{q_\infty} \sin 2\theta \left[\int_{(R_N/R_B)\csc\theta}^{\csc\theta} C_{f_c} \left[\frac{X_c}{R_B} \right] d \left[\frac{X_c}{R_B} \right] \right] \quad (13.25)$$

According to Glover, C_{l_p} and C_A may be related as

$$C_{l_p}/C_A = C/\cos 2\theta \quad (13.26)$$

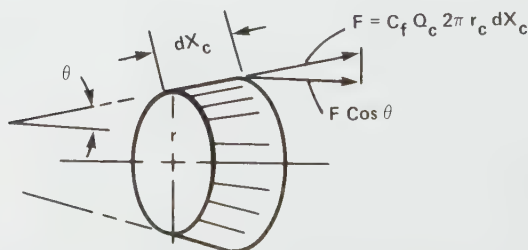


Fig. 13.3(c) Cone area element.

where the parameter C is defined over the range

$$-0.25 \leq C \leq -0.20$$

For R_N/R_B of 0.22 we have $C = -0.225$.

13.3 Pitching Moment Equations-Exponential Atmosphere

Another closed-form solution of interest is that of planar pitching motion. Since we are confining motion to the xz -plane (i.e., $r = p = 0$), we shall use Eq. (13.5c) and Eq. (13.7b). If we neglect any perturbations in the velocity component along the x axis, i.e., $U = \text{const}$, we get from Eq. (13.5c),

$$\dot{\alpha} - q = \frac{\rho V_\infty S}{2m} \left[C_{z_\alpha} \alpha + C_{z_{\dot{\alpha}}} \frac{\dot{\alpha} d}{V_\infty} \right] \quad (13.27)$$

where we have taken V_∞ as approximately equal to U . In replacing w by α to form Eq. (13.27), we made use of Eq. (13.2f). In using Eq. (13.4a), we let $S = \pi R_B^2$. In hypersonic flow we recognized that the flow lag terms $C_{z_{\dot{\alpha}}}$ (and later $C_{m_{\dot{\alpha}}}$) are negligible. Such terms indicate the ability of the flowfield to adjust to cross flow or angle-of-attack changes. The translational speed of the flow is so great relative to the speeds of angular-induced cross that the flowfield adjusts nearly instantaneously to changes in cross flow. Also we shall let $C_{z_\alpha} = -C_{N_\alpha}$ since C_{z_α} is nearly always negative. Thus, the preceding equation becomes

$$\dot{\alpha} - q = \frac{\rho V_\infty S}{2m} \left[-C_{N_\alpha} \alpha \right] \quad (13.28)$$

From Eqs. (13.7b) and (13.4e) we may get for the moment equation,

$$q = \frac{\rho V_\infty^2 S d}{I_y} \left[C_{m_\alpha} \alpha + \frac{q d}{V_\infty} C_{m_q} \right] \quad (13.29)$$

It will be noted that in the above equation we have neglected lag terms and, because the roll-rate p is zero, the Magnus terms are also zero. We can now differentiate Eq. (13.28) to get:

$$\dot{q} = \ddot{\alpha} + \left[\frac{\rho_\infty V_\infty S}{2m} \right] C_{N_\alpha} \dot{\alpha} + \frac{S}{2m} \frac{d}{dt} (\rho_\infty V_\infty) C_{N_\alpha} \alpha \quad (13.30)$$

It will be noted that the third term on the right allows for changes in free-stream conditions of both density and velocity. In some applications (such as in wind-tunnel testing) the product $\rho_\infty V_\infty$ can be regarded as constant.

For the general re-entry body trajectory $\rho_\infty V_\infty$ will undergo significant change. Both q and $\dot{q}$ may be removed from Eq. (13.29) to provide an equation in the variable α . After some manipulation we get

$$\begin{aligned} \alpha + \frac{\rho_\infty V_\infty}{2} \left[\left[\frac{S}{m} \right] C_{N_\alpha} - \left[\frac{S}{m} \right] \left[\frac{d}{K_y} \right]^2 C_{m_q} \right] \alpha \\ + \frac{\rho_\infty V_\infty}{2} \left\{ -V_\infty \left[\frac{S}{m} \right] \left[\frac{d}{K_y^2} \right] C_{m_\alpha} - \left[\frac{\rho_\infty V_\infty}{2} \right] \left[\frac{S}{m} \right]^2 \left[\frac{d}{K_y} \right]^2 C_{m_q} C_{N_\alpha} \right. \\ \left. + \frac{d}{dt} [\ln(\rho_\infty V_\infty)] \left[\frac{S}{m} \right] C_{N_\alpha} \right\} \alpha = 0 \end{aligned} \quad (13.31)$$

Note that we have used the radius of gyration, $K_y^2 = I_y / md$ as a substitution for the moment of inertia. Equation (13.31) is the general equation of planar pitching motion for which we may seek closed-form solutions. There may be some confusion concerning the last term in the bracketed coefficient of α , i.e., $d/dt [\ln(\rho_\infty V_\infty)]$. From Eq. (6.19) we may write

$$\rho \frac{dV}{dt} = -\frac{(\rho V)^2 g}{2\beta} + g \cong -\frac{(\rho V)^2 g}{2\beta} \quad (13.32)$$

where we have omitted the subscript to the velocity magnitude V . In Eq. (13.32) we first assume that V is the vertical velocity, although this can only be an approximation for a steep re-entry trajectory; secondly, we are neglecting the gravitational acceleration in comparison with the aerodynamic deceleration. Note that

$$\frac{d}{dt}(\rho V) = \rho \frac{dV}{dt} + V \frac{d\rho}{dt} = -\frac{(\rho V)^2 g}{2\beta} - V^2 \sin \gamma_E \frac{d\rho}{dZ}$$

but

$$\rho = \rho_0 e^{-Z/H}$$

so

$$\frac{d\rho}{dZ} = -\frac{\rho}{H}$$

Therefore we have:

$$\frac{d}{dt} \ln(\rho V) \equiv \frac{1}{\rho V} \frac{d}{dt}(\rho V) = \frac{-(\rho V)g}{2\beta} + \frac{V}{H} \sin \gamma_E \quad (13.33)$$

The above relationship may be substituted into Eq. (13.31) if desired (and if the assumption that $V \sin \gamma_F$ is the vertical velocity remains valid).

One simplification we might make in Eq. (13.31) is to keep $\rho V = \rho_\infty V_\infty = \text{const}$. This assumption might have validity for horizontal re-entry at high altitudes or over an altitude band where some sort of "average" values might be assigned to $\rho_\infty V_\infty$. As an aside we might consider this assumption of constant $\rho_\infty V_\infty$ over a significant portion of the re-entry trajectory. The reasoning is that while ρ_∞ increases with decreasing altitude, V_∞ decreases with decreasing altitude. Nevertheless, we might check on the constancy of $\rho_\infty V_\infty$ with altitude z by writing

$$\rho_\infty = \rho_0 e^{-Z/H} \quad (2.21) \quad (6.34)$$

$$V_\infty = V_E e^{-a_E e^{-Z/H}}$$

so that

$$\rho_\infty V_\infty = (\rho_0 V_E) \exp \left[- \left[\frac{Z}{H} + a_E e^{-Z/H} \right] \right]$$

taking the derivative of the above expression with respect to Z , we get:

$$\frac{d(\rho_\infty V_\infty)}{dz} = \rho_0 V_E \exp \left[- \left[\frac{Z}{H} + a_E e^{-Z/H} \right] \right] \frac{d}{dZ} \left[- \left[\frac{Z}{H} + a_E e^{-Z/H} \right] \right]$$

If we inquire as to the stationary value of $\rho_\infty V_\infty$ we might let

$$\frac{d}{dZ} \left[- \left[\frac{Z}{H} + a_E e^{-Z/H} \right] \right] = -\frac{1}{H} - a_E e^{-Z/H} \left[-\frac{1}{H} \right] = 0$$

or

$$Z = H \ln a_E$$

Clearly the rate of change of $\rho_\infty V_\infty$ with altitude is zero only at a certain altitude which depends upon a_E [See Eq. (6.33)] and therefore $\rho_\infty V_\infty$ cannot be constant over the entire trajectory.

It can be shown according to Eqs. (2.21) and (6.34) that the ratio of $\rho_\infty V_\infty$ at two different altitudes, (1) and (2), say, may be written as

$$\frac{(\rho_\infty V_\infty)_{(2)}}{(\rho_\infty V_\infty)_{(1)}} = \left\{ \exp \left[- \left[\frac{Z_2}{H} - \frac{Z_1}{H} \right] \right] \right\} \{ \exp(a_E [e^{-Z_1/H} - e^{-Z_2/H}]) \}$$

Let us take altitude (2) as the "final" altitude and altitude (1) as the "initial" altitude: for a numerical example we can take $Z_1 = 45$ km (147 k/ft) and $Z_2 = 15$ km (50 k/ft). For an RV with a ballistic factor β of 6×10^5 Pa, a_E

[from Eq. (6.33)] is about one-tenth. The ratio of $\rho_\infty V_\infty$ at altitude (2) to that at altitude (1) is about 90, or the quantity $\rho_\infty V_\infty$ increases by nearly two orders of magnitude for the particular RV taken as an example. Since a_E varies inversely with the ballistic factor β , or directly with $C_D S/W$, a high-drag, low-weight RV will result in a larger change in $\rho_\infty V_\infty$ over the same altitude range. Even though the assumption of constant $\rho_\infty V_\infty$ is made in the formulation of Eq. (13.34) for many RV's of interest such an assumption is questionable over an altitude range that is a significant portion of the re-entry trajectory.

In spite of the above discussion we shall assume constancy in $\rho_\infty V_\infty$ in the interest of analytic simplicity. In adapting Eq. (13.31) to our needs we shall also ignore the force derivatives as being of less importance than the moment derivatives. Thus Eq. (13.31) may be written as

$$\begin{aligned} \ddot{\alpha} + \frac{\rho_\infty V_\infty}{2} \left[- \left[\frac{S}{m} \right] \left[\frac{d}{K_y^2} \right]^2 C_{m_q} \right] \dot{\alpha} \\ + \frac{\rho_\infty V_\infty}{2} \left[- V_\infty \left[\frac{S}{m} \right] \left[\frac{d}{K_y} \right] C_{m_\alpha} \right] \alpha = 0 \end{aligned} \quad (13.34)$$

From elementary vibration analysis we can show that an oscillatory second-order system may have its differential equation written in terms of the damping ratio ζ_α and undamped natural frequency ω_{n_α} as

$$\ddot{\alpha} + 2\zeta_\alpha \omega_{n_\alpha} \dot{\alpha} + \omega_{n_\alpha}^2 \alpha = 0 \quad (13.35)$$

Comparing terms between Eqs. (10.39) and (10.40) we get:

$$\omega_{n_\alpha} = \sqrt{-\frac{\rho_\infty V_\infty^2}{2} \left[\frac{S}{m} \right] \left[\frac{d}{K_y^2} \right] C_{m_\alpha}} \quad (13.36)$$

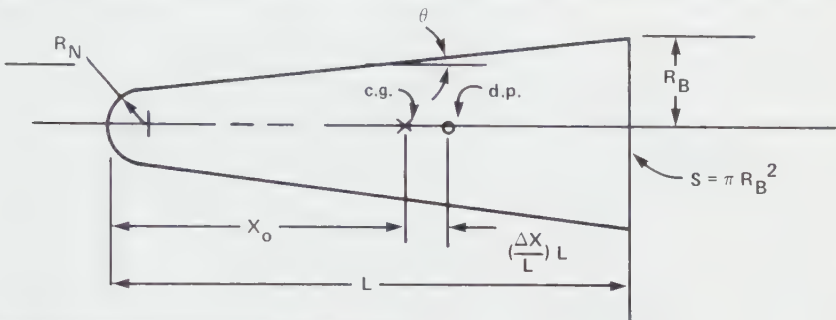
where C_{m_α} must be negative for ω_{n_α} to be real; and

$$\zeta_\alpha = \frac{1}{2\omega_{n_\alpha}} \left[\left[-\frac{S}{m} \right] \left[\frac{d}{K_y} \right]^2 C_{m_q} \right] \frac{\rho_\infty V_\infty}{2} \quad (13.37)$$

We shall now use some representative RV aerodynamic/metrology data to calculate the pitch frequency ω_{n_α} and pitch damping ratio ζ_α . Table (13.1) gives the needed information.

We may now use the data in Table 13.1 to calculate the pitch frequency ω_{n_α} and the pitch damping ratio ζ_α . We shall assume an altitude of 30 km

Table 13.1 Representative RV Characteristics



Nose radius, cm (in.)	1.98 (0.77)
Base radius, cm (in.)	22 (8.67)
Bluntness, R_N/R_B (—)	0.09
Reference length, $d = 2R_B$, cm (in.)	44 (17.34)
Length, L , m (ft)	1.52 (5)
Mass, kg (slugs)	92. (6.3)
Reference area, $S = \pi R_B^2$, m ² (ft ²)	0.152 (1.64)
Roll moment of inertia, I_x , kg/m ² (slug/ft ²)	0.972 (0.72)
Pitch moment of inertia, I_y , kg/m ² (slug/ft ²)	9.32 (6.88)
Yaw moment of inertia, I_z , kg/m ² (slug/ft ²)	9.32 (6.88)
c.g. from nose, X_0/L (—)	0.61
Static margin, $\Delta X_0/L$ (—)	0.071
Pitching moment derivative, C_{m_α} (—)	−0.520
Normal force derivative, C_{N_α} (—)	2.15
Pitch damping force, C_{N_q} (—)	−0.35
Pitch damping moment, $^q C_{m_q}$ (—)	−8.0
Drag coefficient, C_D (—)	0.10
Ballistic factor, β , Pa (lbs/ft ²)	6×10^4 (1254)
Pitch/yaw radius of gyration, $K_y = I_y/M$, m (ft)	0.318 (1.045)
Stability factor, P_s , m/kg (ft/slug)	3.73×10^{-3} (0.179)

(98.4 k/ft) and a velocity magnitude, V_∞ of 3000 m/s (9842 ft/s). The atmospheric density may be calculated from Eq. (2.21) as

$$\rho_\infty = (1.752)e^{-30/6.7} = 2.0 \times 10^{-2} \text{ kg/m}^3$$

Inserting the above value of density, airspeed $V_\infty = 3000$ m/s and the data from Table 13.1

$$\omega_{n_\alpha} = 18.3 \text{ rad/s}$$

or a cyclic frequency of about 2.91 cycles/s or requiring about 1 km per

oscillation. For a few oscillations we may regard $\rho_\infty V_\infty$ as essentially constant. The damping ratio ζ_α becomes from Eq. (13.37)

$$\zeta_\alpha = 0.0326$$

Thus at an altitude of 30 km the angular motion of the RV is lightly damped. We can see from Eq. (13.37) that the damping ratio ζ_α varies with $\rho_\infty^{1/2}$ and is independent of the velocity magnitude V_∞ . At an altitude of 6 km the density is about 0.71 kg/m³. The density ratio between 30 and 6 km is about 36 so the damping ratio will be about five and one half times that calculated at 30 km, or $\zeta_\alpha = 0.18$. It was pointed out in considering Eq. (13.34) that the coefficients are time varying because the term $\rho_\infty V_\infty$ is changing. For a re-entry body the density ρ_∞ will tend to increase and the velocity magnitude V_∞ will tend to decrease. Note in Eq. (13.33) if $\rho_\infty V_\infty$ is constant, then

$$\frac{\rho_\infty g}{2\beta} = \frac{1}{H} \sin \gamma_E \quad (13.38)$$

However since ρ_∞ is continually changing with either altitude or time and H , β , and g are constants, the condition of Eq. (13.38) cannot be true for all altitudes and hence as pointed out earlier there is no ballistic coefficient β which will result in a constant value of $\rho_\infty V_\infty$. If we assume that most of the change in $\rho_\infty V_\infty$ is due to density changes we may approximate V_∞ by either velocity magnitude at entry or at some other convenient point along the re-entry trajectory. Under this assumption we see that changes in $\rho_\infty V_\infty$ reflect the exponential nature of $\rho_\infty = \rho_\infty(z)$. To make the analysis more tractable we shall assume zero aerodynamic damping, i.e., $C_{m_q} = 0$. We justify ignoring aerodynamic damping on the grounds that the aerodynamic damping has been demonstrated to be very low. Thus Eq. (13.34) becomes

$$\ddot{\alpha} - [\rho_\infty V_\infty^2 C_{m_\alpha} S d / 2I_y] \alpha = 0 \quad (13.39)$$

We shall now define the static stability parameter P_s as

$$P_s = -(C_{m_\alpha} S d / I_y) \quad (13.40)$$

where P_s must always be positive for a statically stable (negative C_{m_α}) RV. As pointed out in Eq. (13.36) the pitch frequency ω_{n_α} may be written

$$2\pi f_\alpha = \omega_{n_\alpha} = (\frac{1}{2} P_s)^{1/2} V_\infty (\rho_\infty)^{1/2} = (q_\infty)^{1/2} (P_s)^{1/2} \quad (13.41)$$

With the anticipated goal of transforming Eq. (13.39) to a more tractable form, we introduce the variable ξ where

$$\xi = \frac{2\omega_{n_\alpha} H}{V_\infty \sin \gamma_E} = \left[\frac{2\rho_0 H^2 P_s}{\sin^2 \gamma_E} \right]^{1/2} e^{-Z/2H} \quad (13.42)$$

or

$$\xi = \tilde{\xi}_E e^{-Z/2H}$$

where

$$\tilde{\xi}_E = (2\rho_0 H^2 P_s / \sin^2 \gamma_E)^{1/2}$$

Z is the geometric altitude, and γ_E the flight-path angle at entry. We might note the resemblance between ξ_E and a_E , the trajectory constant [See Eq. (6.33)]. Equation (13.39) might be rewritten as

$$\ddot{\alpha} - [\rho_\infty V_\infty^2 C_{m_\alpha} S d / 2I_y] \alpha = 0$$

$$\ddot{\alpha} + [(\rho_0 V_E^2 P_s / 2) e^{-Z/H}] \alpha = 0 \quad \alpha + \xi^2 [(V_E \sin \gamma_E / 2H)]^2 \alpha = 0 \quad (13.43)$$

where we have manipulated the coefficient of α in Eq. (13.39) using Eq. (13.42) and by setting $V_\infty = V_E$. We can now transform α to $d^2\alpha/d\xi^2$ as follows:

$$\alpha \equiv \frac{d\alpha}{dt} = \frac{d\alpha}{d\xi} \frac{d\xi}{dz} \frac{dz}{dt} = \frac{d\alpha}{d\xi} \left[- \left(\frac{1}{2H} \xi_E e^{-Z/2H} \right) (-V_E \sin \gamma_E) \right]$$

$$\dot{\alpha} \equiv \frac{d\alpha}{dt} = \left[\left(\frac{V_E \sin \gamma_E}{2H} \right) \xi \right] \frac{d\alpha}{d\xi} \quad (13.44a)$$

and for the second derivative

$$\ddot{\alpha} \equiv \frac{d}{dt} \left(\frac{d\alpha}{dt} \right) = \frac{d}{d\xi} \left\{ \left[\left(\frac{V_E \sin \gamma_E}{2H} \right) \xi \right] \frac{d\alpha}{d\xi} \right\} \frac{d\xi}{dz} \frac{dz}{dt}$$

$$\ddot{\alpha} = \left[\left(\frac{V_E \sin \gamma_E}{2H} \right) \frac{d\alpha}{d\xi} + \left(\frac{V_E \sin \gamma_E}{2H} \right) \xi \frac{d^2\alpha}{d\xi^2} \right] \left(-\frac{\xi}{2H} \right) (-V_E \sin \gamma_E)$$

$$\ddot{\alpha} = \left[\frac{V_E \sin \gamma_E}{2H} \right]^2 \left[\xi \frac{d\alpha}{d\xi} + \xi^2 \frac{d^2\alpha}{d\xi^2} \right] \quad (13.44b)$$

Inserting Eq. (13.44b) into Eq. (13.43) we get:

$$\frac{d^2\alpha}{d\xi^2} + \frac{1}{\xi} \frac{d\alpha}{d\xi} + \alpha = 0 \quad (13.45)$$

As an aside, we may take (x, y) to be an independent-dependent set related through Bessel's equation as

$$\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + \left[1 - \frac{v^2}{x^2} \right] y = 0 \quad (13.46)$$

We note the similarity with Eq. (13.45) if $y \rightarrow \alpha$ and $x \rightarrow \xi$ and $v = 0$. Since most advanced calculus texts treat in some depth solutions to Eq. (13.46), it will be sufficient here merely to note the solution:

$$y = C_1 J_v(x) + C_2 J_{-v}(x)$$

if v is *not* an integer or zero, and

$$y = C_1 J_v(x) + C_2 Y_v(x)$$

if v is either an integer or zero. J_v is called a Bessel function of the first kind (of order v) and Y is called a Weber-Bessel function of the second kind (of order v). The reason for introducing the $Y_v(x)$ function is that $J_v(x)$ and $J_{-v}(x)$ are not independent when v is zero or an integer. Since $v = 0$ in Eq. (13.45) we have as its solution:

$$\alpha = C_1 J_0(\xi) + C_2 Y_0(\xi) \quad (13.47a)$$

With a little manipulation and the use of Eq. (13.42), Eq. (13.47a) may be written:

$$\dot{\alpha} = \omega_{n_\alpha} \left[\frac{d\alpha}{d\xi} \right]$$

or equivalently

$$\frac{d\xi}{dt} = \omega_{n_\alpha}$$

Using the above relationship we may write the time derivative of the angle-of-attack $\dot{\alpha}$ from Eq. (13.47a) as

$$\dot{\alpha} = \omega_{n_\alpha} [C_1 J'_0(\xi) + C_2 Y'_0(\xi)] \quad (13.47b)$$

where the prime indicates differentiation with respect to the variable ξ . If we take α_E and $\dot{\alpha}_E$ as initial conditions, we may solve Eqs. (13.47a) and (13.47b) for C_1 and C_2 as

$$C_1 = \frac{\alpha_E Y'_0(\xi_E) - (\dot{\alpha}/\omega_{n_\alpha})_E Y_0(\xi_E)}{J_0(\xi_E) Y'_0(\xi_E) - J'_0(\xi_E) Y_0(\xi_E)} \quad (13.48a)$$

$$C_2 = \frac{(\dot{\alpha}/\omega_{n_\alpha})_E J_0(\xi_E) - \alpha_E J'_0(\xi_E)}{J_0(\xi_E) Y'_0(\xi_E) - J'_0(\xi_E) Y_0(\xi_E)} \quad (13.48b)$$

where the subscript E refers to conditions at re-entry. Let us examine the magnitude of ξ_E . From Eq. (13.42) we have

$$\xi_E = (2\rho_0 H^2 P_s / \sin^2 \gamma_E)^{1/2} e^{-Z_E/2H}$$

From the data in Table 13.1 we calculate P_s as

$$P_s = -\frac{C_{m_\alpha} S d}{I_y} = -\frac{(-0.520)(0.152)(0.44)}{9.32} = 3.73 \times 10^{-3}$$

we can then calculate ξ_E as

$$\xi_E = \left[\frac{(2)(1.752)(6700)^2(3.73 \times 10^{-3})}{\sin^2 15} \right]^{1/2} e^{[-150 - (2)(6.7)]}$$

$$\xi_E = 0.0407$$

Since ξ_E is small compared with unity we may use the small-argument approximations to the Bessel functions $J_0(\xi)$ and $Y_0(\xi)$ and their derivatives, as³

$$J_0(\xi) = 1$$

$$Y_0(\xi) = (2/\pi) \ln(0.891 \xi_E)$$

$$J'_0(\xi) = (\xi_E/2)$$

$$Y'_0(\xi) = \frac{2}{\pi \xi_E} \left[1 + \frac{\xi_E^2}{2} \ln \left[\frac{1.123}{\xi_E} \right] \right] \quad (13.49)$$

The constants of integration may be written from Eqs. (13.48) as

$$C_1 = \alpha_E + \frac{2\alpha_E H}{V_\infty \sin \gamma_E} \ln \left[\frac{1.124}{\xi_E} \right] \quad C_2 = \frac{\pi \alpha_E H}{V_\infty \sin \gamma_E} \quad (13.50)$$

Now it is possible to see just what effect the exponential atmosphere has on the envelope of angle at attack. The simplest example we might take is to let $\alpha_E \neq 0$, $\dot{\alpha}_E = 0$. From Eqs. (13.47b) and (13.50) we get:

$$\alpha = \alpha_E J_0(\xi) \quad (13.51a)$$

$$\dot{\alpha} = \alpha_E \omega_{n_\alpha} J'_0(\xi) = -\alpha_E \omega_{n_\alpha} J_1(\xi) \quad (13.51b)$$

Note that we have made use of a property of the Bessel function of the first kind, viz†

$$-J_1(\xi) = J_0'(\xi) \triangleq dJ_0(\xi)/d\xi \quad (13.52)$$

The function $J_1(\xi)$ is a Bessel function of the first kind, order 1. When the order n is zero or a positive integer a Bessel function of the first kind, order n , can be calculated from the following series,

$$J_n(\xi) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s!(n+s)!} \left[\frac{\xi}{2} \right]^{n+2s} \quad (13.53)$$

The general definition of $Y_n(\xi)$, Bessel function of the second kind, order n , is too complicated to be given here; reference should be made to a standard text on advance calculus, e.g., see Ref. 4. Figure 13.4 indicates the variation in $J_n(\xi)$, $n = 0, 1, 2$ and $Y_n(\xi)$, $n = 0, 1$.

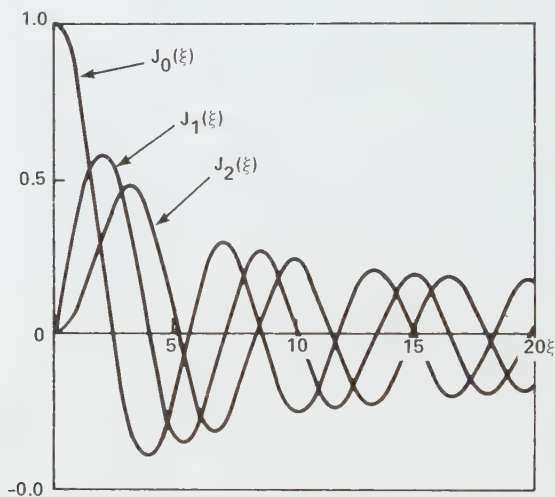


Fig. 13.4(a) Bessel functions of the first kind.

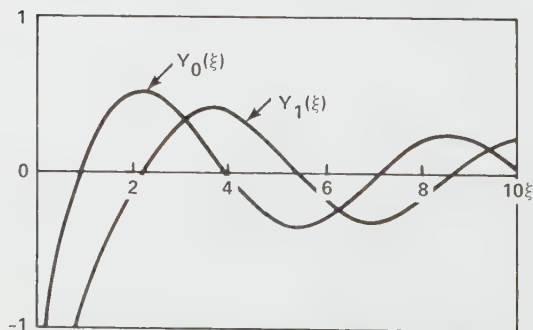


Fig. 13.4(b) Bessel functions of the second kind.

† The general expression is: $\xi J_n'(\xi) = nJ_n(\xi) - \xi J_{n+1}(\xi)$. In Eq. (13.51), $n = 0$, so the result noted in Eq. (13.52) follows.

It will be noted that $J_0(\xi)$ resembles a cosine except that the envelope to the amplitude decreases and the frequency increases with increasing value of the argument.

We shall now return to Eq. (13.51). Since we are interested in the amplitude of α at altitudes below 30 km we might look for approximations to the Bessel functions $J_0(\xi)$ and $J_1(\xi)$. First we note that ξ has the following values at $Z = 100, 60, 30, 15, 10, 5$ km (see Table 13.2). These values should be contrasted with those given earlier for $z = 150$ km (i.e., $\xi = 0.0407$). Whereas the small value of ξ at 150 km justified the small argument approximations of Eq. (13.49), the large values of ξ at low altitudes justifies replacement of Bessel functions by the asymptotic relationships:

$$J_0(\xi) = \left[\frac{2}{\pi\xi} \right]^{1/2} \cos \left[\xi - \pi/4 \right] \quad J_1(\xi) = \left[\frac{2}{\pi\xi} \right]^{1/2} \cos \left[\xi - \frac{3\pi}{4} \right] \quad (13.54)$$

The above relationship seems in accordance with the observation of the graphical presentation of $J_0(\xi)$ and $J_1(\xi)$ given earlier. Thus below 60 km we might rewrite Eqs. (13.49) as

$$\alpha = \alpha_E \left[\frac{2}{\pi\xi} \right]^{1/2} \cos(\xi - \pi/4) \quad \dot{\alpha} = \alpha_E \left[\frac{2}{\pi\xi} \right]^{1/2} \omega_{n_a} \cos \left[\xi - \frac{3\pi}{4} \right] \quad (13.55)$$

From the first of Eqs. (13.55) we note that the envelope of the angle-of-attack oscillations may be written as

$$\alpha_{\text{env}} = \alpha_E \left[\frac{2}{\pi\xi} \right]^{1/2} \quad \alpha_{\text{env}} = \alpha_E \left[\frac{2\sin^2\gamma_E}{\pi^2\rho_0 H^2 P_s} \right]^{1/4} e^{Z/4H} \quad (13.56)$$

It will be recalled that at the beginning of this section the aerodynamic damping was assumed zero, i.e., $C_{m_q} = 0$ [see Eq. (13.39)]. In spite of the absence of aerodynamic damping, it is interesting to note that the angle-of-attack envelope decreases as the altitude decreases and density increases. Subsequently we shall plot $\alpha_{\text{env}}/\alpha_E$ as a function of altitude.

Table 13.2 Variation of Trajectory Variable ξ with Altitude

z (km)	ξ
100	1.69
60	33
30	315
15	966
10	1403
5	2038

Going now to the second of Eqs. (13.55) we can consider the rate of change (rate of growth or decay) of the angle-of-attack envelope.

$$\alpha_{\text{env}} = \alpha_E \omega_{n_\alpha} \left[\frac{2 \sin^2 \gamma_E}{\pi^2 \rho_0 H^2 P_s} \right]^{1/4} e^{+Z/4H} \quad (13.57)$$

Transverse loads on an oscillating structure are proportional to angular acceleration. We can calculate the angular acceleration $\ddot{\alpha}$ from Eq. (13.39) as

$$\ddot{\alpha} = [\rho_\infty V_\infty^2 C_{m_\alpha} S d / 2 I_y] \alpha$$

and using Eq. (13.41) get

$$\ddot{\alpha} = -\omega_{n_\alpha}^2 \alpha$$

Then from Eq. (13.51) and the approximation of Eq. (13.55) we have:

$$\ddot{\alpha} = -\alpha_E \omega_{n_\alpha}^2 J_0(\xi) \quad \ddot{\alpha} = -\alpha_E \omega_{n_\alpha}^2 \left[\frac{2}{\pi \xi} \right]^{1/2} \cos(\xi - \pi/4) \quad (13.58)$$

The maximum acceleration or the acceleration of the angle-of-attack envelope is

$$\ddot{\alpha}_{\text{env}} = -\alpha_E \omega_{n_\alpha}^2 \left[\frac{2}{\pi \xi} \right]^{1/2} \quad \alpha_{\text{env}} = \alpha_E \omega_{n_\alpha}^2 \left[\frac{2 \sin^2 \gamma_E}{\pi^2 \rho_0 H^2 P_s} \right]^{1/4} e^{Z/4H} \quad (13.59)$$

Finally, we note that in Eqs. (13.57) and (13.58) the pitch frequency appears. The pitch frequency ω_{n_α} has been defined in Eq. (13.41), as

$$\omega_{n_\alpha} = (\frac{1}{2} P_s)^{1/2} V_\infty (\rho_\infty)^{1/2} = (\frac{1}{2} P_s)^{1/2} V_\infty \rho_0^{1/2} e^{-Z/2H}$$

The analysis that followed from Eq. (13.41) was based upon the assumption that the density varied exponentially with altitude, but it was assumed that the velocity magnitude remained constant, i.e., $V_\infty = V_k$. Equations (13.56-13.58) were of course derived on the assumption that ρ_∞ varied but V_∞ did not. During re-entry ρ_∞ changes several orders of magnitude while V_∞ changes by a factor of 3. Since we shall require ω_{n_α} subsequently when we discuss first and second resonance we can allow ω_{n_α} to reflect the variation in V_∞ . From Eq. (6.34)

$$V_\infty = V_E e^{-a_E e^{-Z/H}}$$

where

$$a_E = \rho_0 g H / 2 \beta \sin \gamma_E$$

The pitch frequency ω_α then becomes

$$\omega_{n_\alpha} = (1/2 P_s \rho_0)^{1/2} V_E \exp(-a_E e^{-Z/H}) e^{-Z/2H} \quad (13.60)$$

Equations (13.56-13.60) have been calculated for geometric altitude Z below 60 km. Remember none of the above equations, except Eq. (13.60), has much validity above 60 km because such relationships are based upon an asymptotic approximation to Eq. (13.54). A program for calculating the pitch frequency, the angle-of-attack envelope and its first and second derivatives are listed in Table 13.3. The output of this program is given in Table 13.4. All data were taken from Table (13.1) with the entry angle γ_E set at 75 deg.

The output of the above calculations gives $\omega_{n_\alpha} \rightarrow W$, $\alpha_{en}/\alpha_E \rightarrow A1$, $\dot{\alpha}_{en}/\alpha_E \rightarrow A2$, and $\ddot{\alpha}_{en}/\alpha_E \rightarrow A3$. Note that the pitch frequency changes rapidly as the altitude decreases and as the density increases. The increasing density also damps out the angle of attack. For example an RV entering the atmosphere at 60 deg. would have an angle of attack of only 3 deg. at 15 km. Interestingly enough the angle-of-attack rate and angle-of-attack acceleration (envelope) increase rapidly with decreasing altitude. Both $\dot{\alpha}_{en}/\alpha_E$ and

Table 13.3 Program to Calculate Envelopes of Angle-of-Attack and First and Second Derivatives

```

00100 REM THIS PROGRAM CALCULATES THE PITCH FREQUENCY, W, (RAD/SEC),
00110 REM THE ANGLE OF ATTACK ENVELOPE, A1, (RAD), THE FIRST
00120 REM DERIVATIVE OF THE ANGLE OF ATTACK ENVELOPE, A2, (RAD/SEC),
00130 REM THE SECOND DERIVATIVE OF THE ANGLE OF ATTACK ENVELOPE, A3, (RAD/SEC-SQ)
00140 REM ALL AS FUNCTIONS OF ALTITUDE, Z (KM). A1, A2, A3 ARE ALL NORMALIZED
00150 REM BY THE INITIAL ANGLE OF ATTACK. G1 IS THE INITIAL ANGLE AT ENTRY (DEG),
00160 REM B IS THE BALLISTIC FACTOR (PASCALS), P IS THE STABILITY FACTOR,
00170 REM V0 IS THE SPEED AT ENTRY AND D0, H ARE ATMOSPHERIC CONSTANTS.
00180 P = 3.73E-3
00190 V0 = 5000
00200 Z0 = 60
00210 B = 6E4
00220 G1 = 75
00230 G = 9.81
00240 H = 6700
00250 Z9 = 2
00260 D0 = 1.752
00270 Z0 = Z0*1000
00280 Z9 = Z9*1000
00290 G1 = G1*3.14159/180
00300 A = (D0*G*H)/(2*B*SIN(G1))
00310 FOR I = 1 TO 100
00320 Z = Z0 - (I-1)*Z9
00330 IF Z < 0 THEN 00470
00340 Z8 = EXP((-1)*Z/H)
00350 Z8 = (-1)*A*EXP((-1)*Z/H)
00360 W = SQR(P*D0/2)*V0*EXP(Z8)*EXP((-1)*Z/(2*H))
00370 A1 = ((SIN(G1)/(3.14159*H))*(.5))*((2/(D0*P))^(.25))
00380 A1 = A1*EXP(Z/(4*H))
00390 A2 = W*A1
00400 A3 = (W*2)*A1
00410 IF I > 1 THEN 00450
00420 PRINT #ALT.#, #PITCH FREQ.#, #A1#, #A2#, #A3#
00430 PRINT #KM.#, #RAD/SEC.#, # - W, #1/SEC#, #1/SEC-2#
00440 PRINT
00450 PRINT Z/1000, W, A1, A2, A3
00460 NEXT I
00470 END

```

Table 13.4 Angle-of-Attack Envelope and Derivatives vs Altitude
for $V_E = 6000$ m/s and $\beta = 6 \times 10^4$ Pa.

ALT. KM.	PITCH FREQ. RAD/SEC.	A1 —	A2 1/SEC	A3 1/SEC ²
60	3.24652	.265832	.863031	2.80185
58	3.76894	.246716	.929858	3.50458
56	4.31536	.228975	1.00185	4.38343
54	5.07923	.212509	1.07938	5.48244
52	5.89618	.197227	1.16289	6.8586
50	6.84427	.183045	1.25281	8.57455
48	7.9444	.169882	1.34961	10.7218
46	9.22072	.157666	1.45379	13.405
44	10.7011	.146328	1.56587	16.7565
42	12.4176	.135805	1.68638	20.9407
40	14.407	.12604	1.81585	26.1609
38	16.7113	.116976	1.95482	32.6676
36	19.3782	.108564	2.10378	40.7675
34	22.4614	.100757	2.26315	50.8337
32	26.0207	9.35119E-2	2.43324	63.3147
30	30.1213	8.67875E-2	2.61415	78.7417
28	34.8328	8.05465E-2	2.80567	97.7293
26	40.2264	7.47544E-2	3.0071	120.965
24	46.3696	6.93788E-2	3.21707	149.174
22	53.3186	6.43898E-2	3.43317	183.052
20	61.1043	5.97595E-2	3.65156	223.126
18	69.712	5.54622E-2	3.86638	269.533
16	79.0505	5.14739E-2	4.06904	321.66
14	88.9091	4.77724E-2	4.2474	377.633
12	98.8997	.044337	4.38492	433.668
10	108.389	4.11487E-2	4.46006	483.42
8	116.43	3.81897E-2	4.44644	517.7
6	121.734	3.54435E-2	4.31468	525.243
4	122.727	3.28947E-2	4.03707	495.457
2	117.801	3.05293E-2	3.59637	423.655
0	105.833	2.83339E-2	2.99865	317.355

$\ddot{\alpha}_{\text{env}}/\alpha_E$ have peaks, but these occur near impact for the β chosen ($\beta = 6 \times 10^4$ Pa). Figure 13.5 illustrates the altitude variations of pitch frequency ω_n , rate of change of the angle-of-attack envelope $\dot{\alpha}_{\text{env}}/\alpha_E$, for three values of the ballistic coefficient β . It will be noted that peak pitching rate occurs at the highest altitudes for smallest β . The reason is of course that low β means large velocity decrement and hence low dynamic pressure; a larger β on the other hand means that the RV reaches the lower altitude with more velocity magnitude. Angular accelerations are relatively small for the small values of β . However, increasing β both increases the magnitude of the angular acceleration and causes the maximum value of acceleration to occur at lower altitudes and greater dynamic pressure.

The significance of angle of attack is that large values will increase drag and move areas of high heat transfer to off-design positions. Large angles of attack as well as large angular rates could interfere with an endo-atmospheric sense of geographic position if the sense relied upon EM radiation. Large angular accelerations can cause linear accelerations, i.e., tangential accelerations, which in turn can produce bending loads on the RV. For example, referring to Table 13.4, $A3 \rightarrow \ddot{\alpha}_{\text{env}}/\alpha_E$, at altitude z equal to 10 km, equals $483.4/\text{s}^2$. For a point in the RV which is 0.5 m from the axis of rotation, the acceleration normal to the axis of symmetry is

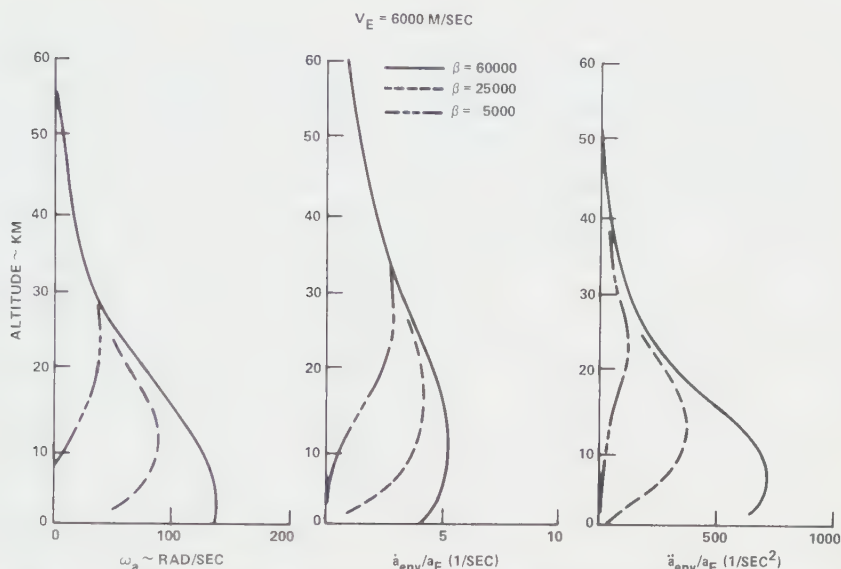


Fig. 13.5 Pitch frequency, angular velocity, and angular acceleration vs altitude.

$$n = (60)(483.4)(0.5)/(57.3)(9.81) = 25.7 \text{ g}$$

where we have assumed $\alpha_E = 60$ deg. Even though endo-maneuvering loads can be an order-of-magnitude larger, such maneuvering loads will cause little bending because they are distributed uniformly across the axis of symmetry. The load distribution due to angular acceleration and the first bending mode are shown in Fig. 13.6.

If the RV enters the atmosphere with some angular rate, i.e., $\dot{\alpha}_E \neq 0$ then there is the possibility of tumbling. if Eqs. (13.50) are inserted in Eqs. (13.47) we get:

$$\alpha = \alpha_E + \frac{2\alpha_E H}{V_\infty \sin \gamma_E} \ln \frac{1.124}{\xi_E} J_0(\xi) + \frac{\pi \dot{\alpha}_E}{V_\infty \sin \gamma_E} Y_0(\xi) \quad (13.61a)$$

$$\begin{aligned} \dot{\alpha} = \omega_{n_\alpha} \left\{ \left[\alpha_E + \frac{2\alpha_E H}{V_\infty \sin \gamma_E} \ln \left[\frac{1.124}{\xi_E} \right] \right] J'_0(\xi) \right. \\ \left. + \left[\frac{\pi \dot{\alpha}_E H}{V_\infty \sin \gamma_E} \right] Y'_0(\xi) \right\} \quad (13.61b) \end{aligned}$$

These equations have some utility in assessing tumbling motion. Even though the RV enters the atmosphere with angular motion, i.e., $\dot{\alpha}_E \neq 0$, we are interested in the altitude at which $\dot{\alpha} = 0$ the first and the second time. We shall assume that below the altitude for which $\dot{\alpha} = 0$ for the second time

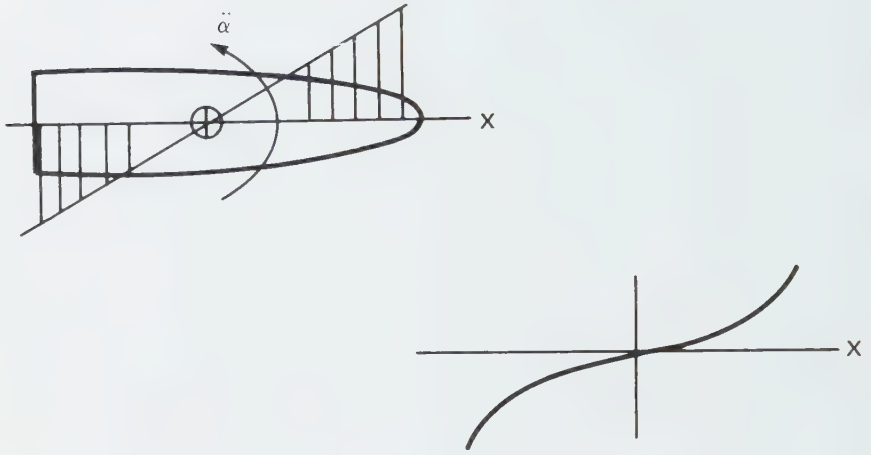


Fig. 13.6 Load distribution due to angular acceleration and the first bending mode.

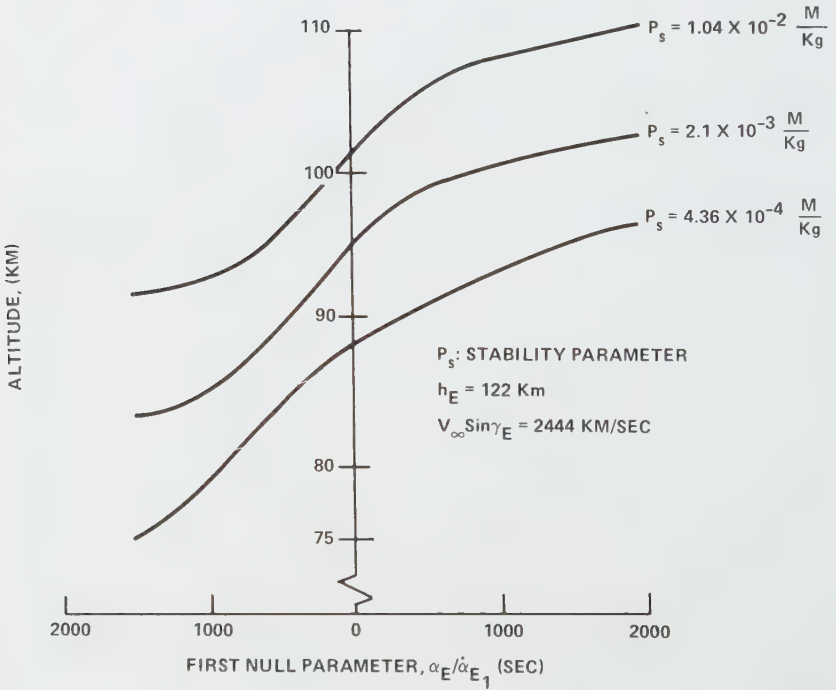


Fig. 13.7(a) Altitude vs the null parameter for first null.

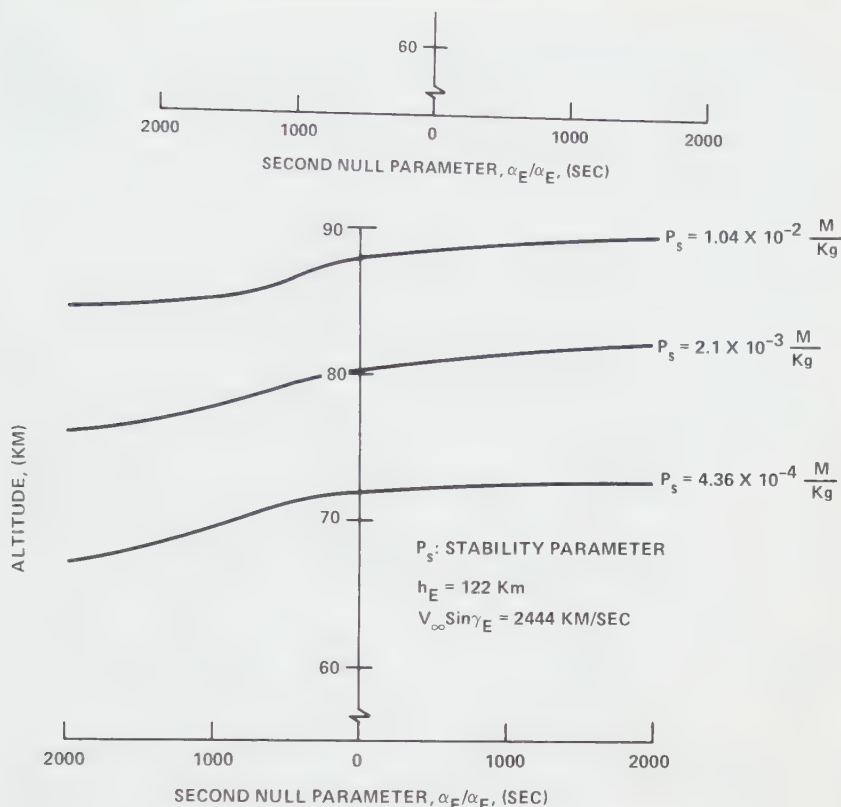


Fig. 13.7(b) Altitude vs the null parameter for second null.

all tumbling motion will have ceased and that the RV is undergoing oscillatory motion. We note that the argument of the Bessel functions J_0, J'_0, Y_0 , and Y'_0 is a function of altitude [see Eq. (13.42)]. Thus we may set $\dot{\alpha} = 0$ in the second of Eqs. (13.61) and get:[‡]

$$\frac{Y'_0(\xi_i)}{J'_0(\xi_i)} + \frac{V_\infty \sin \alpha_E}{\pi H} \frac{\alpha_E}{\dot{\alpha}_E} + \frac{2}{\pi} \ln \left[\frac{1.124}{\xi_E} \right] = 0 \quad (13.62)$$

The subscript i takes on the values (1,2) to indicate altitudes at which $\dot{\alpha}$ equals zero the first (1) and second (2) time. The altitudes of first and second are null. $z = z_i$ ($i = 1,2$) can first be estimated and then followed by an iterative procedure. This is a tedious effort and will not be pursued further here. However the results for certain specific values of the stability parameter P_s have been taken from an article authored by R.L. Norling (Ref. 5) for $V_\infty \sin \gamma_E = 2440$ km/s (8000 ft/s) and a pierce altitude of $z_E = 122$ km (400 k/ft).

[‡] We might note the following identities

$$\frac{dJ_0}{d\xi}(\xi) \equiv J'_0(\xi) \equiv -J_1(\xi) \quad \frac{dY_0}{d\xi}(\xi) \equiv Y'_0(\xi) = -Y_1(\xi)$$

Let us take an example to illustrate the use of Figs. 13.7. Suppose the RV enters the atmosphere at an angular rate of 0.06 deg./s and at an angle of attack of 60 deg. , i.e., $\dot{\alpha}_E = 0.60 \text{ deg./s}$ and $\alpha_E = 60 \text{ deg.}$ The null parameter $\dot{\alpha}_E / \alpha_E = 1000 \text{ s.}$ Assume that the stability parameter P_ϵ equals $1.04 \times 10^{-3} \text{ m/kg.}$ From Fig. 13.7 the altitude at which first null (first time $\dot{\alpha} = 0$) occurs is 108 km ; from Fig. 13.7 the altitude at which second null occurs is 88.5 km . Note that if α_E and $\dot{\alpha}_E$ have opposite signs then first null occurs at a much lower altitude than if both have the same sign. The null altitudes are shown in Fig. 13.8. Further consideration of RV tumbling may be found in Ref. 5.

Our next step might be to return to Eq. (10.38) and retain the normal force derivative C_{N^a} and the pitch damping derivative C_{mq} . Such analysis has been carried out but it is extremely lengthy. As we have noted the presence of an exponential atmosphere “damps” out the angular motion (although we may develop large angular velocities and accelerations). A nonzero value of C_{mq} damps out the angular motion more quickly. Reference may be made to original papers for treatment of this case. The seminal paper in this complicated dynamic problem is that due to Friedrich (Ref. 6).

13.4 Equations of Motion: Pitch, Yaw, and Roll in a Constant Density Atmosphere

In Chapter X we looked at the dynamics of an RV which had “slight” configurational asymmetry. The term “slight” was taken to mean that while we assumed identical aerodynamic characteristics in both yaw and pitch, we

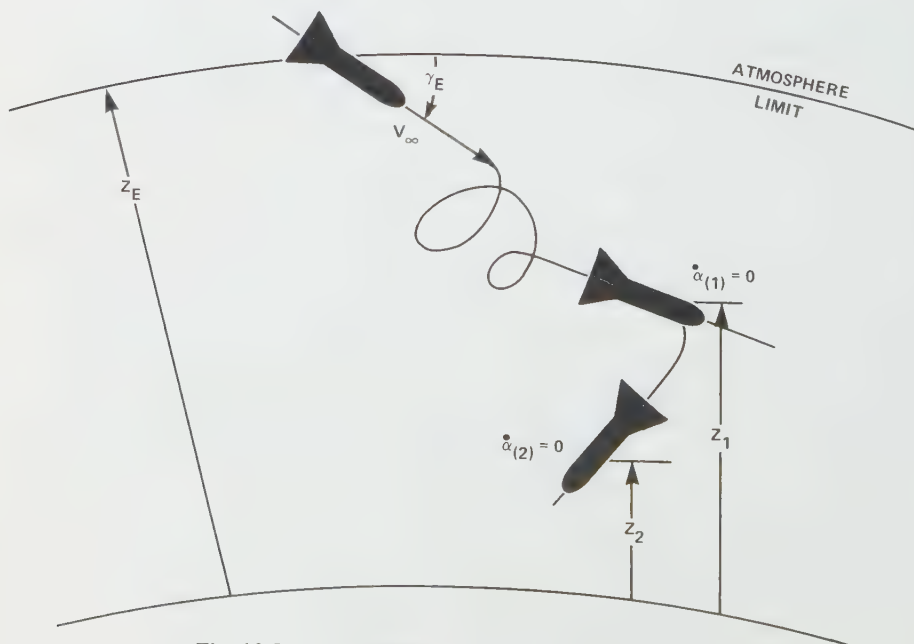


Fig. 13.8 Null altitudes Z_1 and Z_2 at which $\dot{\alpha} = 0$.

admitted the presence of a small body-fixed asymmetry which causes the RV to trim at a nonzero angle of attack. This asymmetry has been represented symbolically by M_δ . The terms M_δ and δ take on separate meaning when δ is associated with a control deflection, δ being the measureable magnitude of such deflection and the derivative M_δ being a measure of the effectiveness of such a control to generate a body moment. Body-fixed asymmetries may have their origins in effects other than control systems. For example there may be nonsymmetric removal of surface material due to ablation. In such cases we shall identify the moment with the product $M_\delta\delta$ and not give separate meaning to each of the terms. The use of the term "slight" implies that the presence of this asymmetry does not detectably alter the aerodynamic derivatives of the otherwise symmetric configuration.

In this section we shall briefly consider the dynamic stability of an RV which has measurably different aerodynamic characteristics in pitch and yaw. In other words the relationships of Eqs. (10.14) which are the conditions of rotational symmetry are no longer valid. We shall assume that the RV has some roll-rate p and confine our attention primarily to the resonance problem.

Earlier, in Chapter IX on aerodynamics, we emphasized certain major events along the trajectory of an RV. Two such events were designated as "first" and "second resonance." To develop an appreciation for the physics surrounding the resonance problem, we need to treat the pitch/yaw/roll motion at the same time. This means that even if we assume a steady-state value for roll-rate p , i.e., $\dot{p} = 0$, we shall have to deal with the Eqs. (13.7b) and (13.7c). For constant freestream density ρ_∞ and velocity magnitude V_∞ , solution of these equations is not difficult since the coefficients are constants. However, if the atmospheric density is varying exponentially with altitude and the velocity is changing significantly due to atmospheric drag, then the equations, while remaining linear, have time varying coefficients. Even if we ignore changes in velocity magnitude V_∞ , the exponential nature of the density ρ_∞ indicates a variable transformation from the time domain to the altitude domain. The resulting equation is of a form that has been identified with Bessel's equation; the solution is expressed in Bessel functions of imaginary order. Solving this equation is fairly straightforward, but it does require some familiarity with this type of Bessel function. Because of the length of this solution and the difficulty in appreciating the essential physics, a simpler approach will be taken here. For the more exact treatment, reference should be made to the original article by Leon (Ref. 7).

In spite of avoiding a confrontation of the moment equations with an exponential (density) atmosphere, it is possible to develop a feeling for the physics of interaction of spin rate and pitch restoring moment by assuming that average values of V_∞ and ρ_∞ prevail for some altitude band. In doing this we can shed some light on the physics of RV resonance. In Chapter X we did uncover a resonance problem for symmetric RV's. It was shown in Eq. (10.43) that when the roll rate is close to the nutational frequency there is an amplification of a trim angle of attack.

Let us assume that the roll-rate p is constant, i.e., $\dot{p} = 0$; this means that Eq. (13.7a) is identically zero. Equations (13.7b) and (13.7c) may then be written as

$$\begin{aligned}\dot{q} + \left[\frac{I_x - I_z}{I_y} \right] pr &= \frac{M}{I_y} = \left[\frac{M_\alpha}{I_y} \right] \alpha + \left[\frac{M_{\dot{\alpha}}}{I_y} \right] \dot{\alpha} + \left[\frac{M_q}{I_y} \right] q \\ \dot{r} + \left[\frac{I_y - I_x}{I_z} \right] qp &= \frac{N}{I_z} = \left[\frac{N_\beta}{I_z} \right] \beta + \left[\frac{N_{\dot{\beta}}}{I_z} \right] \dot{\beta} + \left[\frac{N_r}{I_z} \right] r\end{aligned}\quad (13.63)$$

where we have expanded the pitch and yaw moments, M and N in a Taylor series. In Chapter X we followed the same procedure except that we made the symmetry assumption of Eqs. (10.14). In this chapter we shall *not* assume identical aerodynamic characteristics in pitch and yaw, and thus shall consider the M and N derivatives as numerically distinct. We shall now examine the force equations.

The force equations were given in Eqs. (13.5). The first force equation, Eq. (13.5a), permits us to compute the component of the velocity vector along the body's axis of symmetry. If we assume a steep trajectory we can probably regard U and V (the velocity magnitude) as being roughly the same. The velocity magnitude for steep re-entry has been given in Eq. (6.34) or

$$V = V_E e^{(-a_E e^{-Z/H})} \quad (6.34)$$

where a_E is the trajectory constant given in Eq. (6.33). If we use the small angle approximation,

$$\begin{aligned}\alpha &= \sin^{-1} W/U \cong W/U & \dot{\alpha} &= \dot{W}/U \\ \beta &= \sin^{-1} V/U \cong V/U & \dot{\beta} &= \dot{V}/U\end{aligned}$$

Using the above approximations, Eqs. (13.5b) and (13.5c) become

$$\dot{\beta} + r - p\alpha = 0 \quad (13.64a)$$

$$\dot{\alpha} - q + p\beta = 0 \quad (13.64b)$$

We may rewrite the two moment and two force Eqs. (13.63) and (13.64) as

$$\dot{q} + Cpr = (M_\alpha/I_y)\alpha + (M_{\dot{\alpha}}/I_y)q \quad (13.65a)$$

$$\dot{r} + Dpr = (N_\beta/I_z)\beta + (N_{\dot{\beta}}/I_z)r \quad (13.65b)$$

$$r = -\beta + p\alpha \quad (13.65c)$$

$$q = \alpha + p\beta \quad (13.65d)$$

where we have omitted the $M_{\dot{\alpha}}$ and $N_{\dot{\beta}}$ derivatives as being of no conse-

quence in hypersonic flows. Also note that the moment of inertia ratios in Eq. (13.63) have been replaced by C and D as

$$C = \left[\frac{I_x - I_z}{I_y} \right] \quad D = \left[\frac{I_y - I_x}{I_z} \right] \quad (13.66)$$

Since $I_x < I_z < I_y$, C will always be negative and D will always be positive.

We may now use the third and fourth of Eqs. (13.65), and their derivatives, to eliminate (r, q) from the first two equations. After some manipulation we get

$$\ddot{\alpha} + p\dot{\beta} + Cp(-\dot{\beta} + p\alpha) = \left[\frac{M_\alpha}{I_y} \right] \alpha + \left[\frac{M_q}{I_y} \right] \alpha + \left[\frac{M_q}{I_y} \right] p\beta \quad (13.67a)$$

$$-\ddot{\beta} + p\dot{\alpha} + Dp(\dot{\alpha} + p\beta) = \left[\frac{N_\beta}{I_z} \right] \beta - \left[\frac{N_r}{I_z} \right] \beta + \left[\frac{N_r}{I_z} \right] p\alpha \quad (13.67b)$$

Clearly in Eqs. (13.67) we have two second-order coupled differential equations in the variables (α, β) . In Eq. (13.35) we indicated that we could represent the planar motion of a nonspinning RV in terms of the damping ratio ζ , and undamped natural frequency ω_n . It will be noted that Eqs. (13.67) are coupled only through spin rate p . This coupling however involves both inertial terms, i.e., Cp , Dp , and aerodynamic terms, i.e., $M_q p \beta$, $N_r p \alpha$. In both cases the coupling vanishes when the spin rate p vanishes. Thus, for a nonspinning RV we have two independent second-order differential equations;

$$\ddot{\alpha} - \left[\frac{M_q}{I_y} \right] \dot{\alpha} - \left[\frac{M_\alpha}{I_y} \right] \alpha = 0 \quad \ddot{\beta} - \left[\frac{N_r}{I_z} \right] \dot{\beta} + \left[\frac{N_\beta}{I_y} \right] \beta = 0 \quad (13.68)$$

From Eqs. (13.35) we may write:

$$\ddot{\alpha} + 2\zeta_\alpha \omega_{n_\alpha} \dot{\alpha} + \omega_{n_\alpha}^2 \alpha = 0 \quad \ddot{\beta} + 2\zeta_\beta \omega_{n_\beta} \dot{\beta} + \omega_{n_\beta}^2 \beta = 0 \quad (13.69)$$

where

$$\omega_{n_\alpha} = \sqrt{-\frac{M_\alpha}{I_y}} = \sqrt{-\frac{C_{m_\alpha} q S d}{I_y}}$$

$$\omega_{n_\beta} = \sqrt{-\frac{N_\beta}{I_z}} = \sqrt{-\frac{C_{n_\beta} q S d}{I_z}}$$

(Equation continued on next page.)

$$\zeta_\alpha = -\frac{M_z/I_y}{2\sqrt{-M_\alpha/I_y}} = -\frac{C_{m_q}}{2\sqrt{-C_{m_\alpha}}}(qSd^3)^{1/2}$$

$$\zeta_\beta = -\frac{N_r/I_z}{2\sqrt{N_\beta/I_z}} = -\frac{C_{n_r}}{2\sqrt{C_{n_\beta}}}(qSd^3)^{1/2} \quad (13.70)$$

Using the definitions of Eq. (13.69) we may write Eq. (13.67) in terms of the Laplace transform (with zero initial conditions) as

$$\begin{aligned} &[s^2 + 2\zeta_\alpha \omega_{n_\alpha} s + (Cp^2 + \omega_{n_\alpha}^2)]\alpha(s) \\ &+ [p(1-C)s + 2\zeta_\alpha \omega_{n_\alpha} p]\beta(s) = 0 \\ &[-p(1+D)s - 2\zeta_\beta \omega_{n_\beta} p]\alpha(s) \\ &+ [s^2 + 2\zeta_\beta \omega_{n_\beta} s + (\omega_{n_\beta}^2 - Dp^2)]\beta(s) = 0 \end{aligned} \quad (13.71)$$

where $\alpha(s)$ and $\beta(s)$ indicate transformed variables $\alpha(t)$ and $\beta(t)$. Equations (13.71) have stable solutions only if the roots of the characteristic equations are negative if real or have negative real parts if complex. The characteristic equation is formed from the determinant of Eqs. (13.71) as

$$\begin{vmatrix} [s^2 + 2\zeta_\alpha \omega_{n_\alpha} + (Cp^2 + \omega_{n_\alpha}^2)] & [p(1-C)s + 2\zeta_\alpha \omega_{n_\alpha} p] \\ [-p(1+D)s - 2\zeta_\beta \omega_{n_\beta} p] & [s^2 + 2\zeta_\beta \omega_{n_\beta} - Dp^2] \end{vmatrix} = 0$$

Expanding the above determinant we get the following quartic,

$$\mathbf{A}s^4 + \mathbf{B}s^3 + \mathbf{C}s^2 + \mathbf{D}s + \bar{\mathbf{E}} = 0 \quad (13.72)$$

where

$$\mathbf{A} = 1 \quad (13.73a)$$

$$\mathbf{B} = 2\zeta_\alpha \omega_{n_\alpha} + 2\zeta_\beta \omega_{n_\beta} \quad (13.73b)$$

$$\mathbf{C} = \omega_{n_\beta}^2 + \omega_{n_\alpha}^2 + 4\zeta_\alpha \zeta_\beta \omega_{n_\alpha} \omega_{n_\beta} + p(1-CD) \quad (13.73c)$$

$$\mathbf{D} = 2\zeta_\alpha \omega_{n_\alpha} \omega_\beta^2 + 2\zeta_\beta \omega_{n_\beta} \omega_{n_\alpha}^2 + p^2(2\zeta_\alpha \omega_{n_\alpha} + 2\zeta_\beta \omega_{n_\beta}) \quad (13.73d)$$

$$\mathbf{E} = p^2(\omega_{n_\beta}^2 C - \omega_{n_\alpha}^2 D + 4\zeta_\alpha \zeta_\beta \omega_{n_\alpha} \omega_{n_\beta} - p^2 DC) + \omega_{n_\alpha}^2 \omega_{n_\beta}^2 \quad (13.73e)$$

If we assume that the RV has both longitudinal and lateral oscillatory stability, i.e.,

$$0 \leq \zeta_{\alpha} < 1 \quad 0 \leq \zeta_{\beta} < 1 \quad \omega_{n_{\alpha}}^2 > 0 \quad \omega_{n_{\beta}}^2 > 0$$

then it can be shown that all the coefficients of Eq. (13.72) are positive with the possible exception of $\mathbf{E}$. Thus according to Routh's criterion there can be no root to the characteristic equation that is positive if real or with a positive real part if complex if $\mathbf{E} > 0$, and conversely there will be a real positive root or complex root pair with positive real parts if $\mathbf{E} < 0$. Since the roots of the characteristic equation appear as exponents in the time domain solution of Eqs. (13.67), a positive real root means exponential divergence and hence unstable motion of the RV. Note of course that E is always positive if the roll rate is zero, i.e., $p = 0$. i.e.,

$$\mathbf{E} = \omega_{n_{\alpha}}^2 \omega_{n_{\beta}}^2 \quad (13.74)$$

The culprit then is the roll-rate p and its inertial and aerodynamic coupling of the set of Eqs. (13.67). The boundary between stable and unstable motion, then, is found by setting $\mathbf{E} = 0$.

$$\mathbf{E} = (Cp^2 + \omega_{n_{\alpha}}^2)(\omega_{n_{\beta}}^2 - Dp^2) + 4\zeta_{\alpha}\zeta_{\beta}\omega_{n_{\alpha}}\omega_{n_{\beta}}p^2 = 0 \quad (13.75)$$

where we have factored the previously given expression for $\mathbf{E}$. The typical RV has moments of inertia such that

$$I_x \ll I_z, I_y \quad I_z \cong I_y \quad (13.76)$$

Thus from Eq. (13.66), C will always be negative and greater than but nearly equal to (-1) . D , on the other hand, will be always positive and slightly less than $(+1)$. An approximation to the stability criteria of Eq. (13.75) is to let $\zeta_{\alpha} = \zeta_{\beta} = 0$, e.g., ignore aerodynamic damping and let $C = -1$ and $D = +1$. Thus Eq. (13.75) becomes

$$\mathbf{E} = (\omega_{n_{\alpha}}^2 - p^2)(\omega_{n_{\beta}}^2 - p^2) = 0 \quad (13.77)$$

Clearly then we shall have unstable motion if:

$$\omega_{n_{\alpha}} > p \text{ and } \omega_{n_{\beta}} < p \quad (13.78a)$$

or

$$\omega_{n_{\alpha}} < p \text{ and } \omega_{n_{\beta}} > p \quad (13.78b)$$

In words, the RV will have unstable motion in the presence of spin if the undamped pitch frequency is greater than the roll rate *and* at the same time

the yaw frequency is less than the roll rate. We can also have unstable motion if the pitch frequency is less than the roll rate *and* the yaw frequency is greater than the roll rate. On the other hand we shall have stable motion if the yaw and pitch undamped natural frequencies ω_{n_β} and ω_{n_α} are *both* less than the spin rate or *both* greater than the spin rate. The RV will be unstable in the presence of spin if either, or both, the pitch and yaw frequencies are equal to the roll rate. For the RV having roll symmetry in both mass distribution and configuration, D and C will have the same magnitude ($I_y = I_z$) and $\omega_{n_\alpha} = \omega_{n_\beta}$. For this RV the motion will be unstable in the presence of roll rate if and only if

$$\omega_{n_\alpha} = D^{1/2}p \quad \omega_{n_\alpha} = 1 - [(I_x/I_y)]^{1/2}p \quad (13.79)$$

The above equation is exactly the same as that presented in Eq. (10.44) if we recognize [from Eq. (13.36)] that $\omega_{n_\alpha} = \sqrt{-M_\alpha/I_y}$. This is an important result for RV dynamics. This result indicates that whenever the pitch frequency satisfies the above relationship we might expect resonant conditions; that is, for any driving function whether time varying such as a periodic change in configuration due to control deflection or ablation or whether the driving function remains constant, such as a configurational offset causing a trim moment, we can expect the angle of attack caused by such a forcing function to be amplified. The degree of amplification will depend upon the amount of damping present and the magnitude of the static trim. The singularity will be avoided if any damping is present. Nevertheless, for the low damping ratios associated with RV's, we might expect considerable angle-of-attack amplification whenever the pitch or yaw frequency is in the vicinity of the roll rate. As we have seen in Table 13.4 the pitch frequency ω_{n_α} changes greatly over the re-entry trajectory.

It is possible to plot stability boundaries from Eq. (13.75) by dividing by p^4 to give:

$$\left[\left(\frac{\omega_{n_\alpha}}{p} \right)^2 + C \right] \left[\left(\frac{\omega_{n_\beta}}{p} \right)^2 - D \right] + 4\zeta_\alpha\zeta_\beta \left(\frac{\omega_{n_\alpha}}{p} \right) \left(\frac{\omega_{n_\beta}}{p} \right) = 0 \quad (13.80)$$

The graph in Fig. 13.9 is a plot of the stability boundaries for $\zeta_\alpha = \zeta_\beta = 0.1$ and 0.25 and data taken from Table 13.1. The stability boundaries will be noted for

$$\omega_{n_\alpha}/p = -C \quad \omega_{n_\beta}/p = D \quad (13.81)$$

Provided that there is a finite amount of damping, the two stable regions are connected. However, the damping is generally very low so any pitch/yaw frequency in the vicinity of the roll rate for any length of time is dangerous.

A simple computer program was presented in Table 13.3 to provide us with a rough calculation of the pitch frequency over some altitude band. This program has been exercised for two ballistic factors, $\beta = 6 \times 10^4$ Pa and $\beta = 6 \times 10^3$ Pa. In both cases the entry velocity magnitude V_E is 5 km/s, the

stability factor $P_c = 3.73 \times 10^{-3}$ m/kg, and the entry angle α_E is 75 deg. If we assume a constant roll rate of 18 rad/s, we show in Fig. 13.9 that there may be one or two resonance points. For a ballistic factor of 6×10^4 Pa there is one resonance point which occurs at about 34 km; for a ballistic factor one tenth as large, i.e., 6×10^3 Pa there are two resonance points, a high or first resonance which occurs, as before, at 37 km and a low or second resonance at about 11 km. These results are presented in Fig. 13.10. Resonance is of course defined as occurring when the roll-rate p and the nonrolling pitch frequency ω_{n_a} satisfy Eq. (13.79). Since I_x/I_y may be negligible in comparison with unity, resonance may be defined to occur when the pitch frequency and roll rate are equal. It should be clear that reducing the ballistic factor will increase the likelihood of two resonance points. According to Eq. (13.41), lowering the stability factor will lower the altitude of first resonance and raise the altitude of the second resonance. Such conclusions will have to be modified when there are significant variations in the roll rate over the trajectory.

Even if resonance does not cause drastic changes in the attitude of an RV, it can have an effect on the trajectory after resonance and ultimately on the location of the impact point. The reason is of course that aerodynamic loads

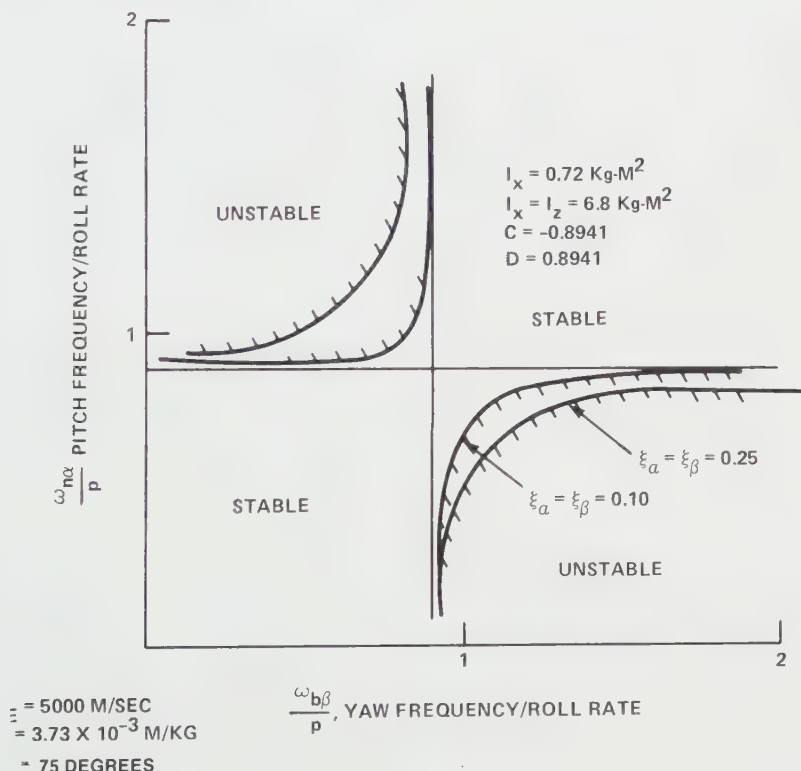


Fig. 13.9 Stability boundaries.

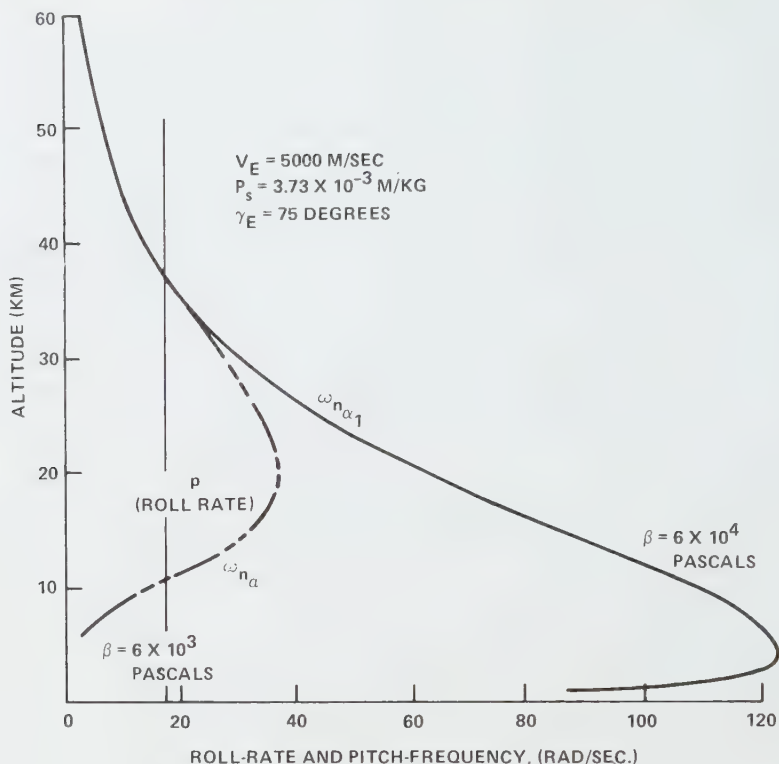


Fig. 13.10 Altitude vs roll rate and pitch frequency.

are generated normal to the local tangent to the trajectory, i.e., the velocity vector, by excursion of the RV's zero-lift-line from the velocity vector.

In Fig. (13.11) we have compared the relative effectiveness of normal loads at first and second resonance. At first resonance the normal loads (for a given angular excursion) are relatively small because of the lower atmospheric density, in comparison to an identical angle of attack at second resonance. However, at first resonance the remaining time of flight is longer giving a relatively small effect more time or distance over which to act.

If resonance is thought of given an angular "kink" to the trajectory, then the smaller change in trajectory slope at first resonance can be as significant as a larger change in trajectory slope at second resonance. As mentioned above, any assessment of a practical situation must account for variations in roll rate over the trajectory. In point of fact there are many phenomena contributing to roll torque, either tending to increase or decrease roll rate; it is unlikely that the roll rate will remain constant once the dynamic pressure increases. Some of the contributions to roll torque were discussed in Sec. 13.2. The physical mechanisms of other significant sources of roll torque are little understood.

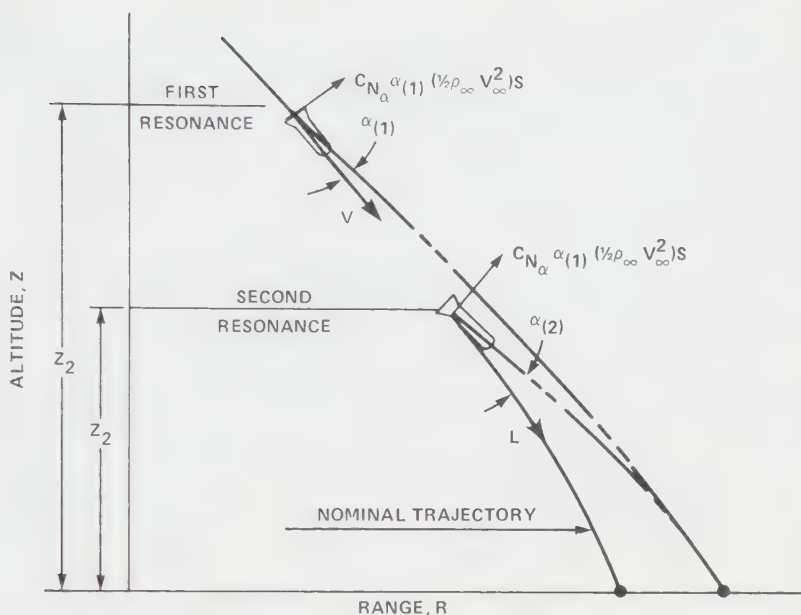


Fig. 13.11 Comparison between impact errors due to first and second resonance.

The preceding discussion has indicated that roll-rate level is a central consideration in exciting pitch/yaw resonance. A final consideration in this chapter is the effect of zero roll rate. Suppose that the RV has “spun down” near zero roll rate due to some unspecified roll torque. From Eq. (13.8b) we can regard the roll acceleration as a constant over the region of small roll rates bracketing the zone where the roll rate is zero. Let,

$$\frac{dp}{dt} \equiv \frac{d^2\phi}{dt^2} = p_0 \quad (13.82)$$

where ϕ is the roll angle and the 0 subscript indicates initial conditions or conditions just entering the zone around $p = 0$. Integrating Eq. (13.82) we get

$$\phi = \phi_0 + p_0 t + \dot{p}_0 t^2 / 2 \quad (13.83)$$

If it is further assumed that if the RV has a small trim angle of attack, there will be a continuous small aerodynamic force normal to the velocity vector. When the RV's roll rate is significantly different from zero, then the effect of this small normal force will be to cause the RV's flight path to be helical. The helix is centered, however, about the flight path that a zero trim RV would fly and with the diameter of the helix small, the trim force is made to be nearly self-cancelling. The situation is considerably different at zero and near zero roll rates. Equation (13.83) is plotted in Fig. 13.12(a). It is obvi-

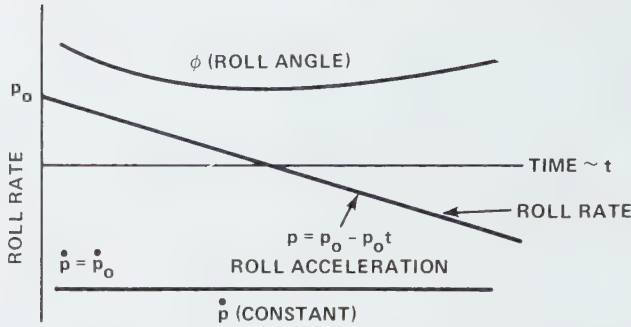


Fig. 13.12(a) Roll angle, roll rate, and roll acceleration in the vicinity of zero roll rate.

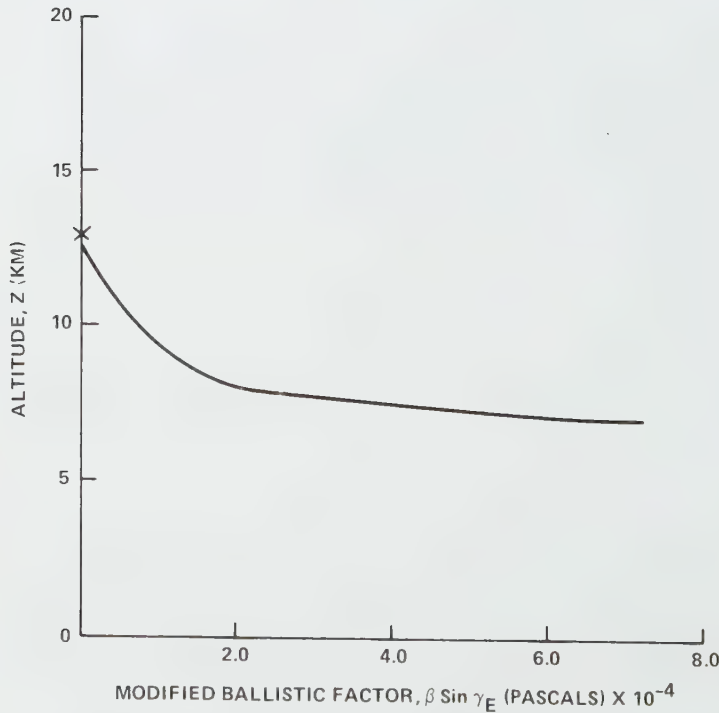


Fig. 13.12(b) Altitude of maximum dispersion vs modified ballistic factor.

ous that because of the parabolic nature of the roll angle function the RV will dwell some time at a nearly constant roll angle. The result is that the trim force, which was effectively "averaged out" at high roll rates, now has a fixed or nearly fixed inertial direction for the period when $p = d\phi/dt = 0$. The result is of course a "kink" in the trajectory which causes a sudden change in the direction of the velocity vector or equivalently a change in the flight-path angle. While the flight-path angle change might be "small" by some standards, if the condition of zero roll rate occurs at a high enough altitude the effect on dispersion at impact is significant. Of course if the zero roll-rate condition occurs shortly after pierce point, where the dynamic pressure is negligible, the aerodynamic sideforce would also be negligible. It would seem that there would be a "worse altitude" for zero roll rate to occur. From a report by Carlson and Louis of the Lockheed Missile and Space Company (Ref. 8), a plot is available giving altitude of maximum dispersion vs the modified ballistic factor $(W/SC_D)\sin\alpha_E$. This altitude at which maximum dispersion occurs lies between about 6 and 13 km, with the higher ballistic coefficient nearer the lower of the altitude limits.

References

- ¹Linnel, R. D. and Bailey, J. Z., "Similarity Rule Estimation Methods for Cones and Parabolic Noses," *Journal of Aerosciences*, No. 23, Aug. 1956, pp. 796-797.
- ²Glover, L., "Effects on Roll Rate of Mass and Aerodynamic Asymmetries for Ballistic Re-Entry Bodies," *Journal of Spacecraft and Rockets*, Vol. 2, March-April 1965.
- ³Jahnke, E. and Emde, F., *Tables of Functions*, Dover Publications, New York, 1945, p. 128-132.
- ⁴I. N. Sneddon, *Special Functions of Mathematical Physics and Chemistry*, University Mathematical Texts, Oliver and Boyd, Edinburgh and London, p. 106.
- ⁵Norling, R. A., "Altitude of Stabilization for Slowly Tumbling Re-entry Vehicles," *ARS Journal*, Dec. 1962, pp. 1867-1870.
- ⁶Friedrich, H. R., et al., "The Dynamic Motion of a Missile Descending Through the Atmosphere," *Journal of Aerospace Sciences*, Vol. 22, Sept. 1955, p. 628.
- ⁷Leon, H. I., "Angle of Attack Convergence of a Spinning Missile Descending Through the Atmosphere," *Journal of Aerospace Sciences*, Vol. 25, Aug. 1958, p. 480.
- ⁸Carlson, R. W. and Louis, C. A. III, "Introduction to Re-entry Flight Dynamics," LMSC Document 805029, Lockheed Missiles and Space Company, Sunnyvale, Calif., March 1968.

XIV. THE INVERSE METHOD

14.1 Introductory Comments

Most of the analysis associated with dynamics involves at least three steps: (a) modeling, (b) solution, and (c) comparison. During the modeling step the analyst represents in differential and algebraic equations what is perceived to be the essential physics associated with the problem. Also during this step the analyst must assign numerical values to the coefficients of the variables as they appear in the various equations. Then during the solution step, these equations must be solved for the state variables. The solution step involves processes of integration and/or matrix inversion, depending upon whether the equations are differential or algebraic. The third, or comparison step, establishes the validity of the modeling and solution efforts, by making comparisons with experimental results or perhaps with other more complete and well-established analytic models. Clearly, it is not feasible to make comparisons between the model and experimental results for each application since the whole point of the modeling effort is to avoid an extensive experimental program.

We might identify the above three-part procedure as the “direct” method of dynamics. In this section, however, we shall consider a somewhat inverse procedure: we shall assume that we have available a “solution” and set as our task, the determination of the numerical values of the coefficients of the equations that we presume adequately represents the phenomenon. In both the direct and inverse method (at least as understood here) we must establish or accept the form of the equations that represents the problem. However, in the direct method, the values of the coefficients were obtained either from measurements or from some other modeling work. In the inverse method we must obtain these coefficients starting with the analytic forms of equations and data which we accept as the solutions of these equations. Since the inverse method appears to be a fitting of data, we must establish some sort of criteria to judge “goodness of fit.” We shall consider these questions subsequently. It would seem that the first question should be the origin of this “solution.”

14.2 Typical Ground-Based Testing Procedures

The data or “solution” as we have been calling it, is a set of dependent variables and the corresponding value of the independent variable. Let us

take a specific example. In Fig. 14.1 we have depicted a wind-tunnel testing procedure for the measurement of the dynamic and static characteristic of an RV configuration. A model is supported so that it is only able to pivot or pitch about the transverse axis (y^b axis)*.

In Fig. 14.1 the attitude of the model relative to the oncoming flow may be defined as the pitch angle or the angle of attack. More precisely, this angle, designated as α , is the angle between some body-fixed line, say x^b , and the flow vector $\mathbf{V}$. We shall presume that there is some value of the angle α for which the aerodynamic moment is zero. For simplicity we shall take $\alpha = 0$ as the angle of attack corresponding to zero moment or a "null" condition. Now if the model is disturbed from null by an angular displacement, then moments will be generated to bring the model back to its null condition. This restoring or static moment will be assumed proportional to the angle α and will act in such a way as to decrease this off-null angle. When the model is released it will develop angular momentum as it returns

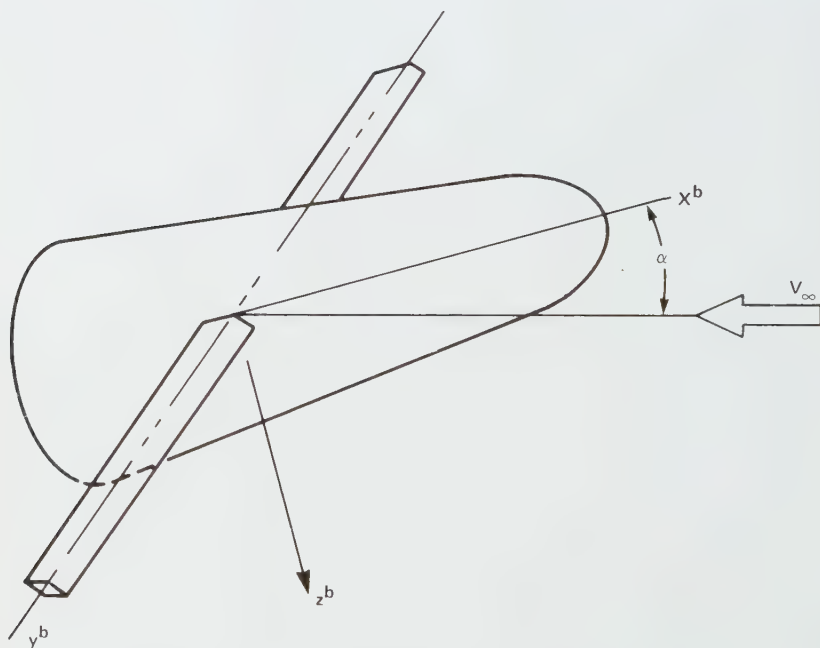


Fig. 14.1 Transverse supported pitch-damping model.

*In an actual test the model would not likely be supported by a transverse rod because of excessive flow interference between the support and the model. In most tests the support takes the form of an axial or "sting" structure. Also, the RV model would likely have some additional degrees of freedom, say, both pitch and yaw, rather than pitch alone. It is also likely that unlimited freedom would be available in roll as well.

to null. A dynamic moment, proportional to angular rate, will oppose angular motion. Because the aerodynamic damping moment is low, the model will overshoot the null condition, defined here as zero angle of attack, and develop an angle of attack in the opposite direction. This combined application of aerodynamic damping and static moments will result in damped oscillatory motion. It is this oscillatory motion that we shall regard as the solution of some differential equation.

For the simple mechanical oscillator illustrated in Fig. 14.1 the appropriate differential equation is easily written by equating the rate of change of angular momentum to the applied aerodynamic moment as

$$I_v \frac{dq}{dt} = I_v \frac{d^2\alpha}{dt^2} = \sum M = \left[\frac{\partial M}{\partial \alpha} \right] \alpha + \left[\frac{\partial M}{\partial q} \right] q \quad (14.1)$$

The above equation has been derived earlier in Eq. (13.29), where α is the angle of attack and q is the pitch rate, i.e., $q = d\theta/dt$. Since the center of gravity of the model is constrained, the force derivative is assumed zero; hence from Eq. (13.28) with the force derivative $C_N = 0$, we note that $d\alpha/dt = q$. Thus, Eq. (13.29) becomes identical to Eq. (14.1) and may be written in terms of the nondimensional derivatives C_{m_α} and C_{m_q} as

$$\frac{d^2\alpha}{dt^2} - \left[\frac{\rho V S d^2}{2I_y} C_{m_q} \right] \frac{d\alpha}{dt} - \left[\frac{\rho V^2 S d}{2I_y} \right] C_{m_\alpha} \alpha = 0 \quad (14.2)$$

Equation (14.2) is essentially Eq. (13.34) except for some differences in lumping parameters. It will be recalled that Eq. (13.34) was obtained from Eq. (13.31) by ignoring the force derivative C_N . As stated earlier we are assuming that the experimentally-obtained damped oscillatory motion is a solution of the above equation. Our goal then is to obtain the coefficients of the differential equation which have been indicated in Eq. (14.2) by the terms in the brackets. Actually it is the nondimensional derivatives that we are trying to measure since the other quantities such as S , d , and I_y will be available from metrology and ρ and V may be computed from wind-tunnel measurements. The method that we shall develop in this chapter will require that an analytic form for the solution of any proposed differential equation be available. It is this analytic solution that we shall fit to the observed "solution." We shall return to Eq. (14.2) after observing some generalities.

The precedent for the inverse problem is the familiar "least squares" technique. A single valued function, i.e., a polynomial, is assumed to relate the dependent and independent variables. In the simplest case of a straight line we set out to obtain two parameters from the fitting process. The "best fit" is obtained by minimizing the sum of the squares of distance between the line and each of the data points.

Unfortunately the least squares technique has limited value when fitting functions that are solutions to differential equations. The reason is that straight lines or even polynomials are not often the solutions of differential

equations that describe dynamic problems. For example, the solution to the linear oscillator of Eq. (14.2) is characterized by an exponentially damped sinusoid. However, we shall make use of an iterative process based upon least squares known as "differential corrections."

Let us assume that we are given data as ordered sequences of dependent variable independent variable pairs as

$$(y_1, x_1), (y_2, x_2), (y_3, x_3), \dots, (y_i, x_i) \dots (y_n, x_n) \quad (14.3)$$

where $x_1 < x_2 < x_3 \dots < x_i < x_n$. Clearly y_i might be replaced by the angle-of-attack α , and x_i by time t_i . In our illustrative problem we shall use these variables. However, to retain some generality in the present discussion, we shall use the symbol y_i to indicate the dependent variable and x_i the independent variable. Since we are assuming a solvable differential equation as a model for the physical phenomenon that produced the above sequence, we shall take the solution to this equation formally as

$$y = f(x, a, b, c) \quad (14.4)$$

where a, b, c , represent a number of undetermined constants. Using the technique to follow we shall determine these constants and then either directly or through algebraic equations find the coefficients of the differential equation for which Eq. (14.4) is the solution.

Let us return to Eq. (14.2). It is fairly straightforward to show that the solution to this equation may be written in terms of exponentially damped sinusoids as

$$\alpha = \left[\frac{\dot{\alpha}_0 - \alpha_0 \lambda}{\omega} \right] e^{\lambda t} \sin \omega t + \alpha_0 e^{\lambda t} \cos \omega t \quad (14.5)$$

where $(\alpha_0, \dot{\alpha}_0)$ are initial conditions in angle of attack and angle-of-attack rate. The constants ω and λ are the frequency of the oscillations and the damping exponents. Clearly we have four constants to determine in fitting Eq. (14.5), the analytic solution, to the experimental data or the experimental "solution" of Eq. (14.2). We may rewrite Eq. (14.5) in the form suggested by Eq. (14.4) as

$$y = ae^{bx} \sin cx + de^{bx} \cos cx \quad (14.6)$$

As mentioned earlier, we are chiefly interested in obtaining numerical values for the nondimensional stability derivatives C_{m_α} and C_{m_q} , although we must obtain the initial conditions as well. The nondimensional derivatives may be easily shown to be related to the b and c constants as

$$C_{m_\alpha} = -(2Iy / \rho V^2 S d)(c^2 + b^2) \quad (14.7a)$$

$$C_{m_q} = (4Iy / \rho V S d^2)b \quad (14.7b)$$

We may consider a second example of ground-based testing which provides us with an experimental "solution" to a differential equation. Assume

that we wish to measure the damping-in-roll derivative C_{l_p} . This derivative was discussed in Sec. 13.2. The experimental procedure might involve fixing the RV model to a sting support which permits unlimited freedom in roll. The model is then spun to some value of roll rate, i.e., p_0 , and then allowed to undergo roll decay or roll damping motion. A tachometer provides the experimentalist with a record of roll rate vs time. The appropriate differential equation was given in Eq. (13.10) and the appropriate solution in Eq. (13.11). The differential equation and solution are given below. We have assumed zero steady-state roll rate, i.e., $p_{ss} = 0$.

$$I_x \frac{dp}{dt} - C_{l_p} (\rho V S d^2 / 4) p = 0 \quad (14.8)$$

$$p = p_0 e^{kt} \quad k = C_{l_p} \rho V S d^2 / 4 I_x \quad (14.9)$$

It will be noted that we have two constants in the solution: a constant of integration p_0 , which is the initial roll rate and the constant k , which is the damping exponent. Again rewriting Eq. (14.9) in the form of Eq. (14.4) we have

$$y = a e^{bx} \quad (14.10)$$

Our purpose in running such a wind-tunnel test is to obtain the damping-in-roll derivative although we must also obtain the initial condition p_0 , which is essentially spurious information. Once the constant b has been obtained, it is a simple matter to obtain the damping-in-roll derivative C_{l_p} as

$$C_{l_p} = (4 I_x / \rho V S d^2) b \quad (14.11)$$

14.3 Method of Differential Corrections

Up to now we have not mentioned how formal solutions to differential equations, such as Eqs. (14.6) and (14.9) may be fitted to experimental data to compute the a , b , c , ... constants. In this section we shall consider a systematic method for doing this, based upon a variation of least squares called "differential corrections."

Let us return to Eq. (14.2) to plot the function $y = f(x; a, b, c)$, the data point y_i and the residual R_i , or the difference between the function y_i and data point y_i , at the same value of the independent variable x_i .

In Fig. 14.2 we have indicated the residual r_i ; clearly, for every data point $(x_i, \bar{y}_i)$ there will be a corresponding residual r_i . In other words there will be a one-to-one correspondence between the data points $(x_i, \bar{y}_i)$ and the residuals r_i , where i takes on the integer values from 1 to n .

Suppose that we fit the data by "eye-balling" an approximation. Mathematically this is equivalent to guessing at the constants a , b , c , ... , etc. If we were fitting a straight line to some fairly linear data then we would be essentially guessing at the two parameters of a straight line, i.e., the slope and the y -intercept. It is important to note in this case that we are *not* guess-

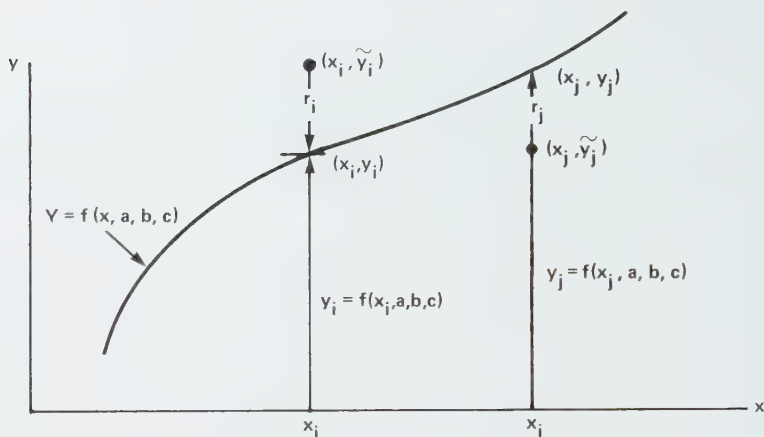


Fig. 14.2 Function and data points.

ing at the functional form, i.e., whether the fitting line is a polynomial, exponential, sinusoid, etc., but only guessing or estimating the constants associated with a particular functional form. Let us designate these initial estimates on $a, b, c, \dots$ by $a_0, b_0, c_0, \dots$.

The fundamental consideration here is to develop some systematic technique to improve these guesses. Let us designate the improvements to our initial estimates by $\alpha, \beta, \gamma, \dots$ or

$$a = a_0 + \alpha \quad b = b_0 + \beta \quad c = c_0 + \gamma \quad (14.12)$$

Now the residual r_i has been defined as the difference between the assumed function and the data point $\tilde{y}_i$, or

$$r_i = \tilde{y}_i - y_i \quad (14.13)$$

Thus we may rewrite Eq. (14.4) using Eq. (14.13).

$$r_i = \tilde{y}_i - y_i = \tilde{y}_i - f(x_i, a_0 + \alpha, b_0 + \beta, c_0 + \gamma) \quad (14.14)$$

The righthand side of the above equation may be expanded in a Taylor series, truncated after the first-order terms as

$$r_i = \tilde{y}_i - f(x_i, a_0, b_0, c_0) - \frac{\partial f}{\partial a} \bigg|_0 \alpha - \frac{\partial f}{\partial b} \bigg|_0 \beta - \frac{\partial f}{\partial c} \bigg|_0 \gamma \quad (14.15)$$

Note that $\partial f / \partial a \big|_0$ is the derivative of the function in Eqs. (14.4) with respect to the parameter a and evaluated at $x = x_i$ and for parametric values $a = a_0, b = b_0, c = c_0$, etc. We shall designate this derivative by the subscripted symbol

$$\frac{\partial f}{\partial a} \bigg|_0 \triangleq f_{ia} \quad (14.16a)$$

and in like manner for the other two derivatives

$$\left. \frac{\partial f}{\partial b} \right|_0 \triangleq f_{i\beta} \quad (14.16b)$$

$$\left. \frac{\partial f}{\partial c} \right|_0 \triangleq f_{i\gamma} \quad (14.16c)$$

Now let the residual from the first estimate be designated as R_i and defined as

$$\tilde{y}_i - f(x_i, a_0, b_0, c_0) = R_i \quad (14.17)$$

Equation (14.17) permits Eq. (14.15) to be rewritten as

$$r_i = -\alpha f_{i\alpha} - \beta f_{i\beta} - \gamma f_{i\gamma} + R_i \quad (14.18)$$

Equation (14.18) gives r_i as a linear expression in the variables α, β, γ . This gives us the opportunity to use the method of least squares. Thus we minimize the square of the residual sum or

$$\sum_{i=1}^{i=n} r_i^2 = g(\alpha, \beta, \gamma) \quad (14.19)$$

Clearly the square of the residuals r_i^2 follows from Eq. (14.18) as

$$\begin{aligned} r_i^2 = & \alpha^2 f_{i\alpha}^2 + \beta^2 f_{i\beta}^2 + \gamma^2 f_{i\gamma}^2 + 2\alpha\beta f_{i\alpha} f_{i\beta} + 2\alpha\gamma f_{i\alpha} f_{i\gamma} \\ & + 2\beta\gamma f_{i\beta} f_{i\gamma} - 2\alpha R_i f_{i\alpha} - 2\beta R_i f_{i\beta} - 2\gamma R_i f_{i\gamma} + R_i^2 \end{aligned} \quad (14.20a)$$

Next we shall take the derivative of the residual squared r_i^2 with respect to each of the parameters α, β, γ in turn, and then set these derivatives equal to zero.

$$\begin{aligned} \frac{\partial}{\partial \alpha} \left[\sum_i r_i^2 \right] &= 0 \\ \frac{\partial}{\partial \beta} \left[\sum_i r_i^2 \right] &= 0 \\ \frac{\partial}{\partial \gamma} \left[\sum_i r_i^2 \right] &= 0 \end{aligned} \quad (14.20b)$$

We now have three equations (the "normal equations") in three unknowns, α, β, γ . Solution of the three normal equations will provide us with the parameters a, b, c . Once we have numerical values for these three parameters we can, according to Eq. (14.12), improve our original estimates. Let us first write the normal equations in terms of the subscripted derivatives [Eqs. (14.16)].

$$\begin{aligned}
\left(\sum_i f_{i\alpha}^2 \right) \alpha + \left(\sum_i f_{i\alpha} f_{i\beta} \right) \beta + \left(\sum_i f_{i\alpha} f_{i\gamma} \right) \gamma &= \left(\sum_i f_{i\alpha} R_i \right) \\
\left(\sum_i f_{i\beta} f_{i\alpha} \right) \alpha + \left(\sum_i f_{i\beta}^2 \right) \beta + \left(\sum_i f_{i\beta} f_{i\gamma} \right) \gamma &= \left(\sum_i f_{i\beta} R_i \right) \\
\left(\sum_i f_{i\gamma} f_{i\alpha} \right) \alpha + \left(\sum_i f_{i\gamma} f_{i\beta} \right) \beta + \left(\sum_i f_{i\gamma}^2 \right) \gamma &= \left(\sum_i f_{i\gamma} R_i \right) \quad (14.21)
\end{aligned}$$

Since the derivatives, $f_{i\alpha}$, $f_{i\beta}$, $f_{i\gamma}$ are analytic functions of x_i , we can define these derivatives at each data point. [These derivatives must also be analytic functions of a , b , c in order for Eq. (14.15) to be acceptable.] With the derivatives defined at each data point x_i , the summations in Eqs. (14.21) produce numbers which are the coefficients in this algebraic equation. Solution of Eq. (14.21) provides us with the correction parameters α , β , γ ; these parameters are then algebraically added to the original (or previous) estimates, a_0 , b_0 , c_0 to update our estimates of these parameters [see Eq. (14.12)]. Obviously we have an iterative process on our hands and if everything converges nicely, we should arrive at the "best" estimates of our three parameters a , b , c . The "best" estimate, of course, is defined in the "least squares" sense. The iterative process terminated when some convergence criteria is satisfied or simply when a preselected number of iterations have been completed.

The above technique has been programmed to provide parameter estimation for a fictitious wind-tunnel pitch-damping test. Earlier, in Eq. (14.2) we showed that a single degree of freedom angular motion may be acceptably described by the following differential equation:

$$\frac{d^2 A}{dt^2} - \left[\frac{\rho V S d^2}{2 I_y} C_{m_q} \right] \frac{dA}{dt} - \left[\frac{\rho V^2 S d}{2 I_y} C_{m_\alpha} \right] A = 0 \quad (14.22)$$

where we have taken A to be the angle-of-attack α . The solution to this equation was given in Eq. (14.6). By using common trigonometric identities, we may rework this solution into the following four-parameter form:

$$A = B e^{Ct} \cos(Dt + E) \quad (14.23)$$

The parameters to be determined are clearly B , C , D , and E . B and E are set by initial conditions imposed on the angular motion, i.e., $A/t = 0$ and $\dot{A}/t = 0$. The parameters C and D contain the static stability derivative C_{m_α} and the pitch damping derivative C_{m_q} . These relationships are:

$$C = \left[\frac{\rho V S d^2}{4 I_y} \right] C_{m_q} \quad (14.24a)$$

$$D = \left[\left[-\frac{\rho V^2 S d}{2I_y} \right] C_{m_\alpha} \right]^{1/2} \left[1 - \left[\frac{\rho S d^2}{8I_y} \right] \left[\frac{C_{m_q}^2}{C_{m_\alpha}} \right] \right]^{1/2}$$

$$D \cong \left[\left[-\frac{\rho V^2 S d}{2I_y} \right] C_{m_\alpha} \right]^{1/2} \quad (14.24b)$$

where the approximation for D in Eq. (14.25) is justified because of the low damping associated with most wind-tunnel free-oscillation tests. Once the parameters C and D are obtained it is a simple matter to calculate C_{m_α} and C_{m_q} . The static and dynamic pitching moment derivatives are easily expressed in terms of C and D . After Eq. (14.7) we have

$$C_{m_\alpha} = -(2I_y / \rho V^2 S d)(D^2 + C^2) \quad (14.25a)$$

$$C_{m_q} = (4I_y / \rho V S d^2) C \quad (14.25b)$$

The normal equations for the four parameters B, C, D, E follow from a development which parallels that given earlier leading up to Eqs. (14.21).

$$\left(\sum_i F_{iB}^2 \right) B + \left(\sum_i F_{iB} F_{iC} \right) C + \left(\sum_i F_{iB} F_{iD} \right) D + \left(\sum_i F_{iB} F_{iE} \right) E = - \left(\sum_i F_{iB} R_i \right)$$

$$\left(\sum_i F_{iC} F_{iB} \right) B + \sum_i F_{iC}^2 C + \sum_i F_{iC} F_{iD} D + \left(\sum_i F_{iC} F_{iE} \right) E = \left(\sum_i F_{iB} R_i \right)$$

$$\left(\sum_i F_{iD} F_{iB} \right) B + \left(\sum_i F_{iD} F_{iC} \right) C + \left(\sum_i F_{iD}^2 \right) D + \left(\sum_i F_{iD} F_{iE} \right) E = \left(\sum_i F_i D R_i \right)$$

$$\left(\sum_i F_{iE} F_{iB} \right) B + \left(\sum_i F_{iE} F_{iC} \right) C + \left(\sum_i F_{iE} F_{iD} \right) D + \left(\sum_i F_{iE}^2 \right) E = \left(\sum_i F_{iE} R_i \right) \quad (14.26)$$

The derivatives used to form the coefficients in the above equations follow from Eq. (14.16) to give

$$\frac{\partial A}{\partial B} \equiv F_{iB} = e^{Ct} \cos(Dt + E) \quad (14.27a)$$

$$\frac{\partial A}{\partial C} \equiv F_{iC} = B t e^{Ct} \cos(Dt + E) \quad (14.27b)$$

$$\frac{\partial A}{\partial D} \equiv F_{iD} = (-1) t B e^{Ct} \sin(Dt + E) \quad (14.27c)$$

$$\frac{\partial A}{\partial E} \equiv F_{iE} = (-1) B e^{Ct} \sin(Dt + E) \quad (14.27d)$$

The preceding analysis involves four constants: B , C , D , E . It should be fairly straightforward to extend the technique to any arbitrary number of constants. In order to make use of matrix subroutines available in most computer languages, it is convenient to redefine the constants and coefficients in the normal equations in terms of subscripted variables. Thus

$$C(I) = \begin{bmatrix} C(1) \\ C(2) \\ C(3) \\ C(4) \end{bmatrix} = \begin{bmatrix} B \\ C \\ D \\ E \end{bmatrix} \quad (14.28a)$$

$$F(I, J) = \begin{bmatrix} \left[\sum_i F_{iB}^2 \right] & \left[\sum_i F_{iB} F_{iE} \right] \\ \left[\sum_i F_{iE} F_{iB} \right] & \left[\sum_i F_{iE}^2 \right] \end{bmatrix} \quad (14.28b)$$

$$R1(I) = \begin{bmatrix} \left[\sum_i F_{iB} R_i \right] \\ \left[\sum_i F_{iC} R_i \right] \\ \left[\sum_i F_{iD} R_i \right] \\ \left[\sum_i F_{iE} R_i \right] \end{bmatrix} \quad (14.28c)$$

The normal equations thus become:

$$F(I, J) C(J) = R1(I) \quad (14.29a)$$

Providing that $F(I, J)$ is nonsingular, we can formally solve for $C(I)$ and hence the four constants B , C , D , E as

$$C(I) = [F(I, J)]^{-1} R1(I) = \begin{bmatrix} B \\ C \\ D \\ E \end{bmatrix} \quad (14.29b)$$

14.4 Computer Program

The accompanying computer program illustrates the application of the inverse method based upon the technique just outlined. In order to have

some "data" available for reduction, Eq. (14.23) was programmed to give $A(t_i)$, i.e., the angle of attack at discrete intervals of time. Since we must specify the four parameters B, C, D, E in order to have numerical values for $A(t_i)$, we can check the data reduction by comparing the reduced values of B, C, D, E with the values originally used to compute the "data" $A(t_i)$. If the method is working at all we would expect perfect reproduction. Next, we shall include some "noise" in the form of a random number generator (RNG) to be superimposed on $A(t_i)$ [or $\mathbf{A}(I), \mathbf{T}(I)$]. The RNG selects an arbitrary number between 0 and 1. We shall multiply the output of the RNG by a constant $E9$, which will take on various powers of 10 such as $10^{-3}, 10^{-2}$, etc. By corrupting the values of $A(t_i)$ to various levels (or significant figures) we may obtain data with noise present. We can then evaluate the data reduction process to reproduce the original constants B, C, D, E in the presence of varying amounts of noise. In actual wind-tunnel tests there would be other, shall we say, spurious effects to contend with in addition to noise. Most significant of these is the departure of the physical situation from an adequate description by the linear differential equation of Eq. (14.22). If the effects due to nonlinear aerodynamics are significant then the simple linear description of the static aerodynamic moment in terms of $C_{m\alpha}$ would not be adequate. In other words fitting to the solution to Eq. (14.22) would not be appropriate.[†]

The computer program for generating "data" and then for reproducing the constants used in the generation of these data is given in Table 14.1.

In line 420 the number of data points per oscillation is set by the number N . Here we have taken N as 20; M is the number of oscillations to be covered so the total number of data points is $N \times M$, given as N again in line 320. The constants B, C, D, E are entered in lines 230-260. The time interval between data points, TI , is set in line 310. The "noise" multiplier $E9$ is set in line 370. This parameter will be printed in the output statement; we shall use values of $10^{-1}, 10^{-2}, 10^{-3}$. The angle-of-attack "data" is generated in lines 360 and 380. Both $\mathbf{A}(I)$ and $\mathbf{T}(I)$ are then available upon completion of the iterative loop, lines 350-460.

Estimates of the four constants B, C, D, E are made initially in lines 420-450. The four functions given in Eqs. (14.27) are defined in lines 490-520. The equivalent to the function $f(x_i; a_0, b_0, c_0)$ of Eq. (14.15), i.e., the initial guess on $\mathbf{A}(I)$ is given in line 570. In other words the angle-of-attack function is computed [for discrete values of time, $\mathbf{T}(I)$] based upon the initial estimates. As the iteration procedure continues, $\mathbf{A}0$ of line 570, will be the n -th estimate on $\mathbf{A}(I)$, based in turn on the n -th estimates on B, C, D, E , as the program begins on the $(n+1)$ -th iteration. The residual R_i is computed

[†] In data from an actual wind-tunnel test there would be some angular bias originating perhaps in flow angularity or drift in the null of the angular transducer. Thus Eq. (14.23) should have an additional additive constant, i.e., c . This five-parameter system has been sidestepped here in the interest of simplicity. If the program were to be used to reduce actual wind-tunnel measurements, i.e., if the data were available from some other external source, then the program would begin at line 420.


```
00100 REM THIS PROGRAM IS AN APPLICATION OF THE METHOD OF DIFFERENTIAL
00110 REM CORRECTIONS TO A SINGLE DEGREE OF FREEDOM WIND TUNNEL PITCH-
00120 REM DAMPING TEST. #DATA# FOR REDUCTION IS PROVIDED BY CALCULATING AN
00130 REM ANGLE OF ATTACK HISTORY,A(I); THE EQUATION FOR A(I) HAS FOUR PARA-
00140 REM METERS: B,C,D,E. N IS THE NUMBER OF DATA POINTS PER OSCILLATION
00150 REM AND M IS THE NUMBER OF PERIODS OF OSCILLATION. THE TOTAL NUMBER OF
00160 REM DATA POINTS IS M X N. A RANDOM NUMBER GENERATOR (RNG)(0-1)
00170 REM PROVIDES NOISE TO THE MEASUREMENTS A(I). E9 IS A MULTIPLIER OF THE RNG.
00180 REM COMPARISONS ARE MADE BETWEEN PARAMETERS B,C,D,F USED IN SETTING
00190 REM A(I) AND ESTIMATES B1,C1,D1,E1. VARIABLE K SETS NUMBER OF ITERATIONS.
00200 BASE 1
00210 N = 20
00220 M = 8
00230 B = 20
00240 C = (-1)*1.571
00250 D = 6.2e31
00260 E = 10
00270 Q = 3.14159/180
00280 B = B*Q
00290 E = E*Q
00300 T = 0
00310 T1 = (2*3.14159/D)/N
00320 N = N*M
00330 DIM A(500),T(500),A0(500),R(500),F(4,4),R1(4)
00340 DIM F1(4,4),C(4)
00350 FOR I = 1 TO N
00360 A(I) = B*EXP(C*T)*COS(D*T+E)
00370 E9 = .1
00380 A(I) = A(I)+RND(0)*E9
00390 T(I) = T
00400 T = T1+T
00410 NEXT I
00420 B0 = 18
00430 C0 = (-1)*1.0
00440 D0 = 4.5
00450 E0 = 5
00460 FOR K = 1 TO 10
00470 B0 = B0*Q
00480 E0 = E0*Q
00490 DEF FNB(T) = EXP(C0*T)*COS(D0*T+E0)
00500 DEF FNC(T) = B0*T*EXP(C0*T)*COS(D0*T+E0)
00510 DEF FND(T) = (-1)*T*B0*EXP(C0*T)*SIN(D0*T+E0)
00520 DEF FNE(T) = (-1)*B0*EXP(C0*T)*SIN(D0*T+E0)
00530 GOSUB 00550
00540 GOTO 00600
00550 FOR I = 1 TO N
00560 T = T(I)
00570 A0(I) = B0*EXP(C0*T)*COS(D0*T+E0)
00580 NEXT I
00590 RETURN
00600 MAT R = A0-A
00610 DATA 0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
00620 MAT READ F(4,4)
00630 RESTORE
00640 DATA 0,0,0,0
00650 MAT READ R1(4)
00660 RESTORE
00670 FOR I = 1 TO N
00680 H = T(I)
00690 F(1,1) = FNB(H)*FNB(H)+F(1,1)
00700 F(1,2) = FNB(H)*FNC(H)+F(1,2)
00710 F(1,3) = FNB(H)*FND(H)+F(1,3)
00720 F(1,4) = FNB(H)*FNE(H)+F(1,4)
00730 F(2,1) = FNC(H)*FNB(H)+F(2,1)
00740 F(2,2) = FNC(H)*FNC(H)+F(2,2)
00750 F(2,3) = FNC(H)*FND(H)+F(2,3)
00760 F(2,4) = FNC(H)*FNE(H)+F(2,4)
00770 F(3,1) = FND(H)*FNB(H)+F(3,1)
00780 F(3,2) = FND(H)*FNC(H)+F(3,2)
00790 F(3,3) = FND(H)*FND(H)+F(3,3)
00800 F(3,4) = FND(H)*FNE(H)+F(3,4)
00810 F(4,1) = FNE(H)*FNB(H)+F(4,1)
00820 F(4,2) = FNE(H)*FNC(H)+F(4,2)
00830 F(4,3) = FNE(H)*FND(H)+F(4,3)
00840 F(4,4) = FNE(H)*FNE(H)+F(4,4)
00850 R1(1) = (-1)*FNB(H)*R(I)+R1(1)
```

(Table continued on next page.)

Table 14.1(cont.) Pitch-Damping Data Reduction Program

```

00860 R1(2) = (-1)*FNC(H)*R(I)+R1(2)
00870 R1(3) = (-1)*FND(H)*R(I)+R1(3)
00880 R1(4) = (-1)*FNE(H)*R(I)+R1(4)
00890 NEXT I
00900 MAT F1 = INV(F)
00910 MAT C = F1*R1
00920 B1 = (B0+C(1))*(1/Q)
00930 C1 = (C0+C(2))
00940 D1 = (D0+C(3))
00950 E1 = (E0+C(4))*(1/Q)
00960 B0 = B1
00970 C0 = C1
00980 D0 = D1
00990 E0 = E1
01000 NEXT K
01010 PRINT #B##,C##,D##,E#
01020 PRINT B/Q,C,D,E/Q
01030 PRINT
01040 PRINT #B1##,C1##,D1##,E1#
01050 PRINT B1,C1,D1,E1
01060 PRINT
01070 PRINT #PARAMETER E9 =##E9
01080 END

```

Table 14.2 Comparisons Between Original and Estimated Parameters

	$E9$	B	C	D	E
Original	...	20.000	-1.571	6.283	10.000
Estimates	0.001	19.98	-1.572	6.289	9.76
	0.01	19.817	-1.587	6.346	7.573
	0.1	19.824	-1.879	6.947	-15.664

in line 600. The coefficients of the normal equation are computed in lines 690-880. The matrix inversion indicated in Eq. (14.29c) is carried out in line 900. The update on estimations of the four parameters is given in two steps in lines 920-990. The variable K is indexed in line 1000 and the process begins another iteration in line 460.

Obviously we may subject the above computer program to a number of trials. We could, for example, examine the effect of the number of data points, the number of oscillations, the presence of varying amounts of noise or even nonlinear or biased data. However, we shall confine our investigation to 8 oscillations with 20 data points per oscillation. We shall then vary the amount of "noise" by changing the multiplier $E9$ of the RNG. In Table 14.2 the results from various trials are given with the parameter $E9$ taking on values of 0 , 10^{-3} , 10^{-2} , 10^{-1} .

We see that with an uncertainty in angle of attack of something like one part in five, the C parameter has an uncertainty of about 20% and the D parameter an uncertainty of about 10%. According to Eqs. (14.24) and (14.25), the presence of uncertainties makes the dynamic derivative C_{mq} twice as uncertain as the static derivative C_{m_α} . We should not be too impressed with such relatively good parameter extraction from such poor data. In an actual experiment data with an uncertainty of one part in five would almost certainly be useless. Some reasons have been hinted at earlier why the parameter recovery is fairly good with "poor data." In an actual

test there will be some error associated with measuring the flow and metrology parameters required to remove C_{m_a} and C_{m_q} from the C and D parameters [See Eqs. (14.24) and (14.25)]. We might expect, though, that the accuracy in the measurement of V , d , I_y should be an order-of-magnitude greater than that of the measurement of a flow parameter or state variable such as the angle of attack. More significant reasons for less favorable recovery of parameters in an actual test will be the inappropriateness of the analytic model and the presence of biases from instruments. (As pointed out earlier, a constant bias might be removed by introducing a fifth constant.) The only point to be made here is that the method of differential corrections is worthy of consideration as a systematic method for reducing experimental data, which is adequately described by a linear differential equation with constant coefficients.

It must be noted that the random number is added to $\mathbf{A}(I)$ after it has been converted to radians (line 360). With B equal to 20 deg., the largest value taken by $\mathbf{A}(I)$ is about 0.34 rad. We should also note that only the parameters C and D are of any interest in reducing the data [see Eqs. (14.24) and (14.25)]. Therefore the process of differential corrections should be judged only on how well it recovers the C and D parameters.

14.5 Generalized Relationships

We can now generalize the preceding development. Let the variable y be the dependent variable, x the independent variable, and $\mathbf{a}$ the vector formed from the system parameters. The goal of the exercise is to assign numerical values to $\mathbf{a}$. We may now write:

$$y = f(x, \mathbf{a}) \quad (14.30)$$

An example of this function was given in Eq. (14.6) where

$$y = y \quad x = x \quad \mathbf{a} = [a, b, c, d]^T$$

As pointed out in Sec. 14.2 the dependent variable y does not always vary linearly with the parameter vector $\mathbf{a}$. In Eq. (14.6) we see that y varies linearly with a and d , but varies according to an exponential relationship with b and a transcendental relationship with c . Because of the nonlinear relationship between the dependent variable and some of the parameters, we expanded the function $f(x, \mathbf{a})$ in a Taylor series in the parameter vector truncating after the first two terms. We then resorted to an iterative procedure to continually improve the estimates of the parameters. Thus, Eq. (14.30) becomes

$$y = f(x, \mathbf{a}_0) + \left[\frac{\partial f}{\partial \mathbf{a}}(x, \mathbf{a}) \Big|_{\mathbf{a}=\mathbf{a}_0} \right]^T \Delta \mathbf{a} \quad (14.31)$$

Suppose we have available n data points and that we represent the i -th such point as:

$$(x_i, \tilde{y}_i)$$

The function given in Eq. (14.31) may be evaluated at the same value of the independent variable x_i to give:

$$y_i = f(x_i, \mathbf{a}_0) + \left[\frac{\partial f}{\partial \mathbf{a}}(x_i, \mathbf{a}) \Big|_{\mathbf{a}=\mathbf{a}_0} \right]^T \Delta \mathbf{a} \quad (14.32)$$

The derivative of a scalar function with respect to a vector is called the gradient and is the vector:

$$\frac{\partial f}{\partial \mathbf{a}} = \left[\frac{\partial f}{\partial a_1}, \frac{\partial f}{\partial a_2}, \dots, \frac{\partial f}{\partial a_m} \right]^T \quad (14.33)$$

where we are assuming m parameters. Since the system we are considering is overdetermined, the number of data points exceeds the number of parameters, i.e., $n > m$. Let us define a row vector as the transpose of the gradient vector of Eq. (14.33) or

$$\begin{aligned} H_i &= \left[\frac{\partial f}{\partial a_1}(x_i, \mathbf{a}_0), \dots, \frac{\partial f}{\partial a_m}(x_i, \mathbf{a}_0) \right]^T \\ &= \left[\frac{\partial f}{\partial \mathbf{a}} \right]^T \end{aligned} \quad (14.34)$$

Equation (14.31) now becomes

$$y_i = f(x_i, \mathbf{a}_0) + H_i \Delta \mathbf{a} \quad (14.35)$$

We should note that in the three parameter system given in Eq. (14.12), the incremental parameter vector $\Delta \mathbf{a}$ is

$$\Delta \mathbf{a} = [\alpha, \beta, \gamma]^T$$

We may now evaluate the residue r_i as the difference between the measured value of y , i.e., $\tilde{y}_i$ and the computed value of y at the same value of x_i , i.e., y_i . Thus

$$r_i = \tilde{y}_i - y_i = \tilde{y}_i - f(x_i, \mathbf{a}_0) - H_i \Delta \mathbf{a} \quad (14.36a)$$

If we let

$$R_i = \tilde{y}_i - f(x_i, \mathbf{a}_0) \quad (14.36b)$$

We get

$$r_i = R_i - H_i \Delta \mathbf{a} \quad (14.36c)$$

Now both r_i and R_i are single numbers. For the n data points, r_i and R_i may be regarded as elements in a column matrix (vector) or

$$\mathbf{r} = [r_1, r_2, \dots, r_i, \dots, r_n]^T$$

$$\mathbf{R} = [R_1, R_2, \dots, R_i, \dots, R_n]^T \quad (14.37a)$$

The row matrix $\mathbf{H}_i$ has the dimension $1 \times m$. For all n data points, $\mathbf{H}_i$ becomes the i -th row in the $n \times m$ $\mathbf{H}$ -matrix as

$$\mathbf{H} = \begin{bmatrix} H_1 \\ H_2 \\ \vdots \\ H_i \\ \vdots \\ H_n \end{bmatrix} \quad (14.37b)$$

Equation (14.36c) becomes

$$\mathbf{r} = \mathbf{R} - \mathbf{H}\Delta\mathbf{a} \quad (14.38)$$

The incremental parameter vector $\Delta\mathbf{a}$ will be assigned numerical values which minimize the sum of the square of the residuals. Taking J as the sum of the squares of the residuals we have

$$J = \sum_{i=1}^n r_i^2 = \mathbf{r}^T \mathbf{r} \quad (14.39)$$

Inserting Eq. (14.38) into Eq. (14.39) we have

$$J = \mathbf{r}^T \mathbf{r} = (\mathbf{R} - \mathbf{H}\Delta\mathbf{a})^T (\mathbf{R} - \mathbf{H}\Delta\mathbf{a}) \quad (14.40)$$

According to the least squares criterion we obtain the m equations to evaluate the m parameters. Thus

$$\frac{\partial J}{\partial \mathbf{a}} = 0 = \frac{\partial}{\partial \mathbf{a}} [(\mathbf{R} - \mathbf{H}\Delta\mathbf{a})^T (\mathbf{R} - \mathbf{H}\Delta\mathbf{a})] \quad (14.41)$$

In carrying out the above vector differentiation we shall make use of two derivative relationships for the vector inner product; these are:

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{z}^T \mathbf{x}) = \mathbf{z} \quad (14.42a)$$

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{z}) = \mathbf{z} \quad (14.42b)$$

We shall also make use of two matrix relationships involving the transpose of the sum (or difference) and the transpose of the product of two (conformable) matrices $\mathbf{A}$ and $\mathbf{B}$. These relationships are:

$$(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T \quad (14.42c)$$

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T \quad (14.42d)$$

Returning to Eq. (14.41) we have the following sequence:

$$\begin{aligned} \frac{\partial}{\partial \mathbf{a}}(J) &= \frac{\partial}{\partial \mathbf{a}}(\mathbf{r}^T \mathbf{r}) \\ &= \frac{\partial}{\partial \mathbf{a}}[(\mathbf{R} - \mathbf{H}\Delta \mathbf{a})^T (\mathbf{R} - \mathbf{H}\Delta \mathbf{a})] \\ &= \frac{\partial}{\partial \mathbf{a}}[(\mathbf{R}^T - (\mathbf{H}\Delta \mathbf{a})^T) (\mathbf{R} - \mathbf{H}\Delta \mathbf{a})] \\ &= \frac{\partial}{\partial \mathbf{a}}[(\mathbf{R}^T \mathbf{R} - \mathbf{R}^T \mathbf{H}\Delta \mathbf{a} - (\mathbf{H}\Delta \mathbf{a})^T \mathbf{R} + (\mathbf{H}\Delta \mathbf{a})^T (\mathbf{H}\Delta \mathbf{a}))] \\ &= \frac{\partial}{\partial \mathbf{a}}[\mathbf{R}^T \mathbf{R} - (\mathbf{H}^T \mathbf{R})^T \Delta \mathbf{a} - \Delta \mathbf{a}^T (\mathbf{R}^T \mathbf{H})^T \\ &\quad + (\mathbf{H}\Delta \mathbf{a})^T (\mathbf{H}\Delta \mathbf{a})] = 0 \end{aligned}$$

where we have made use of Eqs. (14.42c) and (14.42d). Now we shall use the gradient expressions of Eqs. (14.42a) and (14.42b).

$$\begin{aligned} \frac{\partial}{\partial \mathbf{a}}[\mathbf{r}^T \mathbf{r}] &= [\mathbf{0} - \mathbf{H}^T \mathbf{R} - (\mathbf{R}^T \mathbf{H})^T + \frac{\partial}{\partial \mathbf{a}}(\Delta \mathbf{a}^T \mathbf{H}^T \mathbf{H}\Delta \mathbf{a})] \\ &= -2\mathbf{H}^T \mathbf{R} + \mathbf{H}^T \mathbf{H}\Delta \mathbf{a} + (\Delta \mathbf{a}^T \mathbf{H}^T \mathbf{H})^T \\ &= -2\mathbf{H}^T \mathbf{R} + 2\mathbf{H}^T \mathbf{H}\Delta \mathbf{a} = 0 \end{aligned}$$

so solving for the incremental parameter vector $\Delta \mathbf{a}$ we get:

$$\Delta \mathbf{a} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R} = \mathbf{A}^\# \mathbf{R} \quad (14.43)$$

The matrix $\mathbf{H}$ of course cannot have an inverse since $n > m$. The matrix $\mathbf{A}^\#$, defined as,

$$\mathbf{A}^\# = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T$$

is sometimes called the pseudo-inverse for the overdetermined case. The dimensions of $\mathbf{A}^\#$ are $n \times m$, m being the number of parameters and n being the number of data points. We have ignored the important step of assuring ourselves that the values assigned to the incremental parameter vector $\Delta \mathbf{a}$ are those which minimize J . The necessary and sufficient condition for Eq. (14.41) to provide a minimum is that the Hessian of J be posi-

tive semidefinite, i.e.,

$$\frac{\partial^2 J}{\partial \mathbf{a}^2} \geq 0$$

The above relationship can be demonstrated. Equation (14.43) provides us with an improvement in our estimate of the parameter vector $\mathbf{a}$. Our initial estimate was $\mathbf{a}_0$; our next or new estimate is

$$\mathbf{a}_1 = \mathbf{a}_0 + \Delta \mathbf{a} \quad (14.44)$$

If the estimation process outlined above is carried out iteratively then $\mathbf{a}_0$ represents our current best estimate $\Delta \mathbf{a}$, the incremental improvement in this current estimate and $\mathbf{a}_1$, the new estimate. This iterative-type solution is particularly adapted to machine computation. This iterative procedure should cause the parameter estimation vector $\mathbf{a}$ to converge. Once the iterative procedure satisfies some convergence criterion, the process is considered complete and the most recent value of the parameter vector $\mathbf{a}$ is considered the best estimate.

We can next consider applying this iterative differential corrections procedure to systems described by a set of simultaneous equations. For example we might consider an extension of Eq. (14.30) to l simultaneous equations as

$$\begin{aligned} y_{(1)} &= f_{(1)}(\mathbf{x}, \mathbf{a}) \\ y_{(2)} &= f_{(2)}(\mathbf{x}, \mathbf{a}) \\ y_{(l)} &= f_{(l)}(\mathbf{x}, \mathbf{a}) \end{aligned} \quad (14.45)$$

Let us consider the situation where we have two simultaneous equations, i.e., $l = 2$. An example of two simultaneous representing solutions to two differential equations was given in Eqs. (10.47a) and (10.47b). In that case we may note the following relationships:

$$\begin{aligned} y_{(1)} &= \alpha \\ y_{(2)} &= \beta \\ x &= t \\ \mathbf{a} &= [A_1, B_1, A_2, B_2, A_3, B_3, \omega_1, \omega_2, \lambda_1, \lambda_2]^T \end{aligned} \quad (14.46)$$

If we let r_i and q_i be the residuals associated with $y_{(1)}$ and $y_{(2)}$, respectively, then the least square condition requires that

$$\frac{\partial}{\partial \mathbf{a}} [(\mathbf{r}^T \mathbf{r}) + (\mathbf{q}^T \mathbf{q})] = 0 \quad (14.47)$$

The elements in the residual matrices $\mathbf{R}$ and $\mathbf{Q}$ are a simple extension of Eqs. (14.36b):

$$R_i = \tilde{y}_{(1)i} - f_{(1)}(x_i, \mathbf{a}_0) \quad (14.48a)$$

$$Q_i = \hat{y}_{(2)i} - f_{(2)}(x_i, \mathbf{a}_0) \quad (14.48b)$$

and the $\mathbf{R}$ and $\mathbf{Q}$ matrices are of course

$$\mathbf{R} = [R_1, R_2, \dots, R_i, \dots, R_n]^T \quad (14.49a)$$

$$\mathbf{Q} = [Q_1, Q_2, \dots, Q_i, \dots, Q_n]^T \quad (14.49b)$$

The derivative matrices will have rows defined according to Eq. (14.43) as

$$\mathbf{H}_i = \frac{\partial f_{(1)}}{\partial \mathbf{a}_i}(x_i, \mathbf{a}_0), \dots, \frac{\partial f_{(1)}}{\partial \mathbf{a}_m}(x_i, \mathbf{a}_0) \quad (14.50a)$$

$$\mathbf{L}_i = \frac{\partial f_{(2)}}{\partial \mathbf{a}_i}(x_i, \mathbf{a}_0), \dots, \frac{\partial f_{(2)}}{\partial \mathbf{a}_m}(x_i, \mathbf{a}_0) \quad (14.50b)$$

The extension of the idea of two derivative matrices $\mathbf{H}$ and $\mathbf{L}$, rather than just one, i.e., $\mathbf{H}$, should follow from Eq. (14.37b) without difficulty. The equivalent to Eq. (14.38) now becomes

$$\mathbf{r} = \mathbf{R} - \mathbf{H}\Delta\mathbf{a} \quad (14.51a)$$

$$\mathbf{q} = \mathbf{Q} - \mathbf{L}\Delta\mathbf{a} \quad (14.51b)$$

Inserting Eq. (14.51) into Eq. (14.47) and carrying out the indicated differentiation gives the vector equation for the incremental parameter vector corresponding to Eq. (14.43).

$$\Delta\mathbf{a} = [(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T]\mathbf{R} + [(\mathbf{L}^T\mathbf{L})^{-1}\mathbf{L}^T]\mathbf{Q} \quad (14.52)$$

The extension of the above analysis to more than two dependent variables is straightforward.

APPENDIX A. FIRST-ORDER LINEAR DIFFERENTIAL EQUATIONS WITH VARIABLE COEFFICIENTS

A.1 Mathematical Background for Solving General First-Order Linear Differential Equations with Variable Coefficients

We shall consider here the following first-order equation

$$\frac{dy}{dx} + P(x)y = Q(x) \quad (\text{A.1})$$

The above equation is a linear differential equation of the first order, linear because y and dy/dx occur linearly and first order because the highest order of derivative is the first derivative dy/dx . If $Q(x)$ is zero then the equation is said to be homogeneous. The homogeneous equation has the following solution

$$y_h = Ae^{[-\int P(x)dx]} \quad (\text{A.2})$$

where A is a constant of the integration and $\int P(x)dx$ is an indefinite integral. Let us try to find a solution to the nonhomogeneous Eq. (A.1) of the form:

$$y = Z(x)e^{[-\int P(x)dx]} \quad (\text{A.3})$$

where $Z(x)$ is an integrating function. Alternately we may write

$$Z(x) = ye^{[\int P(x)dx]}$$

Differentiating we have:

$$\frac{dZ(x)}{dx} = \frac{dy}{dx}e^{[\int P(x)dx]} + yP(x)e^{[\int P(x)dx]}$$

but from Eq. (A.1) we have:

$$\begin{aligned} \frac{dZ}{dx} &= [Q(x) - P(x)y]e^{[\int P(x)dx]} + P(x)ye^{[\int P(x)dx]} \\ &= Q(x)e^{[\int P(x)dx]} \end{aligned}$$

Integrating we get

$$Z = \int Q(x) e^{[\int P(x) dx]} dx + C$$

Replacing the above expression in Eq. (A.3) we get

$$y(x) = e^{[\int P(x) dx]} \left[Q(x) e^{[\int P(x) dx]} dx + C \right] \quad (\text{A.4})$$

The above equation can be used to examine certain closed-form solutions to various equations of re-entry body motion, particularly Eq. (6.34).

A.2 Finding the Particular Integral of a Second-Order Differential Equation with Variable Coefficients

Given the following second-order differential equation with variable coefficients,

$$\frac{d^2 y}{dt^2} + P(x) \frac{dy}{dx} + Q(x)y = R(x) \quad (\text{A.5})$$

we are required to find the particular solution. The method of variation of parameters sometimes has great utility in finding the particular solution provided we have the homogeneous solution of the equation. Since Eq. (A.5) is second order, it will have two linearly independent homogeneous solutions. These solutions will be designated as $y_1(x)$ and $y_2(x)$. Thus the homogeneous solution may be written as

$$y_h = c_1 y_1(x) + c_2 y_2(x) \quad (\text{A.6})$$

where the constants c_1 and c_2 may be numerically established according to two conditions we may set upon y_h and its derivatives. The method of variation of parameters then states that the particular solution y_p may be written in terms of the two homogeneous solutions $y_1(x)$ and $y_2(x)$ and two, yet-to-be-determined functions $u_1(x)$ and $u_2(x)$ as

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x) \quad (\text{A.7})$$

The goal then is to find $u_1(x)$ and $u_2(x)$. Since $y_1(x)$ and $y_2(x)$ are assumed known, then with $u_1(x)$ and $u_2(x)$ available, we shall have the particular solution y_p according to Eq. (A.7).

Differentiating y_p in Eq. (A.7) gives

$$y_p' = u_1'(x)y_1(x) + u_1(x)y_1'(x) + u_2'(x)y_2(x) + u_2(x)y_2'(x) \quad (\text{A.8})$$

Since we have two conditions on y_p , we can use one of them by arbitrarily setting the following expression equal to zero as

$$u_1'(x)y_1(x) + u_2'(x)y_2(x) = 0 \quad (\text{A.9})$$

The second condition on y_p follows from the obvious requirement that y_p be a solution of the differential Eq. (A.5). Inserting y_p into this original equation gives

$$u_1[y_1'' + P(x)y_1' + Q(x)] + u_2[y_2'' + P(x)y_2' + Q(x)] + u_1'y_1' + u_2'y_2' = R(x) \quad (\text{A.10})$$

Obviously the bracketed terms must vanish since y_1 and y_2 are solutions of the homogeneous equation. We are then left with

$$u_1'y_1' + u_2'y_2' = R(x) \quad (\text{A.11})$$

Rewriting Eq. (A.9) we have:

$$u_1'y_1 + u_2'y_2 = 0 \quad (\text{A.12})$$

A simultaneous solution of Eqs. (A.11) and (A.12) gives

$$u_1 = - \int \frac{R(\tau) y_2^{dr}}{(y_1 y_2' - y_2 y_1')} d\tau \quad \text{and} \quad u_2 = \int \frac{R(\tau) y_1^{dr}}{(y_1 y_2' - y_2 y_1')} d\tau \quad (\text{A.13})$$

Inserting Eqs. (A.13) into (A.7) gives

$$y_p = \int^x \left[\frac{y_2(x) y_1(t) - y_1(x) y_2(t)}{y_1(\tau) y_2'(\tau) - y_1'(\tau) y_2(\tau)} R(\tau) \right] d\tau \quad (\text{A.14})$$

where τ is a dummy variable of integration. Provided that the Wronskian of the functions $y_1(x)$ and $y_2(x)$ does not vanish, Eq. (A.14) gives us a method of obtaining the particular solution of a second-order differential equation. The Wronskian is of course the denominator in the above expression and its nonvanishing is a condition of the linear independence of $y_1(x)$ and $y_2(x)$. The value of Eq. (A.14) in practice is limited owing to the difficulty in carrying out the required integration.

APPENDIX B. RING LASER GYROS AND PENDULOUS ACCELEROMETERS

B.1 Ring Laser Gyros

The gyro function in the earliest IMU's and continuing even down to the present day was and is supplied by various devices with spinning wheel rotors. When the case containing the wheel is rotated, a moment is developed at right angles to the angular momentum vector and the input angular rate vector. Measurement of the angular displacement caused by this moment provides a signal that can be equated to the input angular rate. So pervasive is the spinning rotor device that the word "gyro" has become almost synonymous with spinning rotor, although the word gyro merely means "ring." Today while the spinning rotor gyro continues to have applications, a promising device for measuring rotation rate of a frame is the ring laser gyro (RLG). In Figs. B.1 and B.2, adapted from Ref. 1, we see that the RLG consists of a closed path or ring which contains a dual beam of monochromatic, columnated light. A laser is the source of this light.

A beam splitter divides the light from the laser into a clockwise and a counterclockwise beam. Now let us assume that the case is rotating clockwise as shown. As a consequence of case rotation, the clockwise beam travels through a slightly longer path to return to the beam splitter than would have been the case had the gyro not been rotating. Conversely, for the beam traveling in a direction opposite to the direction of rotation, its apparent path is slightly shortened. The reason is of course that while the photons are traversing the path or ring, the position of the beam splitter (or any other point on the gyro case) has moved. Let us look at a section of the ring, represented in Fig. B.3 by an arbitrary closed path.

The ring encloses an area A whose elementary area ΔA is subtended by the angle $\delta\phi$. The time to traverse the elemental arc length $r\delta\phi$ is

$$\Delta T = r\delta\phi / C \quad (\text{B.1})$$

where C is the speed of light. Now assume that the ring is rotating clockwise about an axis normal to the ring at a rate of ω rad/s. For the beam traveling in the direction of rotation, the apparent elemental path length is increased by $r\delta\phi$, and for the beam traveling in the opposite direction the apparent path length is decreased also by a magnitude $r\delta\phi$. The angle $\delta\phi$ may be written in terms of ω and ΔT as

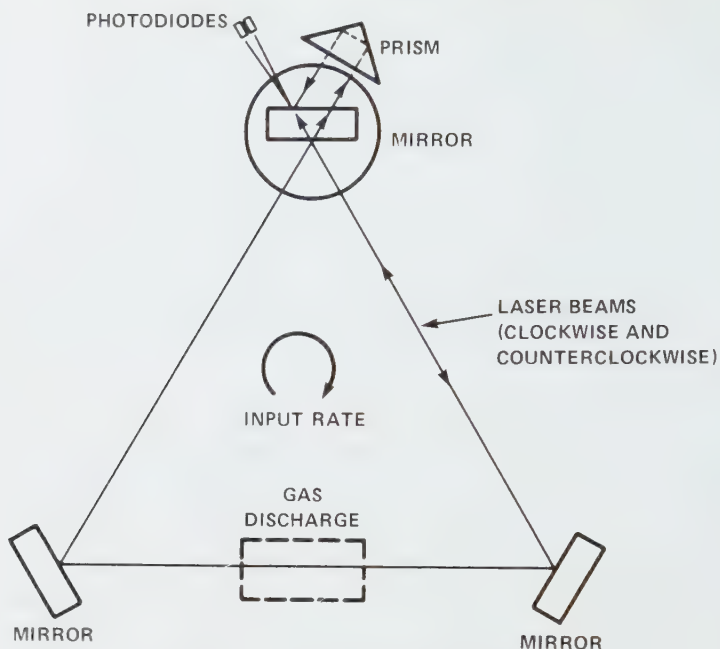


Fig. B.1 Laser operating elements

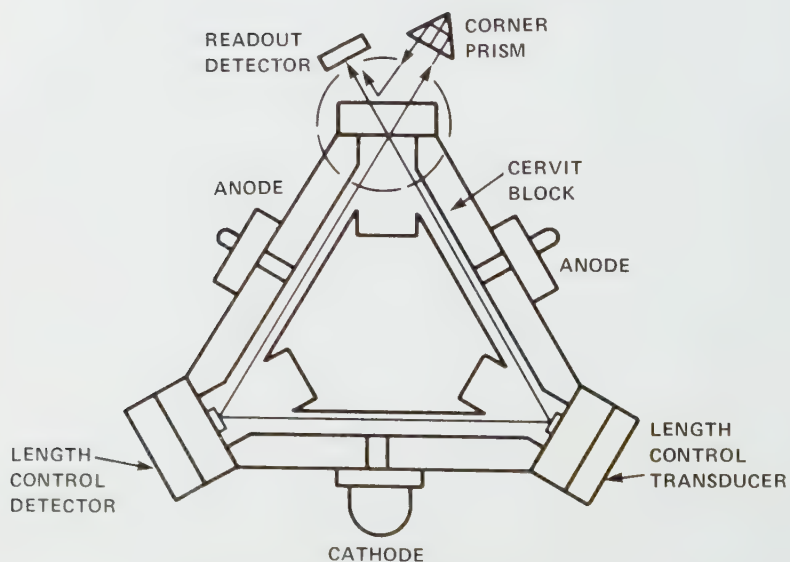


Fig. B.2 Ring laser gyro schematic.

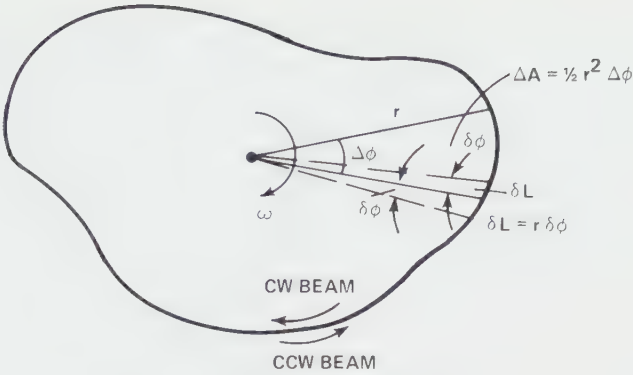


Fig. B.3 Arbitrary closed ring.

$$\delta\phi = \omega\Delta T = \omega r \Delta\phi / C$$

The apparent increase in arc length δL is

$$\delta L = r\delta\phi = r^2\omega\Delta\phi / C = 2\omega\Delta A / C \quad (\text{B.2})$$

The total change in path length of the total ring is ΔL and is given as

$$\Delta L = \int \delta L = \int r\delta\phi \approx \int 2\omega / C \Delta A = 2A\omega / C \quad (\text{B.3})$$

Let us see how much the apparent path length increases (or decreases) as a consequence of case rotation. Below are given the dimensions of the Honeywell GG1328AA gyro used as a strapdown instrument in the AMarV (advanced maneuvering re-entry vehicle) project. The GG1328AA gyro is an equilateral triangle with 0.07 m leg length.



$$l = 0.07 \text{ m}$$

$$A = (\sqrt{3}/4)l^2 = 2.12 \times 10^{-3} \text{ m}^2$$

$$C = 3 \times 10^8 \text{ m/s}$$

$$\omega = 2 \text{ rad/s}$$

From Eq. (B.3) we have for ΔL :

$$\Delta L = \frac{(2)(2.12)(10)^{-3}(2)}{(3) \times (10)^8} = \frac{2.83 \times 10^{11} \text{ m}}{0.283 \text{ A}}$$

It is clear that the measurement of a small distance is required to detect a typical body rate of 2 rad/s using a packageable ring of 0.07 m leg length.

The question is how to carry out such a measurement. Remember that time and frequency are two quantities that can be measured with precision. Assume that we have a triangular ring length L (each side has a length $L/3$) and that we are using a laser light source of wavelength λ . The path length and the wavelength are related as

$$\lambda = L/N$$

where N is the number of waves in the circuit. The change in wavelength $\Delta\lambda$ is therefore:

$$\Delta\lambda = \Delta L/N = 2A\omega/CN \quad (\text{B.4})$$

Now $\lambda f = C$, so

$$(\Delta\lambda)f + \Delta f(\lambda) = \Delta C = 0$$

because the speed of light is invariant with rotation rate. From Eq. (B.4) we get

$$\Delta f = -\frac{f}{\lambda} \Delta\lambda = -\left[\frac{f}{\lambda}\right] \left[\frac{2A\omega}{CN}\right] = -\frac{2A\omega f}{LC}$$

Since $f/C = 1/\lambda$ we have:

$$\Delta f = -(2A\omega/\lambda L) \quad (\text{B.5})$$

The physics underlying Eq. (B.5) should probably be given some emphasis. In arriving at Eq. (B.4) we are stating that for both the CW and CCW beams the number N of waves is constant; further, this constant does not change with rotation rate. Rotation rate *does* change the apparent path length ΔL [see Eq. (D.3)] and consequently the beam having motion in the direction of case rotation has a wavelength increase and the beam in the direction opposing case rotation has a wavelength decrease. Another important physical principle that deserves note is the invariance of the speed of light with beam direction.

Equation (B.5) is the basic relationship for a practical RLG. Again using numbers from the previous example we get

$$\frac{\Delta f}{f} = -\frac{2\omega A}{LC} = -\frac{(2)(2)(2.12 \times 10^{-3})}{(0.21)(3 \times 10^8)} = 1.346 \times 10^{-10}$$

Now let us compare the frequencies of the CW and CCW beams.

$$\delta f = f_{\text{CW}} - f_{\text{CCW}} = (f + \Delta f) - (f - \Delta f) = 2\Delta f = 4A\omega/\lambda L \quad (\text{B.6})$$

where we have defined the CW beam to give a positive frequency shift. The significance of Eqs. (B.6) is that we have dropped out the frequency and can

measure δf directly. Frequency is measured by forming an interference pattern from the CCW and CW beams. Roughly this is done by placing a partially silvered mirror at one of the vertices of the ring (triangular in this case).

In Fig. B.1 given earlier, the circled portion contains a partially silvered mirror which combines the clockwise and counterclockwise beams to form an interference pattern. If the two beams have different frequencies, i.e., the ring is rotating, the fringe pattern moves. The signals from the detectors are digitized so that in essence interference fringes pattern formed by two waves (which have slightly different frequencies) is counted. Since the frequency difference is assumed to be entirely due to ring rotation, then according to Eq. (B.6), we have a measure of rotation rate.

We shall now define $\Delta\phi$ as the phase change in the fringe pattern (cycles) and $\Delta\theta$ as the angle through which the case rotates about the gyro's input axis. The phase change in the fringe pattern may be written as

$$\Delta\phi = \delta f \Delta T = (4A / \lambda L) \omega \Delta T = (4A / \lambda L) \Delta\theta \quad (\text{B.7})$$

Equation (B.7) relates a change in fringe pattern phase to case rotation angle. Thus every time the passage of a fringe pattern has been detected, the case has rotated about the input axis through an angle $\Delta\theta$. Solving Eq. (B.7) for $\Delta\theta$ we get

$$\Delta\theta = (\lambda L / 4A) \Delta\phi \quad (\text{B.8})$$

The quantity $(\lambda L / 4A) \Delta\phi$ is called the *scale factor*. The units are arcs/pulse. The logic circuit from the detectors will produce a pulse every time a fringe pattern passes. Let us calculate a typical magnitude for the scale factor (*SF*) using data from the Honeywell GG1328AA gyro.

$$L = 0.21 \text{ m} \quad A = 2.12 \times 10^{-3} \text{ m}^2 \quad \lambda = 6.328 \times 10^{-7} \text{ m} \quad \Delta\phi = 1/4$$

$$SF = \left[\frac{\lambda L}{4A} \right] \Delta\phi = \frac{(6.328 \times 10^{-7})(0.21)}{(4)(2.12 \times 10^{-3})(4)} = \frac{3.92 \times 10^{-6} \text{ rad/count}}{3.92 \mu\text{-rad/count}}$$

In the actual GG1328AA gyro the scale factor is more like $15 \mu\text{-rad/count}$ or 3.2 arc-s/count . In the above calculations we have assumed a quarter wave separation between detectors, i.e., $\Delta\phi = 1/4$.

According to the calculations we get a pulse every time the case rotates through an angle of $2.6 \times 10^{-6} \text{ rad}$ (0.8 arc-s). Alternately if the case is rotating about the input axis (normal to the ring) at a rate of rad/s we get $1/SF$ or $2.55 \times 10^5 \text{ pulses/s}$.

The RLG has certain advantages as a strapdown rate measuring instrument. Among these are certainly (a) good quantization and (b) no-rotation induced errors. However there are also some problems associated with RLG's. One problem is "lock-in."

The term "lock-in" refers to the coupling between the CW and CCW beams. In practical RLG's, for rotational rates that are below a certain value, the CW and CCW beams lock together and produce no measurable

output. Depending upon the design, lock-in can range from 1 to 500 deg/h rotational rate. One solution to the lock-in phenomenon is to mechanically dither the case about the input axis. Dither amplitude may be about 1500 μ -rad and at a frequency of 20-40 Hz. For example, let us take a dead zone of 500 deg/h, a dither frequency of 30 Hz, and an amplitude of 1500 μ -rad.

$$500 \text{ deg/h} = 0.0024 \text{ rad/s}$$

$$f_d = 30 \text{ Hz} \quad T_d = 0.0333 \text{ s}$$

$$\omega_{nd} = 188.49 \text{ rad/s} \quad \theta_{\max} = \omega_{nd} \theta = (188.49)(1500 \times 10^{-6}) = 0.2827$$

$$\text{Error} = \frac{\text{time in dead zone}}{\text{total period of dither}} = \frac{(4)(4.54 \times 10^{-5})}{0.0333} = 0.00546 = \frac{1}{2}\%$$

thus dither removes all but one-half of one percent of the potential 500 deg/h drift rate.

Another response to lock-in is to use a Faraday cell to induce an apparent shift in frequency. This is essentially a biasing mechanism superimposing a frequency on the beams to remove them from the dead zone of lock-in. Since a strapdown IMU is expected to see rates up to 2 rad/s, the Faraday cell must bias the beam at this rate.

Other considerations in evaluation of RLG's is gyro bias and scale factor stability. Gyro bias, from the performance point of view, is something like the reverse of lock-in. Whereas lock-in was no signal output for a nonzero input, RLG bias is a nonzero output for zero input. Thus the nonrotating gyro will read an apparent input. Physically, bias is caused by circulating flow in the laser cavity that causes forward scattering due to mirror imperfections, etc.

The scale factor was evaluated in Eq. (B.8). Obviously variations in scale factor will result in a quantization error in the measurement of case rotation. Attribution of this error source can come from a variety of phenomena. In this appendix it is important only to indicate scale factor stability requirements for strapdown applications. If gyro case rotates at 10^6 deg/h and performance is set comparable to inertial grade mechanical gyros (0.05 deg/h), then the instrument must be able to measure one part in 20×10^6 . To realize this performance level the RLG must maintain scale factor stability to better than one part in 20 million. After startup transients have passed, RLG scale factor stability has been held to two parts in a million. Unfortunately startup transients which may last from 1 to 2 h are a period of unstable scale factor and hence a practical RLG may require considerable warm-up time. Further discussion of variation of scale factor during startup are given in Ref. 2.

B.2 Accelerometers

Contrary to their name, accelerometers do not measure acceleration. In free fall (at one g acceleration, say) an accelerometer will read nothing. On the other hand on a bench at zero acceleration the accelerometer will mea-

sure 1 g. An accelerometer measures the difference between inertial kinetic acceleration and gravity, (the specific force) as

$$f^i = a^i - G^i \quad (\text{B.9})$$

where f^i is the specific force (contact force per unit mass) and indicated as the difference between inertial acceleration and gravitational (field) acceleration. In free fall $a^i = G^i$ so $f^i = 0$; on a test bench, on the other hand, $a^i = 0$, so $f^i = -G^i$.

Conceptually the simplest accelerometer is a spring/mass/dashpot which will indicate specific force by an off-null measurement. A rebalance-to-null instrument would be more accurate because it would be free of the spring constant but would also be more complicated. Inertial grade accelerometers do use balance-to-null, but the device itself resembles a pendulum more than the spring/mass instrument. A pendulous accelerometer is shown in Fig. B.4 with x -axis input, y -axis output, and z -axis pendulous. An offset center of mass causes rotation of the pendulum about the y -axis. A rebalance torque is generated bringing the pendulum back to null, after a small displacement angle is detected. Clearly, a component of specific force in the positive X_{IRA} direction will cause a lagging motion in the pendulum.

The null direction is Z_{PRA} or the PRA (pendulum reference axis). The X_{IRA} axis is the null input axis or the IRA (input reference axis). The off-null angle is A_g . A tight feedback loop keeps the off-null angle A_g small. The purpose in keeping A_g small is to minimize the cross coupling error which occurs because the IA (input axis) rotates relative to the IRA through the angle A_g . If the angle A_g is small but nonzero, a component of specific force along the PRA or Z_{PRA} axis will also cause rotation about the IA according to the following small angle approximation:

$$\Delta f_{IA} \cong f_{IRA} A_g \quad (\text{B.10})$$

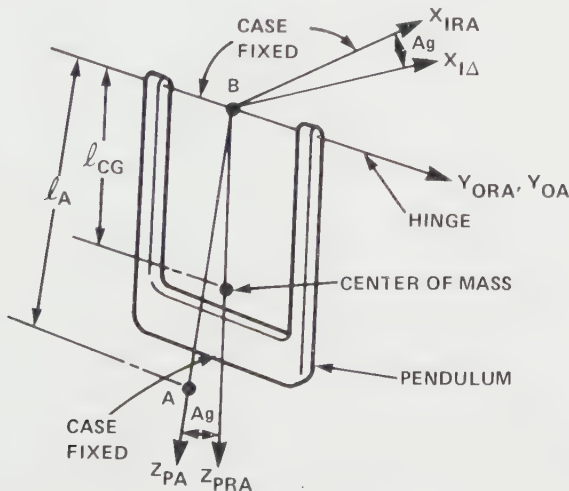


Fig. B.4 Pendulous accelerometer.

In addition to cross coupling errors, accelerometers also have "finite separation" effects. When the strapdown IMU cluster rotates, all instruments are subjected to the same rotation rate. When the cluster then linearly accelerates, not all instruments have the same linear acceleration hence the term "finite separation effects." We can develop equations to get at the physics of this error source.

From Fig. (B.5) we have

$$\mathbf{r}_A = \mathbf{r}_R + \mathbf{r}_{RA}$$

where $\mathbf{r}_R$ is the vector from the arbitrary inertial frame to the center of the cluster, $\mathbf{r}_{RA}$ the vector from the center of the cluster to an accelerometer, and $\mathbf{r}_A$ the vector from the inertial frame to the accelerometer. Differentiating the above expression gives

$$\begin{aligned} \left. \frac{d\mathbf{r}_A}{dt} \right|_I &= \left. \frac{d\mathbf{r}_R}{dt} \right|_I + \left. \frac{d\mathbf{r}_{RA}}{dt} \right|_I = \left. \frac{d\mathbf{r}_R}{dt} \right|_I + \left. \frac{d\mathbf{r}_{RA}}{dt} \right|_M + \boldsymbol{\omega}_{IB} \times \mathbf{r}_{RA} \\ \left. \frac{d\mathbf{r}_A}{dt} \right|_I &= \left. \frac{d\mathbf{r}_R}{dt} \right|_I + \boldsymbol{\omega}_{IB} \times \mathbf{r}_{RA} \end{aligned} \quad (\text{B.11})$$

where the third term, $\left. \frac{d\mathbf{r}_{RA}}{dt} \right|_M = 0$, because we are restricting attention to a rigid body. The acceleration follows from differentiating Eq. (B.11):

$$\left. \frac{d^2\mathbf{r}_A}{dt^2} \right|_I = \left. \frac{d^2\mathbf{r}_R}{dt^2} \right|_I + \frac{d\boldsymbol{\omega}_{IB}}{dt} \times \mathbf{r}_{RA} + \boldsymbol{\omega}_{IB} \times (\boldsymbol{\omega}_{IB} \times \mathbf{r}_{RA}) \quad (\text{B.12})$$

The second and third terms on the right in Eq. (B.13) are the source of "finite separation effects." This error source may be separated further to give:

$$\dot{\boldsymbol{\omega}}_{IB} \times \mathbf{r}_{RA} = \text{tangential acceleration}$$

$$\boldsymbol{\omega}_{IB} \times (\boldsymbol{\omega}_{IB} \times \mathbf{r}_{RA}) = \text{centripetal acceleration}$$

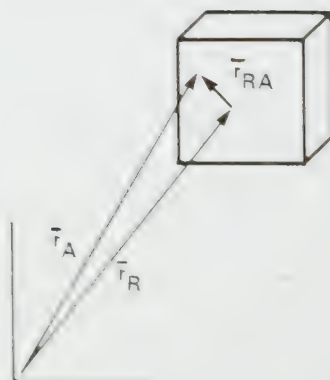


Fig. B.5 Accelerometer separation.

We can estimate both effects using the following typical values

$$\omega_{IB} = 1 \text{ rad/s} = \omega \quad r_{RA} = 5 \text{ cm}$$

For the tangential acceleration we can find its integral to give a velocity error as

$$\begin{aligned} \text{tangential } \int_0^T a dt &= \int_0^t \dot{\omega} r_{RA} dT = r_{RA} \omega \\ &= (5)(1) = 5 \text{ cm/s} = 0.16 \text{ ft/s} \end{aligned}$$

The centripetal error is

$$\omega^2 r = (1)(1)(5) = 5 \text{ cm/s}^2 = 5.1 \times 10^{-3} g \quad (\text{B.13})$$

A discrepancy of $5.1 \times 10^{-3} g$ is significant by inertial navigation standards for re-entry vehicles. For example, if we assume a flight time of 1500 s, the displacement error of the RV at impact could be something like

$$\begin{aligned} S &= \frac{1}{2}(\Delta a)gt^2 \\ &= \frac{1}{2}(5.1 \times 10^{-3})(9.81)(1500)^2 = 56 \text{ km} \end{aligned}$$

Of course such an error as the finite separation effect is easily compensated.

There are various other error sources associated with the pendulous accelerometer. An error model allows us to estimate the uncertainty δf_I associated with making a measurement of specific force:

$$\begin{aligned} \delta f_I &= B + K_{SF}f_I + K_{DF}f_I + K_{2I}f_I^2 + K_{3I}f_I^3 + K_{IP}f_I f_P \\ &\quad + K_{I0}f_I f_0 + \delta_{IP}f_P - \delta_{I0}f_0 \end{aligned} \quad (\text{B.14})$$

The terms may be defined below:

B = bias

K_{SF} = scale factor error

K_{DF} = scale factor dissymmetry

K_{2I} = quadratic scale factor

K_{3I} = cubic scale factor

K_{IP} = I - P coupling

K_{I0} = I -0 coupling

δ_{IP} = pendulous-to-input axes misalignment

δ_{I0} = output-to-input axes misalignment

In the ideal accelerometer the input specific force would be multiplied by a constant (scale factor) to provide an electrical signal equivalent to the input. This scale factor would not change with the specific force input level nor would it be influenced by environmental effects such as temperature.

Since a realizable accelerometer can only approximate this ideal, some or all of the coefficients in Eq. (B.14) are nonzero. For example even if a nonlinear scale factor is calibrated, there is some uncertainty in this calibration.

The terms K_{SP} , K_{PS} , and K_{SS} represent this uncertainty in the scale factor linear nonlinear calibration. To some degree "cross-talk" of components of the specific force vector normal to the input axis, i.e., along the pendulous and output axes, can largely be removed through careful design and ultimately by calibration. Uncertainties in such calibrations will remain, however; they are represented by the K_{IP} and K_{IO} coefficient. The δ_{IP} and δ_{IO} terms represent uncertainties in accelerometer alignment. The error coefficients of Eq. (B.14) also account for "dormancy effects" or the effect of component aging on accelerometer performance.

References

¹Savage, P. G., "Strapdown Sensors," *Strapdown Inertial Systems*, Agard Lecture Series LS-95, May 1978.

²Coccoli, J. D., "An Overview of Laser Gyros," Charles Stark Draper Laboratory, Inc., Rep. P-684, Aug. 1978.

APPENDIX C. CALCULATION OF THE REAL ROOTS OF A POLYNOMIAL

This appendix describes a computer program for the calculation of the real roots of a polynomial (see Table C.1). In this program the coefficients of the polynomial are entered in the DATA statement of line 200 as

$$y = a_0x + a_1x + a_2x^2 + \cdots + a_nx^n \quad (\text{C.1})$$

The domain extent is entered in the INPUT Statement of line 250. The terms $X1$, $X2$, and S are defined in the REM statement. A sample DATA statement, INPUT statement, and output are given after the program listing.

The preceding program computes the zero crossings or roots of some polynomial of interest. The most rudimentary program for computing zero crossings involves the active participation of the operator. The operator merely guesses a possible root, inserts this guess into the program, and the program calculates the value of the polynomial. The operator then checks the value of the sign. When successive guesses indicate sign changes in the polynomial then the domain of the polynomial over which at least one zero crossing exists has been discovered. Further iterations within this interval will uncover this zero crossing to within some arbitrary uncertainty. To remove the operator from the decision-making process is desirable, particularly since it makes the program available as a subroutine in other programs where root solving must take place from time to time. An example in this book is the solution of the eigenvalue and eigenvector problem associated with the inertia tensor.

In the preceding program the function is defined in line 210. The degree of the polynomial has been dimensioned at 5 in line 230, although this is an arbitrary limitation. The coefficients of the polynomial are entered in lines 190-200. Since the sign change of the polynomial was central to the operator's decision making, the derivative of the polynomial will be used to make the first guess at the root after a sign change has been detected. The derivative of the polynomial is defined in line 220. We shall use the index J to count zero crossings (line 240). Next we select a range of values of the independent variable. The coarse step through the range is S . If the program were to be used as a subroutine, then the INPUT statement would be omitted and line 260 would be the first line of the program. Line 290 will throw out any value of the function for which the sign does not change over

Table C.1 Root Extraction Program

```

00100 REM THIS PROGRAM CALCULATES ALL THE REAL ROOTS OF A
00110 REM POLYNOMIAL. AT PRESENT THE DEGREE IS SET AT 6, BUT THIS
00120 REM CAN EASILY BE INCREASED IN THE READ/DATA STATEMENTS.
00130 REM DOMAIN LIMITS ARE SET BY X1(LEFT) AND X2 (RIGHT).
00140 REM THE DOMAIN IS COVERED IN INCREMENTS OF LENGTH S.
00150 REM DOMAIN REGIONS OF SIGN CHANGES IN FUNCTION ARE DETECTED.
00160 REM FINE EXAMINATIONS OF THESE REGIONS ARE EXAMINED USING
00170 REM THE DERIVATIVE AS  $F(X) + (DF/DX)U = 0$ , TO FIND THE
00180 REM NEW INCREMENT, D.
00190 READ A0,A1,A2,A3,A4,A5,A6
00200 DATA -6,11,-6,1,0,0,0
00210 DEF FNY(X) = A0+A1*X+A2*X2+A3*X3+A4*X4+A5*X5+A6*X6
00220 DEF FND(X) = A1+2*A2*X+3*A3*X2+4*A4*X3+5*A5*X4+6*A6*X5
00230 DIM Z(5)
00240 J = 1
00250 INPUT X1,X2,S
00260 FOR X = X1 TO X2 STEP S
00270 Y1 = FNY(X)
00280 Y2 = FNY(X+S)
00290 IF SGN(Y1) = SGN(Y2) THEN 00400
00300 X0 = X
00310 Y1 = FNY(X0)
00320 IF ABS(Y1)<1E-3 THEN 00380
00330 X1 = FND(X0)
00340 IF X1 = 0 THEN 00400
00350 D = (-1)*Y1/X1
00360 X0 = X0+D
00370 GOTO 00310
00380 Z(J) = X0
00390 J = J+1
00400 NEXT X
00410 REM IF THE ABOVE PROGRAM WERE USED AS A PORTION OR SUBROUTINE IN
00420 REM ANOTHER PROGRAM THE FOLLOWING PRINT STATEMENT WOULD NOT
00430 REM NECESSARILY APPEAR SEQUENTIALLY. IT IS INCLUDED HERE
00440 REM ONLY FOR CHECKOUT PURPOSES
00450 FOR I = 1 TO J-1
00460 PRINT Z(I)
00470 NEXT I
00480 END

```

the interval $(X, X+S)$. Thus there will be no zero crossing (root of the polynomial) over this interval; the program only gets to line 300 if the function changes sign.

Starting at line 300 we narrow in on the zero crossing in an iterative manner. We first let $X0 = X$ (line 300) and then use the first (linear) term in a Taylor series as an expansion about $X0$. Then, in line 350 we calculate the displacement D required to reveal the zero crossing if all the other higher order terms in the Taylor series are neglected. We shall more than likely miss on this first attempt, but we shall try anyway. We add D to $X0$ in line 360 and return to 310 for a check to see if we have a zero, according to some criteria of closeness to the zero crossing (line 320). We go through the routine again and again until the criterion is met. We then record the zero crossing, index J , and move on through the $X1 - X2$ interval by increment S . We could also use the above program to calculate maxima and minima. All we would have to do would be to replace FNY in line 210 with FND and then replace FND in line 220 with the second derivative. Further extension of the method can be made to include points of inflection. The included

program is a modification of one in W. R. Bennett's excellent book on using the computer to solve problems in engineering and science.¹ Bennett gives some suggestions for permitting the program to include inflexion points as well in the extrema search.

Reference

¹Bennett, W. R. Jr., *Scientific and Engineering Problem Solving with the Computer*, Prentice-Hall Series in Automatic Computation, Prentice-Hall, Englewood Cliffs, N.J., 1976.

APPENDIX D. QUATERNIONS

In Chapter VIII we developed the direction cosine matrix (DCM) as a method of relating the components of a vector in one frame with the components of the same vector in a second frame. Euler angles were introduced somewhat earlier and it was then shown that the elements in the DCM could be expressed in terms of the three Euler angles. The DCM has nine elements and if each element is to be considered an entity unto itself, then we discovered six redundancy relationships. These redundancy relationships turned out to be six algebraic equations among the various elements. The form of these equations followed as a consequence of the orthogonality of the two frames. Of course we also showed that the nine elements could be expressed as sums and products of trigonometric functions of the three Euler angles. Since the Euler angles are three in number there are no redundancy relationships, but there are the often computationally demanding steps of providing the needed transcendental functions [see Eqs. (8.5) and (8.6)]. If the DCM elements are to be computed or updated directly we can avoid transcendental functions, but we shall require the solution of some combination of algebraic and differential equations, totalling nine. Assuring continual orthogonality in the frame which we hold in computer memory is difficult. While the requirement for orthogonality can be stated simply enough, it is difficult to determine just which elements should be modified and by how much to maintain orthogonality.

In this appendix we shall introduce a new entity: the quaternion, or more accurately perhaps, the four parameters of Euler. The word "quaternion" is justified since these four parameters obey the rules of a rather obscure algebra developed by W. R. Hamilton. The attractiveness of the quaternion is that the quaterion has only four parameters in contrast to the nine elements in the DCM. The advantage of quaternions vis-à-vis Euler angles is that quaternions do not require transcendental functions in their update differential equations.

Central to the development of the quaternions is a theorem of Euler which may be stated as follows: Any physical rotation or three-axis frame rotation can be expressed as a rotation through some finite angle about some fixed axis. We see the four parameters emerging: an angle of rotation about an axis and the three directional cosines or direction angles of the axis. As it turns out the four Euler parameters or quaternions are trigonometric functions of these four angular entities: three direction angles and one rotation angle.

First let us prove the Euler theorem of angular rotation. Consider $\mathbf{C}_a^b$ to be an orthogonal transformation matrix relating components in frame a to components in frame b . If Euler's theorem is true then there must be some vector $\mathbf{r}$ whose components are the same both before rotation and after rotation, if frame a is to be rotated into frame b . If the physics of a problem is such that frames a and b must remain distinct, then there is still an axis (according to Euler's theorem) that in concept would carry a into b .

Regardless of how the geometry is conceived, if the vector $\mathbf{r}$ lies along this axis of rotation, it must have identical components in both frames a and b , i.e.,

$$\mathbf{r}^b = \mathbf{C}_a^b \mathbf{r}^a = \mathbf{r}^a \quad (\text{D.1})$$

or

$$\left[\mathbf{C}_a^b - [1] \right] \mathbf{r}^a = 0 \quad (\text{D.2})$$

where $[1]$ is the identity matrix. Equations (D.2) require that the following determinant be zero,

$$\begin{vmatrix} C_{11} - 1 & C_{12} & C_{13} \\ C_{21} & C_{22} - 1 & C_{23} \\ C_{31} & C_{32} & C_{33} - 1 \end{vmatrix} = 0 \quad (\text{D.3})$$

Euler's theorem will be demonstrated if the statement of Eq. (D.3) can be satisfied. Expansion of Eq. (D.3) gives

$$\begin{aligned} & [(C_{11}C_{22}C_{33} + C_{12}C_{23}C_{31}) + (C_{21}C_{32}C_{13} - C_{31}C_{13}C_{22}) \\ & - (C_{21}C_{12}C_{33} - C_{32}C_{23}C_{11}) - 1] + [(C_{11} - C_{22}C_{33} + C_{23}C_{32}) \\ & + (C_{22} - C_{11}C_{33} + C_{13}C_{31}) + (C_{33} - C_{11}C_{22} + C_{21}C_{12})] = 0 \end{aligned} \quad (\text{D.4})$$

The terms in the second brackets vanish by using the redundancy relationships of Eqs. (8.45). To illustrate that the expression in the first bracket also vanishes first consider the statement of Eq. (D.1),

$$\mathbf{r}^b = \mathbf{C}_a^b \mathbf{r}^a \quad \mathbf{r}^a = (\mathbf{C}_a^b)^T \mathbf{r}^b$$

or

$$\mathbf{r}^a = (\mathbf{C}_a^b)^T \mathbf{C}_a^b \mathbf{r}^a$$

which requires that

$$(\mathbf{C}_a^b)^T (\mathbf{C}_a^b) = [1]$$

Since the determinant of a matrix product is the product of the determinates of each of the matrices, we have:

$$\text{Det}[(\mathbf{C}_a^b)^T \mathbf{C}_a^b] = 1$$

and since $\text{Det}(\mathbf{C}_a^b)^T = \text{Det}(\mathbf{C}_a^b)$ (interchanging rows and columns does not effect a determinant), we have:

$$[\text{Det} \mathbf{C}_a^b]^2 = 1 \quad (\text{D.5})$$

Expansion of the determinant in Eq. (D.5) will indicate that the terms in the first bracket of Eq. (D.4) are also zero.

Now with Euler's theorem as a foundation we can return to Eq. (D.1). Assume that we wish to rotate from frame a to frame b by identifying an axis of rotation by the directional angles α , β , and γ and then rotating through the angle δ about this axis (see Fig. D.1). Notice that the axis of rotation, identified by the vector $\mathbf{r}$, has the same directional angles in both the a frame and b frame. Our goal then is to express the elements of $\mathbf{C}_a^b$ in terms of α , β , γ , and δ .

To carry out this work we shall require an auxiliary system identified as r . We shall require that the X^r axis be along the axis of rotation vector $\mathbf{r}$, and that the Y^r axis lie in the $X^a Y^a$ plane. The rotation, through the angle δ , will be about the X^r axis. First we must go from the a frame to the r frame. The X^r and Y^r axes have been defined and we have designated α , β , and γ as the directional angles of X^r (axis of rotation). Thus we may write:

$$\mathbf{C}_a^r = \begin{bmatrix} \cos\alpha & \cos\beta & \cos\gamma \\ C_{21} & C_{22} & 0 \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \quad (\text{D.6})$$

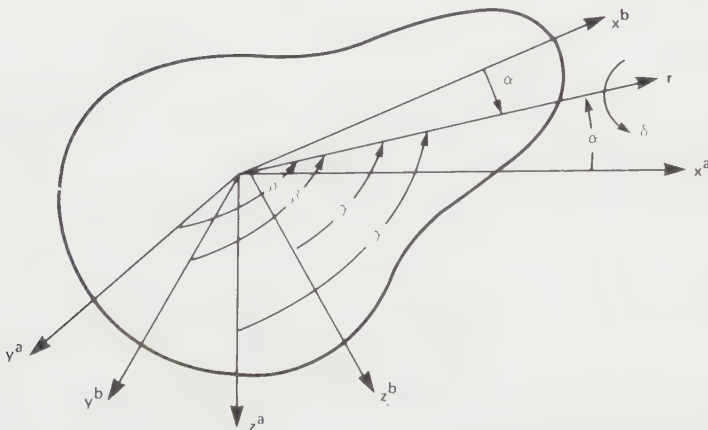


Fig. D.1 Euler rotation axis definition.

The element C_{23} must of course be zero as the angle between the Y^r and Z^a axes must be 90 deg. Using the orthogonality relationships of Eqs. (8.53) completes the elements of the matrix in Eq. (D.6) as

$$\mathbf{C}_a^r = \begin{bmatrix} +\cos\alpha & +\cos\beta & +\cos\gamma \\ \mp \cos\beta\csc\gamma & \pm \cos\alpha\csc\gamma & +0 \\ \mp \cos\alpha\cot\gamma & \mp \cos\beta\cot\gamma & \pm \sin\gamma \end{bmatrix} \quad (\text{D.7})$$

The ambiguity in sign may be omitted as we shall require that if $\alpha = 0$, i.e., X^r, X^a are co-linear, then $\beta = 0$ and $\gamma = 0$; $\mathbf{C}_a^r$ then becomes the identity matrix [1]. The result is a single sign associated with each element:

$$\mathbf{C}_a^r = \begin{bmatrix} \cos\alpha & \cos\beta & \cot\gamma \\ -\cos\beta\csc\gamma & \cos\alpha\csc\gamma & 0 \\ -\cos\alpha\cot\gamma & -\cos\beta\cot\gamma & \sin\gamma \end{bmatrix} \quad (\text{D.8})$$

We must now rotate about the X^r axis through the angle δ . We shall call this rotation matrix $\mathbf{C}_r^{r'}$ and it is easily given as

$$\mathbf{C}_r^{r'} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\delta & \sin\delta \\ 0 & -\sin\delta & \cos\delta \end{bmatrix} \quad (\text{D.9})$$

The third and last rotation must be the inverse of the first (in order that $\mathbf{r}$ has the same directional angle in both frames). Thus, the DCM from a to b can be written as the product:

$$\mathbf{C}_a^b = (\mathbf{C}_a^r)^{-1} \mathbf{C}_r^{r'} \mathbf{C}_r^a \quad (\text{D.10})$$

Inserting values from Eqs. (D.8) and (D.9) gives

$$\mathbf{C}_a^b = \begin{bmatrix} 1 - 2\sin^2 \frac{\delta}{2} \sin^2 \alpha & 2 \left[\sin^2 \frac{\delta}{2} \cos\alpha \cos\beta + \sin \frac{\delta}{2} \cos\gamma \cos \frac{\delta}{2} \right] & 2 \left[\cos\alpha \cos\gamma \sin^2 \frac{\delta}{2} - \sin \frac{\delta}{2} \cos \frac{\delta}{2} \cos\beta \right] \\ 2 \left[\sin^2 \frac{\delta}{2} \cos\alpha \cos\beta - \sin \frac{\delta}{2} \cos \frac{\delta}{2} \cos\gamma \right] & 1 - 2\sin^2 \frac{\delta}{2} \sin^2 \beta & 2 \left[\sin^2 \frac{\delta}{2} \cos\beta \cos\gamma + \sin \frac{\delta}{2} \cos \frac{\delta}{2} \cos\alpha \right] \\ 2 \left[\cos\alpha \cos\gamma \sin^2 \frac{\delta}{2} + \sin \frac{\delta}{2} \cos \frac{\delta}{2} \cos\beta \right] & 2 \left[\sin^2 \frac{\delta}{2} \cos\beta \cos\gamma - \sin \frac{\delta}{2} \cos \frac{\delta}{2} \cos\alpha \right] & 1 - 2\sin^2 \frac{\delta}{2} \sin^2 \gamma \end{bmatrix} \quad (\text{D.11})$$

The above matrix has nine elements as we might have expected. However, the elements are fairly complicated transcendental functions of the four angles α , β , γ , and δ . The first three angles, α , β , and γ give the directional angles of the axis of rotation, and the fourth angle δ gives the angle of rotation about this axis. As noted earlier, the geometry of the process in going from frame a to frame b is based upon Euler's theorem. Unfortunately if we persist in retaining the four angles, α , β , γ , and δ as the descriptive parameters of frame transformation we appear to be at a disadvantage in a comparison with Euler angles. In the case of a DCM based upon Euler angles we have only three angles ψ , θ , and ϕ and a DCM whose elements contain trigonometric functions of no greater complexity [compare Eq. (8.6) with Eq. (D.11)]. However, if we make the following substitutions we can eliminate the trigonometric functions entirely from Eq. (D.11). Let

$$a = \cos\alpha \sin\frac{\delta}{2} \quad b = \cos\beta \sin\frac{\delta}{2} \quad c = \cos\gamma \sin\frac{\delta}{2} \quad d = \cos\frac{\delta}{2} \quad (\text{D.12})$$

Using the relationships of Eq. (D.12) we may express the DCM C_a^b as expressed in Eq. (D.11) as

$$C_a^b = \begin{bmatrix} a^2 + d^2 - b^2 - c^2 & 2(ab + cd) & 2(ac - bd) \\ 2(ab - cd) & b^2 + d^2 - a^2 - c^2 & 2(bc + ad) \\ 2(ac + bd) & 2(bc - ad) & c^2 + d^2 - a^2 - b^2 \end{bmatrix} \quad (\text{D.13})$$

The four parameters a , b , c , and d are called the Euler parameters. Since only three parameters are necessary to uniquely specify the rotation of one frame relative to another, we must expect one redundancy relationship. We first note that the directional angles, α , β , and γ of the axis of rotation $\mathbf{r}$ must satisfy the familiar orthonormal relationship:

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

or

$$a^2 + b^2 + c^2 = \sin^2(\delta/2)$$

Now if we add $\cos^2(\delta/2)$ to both sides we get the orthogonality condition of the Euler four parameter method (subsequently to be identified as "quaternions").

$$a^2 + b^2 + c^2 + d^2 = 1 \quad (\text{D.14})$$

We also note from Eq. (D.12) that in addition to satisfying the above relationship, a , b , c , and d must individually remain within the range ± 1 .

If the four quantities α , β , γ , and δ are given, it is possible to compute the

four Euler parameters a , b , c , and d ; however; this procedure is not of much practical value since we are trying to replace the four angles with the four Euler parameters. Of more interest is the computation of a , b , c , and d when we have available the DCM C_a^b .

It is possible to determine the four Euler parameters from the nine elements of the DCM C_a^b . However, some difficulties arise, centered on the fact that the α , β , γ , and δ angles cannot be determined uniquely from the elements of the DCM. The converse however, is true, i.e., the four elements uniquely determine the elements of the DCM. For example, an identical axis transformation will be realized by four distinct sets of rotation about the $\mathbf{r}$ axis: (a) rotation about the positive $\mathbf{r}$ axis through a positive angle δ ; (b) rotation about the negative $\mathbf{r}$ axis through a (negative) angle $-\delta$; (c) rotation about the $\mathbf{r}$ axis through the angle $-(2\pi - \delta)$; and (d) rotation about the negative $\mathbf{r}$ axis through the angle $(2\pi - \delta)$. The first two rotations are described by identical values of a , b , c , and d , while the final two rotations, although retaining the same magnitudes in a , b , c , and d reverse their signs.

In any event it may be shown that the Euler parameters are related to the elements of the DCM C_a^b by the following two sets of relationships. The first of these are:

$$\begin{aligned} 4a^2 &= 1 + C_{11} - C_{22} - C_{33} \\ 4b^2 &= 1 + C_{22} - C_{11} - C_{33} \\ 4c^2 &= 1 + C_{33} - C_{11} - C_{22} \\ 4d^2 &= 1 + C_{11} - C_{22} - C_{33} \end{aligned} \quad (D.15a)$$

and the second are:

$$4ad = C_{23} - C_{32} \quad 4bd = C_{31} - C_{13} \quad 4cd = C_{12} - C_{21} \quad (D.15b)$$

Equations (D.15a) will determine the magnitude of the parameters, although there remains an ambiguity regarding sign. If d is assumed always to be positive, then Eqs. (D.15b) may be used to determine the sign of a , b , and c .

Up to now the development could be termed the Euler four parameter method. The justification will now be made for the term "quaternion." In 1843, W. R. Hamilton developed an algebra for an entity he termed a "quaternion." The quaternion has four parts q_0 , q_1 , q_2 , and q_3 which are combined as a scalar and vector as shown,

$$Q = q_0 + \mathbf{i}q_1 + \mathbf{j}q_2 + \mathbf{k}q_3 \quad (D.16)$$

An analogy comes to mind in the description of the axis transformation just discussed since we used a vector to locate the axis of rotation $\mathbf{r}$ and a scalar to describe the rotation about that axis. Without pressing this analogy further, we accept the following rules of quaternion algebra:

$$\begin{aligned}
\mathbf{i}^2 = \mathbf{ii} &= -1 & \mathbf{i} \cdot \mathbf{j} &= -\mathbf{j} \cdot \mathbf{i} = \mathbf{k} \\
\mathbf{j}^2 = \mathbf{jj} &= -1 & \mathbf{j} \cdot \mathbf{k} &= -\mathbf{k} \cdot \mathbf{j} = \mathbf{i} \\
\mathbf{k}^2 = \mathbf{kk} &= -1 & \mathbf{k} \cdot \mathbf{i} &= -\mathbf{i} \cdot \mathbf{k} = \mathbf{j}
\end{aligned} \tag{D.17}$$

These relationships are reminiscent of the rules of vector analysis where the inner product of unit vectors would be +1 and the cross product, $\mathbf{i} \times \mathbf{j} = \mathbf{k}$. Continuing, we define the conjugate quaternion, Q^* as

$$Q^* = q_0 - \mathbf{i}q_1 - \mathbf{j}q_2 - \mathbf{k}q_3 \tag{D.18}$$

It follows then that

$$\text{norm} Q = QQ^* = q_0^2 + q_1^2 + q_2^2 + q_3^2 \tag{D.19a}$$

If the norm is unity

$$QQ^* = 1 \tag{D.19b}$$

then the unimodular quaternion has the special name of “versor.” If we define quaternions P and R in a manner identical to that of Q [Eq. (D.16)] and take m and n as unspecified constants we can easily demonstrate the following algebraic identities.

$$Q + P = P + Q$$

$$Q + (P + R) = (Q + P) + R$$

$$m(P + Q) = mP + mQ$$

$$(m + n)Q = mQ + nQ$$

$$\begin{aligned}
QP &= (Q_0P_0 - P_1Q_1 - P_2Q_2 - P_3Q_3) + \mathbf{i}(P_1Q_0 + P_0Q_1 + P_2Q_3 - P_3Q_2) \\
&\quad + \mathbf{j}(P_2Q_0 + P_0Q_2 + P_3Q_1 - P_1Q_3) + \mathbf{k}(P_3Q_0 + P_0Q_3 + P_1Q_2 - P_2Q_1)
\end{aligned}$$

Apparently the product of two quaternions is also a quaternion. Note that the quaternion product is not commutative.

$$QP \neq PQ$$

In Eq. (D.18) we defined the quaternion conjugate; since the quaternion product is also a quaternion, we can easily demonstrate validity of conjugate of a quaternion product in terms of its components as

$$(QP)^* = P^*Q^* \tag{D.20}$$

Let vector V be written in terms of its component in a frame having unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

$$\mathbf{V} = X\mathbf{i} + Y\mathbf{j} + Z\mathbf{k}$$

We can show that the product $Q^* \mathbf{V} Q$ is also a vector $\mathbf{V}$ as

$$\begin{aligned} \mathbf{V}' &= Q^* \mathbf{V} Q \\ &= \mathbf{i}[X(q_0^2 + q_1^2 - q_2^2 - q_3^2) + Y(2q_3q_0 + 2q_1q_2) + Z(2q_1q_3 - 2q_0q_2)] \\ &\quad + \mathbf{j}[X(2q_1q_2 - 2q_3q_0) + Y(q_0^2 - q_1^2 - q_2^2 - q_3^2) + Z(2q_1q_0 + 2q_3q_1)] \\ &\quad + \mathbf{k}[X(2q_0q_2 + 2q_1q_3) + Y(2q_2q_3 - 2q_0q_1) + Z(q_0^2 - q_1^2 - q_2^2 + q_3^2)] \quad (\text{D.21}) \end{aligned}$$

Obviously if we let

$$q_0 = d \quad q_1 = a \quad q_2 = b \quad q_3 = c \quad (\text{D.22})$$

then the quaternion operation in Eq. (D.20) is equivalent to transforming a vector having components available in one frame to components in a second frame. The operation is identical to premultiplying the vector $\mathbf{V}$ by a DCM if the elements of the DCM are expressed in terms of the four quaternions. The DCM operation analogous to Eq. (D.20) is

$$\mathbf{V}^b = \mathbf{C}_a^b \mathbf{V}^a \quad (\text{D.23})$$

where $\mathbf{V}^a = \mathbf{V}$ and $\mathbf{C}_a^b$ has elements expressed in terms of a, b, c , and d as in Eq. (D.13). Successive rotations are easily handled. Let

$$\mathbf{V}' = Q_1^* \mathbf{V} Q_1$$

and

$$\mathbf{V}'' = Q_2^* \mathbf{V}' Q_2$$

or

$$\mathbf{V}'' = Q_2^* Q_1^* \mathbf{V} Q_1 Q_2$$

If we define the quaternion Q_3 as $Q_3 = Q_1 Q_2$, it is easy to show, using Eq. (D.20) that

$$\mathbf{V}'' = Q_3^* \mathbf{V} Q_3 \quad (\text{D.24})$$

indicating that two successive rotations may be conceived as a single rotation. The product of the quaternions associated with each rotation produces

the quaternion associated with the single rotation. The analogous operation using DCM's is if

$$\mathbf{V} = \mathbf{V}^a \quad , \quad \mathbf{V}' = \mathbf{V}^b \quad \mathbf{V}'' = \mathbf{V}^c$$

then

$$\mathbf{V}^c = \mathbf{C}_b^c \mathbf{C}_a^b \mathbf{V}^a = \mathbf{C}_a^c \mathbf{V}^a \quad (\text{D.25})$$

Equations (D.24) and (D.25) are alternate ways of expressing vector transformations, the former using the quaternion product the latter the matrix product.

Before considering the question of quaternion updating, we should note the equivalence among the elements of the DCM, the Euler angles, and quaternions. The equivalence between the elements of the DCM and the Euler angles was discussed in an example in Sec. (8.3). The equivalent relationships between the elements of the DCM and the four quaternions a , b , c , and d has been given in Eqs. (D.15). Equivalent relationships may also be written between the Euler angles ϕ , θ , ψ , and the four quaternions.

It is fairly straightforward to go from quaternions to Euler angles. This is simply done if we regard the matrix D in Eq. (8.5) as the DCM $\mathbf{C}_i^b$, i.e., the DCM which transforms vector components in an inertial or i frame to components in a body or b frame. We can then make a term by term comparison between Eq. (8.5) and the matrix given in Eq. (D.13), if we regard the matrix $\mathbf{C}_a^b$ as $\mathbf{C}_i^b$. The pitch angle θ is easy to write as

$$-\sin\theta = C_{13} = 2(ac - bd)$$

or

$$\theta = \sin^{-1}[-2(ac - bd)] \quad (\text{D.26a})$$

Again comparing the two matrices we see that,

$$\tan\psi = C_{12}/C_{11} = 2(ab + cd)/(a^2 + d^2 - (b^2 + c^2))$$

or

$$\psi = \tan^{-1}2(ab + cd)/[2(a^2 + d^2) - 1] \quad (\text{D.26b})$$

where we have made use of the orthogonality relationship of Eq. (D.14). In like manner we have

$$\tan\phi = C_{23}/C_{33} = 2(bc + ad)/[c^2 + d^2 - (a^2 + b^2)]$$

or

$$\phi = \tan^{-1}2(bc + ad)/[2(c^2 + d^2) - 1] \quad (\text{D.26c})$$

It is a simple matter to obtain the quaternion in terms of the Euler angles. First we observe that the trace of the DCM C_i^b [Eqs. (8.5) and (D.13)] is

$$4d^2 = \cos\theta\cos\phi + \sin\psi\sin\theta\sin\phi + \cos\psi\cos\phi + \cos\phi\cos\theta$$

or

$$d^2 = \frac{1}{4}\text{trace}(D)$$

From Eqs. (D.15) we may write the remaining quaternions as

$$a^2 = \frac{1}{2}[1 + \cos\psi\cos\theta] - d^2 \quad (\text{D.27a})$$

$$b^2 = \frac{1}{2}[1 + \cos\psi\cos\theta + \sin\psi\sin\theta\sin\phi] - d^2 \quad (\text{D.27b})$$

$$c^2 = \frac{1}{2}[1 + \cos\phi\cos\theta] - d^2 \quad (\text{D.27c})$$

The one remaining topic we shall consider in this subject of quaternions is the problem of updating.

Directional cosine matrix updating was discussed in Sec. 8.4. Equations (8.90) indicated a simple method for updating the DCM in terms of the elements C_{11} and C_{23} taken as representative examples. Equations (8.65) indicate nine first-order differential equations for the update of the DCM elements, although the six redundancy relationships help to reduce this number, replacing differential equations with algebraic equations. Nevertheless computational errors cause the continually updated DCM to drift from orthonormality. Unfortunately the many degrees of freedom represented by the redundancy equations do not allow these orthonormality constraints to uniquely define a solution. The quaternion has a single orthonormal constraint [see Eq. (D.14)]. We shall now outline the development of a set of update equations for the quaternions analogous to those developed for the Euler angles [Eq. (8.16)] and the direction cosine matrix [Eq. (8.68)].

From Eq. (D.15a) we have

$$8a\dot{a} = \dot{C}_{11} - \dot{C}_{22} - \dot{C}_{33}$$

From Eqs. (8.65) we may replace the derivatives of the DCM elements to give

$$8a\dot{a} = (C_{12}r - C_{13}q) + (C_{21}r - C_{23}p) + (C_{23}p - C_{31}q)$$

$$8a\dot{a} = (C_{21} + C_{12})r - (C_{13} + C_{31})q + (C_{32} - C_{23})p$$

Replacing the elements as appropriate from the matrix in Eq. (D.13), we get

$$8a\dot{a} = 4abr - 4acq + 4dap$$

or

$$\dot{a} = \frac{1}{2}[br - cq + dp] \quad (\text{D.28})$$

By similar reasoning it is possible to develop expressions for b, c, d in terms of the four quaternions and the components of the angular velocity in a body frame. If the quaternions a, b, c , and d described a transformation from the body-to-the-inertial frame we may express the equations for update of the quaternions as

$$\begin{bmatrix} \dot{a} \\ \dot{b} \\ \dot{c} \\ \dot{d} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & r & -q & p \\ -r & 0 & p & q \\ q & -p & 0 & r \\ -p & -q & -r & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \quad (\text{D.29})$$

Equations (D.29) are the matrix form of the update equations of the quaternions a, b, c , and d . These equations are the equivalent of the direction cosine update equations given earlier [Eqs. (8.68)]. We may also write Eq. (D.29) in a form more useful for simulation studies:

$$\begin{bmatrix} \dot{a} \\ \dot{b} \\ \dot{c} \\ \dot{d} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} d & -c & b \\ c & d & -a \\ -b & a & d \\ -a & -b & -c \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} \quad (\text{D.30})$$

The above matrix equation relates the time derivatives of the quaternions to the components of the angular velocity of the RV (relative to inertial space) in a body frame. The corresponding equation for the Euler angles (ψ, θ, ϕ related to p, q, r) was given in Eq. (8.12).

A. Van Bronkhurst suggests modifying Eq. (D.29) in an updating process to maintain orthonormality (Ref. 1). From Eq. (D.14) we may define a parameter k_n as

$$k_n = n(a^2 + b^2 + c^2 + d^2 - 1) \quad (\text{D.31})$$

where n is some unspecified number. If the DCM C_t^b [Eq. (D.13)] is orthogonal and the quaternions are normalized [Eq. (D.19b)], k_n will be zero. The updating process Eq. (D.29) can be maintained orthogonal if we rewrite it as

$$\begin{bmatrix} \dot{a} \\ \dot{b} \\ \dot{c} \\ \dot{d} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2-k_n & r & q & p \\ -r & 2-k_n & p & q \\ q & -p & 2-k_n & r \\ -p & -q & -r & 2-k_n \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \quad (\text{D.32})$$

In Sec. 8.6 we outlined an updating procedure for DCM's. We can now consider a simple algorithm for quaternion updating, after Van Bronkhurst (Ref. 1). First let us define the incremental angles in roll, pitch, and yaw as

$$\Delta\phi = \int_{T_n}^{T_{n+1}} p \, dt \quad \Delta\theta = \int_{T_n}^{T_{n+1}} q \, dt \quad \Delta\psi = \int_{T_n}^{T_{n+1}} r \, dt \quad (\text{D.33})$$

From Eq. (D.29) we may define the matrix $[\Delta\alpha]$ as

$$[\Delta\alpha] = \begin{bmatrix} 0 & \Delta\psi & -\Delta\theta & \Delta\phi \\ -\Delta\psi & 0 & \Delta\phi & \Delta\theta \\ \Delta\theta & -\Delta\phi & 0 & \Delta\psi \\ -\Delta\phi & -\Delta\theta & -\Delta r & 0 \end{bmatrix} \quad (\text{D.34})$$

The integral of Eq. (D.29) over the interval T_n to T_{n+1} then becomes

$$\mathbf{q}_{n+1} = \mathbf{q}_n e^{[\Delta\alpha]} \quad (\text{D.35})$$

The first three terms in the exponential expansion gives

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}_{n+1} = \left[[\mathbf{1}] + [\Delta\alpha] - \frac{1}{4}(\Delta\alpha^2)[\mathbf{1}] - \frac{1}{6}(\Delta\alpha^2)[\mathbf{1}][\Delta\alpha] + \cdots \right] \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}_n \quad (\text{D.36})$$

where $[\mathbf{1}]$ is the unit matrix and where,

$$\Delta\alpha^2 = \Delta\theta^2 + \Delta\phi^2 + \Delta\psi^2 \quad (\text{D.37})$$

There are two obvious ways of improving the updating algorithm. One is to increase the order or the number of terms in the series; the second is to decrease the integration time interval. Both will increase the computation loads and ultimately a point of diminishing returns is reached.

Reference

Van Bronkhorst, A., "Strapdown System Algorithms," Strapdown Inertial Systems, AGARD Lecture Series, May 1978, LS-95, pp. 3.1-3.22.

APPENDIX E. PROGRAM FOR COMPUTING α - β PETAL PLOTS

In Chapter X we considered the equations of motion of an RV with identical aerodynamic characteristics for excursions in angle-of-attack α and angle of sideslip β . The closed-form integrals of these equations of motion were given in Eqs. (10.47). For reference these equations are repeated below:

$$\alpha = [A_1 \cos \omega_1 t + B_1 \sin \omega_1 t] e^{\lambda_1 t} + [A_2 \cos \omega_2 t + B_2 \sin \omega_2 t] e^{\lambda_2 t} + A_3 \cos pt + B_3 \sin pt \quad (\text{E.1a})$$

$$\beta = [B_1 \cos \omega_1 t - A_1 \sin \omega_1 t] e^{\lambda_1 t} + [B_2 \cos \omega_2 t - A_2 \sin \omega_2 t] e^{\lambda_2 t} + B_3 \cos pt - A_3 \sin pt \quad (\text{E.1b})$$

In the above equations it will be noted that there are ten constants of integration A_1 , B_1 , A_2 , B_2 , A_3 , and B_3 given in Eqs. (10.48); two frequencies ω_1 (nutational) and ω_2 (precessional) given in Eqs. (10.37); two exponential damping terms λ_1 (nutational) and λ_2 (precessional) given in Eqs. (10.39). The roll rate p appears in Eqs. (F.1); however, if ω_1 and ω_2 are given, then the roll rate is not independent, but rather can be expressed in terms of ω_1 and ω_2 according to Eqs. (10.58a) as

$$p = (I/I_x)(\omega_1 + \omega_2) \quad (\text{E.2})$$

The equations for the ten constants of integration in terms of aerodynamic derivatives, mass properties, and initial conditions will not be given. The procedure here will be to simply indicate that these ten constants must be computed and then inserted as appropriate into Eq. (E.1). Programming of Eq. (E.1) will give α and β as time functions; a plotting routine will provide petal plots. Examples of such petal plots were given in Fig. (10.4).

The program and sample output are provided in Table E.1. Most of the variables and constants used in the program have been defined in the REM section. The moments of inertia are entered in lines 280-290. Atmospheric temperature and pressure ratio are entered in lines 310 and 340. Reference length D (body diameter) and overall body length L are entered in lines 320 and 330. Aerodynamic derivatives C_{m_n} , C_{m_q} , $C_{m_{p\beta}}$ are entered in lines 370-390; velocity magnitude and roll rate are entered in lines 300 and 410.

Table E.1 Listing of α - β Plotting Program

```

00100 REM I IS THE TRANSVERSE MOMENT OF INERTIA(SLUG-FT-SQ)
00110 REM I1 IS THE AXIAL MOMENT OF INERTIA (SLUG-FT-SQ)
00120 REM D IS THE MODEL DIAMETER (FT); L IS THE MODEL LENGTH (FT)
00130 REM M IS THE MACH NUMBER
00140 REM P0 IS THE NUMBER OF ATMOSPHERES (TOTAL)
00150 REM P IS THE ROLL RATE (RAD/SEC)
00160 REM T0 IS THE TOTAL TEMPERATURE (DEG-FARENHEIT)
00170 REM T2 IS THE PRINT-OUT FREQUENCY ( SECONDS)
00180 REM T3 IS THE TIME FOR START OF TRIM ANGLE OF ATTACK (SECONDS)
00190 REM T4 IS THE TIME DURATION OF TRIM (SECONDS)
00200 REM C1,C2,C3 ARE STATIC ,PITCH DAMPING, MAGNUS DERIVATIVES
00210 REM A0,B0 ARE INITIAL ANGLES OF ATTACK AND SIDESLIP (DEGREES)
00220 REM A9,B9 ARE INITIAL RATES IN A0,B0 (DEG/SEC)
00230 REM A8 IS THE TRIM ANGLE OF ATTACK (DEG).
00240 DIM A(100),B(100),T(100)
00250 FILE?1=*TAPE1#
00260 T2 = .01
00270 I1 = 6.3131E-4
00280 I = 5.55E-3
00290 T0 = 3070.4
00300 D = 5.5
00310 L = 22.39
00320 M = 14
00330 P0 = 1496.59
00340 T3 = 0
00350 T4 = 10
00360 C1 = (-1)*.312
00370 C2 = (-1)*4.0
00380 C3 = (-1)*1.2
00390 E1 = 0.5
00400 P = 60
00410 A0 = 12
00420 B0 = 0
00430 A9 = 0
00440 B9 = 0
00450 A8 = 0
00460 D = D/12
00470 S = 3.14159*D*D/4
00480 L = L/12
00490 R9 = 180/3.14159
00500 A0 = A0/R9
00510 B0 = B0/R9
00520 A9 = A9/R9
00530 B9 = B9/R9
00540 A8 = A8/R9
00550 P0 = P0*2116.8
00560 Q = 0.7*(M*2)*((1+((M*2)/5)))*((-1)*3.5))*P0
00570 V = M*SQR(1/(1+((M*2)/5)))*49.014*SQR(T0+459.6)
00580 M1 = C1*Q*S*D
00590 M2 = C2*Q*S*D*D/(2*V)
00600 M3 = C3*Q*S*D*D/(2*V)
00610 W3 = (P*I1/(2*I))
00620 W2 = SQR((W3*2)-(M1/I))
00630 T1 = W3/W2
00640 W1 = W3+W2
00650 W2 = W3-W2
00660 L1 = (M2/(2*I))*((1+T1)+(M3*T1/I1)
00670 L2 = (M2/(2*I))*((1-T1)-(M3*T1/I1)
00680 FOR I = 1 TO 71
00690 T = T2*(I-1)
00700 IF T<T3 THEN 00750
00710 IF T>= (T3+T4) THEN 00750
00720 A7 = A8
00730 GOSUB 00780
00740 GOT0 00950
00750 A7 = 0
00760 GOSUB 00780

```

(Table continued on next page)

Table E.1 (cont.) Listing of α - β Plotting Program

```

00770 GOTO 00950
00780 Z8 = L2*(W1-P)+L1*(W2-P)
00790 Z7 = W2*(P-W1)+P*(W1-P)+L1*L2
00800 Z6 = (Z8*2)+(Z7*2)
00810 A3 = (-1)*(M1*A7/I)*(W2*(P-W1)+P*(W1-P)+L1*L2)
00820 A3 = A3/Z6
00830 B3 = (-1)*(M1*A7/I)*(L2*(W1-P)+L1*(W2-P))
00840 B3 = B3/Z6
00850 Z9 = ((W1-W2)*2)+((L1-L2)*2)
00860 Y1 = A9-L2*A0-W2*B0+L2*A3-(P-W2)*B3
00870 Y2 = B9-L2*B0+W2*A0+(P-W2)*A3+L2*B3
00880 Y3 = A9-L1*A0-W1*B0+L1*A3-(P-W2)*B3
00890 Y4 = B9-L1*B0+W1*A0+(P-W1)*A3+L1*B3
00900 A1 = ((Y1*(L1-L2))-(Y2*(W1-W2)))/Z9
00910 B1 = ((Y1*(W1-W2))+(Y2*(L1-L2)))/Z9
00920 A2 = ((Y3*(L2-L1))-(Y4*(W2-W1)))/Z9
00930 B2 = ((Y3*(W2-W1))+(Y4*(L2-L1)))/Z9
00940 RETURN
00950 A = (A1*COS(W1*T)+B1*SIN(W1*T))*EXP(L1*T)
00960 A = A+(A2*COS(W2*T)+B2*SIN(W2*T))*EXP(L2*T)
00970 A = A+A3*COS(P*T)+B3*SIN(P*T)
00980 B = (B1*COS(W1*T)-A1*SIN(W1*T))*EXP(L1*T)
00990 B = B+(B2*COS(W2*T)-A2*SIN(W2*T))*EXP(L2*T)
01000 B = B+B3*COS(P*T)-A3*SIN(P*T)
01010 A = A*R9+E1*NRND(0)
01020 B = B*R9+E1*NRND(0)
01030 A(I) = A
01040 B(I) = B
01050 T(I) = T
01060 IF I>1 THEN 01130
01070 PRINT #NU FREQ#, #PREC FREQ#, #NU DAMP#, #PREC DAMP#, #TAU#
01080 PRINT # (1/SEC) #, # (1/SEC) #, # --- #, # --- #, # --- #
01090 PRINT W1, W2, L1, L2, T1
01100 PRINT
01110 PRINT #ANG ATTACK#, #ANG SIDE SLP#, #TIME#, #TRIM ANG#
01120 PRINT # (DEG) #, # (DEG) #, # (SEC) #, # (DEG) #
01130 PRINT A(I), B(I), T(I), A7*R9
01140 PRINT ?1 USING 1150, A(I), B(I), T(I)
01150 :????,???? ???? ???? ??.??
01160 NEXT I
01170 END

```

Initial conditions in angle and angular rate are entered in lines 420-460. Nutational and precessional frequencies are calculated in lines 630-640 and the corresponding exponential damping terms in lines 650-660. The six constants of integration $A_1 \dots B_3$ are computed in lines 770-920. Finally in lines 910-1010 the angles of attack and sideslip are calculated in time increments $T2$ (line 270). The two angles and time, i.e., $\alpha(t_i)$, $\beta(t_i)$, t_i are stored as column matrices in lines 1020-1040, according to the format of line 1140 on File #1 (line 250).

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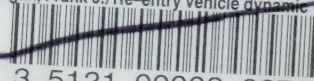
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